

```
# setwd("C:/Users/amhasan1/OneDrive - North Carolina A&T State University/Desktop/Soil & Dust/AHasan Soil & Dust Analysis")
# setwd("C:/Users/amhasan1/OneDrive - North Carolina A&T State University/Desktop/Soil & Dust/Updated Sep analysis")
setwd("C:/Users/amhasan1/OneDrive - North Carolina A&T State University/Desktop/Soil & Dust/FINAL ANALYSIS")
```

Global Error Options

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(janitor)
```

```
##
## Attaching package: 'janitor'
```

```
## The following objects are masked from 'package:stats':
##
##   chisq.test, fisher.test
```

```
library(ggplot2)
library(rlang)
library(ggmosaic)
library(tidyverse)
```

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ forcats   1.0.0   ✓ stringr   1.5.1
## ✓ lubridate 1.9.3   ✓ tibble    3.2.1
## ✓ purrr     1.0.2   ✓ tidyr     1.3.1
## ✓ readr     2.1.5
```

```
## — Conflicts — tidyverse_conflicts() —
## X purrr::%@%()      masks rlang::%@%()
## X dplyr::filter()   masks stats::filter()
## X purrr::flatten()  masks rlang::flatten()
## X purrr::flatten_chr() masks rlang::flatten_chr()
## X purrr::flatten_dbl() masks rlang::flatten_dbl()
## X purrr::flatten_int() masks rlang::flatten_int()
## X purrr::flatten_lgl() masks rlang::flatten_lgl()
## X purrr::flatten_raw() masks rlang::flatten_raw()
## X purrr::invoke()   masks rlang::invoke()
## X dplyr::lag()       masks stats::lag()
## X purrr::splice()    masks rlang::splice()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
Children_DF <- read.csv( "Children_DF.csv")
# attach(ChildrenADF)
# glimpse(Children_DF)
# names(Children_DF)
```

Removing Age Groups 0-6 months, 72 months and NA:

```
## Cleaning up children age variables "age_total_months", age_group
## Child Age-group
```

```
Children_DF %>% tabyl(age_group)
```

```
## age_group    n      percent valid_percent
##           0  11 0.018363940    0.01851852
##           1  76 0.126878130    0.12794613
##           2 123 0.205342237    0.20707071
##           3  84 0.140233723    0.14141414
##           4 259 0.432387312    0.43602694
##           5  41 0.068447412    0.06902357
##          NA   5 0.008347245           NA
```

```
summary(Children_DF$age_total_months)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      2.00   19.75   37.00   38.81   58.00  128.00      11
```

*# are our final age groups: 6 to <12 months; 1 to <2 years; 2 to <3. years; 3 to <6 years;
 # This means include 6 month-old in the first group and exclude 72 months from the last group.*

*# 0: 0 months to less than 6 months (under 6 months)
 # 1: 6 months to 11 months (6 months to 1 yr)
 # 2: 12 months to 23 months (1 yr to 2 yrs)
 # 3; 24 months to 35 months (2 yrs to 3yrs)
 # 4: 36 months to 71 months (3yrs to 6yrs)
 # 5: above or equal 72 months (above 6yrs)*

```
Children_DF = Children_DF %>%
  mutate(
    Age_Category = case_when(
      age_total_months < 6 ~ "0",
      age_total_months >= 6 & age_total_months <= 11 ~ "1",
      age_total_months >= 12 & age_total_months <= 23 ~ "2",
      age_total_months >= 24 & age_total_months <= 35 ~ "3",
      age_total_months >= 36 & age_total_months <= 71 ~ "4",
      age_total_months >= 72 ~ "5"
    )
  )
```

*# Under 6 months
 # 6 months to 1 year
 # 1 year to 2 years
 # 2 years to 3 years
 # 3 years to 6 years
 # Above 6 years*

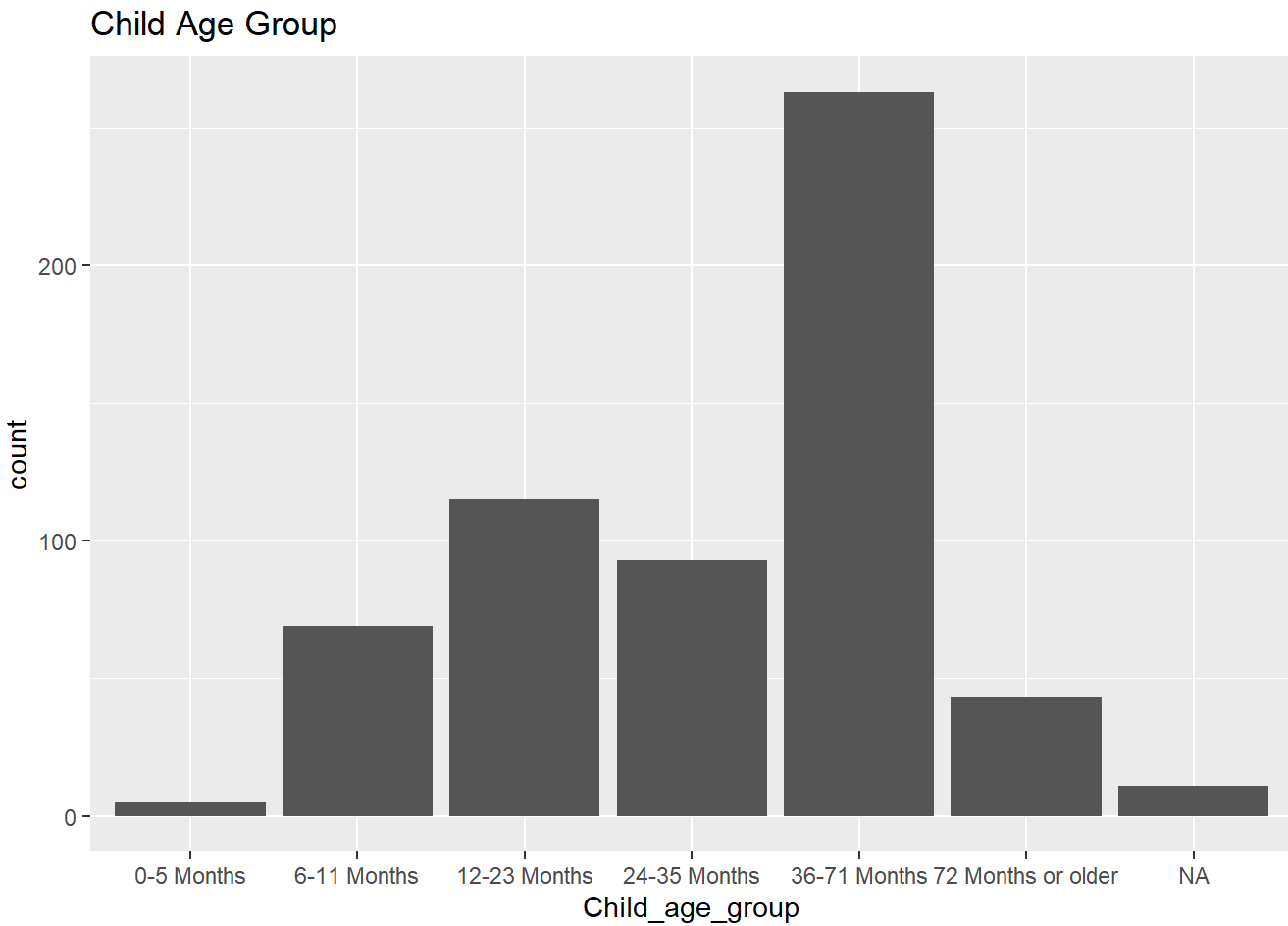
```
Children_DF = Children_DF %>% mutate(
  Child_age_group= case_when(
    Age_Category== "0"~ "0-5 Months",
    Age_Category== "1"~ "6-11 Months",
    Age_Category== "2"~ "12-23 Months",
    Age_Category== "3"~ "24-35 Months",
    Age_Category== "4"~ "36-71 Months",
    Age_Category== "5"~ "72 Months or older"
  )
)
```

```
Children_DF$Child_age_group <- factor(Children_DF$Child_age_group,
  levels = c("0-5 Months","6-11 Months","12-23 Months",
    "24-35 Months","36-71 Months","72 Months or older"))
Children_DF %>% tabyl(Child_age_group)
```

##	Child_age_group	n	percent	valid_percent
##	0-5 Months	5	0.008347245	0.008503401
##	6-11 Months	69	0.115191987	0.117346939
##	12-23 Months	115	0.191986644	0.195578231
##	24-35 Months	93	0.155258765	0.158163265
##	36-71 Months	263	0.439065109	0.447278912
##	72 Months or older	43	0.071786311	0.073129252
##	<NA>	11	0.018363940	NA

```
#
#   Child_age_group    n    percent valid_percent
#     0-5 Months      5 0.008347245 0.008503401
#     6-11 Months     69 0.115191987 0.117346939
#    12-23 Months    115 0.191986644 0.195578231
#    24-35 Months     93 0.155258765 0.158163265
#    36-71 Months    263 0.439065109 0.447278912
# 72 Months or older   43 0.071786311 0.073129252
#           <NA>      11 0.018363940          NA
```

```
Children_DF %>% ggplot(aes(Child_age_group))+
  geom_bar()+
  ggtitle("Child Age Group")
```




```
# Filtering out rows where Child_age_group is "0-6 Months", "over 72 Months", or "NA"
```

```
ChildrenADF <- Children_DF %>%  
filter((Child_age_group %in% c( "6-11 Months", "12-23 Months", "24-35 Months","36-71 Months")))  
# Rename the column  
ChildrenADF <- ChildrenADF %>%  
  rename(Child_Age_Gp = Child_age_group)  
  
write.csv(ChildrenADF,"ChildrenADF_6-71_Months_Nov28")
```

```
## Sample size is reduced from 599 to 540 with this filter
```

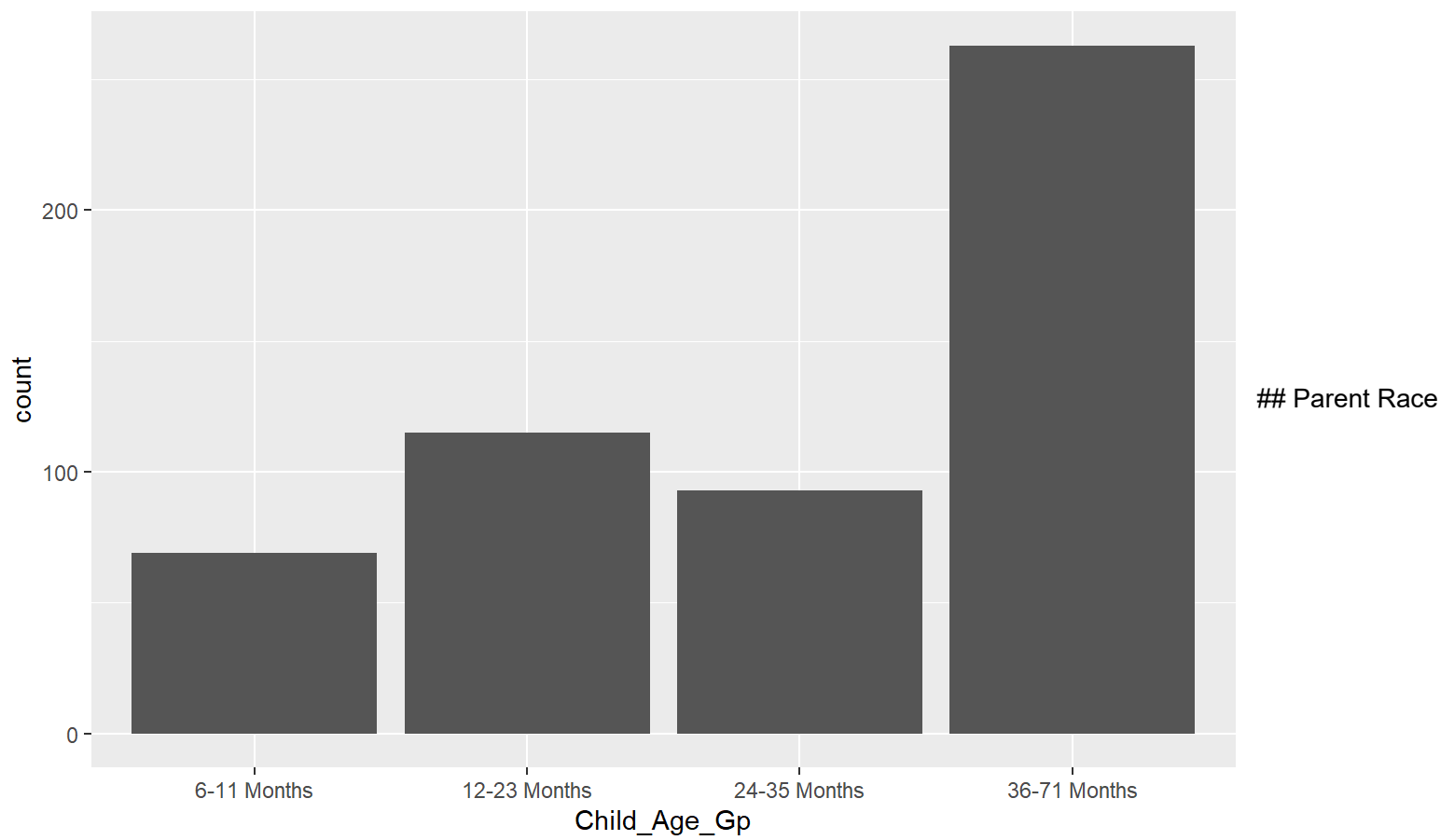
```
ChildrenADF$Child_Age_Gp <- factor(ChildrenADF$Child_Age_Gp,  
  levels = c("6-11 Months", "12-23 Months", "24-35 Months","36-71 Months"))  
ChildrenADF %>% tabyl(Child_Age_Gp)
```

```
## Child_Age_Gp    n    percent  
##    6-11 Months    69 0.1277778  
##   12-23 Months   115 0.2129630  
##   24-35 Months    93 0.1722222  
##   36-71 Months   263 0.4870370
```

```
# ##### Temp Fix  
# ChildrenADF$Child_Age_Gp <- factor(Children_DF$Child_age_group,  
#   Levels = c("6-11 Months", "12-23 Months", "24-35 Months","36-71 Months"))  
# ChildrenADF %>% tabyl(Child_Age_Gp)
```

```
ChildrenADF %>% ggplot(aes(Child_Age_Gp))+  
  geom_bar()+  
  ggtitle("Child Age Group")
```

Child Age Group



Var

```
#####
ChildrenADF$P1_Race<- factor(ChildrenADF$Parent1_Race,
                             levels = c("White", "Black", "Asian", "Mixed", "Other/Unknown"))
tabyl(ChildrenADF$P1_Race)
```

```
## ChildrenADF$P1_Race  n    percent
##           White 288 0.53333333
##           Black 158 0.29259259
##           Asian  25 0.04629630
##           Mixed  34 0.06296296
##           Other/Unknown 35 0.06481481
```

```
# Asian  28 0.04674457
#           Black 176 0.29382304
#           Mixed  36 0.06010017
#           Other/Unknown 44 0.07345576
#           White 315 0.52587646
```

```
#####
## Creating a race variable for the child
##### Here are the childrace names
# "race__1"                "race__2"
# [136] "race__3"            "race__4"            "race__5"
# [139] "race__6"            "race__89"          "race__99"
# [142] "race_other2"        "race_other3"        "race_other4"
# #####
# tabyl(ChildrenADF$race_other4)

# ChildrenADF$race_other4  n      percent valid_percent
# .   1 0.001669449      0.01960784
# Arab   1 0.001669449      0.01960784
# Hispanic 21 0.035058431      0.41176471
# Hispanic/Latin  1 0.001669449      0.01960784
# Hispanic/Latino  1 0.001669449      0.01960784
# Jamaica   2 0.003338898      0.03921569
# Jamaican   1 0.001669449      0.01960784
# Latina    1 0.001669449      0.01960784
# Latina, Indigenous  1 0.001669449      0.01960784
# Latino    4 0.006677796      0.07843137
# Mexican   3 0.005008347      0.05882353
# Mexican American  1 0.001669449      0.01960784
# Middle Eastern  1 0.001669449      0.01960784
# Middle Easterner  1 0.001669449      0.01960784
# Mixed    1 0.001669449      0.01960784
# Multi Racial  1 0.001669449      0.01960784
# Native American, Mexican  1 0.001669449      0.01960784
# biracial  1 0.001669449      0.01960784
# Latino    2 0.003338898      0.03921569
# Latinoamericano  1 0.001669449      0.01960784
# mexican american  1 0.001669449      0.01960784
# mix      2 0.003338898      0.03921569
# moreno   1 0.001669449      0.01960784
# <NA> 548 0.914858097          NA

# #####
#
# ## Rename the child race columns to make them distinguishable from parent race columns
ChildrenADF= ChildrenADF %>%
rename(
  Childrace__1= race__1,
  Childrace__2= race__2,
  Childrace__3 = race__3,
  Childrace__4 = race__4,
  Childrace__5 = race__5,
  Childrace__6 = race__6,
  Childrace__89 = race__89,
  Childrace__99 = race__99,
  Childrace_other2 =race_other2,
  Childrace_other4 =race_other4
)

#names(ChildrenADF)
#tabyl(ChildrenADF$Childrace_other4)
#
```

```

##
create_race_category <- function(row) {
  if ( !is.na(row["Childrace_other4"]) && row["Childrace_other4"] %in% c("mix", "Multi Racial", "Mixed","bi
racial")) {
    return("Mixed")
  } else if (!is.na(row["Childrace_other4"]) && row["Childrace_other4"] %in% c("Arab", "Middle Eastern", "M
iddle Easterner")) {
    return("White")
  } else {
    Childrace_cols <- grep( "Childrace___[1-6]", names(ChildrenADF), value = TRUE)
    if (sum(Childrace_cols = 1) > 1) {
      return("Mixed")
    } else
    if (sum(Childrace_cols = 1)== 1) {
      if (row["Childrace___1"] == 1) return("White")
      if (row["Childrace___2"] == 1) return("Black")
      if (row["Childrace___3"] == 1) return("American Indian")
      if (row["Childrace___4"] == 1) return("Asian")
      if (row["Childrace___5"] == 1) return("Hawaiian/Pacific Islander")
      if (row["Childrace___6"] == 1) return("Some other race")
      if (row["Childrace___89"] == 1 || row["Childrace___99"] == 1) return("Unknown")
    }
    return("Unknown")
  }
}

## Apply the function to the data
ChildrenADF$Child_Race <- apply(ChildrenADF, 1, create_race_category)

# Convert to factor with specified levels
ChildrenADF$Child_Race <- factor(ChildrenADF$Child_Race, levels = c("White", "Black",
  "Asian", "American Indian", "Mixed",
  "Hawaiian/Pacific Islander", "Some other race", "Unknown"))

# #####
# ## I got Mixed 0 Check why is this not working for the children also having the two parents same race
# is probably wrong!

## Get table summary so far:
tabyl(ChildrenADF$Child_Race)

```

```

##      ChildrenADF$Child_Race    n    percent
##                White  327 0.6055555556
##                Black  161 0.298148148
##                Asian   19 0.035185185
##      American Indian    2 0.003703704
##                Mixed    3 0.005555556
## Hawaiian/Pacific Islander  1 0.001851852
##      Some other race   21 0.038888889
##                Unknown    6 0.011111111

```

```

ChildrenADF <- ChildrenADF %>%
  mutate(Child_Race4 = fct_collapse(Child_Race,
                                     "Otherwise" = c("American Indian", "Mixed" ,"Hawaiian/Pacific Islander",
               "Some other race", "Unknown" ) ))
ChildrenADF$Child_Race4 = factor(ChildrenADF$Child_Race4,
                                 levels=c( "White", "Black", "Asian", "Otherwise"))

ChildrenADF %>% tabyl(Child_Race4)

```

```

## Child_Race4  n    percent
##           White 327 0.60555556
##           Black 161 0.29814815
##           Asian  19 0.03518519
##           Otherwise 33 0.06111111

```

```

tabyl(ChildrenADF,Child_Race, Parent1_Race)

```

```

##           Child_Race Asian Black Mixed Other/Unknown White
##           White      5      7    26          12    277
##           Black      1    150     4           3      3
##           Asian     19      0     0           0      0
##           American Indian  0      0     0           1      1
##           Mixed      0      0     1           1      1
##           Hawaiian/Pacific Islander  0      0     0           1      0
##           Some other race  0      0     3          13      5
##           Unknown     0      1     0           4      1

```

```

tabyl(ChildrenADF,State,Parent1_Race)

```

```

##           State Asian Black Mixed Other/Unknown White
##           Arizona   7    20    7          16    130
##           Florida   15    22   11          10    116
##           North Carolina  3   116   16           9     42

```

```

tabyl(ChildrenADF,State,Child_Race)

```

```

##           State White Black Asian American Indian Mixed
##           Arizona  140   24    5           2     0
##           Florida  125   22   11           0     2
##           North Carolina  62  115    3           0     1
##           Hawaiian/Pacific Islander Some other race Unknown
##           1           4     4
##           0           12    2
##           0           5     0

```

Get Percentages distn of race for each state

```
ChildrenADF %>% tabyl(State,Child_Race) %>%
  adorn_percentages(denominator = "row") %>%
  adorn_totals("col") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

```
##           State      White      Black      Asian American Indian      Mixed
##           Arizona 77.8% (140) 13.3% (24) 2.8% (5)           1.1% (2) 0.0% (0)
##           Florida 71.8% (125) 12.6% (22) 6.3% (11)           0.0% (0) 1.1% (2)
## North Carolina 33.3% (62) 61.8% (115) 1.6% (3)           0.0% (0) 0.5% (1)
## Hawaiian/Pacific Islander Some other race Unknown           Total
##           0.6% (1)           2.2% (4) 2.2% (4) 100.0% (180)
##           0.0% (0)           6.9% (12) 1.1% (2) 100.0% (174)
##           0.0% (0)           2.7% (5) 0.0% (0) 100.0% (186)
```

Child_Race Vs Parent Race

```
ChildrenADF %>% tabyl(Child_Race,Parent1_Race) %>%
  adorn_percentages(denominator = "row") %>%
  adorn_totals("col") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

```
##           Child_Race      Asian      Black      Mixed Other/Unknown
##           White 1.5% (5) 2.1% (7) 8.0% (26) 3.7% (12)
##           Black 0.6% (1) 93.2% (150) 2.5% (4) 1.9% (3)
##           Asian 100.0% (19) 0.0% (0) 0.0% (0) 0.0% (0)
## American Indian 0.0% (0) 0.0% (0) 0.0% (0) 50.0% (1)
##           Mixed 0.0% (0) 0.0% (0) 33.3% (1) 33.3% (1)
## Hawaiian/Pacific Islander 0.0% (0) 0.0% (0) 0.0% (0) 100.0% (1)
##           Some other race 0.0% (0) 0.0% (0) 14.3% (3) 61.9% (13)
##           Unknown 0.0% (0) 16.7% (1) 0.0% (0) 66.7% (4)
##           White      Total
## 84.7% (277) 100.0% (327)
## 1.9% (3) 100.0% (161)
## 0.0% (0) 100.0% (19)
## 50.0% (1) 100.0% (2)
## 33.3% (1) 100.0% (3)
## 0.0% (0) 100.0% (1)
## 23.8% (5) 100.0% (21)
## 16.7% (1) 100.0% (6)
```

Child Behavior Variable #1 :pacifier

```
#####  
## Child Behavior Variable #1 :pacifier  
## Documentation: 13. Does your child use a pacifier?  
#  
# 0 No / No  
# 1 Yes / Sí  
# 89 Don't know / No sabe  
# 99 Refused / Se negó a responder  
#  
ChildrenADF %>% tabyl(pacifier)
```

```
## pacifier    n    percent  
##          0 450 0.8333333  
##          1  90 0.1666667
```

```
ChildrenADF = ChildrenADF %>% mutate(  
  Pacifier= case_when(  
    pacifier== "0"~ "No",  
    pacifier== "1"~ "Yes",  
    pacifier== "89"~ "Don't know",  
    pacifier== "99"~ "Refused "  
  ))  
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts  
  
ChildrenADF$Pacifier <- factor(ChildrenADF$Pacifier,  
                              levels = c( "Yes","No"))  
  
ChildrenADF %>% tabyl(Pacifier)
```

```
## Pacifier    n    percent  
##       Yes  90 0.1666667  
##       No 450 0.8333333
```

```
# Generate a two-way table with counts and relative frequencies  
(table_counts <- table(ChildrenADF$Child_Race4,ChildrenADF$Pacifier))
```

```
##  
##           Yes  No  
## White      58 269  
## Black      30 131  
## Asian       0  19  
## Otherwise   2  31
```

```
(table_rel_freq <- prop.table(table_counts))
```

```
##
##           Yes           No
## White    0.107407407 0.498148148
## Black    0.055555556 0.242592593
## Asian    0.000000000 0.035185185
## Otherwise 0.003703704 0.057407407
```

```
# Test for significance of dependence between the two variables
(chi_sq_test <- chisq.test(table_counts))
```

```
## Warning in stats::chisq.test(x, y, ...): Chi-squared approximation may be
## incorrect
```

```
##
## Pearson's Chi-squared test
##
## data:  table_counts
## X-squared = 7.1909, df = 3, p-value = 0.06606
```

```
(Fisher_sq_test <- fisher.test(table_counts))
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  table_counts
## p-value = 0.05107
## alternative hypothesis: two.sided
```

```
#
ChildrenADF %>% tabyl(Child_Race4,Pacifier) %>%
  adorn_percentages(denominator = "row") %>%
  adorn_totals("col") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

```
## Child_Race4      Yes      No      Total
##      White 17.7% (58) 82.3% (269) 100.0% (327)
##      Black 18.6% (30) 81.4% (131) 100.0% (161)
##      Asian  0.0% (0) 100.0% (19) 100.0% (19)
##      Otherwise 6.1% (2) 93.9% (31) 100.0% (33)
```


Child Gender variable

```
ChildrenADF = ChildrenADF %>%
mutate(
  Child_Sex = case_when(
    Child_Gender == "1" ~ "Male",
    Child_Gender == "2" ~ "Female",
    Child_Gender == "3" ~ "Other"
  )
)

ChildrenADF$Child_Sex <- factor(ChildrenADF $Child_Sex,
                               levels = c( "Male","Female","Other"))

ChildrenADF %>% tabyl(Child_Sex)
```

```
##   Child_Sex    n    percent
##      Male  299 0.553703704
##   Female  240 0.444444444
##      Other   1 0.001851852
```

Add City variable

```
ChildrenADF = ChildrenADF %>%
mutate(
  City = case_when(
    city=="G" ~ "Greensboro",
    city=="M" ~ "Miami",
    city=="T" ~ "Tucson",
  )
)
ChildrenADF %>% tabyl(City)
```

```
##      City    n    percent
## Greensboro 186 0.3444444
##      Miami 174 0.3222222
##      Tucson 180 0.3333333
```

```
ChildrenADF %>% tabyl( Child_Sex, Child_Age_Gp ,City) %>%
# adorn_percentages(denominator = "row") %>%
  adorn_totals("col") %>%
  adorn_totals("row") %>%
# adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

```
## $Greensboro
## Child_Sex 6-11 Months 12-23 Months 24-35 Months 36-71 Months      Total
##      Male      11 (11)      26 (26)      16 (16)      48 (48) 101 (101)
##      Female     14 (14)     16 (16)     15 (15)     40 (40) 85 (85)
##      Other       0 (0)       0 (0)       0 (0)       0 (0)  0 (0)
##      Total      25 (25)     42 (42)     31 (31)     88 (88) 186 (186)
##
## $Miami
## Child_Sex 6-11 Months 12-23 Months 24-35 Months 36-71 Months      Total
##      Male      19 (19)     19 (19)     11 (11)     52 (52) 101 (101)
##      Female       8 (8)     16 (16)     16 (16)     33 (33) 73 (73)
##      Other       0 (0)       0 (0)       0 (0)       0 (0)  0 (0)
##      Total      27 (27)     35 (35)     27 (27)     85 (85) 174 (174)
##
## $Tucson
## Child_Sex 6-11 Months 12-23 Months 24-35 Months 36-71 Months      Total
##      Male       5 (5)      20 (20)     20 (20)     52 (52) 97 (97)
##      Female     12 (12)     17 (17)     15 (15)     38 (38) 82 (82)
##      Other       0 (0)       1 (1)       0 (0)       0 (0)  1 (1)
##      Total      17 (17)     38 (38)     35 (35)     90 (90) 180 (180)
```

```
ChildrenADF %>% tabyl(Parent1_latino)
```

```
## Parent1_latino  n    percent
##              1 351 0.65000000
##              2  10 0.01851852
##              3  28 0.05185185
##              4  64 0.11851852
##              5  87 0.16111111
```

```
ChildrenADF=ChildrenADF %>%
  mutate(
    P1_Ethnicity = case_when(
      Parent1_latino == "1" ~ "Not Hispanic",
      Parent1_latino == "89" ~ "Don't Know",
      Parent1_latino == "99" ~ "Refused",
      .default = "Hispanic"
      ## Unknown includes Mixed, Refused, Don't Know
    )
  )

ChildrenADF %>% tabyl(P1_Ethnicity)
```

```
## P1_Ethnicity  n percent
##      Hispanic 189     0.35
##    Not Hispanic 351     0.65
```

Table 2

Table 2

```
ChildrenADF %>% tabyl( P1_Ethnicity, Child_Race,City) %>%
  adorn_percentages(denominator = "row") %>%
  # adorn_totals("row") %>%
  adorn_totals("col") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

```
## $Greensboro
##   P1_Ethnicity      White      Black      Asian American Indian      Mixed
##   Hispanic 60.9% (14) 17.4%   (4) 0.0% (0)      0.0% (0) 0.0% (0)
##   Not Hispanic 29.4% (48) 68.1% (111) 1.8% (3)      0.0% (0) 0.6% (1)
##   Hawaiian/Pacific Islander Some other race Unknown      Total
##           0.0% (0)      21.7% (5) 0.0% (0) 100.0% (23)
##           0.0% (0)      0.0% (0) 0.0% (0) 100.0% (163)
##
## $Miami
##   P1_Ethnicity      White      Black      Asian American Indian      Mixed
##   Hispanic 83.3% (90) 4.6%   (5) 0.0% (0)      0.0% (0) 1.9% (2)
##   Not Hispanic 53.0% (35) 25.8% (17) 16.7% (11)      0.0% (0) 0.0% (0)
##   Hawaiian/Pacific Islander Some other race Unknown      Total
##           0.0% (0)      8.3% (9) 1.9% (2) 100.0% (108)
##           0.0% (0)      4.5% (3) 0.0% (0) 100.0% (66)
##
## $Tucson
##   P1_Ethnicity      White      Black      Asian American Indian      Mixed
##   Hispanic 79.3% (46) 5.2%   (3) 0.0% (0)      1.7% (1) 0.0% (0)
##   Not Hispanic 77.0% (94) 17.2% (21) 4.1% (5)      0.8% (1) 0.0% (0)
##   Hawaiian/Pacific Islander Some other race Unknown      Total
##           1.7% (1)      5.2% (3) 6.9% (4) 100.0% (58)
##           0.0% (0)      0.8% (1) 0.0% (0) 100.0% (122)
```

```
ChildrenADF %>% tabyl( Child_Race,City) %>%
  adorn_percentages(denominator = "col") %>%
  adorn_totals("row") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

```
##           Child_Race  Greensboro      Miami      Tucson
##           White 33.3% (62) 71.8% (125) 77.8% (140)
##           Black 61.8% (115) 12.6% (22) 13.3% (24)
##           Asian 1.6% (3) 6.3% (11) 2.8% (5)
##           American Indian 0.0% (0) 0.0% (0) 1.1% (2)
##           Mixed 0.5% (1) 1.1% (2) 0.0% (0)
##   Hawaiian/Pacific Islander 0.0% (0) 0.0% (0) 0.6% (1)
##           Some other race 2.7% (5) 6.9% (12) 2.2% (4)
##           Unknown 0.0% (0) 1.1% (2) 2.2% (4)
##           Total 100.0% (186) 100.0% (174) 100.0% (180)
```

```
ChildrenADF %>% tabyl(Workstatus_P1)
```

```
##      Workstatus_P1    n    percent valid_percent
##      Going_to_school  35 0.064814815    0.066539924
##      Keeping house   143 0.264814815    0.271863118
##      Looking for work  20 0.037037037    0.038022814
##      Unknown/Refused   2 0.003703704    0.003802281
##      Working_full_time 238 0.440740741    0.452471483
##      Working_part_time  88 0.162962963    0.167300380
##      <NA>             14 0.025925926            NA
```

```
ChildrenADF <- ChildrenADF %>%
  mutate(P1_Workstatus = fct_collapse(Workstatus_P1,
    "Not_Working/Refused" = c("Looking for work", "Unknown/Refused", "Keeping house")))
```

```
ChildrenADF$P1_Workstatus <- factor(ChildrenADF$P1_Workstatus,
  levels = c("Working_full_time", "Working_part_time", "Going_to_school",
    "Not_Working/Refused" ))
```

```
# # 27/449
```

```
# ## 7 are self employed/ indep contractors the rest are unemployed
```

```
ChildrenADF %>% tabyl(P1_Workstatus)
```

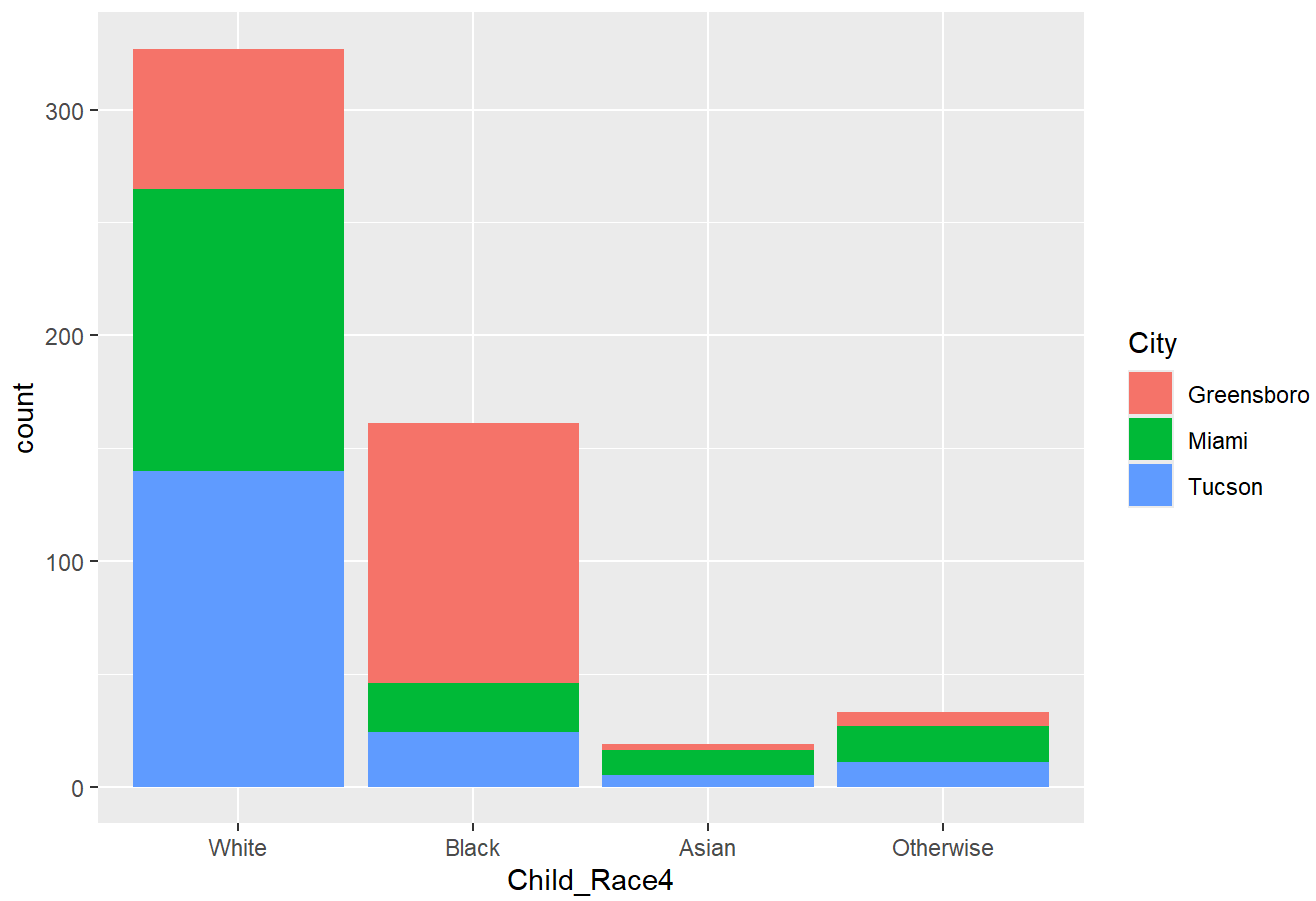
```
##      P1_Workstatus    n    percent valid_percent
##      Working_full_time 238 0.44074074    0.45247148
##      Working_part_time  88 0.16296296    0.16730038
##      Going_to_school   35 0.06481481    0.06653992
##      Not_Working/Refused 165 0.30555556    0.31368821
##      <NA>              14 0.02592593            NA
```

```
summary(ChildrenADF$Child_Race)
```

```
##      White      Black      Asian
##      327      161      19
##      American Indian  Mixed Hawaiian/Pacific Islander
##      2      3      1
##      Some other race  Unknown
##      21      6
```

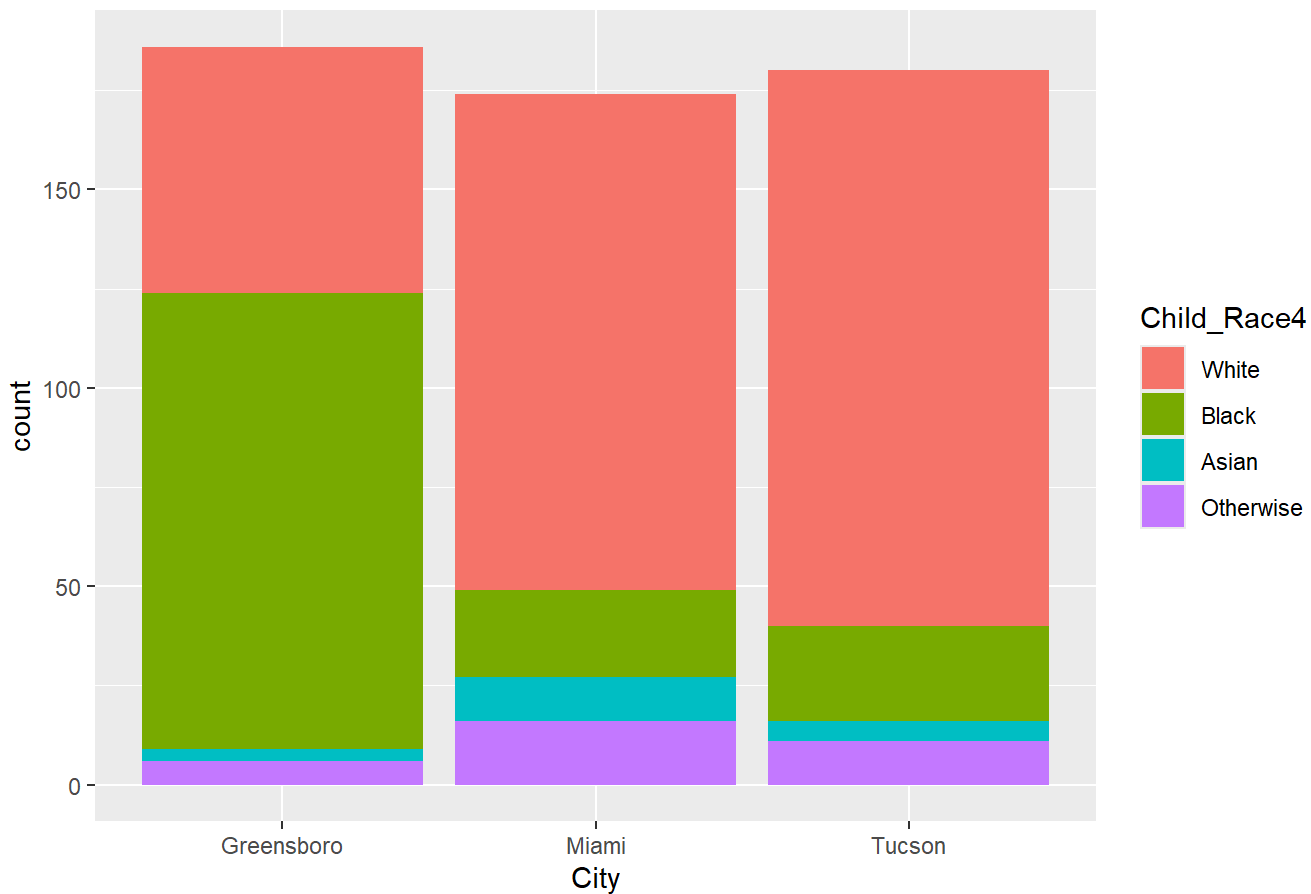
```
ggplot(data=ChildrenADF, aes(x=Child_Race4, fill=City))+
  geom_bar()+
  ggtitle("Child Race Vs. City")
```

Child Race Vs. City



```
ggplot(data=ChildrenADF, aes(x=City, fill=Child_Race4))+  
  geom_bar()+  
  ggtitle("Child Race Vs. City")
```

Child Race Vs. City



#

Adding parent age categorized & Household income

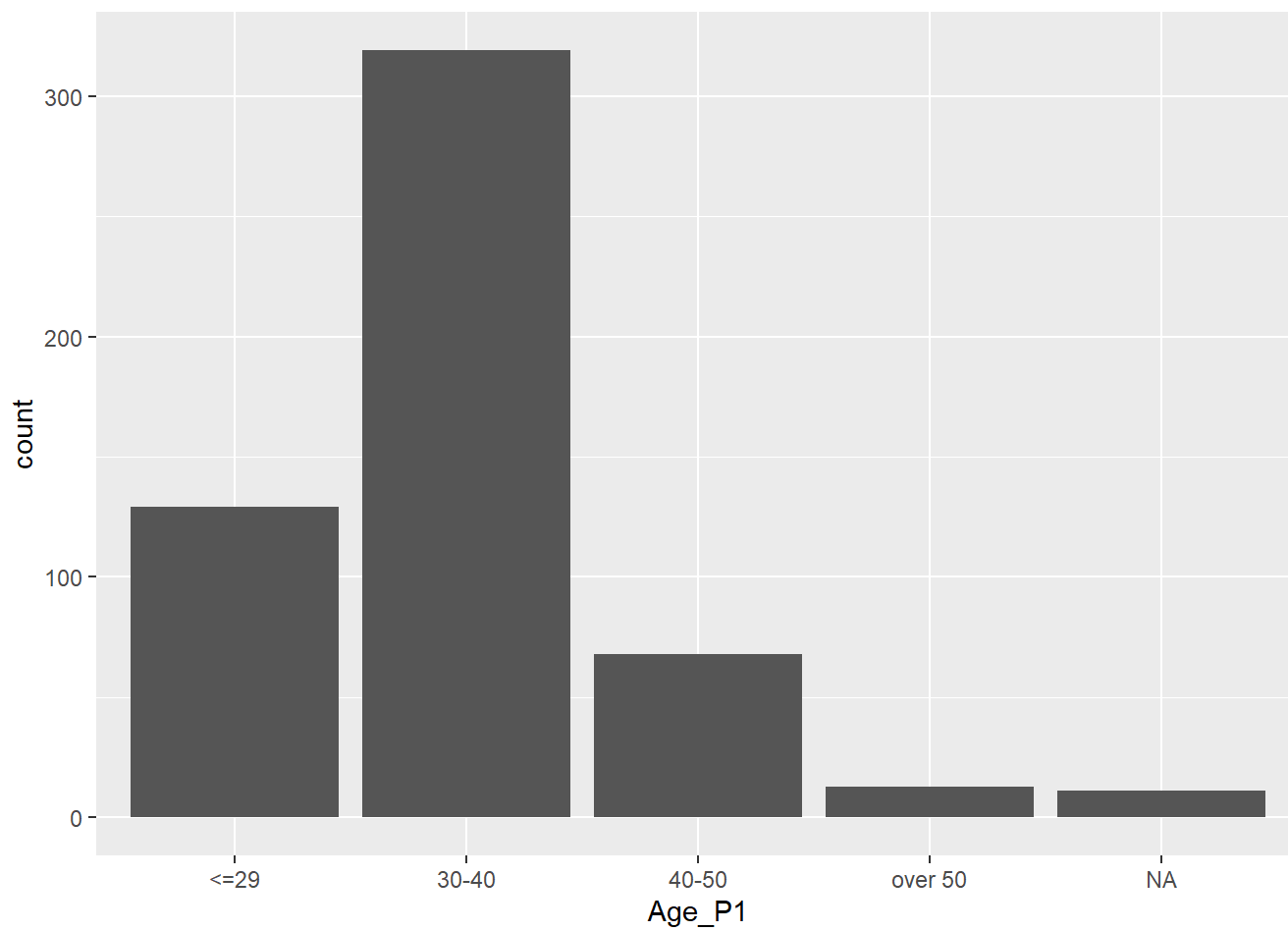
```
# hist( ChildrenADF$Parent1_Age)
# summary( ChildrenADF$Parent1_Age)

# Rename the column and convert it to a categorical variable
ChildrenADF <- ChildrenADF %>%
  rename(Age_P1 = Parent1_Age ) %>%
  mutate(
    Age_P1 = cut(
      Age_P1,
      breaks = c(-Inf, 29, 40, 50, Inf),
      labels = c("<=29", "30-40", "40-50", "over 50"),
      right = TRUE
    )
  )

ChildrenADF %>% tabyl(Age_P1)
```

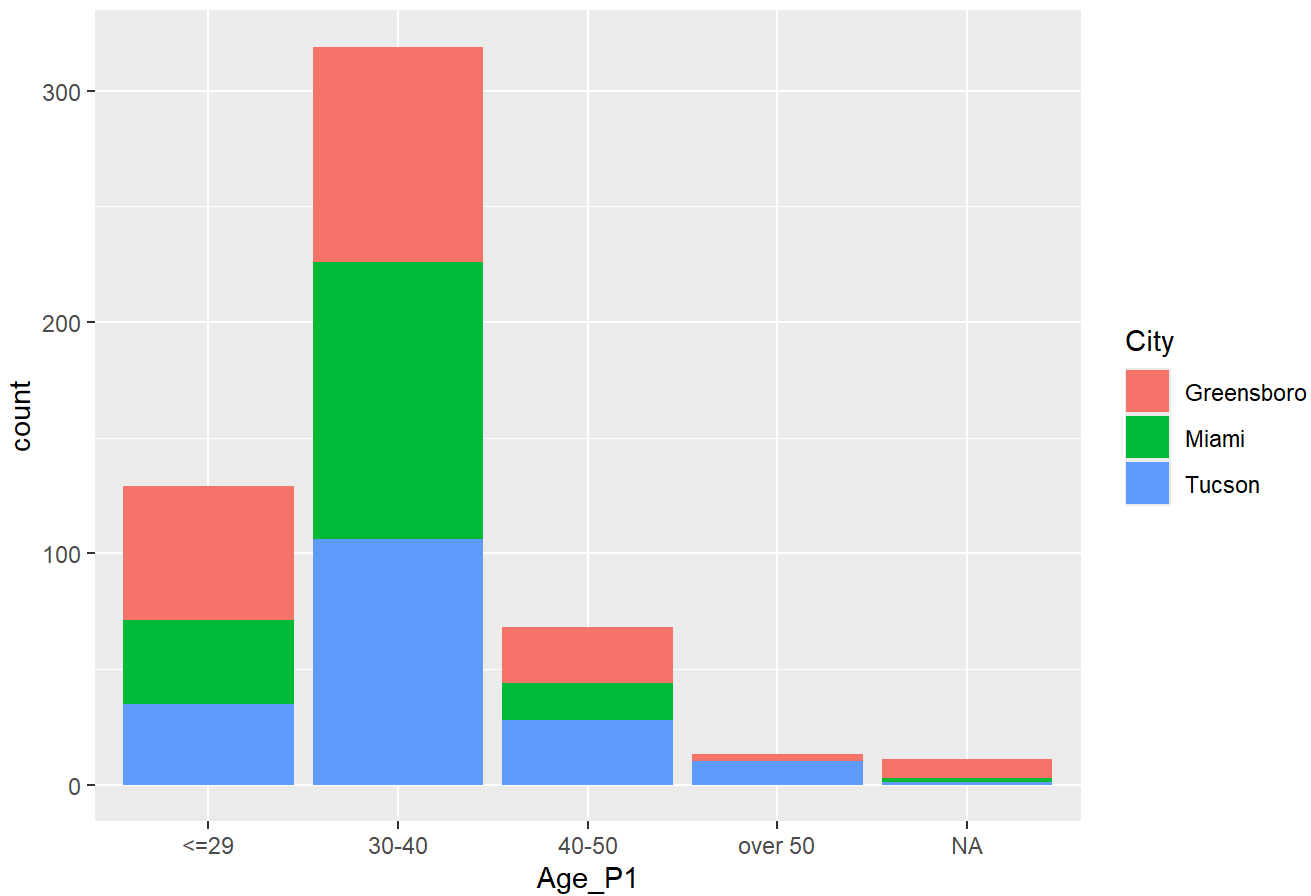
```
##   Age_P1    n  percent valid_percent
##   <=29  129  0.23888889    0.24385633
##   30-40  319  0.59074074    0.60302457
##   40-50   68  0.12592593    0.12854442
##   over 50  13  0.02407407    0.02457467
##   <NA>   11  0.02037037           NA
```

```
ggplot(data=ChildrenADF, aes(x=Age_P1))+  
  geom_bar()
```



```
ggplot(data=ChildrenADF, aes(x=Age_P1, fill=City))+  
  geom_bar()+  
  ggtitle("Parent 1's Age Group Vs. City")
```

Parent 1's Age Group Vs. City



```
#
# ## I accidentally replaced over 50 by NA, quick fix here:
# Replace NA values in Age_P1 with "over 50"
# ChildrenADF$Age_P1[is.na(ChildrenADF$Age_P1)] <- "over 50"

ChildrenADF$Age_P1 <- factor(ChildrenADF$Age_P1,
                             levels = c("<=29", "30-40", "40-50", "over 50"))

ChildrenADF <- ChildrenADF %>%
  rename(P1_Age= Age_P1 )

ChildrenADF$P1_Age <- factor(ChildrenADF$P1_Age,
                             levels = c("<=29", "30-40", "40-50", "over 50"))
ChildrenADF %>% tabyl (P1_Age)
```

```
##   P1_Age    n   percent valid_percent
##   <=29  129  0.23888889    0.24385633
##   30-40  319  0.59074074    0.60302457
##   40-50   68  0.12592593    0.12854442
## over 50   13  0.02407407    0.02457467
##   <NA>   11  0.02037037             NA
```

Creating a household income variable:

```
ChildrenADF%>%tabyl(Parent1_income )
```



```
## Parent1_income    n    percent
##                0 171 0.31666667
##                1 110 0.20370370
##                2  90 0.16666667
##                3  60 0.11111111
##                4  86 0.15925926
##               89  15 0.02777778
##               99   8 0.01481481
```

```
ChildrenADF%>%tabyl(Parent2_income )
```

```
## Parent2_income    n    percent valid_percent
##                0  53 0.09814815    0.13216958
##                1  46 0.08518519    0.11471322
##                2  81 0.15000000    0.20199501
##                3  62 0.11481481    0.15461347
##                4 128 0.23703704    0.31920200
##               89  22 0.04074074    0.05486284
##               99   9 0.01666667    0.02244389
##               NA 139 0.25740741           NA
```

```
# Convert income categories to numeric midpoints
ChildrenADF <- ChildrenADF %>%
  mutate(
    Income_p1_num = case_when(
      Parent1_income == "0" ~ 9500,    # Midpoint of 0-19K
      Parent1_income == "1" ~ 29500,   # Midpoint of 20K-39K
      Parent1_income == "2" ~ 49500,   # Midpoint of 40K-59K
      Parent1_income == "3" ~ 69500,   # Midpoint of 60K-79K
      Parent1_income == "4" ~ 90000,   # More than 80K, using 90K as a proxy
      TRUE ~ NA_real_                  # For "Don't know" or "Refused"
    ),
    Income_p2_num = case_when(
      Parent2_income == "0" ~ 9500,    # Same for second parent
      Parent2_income == "1" ~ 29500,
      Parent2_income == "2" ~ 49500,
      Parent2_income == "3" ~ 69500,
      Parent2_income == "4" ~ 90000,
      TRUE ~ NA_real_
    )
  ) %>%
  # Create Household Income variable
  mutate(
    Household_income = case_when(
      !is.na(Income_p2_num) ~ Income_p1_num + Income_p2_num, # Sum both incomes if available
      is.na(Income_p2_num) ~ Income_p1_num                    # Use only Income_p1 if Income_p2 is not available
    )
  )

summary(ChildrenADF$Household_income)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	9500	39000	79000	82426	119000	180000	23

```
## Converting Household income to categorical variable:
```

```
ChildrenADF <- ChildrenADF %>%
```

```
  mutate(
    Household_income = cut(
      Household_income,
      breaks = c(-Inf, 20000, 40000, 60000, 80000, 100000, Inf),
      labels = c("<=20K", "20K-40K", "40K-60K", "60K-80K", "80K-100K", "over 100K"),
      right = TRUE
    )
  )
```

```
ChildrenADF$Household_income <- factor(ChildrenADF$Household_income,
levels = c("<=20K", "20K-40K", "40K-60K", "60K-80K", "80K-100K", "over 100K"))
```

```
## Household Income By City
```

```
Table_2b=
```

```
ChildrenADF %>% tabyl(City, Household_income) %>%
  adorn_percentages(denominator = "row") %>%
  adorn_totals("col") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

```
#####
```

```
ChildrenADF %>% tabyl(Household_income)
```

##	Household_income	n	percent	valid_percent
##	<=20K	63	0.11666667	0.1218569
##	20K-40K	90	0.16666667	0.1740812
##	40K-60K	68	0.12592593	0.1315280
##	60K-80K	66	0.12222222	0.1276596
##	80K-100K	96	0.17777778	0.1856867
##	over 100K	134	0.24814815	0.2591876
##	<NA>	23	0.04259259	NA

```
ChildrenADF %>% tabyl(Household_income, Pacifier)
```

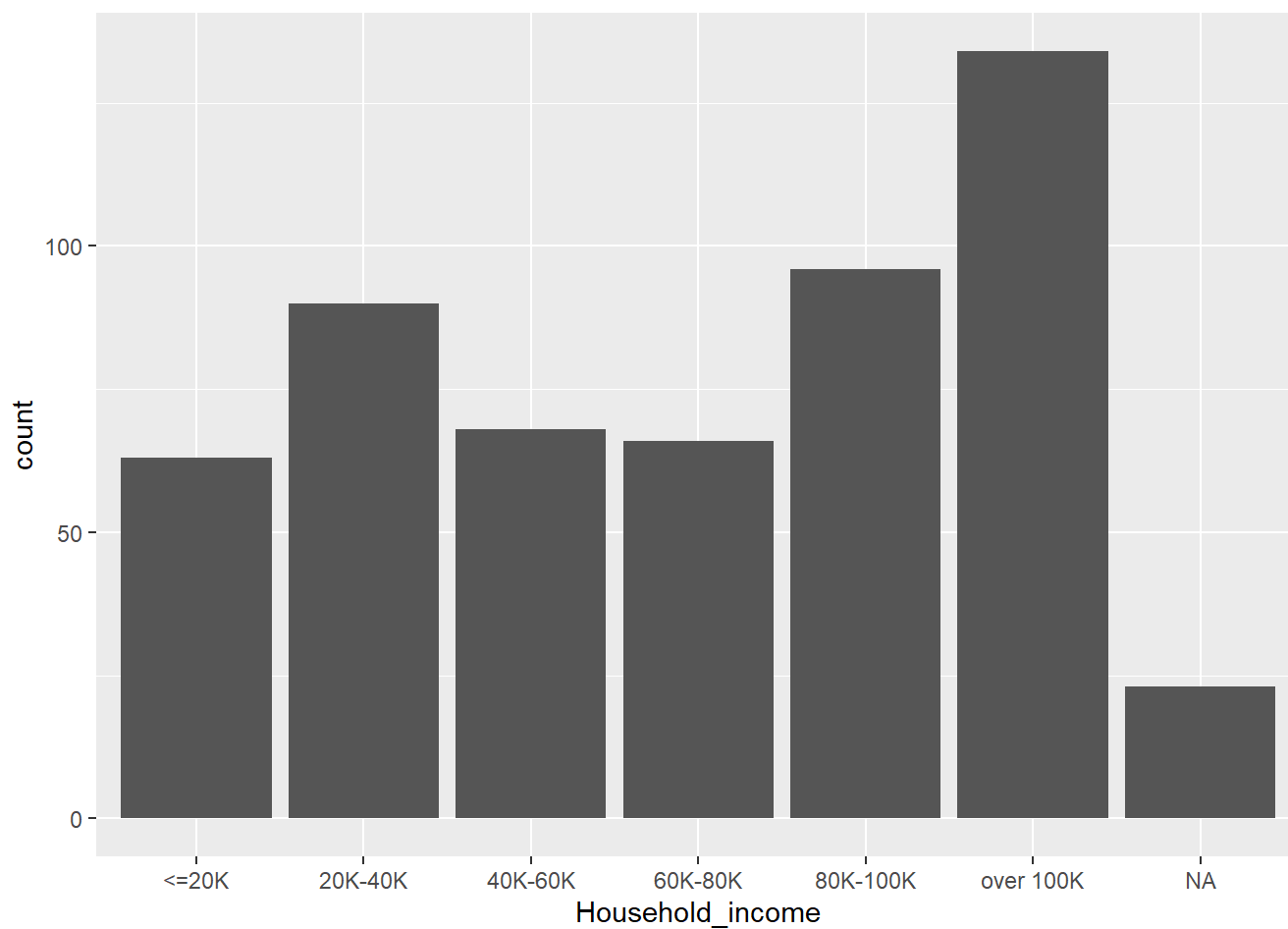
##	Household_income	Yes	No
##	<=20K	11	52
##	20K-40K	16	74
##	40K-60K	11	57
##	60K-80K	9	57
##	80K-100K	12	84
##	over 100K	26	108
##	<NA>	5	18

```
ChildrenADF %>% tabyl(Household_income, Pacifier) %>%
```

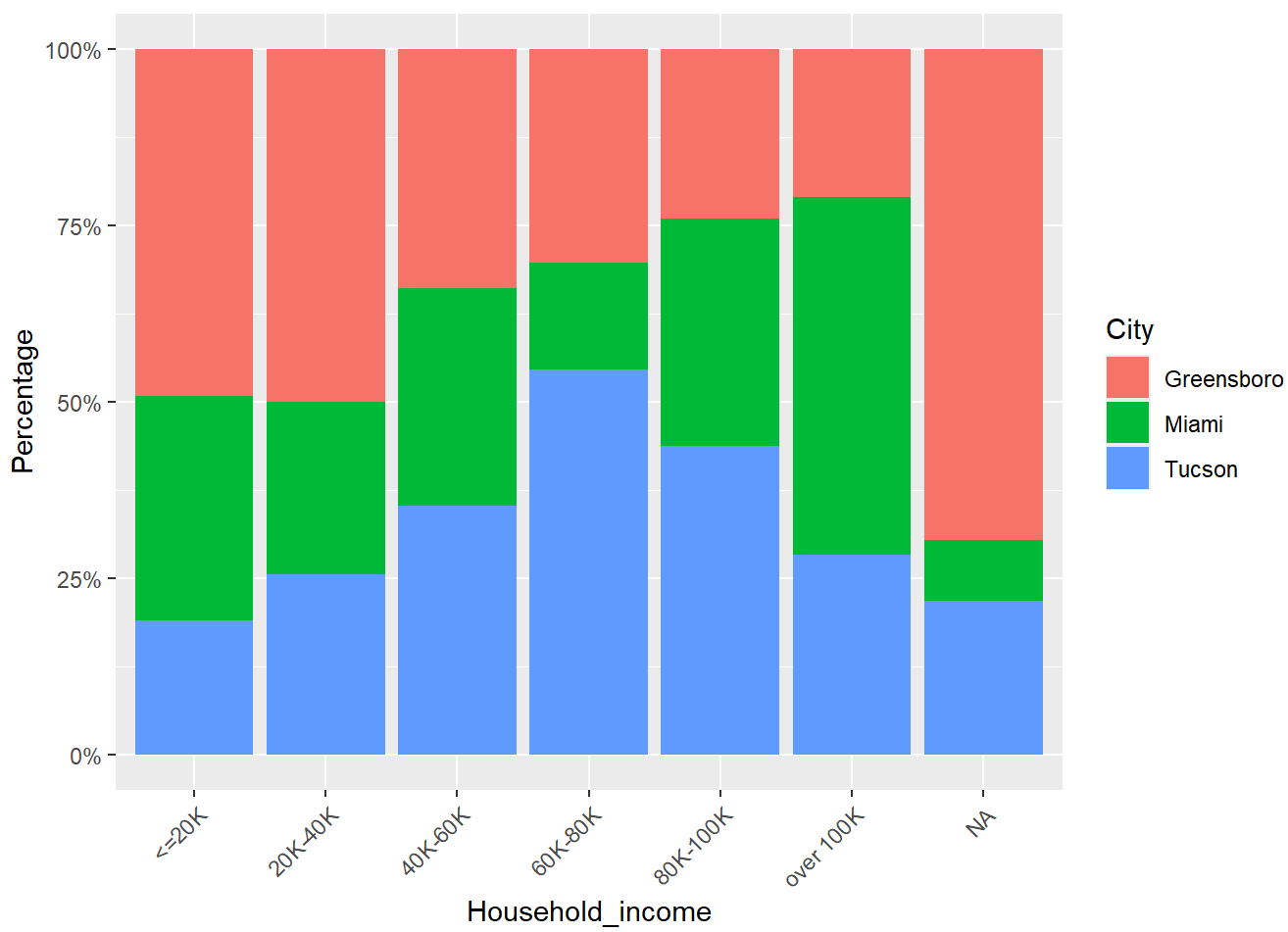
```
  adorn_percentages(denominator = "row") %>%
  adorn_totals("col") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

##	Household_income	Yes	No	Total
##	<=20K	17.5% (11)	82.5% (52)	100.0% (63)
##	20K-40K	17.8% (16)	82.2% (74)	100.0% (90)
##	40K-60K	16.2% (11)	83.8% (57)	100.0% (68)
##	60K-80K	13.6% (9)	86.4% (57)	100.0% (66)
##	80K-100K	12.5% (12)	87.5% (84)	100.0% (96)
##	over 100K	19.4% (26)	80.6% (108)	100.0% (134)
##	<NA>	21.7% (5)	78.3% (18)	100.0% (23)

```
ggplot(data=ChildrenADF, aes(Household_income))+
  geom_bar()
```



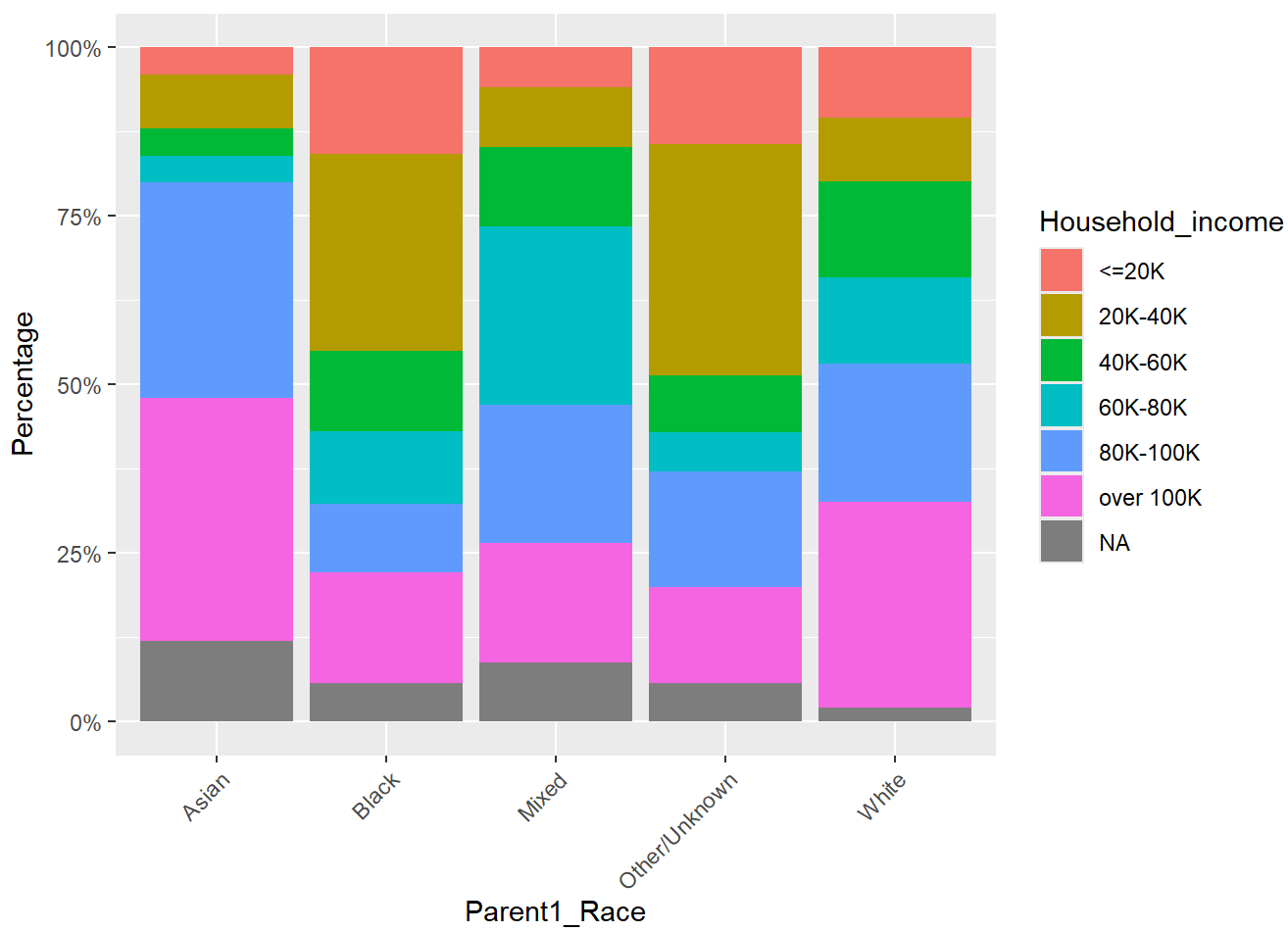
```
ggplot(data=ChildrenADF, aes(Household_income, fill=City))+
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(y = "Percentage")
```



```
ggtitle('Houshold Income Vs City')
```

```
## $title
## [1] "Houshold Income Vs City"
##
## attr(,"class")
## [1] "labels"
```

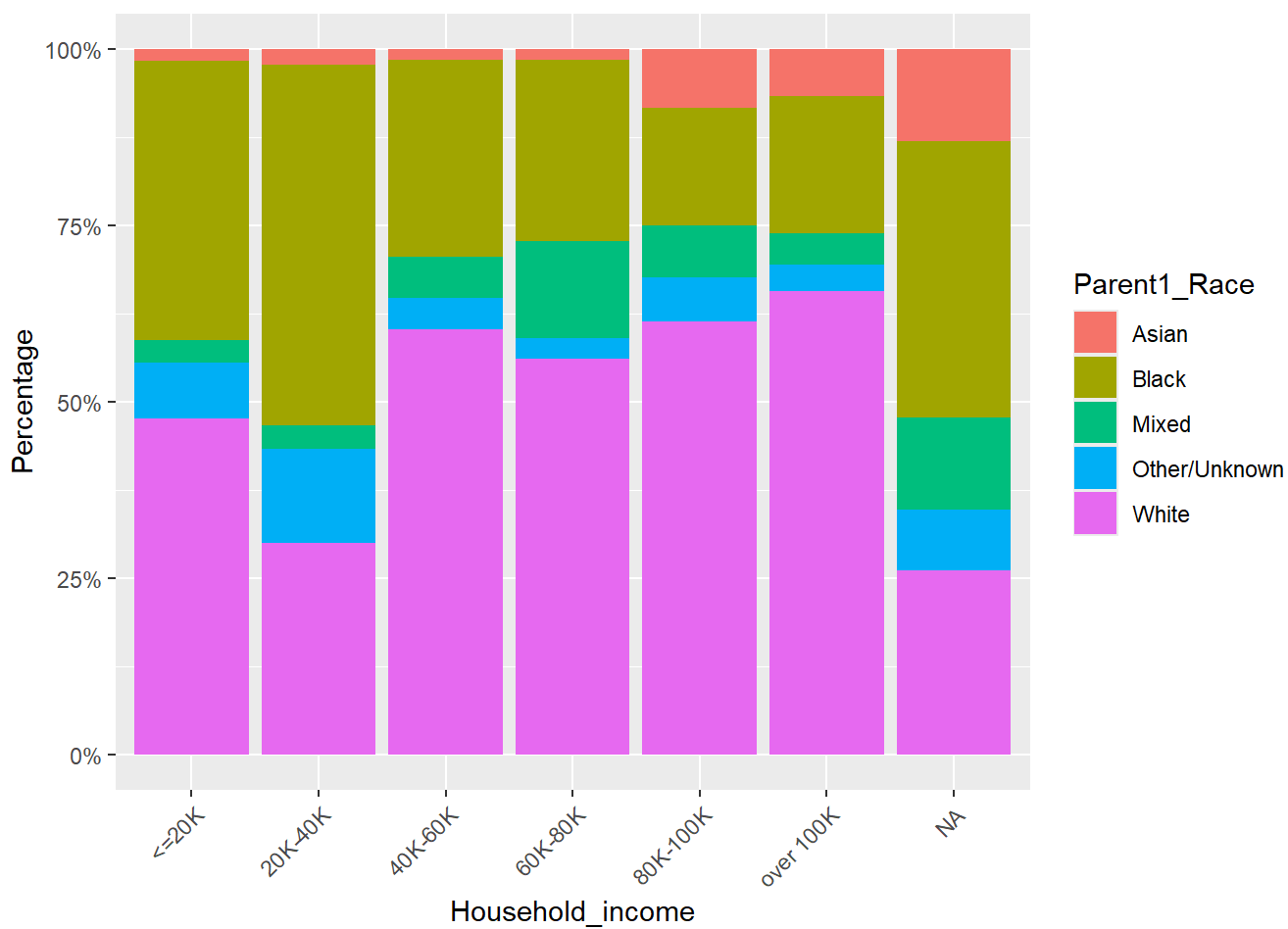
```
ggplot(data=ChildrenADF, aes(Parent1_Race , fill= Household_income))+
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(y = "Percentage")
```



```
ggtitle('Houshold Income Vs P1 Race')
```

```
## $title
## [1] "Houshold Income Vs P1 Race"
##
## attr(,"class")
## [1] "labels"
```

```
ggplot(data=ChildrenADF, aes(Household_income , fill= Parent1_Race ))+
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(y = "Percentage")
```



```
ggtitle('Houshold Income Vs P1 Race')
```

```
## $title
## [1] "Houshold Income Vs P1 Race"
##
## attr(,"class")
## [1] "labels"
```

```
### Number of parents reporting income for P2
53+46+81+62+128
```

```
## [1] 370
```

P1_Degree (education variable)

```

# degree_p1
# ]
# 7. What is your highest academic degree?
# ¿Cuál es su grado académico más alto?
# radio, Required
# 0
# None / Ninguno
# 1
# High school or GED / Escuela Preparatoria o GED
# 2
# Certification or professional licensing /Certifi cación o licencia profesional
# 3
# Associate Degree / Grado asociado
# 4
# Bachelor's degree / Licenciatura oBachillerato
# 5
# Master's degree / Maestría
# 6
# Doctoral degree / Doctorado
# 89
# Don't know / No sabe
# 99
# Refused / Se negó a responder

```

```

ChildrenADF %>% tabyl(degree_P1 )

```

```

##              degree_P1    n    percent
##      Associate_Degree   55 0.10185185
##      Bachelor's_degree 170 0.31481481
##      Certification/Professional_licensing  10 0.01851852
##      Doctoral_degree   37 0.06851852
##      High_school/GED   115 0.21296296
##      Master's_degree   141 0.26111111
##      None              12 0.02222222

```

```

ChildrenADF <- ChildrenADF %>%
  mutate(
    P1_Degree = case_when(
      Parent1_degree == "0" ~ "None",
      Parent1_degree == "1" ~ "High_school/GED",
      Parent1_degree == "2" ~ "Certification/Licensing",
      Parent1_degree == "3" ~ "Associate_Degree",
      Parent1_degree == "4" ~ "Bachelor's_degree",
      Parent1_degree == "5" ~ "Master's_degree",
      Parent1_degree == "6" ~ "Doctoral_degree"
    )
  )

ChildrenADF <- ChildrenADF %>%
  mutate(P1_Degree = fct_collapse(P1_Degree,
    "Associate/Certification" = c("Associate_Degree", "Certification/Licensing")))

ChildrenADF$P1_Degree<- factor(ChildrenADF$P1_Degree ,
  levels = c( "None", "High_school/GED", "Associate/Certification",
    "Bachelor's_degree", "Master's_degree", "Doctoral_degree"))

ChildrenADF %>% tabyl(P1_Degree)

```

```

##           P1_Degree    n   percent
##           None      12  0.02222222
##   High_school/GED  115  0.21296296
## Associate/Certification  65  0.12037037
##   Bachelor's_degree  170  0.31481481
##   Master's_degree   141  0.26111111
##   Doctoral_degree    37  0.06851852

```

```

ChildrenADF %>% tabyl(P1_Degree ,city)

```

```

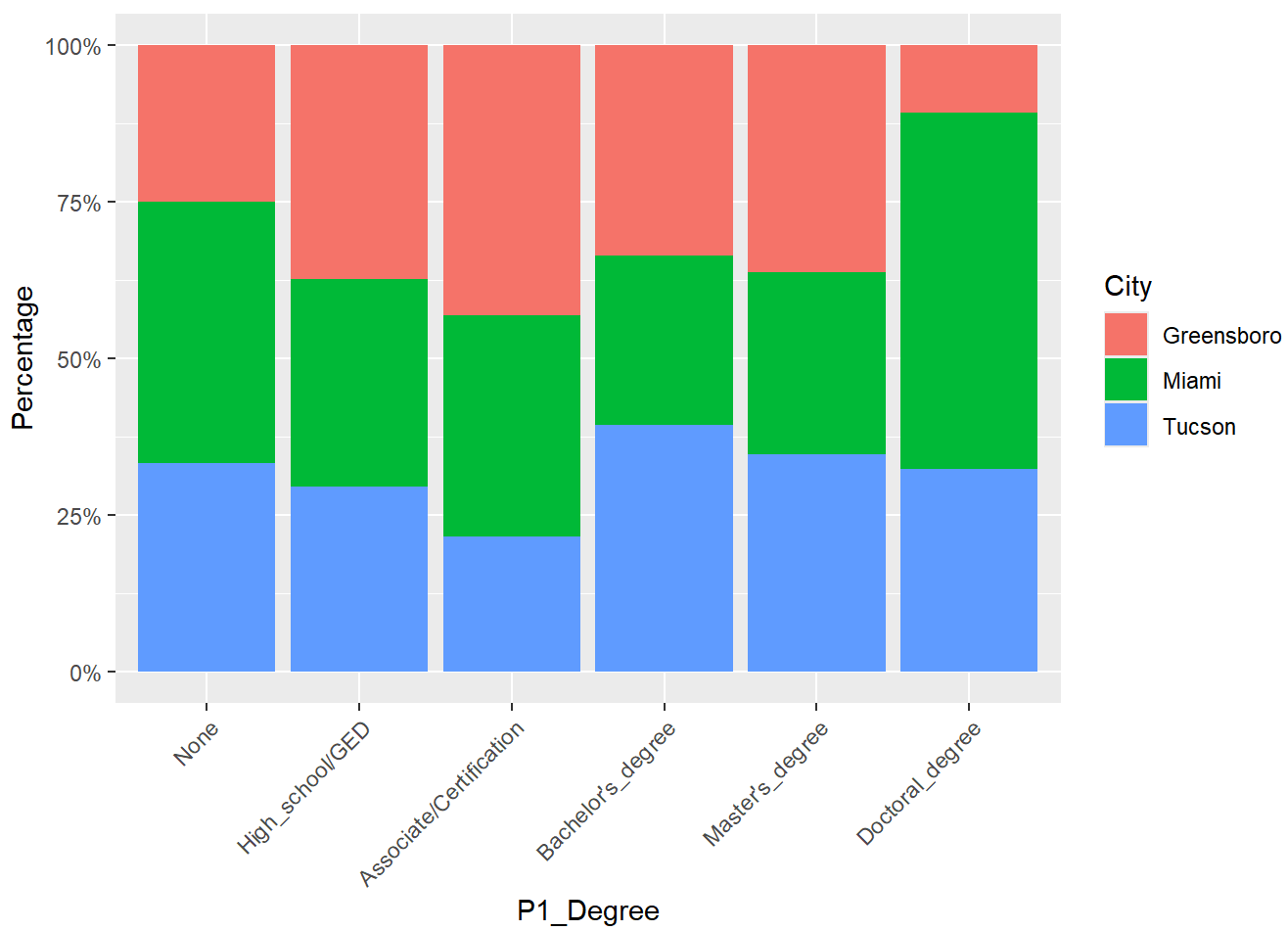
##           P1_Degree    G    M    T
##           None       3    5    4
##   High_school/GED   43   38   34
## Associate/Certification  28  23  14
##   Bachelor's_degree   57  46  67
##   Master's_degree     51  41  49
##   Doctoral_degree     4   21  12

```

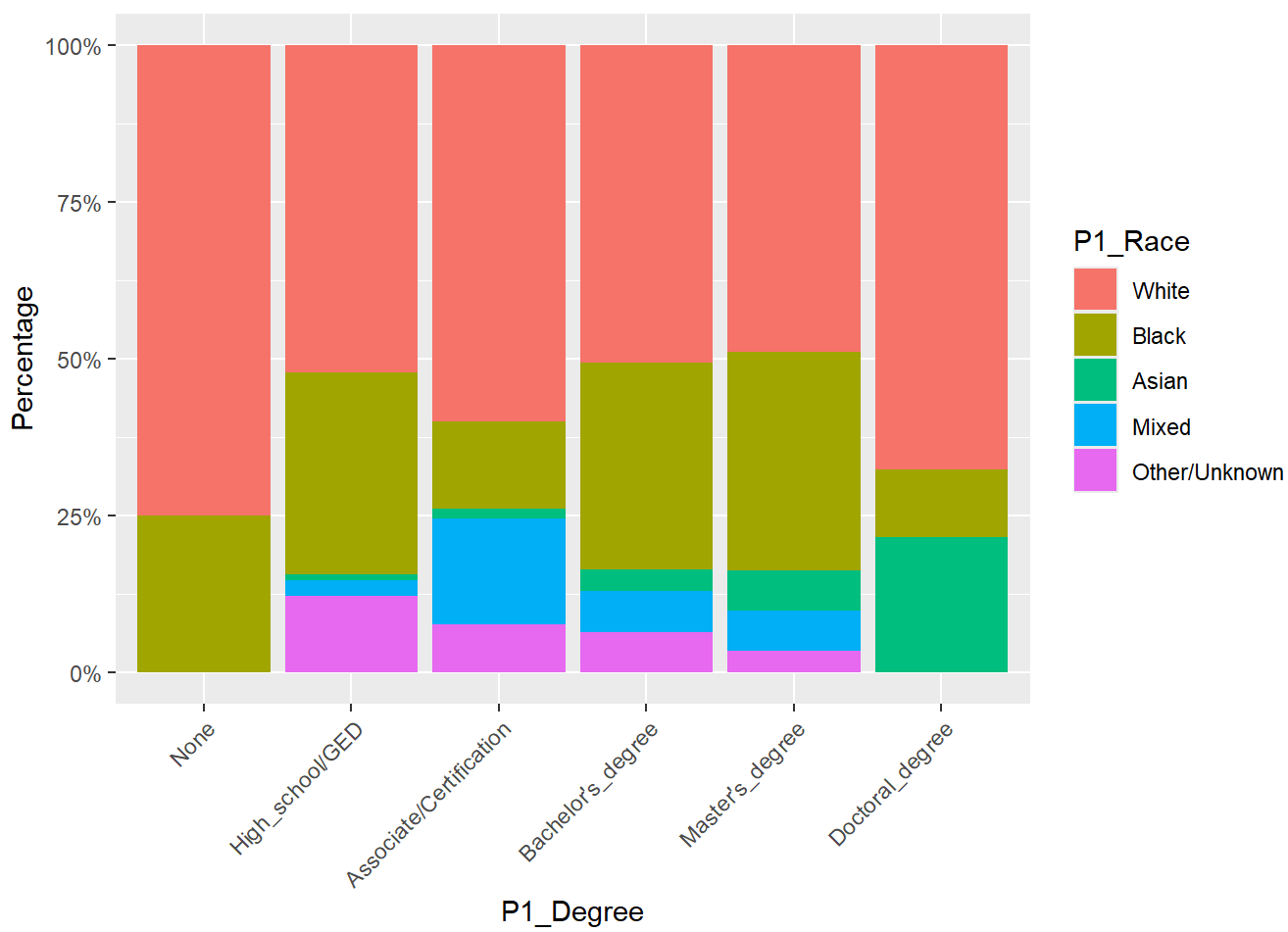
```

ggplot(data=ChildrenADF, aes(P1_Degree, fill=City))+
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(y = "Percentage")

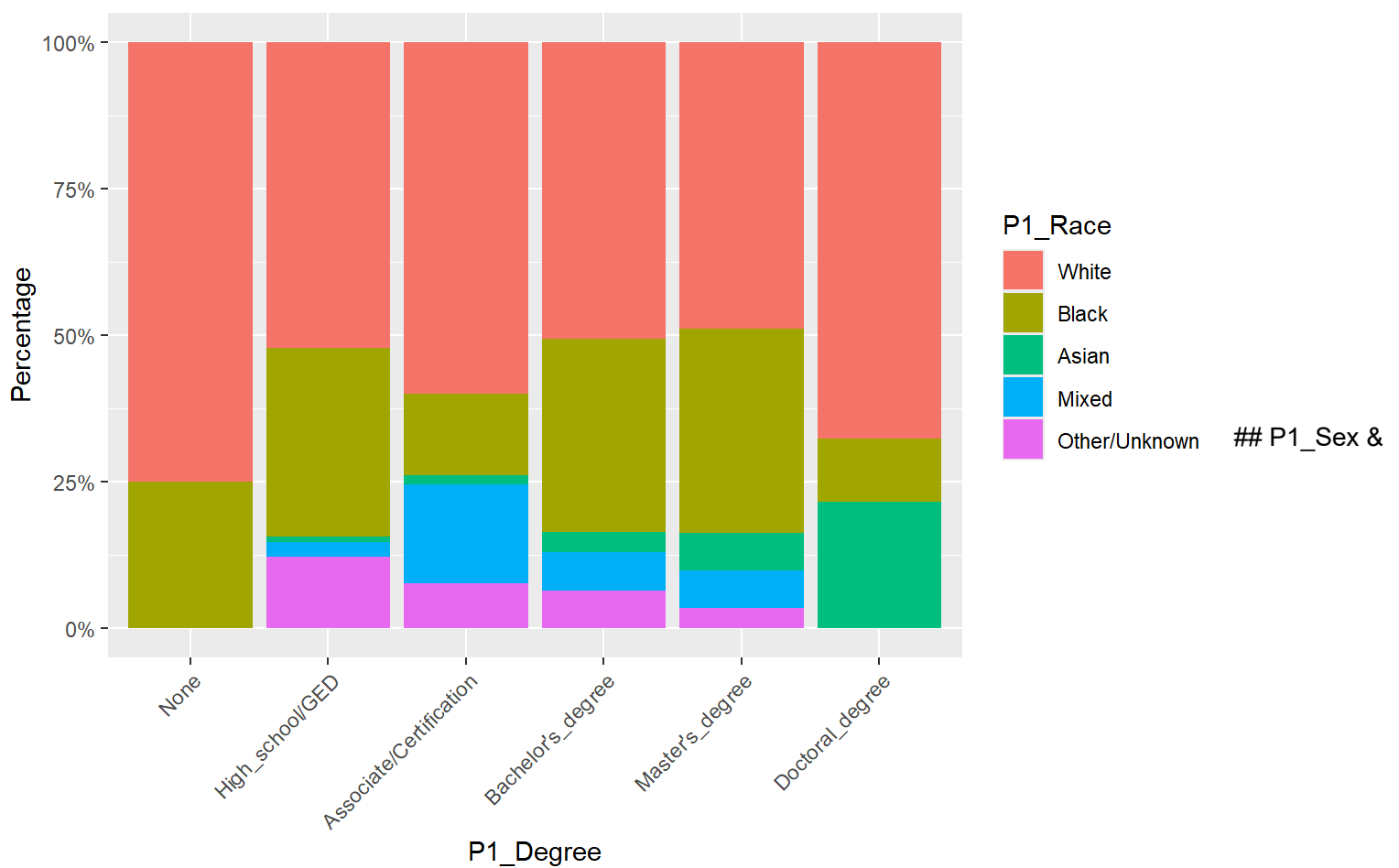
```

```
ggplot(data=ChildrenADF, aes(P1_Degree , fill=P1_Race ))+  
  geom_bar(position = "fill") +  
  scale_y_continuous(labels = scales::percent_format()) +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  
  labs(y = "Percentage")
```



```
ggplot(data = ChildrenADF, aes(x = P1_Degree, fill = P1_Race)) +  
  geom_bar(position = "fill") +  
  scale_y_continuous(labels = scales::percent_format()) +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  
  labs(y = "Percentage")
```



Single_Parent

```
ChildrenADF$P1_Sex <- factor(ChildrenADF$Parent1_Gender,
  levels = c(" Male","Female"))

ChildrenADF$Single_Parent <- factor(ChildrenADF$Single_Parent,
  levels = c("Single_Parent","Two-Parents"))
tabyl(ChildrenADF$P1_Sex)
```

```
## ChildrenADF$P1_Sex    n  percent
##           Male    61 0.112963
##           Female  479 0.887037
```

```
tabyl(ChildrenADF$Single_Parent )
```

```
## ChildrenADF$Single_Parent    n  percent
##           Single_Parent  139 0.2574074
##           Two-Parents  401 0.7425926
```

Order the levels of each categorical variable

```
ChildrenADF$P1_Income <- factor(ChildrenADF$Parent1_income ,
  levels = c( "0-19K","20K-39K","40K-59K", "60K-79K",
    "More than 80K","Unknown"))

tabyl(ChildrenADF$Child_Race4)
```

```
## ChildrenADF$Child_Race4    n    percent
##               White 327 0.60555556
##               Black 161 0.29814815
##               Asian  19 0.03518519
##               Otherwise 33 0.06111111
```

```
ChildrenADF$Child_Race <- factor(ChildrenADF$Child_Race,
  levels = c("White", "Black", "Asian", "American Indian", "Mixed", "Hawaiian/Pacific Islander", "Some other race", "Unknown"))
```

```
ChildrenADF$Child_Race4 = factor(ChildrenADF$Child_Race4,
  levels=c("White", "Black", "Asian", "Otherwise"))
```

```
tabyl(ChildrenADF$Child_Race4)
```

```
## ChildrenADF$Child_Race4    n    percent
##               White 327 0.60555556
##               Black 161 0.29814815
##               Asian  19 0.03518519
##               Otherwise 33 0.06111111
```

```
ChildrenADF$P1_Ethnicity= factor(ChildrenADF$P1_Ethnicity,
  levels=c("Not Hispanic", "Hispanic"))
```

```
tabyl(ChildrenADF$P1_Race)
```

```
## ChildrenADF$P1_Race    n    percent
##               White 288 0.53333333
##               Black 158 0.29259259
##               Asian  25 0.04629630
##               Mixed  34 0.06296296
##               Other/Unknown 35 0.06481481
```

```
ChildrenADF <- ChildrenADF %>%
  mutate(P1_Race4 = fct_collapse(P1_Race,
    "Otherwise" = c("Mixed", "Other/Unknown")))
```

```
ChildrenADF$P1_Race4 = factor(ChildrenADF$P1_Race4,
  levels=c("White", "Black", "Asian", "Otherwise"))
```

```
tabyl(ChildrenADF$P1_Race4)
```

```
## ChildrenADF$P1_Race4    n    percent
##               White 288 0.53333333
##               Black 158 0.2925926
##               Asian  25 0.0462963
##               Otherwise 69 0.1277778
```

Child Behavior Variable #0: Mouthing

```
## Documentation: 13.37. In general, does your child put items, such as toys,
# in his or her mouth, not including fi ngers and toes?
# 37. En lo general, ¿su hijo/a se pone artículos, comojuguetes, en la boca, sin # incluir los dedos de las
# manos ylos pies?
# radio, Required
# 0
# No / No
# 1
# Yes / Sí
# 89
# Don't know / No sabe
# 99
# Refused / Se negó a responder
ChildrenADF %>% tabyl(mouthing)
```

```
## mouthing    n      percent
##           0 148 0.274074074
##           1 390 0.722222222
##           89   2 0.003703704
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Mouthing= case_when(
    mouthing== "0"~ "No",
    mouthing== "1"~ "Yes",
    mouthing== "89"~ "Don't know",
    mouthing== "99"~ "Refused "

  ))
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Mouthing <- factor(ChildrenADF$Mouthing,
                              levels = c( "Yes","No" ))

ChildrenADF %>% tabyl(Mouthing)
```

```
## Mouthing    n      percent valid_percent
##       Yes 390 0.722222222      0.7249071
##       No 148 0.274074074      0.2750929
##      <NA>   2 0.003703704             NA
```

```
## since 72% of the valid responses are yes, we can replace the dont know by yes
```

Exploring Mouthing non-food items

```
# Old names and new names
old_names <- c(
  "mouthing2__1", "mouthing2__2", "mouthing2__3", "mouthing2__4",
  "mouthing2__5", "mouthing2__6", "mouthing2__7", "mouthing2__8",
  "mouthing2__9", "mouthing2__10"
)

new_suffixes <- c(
  "Dirt", "Paint_chips", "Toys", "Jewelry",
  "Vinyl_blinds", "Chalk", "Pottery", "Shoes", "Pets", "Other"
)

# # Create the new names by appending the suffix to "Mouthing_"
new_names <- paste0("Mouthing_", new_suffixes)
#

ChildrenADF <- ChildrenADF %>%
  rename_with(~ new_names, .cols = old_names)
```

```
## Warning: Using an external vector in selections was deprecated in tidysselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(old_names)
##
##   # Now:
##   data %>% select(all_of(old_names))
##
## See <https://tidysselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
# # Print the mapping (optional, for verification)
# mapping <- data.frame(Old = old_names, New = new_names)
# print(mapping)
#
# # If renaming columns in a data frame:
# # Example data frame
# df <- data.frame(matrix(ncol = length(old_names)))
# colnames(df) <- old_names
#
# # Rename the columns
# colnames(df) <- new_names
#
# # View the updated data frame
# print(df)
# names(ChildrenADF)

ChildrenADF %>% tabyl(Mouthing_Other)
```

```
## Mouthing_Other    n    percent
##                0 364 0.6740741
##                1 176 0.3259259
```

```
# Load necessary library
```

```
library(dplyr)
```

```
# List of target variables
```

```
mouthings_vars <- c(
  "Mouthing_Dirt", "Mouthing_Paint_chips", "Mouthing_Toys",
  "Mouthing_Jewelry", "Mouthing_Vinyl_blinds", "Mouthing_Chalk",
  "Mouthing_Pottery", "Mouthing_Shoes", "Mouthing_Pets",
  "Mouthing_Other"
)
```

```
# Compute the proportion of ones (assuming binary) or mean (if continuous)
```

```
proportions <- colMeans(ChildrenADF[, mouthings_vars], na.rm = TRUE)
```

```
# Create a two-row table
```

```
mouthings_table <- cbind(Variable = mouthings_vars,
                          Percentage = round(proportions*100, 3))
```

```
# # Reorder levels for only the specified variables in your dataset
```

```
# ChildrenADF[mouthing_vars] <- lapply(ChildrenADF[mouthing_vars], function(x) {
#   if (is.factor(x) && all(levels(x) %in% c("Yes", "No"))) {
#     factor(x, levels = c("Yes", "No")) # Reorder levels
#   } else {
#     x # Keep non-factor variables unchanged
#   }
# })
```

```
# Display the table
```

```
write.csv(mouthings_table, "Mouthing_Table.csv")
```

Child Behavior Variable #1: Pacifier

```
## Documentation: 13. Does your child use a pacifier?
#      #      0 No / No
#      1 Yes / Sí
#      89 Don't know / No sabe
#      99 Refused / Se negó a responder

# ChildrenADF1 1 %>% tabyl(pacifier)
ChildrenADF = ChildrenADF %>% mutate(
  Pacifier= case_when(
    pacifier== "0"~ "No",
    pacifier== "1"~ "Yes",
    pacifier== "89"~ "Don't know",
    pacifier== "99"~ "Refused "

  ))

## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Pacifier <- factor(ChildrenADF $Pacifier,
                              levels = c( "Yes","No"))

ChildrenADF %>% tabyl(Pacifier)
```

```
## Pacifier    n    percent
##          Yes  90 0.1666667
##          No 450 0.8333333
```

Child Behavior Variable #2: Pacifier use

```
## Documentation:
# pacifier_use_c1:
# Show the field ONLY if:[pacifier_c1]='1'
# 14. How often does your child use a pacifier?
# 1 Very often (multiple times an hour)
# 2 Often (multiple times a day)
# 3 Sometimes (about once a day)
# 4 Rarely (less than once a day)
# 5 Never
# 89 Don't know
# 99 Refused
ChildrenADF %>% tabyl(pacifier_use)
```

```
## pacifier_use    n    percent valid_percent
##           1   24 0.044444444    0.26666667
##           2   42 0.077777778    0.46666667
##           3   19 0.035185185    0.21111111
##           4    4 0.007407407    0.04444444
##          99    1 0.001851852    0.01111111
##          NA 450 0.833333333           NA
```



```

ChildrenADF = ChildrenADF %>% mutate(
  Pacifier_Use= case_when(
    pacifier_use == "1"~ "Very often",
    pacifier_use == "2"~ "Often",
    pacifier_use == "3"~ "Sometimes",
    pacifier_use == "4"~ "Rarely",
#   pacifier_use == "5"~ "Never",
#   pacifier_use == "89"~ "Don't know"
    pacifier_use == "99"~ "Refused",
    is.na(pacifier_use )~"No Pacifier",
  ))
## Ignoring "Never" & "Don't Know" from here on because they had zero counts

ChildrenADF$Pacifier_Use<- factor(ChildrenADF$Pacifier_Use,
  levels = c( "Very often","Often", "Sometimes","Rarely","No Pacifier","Refused"))
## Dropped , "Never" zero frequency

## Collapse Never and
ChildrenADF <- ChildrenADF %>%
mutate(Pacifier_Use = fct_collapse(Pacifier_Use,
  "Rarely/Refused" = c( "Rarely", "Refused") ))

ChildrenADF %>% tabyl(Pacifier_Use)

```

```

##   Pacifier_Use    n    percent
##      Very often  24 0.044444444
##         Often   42 0.077777778
##      Sometimes  19 0.035185185
## Rarely/Refused   5 0.009259259
##      No Pacifier 450 0.833333333

```

Child Behavior Variable #3: pacifier_wash

```

## Documentation:
#   Show the field ONLY if:
# [pacifier_c1]='1'
# 15. How often do you wash the pacifier?
# 1 Very often (multiple times an hour)
# 2 Often (multiple times a day)
# 3 Sometimes (about once a day)
# 4 Rarely (less than once a day)
# 5 Never
# 89 Don't know
# 99 Refused

ChildrenADF %>% tabyl(pacifier_wash)

```

```
##   pacifier_wash    n      percent valid_percent
##         1  21 0.038888889    0.23333333
##         2  26 0.048148148    0.28888889
##         3  26 0.048148148    0.28888889
##         4  16 0.029629630    0.17777778
##         5   1 0.001851852    0.01111111
##        NA 450 0.833333333          NA
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Pacifier_Wash= case_when(
    pacifier_wash == "1"~ "Very often",
    pacifier_wash == "2"~ "Often",
    pacifier_wash == "3"~ "Sometimes",
    pacifier_wash == "4"~ "Rarely",
    pacifier_wash == "5"~ "Never",
    # pacifier_wash == "89"~ "Don't know",
    # pacifier_wash == "99"~ "Refused",
    is.na(pacifier_wash)~"No Pacifier"
  ))
## Ignoring "Refused" & "Don't Know" from here on because they had zero counts
## Collapsing Rarely/Never
ChildrenADF <- ChildrenADF %>%
  mutate(Pacifier_Wash = fct_collapse(Pacifier_Wash,
    "Rarely/Never" = c( "Rarely", "Never") ))

ChildrenADF$Pacifier_Wash<- factor(ChildrenADF$Pacifier_Wash,
  levels = c( "Rarely/Never", "Sometimes", "Often",
    "Very often", "No Pacifier"))

ChildrenADF %>% tabyl(Pacifier_Wash)
```

```
##   Pacifier_Wash    n      percent
##   Rarely/Never  17 0.03148148
##      Sometimes  26 0.04814815
##         Often  26 0.04814815
##    Very often  21 0.03888889
##    No Pacifier 450 0.83333333
```

Child Behavior Variable #4: blanket

```
## Documentation:
# 16. Does your child have a favorite blanket he or she uses
# often?
# 0 No / No
# 1 Yes / Sí
# 89 Don't know / No sabe
# 99 Refused / Se negó a responder

# ChildrenADF %>% tabyl(blanket)
ChildrenADF = ChildrenADF %>% mutate(
  Blanket= case_when(
    blanket== "0"~ "No",
    blanket== "1"~ "Yes",
    blanket== "89"~ "Don't know",
    blanket== "99"~ "Refused "
  ))
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Blanket <- factor(ChildrenADF$Blanket,
                             levels = c( "Yes","No"))

ChildrenADF %>% tabyl(Blanket)
```

```
## Blanket    n    percent
##      Yes 226 0.4185185
##      No 314 0.5814815
```

Child Behavior Variable #5: blanket_wash

```
## check why this is not working

## Documentation:
# [blanket_wash_c1]
# Show the field ONLY if:
# [blanket_c1]='1'
# 17. How often do you wash your child's favorite blanket?
# 1 At least once a day
# 2 At least once a week
# 3 At least once a month
# 4 At least once every few months
# 5 At least once a year
# 6 Less than once a year, but at least once in two years
# 0 Never
# 89 Don't know
# 99 Refused

ChildrenADF %>% tabyl(blanket_wash )
```

```
## blanket_wash    n      percent valid_percent
##           0    1 0.001851852    0.004424779
##           1    5 0.009259259    0.022123894
##           2 152 0.281481481    0.672566372
##           3   61 0.112962963    0.269911504
##           4    5 0.009259259    0.022123894
##           5    2 0.003703704    0.008849558
##           NA 314 0.581481481             NA
```

```
str(ChildrenADF$blanket_wash)
```

```
## int [1:540] NA NA 2 NA NA NA NA NA NA 3 ...
```

```
# blanket_wash    n      percent valid_percent
#           0    1 0.001851852    0.004424779
#           1    5 0.009259259    0.022123894
#           2 152 0.281481481    0.672566372
#           3   61 0.112962963    0.269911504
#           4    5 0.009259259    0.022123894
#           5    2 0.003703704    0.008849558
#           NA 314 0.581481481             NA
ChildrenADF = ChildrenADF %>% mutate(
  Blanket_Wash = case_when(
    blanket_wash == "1"~ "Daily",
    blanket_wash == "2"~ "Weekly",
    blanket_wash == "3"~ "Monthly",
    blanket_wash == "4"~ "Every few Months",
    blanket_wash == "5"~ "Annually",
    # blanket_wash == "6"~ "Every 12-24 Months", ## 0 observed
    blanket_wash == "0"~ "Never",
    # blanket_wash == "89"~ "Don't know",
    # blanket_wash == "99"~ "Refused",
    is.na(blanket_wash)~"No Fav. Blanket"
  ))
## Ignoring "Never" & "Don't Know" from here on because they had zero counts

ChildrenADF %>% tabyl(Blanket_Wash )
```

```
## Blanket_Wash    n      percent
## Annually      2 0.003703704
## Daily         5 0.009259259
## Every few Months 5 0.009259259
## Monthly      61 0.112962963
## Never        1 0.001851852
## No Fav. Blanket 314 0.581481481
## Weekly     152 0.281481481
```

```
# ChildrenADF %>% tabyl(Blanket_wash )
# Every few months    5 0.009259259
#           Annually    2 0.003703704
# Every 12-24 Months    0 0.000000000
#           Never      1 0.001851852
#           Refused    0 0.000000000

# Collapsing the Levels "Every few months", "Annually", "Every 12-24 Months", "Never", "Refused", as they total
8 individuals only

ChildrenADF <- ChildrenADF %>%
mutate(Blanket_Wash = fct_collapse(as.factor(Blanket_Wash),
  "Rarely" = c("Every few Months", "Annually", "Never")
))

ChildrenADF$Blanket_Wash <- factor(ChildrenADF$Blanket_Wash,
  levels = c( "Rarely", "Monthly", "Weekly", "Daily",
    "No Fav. Blanket"))

ChildrenADF %>% tabyl(Blanket_Wash )
```

```
##      Blanket_Wash    n      percent
##           Rarely     8 0.014814815
##           Monthly   61 0.112962963
##           Weekly   152 0.281481481
##           Daily     5 0.009259259
## No Fav. Blanket  314 0.581481481
```

Child Behavior Variable #6: Stuffedanimal

```
# [stuffedanimal_c1] 18. Does your child have a favorite stuffed animal he or
# she uses often?
# ¿Su hijo/a tiene un peluche favorito que usa con
# frecuencia?
# radio, Required
# 0 No / No
# 1 Yes / Sí
# 89 Don't know / No sabe
# 99 Refused / Se negó a responder
```

```
ChildrenADF %>% tabyl(stuffedanimal)
```

```
## stuffedanimal    n      percent
##           0  314 0.581481481
##           1  225 0.416666667
##           89    1 0.001851852
```

```

ChildrenADF = ChildrenADF %>% mutate(
  Stuffedanimal = case_when(
    stuffedanimal == "0"~ "No",
    stuffedanimal == "1"~ "Yes",
    stuffedanimal == "89"~ "No", ## replaced Dont know by No only one case
    stuffedanimal == "99"~ "Refused "
  ))
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Stuffed_Animal <- factor(ChildrenADF$Stuffedanimal,
  levels = c( "Yes","No"))
## Replace Dont know by No from here on to simplify only 1 case of dont know
ChildrenADF %>% tabyl(Stuffed_Animal)

```

```

## Stuffed_Animal    n    percent
##                Yes 225 0.4166667
##                No 315 0.5833333

```

Child Behavior Variable #7: Stuffed_Animal_Wash

```

# [stuffedanimal_wash_c1]
# Show the field ONLY if:
# [stuffedanimal_c1]='1'
# 19. How often do you wash your child's favorite stuffed
# animal?
#
# 1 At least once a day
# 2 At least once a week
# 3 At least once a month
# 4 At least once every few months
# 5 At least once a year
# 6 Less than once a year, but at least once in two years
# 0 Never
# 89 Don't know
# 99 Refused

```

```
ChildrenADF %>% tabyl(stuffedanimal_wash )
```

```

## stuffedanimal_wash    n    percent valid_percent
##                0  42 0.077777778    0.186666667
##                1   3 0.005555556    0.013333333
##                2  72 0.133333333    0.320000000
##                3  69 0.127777778    0.306666667
##                4  24 0.044444444    0.106666667
##                5  10 0.018518519    0.044444444
##                6   4 0.007407407    0.017777778
##               89   1 0.001851852    0.004444444
##               NA 315 0.583333333             NA

```

```
ChildrenADF = ChildrenADF %>% mutate(
  Stuffed_Animal_Wash = case_when(
    stuffedanimal_wash == "1"~ "Daily",
    stuffedanimal_wash == "2"~ "Weekly",
    stuffedanimal_wash == "3"~ "Monthly",
    stuffedanimal_wash == "4"~ "Every few months",
    stuffedanimal_wash == "5"~ "Annually",
    stuffedanimal_wash == "6"~ "Every 12-24 Months",
    stuffedanimal_wash == "0"~ "Never",
    stuffedanimal_wash == "89"~ "Don't know",
    stuffedanimal_wash == "99"~ "Refused",
    is.na(stuffedanimal_wash)~"No Fav Stuffie"
  ))
##
```

```
# Stuffed_Animal_Wash      n      percent
#      Annually    10 0.018518519
#      Daily       3 0.005555556
#      Don't know   1 0.001851852
# Every 12-24 Months  4 0.007407407
# Every few months  24 0.044444444
#      Monthly    69 0.127777778
#      Never     42 0.077777778
# No Fav. Stuffie 315 0.583333333
#      Weekly    72 0.133333333
```

```
ChildrenADF <- ChildrenADF %>%
  mutate(Stuffed_Animal_Wash = fct_collapse(Stuffed_Animal_Wash,
    "Rarely" = c( "Every 12-24 Months",
      "Every few months", "Don't know","Annually") ))
```

```
ChildrenADF <- ChildrenADF %>%
  mutate(Stuffed_Animal_Wash = fct_collapse(Stuffed_Animal_Wash,
    "Daily/Weekly" = c("Daily","Weekly") ))
```

```
ChildrenADF$Stuffed_Animal_Wash <- factor(ChildrenADF$Stuffed_Animal_Wash ,
  levels = c("Never", "Rarely", "Monthly","Daily/Weekly",
    "No Fav Stuffie"))
```

```
ChildrenADF %>% tabyl(Stuffed_Animal_Wash)
```

```
## Stuffed_Animal_Wash      n      percent
##      Never    42 0.07777778
##      Rarely   39 0.07222222
##      Monthly  69 0.12777778
## Daily/Weekly  75 0.13888889
## No Fav Stuffie 315 0.58333333
```

Child Behavior Variable #8: sucking

```
# [sucking_c1] 20. How often does your child suck his or her fingers,  
# toes, or thumbs?  
# 1 Very often (multiple times an hour)  
# 2 Often (multiple times a day)  
# 3 Sometimes (about once a day)  
# 4 Rarely (less than once a day)  
# 5 Never  
# 89 Don't know  
# 99 Refused
```

```
ChildrenADF %>% tabyl(sucking)
```

```
##   sucking    n  percent  
##      1  58 0.1074074  
##      2  95 0.1759259  
##      3  80 0.1481481  
##      4  84 0.1555556  
##      5 223 0.4129630
```

```
ChildrenADF = ChildrenADF %>% mutate(  
  Sucking = case_when(  
    sucking == "1"~ "Very Often",  
    sucking == "2"~ "Often",  
    sucking == "3"~ "Sometimes",  
    sucking == "4"~ "Rarely",  
    sucking == "5"~ "Never",  
    sucking == "89"~ "Don't know",  
    sucking == "99"~ "Refused "  
  ))  
  
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts  
  
ChildrenADF$Sucking <- factor(ChildrenADF$Sucking,  
  levels = c( "Very Often", "Often", "Sometimes", "Rarely", "Never"))  
  
ChildrenADF %>% tabyl(Sucking)
```

```
##   Sucking    n  percent  
## Very Often  58 0.1074074  
##      Often  95 0.1759259  
## Sometimes  80 0.1481481  
##      Rarely  84 0.1555556  
##      Never 223 0.4129630
```


Child Behavior Variable #9: Handwash

```
# [handwash_c1] 21. Are your child's hands washed before meals or
# snacks?
# 0 No / No
# 1 Yes / Sí
# 2 Sometimes / A veces
# 89 Don't know / No sabe
# 99 Refused / Se negó a responde
```

```
ChildrenADF %>% tabyl(handwash)
```

```
##   handwash    n   percent
##         0   52 0.0962963
##         1  330 0.6111111
##         2  158 0.2925926
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Handwash = case_when(
    handwash == "0" ~ "No",
    handwash == "1" ~ "Yes",
    handwash == "2" ~ "Sometimes",
    handwash == "89" ~ "Don't know",
    handwash == "99" ~ "Refused "
  ))
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Handwash <- factor(ChildrenADF$Handwash,
  levels = c( "No","Sometimes","Yes"))

ChildrenADF %>% tabyl(Handwash)
```

```
##   Handwash    n   percent
##         No   52 0.0962963
##   Sometimes 158 0.2925926
##         Yes  330 0.6111111
```

Child Behavior Variable # 10:away_weekday

```
# 3 [away_weekday_c1] 22. How many hours per day is your child away from
# home on a typical weekday?
# 1 0-4 hours per day
# 2 5-10 hours per day
# 3 11-16 hours per day
# 4 Greater than 16 hours per day
# 89 Don't know
# 99 Refused
```

```
ChildrenADF %>% tabyl(away_weekday)
```

```
## away_weekday    n      percent
##              1 268 0.496296296
##              2 257 0.475925926
##              3  12 0.022222222
##              4   3 0.005555556
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Hours_Away_weekday = case_when(
    away_weekday == "1"~ "0-4 Hours",
    away_weekday == "2"~ "5-10 Hours",
    away_weekday == "3"~ "11-16 Hours",
    away_weekday == "4"~ ">16 Hours",
    away_weekday == "89"~ "Don't know",
    away_weekday == "99"~ "Refused "
  ))
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Hours_Away_weekday<- factor(ChildrenADF$Hours_Away_weekday,
  levels = c("0-4 Hours","5-10 Hours","11-16 Hours", ">16 Hours"))

ChildrenADF %>% tabyl(Hours_Away_weekday)
```

```
## Hours_Away_weekday    n      percent
##           0-4 Hours 268 0.496296296
##           5-10 Hours 257 0.475925926
##          11-16 Hours  12 0.022222222
##           >16 Hours   3 0.005555556
```

Hours_Away_weekend

```
# [away_weekend_c1] 23. How many hours per day is your child away from
# home on a typical weekend day?
# 1 0-4 hours per day
# 2 5-10 hours per day
# 3 11-16 hours per day
# 4 Greater than 16 hours per day
# 89 Don't know
# 99 Refused
```

```
ChildrenADF %>% tabyl(away_weekend)
```

```
## away_weekend    n      percent
##              1 418 0.774074074
##              2 102 0.188888889
##              3  10 0.018518519
##              4   9 0.016666667
##             89   1 0.001851852
```

```

ChildrenADF = ChildrenADF %>% mutate(
  Hours_Away_weekend = case_when(
    away_weekend == "1"~ "0-4 Hours",
    away_weekend == "2"~ "5-10 Hours",
    away_weekend == "3"~ "11-16 Hours",
    away_weekend == "4"~ ">16 Hours",
    away_weekend == "89"~ "Don't know",
    away_weekend == "99"~ "Refused "
  ))
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Hours_Away_weekend <- factor(ChildrenADF$Hours_Away_weekend,
  levels = c("0-4 Hours", "5-10 Hours", "11-16 Hours",
    ">16 Hours", "Don't know"))

ChildrenADF %>% tabyl(Hours_Away_weekend)

```

```

##   Hours_Away_weekend    n    percent
##           0-4 Hours 418 0.774074074
##           5-10 Hours 102 0.188888889
##          11-16 Hours  10 0.018518519
##           >16 Hours   9 0.016666667
##           Don't know   1 0.001851852

```

Hours_Outdoors_Home

```

# [playoutsidehome_c1] 24. How much time per week does the child play
# outdoors at home??
# 1 Less than one hour
# 2 1-5 hours
# 3 6-10 hours
# 4 More than 10 hours
# 89 Don't know
# 99 Refused

```

```

ChildrenADF %>% tabyl(playoutsidehome)

```

```

##   playoutsidehome    n    percent
##           1 118 0.218518519
##           2 249 0.461111111
##           3 100 0.185185185
##           4  72 0.133333333
##          89   1 0.001851852

```

```

ChildrenADF = ChildrenADF %>% mutate(
  Hours_Outdoors_Home = case_when(
    playoutsidehome == "1"~ "<1 Hour",
    playoutsidehome == "2"~ "1-5 Hours",
    playoutsidehome == "3"~ "6-10 Hours",
    playoutsidehome == "4"~ ">10 Hours",
    playoutsidehome == "89"~ "Don't know",
    playoutsidehome == "99"~ "Refused"
  ))
## Ignoring "Don't Know" & "Refused" from here on because they had zero counts

ChildrenADF$Hours_Outdoors_Home <- factor(ChildrenADF$Hours_Outdoors_Home,
  levels = c("<1 Hour", "1-5 Hours", "6-10 Hours",
    ">10 Hours", "Don't know"))

ChildrenADF %>% tabyl(Hours_Outdoors_Home)

```

```

## Hours_Outdoors_Home  n    percent
##           <1 Hour 118 0.218518519
##           1-5 Hours 249 0.461111111
##           6-10 Hours 100 0.185185185
##           >10 Hours  72 0.133333333
##           Don't know   1 0.001851852

```

Child Behavior Variable # 13: playoutsidenothome

```

# playoutsidenothome_c1] 25. Where else does your child often spend time
# playing outdoors other than at home?
# 1 playoutsidenothome_c1__1 Park
# 2 playoutsidenothome_c1__2 Daycare
# 3 playoutsidenothome_c1__3 Babysitter's home
# 4 playoutsidenothome_c1__4 Relative's home
# 5 playoutsidenothome_c1__5 Other (please s
# 79 playoutsidenothome_c1__79 Not applicable
# 89 playoutsidenothome_c1__89 Don't know
# 99 playoutsidenothome_c1__99 Refused

```

```

ChildrenADF %>% tabyl(playoutsidenothome__1)

```

```

## playoutsidenothome__1  n    percent
##           0 168 0.3111111
##           1 372 0.6888889

```

```

ChildrenADF = ChildrenADF %>% mutate(
  Play_Park = case_when(
    playoutsidenothome__1 == "0"~ "No",
    playoutsidenothome__1 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Play_Park)

```

```
## Play_Park    n    percent
##           No 168 0.3111111
##           Yes 372 0.6888889
```

```
ChildrenADF %>% tabyl(playoutsidenothome__2)
```

```
## playoutsidenothome__2    n    percent
##                   0 390 0.7222222
##                   1 150 0.2777778
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Play_Daycare= case_when(
    playoutsidenothome__2 == "0"~ "No",
    playoutsidenothome__2 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Play_Daycare)
```

```
## Play_Daycare    n    percent
##           No 390 0.7222222
##           Yes 150 0.2777778
```

```
ChildrenADF %>% tabyl(playoutsidenothome__3)
```

```
## playoutsidenothome__3    n    percent
##                   0 515 0.9537037
##                   1  25 0.0462963
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Play_Babysitter_Home= case_when(
    playoutsidenothome__3 == "0"~ "No",
    playoutsidenothome__3 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Play_Babysitter_Home)
```

```
## Play_Babysitter_Home    n    percent
##           No 515 0.9537037
##           Yes  25 0.0462963
```

```
ChildrenADF %>% tabyl(playoutsidenothome__4)
```

```
## playoutsidenothome__4    n    percent
##                   0 375 0.6944444
##                   1 165 0.3055556
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Play_Relative_Home= case_when(
    playoutsidenothome__4 == "0"~ "No",
    playoutsidenothome__4 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Play_Relative_Home)
```

```
## Play_Relative_Home    n    percent
##                      No 375 0.6944444
##                      Yes 165 0.3055556
```

```
ChildrenADF %>% tabyl(playoutsidenothome__5)
```

```
## playoutsidenothome__5    n    percent
##                      0 388 0.7185185
##                      1 152 0.2814815
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Play_Other= case_when(
    playoutsidenothome__5 == "0"~ "No",
    playoutsidenothome__5 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Play_Other)
```

```
## Play_Other    n    percent
##             No 388 0.7185185
##             Yes 152 0.2814815
```

```
## playoutsidenothome_othr Check if the original file had this variable
# Define the variable names
PlayVars <- c("Hours_Outdoors_Home", "Play_Park", "Play_Daycare",
              "Play_Babysitter_Home", "Play_Relative_Home", "Play_Other")
```

```
# Reorder levels for only the specified variables in your dataset
ChildrenADF[PlayVars] <- lapply(ChildrenADF[PlayVars], function(x) {
  if (is.factor(x) || is.character(x)) {
    factor(x, levels = c("Yes", "No"))
  } else {
    x # Keep non-factor variables unchanged
  }
})
```

Sandbox Variables

```
# [sandbox_c1] 26. Does your child play in a sandbox at any of the
# following locations?
# 1 sandbox_c1__1 Home
# 2 sandbox_c1__2 Park
# 3 sandbox_c1__3 Daycare
# 4 sandbox_c1__4 Babysitter's home /
# 5 sandbox_c1__5 Relative's home
# 6 sandbox_c1__6 Other (please specify) /
# {sandbox_other_c1}
# 79 sandbox_c1__79 Not applicable
# 89 sandbox_c1__89 Don't know
# 99 sandbox_c1__99 Refused

# "Sandbox_Home"
# [157] "Sandbox_Park"          "Sandbox_Daycare"          "Sandbox_Babysitter"      "Sandbox_Relative"
# [161] "Sandbox_Other"

ChildrenADF %>% tabyl(sandbox__1)
```

```
## sandbox__1    n    percent
##           0 457 0.8462963
##           1  83 0.1537037
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Sandbox_Home = case_when(
    sandbox__1== "0"~ "No",
    sandbox__1 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Sandbox_Home)
```

```
## Sandbox_Home    n    percent
##           No 457 0.8462963
##           Yes  83 0.1537037
```

```
ChildrenADF %>% tabyl(sandbox__2)
```

```
## sandbox__2    n    percent
##           0 443 0.8203704
##           1  97 0.1796296
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Sandbox_Park = case_when(
    sandbox__2== "0"~ "No",
    sandbox__2 == "1"~ "Yes",
  ))
ChildrenADF%>% tabyl(Sandbox_Park)
```

```
## Sandbox_Park    n    percent
##                No 443 0.8203704
##                Yes  97 0.1796296
```

```
ChildrenADF %>% tabyl(sandbox__3)
```

```
## sandbox__3    n    percent
##           0 494 0.91481481
##           1  46 0.08518519
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Sandbox_Daycare = case_when(
    sandbox__3== "0"~ "No",
    sandbox__3 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Sandbox_Daycare)
```

```
## Sandbox_Daycare    n    percent
##                No 494 0.91481481
##                Yes  46 0.08518519
```

```
ChildrenADF %>% tabyl(sandbox__4)
```

```
## sandbox__4    n    percent
##           0 538 0.996296296
##           1   2 0.003703704
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Sandbox_Babysitter = case_when(
    sandbox__4== "0"~ "No",
    sandbox__4 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Sandbox_Babysitter)
```

```
## Sandbox_Babysitter    n    percent
##                No 538 0.996296296
##                Yes   2 0.003703704
```

```
ChildrenADF %>% tabyl(sandbox__5)
```

```
## sandbox__5    n    percent
##           0 516 0.955555556
##           1  24 0.044444444
```



```
ChildrenADF = ChildrenADF %>% mutate(
  Sandbox_Relative= case_when(
    sandbox__5== "0"~ "No",
    sandbox__5 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Sandbox_Relative)
```

```
##  Sandbox_Relative    n    percent
##                No 516 0.95555556
##                Yes  24 0.04444444
```

```
ChildrenADF %>% tabyl(sandbox__6)
```

```
##  sandbox__6    n    percent
##           0 496 0.91851852
##           1  44 0.08148148
```

```
ChildrenADF = ChildrenADF %>% mutate(
  Sandbox_Other= case_when(
    sandbox__6== "0"~ "No",
    sandbox__6 == "1"~ "Yes",
  ))
ChildrenADF %>% tabyl(Sandbox_Other)
```

```
##  Sandbox_Other    n    percent
##                No 496 0.91851852
##                Yes  44 0.08148148
```

```
ChildrenADF <- ChildrenADF %>%
  mutate(across(c("Sandbox_Home", "Sandbox_Park", "Sandbox_Daycare",
    "Sandbox_Babysitter", "Sandbox_Relative", "Sandbox_Other"), as.factor))

SandboxVars <- c( "Sandbox_Home",
"Sandbox_Park","Sandbox_Daycare","Sandbox_Babysitter","Sandbox_Relative")

# Reorder Levels for only the specified variables in your dataset
ChildrenADF[SandboxVars] <- lapply(ChildrenADF[SandboxVars], function(x) {
  if (is.factor(x) || is.character(x)) {
    factor(x, levels = c("Yes", "No"))
  } else {
    x # Keep non-factor variables unchanged
  }
})
```

```

write.csv(ChildrenADF, "Child_Clean_March_22_24.csv")
# Step 2: Select relevant columns based on the list of features
# selected_columns <- c('Parent1_gender', 'Parent1_Latino', 'Parent1_Mixed_Race', 'Parent1_Race',
#                        'Race_P1', 'Parent1_income', 'Race_P2', 'ChildRank', 'age_group',
#                        'gender', 'Child_Race4', 'Child_age_group', 'Child_Sex',
#                        'Income_p1_num', 'Income_p2_num', 'Household_income', 'Pacifier')
#
# # Filter the dataset to include only the selected columns
# data <- data %>% select(all_of(selected_columns))
# summary(data)
# Create a subset of the data by keeping only the specified columns
#
# data_subset <- ChildrenADF %>%
#   select( record_id, householdid, Age_P1, Parent1_degree, Parent1_Race, Race_P1, Parent1_Latino, P1_Workst
# atus, Child_Age_Group,
#   Parent1_Income, Race_P2, Income_p1_num, Income_p2_num, Household_income, Workstatus_P1,
#   Child_Sex, City, Child_Race, Child_Race4,
#   Pacifier, Pacifier_Use, Pacifier_Wash,
#   Blanket, Blanket_wash, Stuffedanimal, Stuffedanimal_Wash,
#   Sucking, Handwash, Away_weekday, Away_weekday1, Away_weekend, Playoutsidehome,
#   Park, Daycare, Babysitter_home, Relativehome, PlayOther, SandboxHome, SandboxPar
# k,
#   SandboxDaycare, SandboxBabysitter, SandboxRelative, SandboxOther
#   )
#
# write.csv(data_subset, "Child_Clean430ct13.csv")

# ## Using the name data for the analysis dataset from here on
# data <- data_subset
# # Fill missing numerical values with the median of each column
# numeric_cols <- sapply(data, is.numeric)
# data[numeric_cols] <- lapply(data[numeric_cols], function(x) ifelse(is.na(x), median(x, na.rm=TRUE), x))
#
# # Fill missing categorical values with the mode (most frequent value)
# categorical_cols <- sapply(data, is.factor)
# data[categorical_cols] <- lapply(data[categorical_cols], function(x) {
#   mode_value <- names(sort(table(x), decreasing=TRUE))[1] # Get the mode
#   ifelse(is.na(x), mode_value, x)
# })
#
# sum(is.na(data))
# summary(data)

```

```
data= ChildrenADF
```

Collapsing P1-Race into 4 levels

```

data <- data %>%
  mutate(P1_Race4 = fct_collapse(P1_Race,
    "Mixed/Other" = c( "Mixed" , "Other/Unknown") ))
data$P1_Race4 = factor(data$P1_Race4, levels=c( "White", "Black", "Asian", "Mixed/Other"))

tabyl(data$P1_Race4)

```

```
## data$P1_Race4    n    percent
##           White 288 0.5333333
##           Black 158 0.2925926
##           Asian  25 0.0462963
##           Mixed/Other 69 0.1277778
```

```
## r staging variable names
```

```
# names(data)
```

```
# glimpse(data)
```

```
DemoVars <- c( "P1_Race4","P1_Age","P1_Sex","P1_Ethnicity", "P1_Degree","P1_Workstatus", "Single_Parent", "Household_income", "City", "Child_Age_Gp", "Child_Sex", "Child_Race4")
```

```
MouthingVars= c("Mouthing", "Mouthing_Toys", "Mouthing_Shoes",  
  "Mouthing_Jewelry", "Mouthing_Dirt")
```

```
PaciVars <- c( "Pacifier")
```

```
PaciVars2 <- c("Pacifier_Use" , "Pacifier_Wash")
```

```
BlanketVars<- c( "Blanket", "Blanket_Wash")
```

```
StuffyVars <- c( "Stuffed_Animal", "Stuffed_Animal_Wash")
```

```
HabitsVars <- c( "Sucking", "Handwash")
```

```
## Review from here on!!
```

```
AwayVars <- c( "Hours_Away_weekend", "Hours_Away_weekday")
```

```
PlayVars <- c("Hours_Outdoors_Home", "Play_Park",
```

```
  "Play_Daycare", "Play_Babysitter_Home", "Play_Relative_Home", "Play_Other")
```

```
SandboxVars <- c( "Sandbox_Home",
```

```
"Sandbox_Park", "Sandbox_Daycare", "Sandbox_Babysitter", "Sandbox_Relative")
```

Use the updates function with ggsave and rank variables

```
library(dplyr)
library(janitor)
library(ggplot2)
library(rlang)
library(ggmosaic)
library(DescTools) # for Cramér's V

# Define the function
Analyze_Bivarite <- function(data, var1, var2) {
  # The check for categorical vars is temporarily disabled
  # # Check if the variables are categorical
  var1_class <- ifelse(is.factor(data[[var1]]) || is.character(data[[var1]]), "categorical", "quantitative")
  var2_class <- ifelse(is.factor(data[[var2]]) || is.character(data[[var2]]), "categorical", "quantitative")

  if (var1_class == "quantitative" || var2_class == "quantitative") {
    stop("Both variables need to be categorical for this analysis.")
  }

  # Generate frequency and relative frequency tables with totals using tabyl
  ## Modify the total by column
  freq_table <- data %>% tabyl(!!sym(var1), !!sym(var2))

  relative_freq_table_col <- freq_table %>%
    # adorn_totals("row") %>% ## commented Feb 24
    adorn_totals("col") %>%
    adorn_percentages("col") %>%
    adorn_pct_formatting()

  # Print the analysis title
  cat("Bivariate Analysis of", var1, "and", var2, "\n\n")

  # Print the tables
  cat("Frequency Table:\n")
  print(freq_table)

  cat("\nRelative Frequency Table:\n")
  print(relative_freq_table_col)

  # # Apply Fisher's exact test and handle errors or warnings
  # fisher_test_result <- tryCatch({
  #   fisher.test(as.matrix(freq_table[, -1]))
  # }, warning = function(w) {
  #   message("Warning in Fisher's exact test: ", conditionMessage(w))
  #   return(NULL)
  # }, error = function(e) {
  #   message("Error in Fisher's exact test: ", conditionMessage(e))
  #   return(NULL)
  # })
  #
  # if (!is.null(fisher_test_result)) {
```

```

#   cat("\nFisher's Exact Test Result:\n")
#   print(fisher_test_result)
# }

## Apply Fisher's test with simulations enabled
fisher_test_result <- tryCatch({
  # Perform Fisher's Exact Test with simulation enabled
  fisher.test(as.matrix(freq_table[,-1]), simulate.p.value = TRUE, B = 10000)
}, error = function(e) {
  # Handle errors gracefully and provide meaningful feedback
  message("Error in Fisher's Exact Test: ", e$message)
  NULL # Return NULL or another placeholder if the test fails
})

# Calculate Cramér's V
cramers_v <- tryCatch({
  CramerV(as.matrix(freq_table[,-1]), correct = TRUE)
}, error = function(e) {
  message("Error in Cramér's V calculation: ", conditionMessage(e))
  return(NULL)
})

if (!is.null(cramers_v)) {
  cat("\nCramér's V: ", crammers_v, "\n")
}

# Function to save plot
save_plot <- function(plot, title) {
  filename <- gsub(" ", "_", title)
  ggsave(paste0(filename, ".jpeg"), plot = plot, device = "jpeg")
  # ggsave(paste0(filename, ".eps"), plot = plot, device = "eps")
}

# Drop NAs temporarily from the column for the plot

# Plot mosaic plot
tryCatch({
  mosaic_title <- paste("Mosaic Plot of", var2, "and", var1)
  mosaic_plot <- ggplot(data) +
    geom_mosaic(aes(weight = 1, x = product(!sym(var2)), fill = !sym(var1))) +
    labs(title = mosaic_title, x = var2, y = var1) +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))

  print(mosaic_plot)
  save_plot(mosaic_plot, mosaic_title)
}, error = function(e) {
  message("Error in mosaic plot: ", conditionMessage(e))
})

# Return the tables and test results as a list, including Cramér's V
return(list(counts = freq_table,
            relative_frequencies = relative_freq_table_col,

```

```

    fisher_test = fisher_test_result,
    cramer_v = cramers_v))
}

```

Exploring Mouthing Variables

```

data= ChildrenADF
list1= c("Mouthing")
list2= DemoVars

# Loop through each combination of var1 and var2 and calculate Cramér's V for ranking
cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(), stringsAsFactors = FALSE)
results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = "_vs_")]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values, data.frame(var1 = var1,
        var2 = var2, cramer_v = round(result$cramer_v, digits=3)
      ))
    }
  }
}

```

```

##
##
## ----- Analysis for Mouthing and P1_Race4 -----
## Bivariate Analysis of Mouthing and P1_Race4
##
## Frequency Table:
##   Mouthing White Black Asian Otherwise
##      Yes   212   113   14      51
##      No    75    45   11      17
##      <NA>    1     0    0       1
##
## Relative Frequency Table:
##   Mouthing White Black Asian Otherwise Total
##      Yes 73.6% 71.5% 56.0%   73.9% 72.2%
##      No 26.0% 28.5% 44.0%   24.6% 27.4%
##      <NA>  0.3%  0.0%  0.0%    1.4%  0.4%
##
## Cramér's V:  0.02781221

```

```

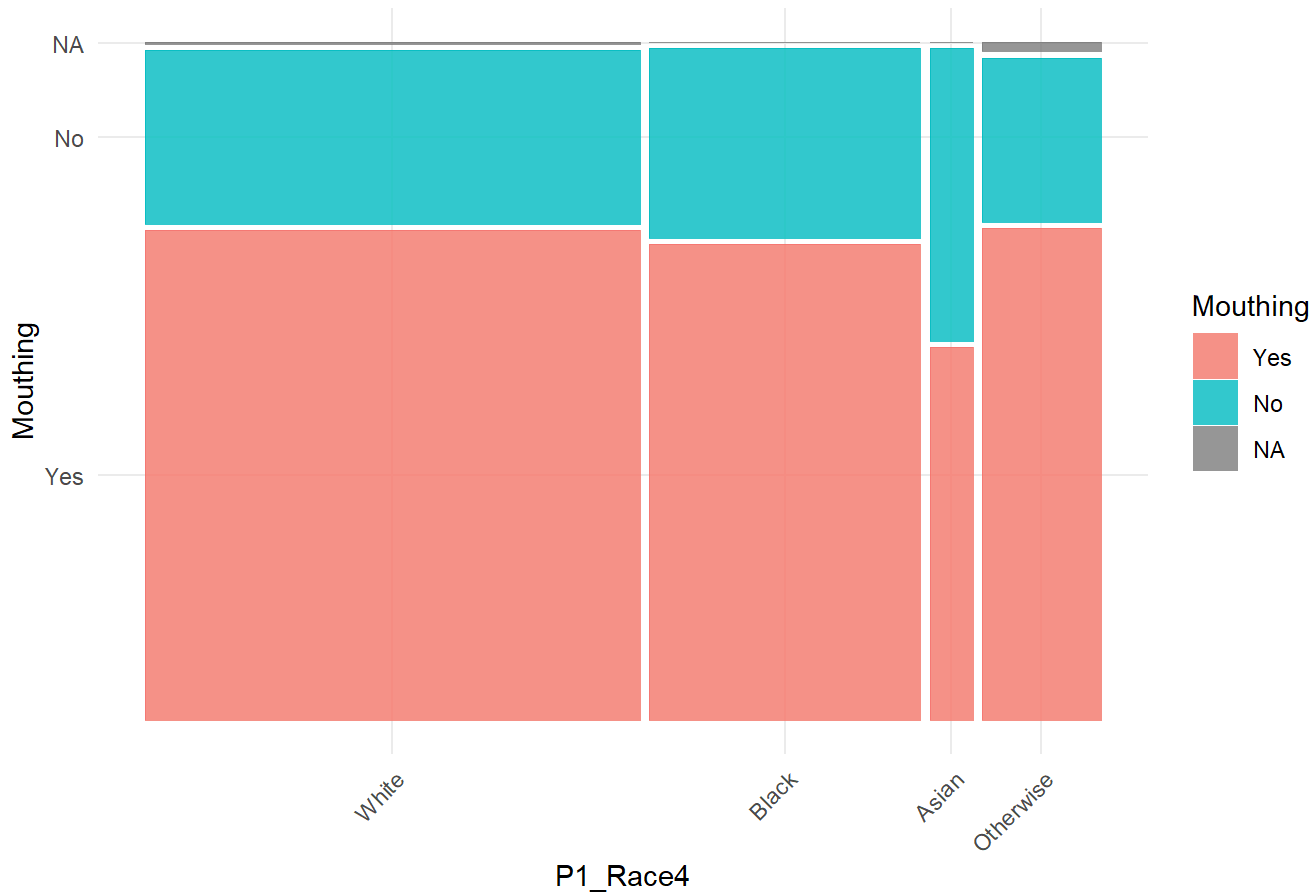
## Warning: The `scale_name` argument of `continuous_scale()` is deprecated as of ggplot2
## 3.5.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```
## Warning: The `trans` argument of `continuous_scale()` is deprecated as of ggplot2 3.5.0.  
## i Please use the `transform` argument instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```

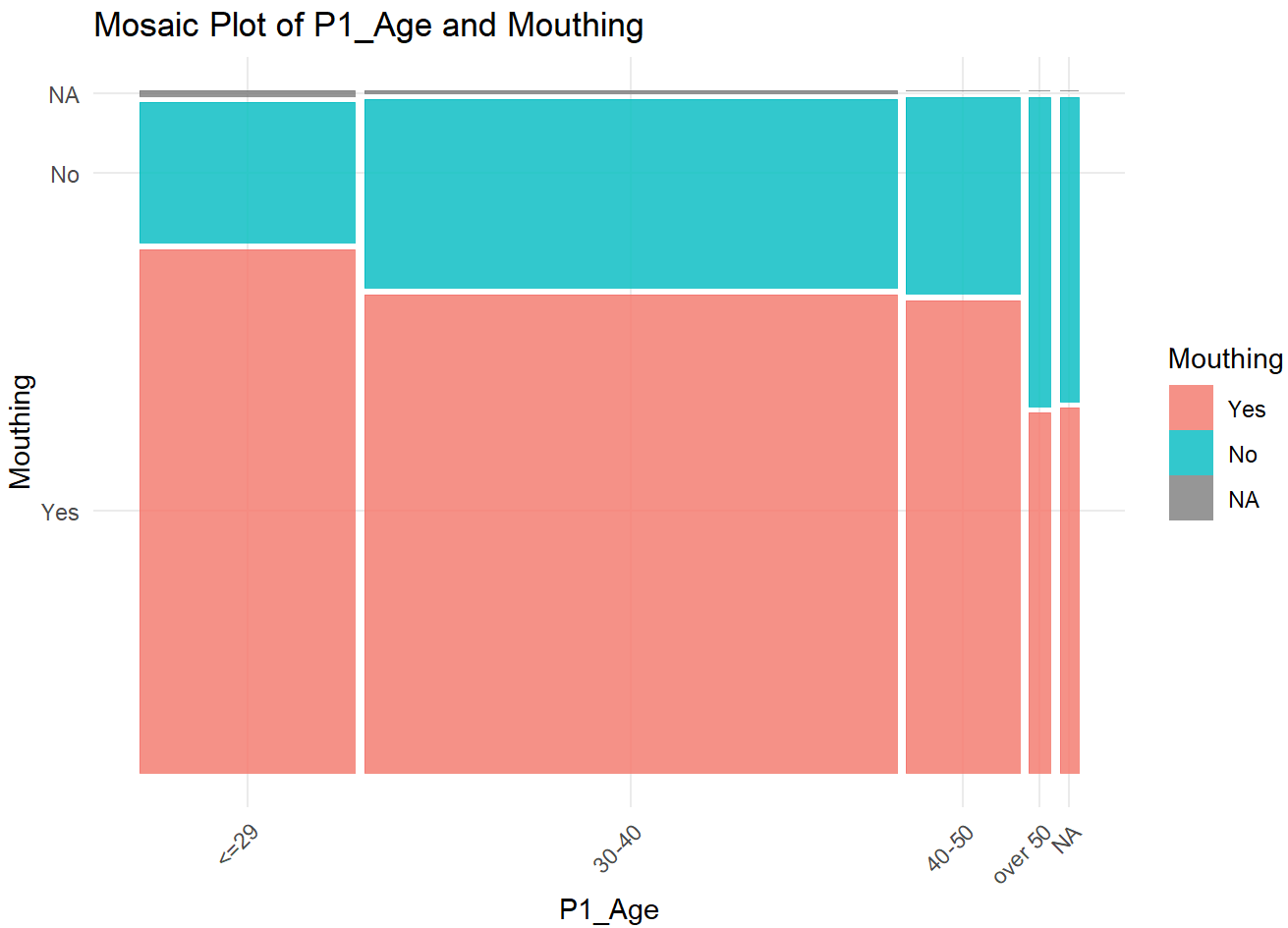
```
## Warning: `unite_()` was deprecated in tidyr 1.2.0.  
## i Please use `unite()` instead.  
## i The deprecated feature was likely used in the ggmosaic package.  
## Please report the issue at <https://github.com/haleyjeppson/ggmosaic>.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```

Mosaic Plot of P1_Race4 and Mouthing



```
## Saving 7 x 5 in image
```

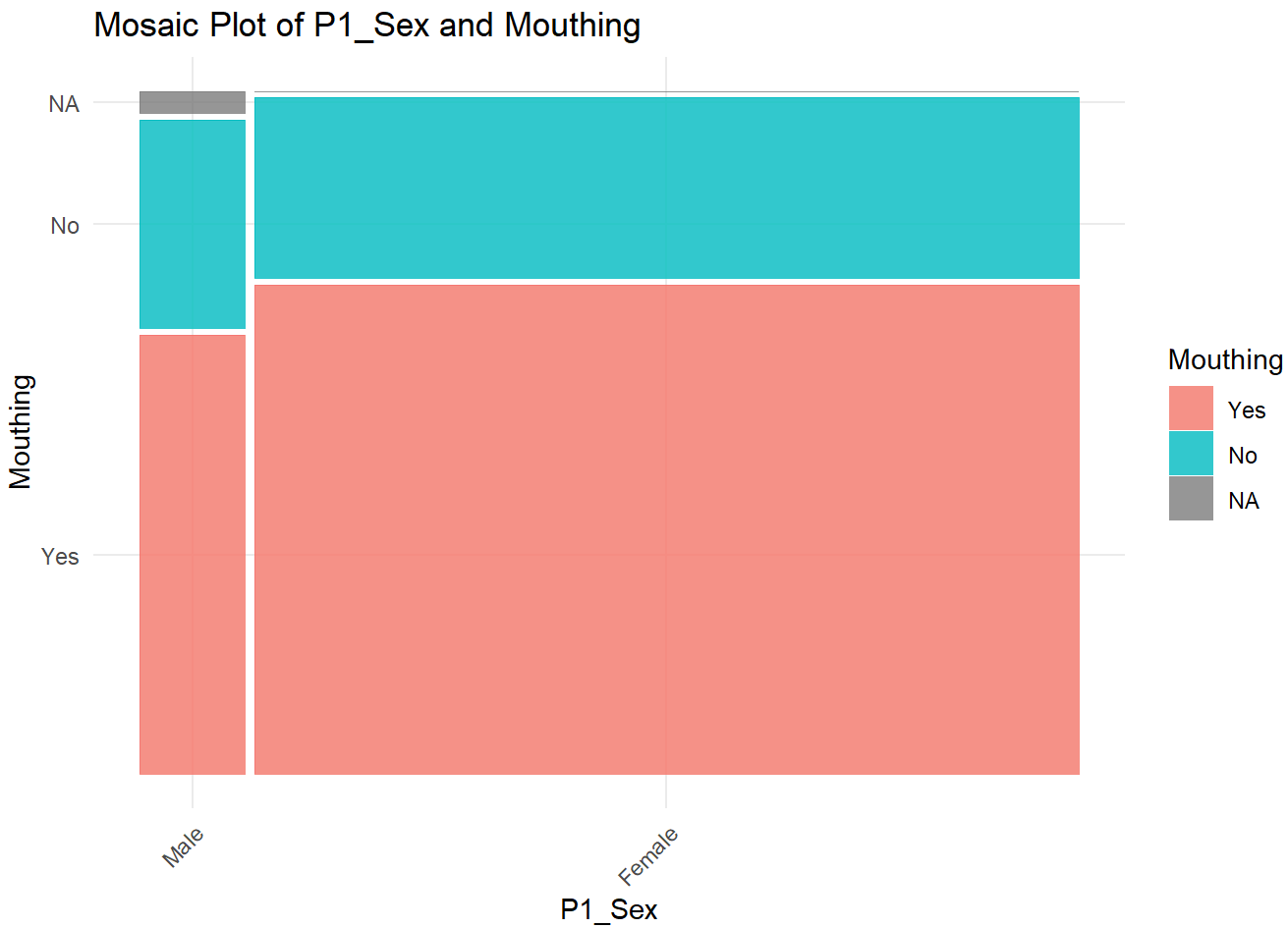
```
##
##
## ----- Analysis for Mouthing and P1_Age -----
## Bivariate Analysis of Mouthing and P1_Age
##
## Frequency Table:
## Mouthing <=29 30-40 40-50 over 50 NA_
##      Yes  101   228    48     7    6
##      No   27    90    20     6    5
##      <NA>   1     1     0     0    0
##
## Relative Frequency Table:
## Mouthing <=29 30-40 40-50 over 50  NA_ Total
##      Yes 78.3% 71.5% 70.6%  53.8% 54.5% 72.2%
##      No 20.9% 28.2% 29.4%  46.2% 45.5% 27.4%
##      <NA> 0.8% 0.3% 0.0%   0.0% 0.0% 0.4%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

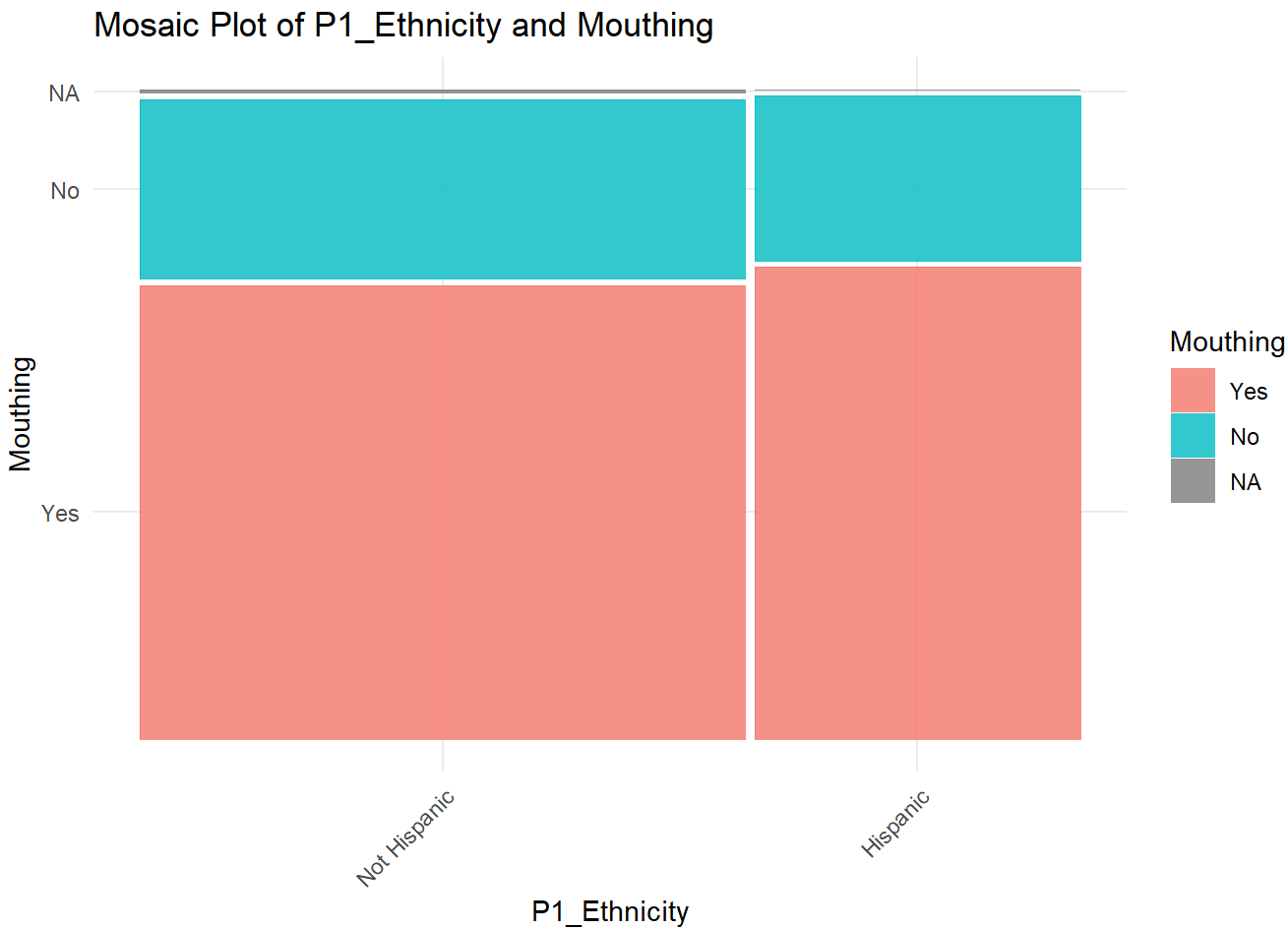


```
##
##
## ----- Analysis for Mouthing and P1_Sex -----
## Bivariate Analysis of Mouthing and P1_Sex
##
## Frequency Table:
##   Mouthing   Male Female
##       Yes     40     350
##       No      19     129
##       <NA>     2       0
##
## Relative Frequency Table:
##   Mouthing   Male Female Total
##       Yes  65.6%  73.1%  72.2%
##       No   31.1%  26.9%  27.4%
##       <NA>   3.3%   0.0%   0.4%
##
## Cramér's V:  0.1638652
```



```
## Saving 7 x 5 in image
```

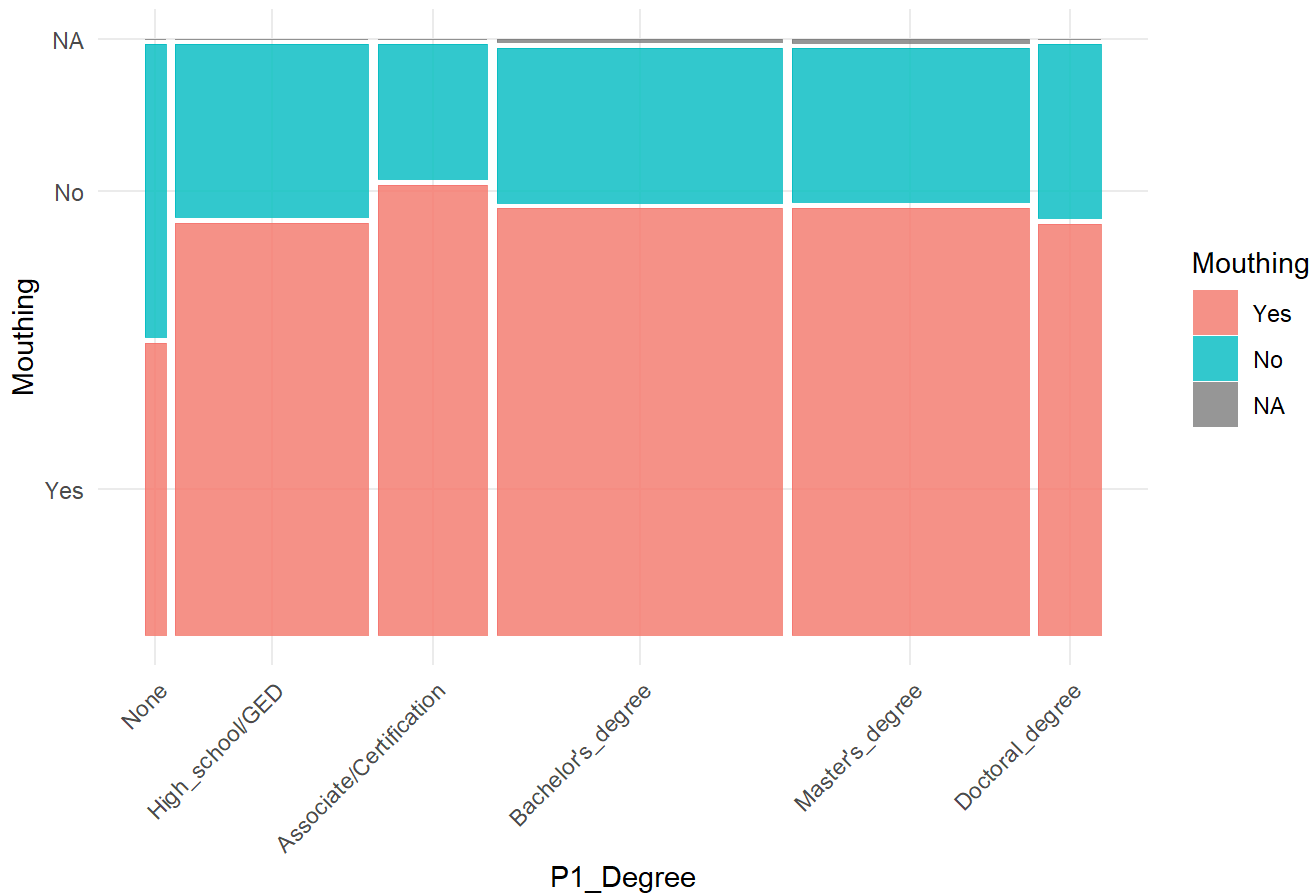
```
##
##
## ----- Analysis for Mouthing and P1_Ethnicity -----
## Bivariate Analysis of Mouthing and P1_Ethnicity
##
## Frequency Table:
##   Mouthing Not Hispanic Hispanic
##       Yes       250       140
##       No        99        49
##       <NA>        2         0
##
## Relative Frequency Table:
##   Mouthing Not Hispanic Hispanic Total
##       Yes       71.2%       74.1% 72.2%
##       No       28.2%       25.9% 27.4%
##       <NA>       0.6%       0.0%  0.4%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Mouthing and P1_Degree -----
## Bivariate Analysis of Mouthing and P1_Degree
##
## Frequency Table:
##   Mouthing None High_school/GED Associate/Certification Bachelor's_degree
##       Yes      6              81              50              124
##       No       6              34              15              45
##       <NA>     0              0              0              1
##   Master's_degree Doctoral_degree
##       103              26
##       37              11
##       1              0
##
## Relative Frequency Table:
##   Mouthing None High_school/GED Associate/Certification Bachelor's_degree
##       Yes 50.0%              70.4%              76.9%              72.9%
##       No  50.0%              29.6%              23.1%              26.5%
##       <NA> 0.0%              0.0%              0.0%              0.6%
##   Master's_degree Doctoral_degree Total
##       73.0%              70.3% 72.2%
##       26.2%              29.7% 27.4%
##       0.7%              0.0% 0.4%
##
## Cramér's V: 0
```

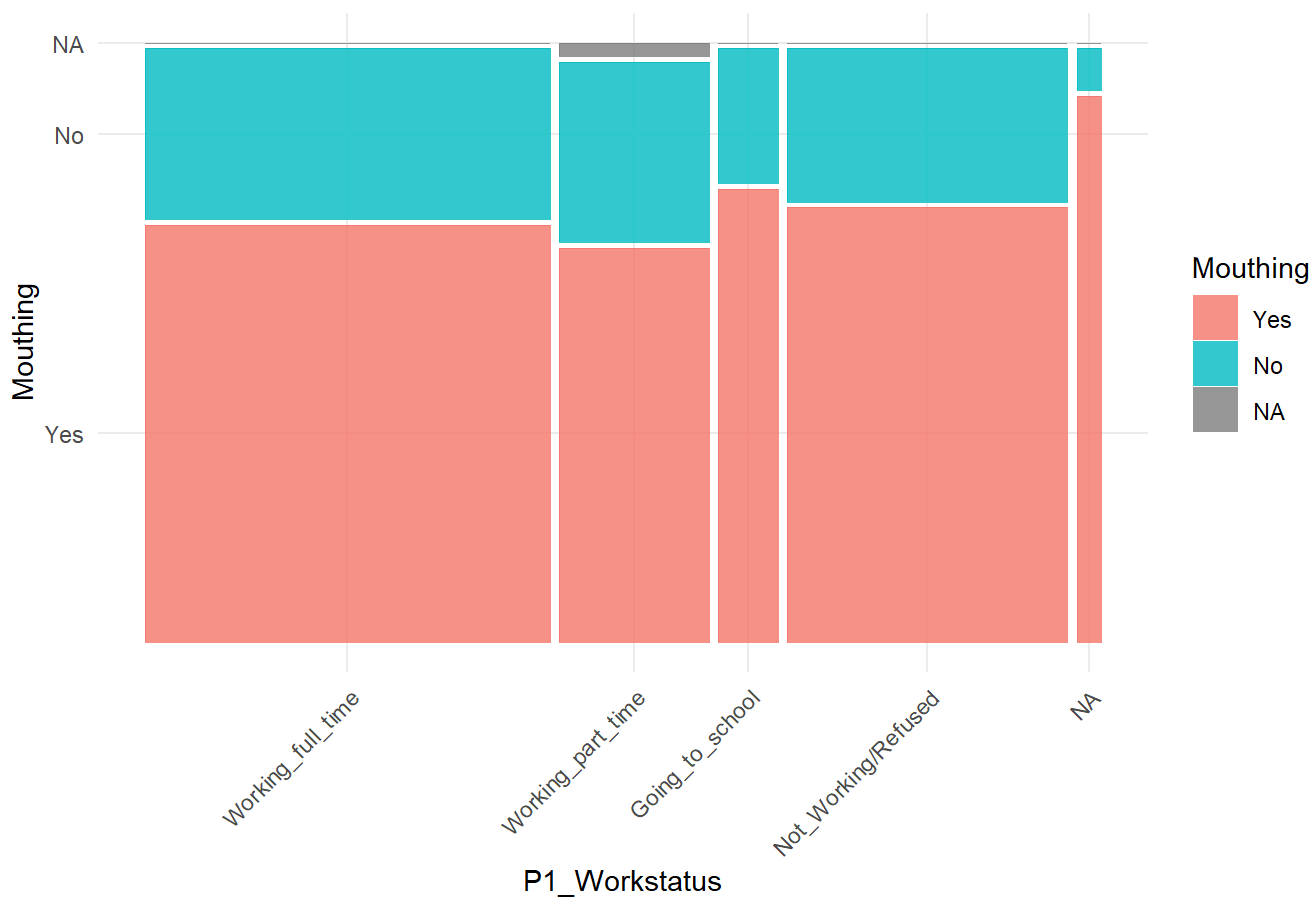
Mosaic Plot of P1_Degree and Mouthing



```
## Saving 7 x 5 in image
```

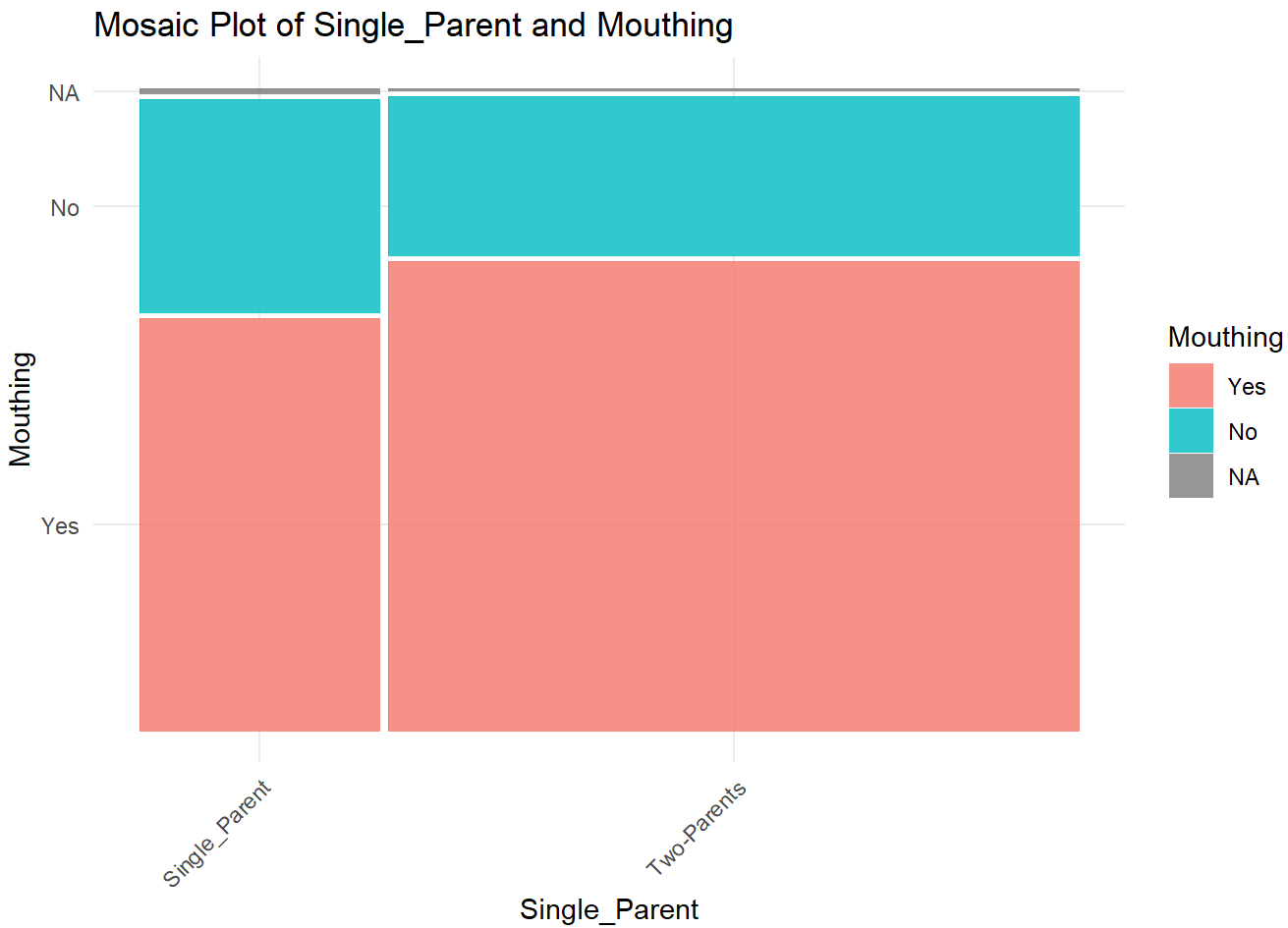
```
##
##
## ----- Analysis for Mouthing and P1_Workstatus -----
## Bivariate Analysis of Mouthing and P1_Workstatus
##
## Frequency Table:
## Mouthing Working_full_time Working_part_time Going_to_school
##      Yes      169      59      27
##      No       69      27      8
##      <NA>      0       2      0
## Not_Working/Refused NA_
##      122  13
##      43   1
##      0   0
##
## Relative Frequency Table:
## Mouthing Working_full_time Working_part_time Going_to_school
##      Yes      71.0%      67.0%      77.1%
##      No      29.0%      30.7%      22.9%
##      <NA>      0.0%      2.3%      0.0%
## Not_Working/Refused  NA_ Total
##      73.9% 92.9% 72.2%
##      26.1% 7.1% 27.4%
##      0.0% 0.0% 0.4%
##
## Cramér's V: 0.07877176
```

Mosaic Plot of P1_Workstatus and Mouthing



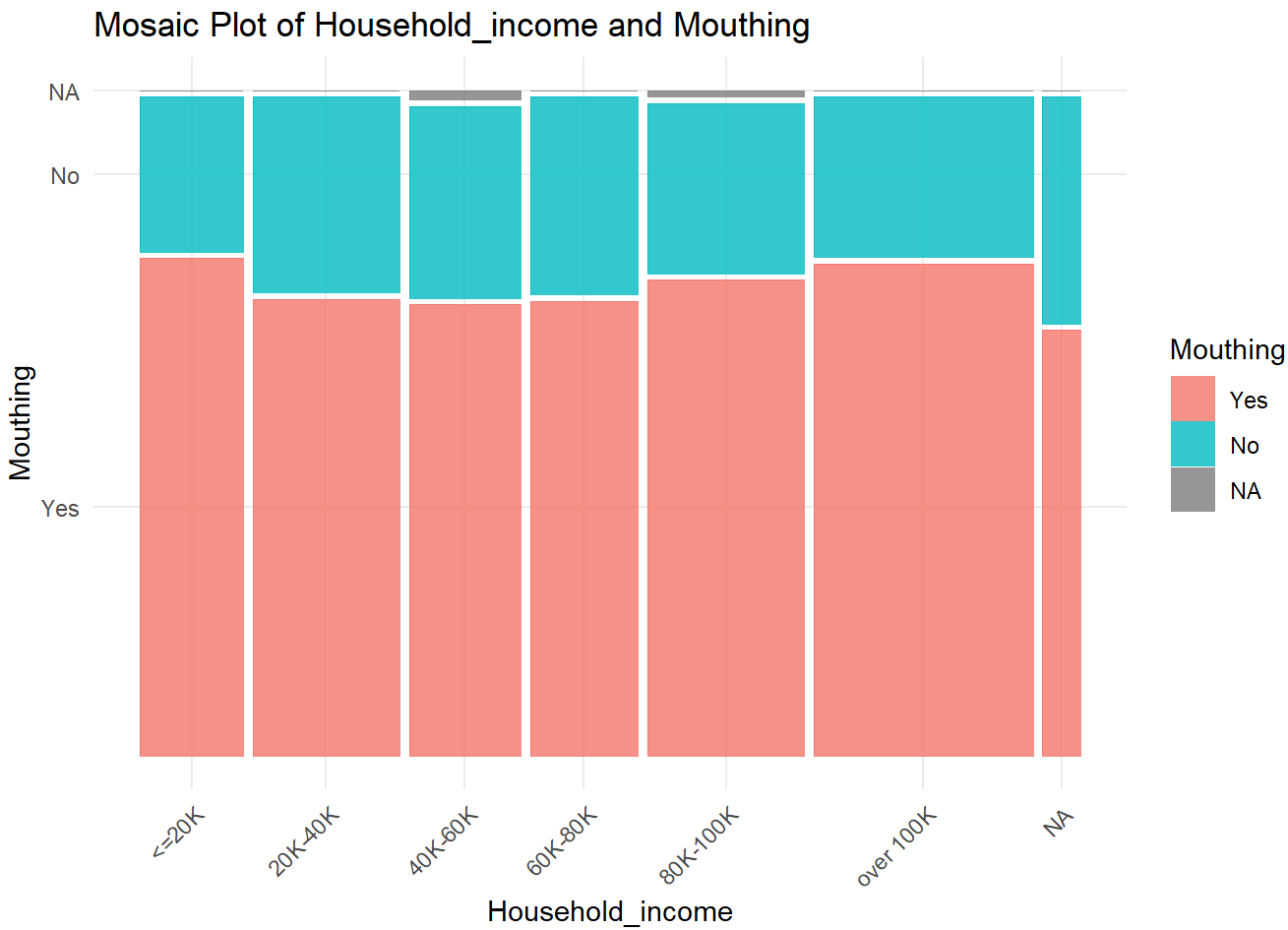
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Mouthing and Single_Parent -----
## Bivariate Analysis of Mouthing and Single_Parent
##
## Frequency Table:
##   Mouthing Single_Parent Two-Parents
##       Yes           91           299
##       No            47           101
##       <NA>           1            1
##
## Relative Frequency Table:
##   Mouthing Single_Parent Two-Parents Total
##       Yes           65.5%           74.6% 72.2%
##       No            33.8%           25.2% 27.4%
##       <NA>           0.7%            0.2% 0.4%
##
## Cramér's V:  0.06941037
```



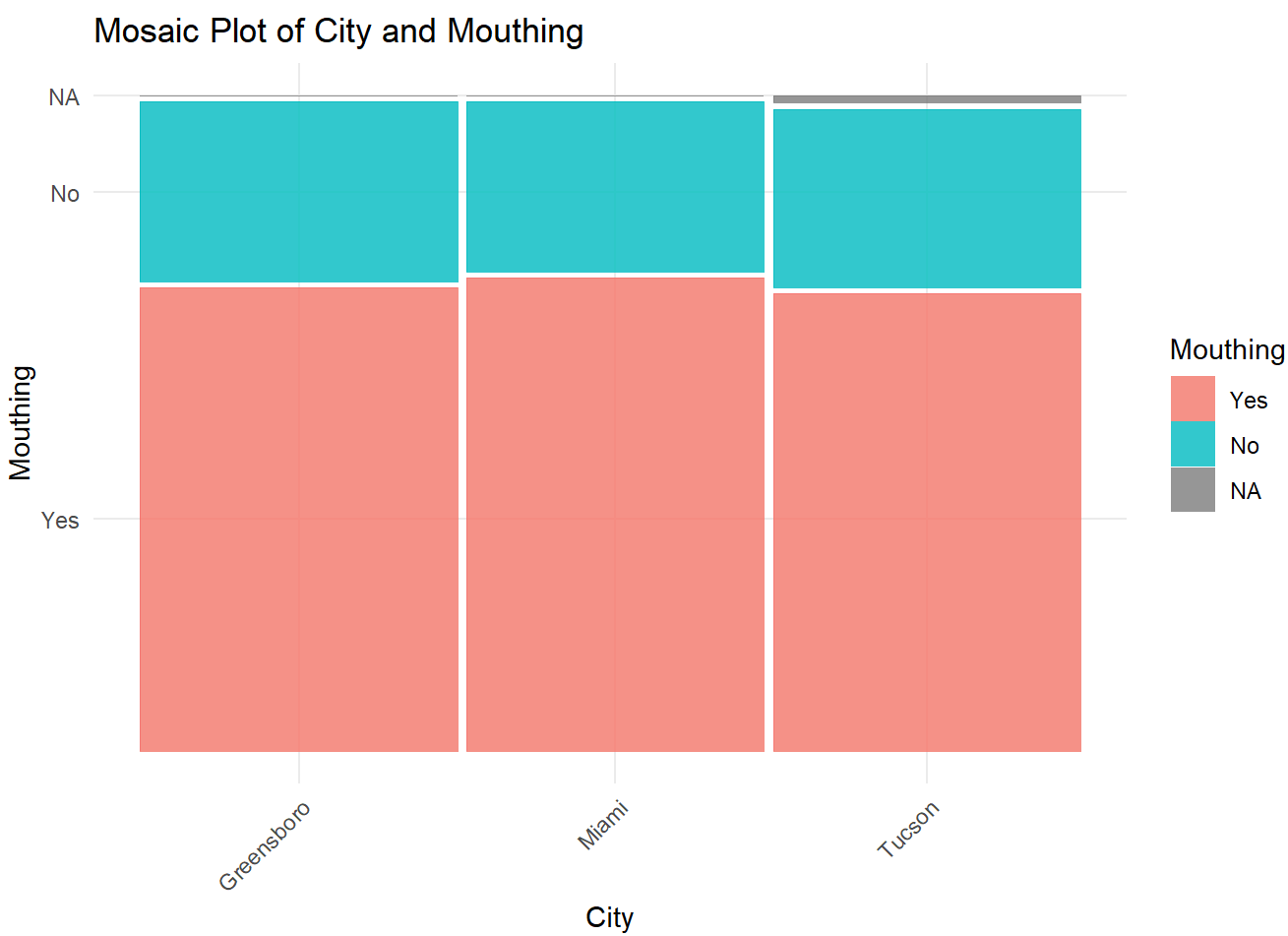
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Mouthing and Household_income -----
## Bivariate Analysis of Mouthing and Household_income
##
## Frequency Table:
## Mouthing <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##      Yes      48      63      47      46      70      101  15
##      No       15      27      20      20      25      33   8
##      <NA>      0       0       1       0       1       0   0
##
## Relative Frequency Table:
## Mouthing <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##      Yes 76.2%  70.0%  69.1%  69.7%  72.9%  75.4% 65.2% 72.2%
##      No  23.8%  30.0%  29.4%  30.3%  26.0%  24.6% 34.8% 27.4%
##      <NA>  0.0%   0.0%   1.5%   0.0%   1.0%   0.0%  0.0%  0.4%
##
## Cramér's V:  0
```



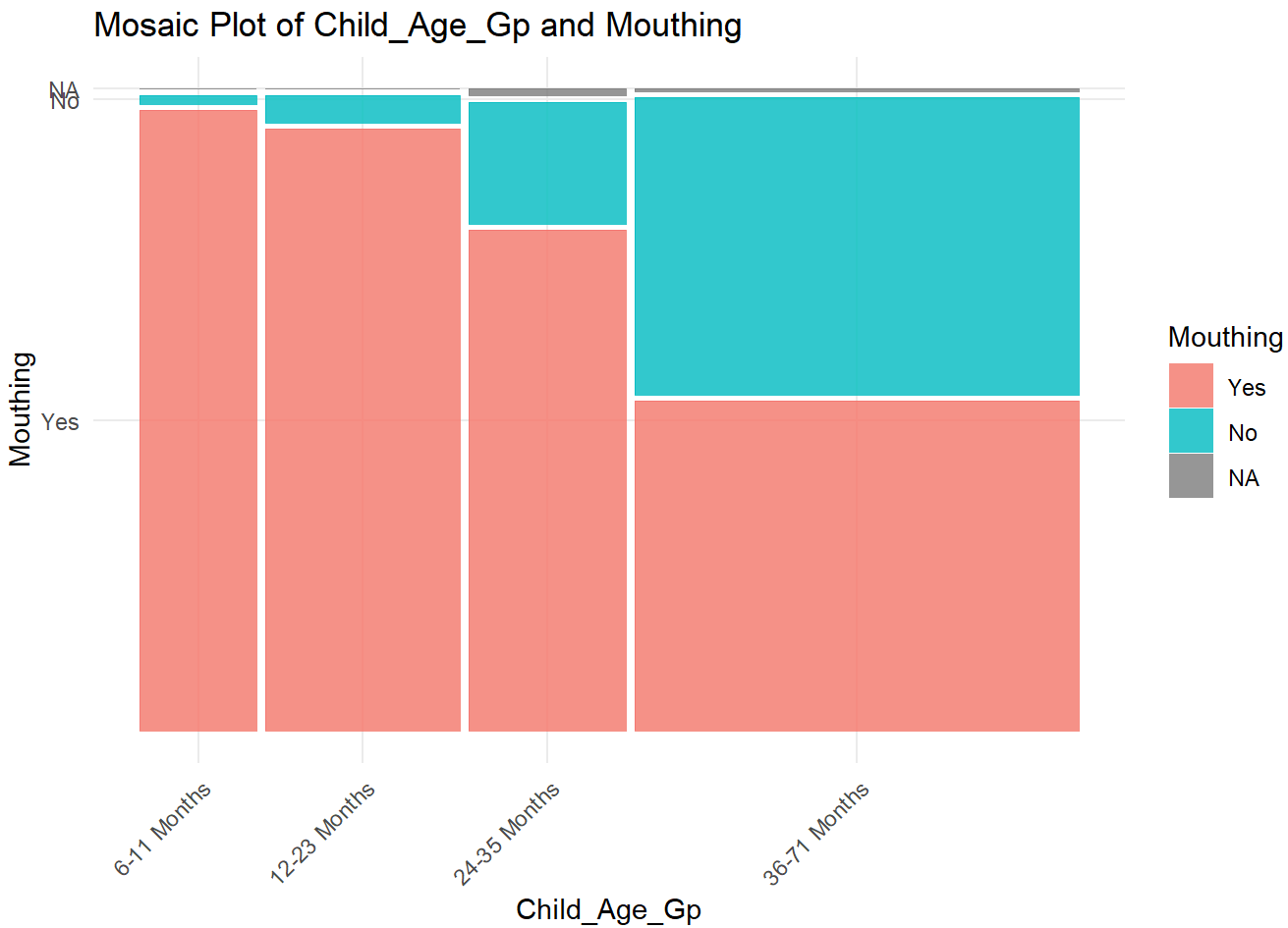
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Mouthing and City -----
## Bivariate Analysis of Mouthing and City
##
## Frequency Table:
##   Mouthing Greensboro Miami Tucson
##      Yes      134    128    128
##      No       52     46     50
##      <NA>       0      0      2
##
## Relative Frequency Table:
##   Mouthing Greensboro Miami Tucson Total
##      Yes     72.0% 73.6% 71.1% 72.2%
##      No     28.0% 26.4% 27.8% 27.4%
##      <NA>     0.0%  0.0%  1.1%  0.4%
##
## Cramér's V:  0.0120724
```



```
## Saving 7 x 5 in image
```

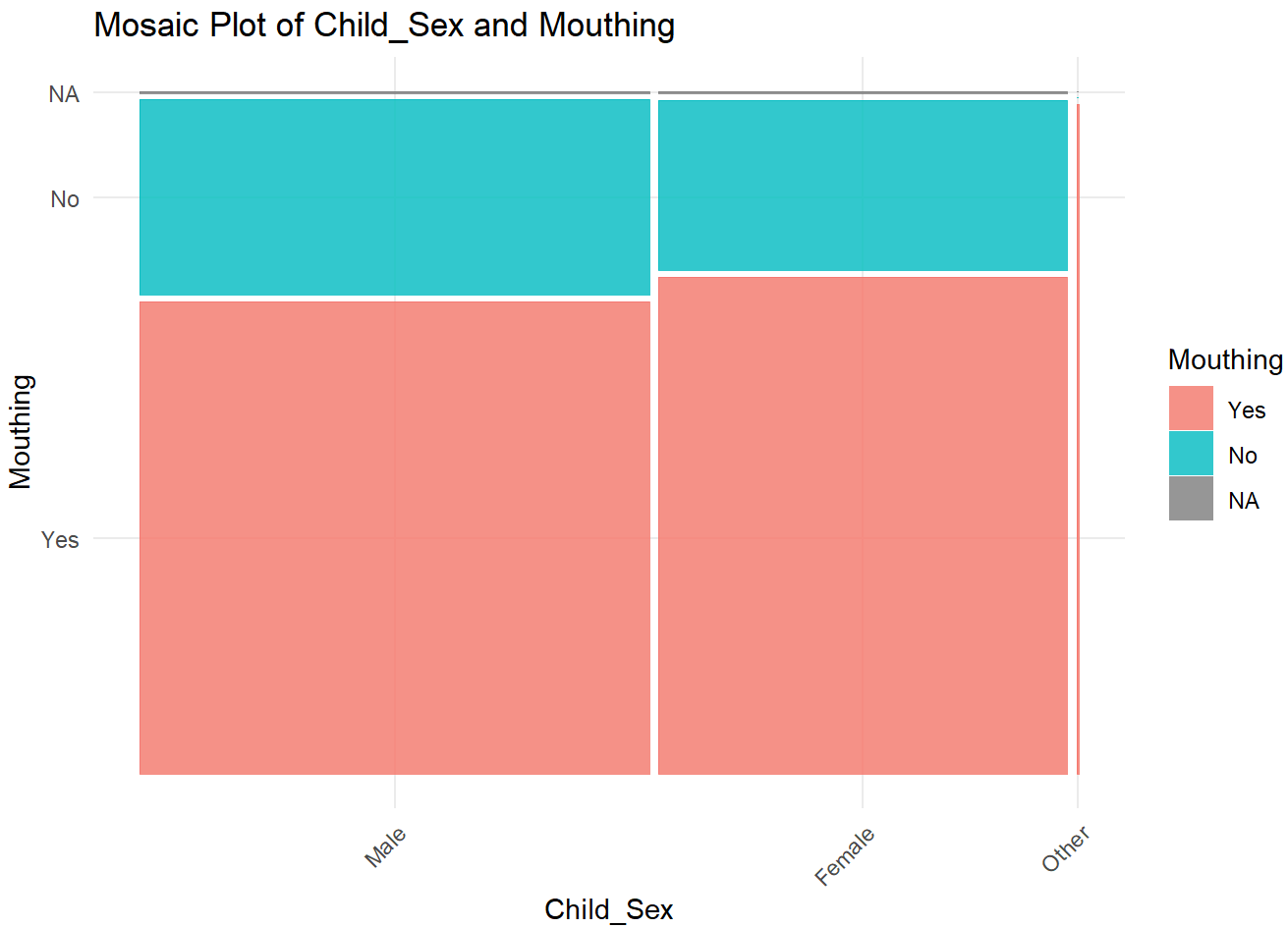
```
##
##
## ----- Analysis for Mouthing and Child_Age_Gp -----
## Bivariate Analysis of Mouthing and Child_Age_Gp
##
## Frequency Table:
##   Mouthing 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##       Yes           68           110           74           138
##       No            1            5           18           124
##       <NA>          0            0            1            1
##
## Relative Frequency Table:
##   Mouthing 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##       Yes           98.6%          95.7%          79.6%          52.5% 72.2%
##       No            1.4%           4.3%          19.4%          47.1% 27.4%
##       <NA>          0.0%           0.0%           1.1%           0.4% 0.4%
##
## Cramér's V:  0.3124662
```



```
## Saving 7 x 5 in image
```

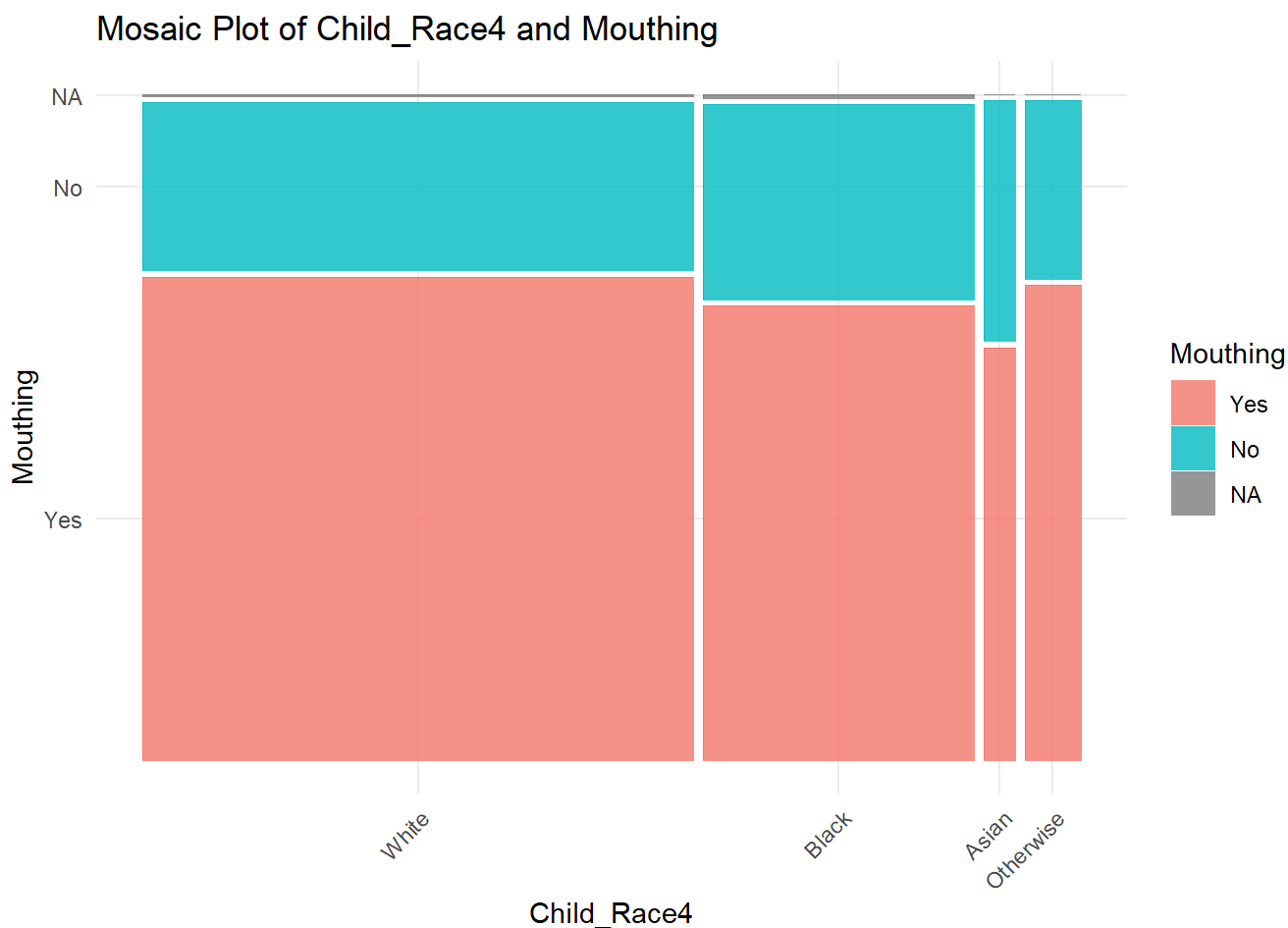


```
##
##
## ----- Analysis for Mouthing and Child_Sex -----
## Bivariate Analysis of Mouthing and Child_Sex
##
## Frequency Table:
##   Mouthing Male Female Other
##      Yes   211    178     1
##      No    87     61     0
##      <NA>    1      1     0
##
## Relative Frequency Table:
##   Mouthing  Male Female  Other Total
##      Yes  70.6%  74.2% 100.0%  72.2%
##      No   29.1%  25.4%   0.0%  27.4%
##      <NA>   0.3%   0.4%   0.0%   0.4%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Mouthing and Child_Race4 -----
## Bivariate Analysis of Mouthing and Child_Race4
##
## Frequency Table:
##   Mouthing White Black Asian Otherwise
##      Yes    242   112    12      24
##      No     84    48     7      9
##      <NA>     1     1     0      0
##
## Relative Frequency Table:
##   Mouthing White Black Asian Otherwise Total
##      Yes  74.0% 69.6% 63.2%    72.7% 72.2%
##      No   25.7% 29.8% 36.8%    27.3% 27.4%
##      <NA>  0.3%  0.6%  0.0%     0.0%  0.4%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

```
#
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```

```
##          var1          var2 cramer_v
## 1 Mouthing   Child_Age_Gp    0.312
## 2 Mouthing          P1_Sex    0.164
## 3 Mouthing   P1_Workstatus    0.079
## 4 Mouthing   Single_Parent    0.069
## 5 Mouthing          P1_Race4    0.028
## 6 Mouthing          City      0.012
## 7 Mouthing          P1_Age     0.000
## 8 Mouthing   P1_Ethnicity     0.000
## 9 Mouthing   P1_Degree        0.000
## 10 Mouthing Household_income    0.000
## 11 Mouthing   Child_Sex       0.000
## 12 Mouthing   Child_Race4     0.000
```

Exploring Other Mouthing Variables

```
data= ChildrenADF
list1=c("mouthing", "Mouthing_Toys", "Mouthing Shoes",
        "Mouthing Jewelry","Mouthing Dirt")
## Need to code all binary vars as factors
```

```
cor_matrix <- cor(ChildrenADF[, c("mouthing", "Mouthing_Toys", "Mouthing_Shoes",   "Mouthing_Jewelry", "Mou
thing_Dirt")],
                  use = "pairwise.complete.obs")

print(round(cor_matrix, digits=3))
```

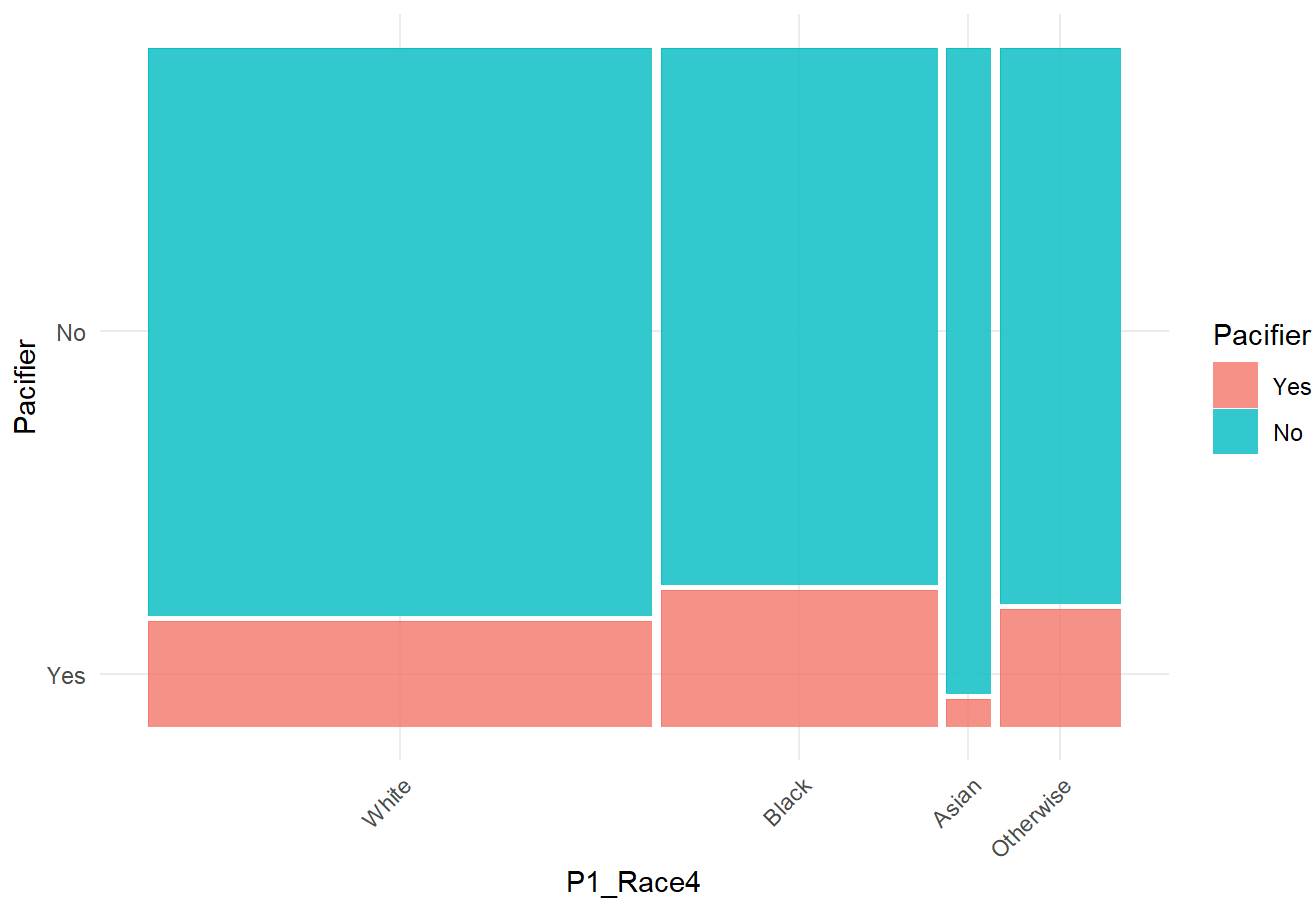
```
##          mouthing Mouthing_Toys Mouthing_Shoes Mouthing_Jewelry
## mouthing          1.000        -0.015        -0.005        -0.005
## Mouthing_Toys     -0.015         1.000         0.340         0.301
## Mouthing_Shoes     -0.005         0.340         1.000         0.377
## Mouthing_Jewelry   -0.005         0.301         0.377         1.000
## Mouthing_Dirt      -0.004         0.281         0.358         0.358
##
##          Mouthing_Dirt
## mouthing          -0.004
## Mouthing_Toys       0.281
## Mouthing_Shoes       0.358
## Mouthing_Jewelry     0.358
## Mouthing_Dirt        1.000
```

Exploring Pacifier Use & Wash Variables

```
PaciVars <- c( "Pacifier","Pacifier_Use" , "Pacifier_Wash")
# data =data
list1 = PaciVars
list2=DemoVars
# Loop through each combination of var1 and var2 and calculate Cramér's V for ranking
cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(), stringsAsFactors = FALSE)
results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = "_vs_")]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values, data.frame(var1=var1, var2 = var2,
        cramer_v = result$cramer_v))
    }
  }
}
```

```
##
##
## ----- Analysis for Pacifier and P1_Race4 -----
## Bivariate Analysis of Pacifier and P1_Race4
##
## Frequency Table:
##   Pacifier White Black Asian Otherwise
##      Yes    45    32     1      12
##      No   243   126    24     57
##
## Relative Frequency Table:
##   Pacifier White Black Asian Otherwise Total
##      Yes 15.6% 20.3%  4.0%    17.4% 16.7%
##      No 84.4% 79.7% 96.0%    82.6% 83.3%
##
## Cramér's V:  0.05442937
```

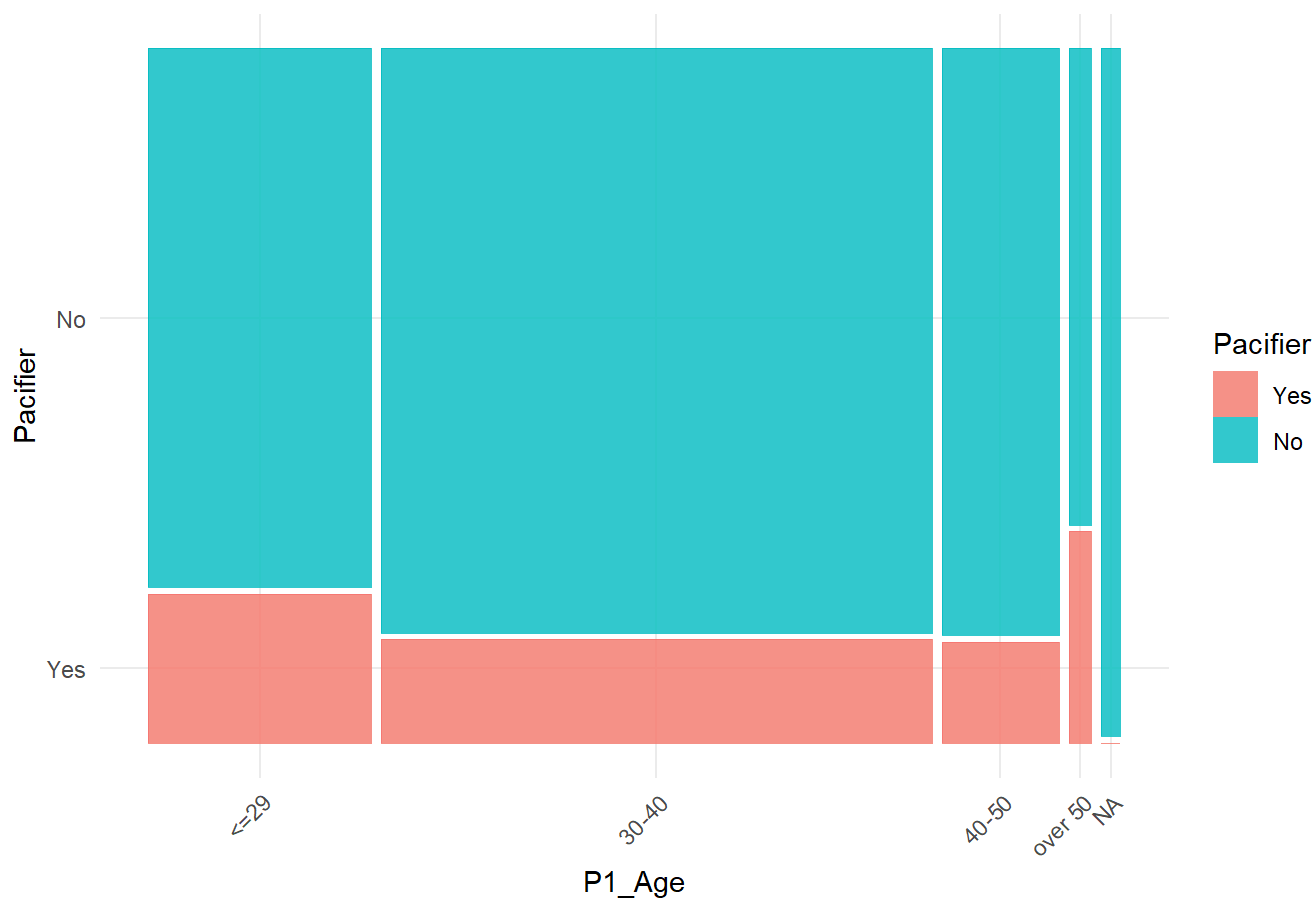
Mosaic Plot of P1_Race4 and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and P1_Age -----
## Bivariate Analysis of Pacifier and P1_Age
##
## Frequency Table:
## Pacifier <=29 30-40 40-50 over 50 NA_
##      Yes    28    48    10     4    0
##      No   101   271    58     9   11
##
## Relative Frequency Table:
## Pacifier <=29 30-40 40-50 over 50    NA_ Total
##      Yes 21.7% 15.0% 14.7%  30.8%   0.0% 16.7%
##      No 78.3% 85.0% 85.3%  69.2% 100.0% 83.3%
##
## Cramér's V:  0.07708765
```

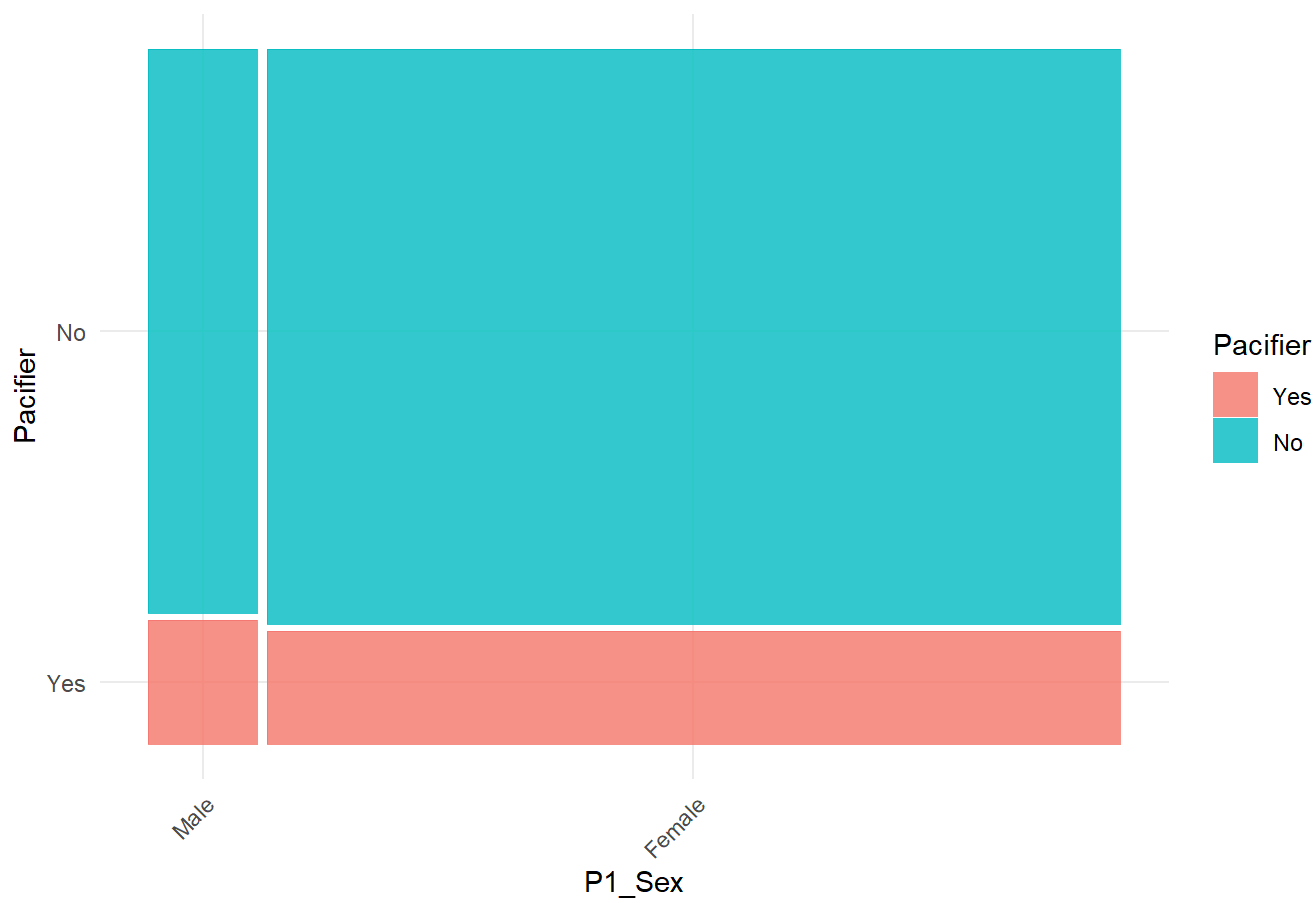
Mosaic Plot of P1_Age and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and P1_Sex -----
## Bivariate Analysis of Pacifier and P1_Sex
##
## Frequency Table:
##   Pacifier  Male Female
##      Yes    11    79
##      No     50   400
##
## Relative Frequency Table:
##   Pacifier  Male Female Total
##      Yes  18.0%  16.5%  16.7%
##      No   82.0%  83.5%  83.3%
##
##
## Cramér's V:  0
```

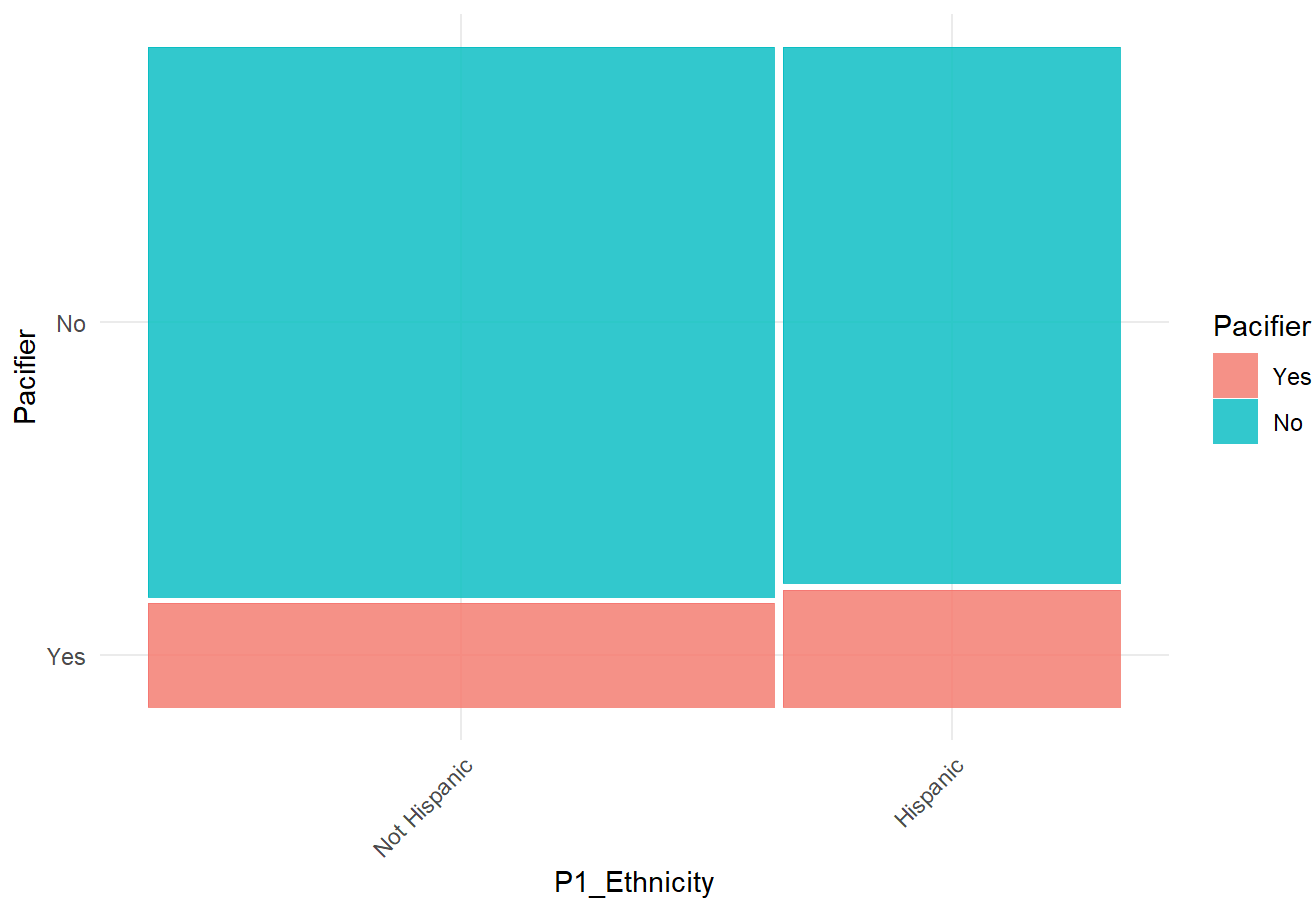
Mosaic Plot of P1_Sex and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and P1_Ethnicity -----
## Bivariate Analysis of Pacifier and P1_Ethnicity
##
## Frequency Table:
##   Pacifier Not Hispanic Hispanic
##      Yes      56      34
##      No     295     155
##
## Relative Frequency Table:
##   Pacifier Not Hispanic Hispanic Total
##      Yes     16.0%    18.0% 16.7%
##      No     84.0%    82.0% 83.3%
##
## Cramér's V:  0
```

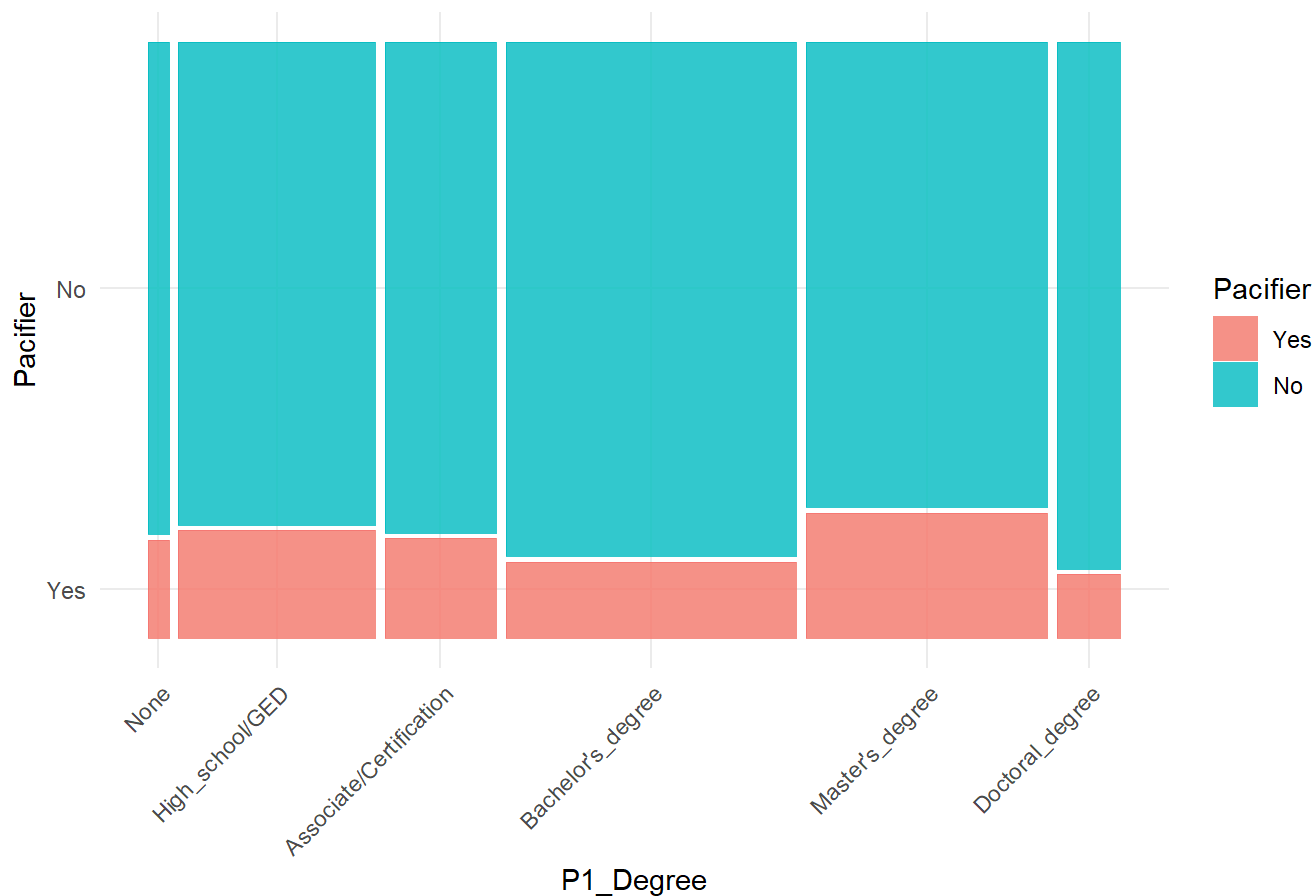
Mosaic Plot of P1_Ethnicity and Pacifier



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier and P1_Degree -----
## Bivariate Analysis of Pacifier and P1_Degree
##
## Frequency Table:
##   Pacifier None High_school/GED Associate/Certification Bachelor's_degree
##       Yes    2         21             11         22
##       No   10         94             54         148
##   Master's_degree Doctoral_degree
##           30         4
##           111        33
##
## Relative Frequency Table:
##   Pacifier None High_school/GED Associate/Certification Bachelor's_degree
##       Yes 16.7%      18.3%          16.9%      12.9%
##       No  83.3%      81.7%          83.1%      87.1%
##   Master's_degree Doctoral_degree Total
##           21.3%      10.8% 16.7%
##           78.7%      89.2% 83.3%
##
## Cramér's V:  0
```

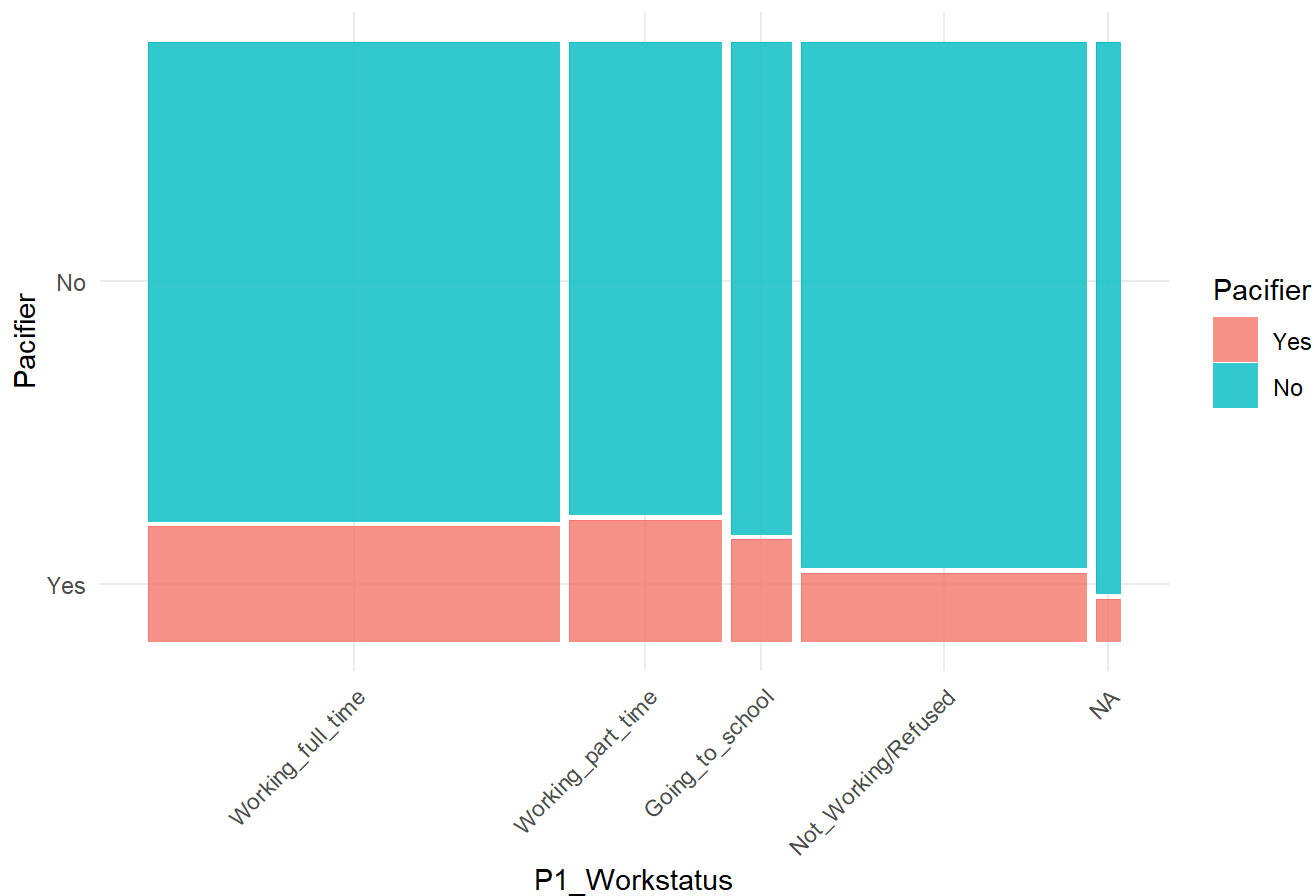

Mosaic Plot of P1_Degree and Pacifier



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier and P1_Workstatus -----
## Bivariate Analysis of Pacifier and P1_Workstatus
##
## Frequency Table:
## Pacifier Working_full_time Working_part_time Going_to_school
##      Yes           46           18           6
##      No           192           70          29
## Not_Working/Refused NA_
##           19    1
##           146   13
##
## Relative Frequency Table:
## Pacifier Working_full_time Working_part_time Going_to_school
##      Yes           19.3%           20.5%           17.1%
##      No           80.7%           79.5%           82.9%
## Not_Working/Refused  NA_ Total
##           11.5%   7.1% 16.7%
##           88.5% 92.9% 83.3%
##
## Cramér's V:  0.06371099
```

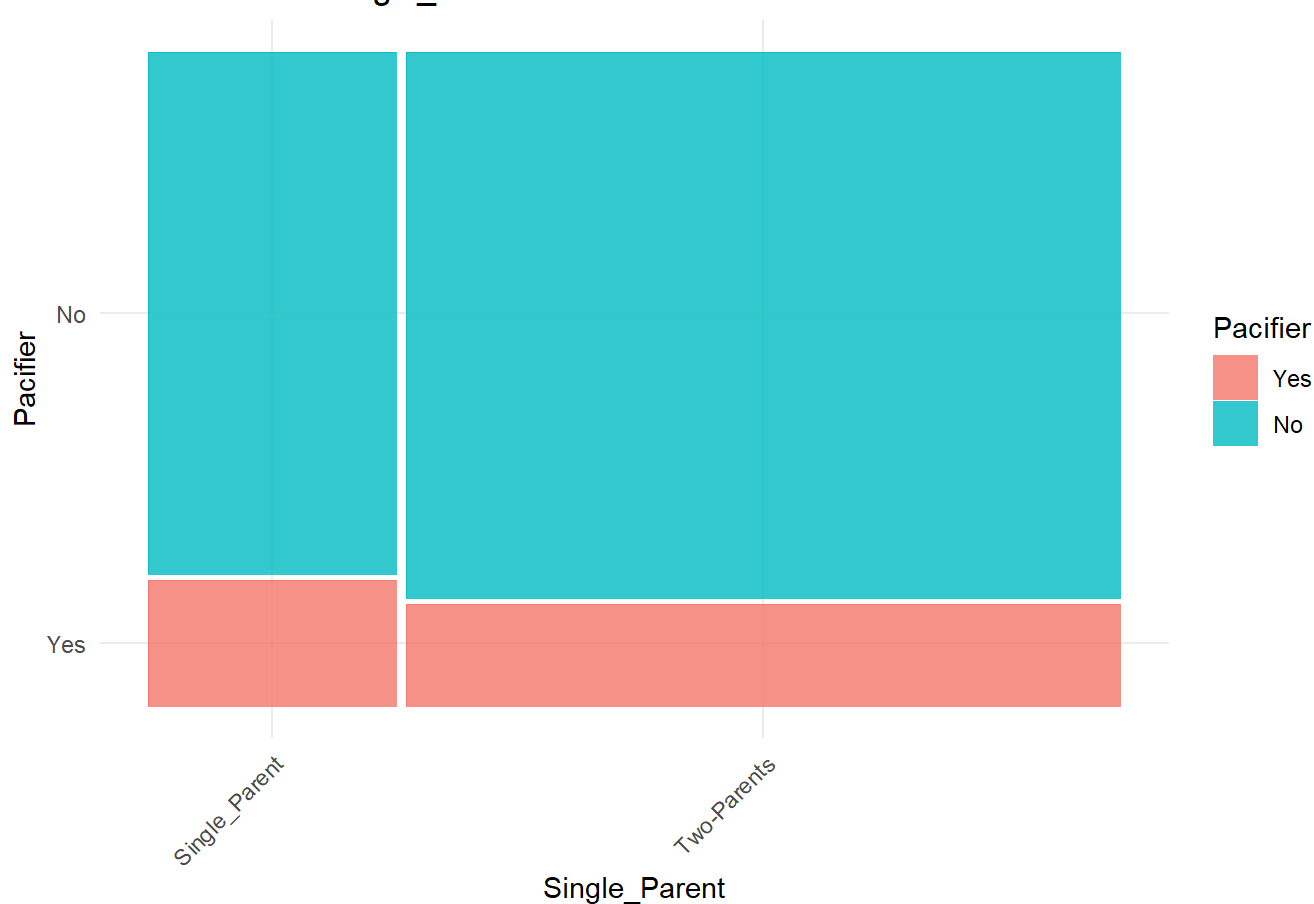
Mosaic Plot of P1_Workstatus and Pacifier



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier and Single_Parent -----
## Bivariate Analysis of Pacifier and Single_Parent
##
## Frequency Table:
##   Pacifier Single_Parent Two-Parents
##   Yes      27          63
##   No      112         338
##
## Relative Frequency Table:
##   Pacifier Single_Parent Two-Parents Total
##   Yes      19.4%      15.7% 16.7%
##   No      80.6%      84.3% 83.3%
##
## Cramér's V: 0.006551674
```

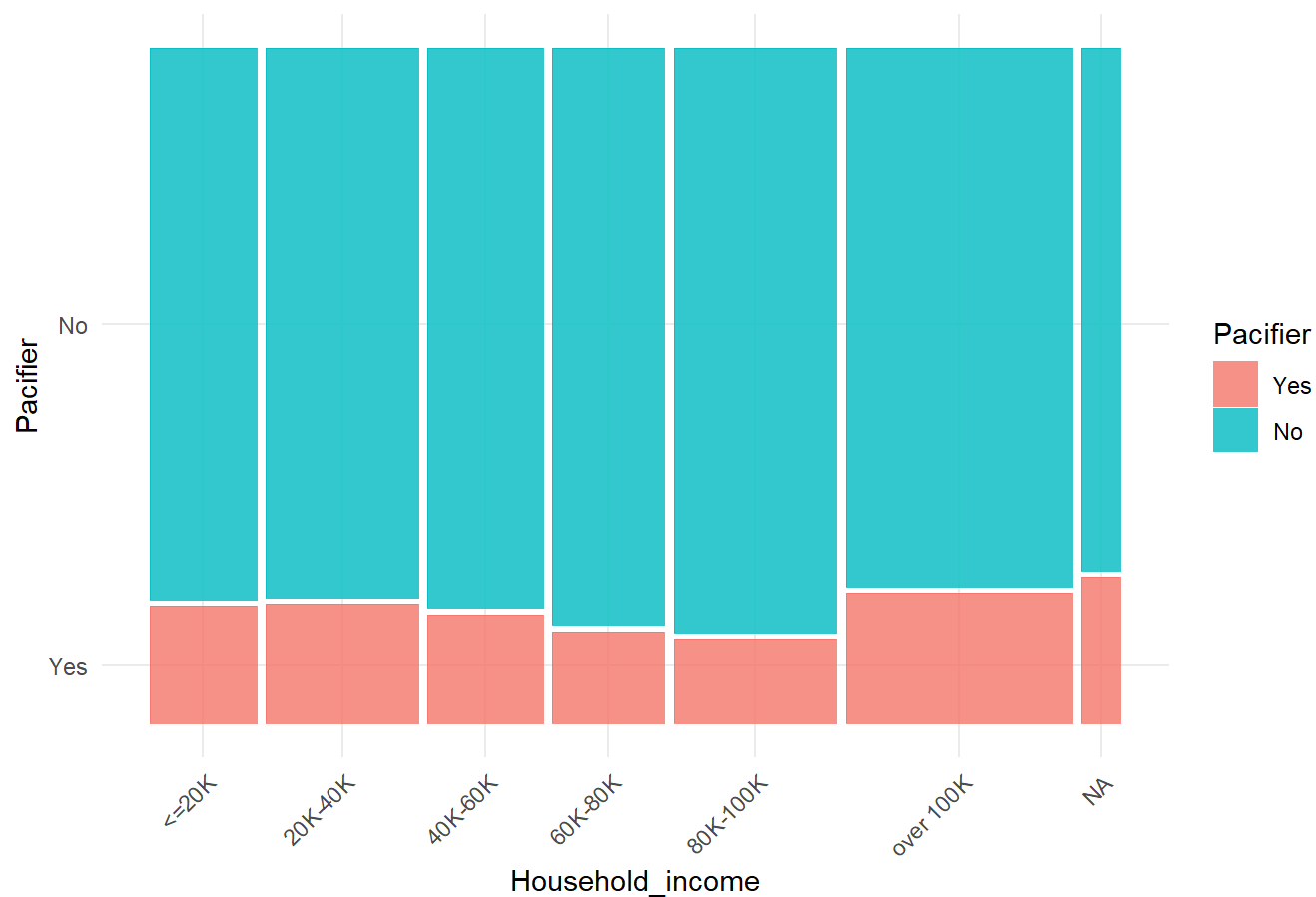
Mosaic Plot of Single_Parent and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and Household_income -----
## Bivariate Analysis of Pacifier and Household_income
##
## Frequency Table:
## Pacifier <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##      Yes      11      16      11       9      12      26   5
##      No       52      74      57      57      84     108  18
##
## Relative Frequency Table:
## Pacifier <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##      Yes 17.5%  17.8%  16.2%  13.6%  12.5%  19.4% 21.7% 16.7%
##      No 82.5%  82.2%  83.8%  86.4%  87.5%  80.6% 78.3% 83.3%
##
## Cramér's V:  0
```

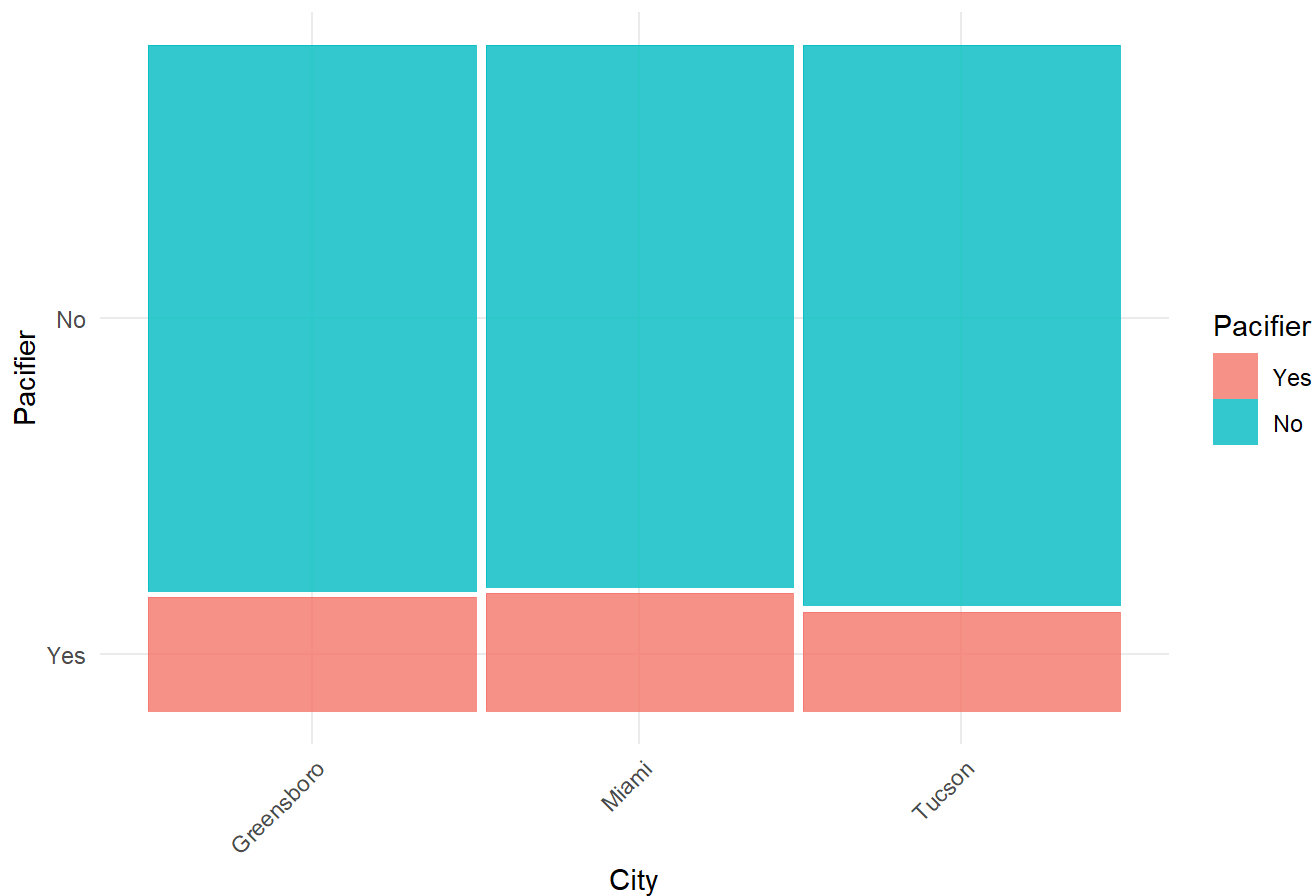
Mosaic Plot of Household_income and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and City -----
## Bivariate Analysis of Pacifier and City
##
## Frequency Table:
##   Pacifier Greensboro Miami Tucson
##      Yes         32      31      27
##      No         154     143     153
##
## Relative Frequency Table:
##   Pacifier Greensboro Miami Tucson Total
##      Yes      17.2%  17.8%  15.0%  16.7%
##      No       82.8%  82.2%  85.0%  83.3%
##
## Cramér's V:  0
```

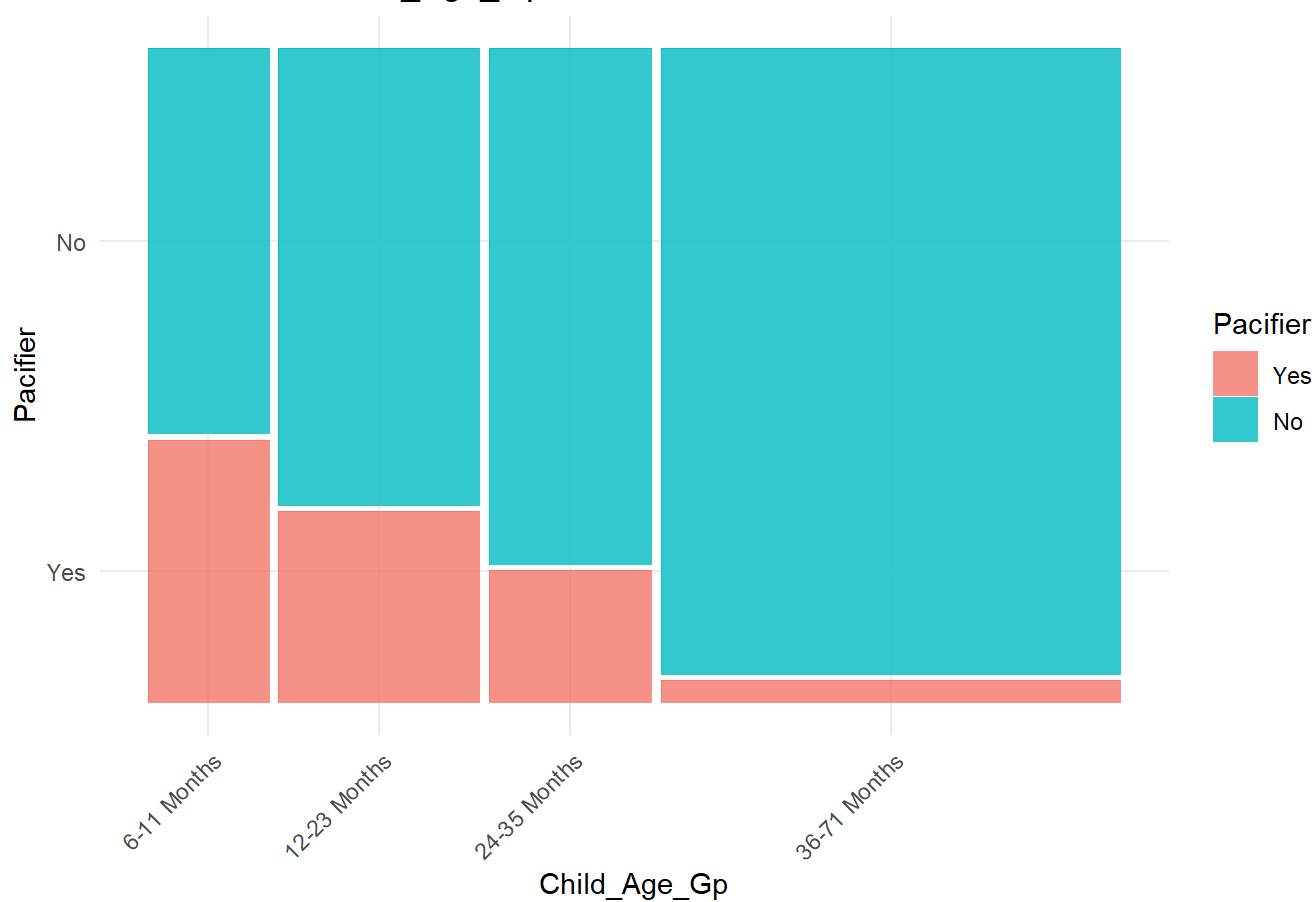
Mosaic Plot of City and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and Child_Age_Gp -----
## Bivariate Analysis of Pacifier and Child_Age_Gp
##
## Frequency Table:
##   Pacifier 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##       Yes         28         34         19         9
##       No          41         81         74        254
##
## Relative Frequency Table:
##   Pacifier 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##       Yes         40.6%         29.6%         20.4%         3.4% 16.7%
##       No          59.4%         70.4%         79.6%        96.6% 83.3%
##
## Cramér's V:  0.3688847
```

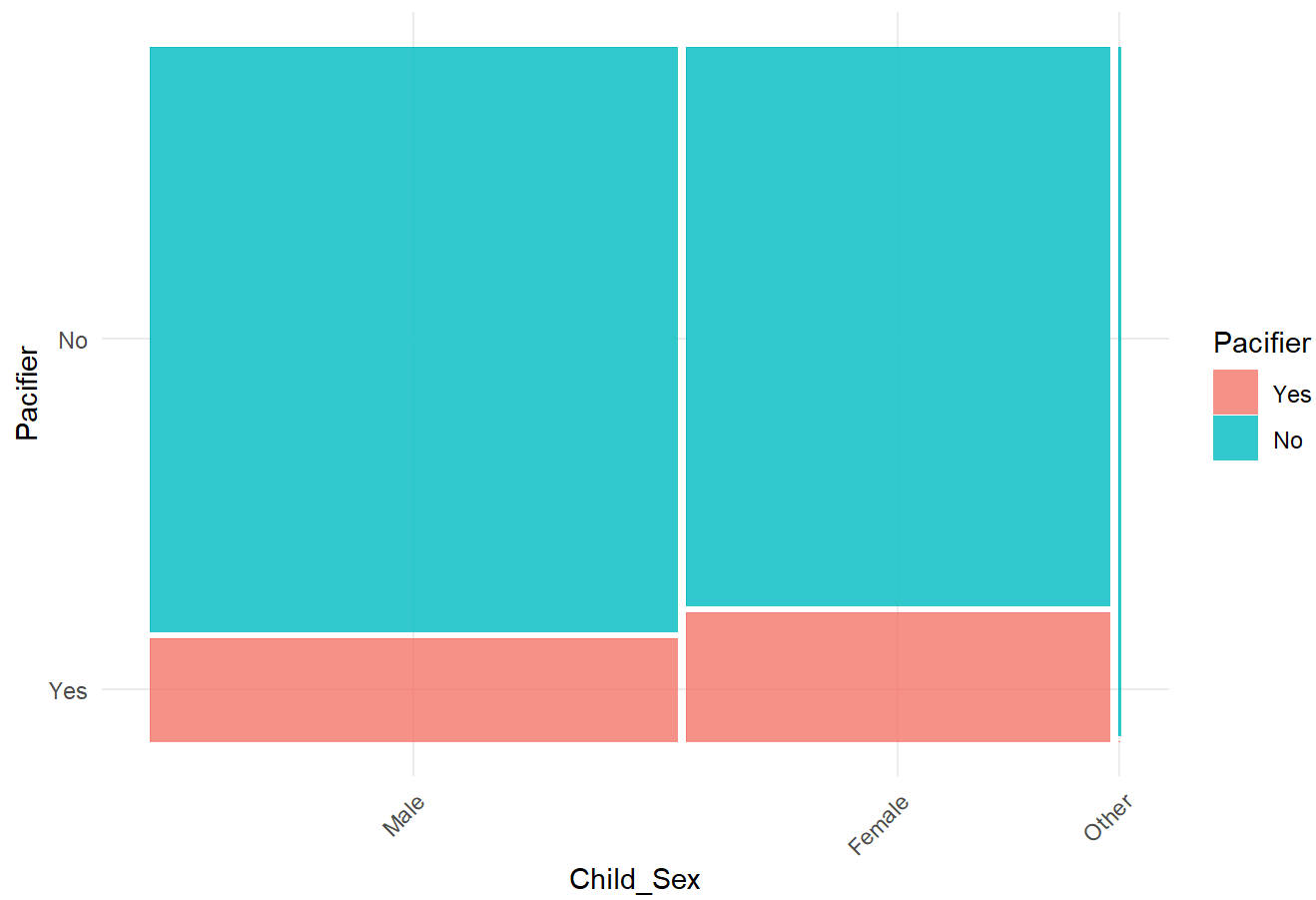
Mosaic Plot of Child_Age_Gp and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and Child_Sex -----
## Bivariate Analysis of Pacifier and Child_Sex
##
## Frequency Table:
##   Pacifier Male Female Other
##      Yes   45    45    0
##      No  254   195    1
##
## Relative Frequency Table:
##   Pacifier Male Female Other Total
##      Yes 15.1% 18.8%  0.0% 16.7%
##      No 84.9% 81.2% 100.0% 83.3%
##
## Cramér's V:  0
```

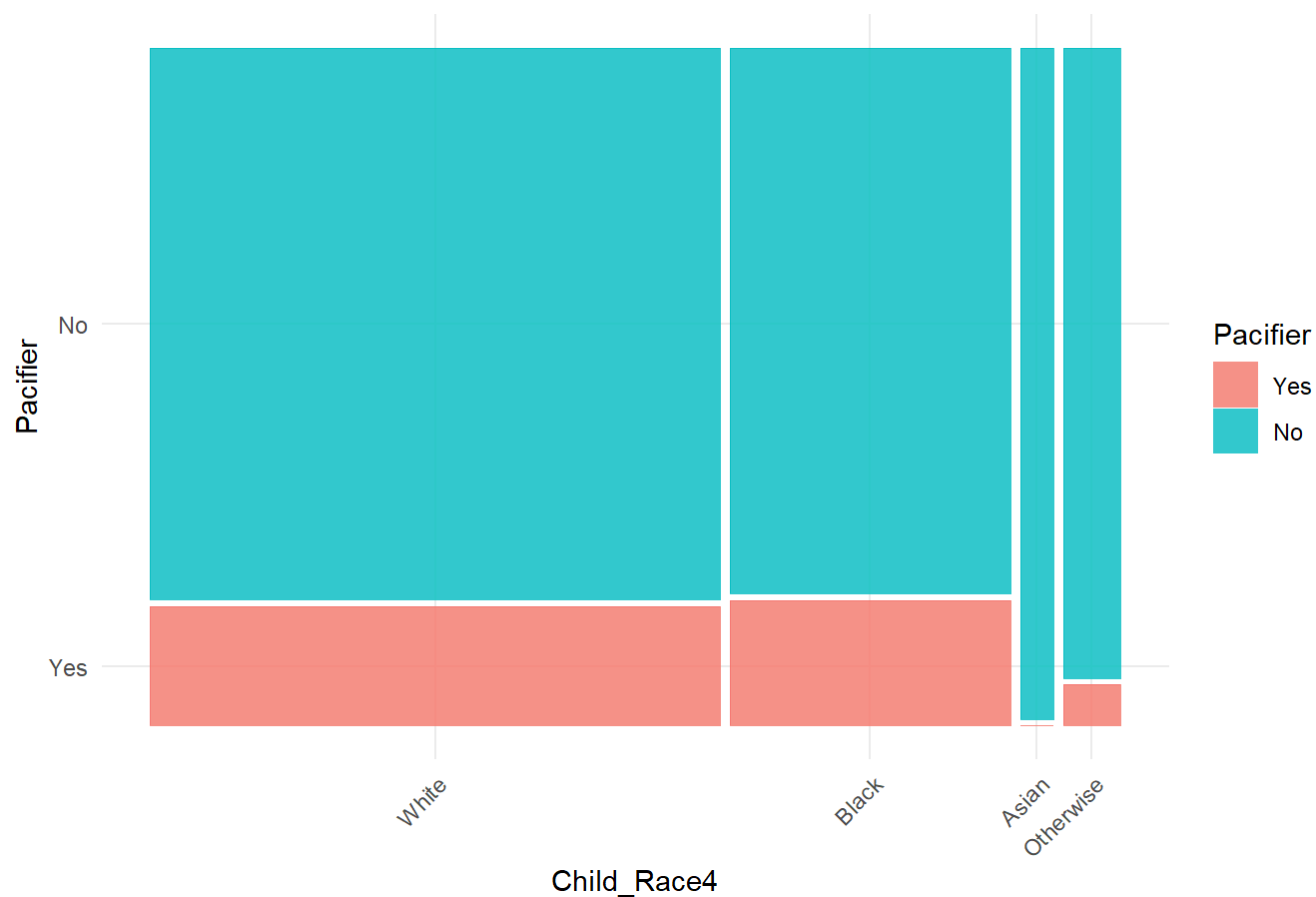
Mosaic Plot of Child_Sex and Pacifier



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier and Child_Race4 -----
## Bivariate Analysis of Pacifier and Child_Race4
##
## Frequency Table:
## Pacifier White Black Asian Otherwise
##      Yes    58    30     0         2
##      No   269   131    19        31
##
## Relative Frequency Table:
## Pacifier White Black Asian Otherwise Total
##      Yes 17.7% 18.6%  0.0%     6.1% 16.7%
##      No 82.3% 81.4% 100.0%    93.9% 83.3%
##
## Cramér's V:  0.08811937
```

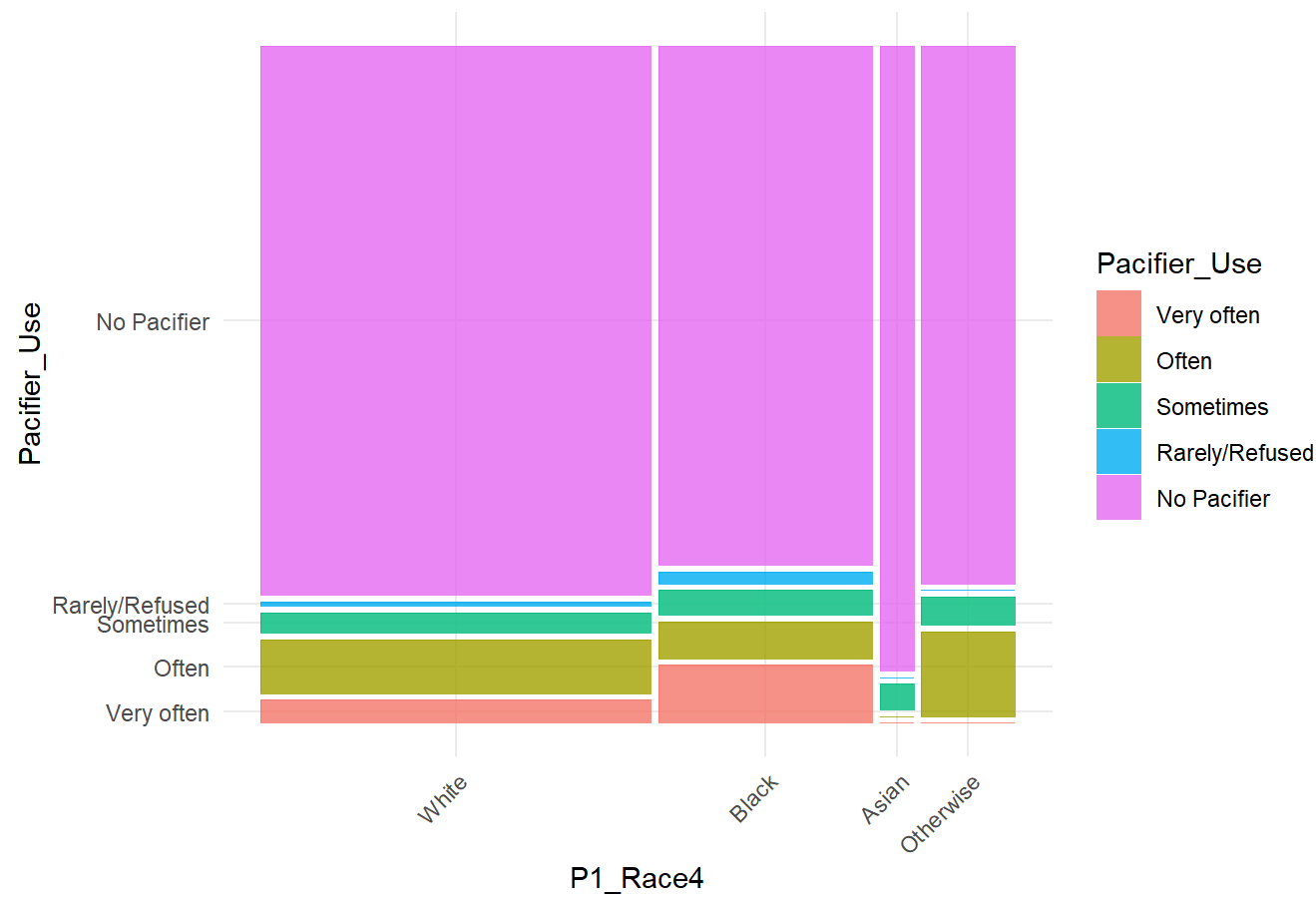
Mosaic Plot of Child_Race4 and Pacifier



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and P1_Race4 -----
## Bivariate Analysis of Pacifier_Use and P1_Race4
##
## Frequency Table:
##   Pacifier_Use White Black Asian Otherwise
##   Very often    10    14     0         0
##   Often        24     9     0         9
##   Sometimes     9     6     1         3
##   Rarely/Refused  2     3     0         0
##   No Pacifier   243   126    24        57
##
## Relative Frequency Table:
##   Pacifier_Use White Black Asian Otherwise Total
##   Very often   3.5%  8.9%  0.0%      0.0%  4.4%
##   Often        8.3%  5.7%  0.0%     13.0%  7.8%
##   Sometimes    3.1%  3.8%  4.0%      4.3%  3.5%
##   Rarely/Refused 0.7%  1.9%  0.0%      0.0%  0.9%
##   No Pacifier  84.4% 79.7% 96.0%     82.6% 83.3%
##
## Cramér's V: 0.07404469
```

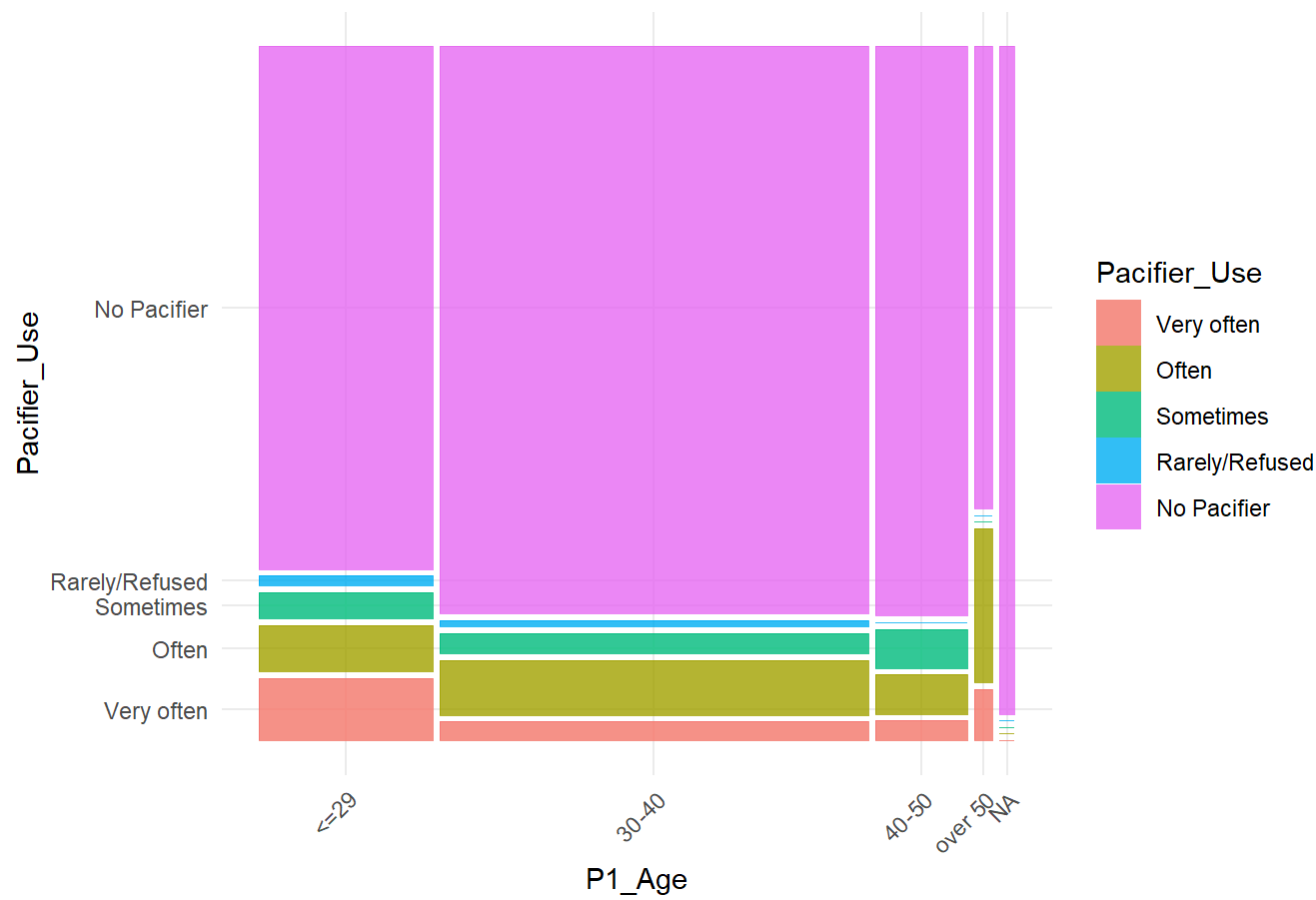

Mosaic Plot of P1_Race4 and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and P1_Age -----
## Bivariate Analysis of Pacifier_Use and P1_Age
##
## Frequency Table:
##   Pacifier_Use  <=29  30-40  40-50  over 50  NA_
##   Very often    12     9     2     1     0
##   Often         9    26     4     3     0
##   Sometimes     5    10     4     0     0
##   Rarely/Refused  2     3     0     0     0
##   No Pacifier  101   271    58     9    11
##
## Relative Frequency Table:
##   Pacifier_Use  <=29  30-40  40-50  over 50   NA_ Total
##   Very often   9.3%  2.8%  2.9%   7.7%  0.0%  4.4%
##   Often        7.0%  8.2%  5.9%  23.1%  0.0%  7.8%
##   Sometimes    3.9%  3.1%  5.9%   0.0%  0.0%  3.5%
##   Rarely/Refused 1.6%  0.9%  0.0%   0.0%  0.0%  0.9%
##   No Pacifier  78.3% 85.0% 85.3%  69.2% 100.0% 83.3%
##
## Cramér's V:  0.04210886
```

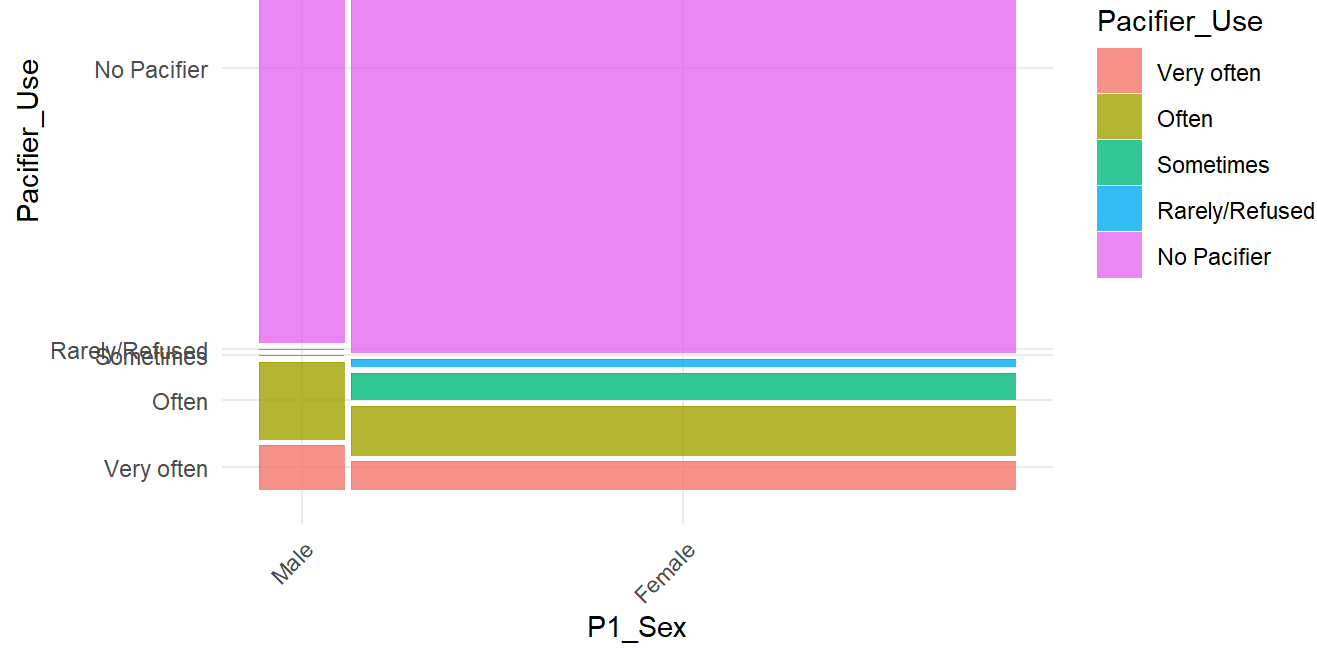
Mosaic Plot of P1_Age and Pacifier_Use



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier_Use and P1_Sex -----
## Bivariate Analysis of Pacifier_Use and P1_Sex
##
## Frequency Table:
##   Pacifier_Use  Male Female
##   Very often     4     20
##   Often          7     35
##   Sometimes      0     19
##   Rarely/Refused  0      5
##   No Pacifier    50    400
##
## Relative Frequency Table:
##   Pacifier_Use  Male Female Total
##   Very often   6.6%   4.2%  4.4%
##   Often       11.5%   7.3%  7.8%
##   Sometimes    0.0%   4.0%  3.5%
##   Rarely/Refused 0.0%   1.0%  0.9%
##   No Pacifier  82.0%  83.5% 83.3%
##
## Cramér's V:  0.04229073
```

Mosaic Plot of P1_Sex and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and P1_Ethnicity -----
## Bivariate Analysis of Pacifier_Use and P1_Ethnicity
##
## Frequency Table:
##   Pacifier_Use Not Hispanic Hispanic
##   Very often      18         6
##   Often          24        18
##   Sometimes       10         9
##   Rarely/Refused   4         1
##   No Pacifier     295       155
##
## Relative Frequency Table:
##   Pacifier_Use Not Hispanic Hispanic Total
##   Very often      5.1%      3.2%  4.4%
##   Often          6.8%      9.5%  7.8%
##   Sometimes       2.8%      4.8%  3.5%
##   Rarely/Refused  1.1%      0.5%  0.9%
##   No Pacifier     84.0%     82.0% 83.3%
##
## Cramér's V:  0.006154184
```

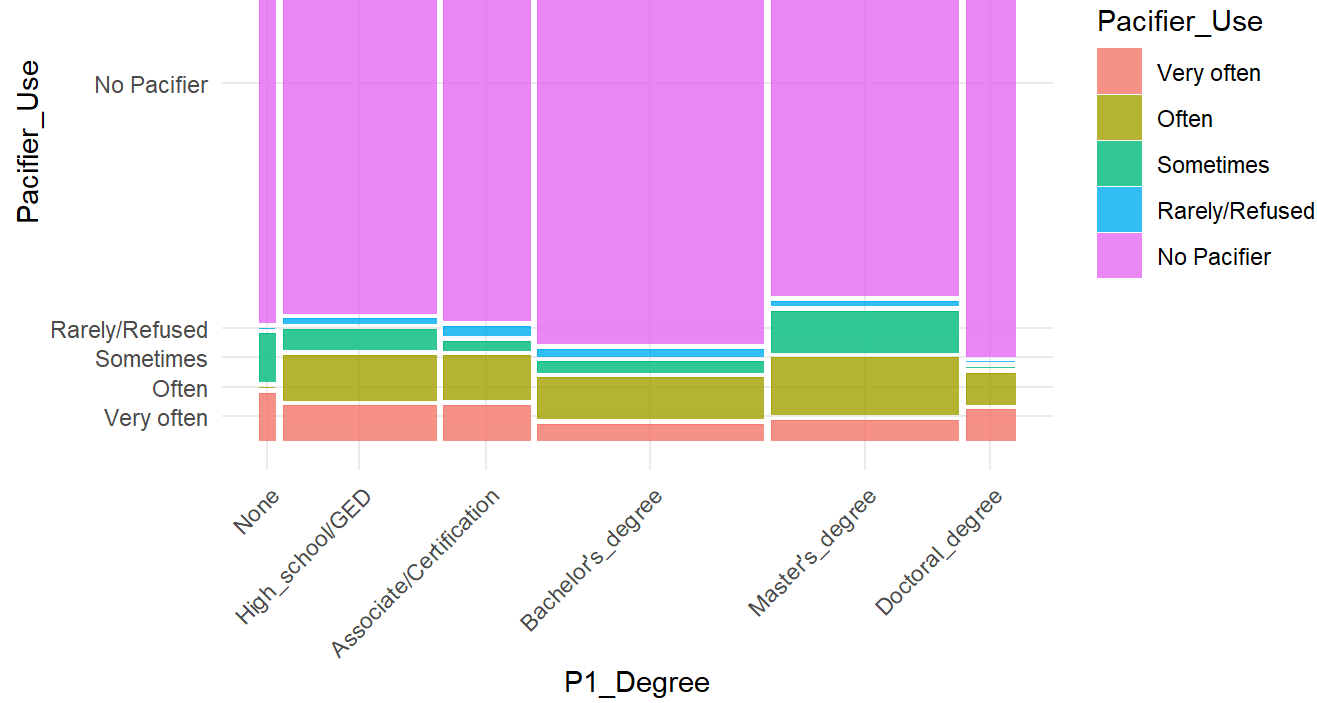
Mosaic Plot of P1_Ethnicity and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and P1_Degree -----
## Bivariate Analysis of Pacifier_Use and P1_Degree
##
## Frequency Table:
##      Pacifier_Use None High_school/GED Associate/Certification Bachelor's_degree
##      Very often    1          7          4          5
##      Often         0          9          5         12
##      Sometimes     1          4          1          3
##      Rarely/Refused 0          1          1          2
##      No Pacifier   10         94         54         148
##      Master's_degree Doctoral_degree
##      5             2
##      14            2
##      10            0
##      1             0
##      111           33
##
## Relative Frequency Table:
##      Pacifier_Use None High_school/GED Associate/Certification Bachelor's_degree
##      Very often   8.3%          6.1%          6.2%          2.9%
##      Often        0.0%          7.8%          7.7%          7.1%
##      Sometimes    8.3%          3.5%          1.5%          1.8%
##      Rarely/Refused 0.0%          0.9%          1.5%          1.2%
##      No Pacifier  83.3%          81.7%          83.1%          87.1%
##      Master's_degree Doctoral_degree Total
##      3.5%          5.4%  4.4%
##      9.9%          5.4%  7.8%
##      7.1%          0.0%  3.5%
##      0.7%          0.0%  0.9%
##      78.7%          89.2% 83.3%
##
## Cramér's V: 0
```

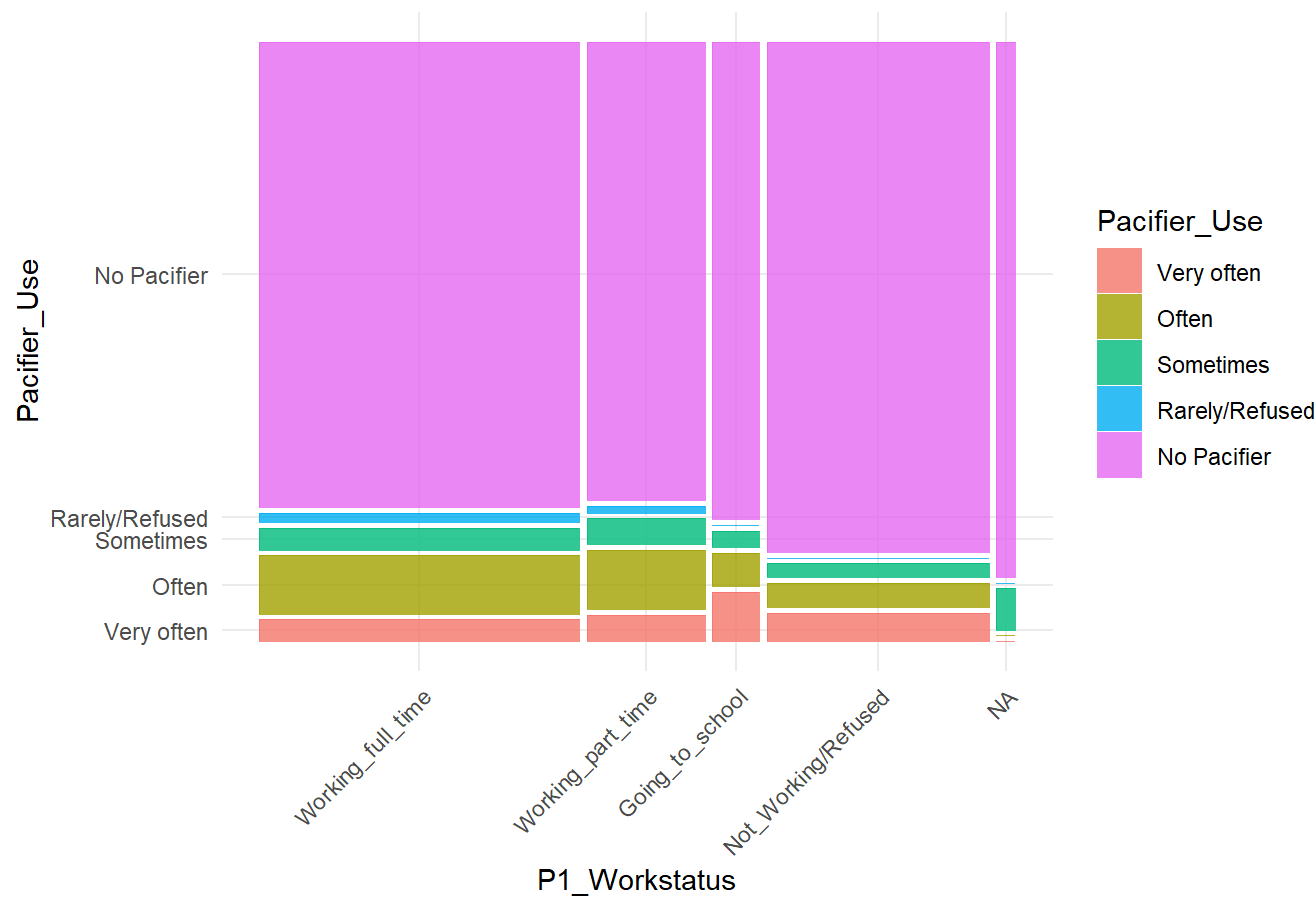
Mosaic Plot of P1_Degree and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and P1_Workstatus -----
## Bivariate Analysis of Pacifier_Use and P1_Workstatus
##
## Frequency Table:
##   Pacifier_Use Working_full_time Working_part_time Going_to_school
##   Very often      9              4              3
##   Often          24              9              2
##   Sometimes       9              4              1
##   Rarely/Refused  4              1              0
##   No Pacifier     192            70             29
##   Not_Working/Refused NA_
##               8    0
##               7    0
##               4    1
##               0    0
##            146  13
##
## Relative Frequency Table:
##   Pacifier_Use Working_full_time Working_part_time Going_to_school
##   Very often      3.8%           4.5%           8.6%
##   Often          10.1%          10.2%           5.7%
##   Sometimes       3.8%           4.5%           2.9%
##   Rarely/Refused  1.7%           1.1%           0.0%
##   No Pacifier     80.7%          79.5%          82.9%
##   Not_Working/Refused NA_ Total
##           4.8% 0.0% 4.4%
##           4.2% 0.0% 7.8%
##           2.4% 7.1% 3.5%
##           0.0% 0.0% 0.9%
##           88.5% 92.9% 83.3%
##
## Cramér's V:  0
```

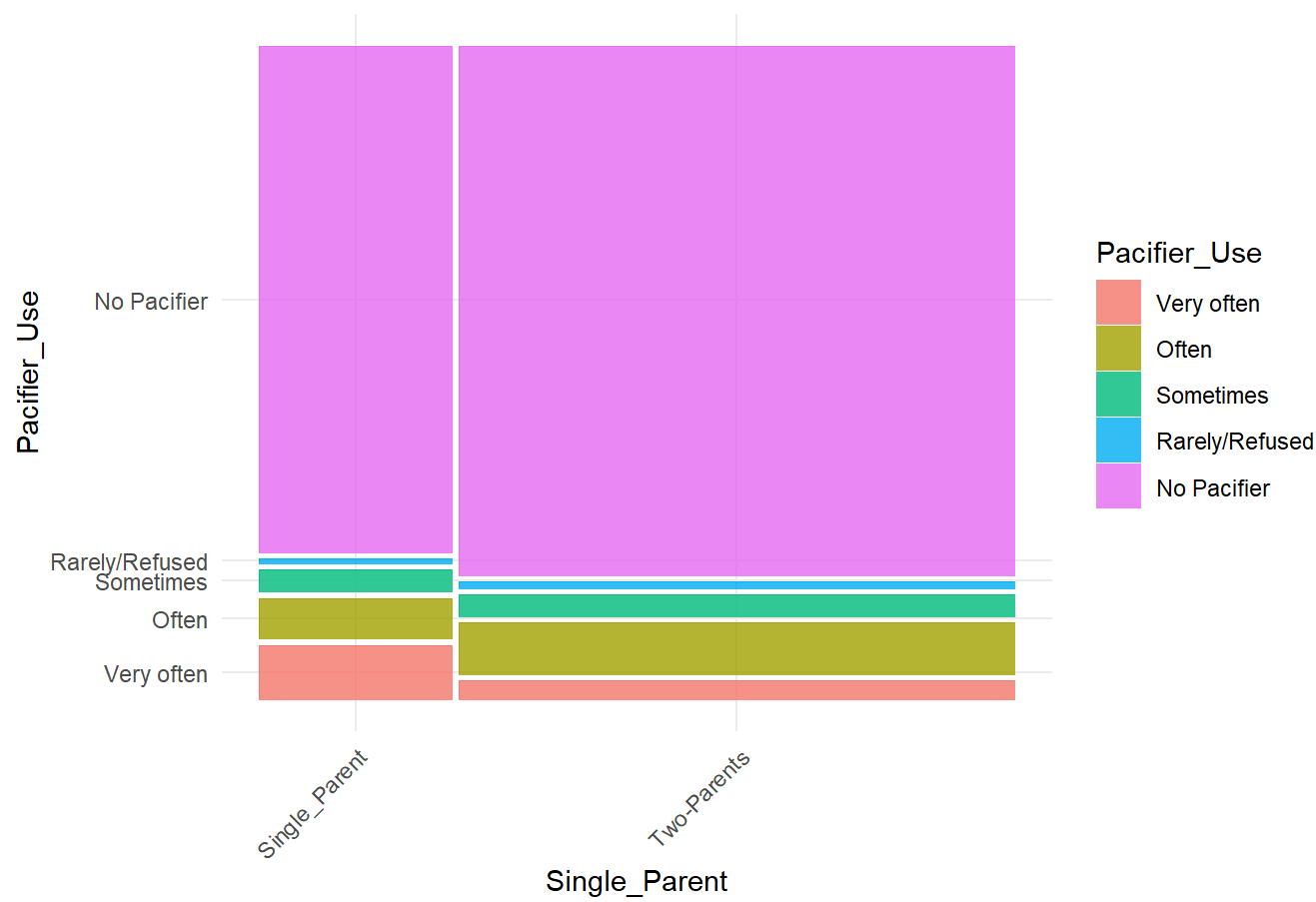
Mosaic Plot of P1_Workstatus and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and Single_Parent -----
## Bivariate Analysis of Pacifier_Use and Single_Parent
##
## Frequency Table:
##   Pacifier_Use Single_Parent Two-Parents
##   Very often      12          12
##   Often           9          33
##   Sometimes       5          14
##   Rarely/Refused  1           4
##   No Pacifier     112         338
##
## Relative Frequency Table:
##   Pacifier_Use Single_Parent Two-Parents Total
##   Very often      8.6%        3.0%  4.4%
##   Often           6.5%        8.2%  7.8%
##   Sometimes       3.6%        3.5%  3.5%
##   Rarely/Refused  0.7%        1.0%  0.9%
##   No Pacifier     80.6%       84.3%  83.3%
##
## Cramér's V:  0.08668972
```

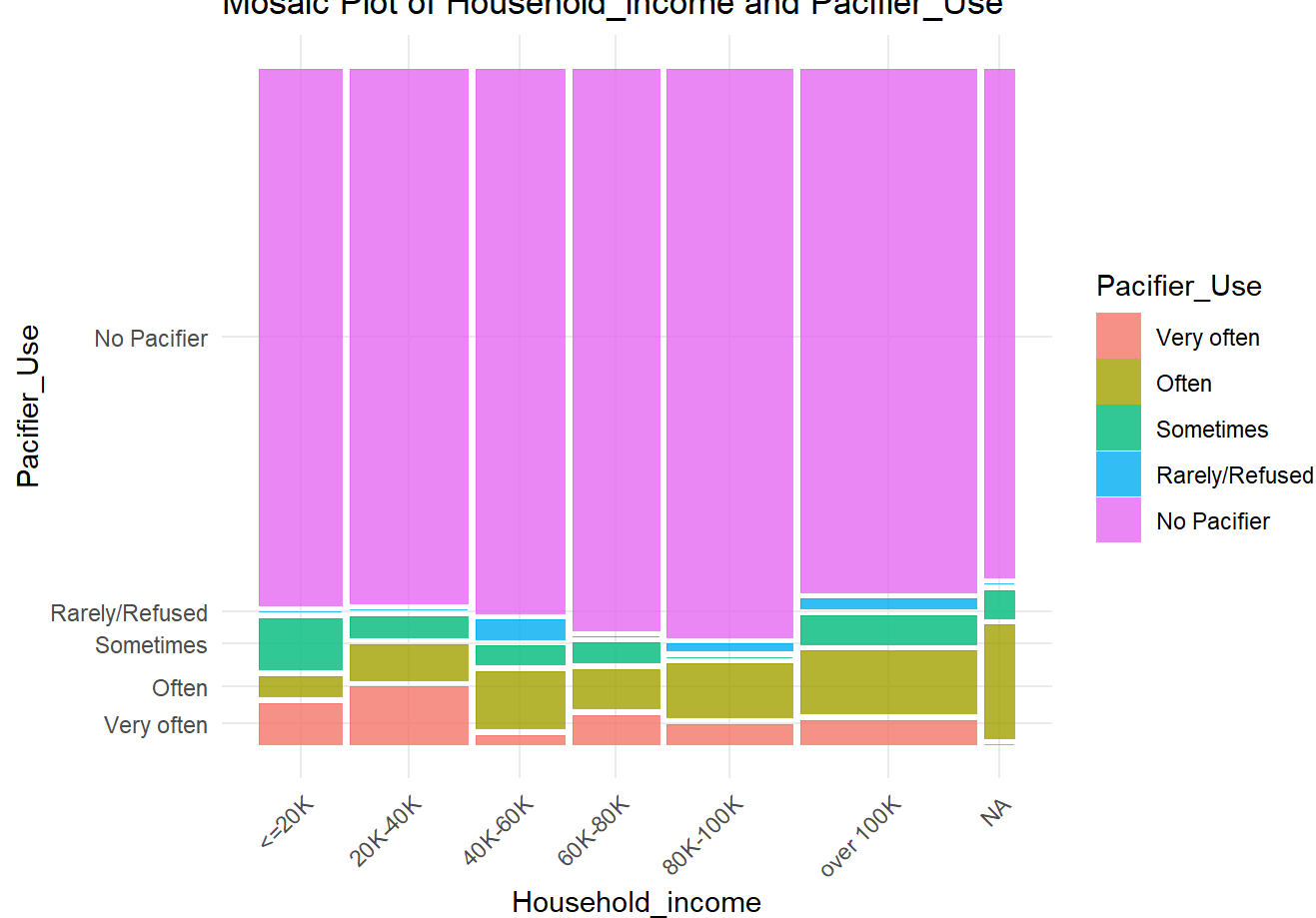

Mosaic Plot of Single_Parent and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and Household_income -----
## Bivariate Analysis of Pacifier_Use and Household_income
##
## Frequency Table:
##   Pacifier_Use  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_
##   Very often      4      8      1      3      3      5      0
##   Often           2      5      6      4      8     13      4
##   Sometimes       5      3      2      2      0      6      1
##   Rarely/Refused  0      0      2      0      1      2      0
##   No Pacifier     52     74     57     57     84     108    18
##
## Relative Frequency Table:
##   Pacifier_Use  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_ Total
##   Very often   6.3%   8.9%   1.5%   4.5%   3.1%   3.7%  0.0%  4.4%
##   Often        3.2%   5.6%   8.8%   6.1%   8.3%   9.7% 17.4%  7.8%
##   Sometimes    7.9%   3.3%   2.9%   3.0%   0.0%   4.5%  4.3%  3.5%
##   Rarely/Refused 0.0%   0.0%   2.9%   0.0%   1.0%   1.5%  0.0%  0.9%
##   No Pacifier  82.5%  82.2%  83.8%  86.4%  87.5%  80.6% 78.3% 83.3%
##
## Cramér's V:  0.0373447
```

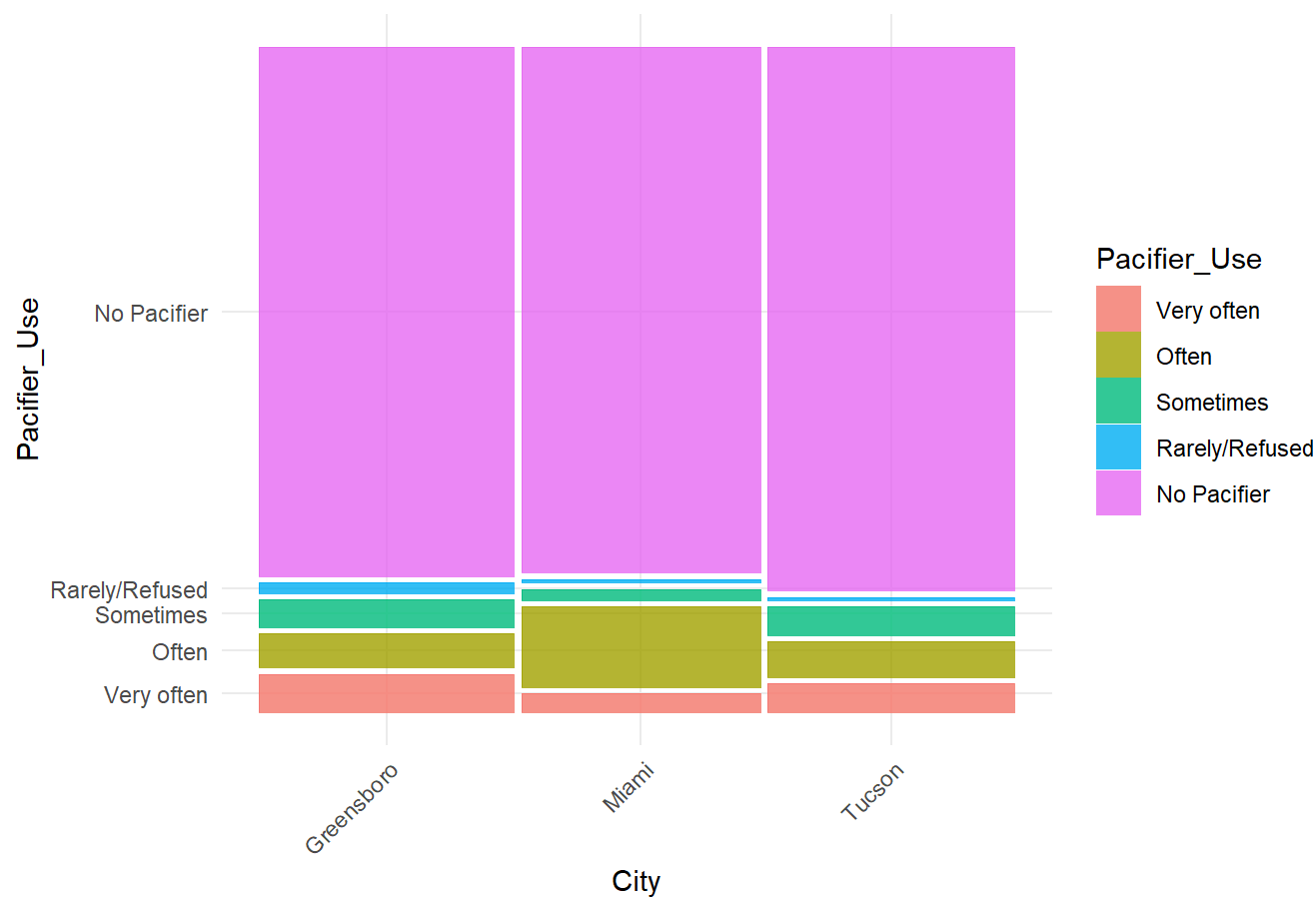
Mosaic Plot of Household_income and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and City -----
## Bivariate Analysis of Pacifier_Use and City
##
## Frequency Table:
##   Pacifier_Use Greensboro Miami Tucson
##   Very often      11      5      8
##   Often          10     22     10
##   Sometimes       8      3      8
##   Rarely/Refused   3      1      1
##   No Pacifier     154    143    153
##
## Relative Frequency Table:
##   Pacifier_Use Greensboro Miami Tucson Total
##   Very often      5.9%  2.9%  4.4%  4.4%
##   Often          5.4% 12.6%  5.6%  7.8%
##   Sometimes       4.3%  1.7%  4.4%  3.5%
##   Rarely/Refused  1.6%  0.6%  0.6%  0.9%
##   No Pacifier     82.8% 82.2% 85.0% 83.3%
##
## Cramér's V:  0.07194223
```

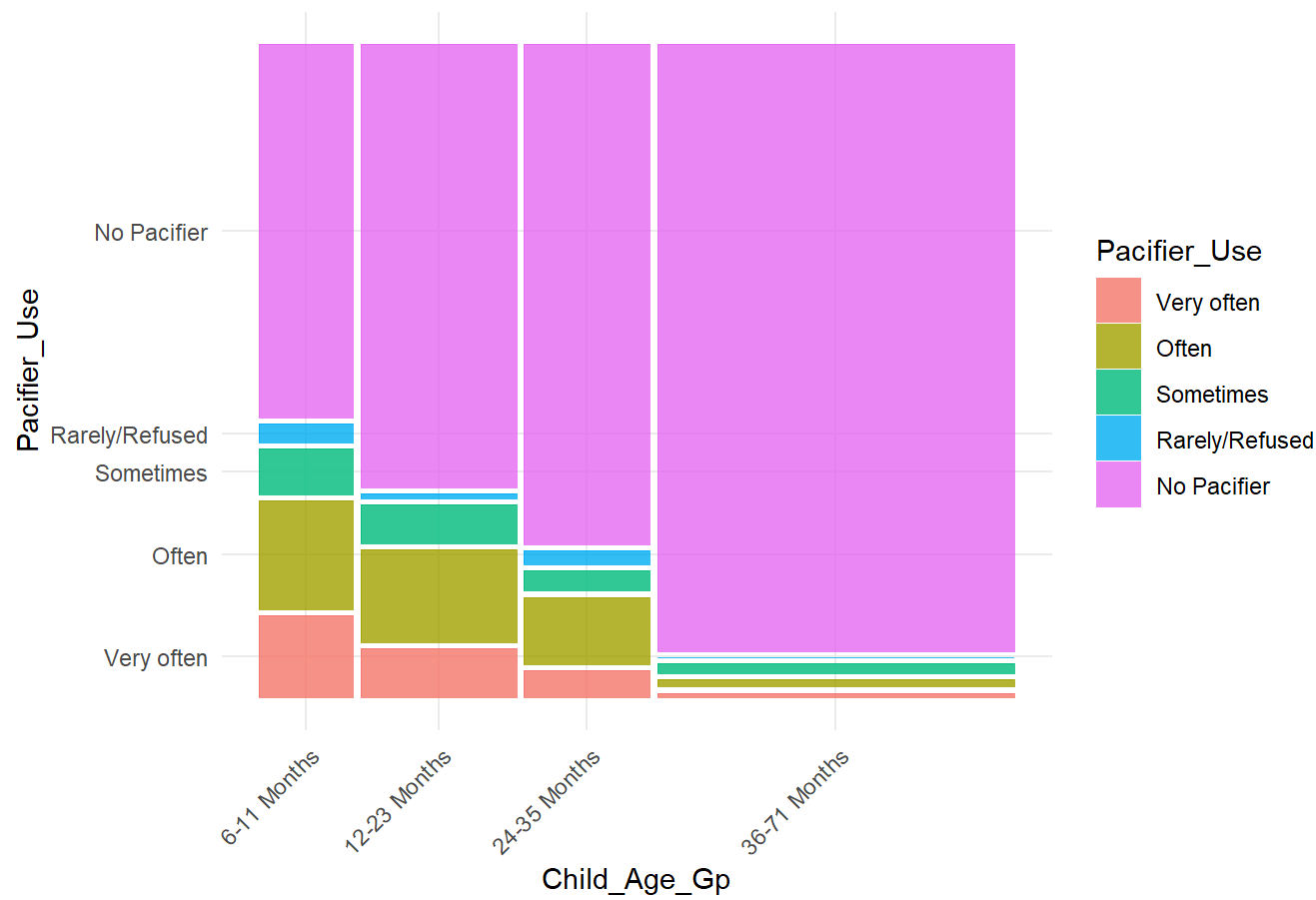
Mosaic Plot of City and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and Child_Age_Gp -----
## Bivariate Analysis of Pacifier_Use and Child_Age_Gp
##
## Frequency Table:
##   Pacifier_Use 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##   Very often      9          9          4          2
##   Often          12         17         10          3
##   Sometimes       5          7          3          4
##   Rarely/Refused  2          1          2          0
##   No Pacifier     41         81         74         254
##
## Relative Frequency Table:
##   Pacifier_Use 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##   Very often    13.0%      7.8%      4.3%      0.8%  4.4%
##   Often         17.4%     14.8%     10.8%     1.1%  7.8%
##   Sometimes     7.2%      6.1%      3.2%     1.5%  3.5%
##   Rarely/Refused 2.9%      0.9%      2.2%     0.0%  0.9%
##   No Pacifier   59.4%     70.4%     79.6%    96.6% 83.3%
##
## Cramér's V: 0.2074306
```

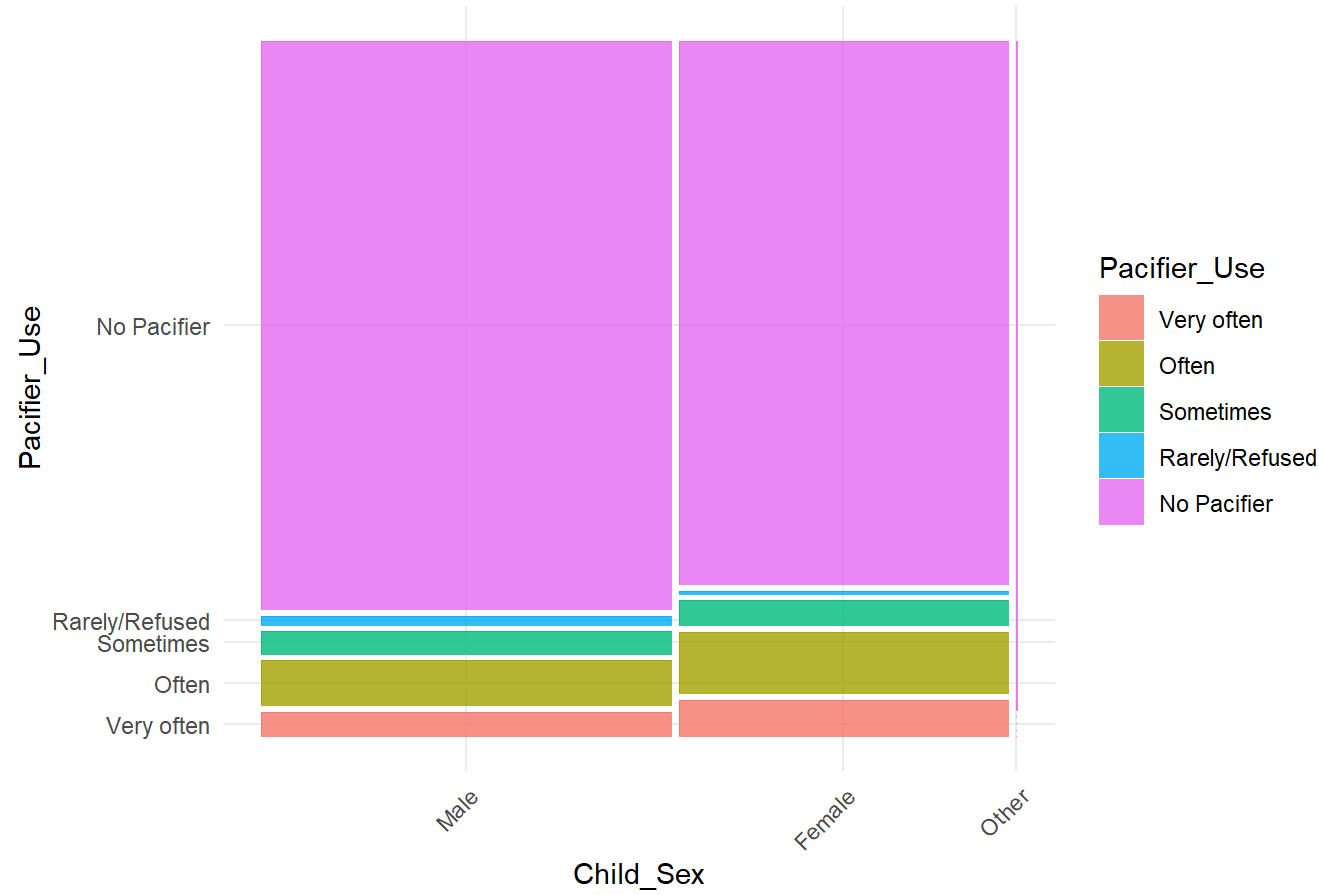
Mosaic Plot of Child_Age_Gp and Pacifier_Use



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Use and Child_Sex -----
## Bivariate Analysis of Pacifier_Use and Child_Sex
##
## Frequency Table:
##   Pacifier_Use Male Female Other
##   Very often   11    13     0
##   Often        20    22     0
##   Sometimes    10     9     0
##   Rarely/Refused  4     1     0
##   No Pacifier  254   195     1
##
## Relative Frequency Table:
##   Pacifier_Use Male Female Other Total
##   Very often   3.7%  5.4%  0.0%  4.4%
##   Often        6.7%  9.2%  0.0%  7.8%
##   Sometimes    3.3%  3.8%  0.0%  3.5%
##   Rarely/Refused 1.3%  0.4%  0.0%  0.9%
##   No Pacifier  84.9% 81.2% 100.0% 83.3%
##
## Cramér's V:  0
```

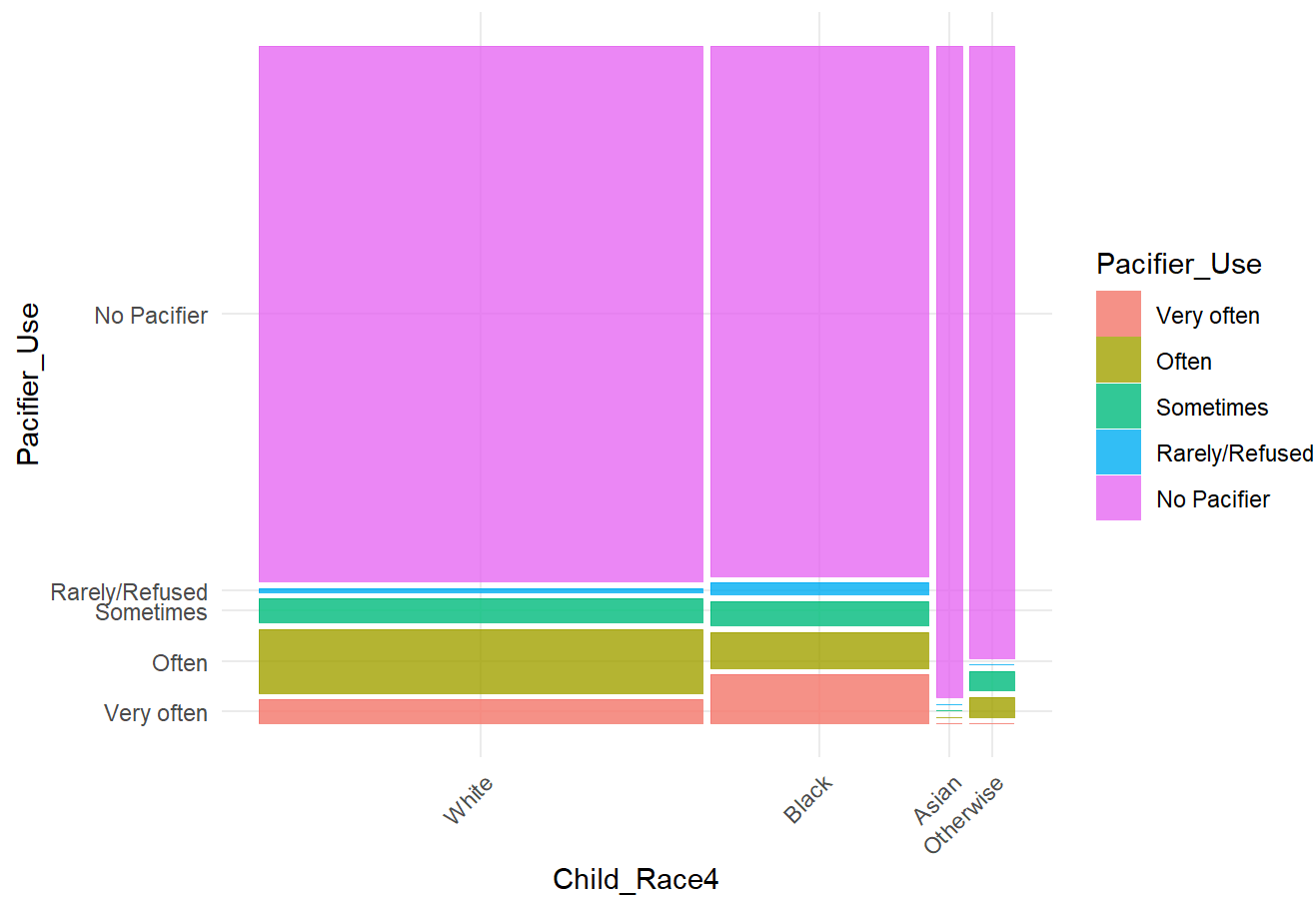
Mosaic Plot of Child_Sex and Pacifier_Use



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier_Use and Child_Race4 -----
## Bivariate Analysis of Pacifier_Use and Child_Race4
##
## Frequency Table:
##   Pacifier_Use White Black Asian Otherwise
##   Very often   12   12    0         0
##   Often        32    9    0         1
##   Sometimes    12    6    0         1
##   Rarely/Refused  2    3    0         0
##   No Pacifier  269  131   19        31
##
## Relative Frequency Table:
##   Pacifier_Use White Black Asian Otherwise Total
##   Very often   3.7%  7.5%  0.0%    0.0%  4.4%
##   Often        9.8%  5.6%  0.0%    3.0%  7.8%
##   Sometimes    3.7%  3.7%  0.0%    3.0%  3.5%
##   Rarely/Refused 0.6%  1.9%  0.0%    0.0%  0.9%
##   No Pacifier  82.3% 81.4% 100.0%   93.9% 83.3%
##
## Cramér's V:  0.04607023
```

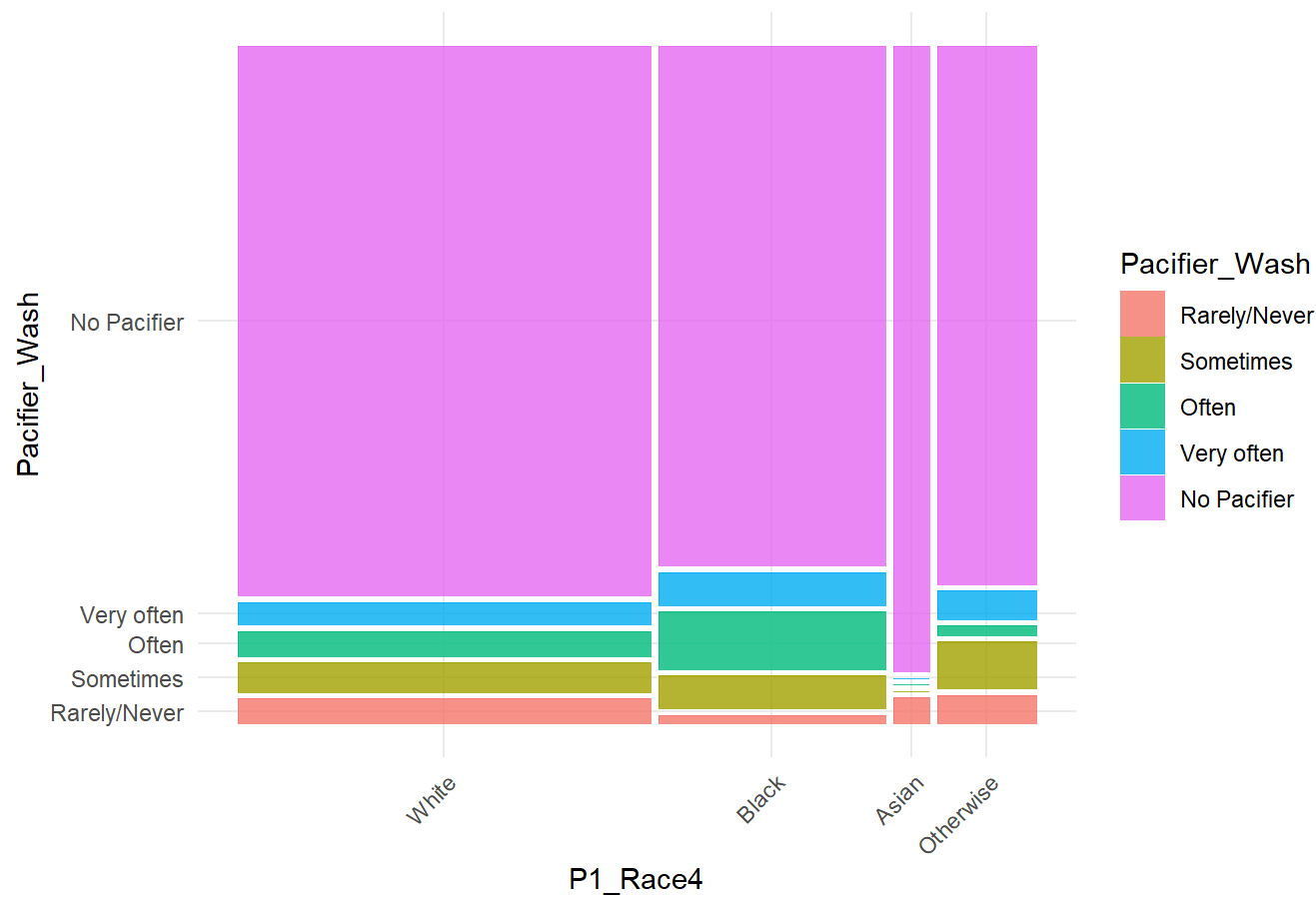
Mosaic Plot of Child_Race4 and Pacifier_Use



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier_Wash and P1_Race4 -----
## Bivariate Analysis of Pacifier_Wash and P1_Race4
##
## Frequency Table:
## Pacifier_Wash White Black Asian Otherwise
## Rarely/Never 11 2 1 3
## Sometimes 13 8 0 5
## Often 11 14 0 1
## Very often 10 8 0 3
## No Pacifier 243 126 24 57
##
## Relative Frequency Table:
## Pacifier_Wash White Black Asian Otherwise Total
## Rarely/Never 3.8% 1.3% 4.0% 4.3% 3.1%
## Sometimes 4.5% 5.1% 0.0% 7.2% 4.8%
## Often 3.8% 8.9% 0.0% 1.4% 4.8%
## Very often 3.5% 5.1% 0.0% 4.3% 3.9%
## No Pacifier 84.4% 79.7% 96.0% 82.6% 83.3%
##
## Cramér's V: 0.04935144
```

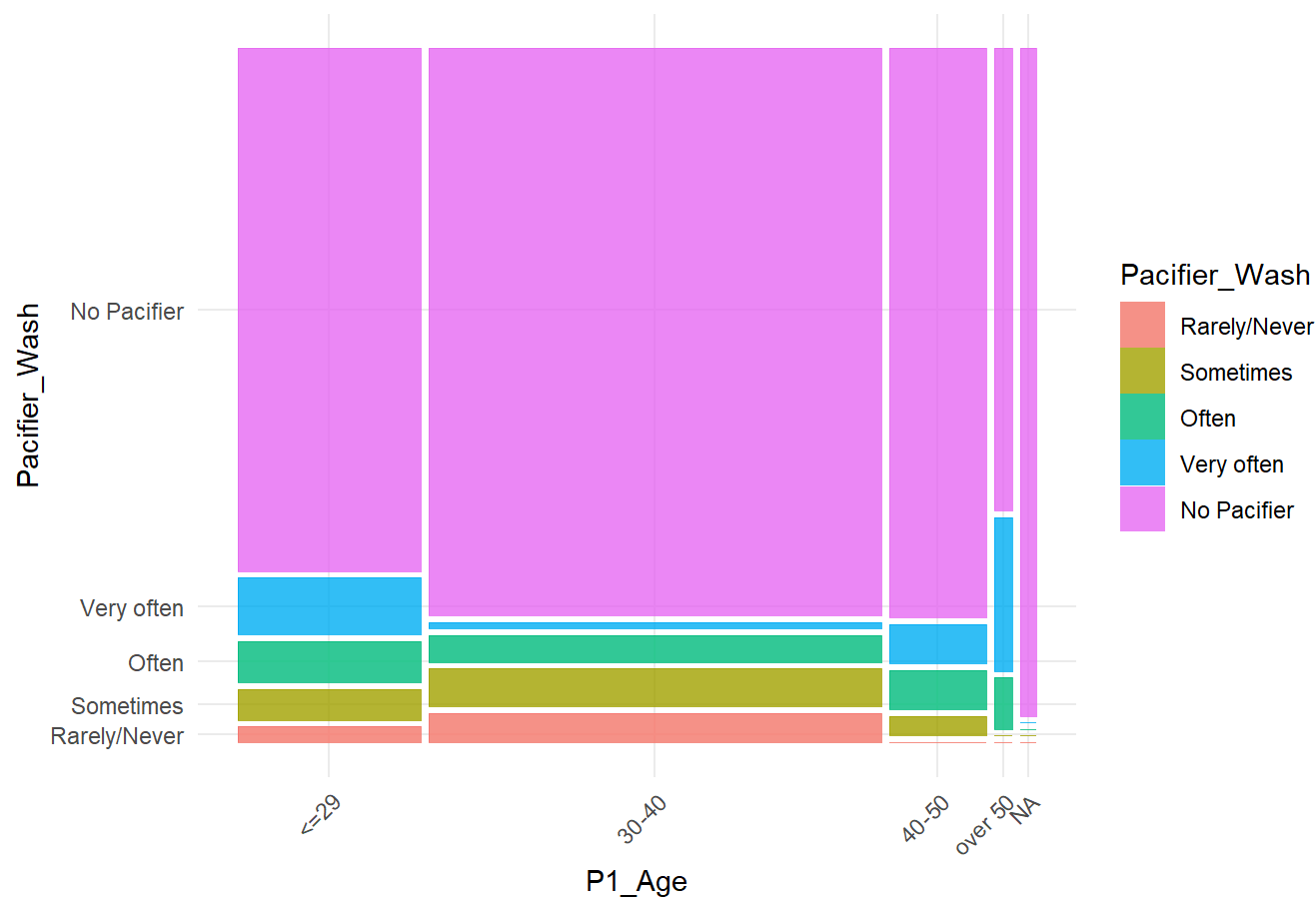
Mosaic Plot of P1_Race4 and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and P1_Age -----
## Bivariate Analysis of Pacifier_Wash and P1_Age
##
## Frequency Table:
## Pacifier_Wash <=29 30-40 40-50 over 50 NA_
## Rarely/Never      3    14     0     0  0
## Sometimes         6    18     2     0  0
## Often             8    13     4     1  0
## Very often        11     3     4     3  0
## No Pacifier       101   271    58     9  11
##
## Relative Frequency Table:
## Pacifier_Wash <=29 30-40 40-50 over 50    NA_ Total
## Rarely/Never  2.3%  4.4%  0.0%   0.0%  0.0%  3.1%
## Sometimes    4.7%  5.6%  2.9%   0.0%  0.0%  4.8%
## Often        6.2%  4.1%  5.9%   7.7%  0.0%  4.8%
## Very often   8.5%  0.9%  5.9%  23.1%  0.0%  3.9%
## No Pacifier  78.3% 85.0% 85.3%  69.2% 100.0% 83.3%
##
## Cramér's V:  0.100157
```

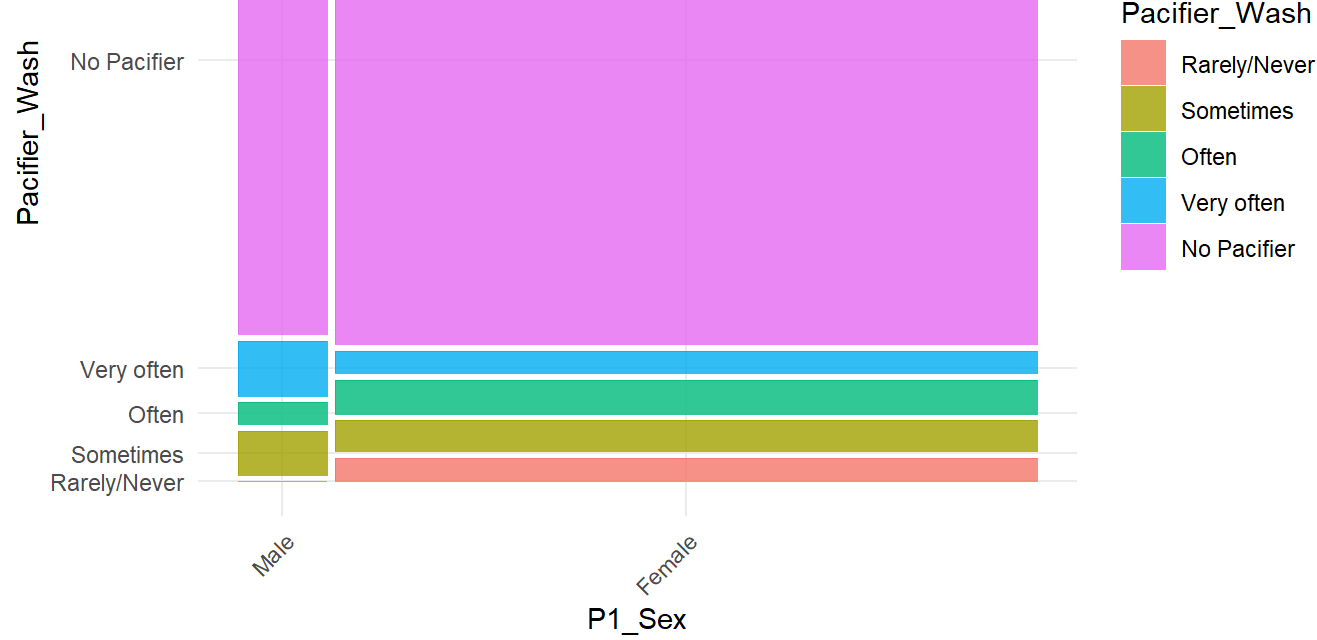
Mosaic Plot of P1_Age and Pacifier_Wash



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier_Wash and P1_Sex -----
## Bivariate Analysis of Pacifier_Wash and P1_Sex
##
## Frequency Table:
## Pacifier_Wash Male Female
## Rarely/Never 0 17
## Sometimes 4 22
## Often 2 24
## Very often 5 16
## No Pacifier 50 400
##
## Relative Frequency Table:
## Pacifier_Wash Male Female Total
## Rarely/Never 0.0% 3.5% 3.1%
## Sometimes 6.6% 4.6% 4.8%
## Often 3.3% 5.0% 4.8%
## Very often 8.2% 3.3% 3.9%
## No Pacifier 82.0% 83.5% 83.3%
##
## Cramér's V: 0.06425243
```

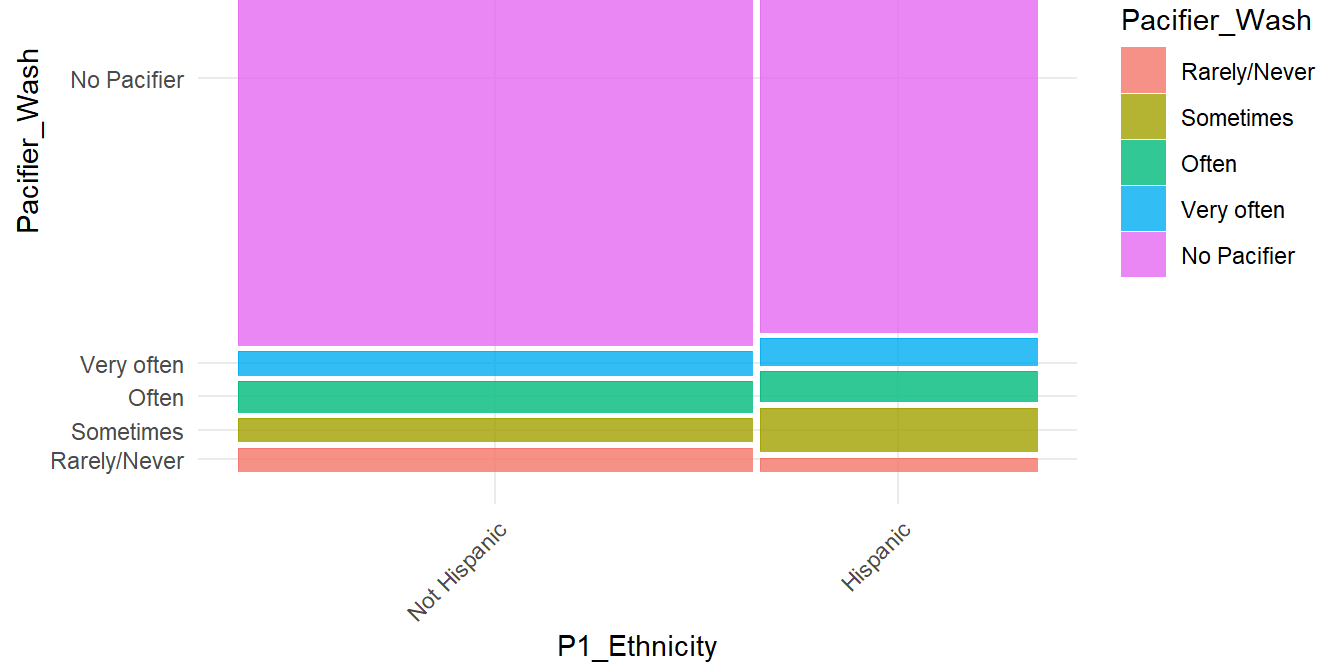

Mosaic Plot of P1_Sex and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and P1_Ethnicity -----
## Bivariate Analysis of Pacifier_Wash and P1_Ethnicity
##
## Frequency Table:
## Pacifier_Wash Not Hispanic Hispanic
## Rarely/Never      13      4
## Sometimes        13     13
## Often            17      9
## Very often        13      8
## No Pacifier       295    155
##
## Relative Frequency Table:
## Pacifier_Wash Not Hispanic Hispanic Total
## Rarely/Never      3.7%    2.1%  3.1%
## Sometimes         3.7%    6.9%  4.8%
## Often             4.8%    4.8%  4.8%
## Very often        3.7%    4.2%  3.9%
## No Pacifier       84.0%   82.0% 83.3%
##
## Cramér's V:  0
```

Mosaic Plot of P1_Ethnicity and Pacifier_Wash



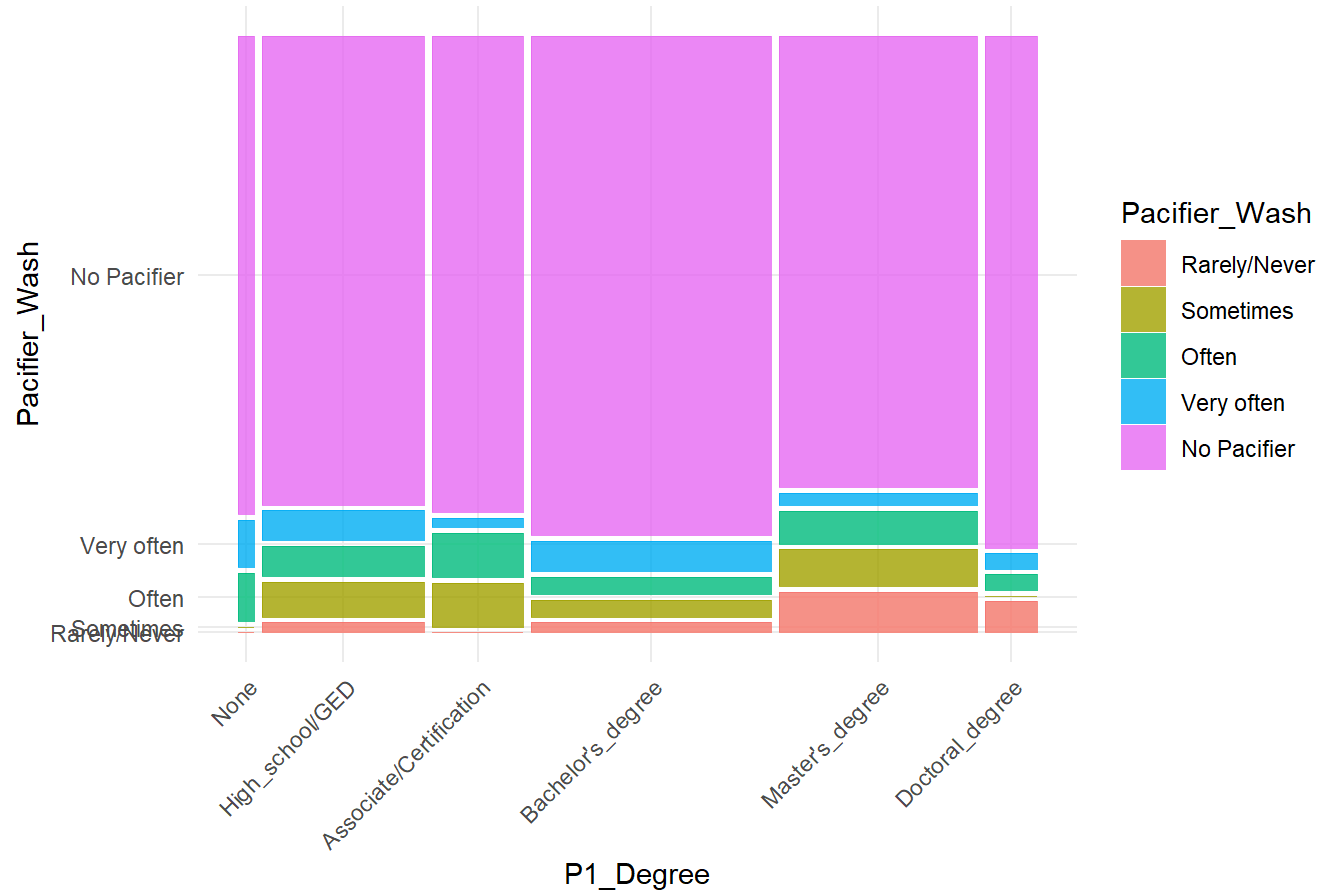
Saving 7 x 5 in image

```

##
##
## ----- Analysis for Pacifier_Wash and P1_Degree -----
## Bivariate Analysis of Pacifier_Wash and P1_Degree
##
## Frequency Table:
## Pacifier_Wash None High_school/GED Associate/Certification Bachelor's_degree
## Rarely/Never 0 2 0 3
## Sometimes 0 7 5 5
## Often 1 6 5 5
## Very often 1 6 1 9
## No Pacifier 10 94 54 148
## Master's_degree Doctoral_degree
## 10 2
## 9 0
## 8 1
## 3 1
## 111 33
##
## Relative Frequency Table:
## Pacifier_Wash None High_school/GED Associate/Certification Bachelor's_degree
## Rarely/Never 0.0% 1.7% 0.0% 1.8%
## Sometimes 0.0% 6.1% 7.7% 2.9%
## Often 8.3% 5.2% 7.7% 2.9%
## Very often 8.3% 5.2% 1.5% 5.3%
## No Pacifier 83.3% 81.7% 83.1% 87.1%
## Master's_degree Doctoral_degree Total
## 7.1% 5.4% 3.1%
## 6.4% 0.0% 4.8%
## 5.7% 2.7% 4.8%
## 2.1% 2.7% 3.9%
## 78.7% 89.2% 83.3%
##
## Cramér's V: 0.05203608

```

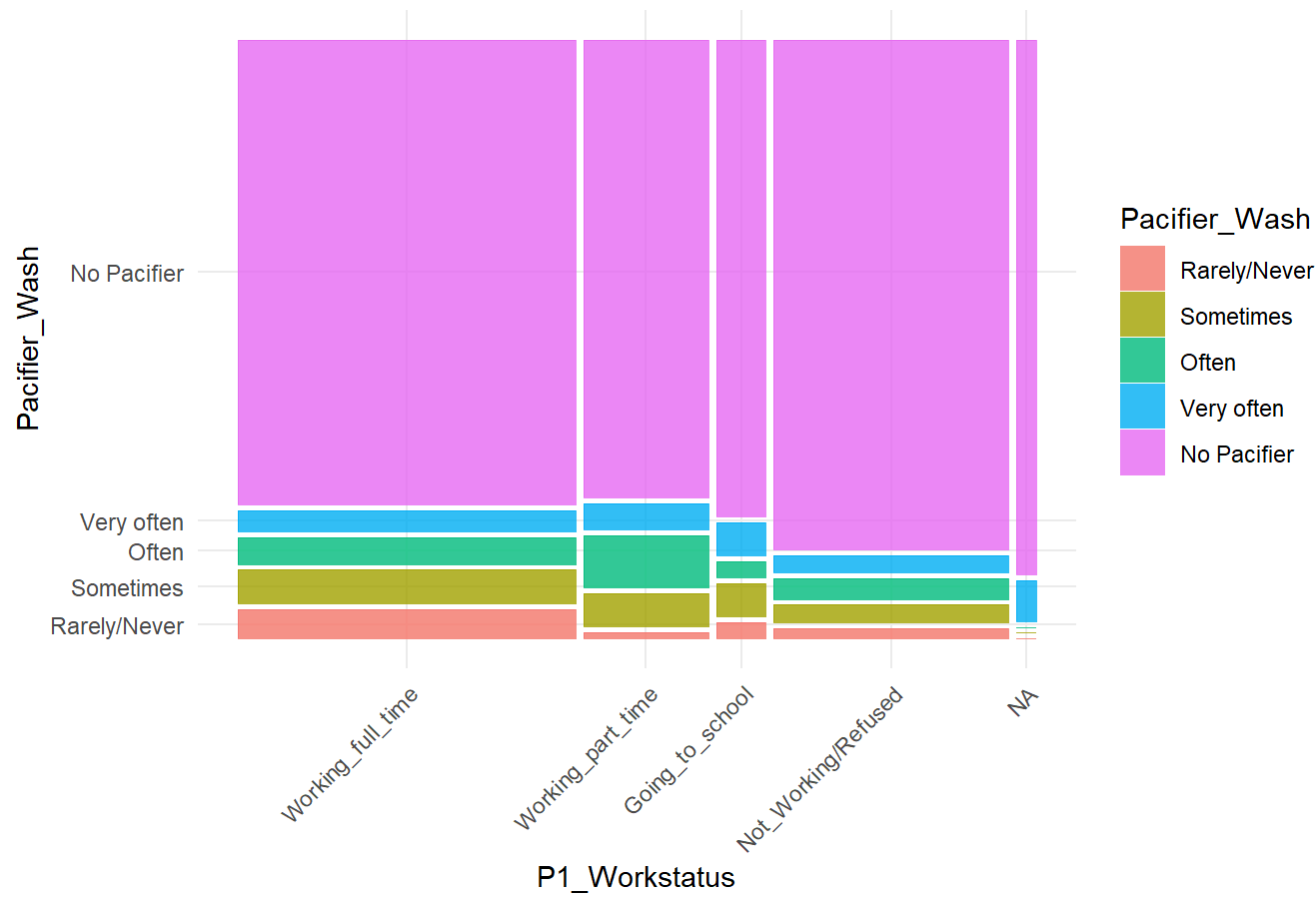
Mosaic Plot of P1_Degree and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and P1_Workstatus -----
## Bivariate Analysis of Pacifier_Wash and P1_Workstatus
##
## Frequency Table:
## Pacifier_Wash Working_full_time Working_part_time Going_to_school
## Rarely/Never          12           1           1
## Sometimes             14           5           2
## Often                 11           8           1
## Very often            9           4           2
## No Pacifier          192          70          29
## Not_Working/Refused NA_
##                3    0
##                5    0
##                6    0
##                5    1
##             146  13
##
## Relative Frequency Table:
## Pacifier_Wash Working_full_time Working_part_time Going_to_school
## Rarely/Never          5.0%          1.1%          2.9%
## Sometimes             5.9%          5.7%          5.7%
## Often                 4.6%          9.1%          2.9%
## Very often            3.8%          4.5%          5.7%
## No Pacifier          80.7%          79.5%          82.9%
## Not_Working/Refused  NA_ Total
##             1.8% 0.0% 3.1%
##             3.0% 0.0% 4.8%
##             3.6% 0.0% 4.8%
##             3.0% 7.1% 3.9%
##             88.5% 92.9% 83.3%
##
## Cramér's V: 0
```

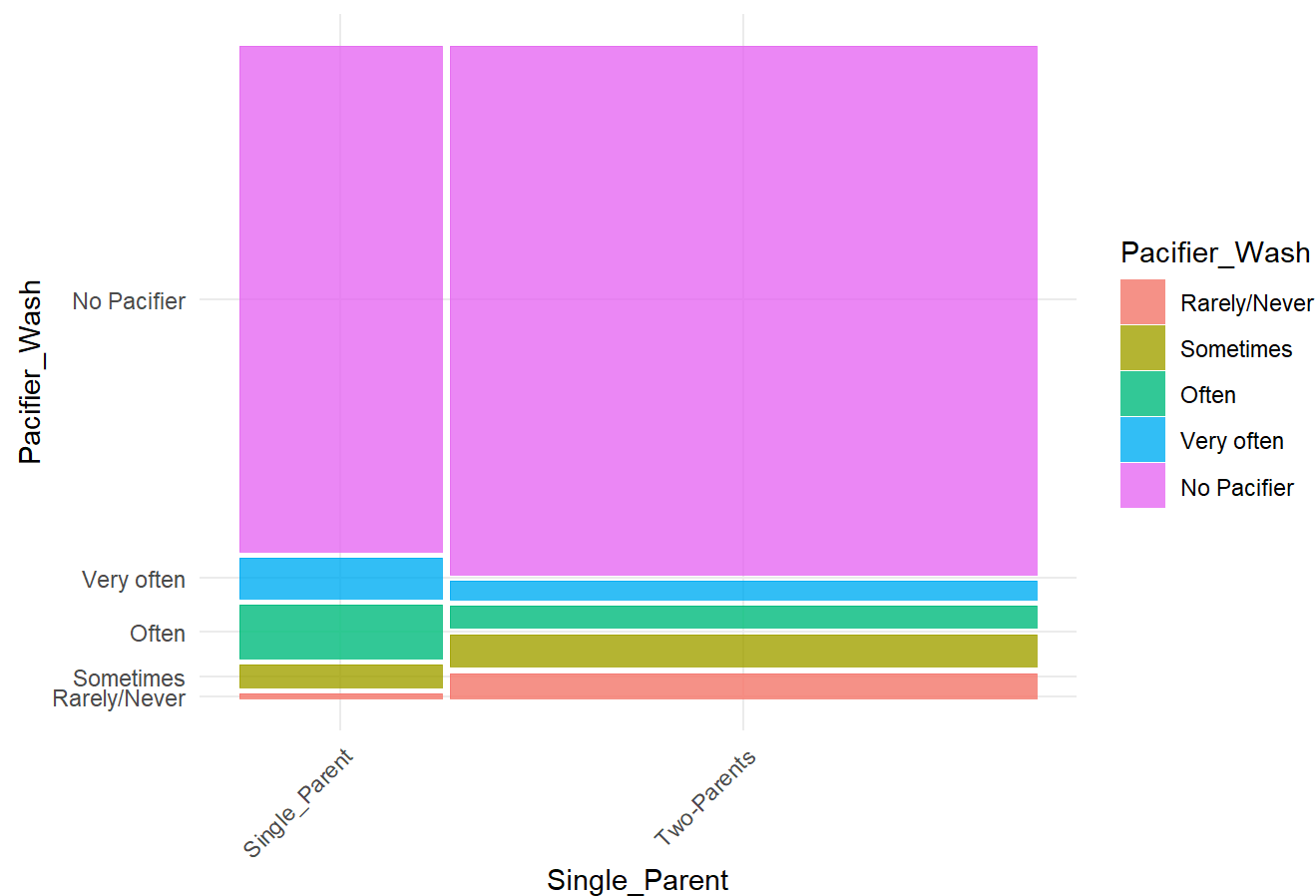
Mosaic Plot of P1_Workstatus and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and Single_Parent -----
## Bivariate Analysis of Pacifier_Wash and Single_Parent
##
## Frequency Table:
## Pacifier_Wash Single_Parent Two-Parents
## Rarely/Never      1      16
## Sometimes         5      21
## Often            12      14
## Very often        9      12
## No Pacifier      112     338
##
## Relative Frequency Table:
## Pacifier_Wash Single_Parent Two-Parents Total
## Rarely/Never      0.7%      4.0%  3.1%
## Sometimes         3.6%      5.2%  4.8%
## Often            8.6%      3.5%  4.8%
## Very often        6.5%      3.0%  3.9%
## No Pacifier      80.6%     84.3% 83.3%
##
## Cramér's V:  0.130175
```

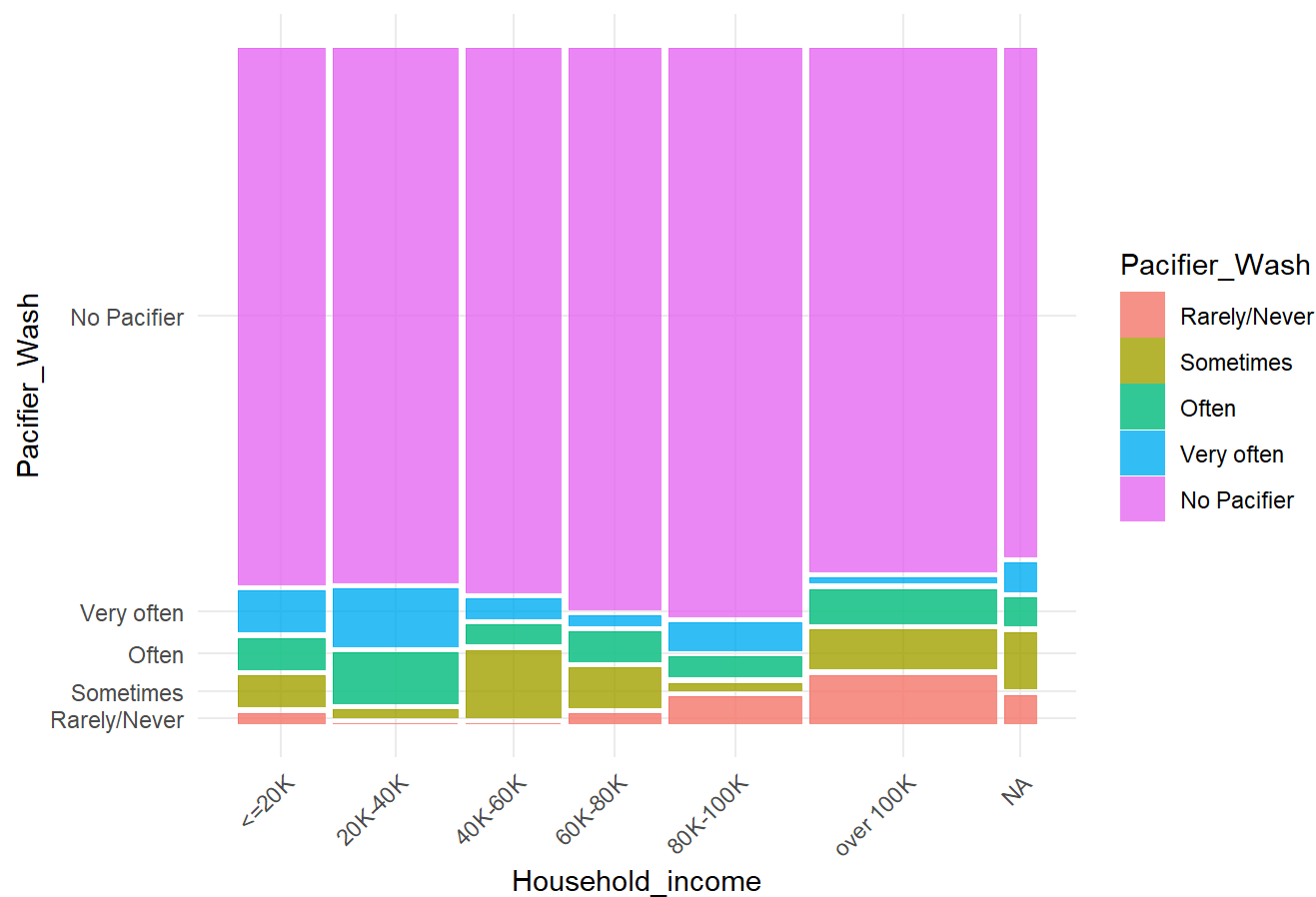
Mosaic Plot of Single_Parent and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and Household_income -----
## Bivariate Analysis of Pacifier_Wash and Household_income
##
## Frequency Table:
## Pacifier_Wash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
## Rarely/Never      1      0      0      1      4      10  1
## Sometimes         3      1      7      4      1      8  2
## Often             3      7      2      3      3      7  1
## Very often        4      8      2      1      4      1  1
## No Pacifier       52     74     57     57     84     108 18
##
## Relative Frequency Table:
## Pacifier_Wash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
## Rarely/Never  1.6%    0.0%    0.0%    1.5%    4.2%    7.5% 4.3% 3.1%
## Sometimes    4.8%    1.1%   10.3%    6.1%    1.0%    6.0% 8.7% 4.8%
## Often        4.8%    7.8%    2.9%    4.5%    3.1%    5.2% 4.3% 4.8%
## Very often   6.3%    8.9%    2.9%    1.5%    4.2%    0.7% 4.3% 3.9%
## No Pacifier  82.5%   82.2%   83.8%   86.4%   87.5%   80.6% 78.3% 83.3%
##
## Cramér's V:  0.08595394
```

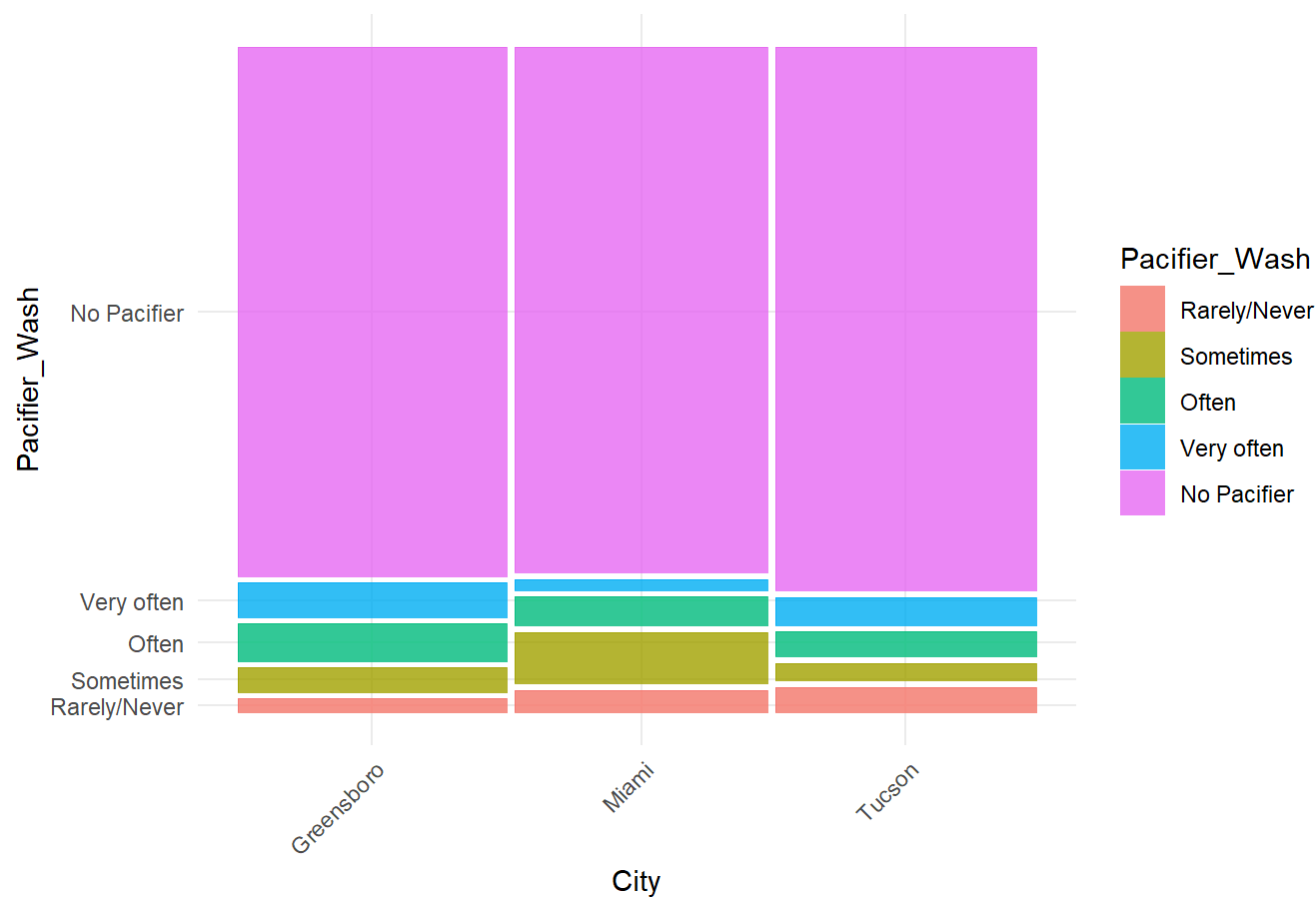
Mosaic Plot of Household_income and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and City -----
## Bivariate Analysis of Pacifier_Wash and City
##
## Frequency Table:
## Pacifier_Wash Greensboro Miami Tucson
## Rarely/Never      4      6      7
## Sometimes         7     14      5
## Often            11      8      7
## Very often        10      3      8
## No Pacifier       154    143    153
##
## Relative Frequency Table:
## Pacifier_Wash Greensboro Miami Tucson Total
## Rarely/Never      2.2%  3.4%  3.9%  3.1%
## Sometimes         3.8%  8.0%  2.8%  4.8%
## Often            5.9%  4.6%  3.9%  4.8%
## Very often        5.4%  1.7%  4.4%  3.9%
## No Pacifier       82.8% 82.2% 85.0% 83.3%
##
## Cramér's V:  0.05177245
```

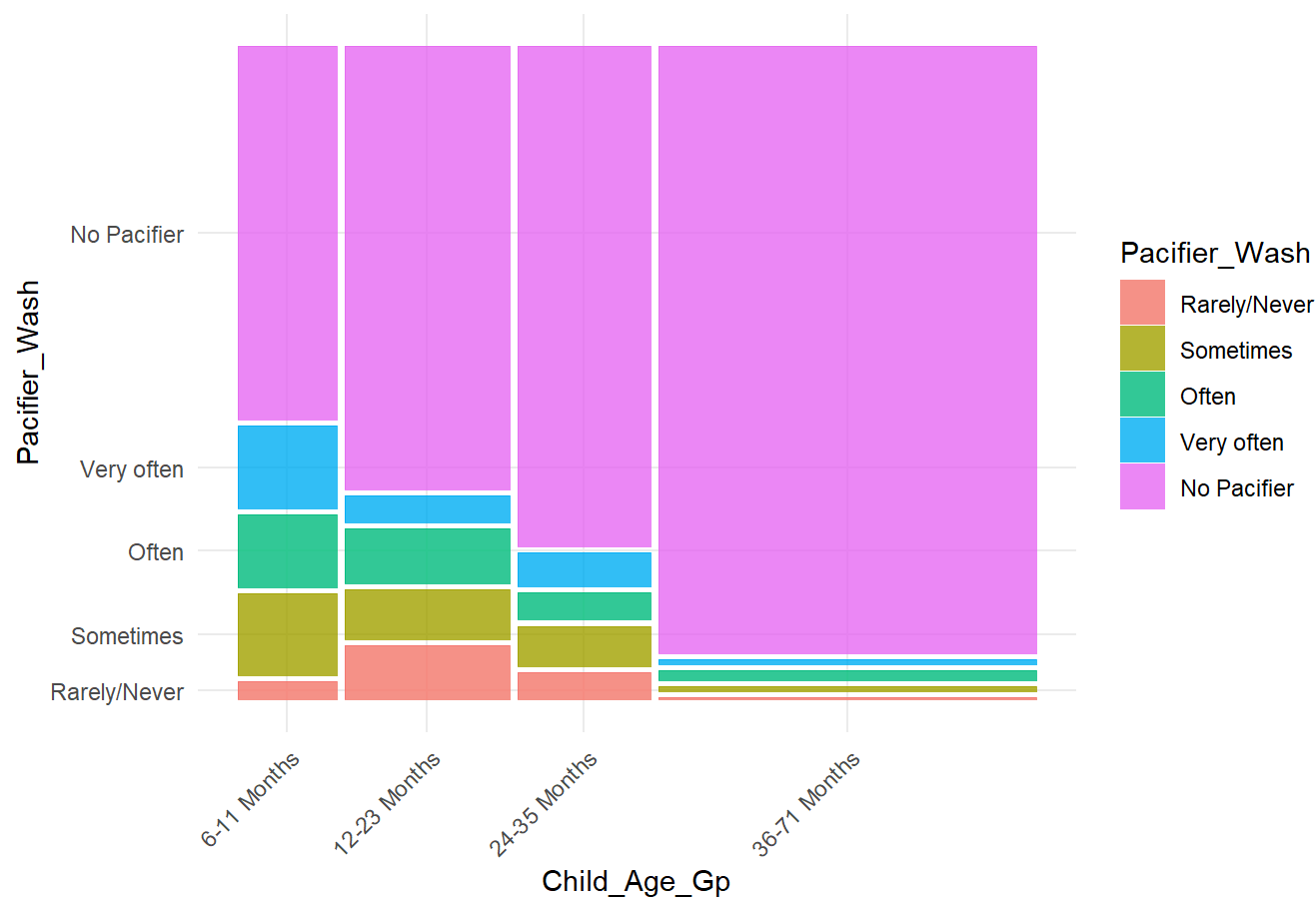

Mosaic Plot of City and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and Child_Age_Gp -----
## Bivariate Analysis of Pacifier_Wash and Child_Age_Gp
##
## Frequency Table:
## Pacifier_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months
## Rarely/Never 2 10 4 1
## Sometimes 9 9 6 2
## Often 8 10 4 4
## Very often 9 5 5 2
## No Pacifier 41 81 74 254
##
## Relative Frequency Table:
## Pacifier_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
## Rarely/Never 2.9% 8.7% 4.3% 0.4% 3.1%
## Sometimes 13.0% 7.8% 6.5% 0.8% 4.8%
## Often 11.6% 8.7% 4.3% 1.5% 4.8%
## Very often 13.0% 4.3% 5.4% 0.8% 3.9%
## No Pacifier 59.4% 70.4% 79.6% 96.6% 83.3%
##
## Cramér's V: 0.2205445
```

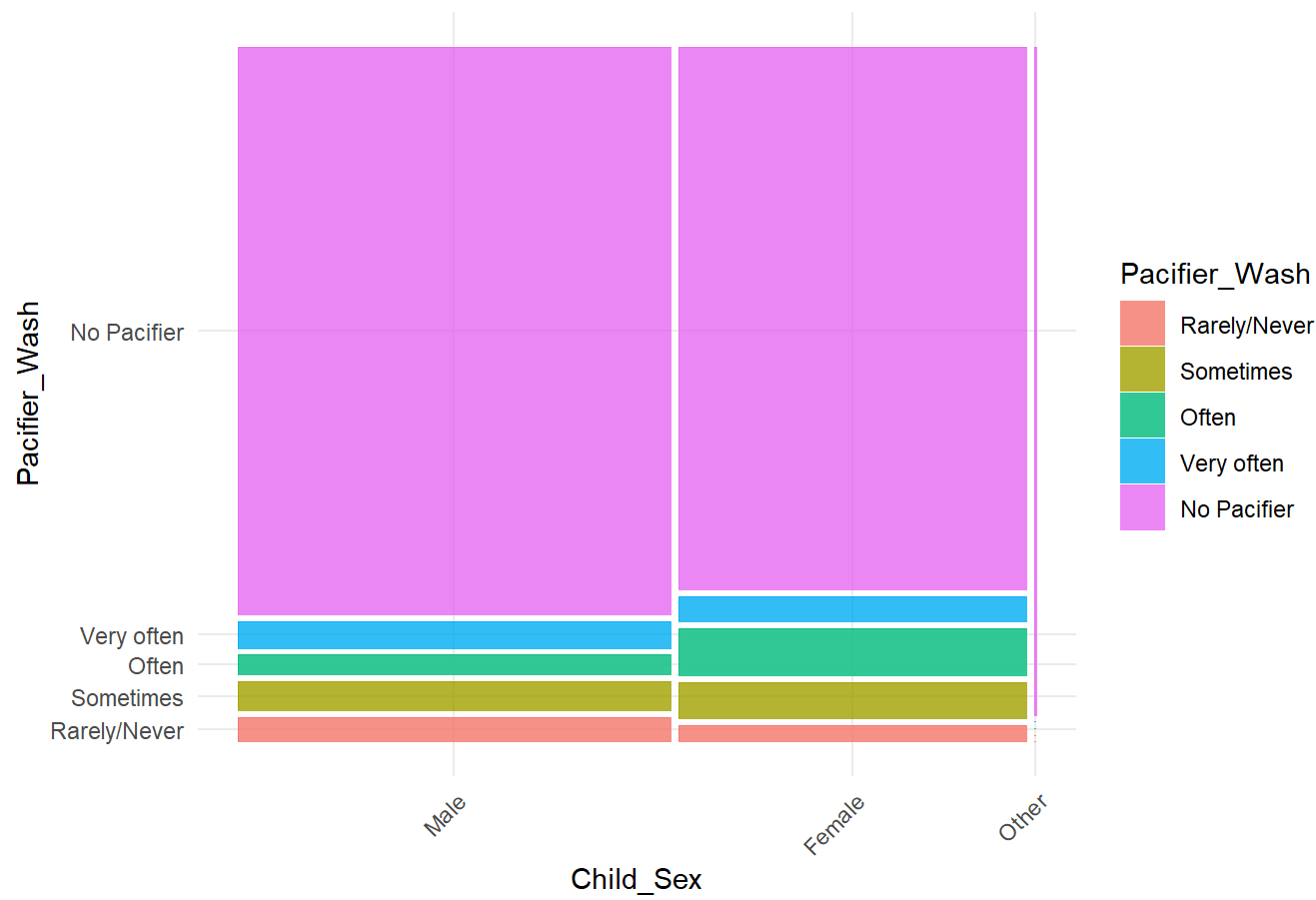
Mosaic Plot of Child_Age_Gp and Pacifier_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Pacifier_Wash and Child_Sex -----
## Bivariate Analysis of Pacifier_Wash and Child_Sex
##
## Frequency Table:
## Pacifier_Wash Male Female Other
## Rarely/Never 11 6 0
## Sometimes 13 13 0
## Often 9 17 0
## Very often 12 9 0
## No Pacifier 254 195 1
##
## Relative Frequency Table:
## Pacifier_Wash Male Female Other Total
## Rarely/Never 3.7% 2.5% 0.0% 3.1%
## Sometimes 4.3% 5.4% 0.0% 4.8%
## Often 3.0% 7.1% 0.0% 4.8%
## Very often 4.0% 3.8% 0.0% 3.9%
## No Pacifier 84.9% 81.2% 100.0% 83.3%
##
## Cramér's V: 0
```

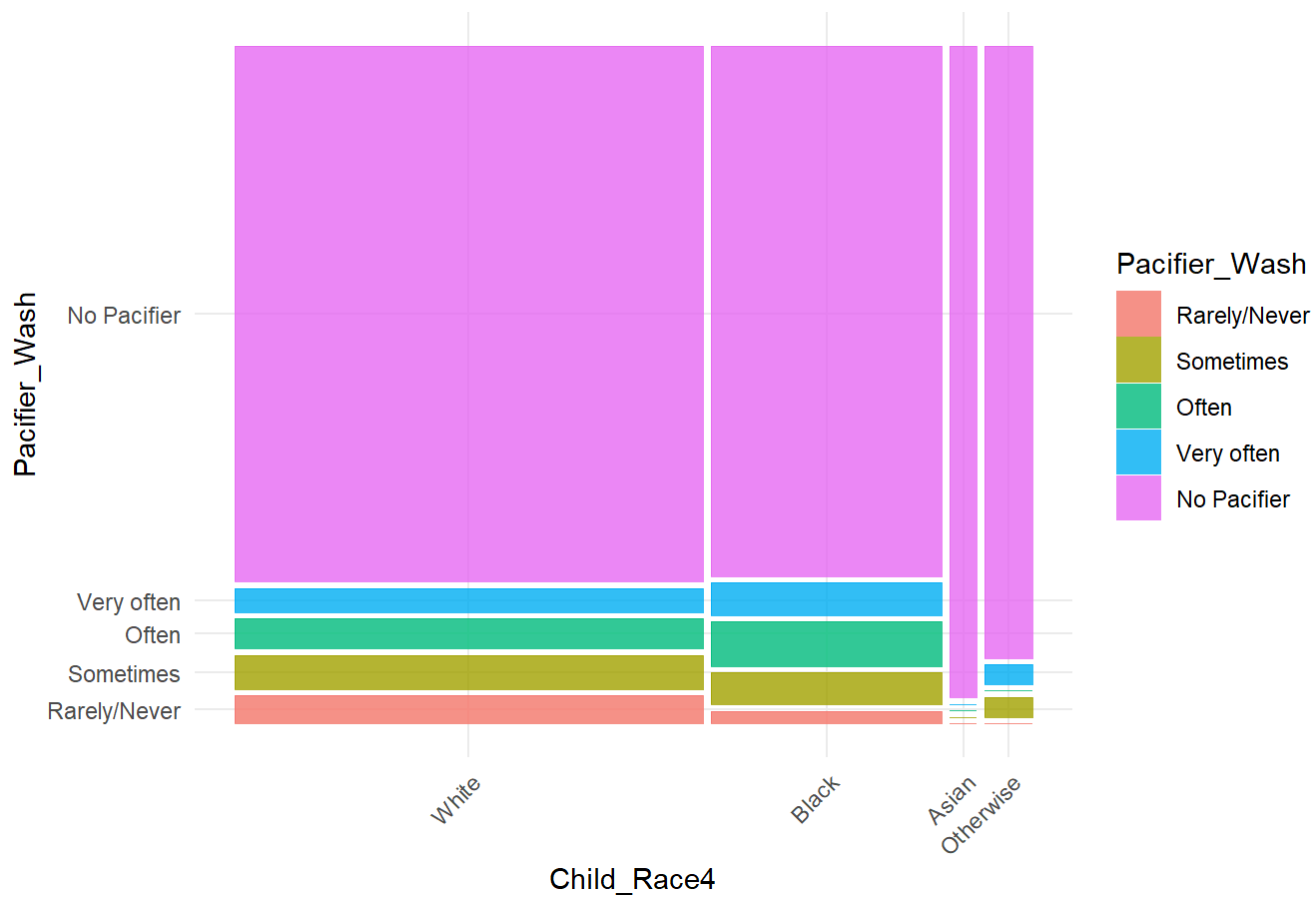
Mosaic Plot of Child_Sex and Pacifier_Wash



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier_Wash and Child_Race4 -----
## Bivariate Analysis of Pacifier_Wash and Child_Race4
##
## Frequency Table:
## Pacifier_Wash White Black Asian Otherwise
## Rarely/Never 14 3 0 0
## Sometimes 17 8 0 1
## Often 15 11 0 0
## Very often 12 8 0 1
## No Pacifier 269 131 19 31
##
## Relative Frequency Table:
## Pacifier_Wash White Black Asian Otherwise Total
## Rarely/Never 4.3% 1.9% 0.0% 0.0% 3.1%
## Sometimes 5.2% 5.0% 0.0% 3.0% 4.8%
## Often 4.6% 6.8% 0.0% 0.0% 4.8%
## Very often 3.7% 5.0% 0.0% 3.0% 3.9%
## No Pacifier 82.3% 81.4% 100.0% 93.9% 83.3%
##
## Cramér's V: 0
```

Mosaic Plot of Child_Race4 and Pacifier_Wash



```
## Saving 7 x 5 in image
```

```
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```

##	var1	var2	cramer_v
## 1	Pacifier	Child_Age_Gp	0.368884675
## 2	Pacifier_Wash	Child_Age_Gp	0.220544538
## 3	Pacifier_Use	Child_Age_Gp	0.207430560
## 4	Pacifier_Wash	Single_Parent	0.130174986
## 5	Pacifier_Wash	P1_Age	0.100157006
## 6	Pacifier	Child_Race4	0.088119372
## 7	Pacifier_Use	Single_Parent	0.086689716
## 8	Pacifier_Wash	Household_income	0.085953937
## 9	Pacifier	P1_Age	0.077087655
## 10	Pacifier_Use	P1_Race4	0.074044694
## 11	Pacifier_Use	City	0.071942229
## 12	Pacifier_Wash	P1_Sex	0.064252426
## 13	Pacifier	P1_Workstatus	0.063710994
## 14	Pacifier	P1_Race4	0.054429368
## 15	Pacifier_Wash	P1_Degree	0.052036082
## 16	Pacifier_Wash	City	0.051772450
## 17	Pacifier_Wash	P1_Race4	0.049351440
## 18	Pacifier_Use	Child_Race4	0.046070228
## 19	Pacifier_Use	P1_Sex	0.042290735
## 20	Pacifier_Use	P1_Age	0.042108865
## 21	Pacifier_Use	Household_income	0.037344696
## 22	Pacifier	Single_Parent	0.006551674
## 23	Pacifier_Use	P1_Ethnicity	0.006154184
## 24	Pacifier	P1_Sex	0.000000000
## 25	Pacifier	P1_Ethnicity	0.000000000
## 26	Pacifier	P1_Degree	0.000000000
## 27	Pacifier	Household_income	0.000000000
## 28	Pacifier	City	0.000000000
## 29	Pacifier	Child_Sex	0.000000000
## 30	Pacifier_Use	P1_Degree	0.000000000
## 31	Pacifier_Use	P1_Workstatus	0.000000000
## 32	Pacifier_Use	Child_Sex	0.000000000
## 33	Pacifier_Wash	P1_Ethnicity	0.000000000
## 34	Pacifier_Wash	P1_Workstatus	0.000000000
## 35	Pacifier_Wash	Child_Sex	0.000000000
## 36	Pacifier_Wash	Child_Race4	0.000000000

Exploring Blanket Variables

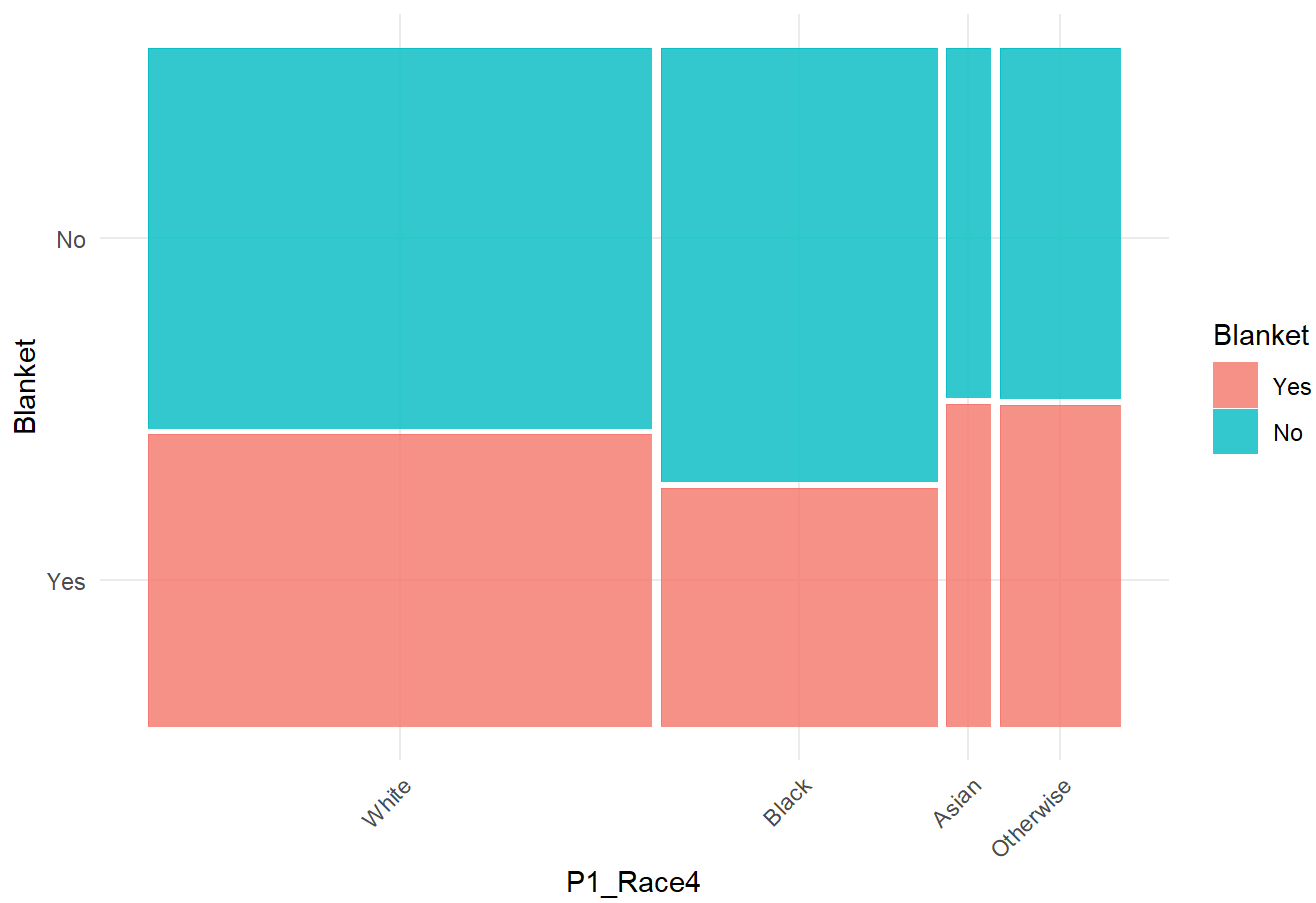
```
## Exploring Blanket Variables
#data=data
list1 = BlanketVars
list2 = DemoVars

cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(),
                              stringsAsFactors = TRUE)

results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = "_vs_")]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values, data.frame(var1= var1, var2 = var2,
                                                            cramer_v = result$cramer_v))
    }
  }
}
```

```
##
##
## ----- Analysis for Blanket and P1_Race4 -----
## Bivariate Analysis of Blanket and P1_Race4
##
## Frequency Table:
##   Blanket White Black Asian Otherwise
##      Yes   125    56   12         33
##      No    163   102   13         36
##
## Relative Frequency Table:
##   Blanket White Black Asian Otherwise Total
##      Yes 43.4% 35.4% 48.0%    47.8% 41.9%
##      No 56.6% 64.6% 52.0%    52.2% 58.1%
##
## Cramér's V:  0.04997199
```

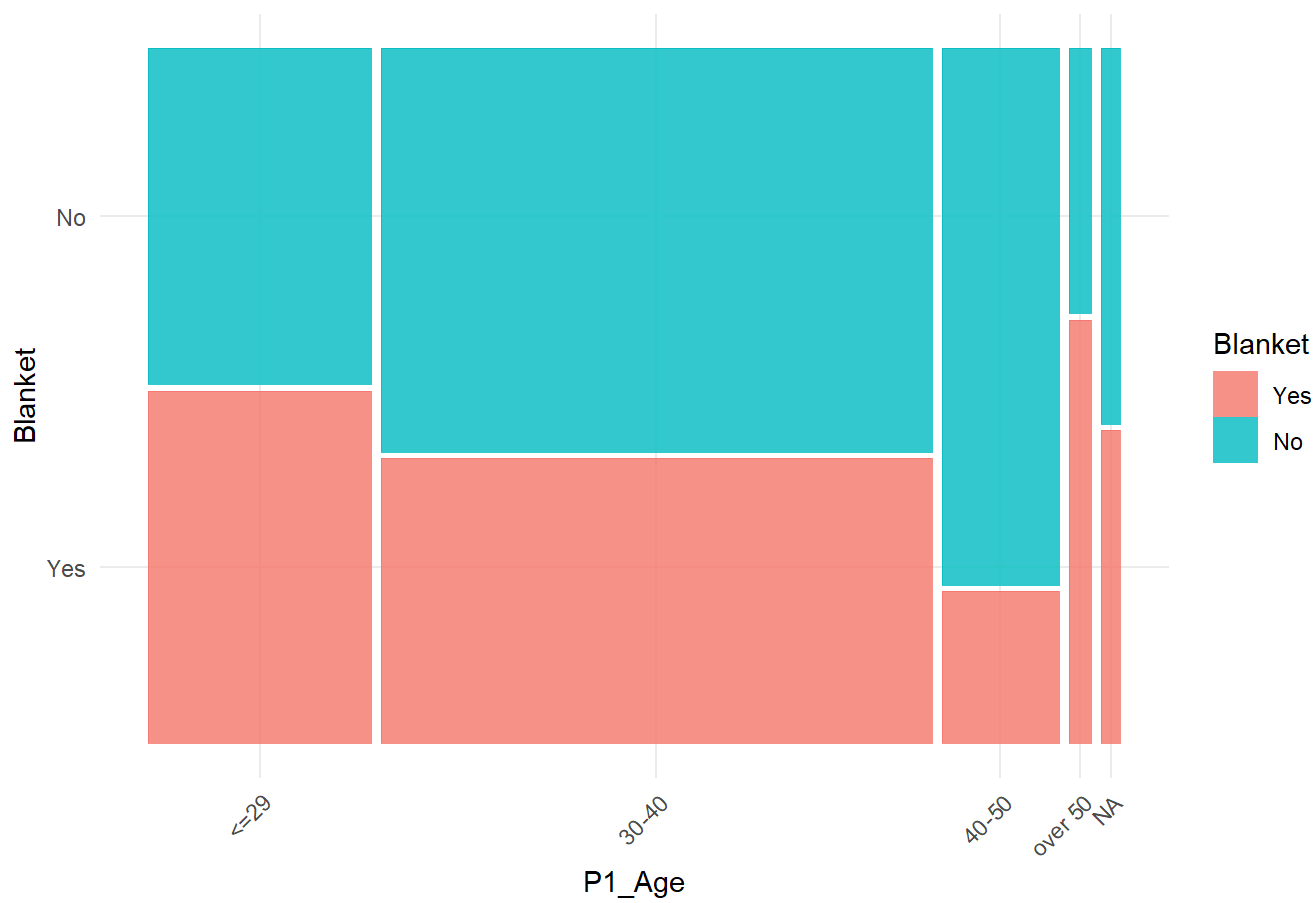
Mosaic Plot of P1_Race4 and Blanket



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket and P1_Age -----
## Bivariate Analysis of Blanket and P1_Age
##
## Frequency Table:
## Blanket <=29 30-40 40-50 over 50 NA_
##      Yes    66   132   15      8    5
##      No     63   187   53      5    6
##
## Relative Frequency Table:
## Blanket <=29 30-40 40-50 over 50  NA_ Total
##      Yes 51.2% 41.4% 22.1%   61.5% 45.5% 41.9%
##      No 48.8% 58.6% 77.9%   38.5% 54.5% 58.1%
##
## Cramér's V:  0.1593877
```

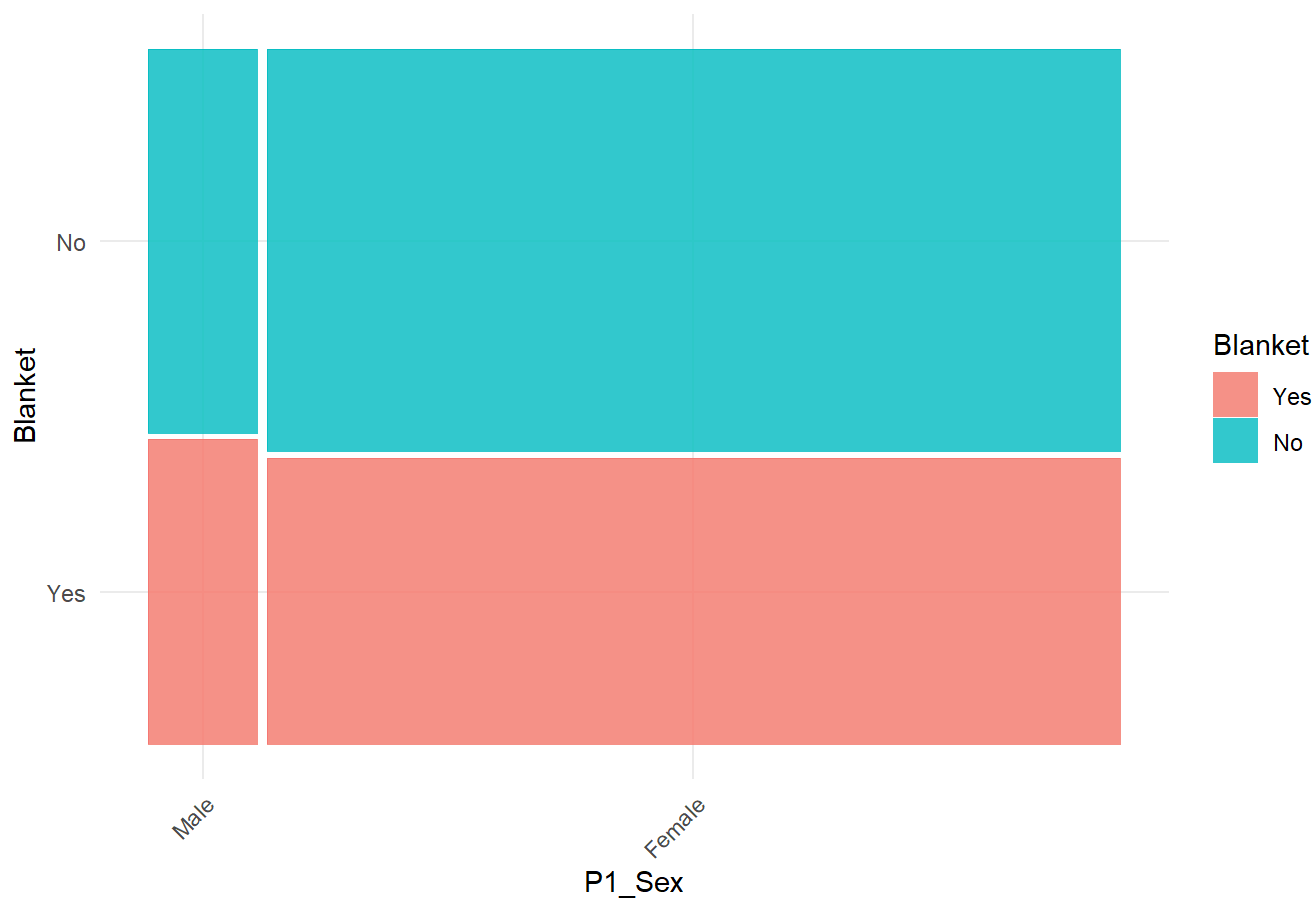
Mosaic Plot of P1_Age and Blanket



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket and P1_Sex -----
## Bivariate Analysis of Blanket and P1_Sex
##
## Frequency Table:
## Blanket Male Female
##      Yes    27    199
##      No     34    280
##
## Relative Frequency Table:
## Blanket Male Female Total
##      Yes 44.3%  41.5% 41.9%
##      No  55.7%  58.5% 58.1%
##
## Cramér's V:  0
```

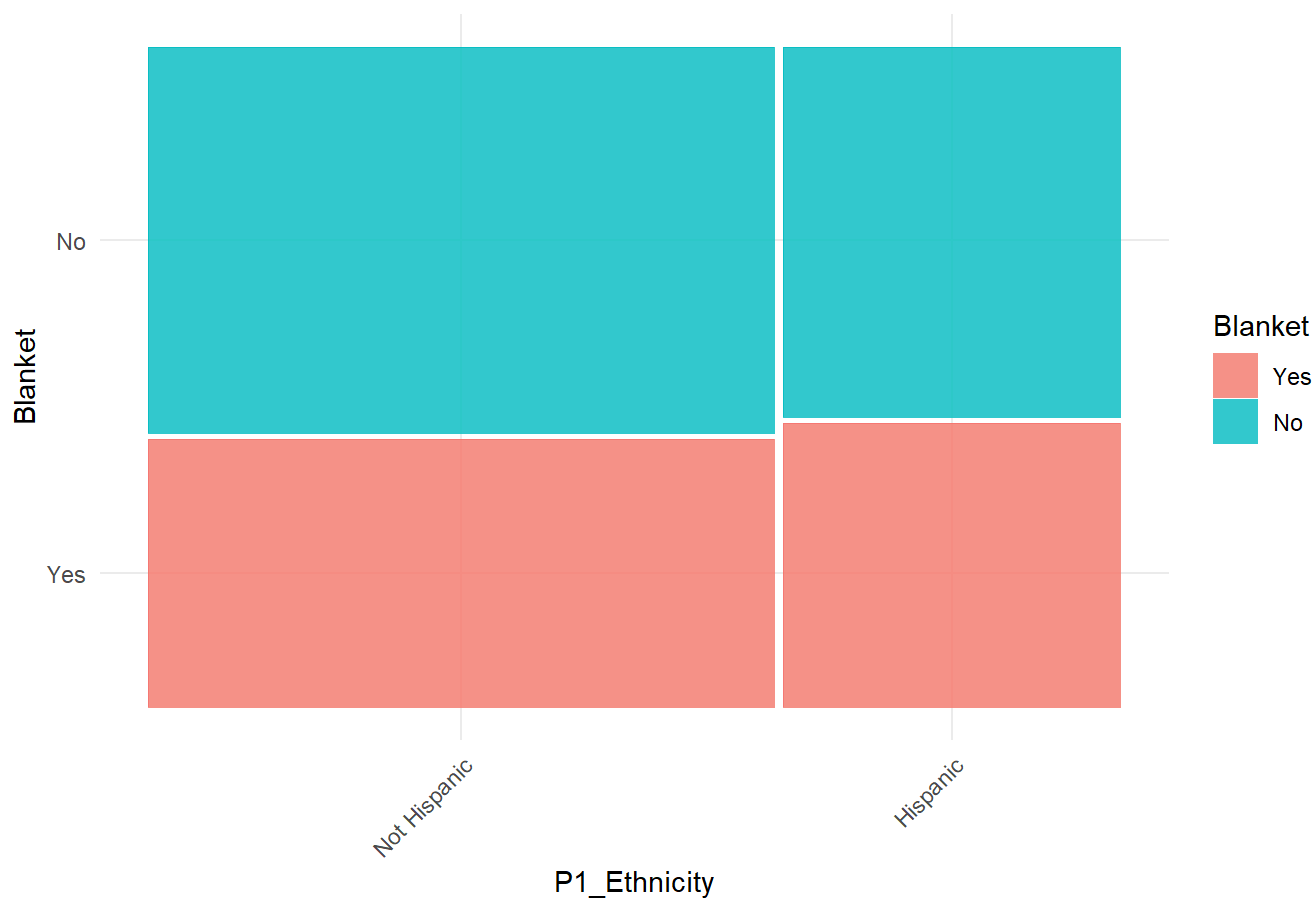

Mosaic Plot of P1_Sex and Blanket



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket and P1_Ethnicity -----
## Bivariate Analysis of Blanket and P1_Ethnicity
##
## Frequency Table:
## Blanket Not Hispanic Hispanic
##   Yes      144      82
##   No      207     107
##
## Relative Frequency Table:
## Blanket Not Hispanic Hispanic Total
##   Yes      41.0%   43.4% 41.9%
##   No      59.0%   56.6% 58.1%
##
## Cramér's V:  0
```

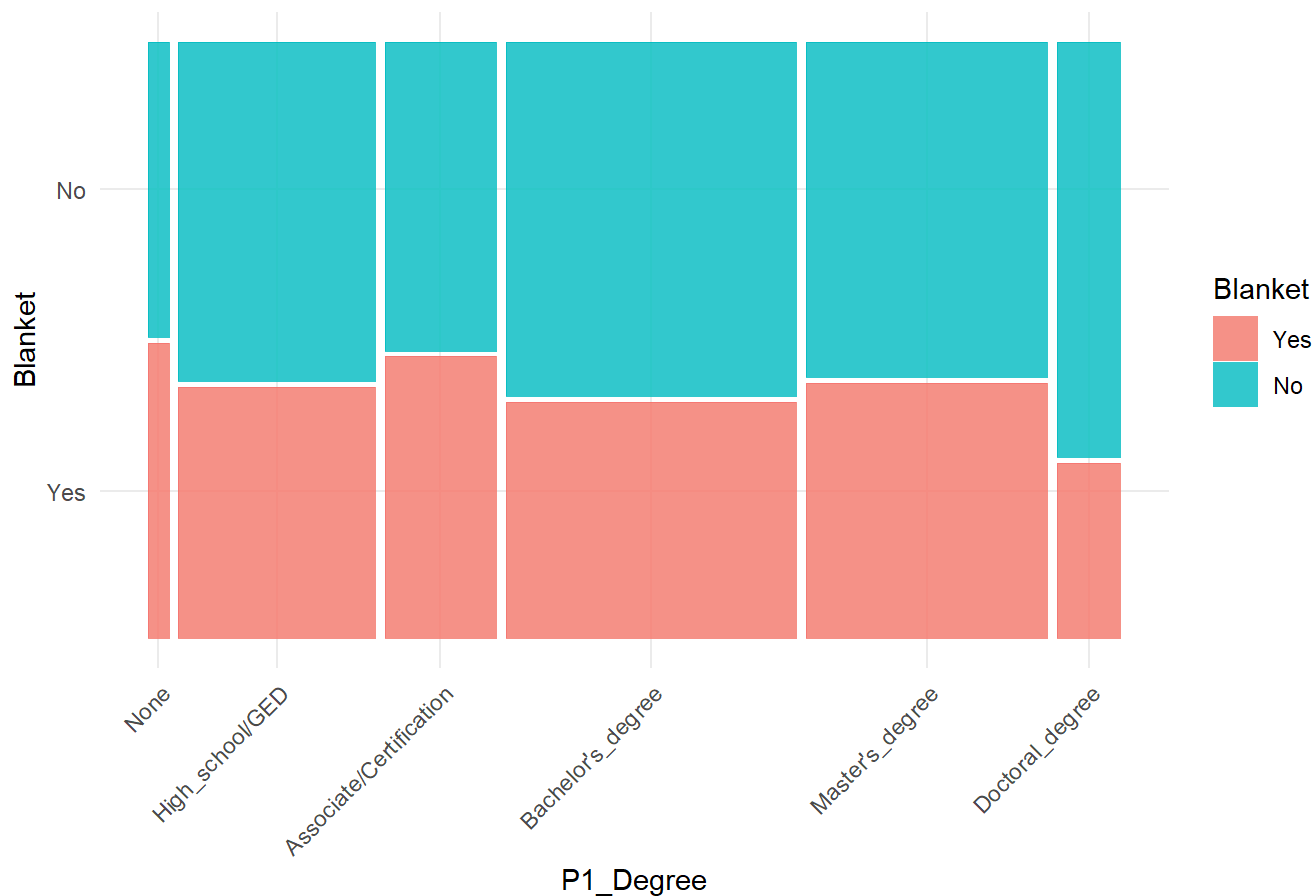
Mosaic Plot of P1_Ethnicity and Blanket



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket and P1_Degree -----
## Bivariate Analysis of Blanket and P1_Degree
##
## Frequency Table:
## Blanket None High_school/GED Associate/Certification Bachelor's_degree
##      Yes    6          49          31          68
##      No     6          66          34          102
## Master's_degree Doctoral_degree
##           61          11
##           80          26
##
## Relative Frequency Table:
## Blanket  None High_school/GED Associate/Certification Bachelor's_degree
##      Yes 50.0%          42.6%          47.7%          40.0%
##      No  50.0%          57.4%          52.3%          60.0%
## Master's_degree Doctoral_degree Total
##           43.3%          29.7% 41.9%
##           56.7%          70.3% 58.1%
##
## Cramér's V:  0
```

Mosaic Plot of P1_Degree and Blanket



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket and P1_Workstatus -----
## Bivariate Analysis of Blanket and P1_Workstatus
##
## Frequency Table:
## Blanket Working_full_time Working_part_time Going_to_school
##      Yes           96           36           16
##      No           142           52           19
## Not_Working/Refused NA_
##                68  10
##                97   4
##
## Relative Frequency Table:
## Blanket Working_full_time Working_part_time Going_to_school
##      Yes           40.3%           40.9%           45.7%
##      No           59.7%           59.1%           54.3%
## Not_Working/Refused  NA_ Total
##                41.2% 71.4% 41.9%
##                58.8% 28.6% 58.1%
##
## Cramér's V:  0.05317616
```

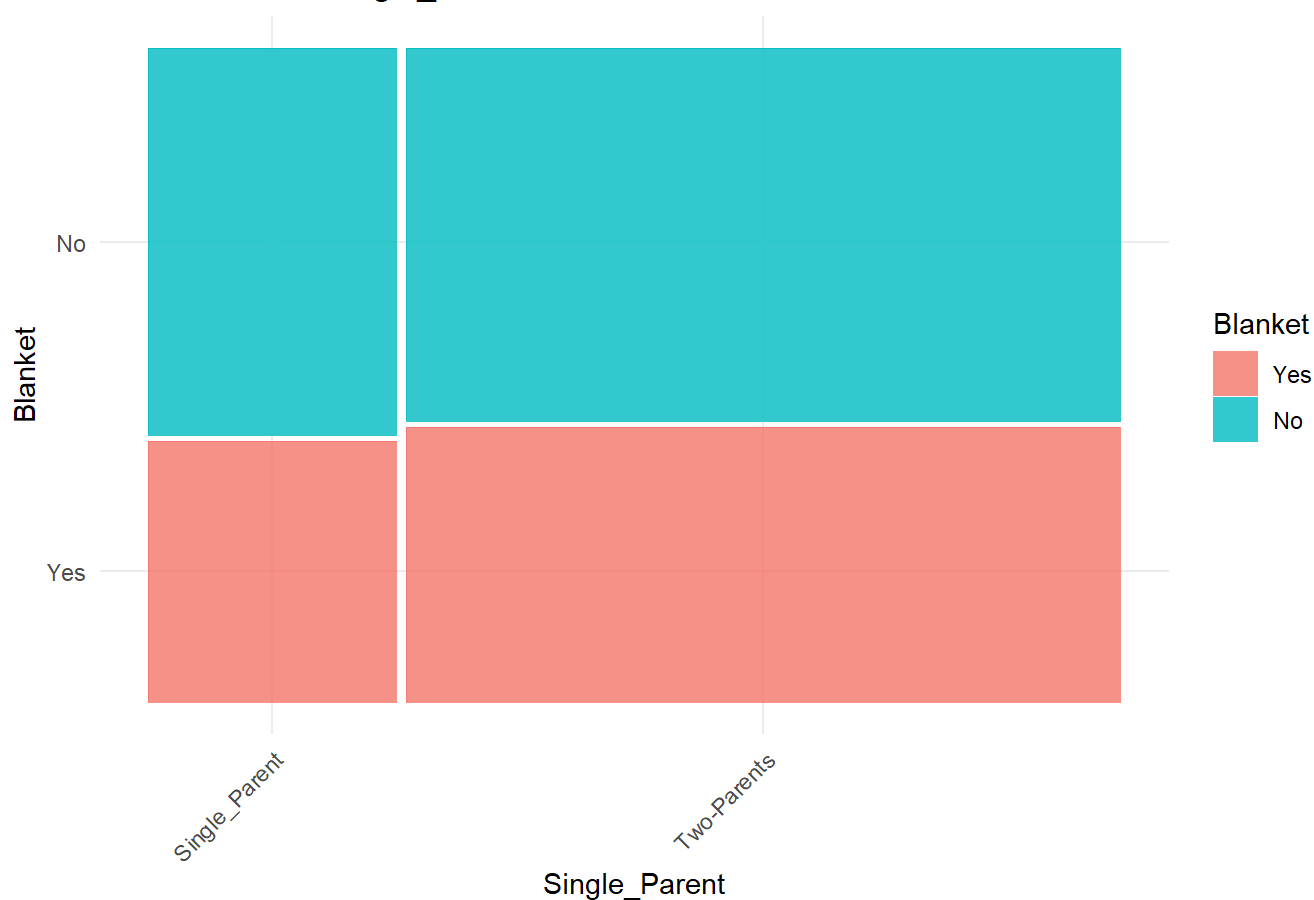
Mosaic Plot of P1_Workstatus and Blanket



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket and Single_Parent -----
## Bivariate Analysis of Blanket and Single_Parent
##
## Frequency Table:
##   Blanket Single_Parent Two-Parents
##      Yes      56      170
##      No       83      231
##
## Relative Frequency Table:
##   Blanket Single_Parent Two-Parents Total
##      Yes      40.3%      42.4% 41.9%
##      No       59.7%      57.6% 58.1%
##
## Cramér's V:  0
```

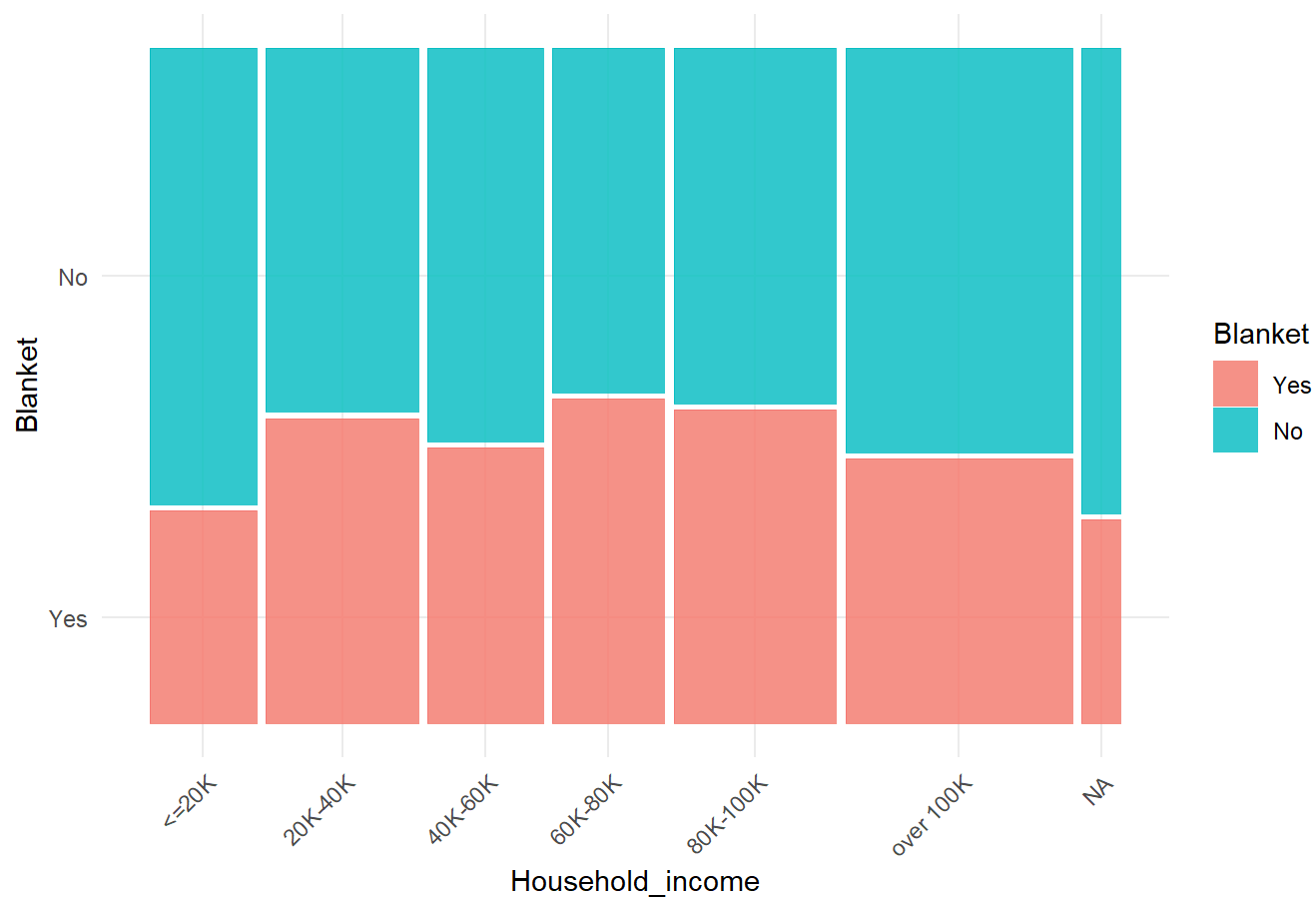
Mosaic Plot of Single_Parent and Blanket



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket and Household_income -----
## Bivariate Analysis of Blanket and Household_income
##
## Frequency Table:
## Blanket <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##      Yes      20      41      28      32      45      53      7
##      No       43      49      40      34      51      81     16
##
## Relative Frequency Table:
## Blanket <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##      Yes 31.7%  45.6%  41.2%  48.5%  46.9%   39.6% 30.4% 41.9%
##      No 68.3%  54.4%  58.8%  51.5%  53.1%   60.4% 69.6% 58.1%
##
## Cramér's V: 0.04004632
```

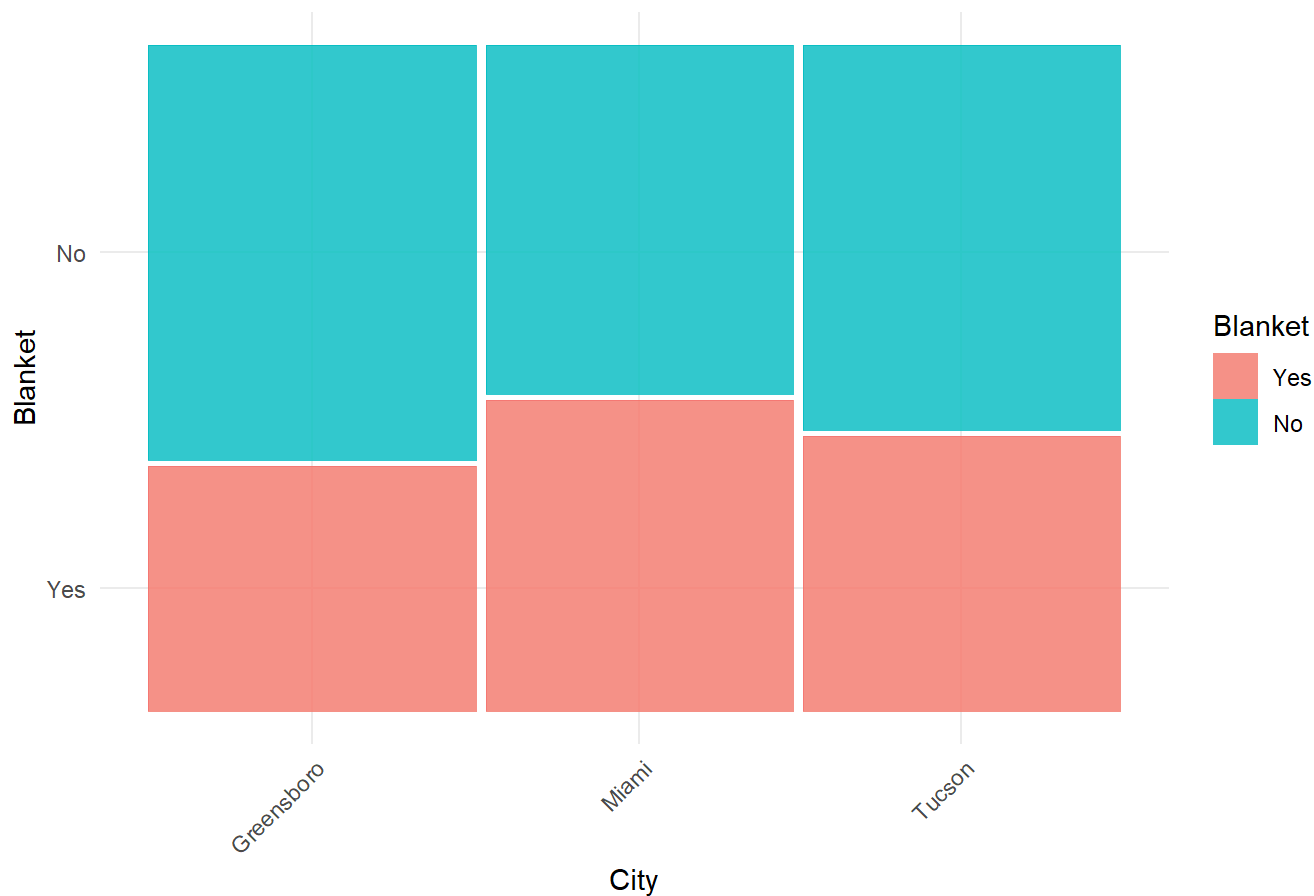
Mosaic Plot of Household_income and Blanket



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket and City -----
## Bivariate Analysis of Blanket and City
##
## Frequency Table:
## Blanket Greensboro Miami Tucson
##      Yes      69      82      75
##      No      117      92     105
##
## Relative Frequency Table:
## Blanket Greensboro Miami Tucson Total
##      Yes    37.1% 47.1% 41.7% 41.9%
##      No     62.9% 52.9% 58.3% 58.1%
##
## Cramér's V: 0.05642655
```

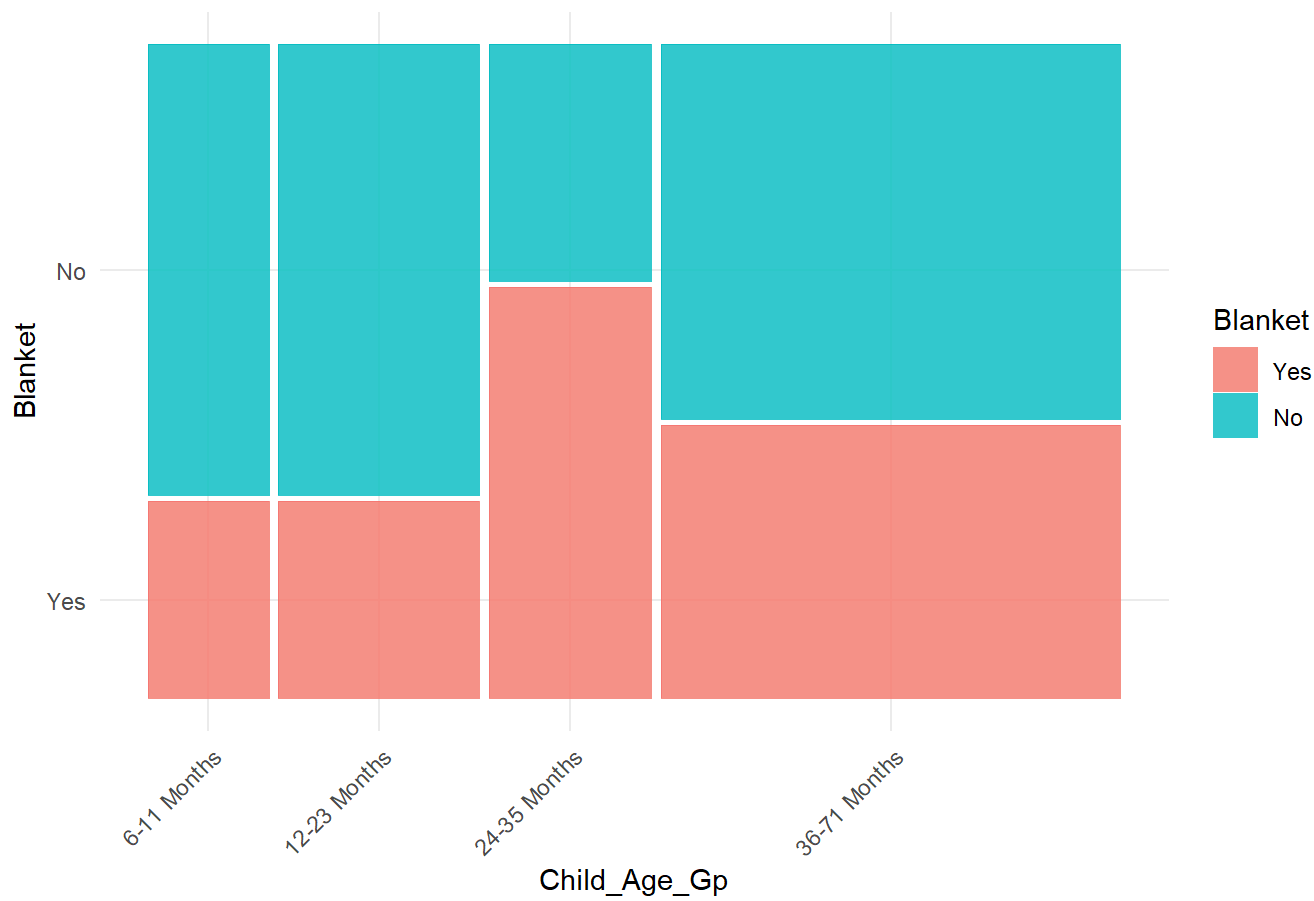
Mosaic Plot of City and Blanket



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket and Child_Age_Gp -----
## Bivariate Analysis of Blanket and Child_Age_Gp
##
## Frequency Table:
## Blanket 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##      Yes      21      35      59      111
##      No       48      80      34      152
##
## Relative Frequency Table:
## Blanket 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##      Yes      30.4%      30.4%      63.4%      42.2% 41.9%
##      No       69.6%      69.6%      36.6%      57.8% 58.1%
##
## Cramér's V:  0.2139602
```

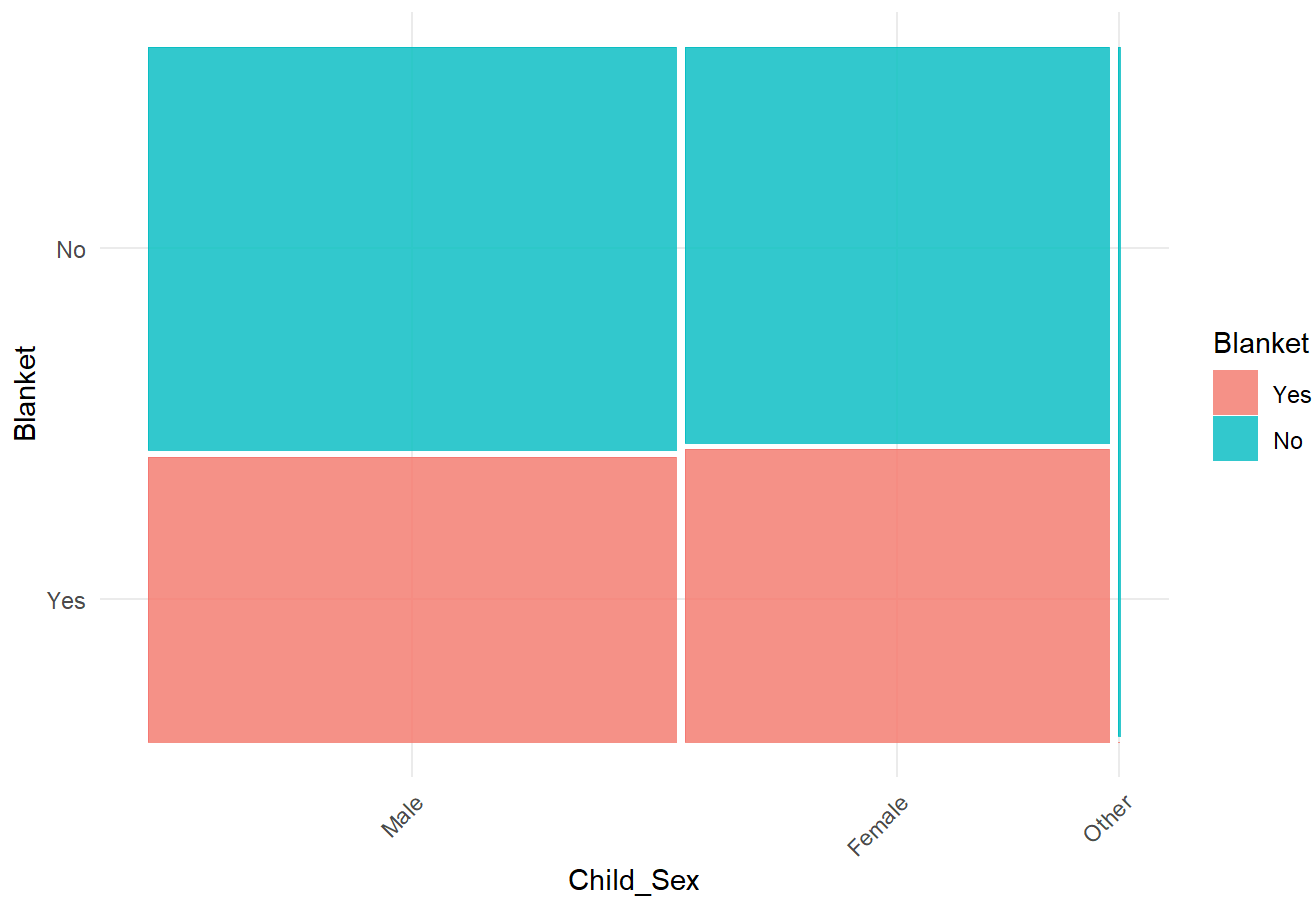
Mosaic Plot of Child_Age_Gp and Blanket



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket and Child_Sex -----
## Bivariate Analysis of Blanket and Child_Sex
##
## Frequency Table:
##   Blanket Male Female Other
##      Yes  124   102    0
##      No   175   138    1
##
## Relative Frequency Table:
##   Blanket  Male Female  Other Total
##      Yes 41.5%  42.5%   0.0% 41.9%
##      No  58.5%  57.5% 100.0% 58.1%
##
## Cramér's V:  0
```

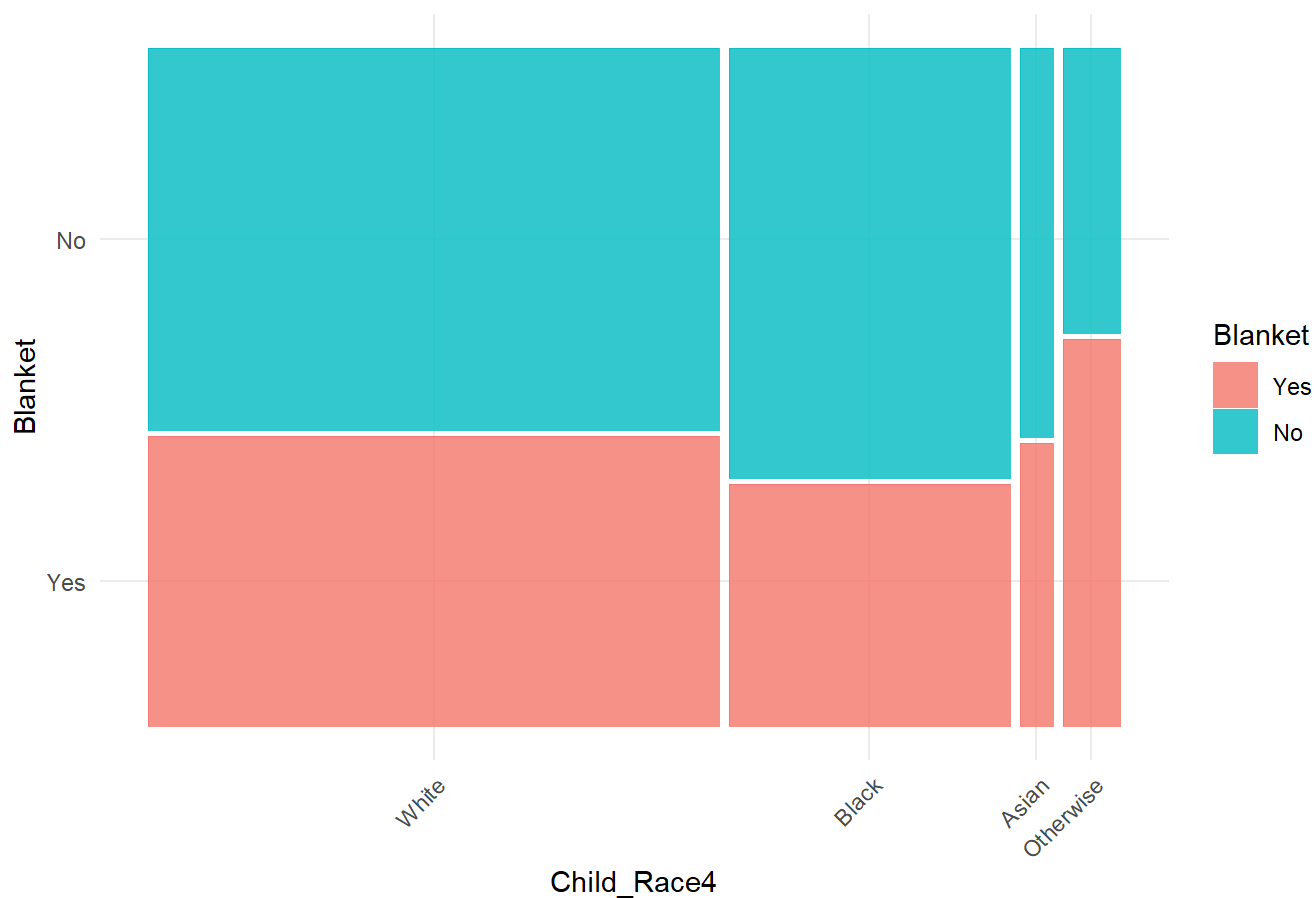

Mosaic Plot of Child_Sex and Blanket



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket and Child_Race4 -----
## Bivariate Analysis of Blanket and Child_Race4
##
## Frequency Table:
## Blanket White Black Asian Otherwise
##      Yes   141    58     8      19
##      No   186   103    11      14
##
## Relative Frequency Table:
## Blanket White Black Asian Otherwise Total
##      Yes 43.1% 36.0% 42.1%    57.6% 41.9%
##      No 56.9% 64.0% 57.9%    42.4% 58.1%
##
## Cramér's V:  0.07219972
```

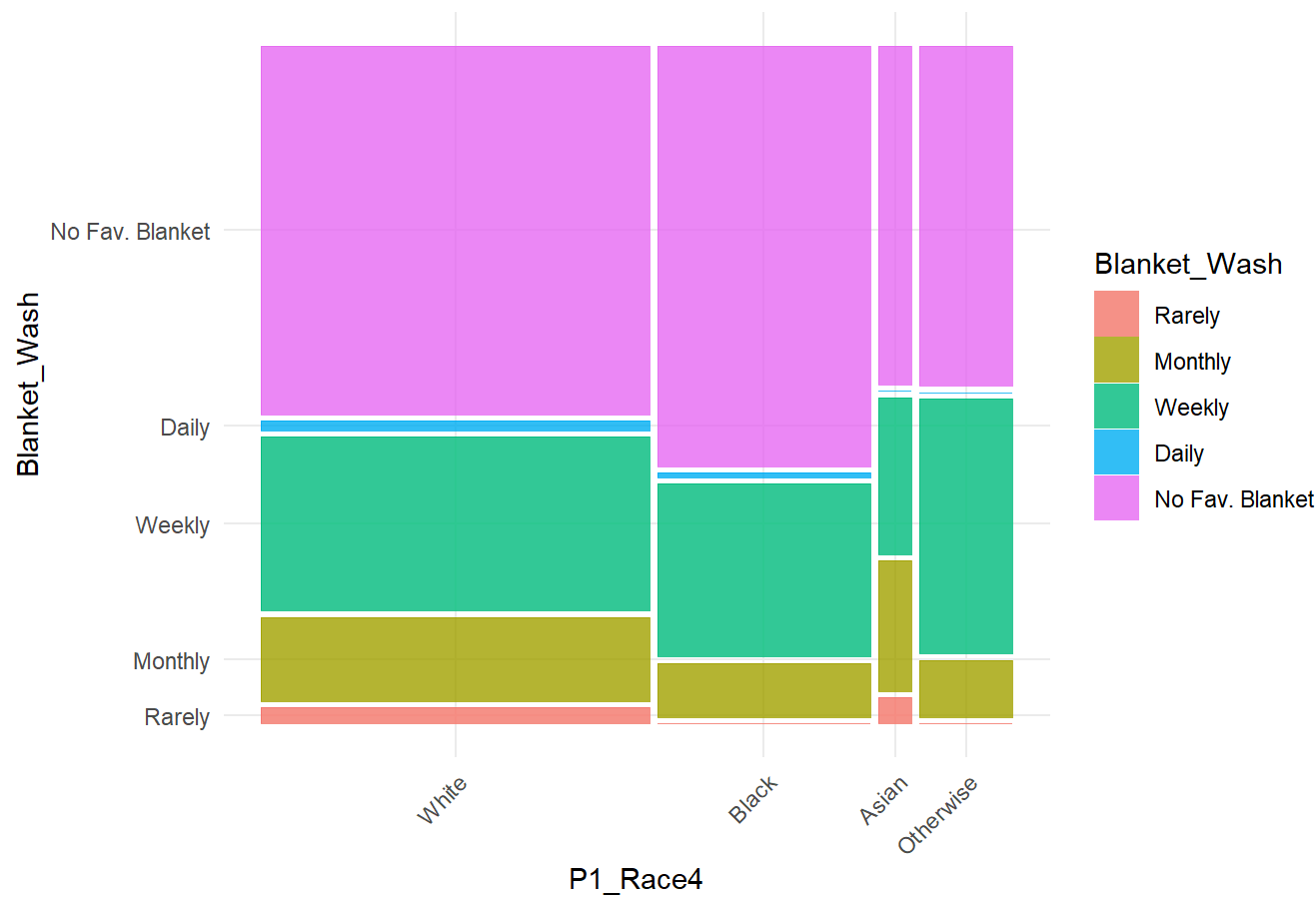
Mosaic Plot of Child_Race4 and Blanket



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket_Wash and P1_Race4 -----
## Bivariate Analysis of Blanket_Wash and P1_Race4
##
## Frequency Table:
##      Blanket_Wash White Black Asian Otherwise
##      Rarely       7      0      1          0
##      Monthly     37     13      5          6
##      Weekly      77     42      6         27
##      Daily        4      1      0          0
## No Fav. Blanket 163    102    13         36
##
## Relative Frequency Table:
##      Blanket_Wash White Black Asian Otherwise Total
##      Rarely    2.4%  0.0%  4.0%      0.0%  1.5%
##      Monthly   12.8%  8.2% 20.0%      8.7% 11.3%
##      Weekly    26.7% 26.6% 24.0%    39.1% 28.1%
##      Daily     1.4%  0.6%  0.0%      0.0%  0.9%
## No Fav. Blanket 56.6% 64.6% 52.0%    52.2% 58.1%
##
## Cramér's V:  0.05647268
```

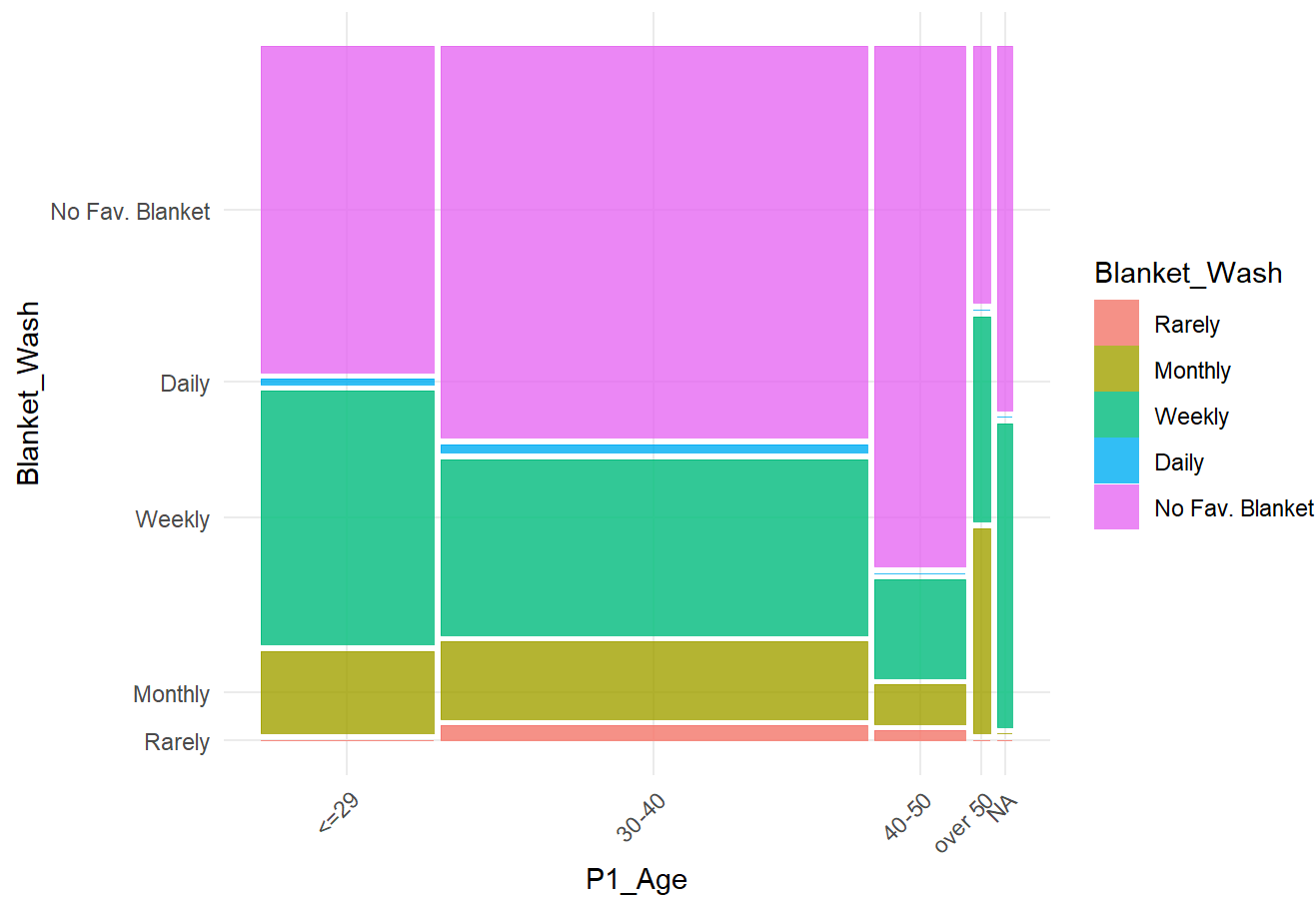
Mosaic Plot of P1_Race4 and Blanket_Wash



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Blanket_Wash and P1_Age -----
## Bivariate Analysis of Blanket_Wash and P1_Age
##
## Frequency Table:
##   Blanket_Wash  <=29  30-40  40-50  over 50  NA_
##   Rarely       0      7      1      0      0
##   Monthly     16     37      4      4      0
##   Weekly      49     84     10      4      5
##   Daily        1      4      0      0      0
##   No Fav. Blanket 63    187    53      5      6
##
## Relative Frequency Table:
##   Blanket_Wash  <=29  30-40  40-50  over 50  NA_ Total
##   Rarely       0.0%  2.2%  1.5%   0.0%  0.0%  1.5%
##   Monthly     12.4% 11.6%  5.9%  30.8%  0.0% 11.3%
##   Weekly      38.0% 26.3% 14.7%  30.8% 45.5% 28.1%
##   Daily        0.8%  1.3%  0.0%   0.0%  0.0%  0.9%
##   No Fav. Blanket 48.8% 58.6% 77.9%  38.5% 54.5% 58.1%
##
## Cramér's V:  0.08053383
```

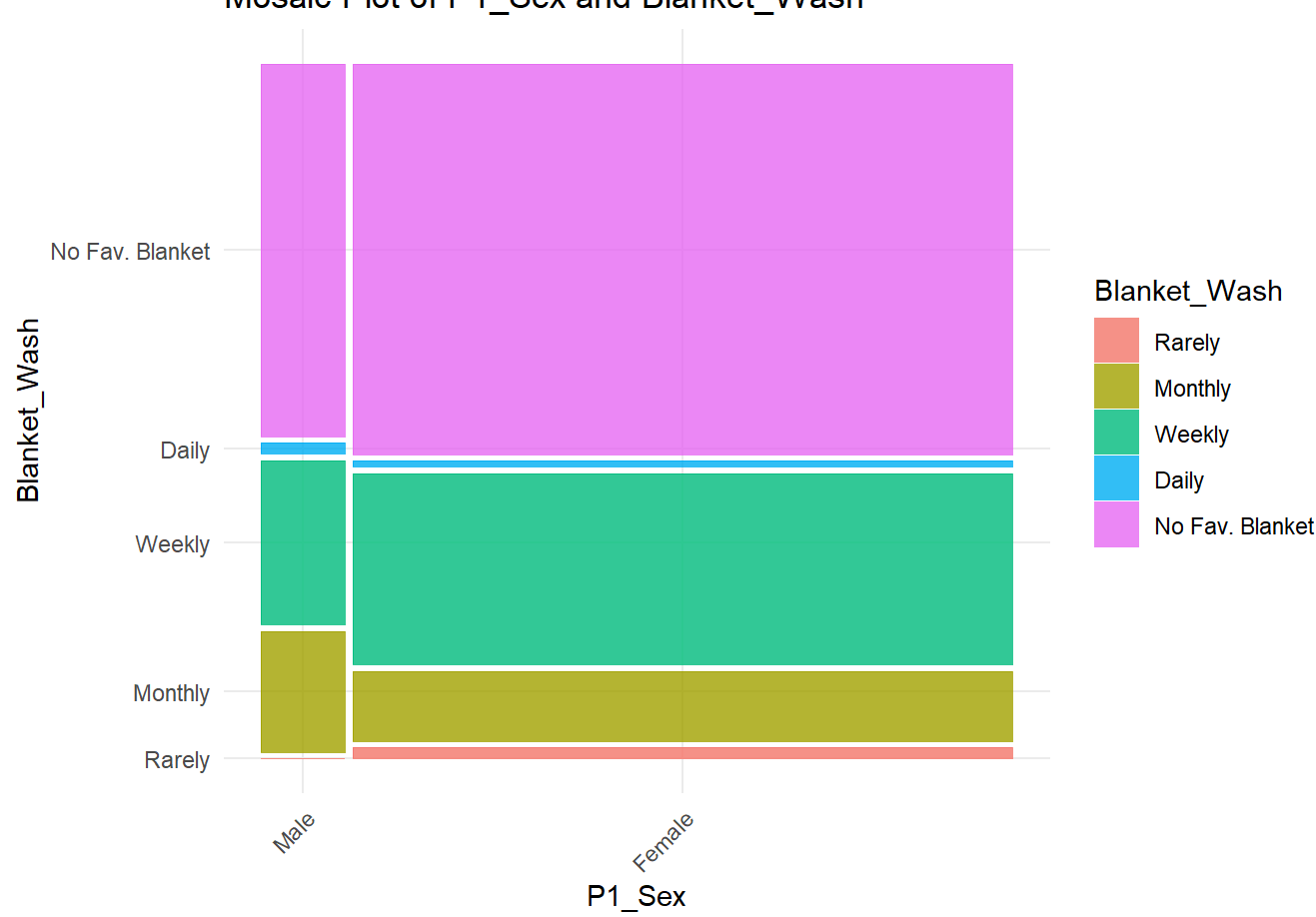
Mosaic Plot of P1_Age and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and P1_Sex -----
## Bivariate Analysis of Blanket_Wash and P1_Sex
##
## Frequency Table:
##   Blanket_Wash  Male  Female
##   Rarely       0      8
##   Monthly      11     50
##   Weekly       15    137
##   Daily         1      4
##   No Fav. Blanket  34   280
##
## Relative Frequency Table:
##   Blanket_Wash  Male  Female  Total
##   Rarely       0.0%   1.7%   1.5%
##   Monthly      18.0%  10.4%  11.3%
##   Weekly       24.6%  28.6%  28.1%
##   Daily         1.6%   0.8%   0.9%
##   No Fav. Blanket 55.7%  58.5% 58.1%
##
## Cramér's V: 0.03135696
```

Mosaic Plot of P1_Sex and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and P1_Ethnicity -----
## Bivariate Analysis of Blanket_Wash and P1_Ethnicity
##
## Frequency Table:
##   Blanket_Wash Not Hispanic Hispanic
##   Rarely      8      0
##   Monthly     43     18
##   Weekly      90     62
##   Daily        3      2
##   No Fav. Blanket 207    107
##
## Relative Frequency Table:
##   Blanket_Wash Not Hispanic Hispanic Total
##   Rarely      2.3%    0.0%  1.5%
##   Monthly     12.3%    9.5% 11.3%
##   Weekly      25.6%   32.8% 28.1%
##   Daily        0.9%    1.1%  0.9%
##   No Fav. Blanket 59.0%   56.6% 58.1%
##
## Cramér's V: 0.08082466
```

Mosaic Plot of P1_Ethnicity and Blanket_Wash

Blanket_Wash

No Fav. Blanket

Daily

Weekly

Monthly

Rarely

Not Hispanic

Hispanic

P1_Ethnicity

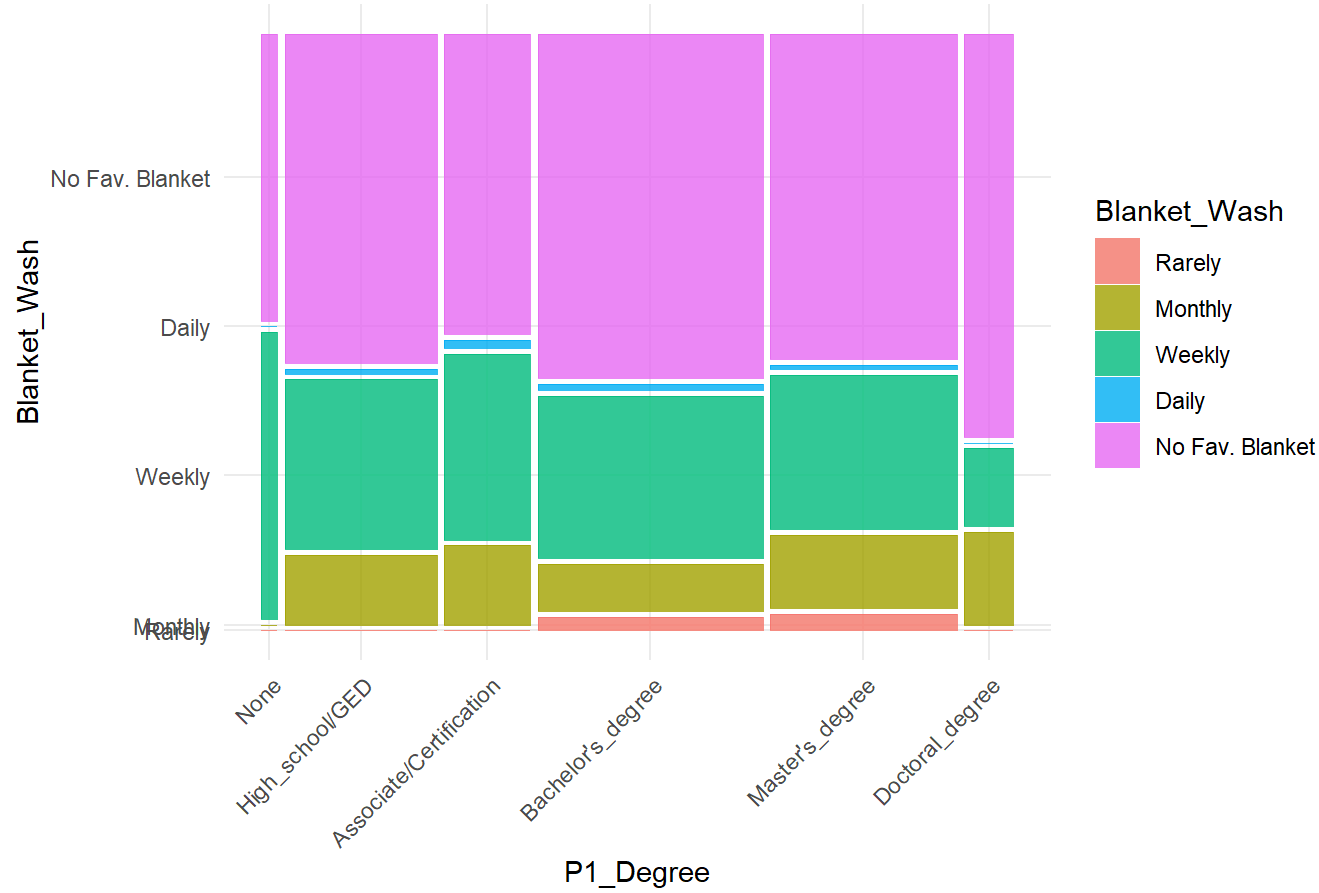
Blanket_Wash

- Rarely
- Monthly
- Weekly
- Daily
- No Fav. Blanket

Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and P1_Degree -----
## Bivariate Analysis of Blanket_Wash and P1_Degree
##
## Frequency Table:
##      Blanket_Wash None High_school/GED Associate/Certification Bachelor's_degree
##      Rarely      0           0           0           4
##      Monthly     0          14           9          14
##      Weekly      6          34          21          48
##      Daily       0           1           1           2
## No Fav. Blanket  6          66          34          102
## Master's_degree Doctoral_degree
##      4           0
##      18          6
##      38          5
##      1           0
##      80          26
##
## Relative Frequency Table:
##      Blanket_Wash None High_school/GED Associate/Certification
##      Rarely      0.0%           0.0%           0.0%
##      Monthly     0.0%          12.2%          13.8%
##      Weekly      50.0%          29.6%          32.3%
##      Daily       0.0%           0.9%           1.5%
## No Fav. Blanket 50.0%          57.4%          52.3%
## Bachelor's_degree Master's_degree Doctoral_degree Total
##      2.4%          2.8%           0.0%  1.5%
##      8.2%          12.8%          16.2% 11.3%
##      28.2%         27.0%          13.5% 28.1%
##      1.2%           0.7%           0.0%  0.9%
##      60.0%         56.7%          70.3% 58.1%
##
## Cramér's V:  0
```

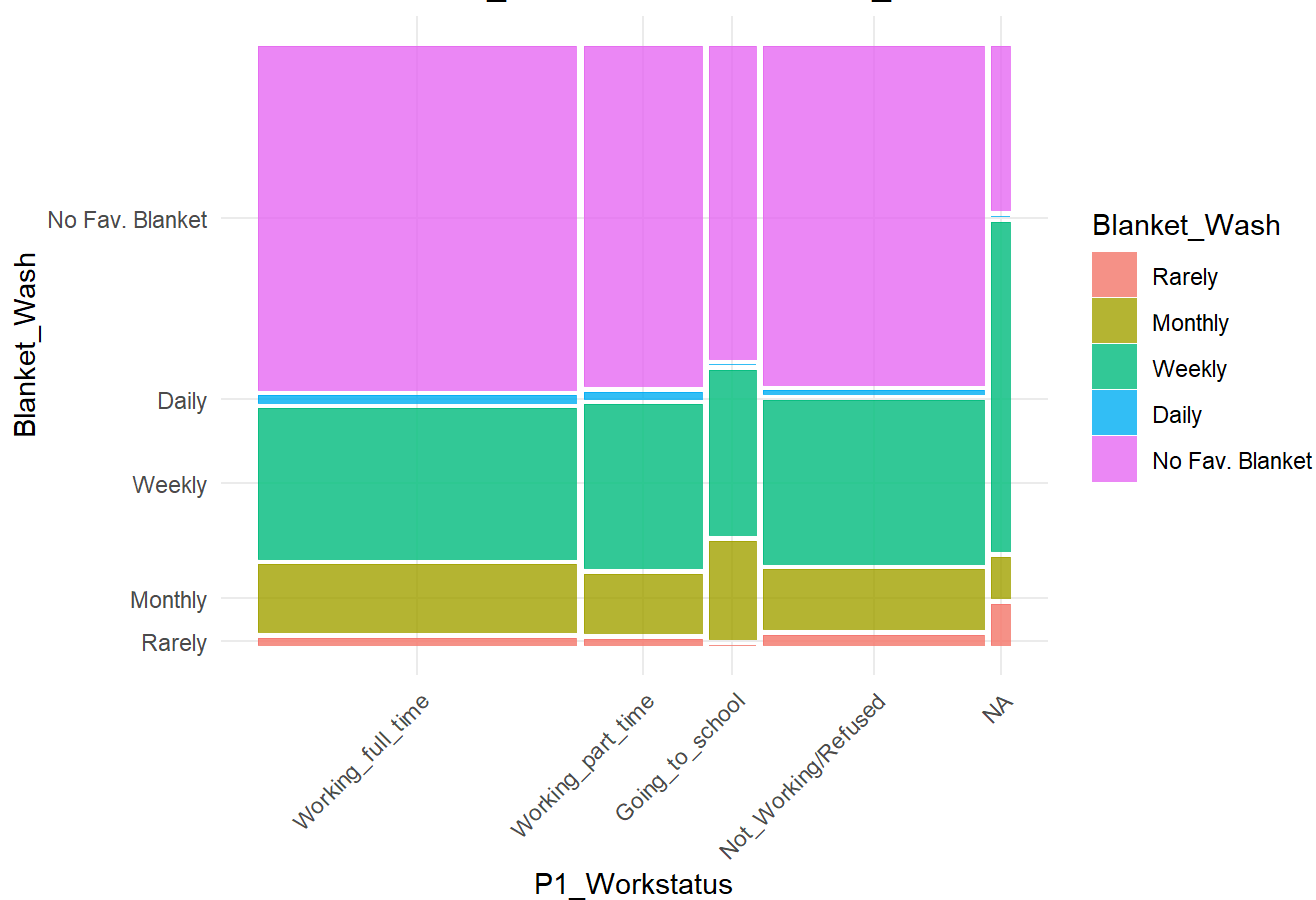
Mosaic Plot of P1_Degree and Blanket_Wash



Saving 7 x 5 in image


```
##
##
## ----- Analysis for Blanket_Wash and P1_Workstatus -----
## Bivariate Analysis of Blanket_Wash and P1_Workstatus
##
## Frequency Table:
##      Blanket_Wash Working_full_time Working_part_time Going_to_school
##      Rarely      3      1      0
##      Monthly     28      9      6
##      Weekly      62     25     10
##      Daily       3      1      0
## No Fav. Blanket      142     52     19
## Not_Working/Refused NA_
##      3      1
##     17     1
##     47     8
##      1     0
##     97     4
##
## Relative Frequency Table:
##      Blanket_Wash Working_full_time Working_part_time Going_to_school
##      Rarely      1.3%      1.1%      0.0%
##      Monthly     11.8%     10.2%     17.1%
##      Weekly      26.1%     28.4%     28.6%
##      Daily       1.3%      1.1%      0.0%
## No Fav. Blanket     59.7%     59.1%     54.3%
## Not_Working/Refused  NA_ Total
##      1.8%  7.1%  1.5%
##     10.3%  7.1% 11.3%
##     28.5% 57.1% 28.1%
##      0.6%  0.0%  0.9%
##     58.8% 28.6% 58.1%
##
## Cramér's V:  0
```

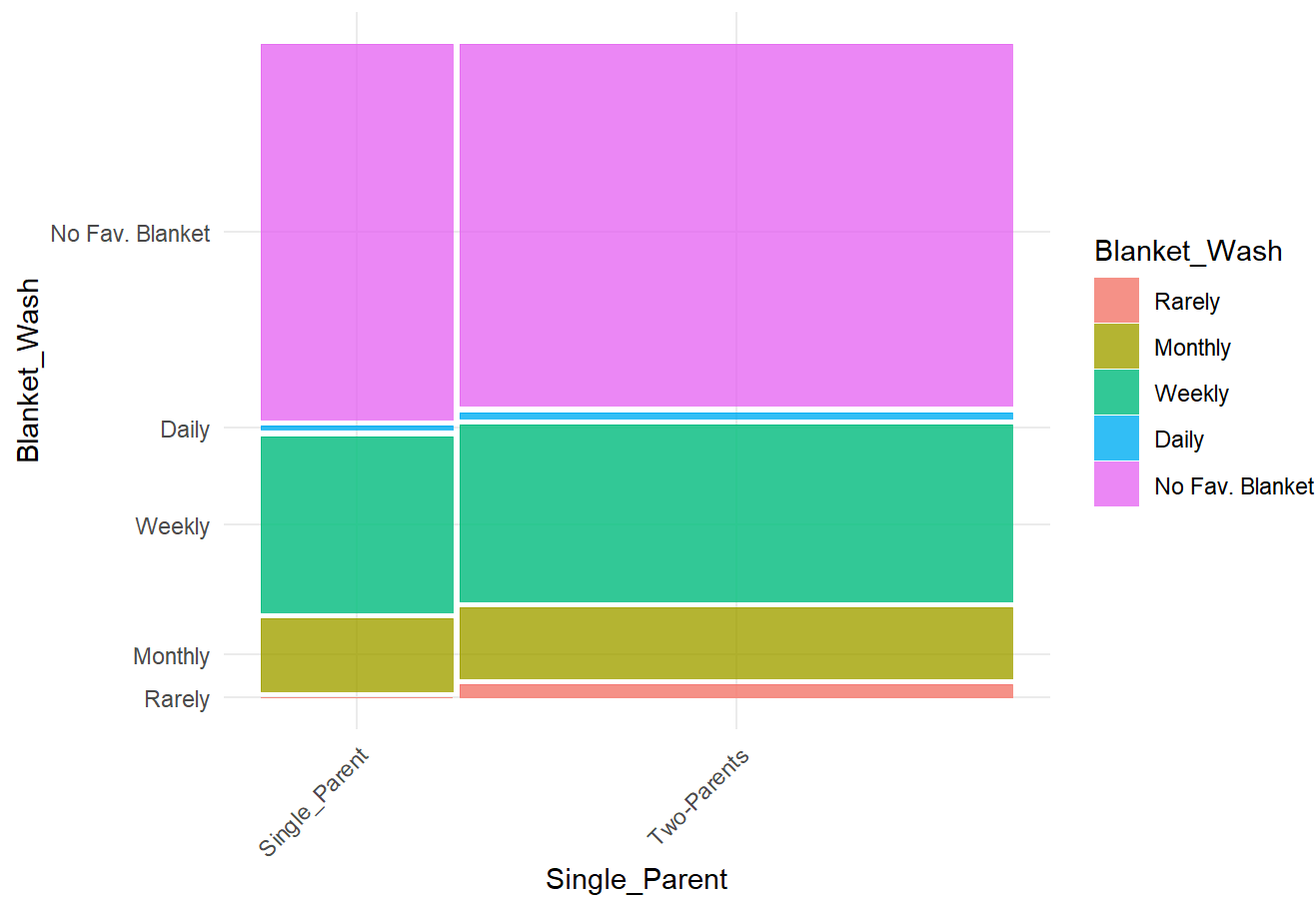
Mosaic Plot of P1_Workstatus and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and Single_Parent -----
## Bivariate Analysis of Blanket_Wash and Single_Parent
##
## Frequency Table:
##   Blanket_Wash Single_Parent Two-Parents
##   Rarely      0           8
##   Monthly     16          45
##   Weekly      39         113
##   Daily        1           4
##   No Fav. Blanket 83        231
##
## Relative Frequency Table:
##   Blanket_Wash Single_Parent Two-Parents Total
##   Rarely      0.0%         2.0%  1.5%
##   Monthly     11.5%        11.2% 11.3%
##   Weekly      28.1%        28.2% 28.1%
##   Daily        0.7%         1.0%  0.9%
##   No Fav. Blanket 59.7%        57.6% 58.1%
##
## Cramér's V:  0
```

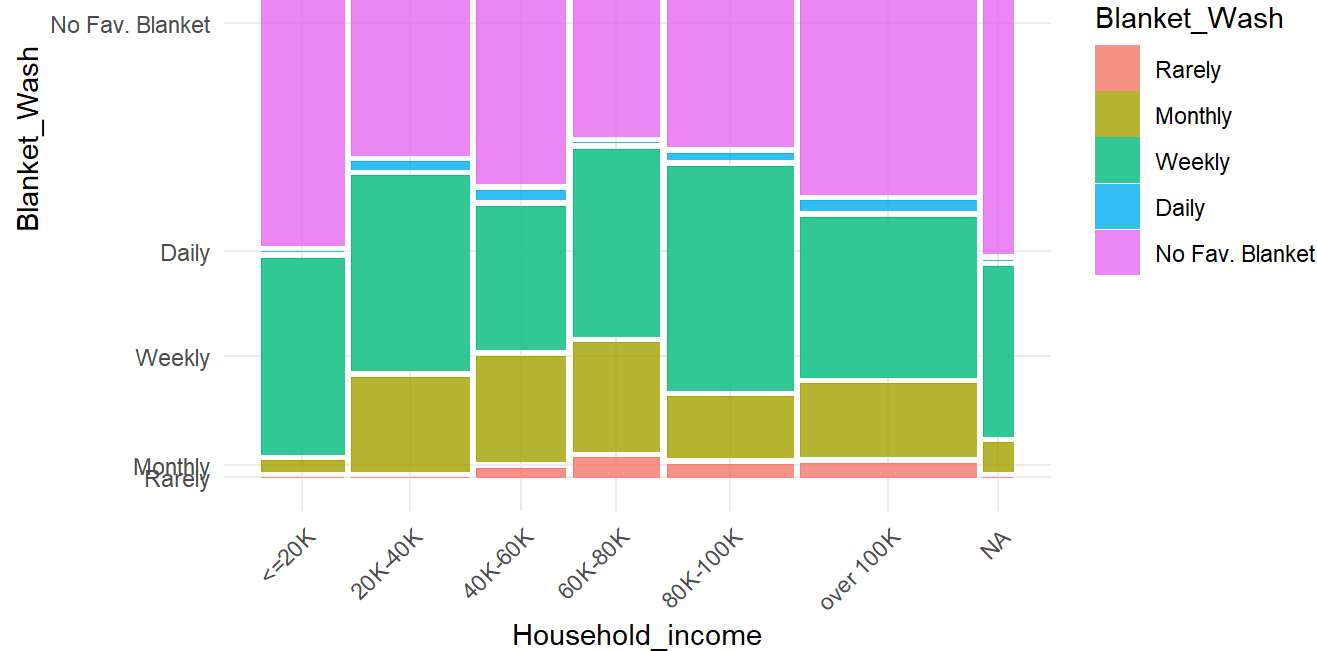
Mosaic Plot of Single_Parent and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and Household_income -----
## Bivariate Analysis of Blanket_Wash and Household_income
##
## Frequency Table:
##   Blanket_Wash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##   Rarely      0      0      1      2      2      3      0
##   Monthly     1     13     11     11     9     15     1
##   Weekly     19     27     15     19     33     33     6
##   Daily       0      1      1      0      1      2      0
## No Fav. Blanket 43     49     40     34     51     81    16
##
## Relative Frequency Table:
##   Blanket_Wash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##   Rarely    0.0%    0.0%    1.5%    3.0%    2.1%    2.2%  0.0%  1.5%
##   Monthly   1.6%   14.4%   16.2%   16.7%    9.4%   11.2%  4.3%  11.3%
##   Weekly   30.2%   30.0%   22.1%   28.8%   34.4%   24.6% 26.1%  28.1%
##   Daily     0.0%    1.1%    1.5%    0.0%    1.0%    1.5%  0.0%   0.9%
## No Fav. Blanket 68.3%   54.4%   58.8%   51.5%   53.1%   60.4% 69.6%  58.1%
##
## Cramér's V:  0
```

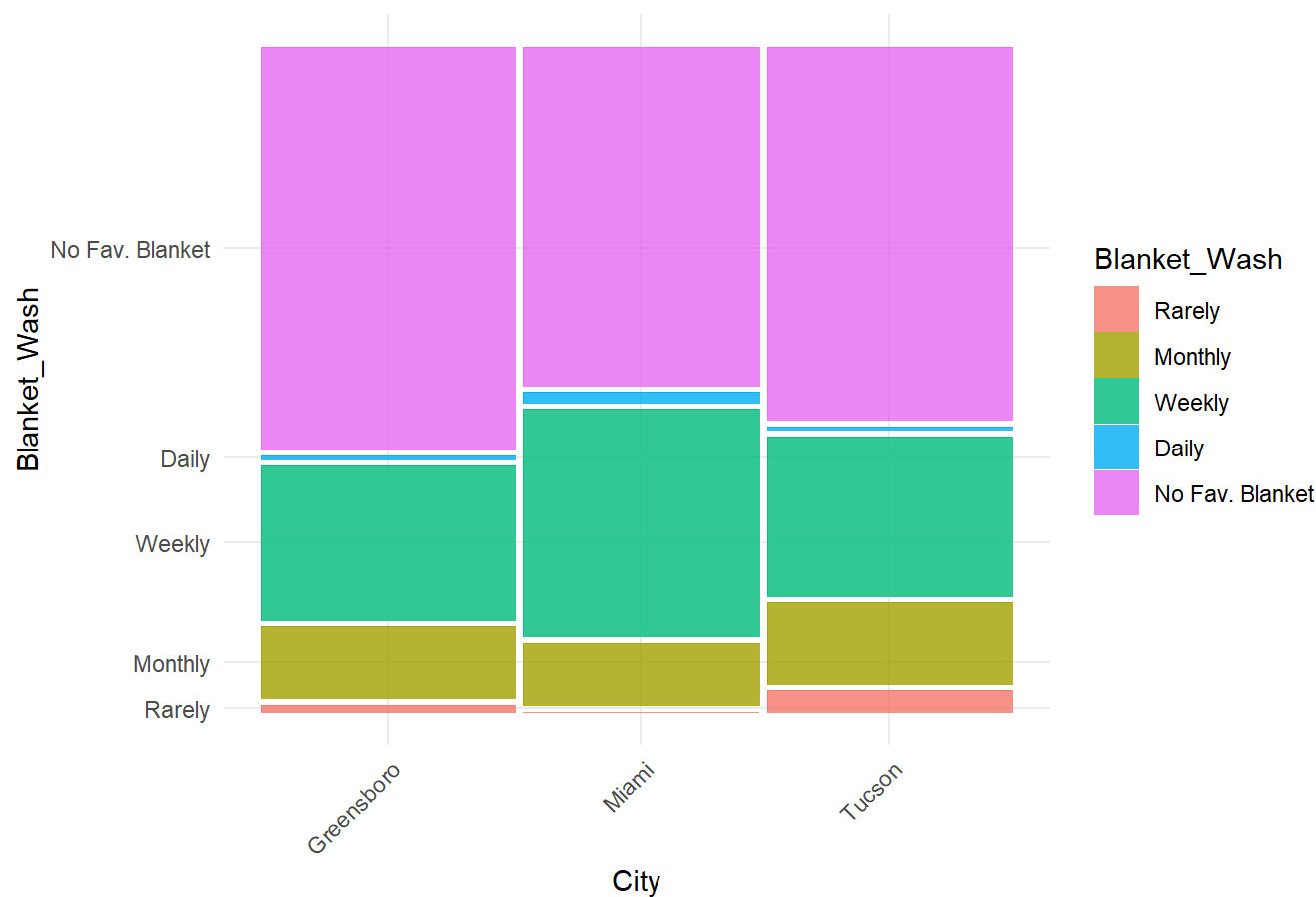
Mosaic Plot of Household_income and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and City -----
## Bivariate Analysis of Blanket_Wash and City
##
## Frequency Table:
##   Blanket_Wash Greensboro Miami Tucson
##   Rarely       2      0      6
##   Monthly      21     17     23
##   Weekly       45     62     45
##   Daily        1      3      1
##   No Fav. Blanket 117    92    105
##
## Relative Frequency Table:
##   Blanket_Wash Greensboro Miami Tucson Total
##   Rarely       1.1%  0.0%   3.3%  1.5%
##   Monthly      11.3%  9.8%  12.8%  11.3%
##   Weekly       24.2% 35.6%  25.0%  28.1%
##   Daily        0.5%  1.7%   0.6%  0.9%
##   No Fav. Blanket 62.9% 52.9%  58.3% 58.1%
##
## Cramér's V: 0.08675554
```

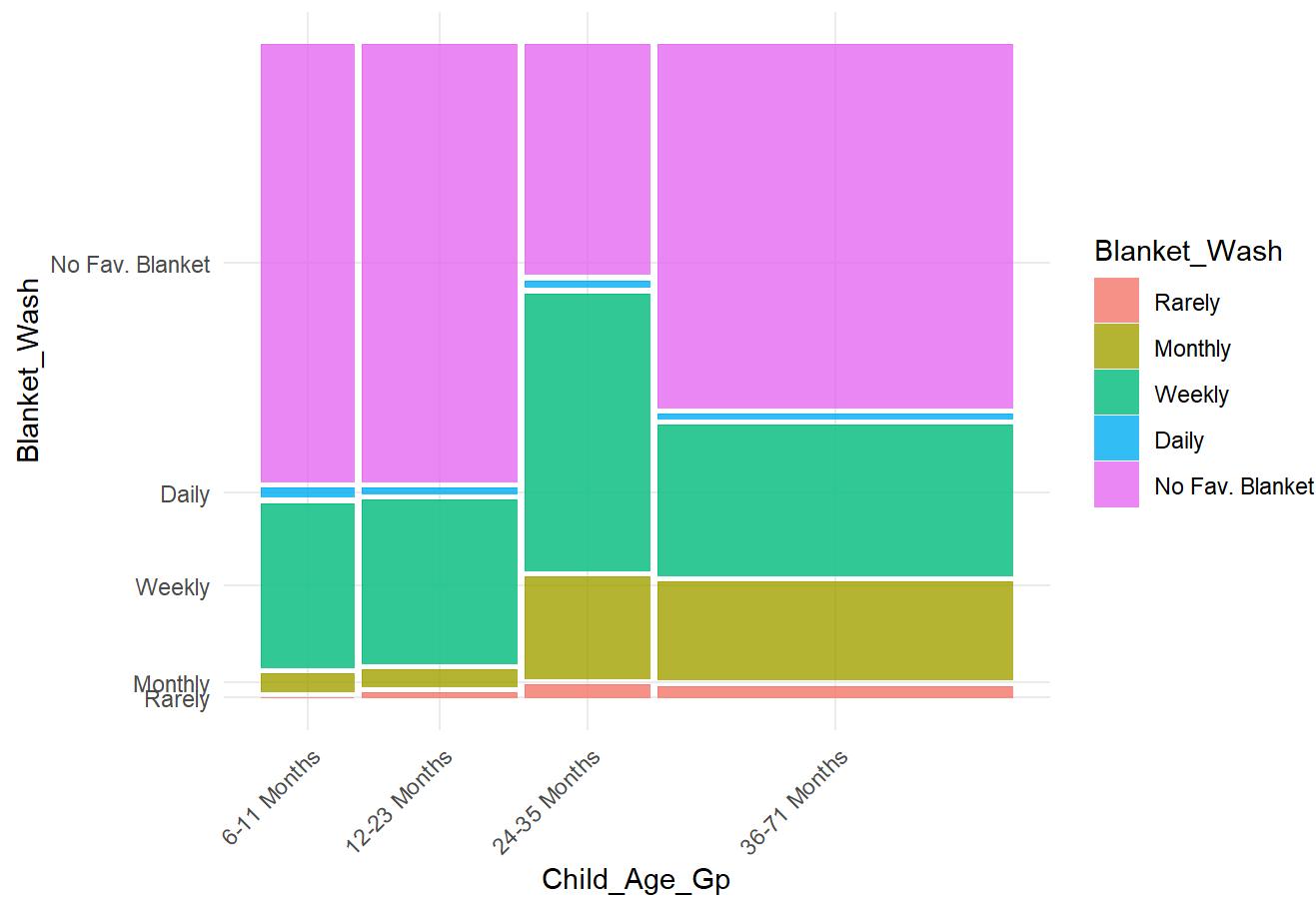
Mosaic Plot of City and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and Child_Age_Gp -----
## Bivariate Analysis of Blanket_Wash and Child_Age_Gp
##
## Frequency Table:
##   Blanket_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##   Rarely         0           1           2           5
##   Monthly        2           3          15          41
##   Weekly         18          30          41          63
##   Daily          1           1           1           2
##   No Fav. Blanket 48          80          34         152
##
## Relative Frequency Table:
##   Blanket_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##   Rarely         0.0%       0.9%       2.2%       1.9%  1.5%
##   Monthly        2.9%       2.6%      16.1%      15.6% 11.3%
##   Weekly         26.1%      26.1%      44.1%      24.0% 28.1%
##   Daily          1.4%       0.9%       1.1%       0.8%  0.9%
##   No Fav. Blanket 69.6%      69.6%      36.6%      57.8% 58.1%
##
## Cramér's V:  0.137134
```

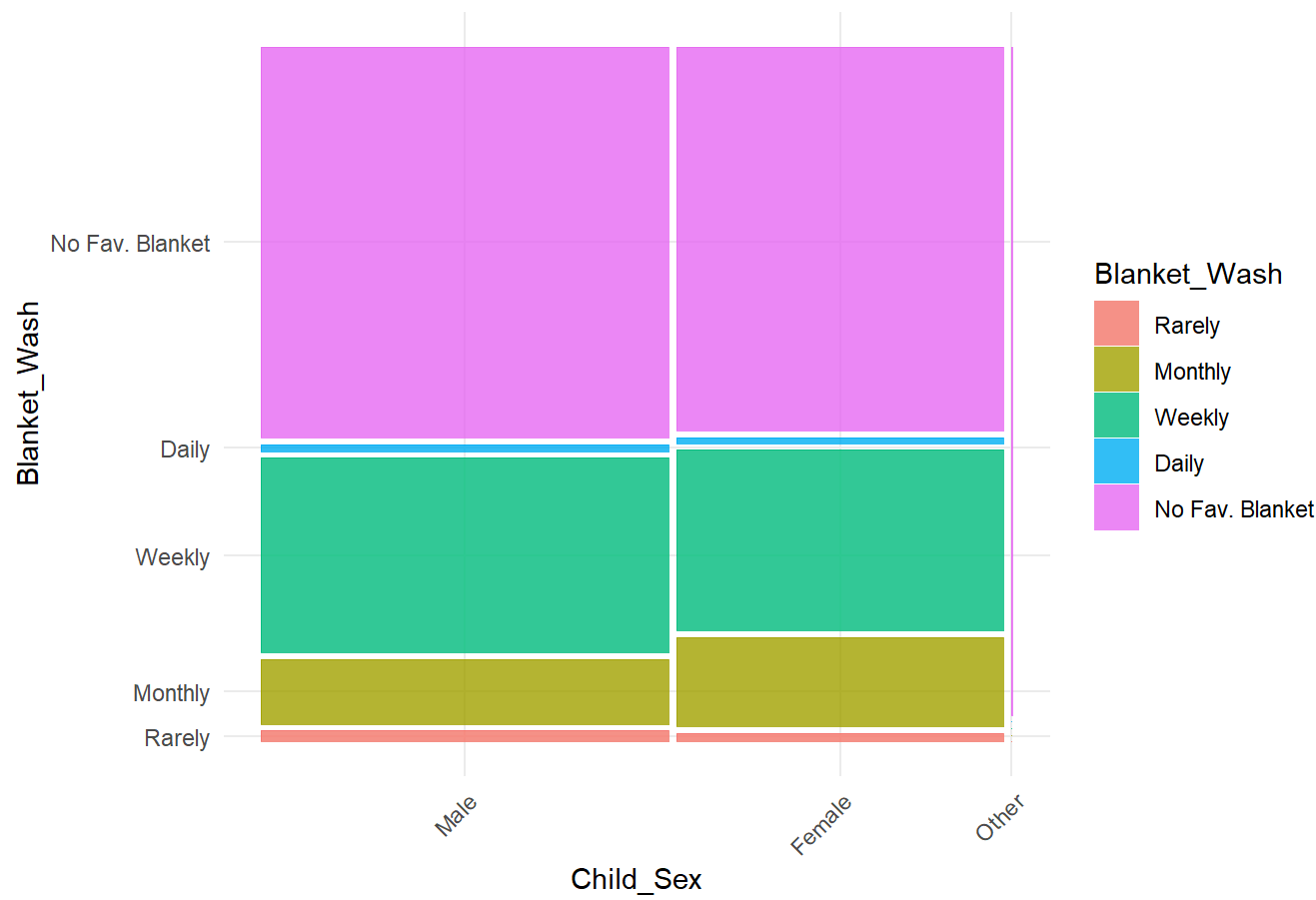
Mosaic Plot of Child_Age_Gp and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and Child_Sex -----
## Bivariate Analysis of Blanket_Wash and Child_Sex
##
## Frequency Table:
##   Blanket_Wash Male Female Other
##   Rarely      5      3      0
##   Monthly     29     32      0
##   Weekly      87     65      0
##   Daily        3      2      0
##   No Fav. Blanket 175   138    1
##
## Relative Frequency Table:
##   Blanket_Wash Male Female Other Total
##   Rarely      1.7%   1.2%   0.0%  1.5%
##   Monthly      9.7%  13.3%   0.0% 11.3%
##   Weekly      29.1%  27.1%   0.0% 28.1%
##   Daily        1.0%   0.8%   0.0%  0.9%
##   No Fav. Blanket 58.5%  57.5% 100.0% 58.1%
##
## Cramér's V:  0
```

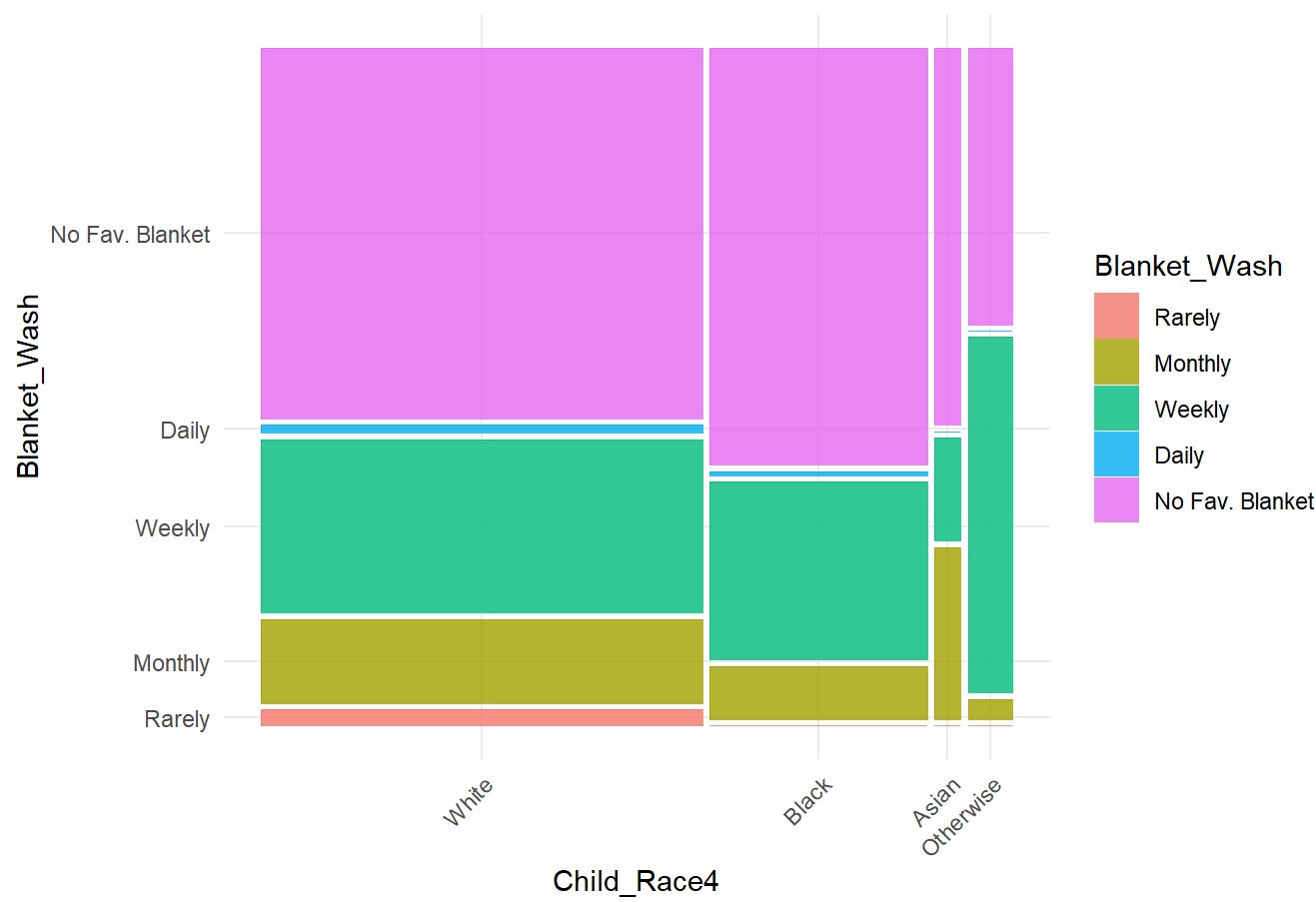
Mosaic Plot of Child_Sex and Blanket_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Blanket_Wash and Child_Race4 -----
## Bivariate Analysis of Blanket_Wash and Child_Race4
##
## Frequency Table:
##   Blanket_Wash White Black Asian Otherwise
##   Rarely      8      0      0          0
##   Monthly     42     13      5          1
##   Weekly      87     44      3         18
##   Daily        4      1      0          0
## No Fav. Blanket 186    103     11         14
##
## Relative Frequency Table:
##   Blanket_Wash White Black Asian Otherwise Total
##   Rarely    2.4%  0.0%  0.0%    0.0%   1.5%
##   Monthly   12.8%  8.1% 26.3%    3.0%  11.3%
##   Weekly    26.6% 27.3% 15.8%   54.5%  28.1%
##   Daily     1.2%  0.6%  0.0%    0.0%   0.9%
## No Fav. Blanket 56.9% 64.0% 57.9%   42.4%  58.1%
##
## Cramér's V:  0.09340803
```

Mosaic Plot of Child_Race4 and Blanket_Wash



```
## Saving 7 x 5 in image
```

```
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```



```
##          var1          var2  cramer_v
## 1    Blanket    Child_Age_Gp 0.21396019
## 2    Blanket          P1_Age 0.15938768
## 3 Blanket_Wash    Child_Age_Gp 0.13713396
## 4 Blanket_Wash      Child_Race4 0.09340803
## 5 Blanket_Wash          City 0.08675554
## 6 Blanket_Wash    P1_Ethnicity 0.08082466
## 7 Blanket_Wash          P1_Age 0.08053383
## 8    Blanket      Child_Race4 0.07219972
## 9 Blanket_Wash      P1_Race4 0.05647268
## 10   Blanket          City 0.05642655
## 11   Blanket    P1_Workstatus 0.05317616
## 12   Blanket      P1_Race4 0.04997199
## 13   Blanket Household_income 0.04004632
## 14 Blanket_Wash      P1_Sex 0.03135696
## 15   Blanket      P1_Sex 0.00000000
## 16   Blanket    P1_Ethnicity 0.00000000
## 17   Blanket      P1_Degree 0.00000000
## 18   Blanket    Single_Parent 0.00000000
## 19   Blanket      Child_Sex 0.00000000
## 20 Blanket_Wash      P1_Degree 0.00000000
## 21 Blanket_Wash    P1_Workstatus 0.00000000
## 22 Blanket_Wash    Single_Parent 0.00000000
## 23 Blanket_Wash Household_income 0.00000000
## 24 Blanket_Wash      Child_Sex 0.00000000
```

Exploring Stuffed Animal Variables

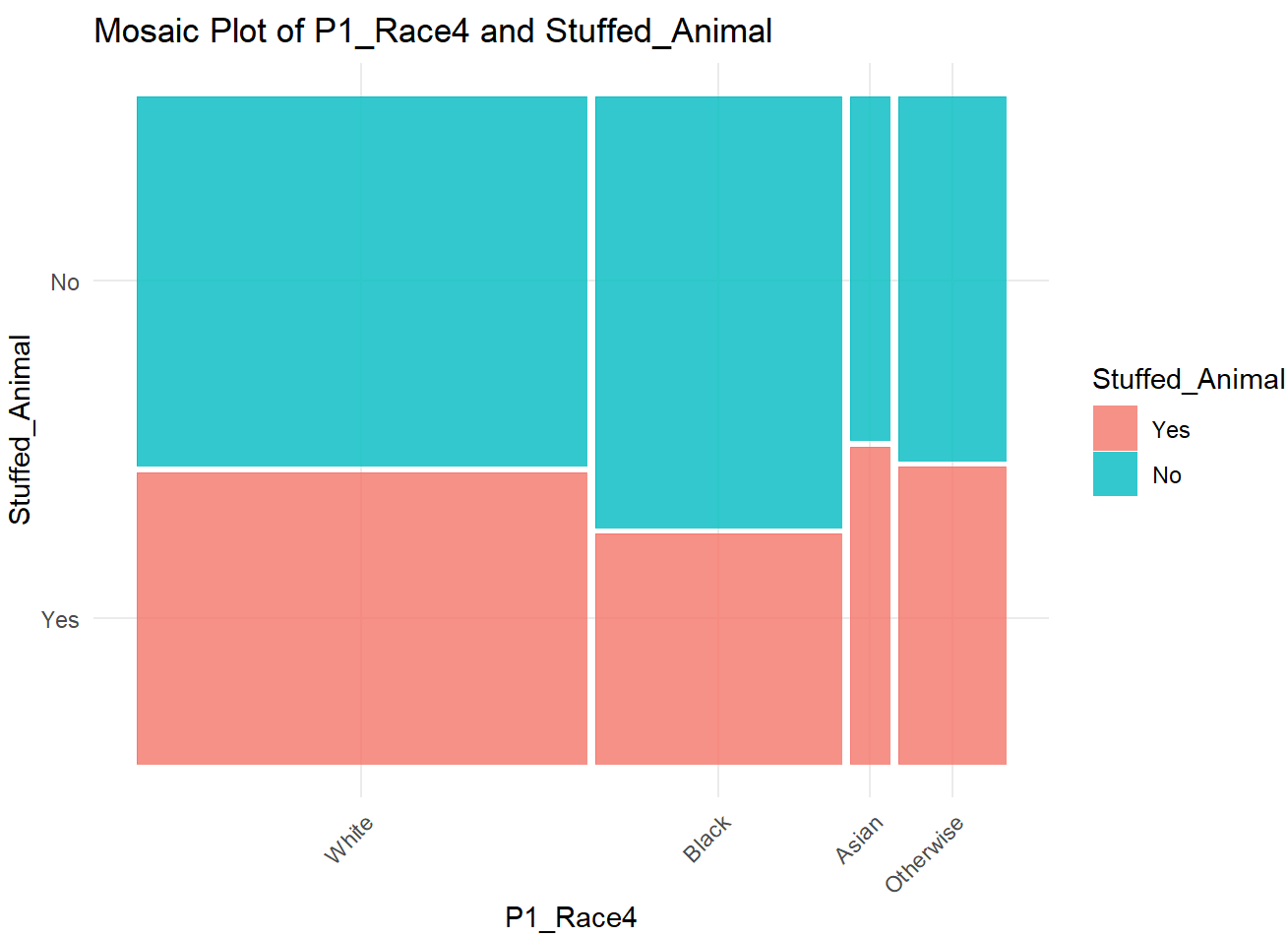
```
## Exploring Blanket Variables
```

```
list1= StuffyVars
list2= DemoVars
```

```
cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(),
                              stringsAsFactors = TRUE)

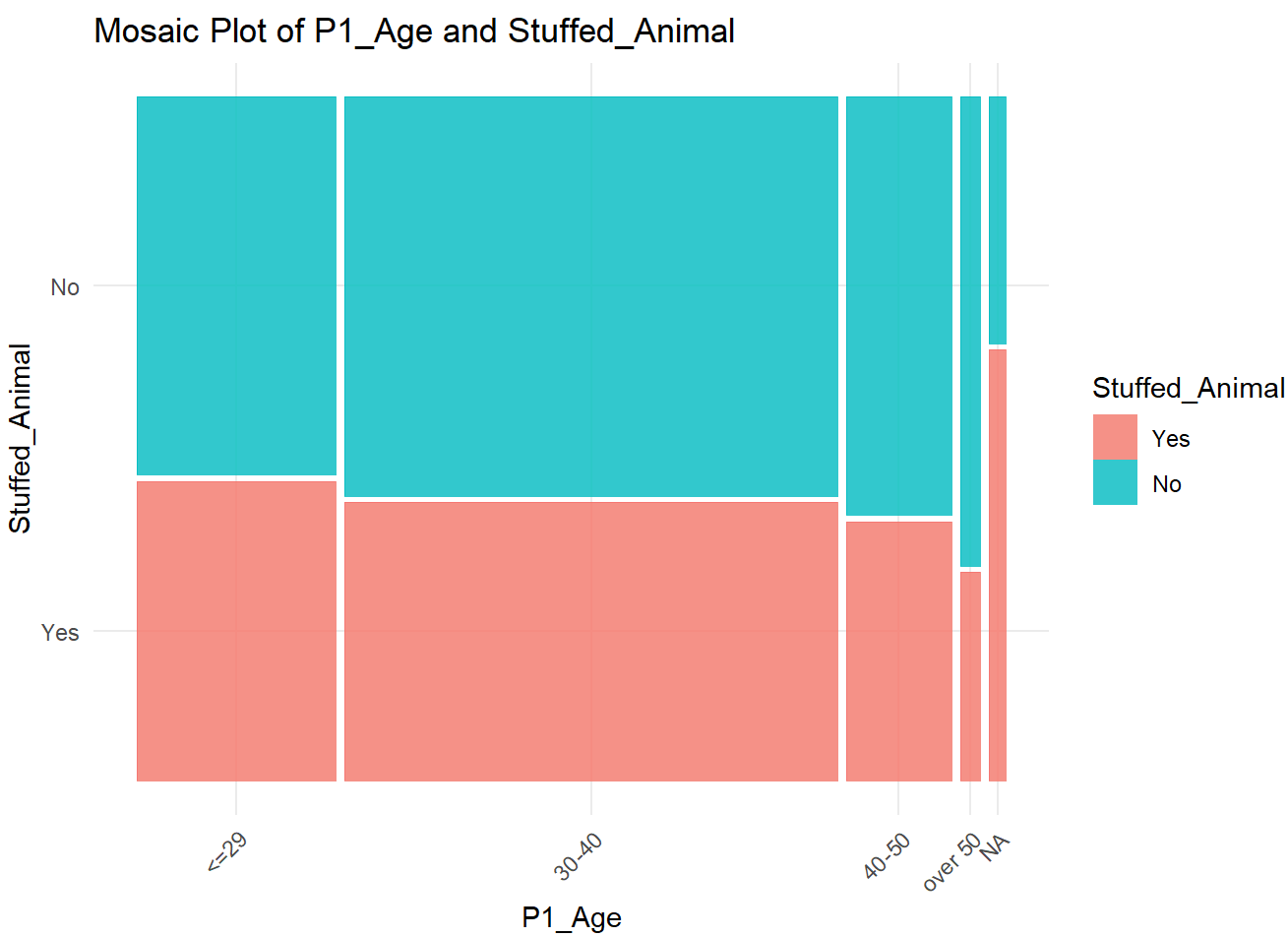
results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = "_vs_")]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values,
                              data.frame(var1 = var1, var2 = var2, cramer_v = result$cramer_v))
    }
  }
}
```

```
##
##
## ----- Analysis for Stuffed_Animal and P1_Race4 -----
## Bivariate Analysis of Stuffed_Animal and P1_Race4
##
## Frequency Table:
##   Stuffed_Animal White Black Asian Otherwise
##           Yes    127    55    12        31
##           No     161   103    13        38
##
## Relative Frequency Table:
##   Stuffed_Animal White Black Asian Otherwise Total
##           Yes  44.1% 34.8% 48.0%    44.9% 41.7%
##           No  55.9% 65.2% 52.0%    55.1% 58.3%
##
## Cramér's V:  0.05213319
```



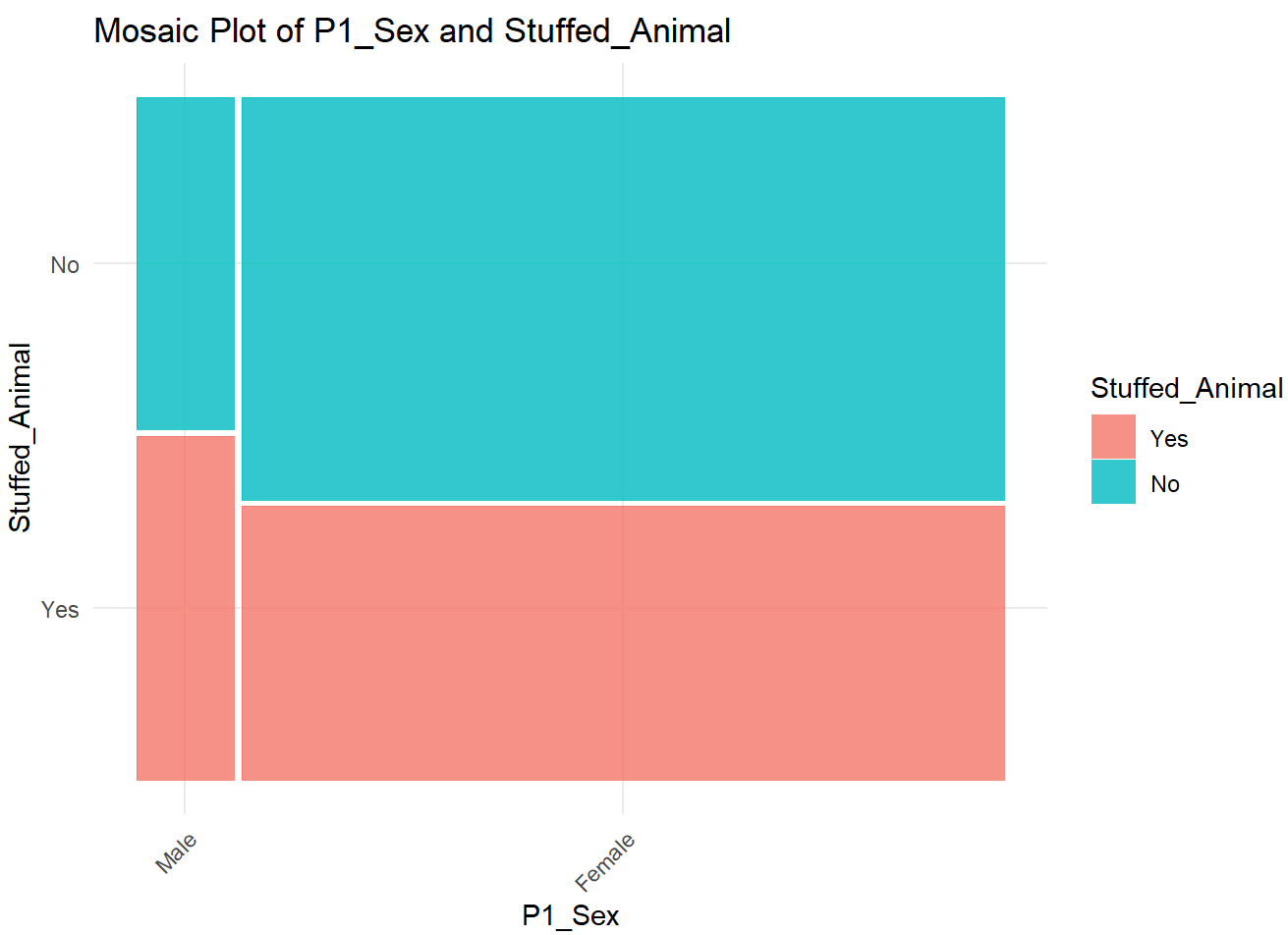
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal and P1_Age -----
## Bivariate Analysis of Stuffed_Animal and P1_Age
##
## Frequency Table:
##   Stuffed_Animal  <=29  30-40  40-50  over 50  NA_
##                   Yes    57    131    26      4    7
##                   No    72    188    42     9    4
##
## Relative Frequency Table:
##   Stuffed_Animal  <=29  30-40  40-50  over 50  NA_ Total
##                   Yes  44.2%  41.1%  38.2%   30.8%  63.6%  41.7%
##                   No  55.8%  58.9%  61.8%   69.2%  36.4%  58.3%
##
## Cramér's V:  0
```



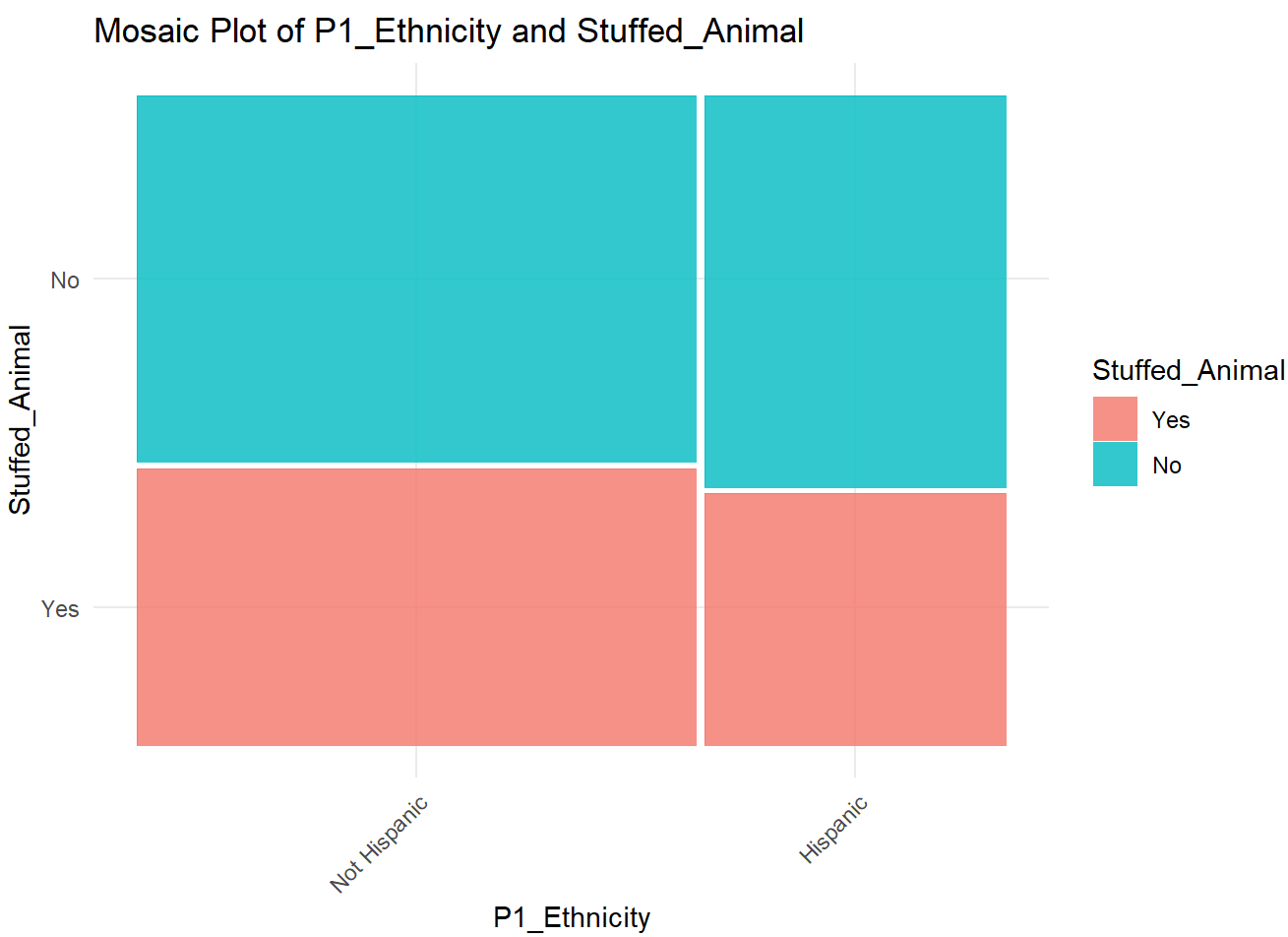
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal and P1_Sex -----
## Bivariate Analysis of Stuffed_Animal and P1_Sex
##
## Frequency Table:
##   Stuffed_Animal  Male Female
##             Yes    31   194
##             No    30   285
##
## Relative Frequency Table:
##   Stuffed_Animal  Male Female Total
##             Yes 50.8%  40.5% 41.7%
##             No 49.2%  59.5% 58.3%
##
## Cramér's V:  0.05038784
```



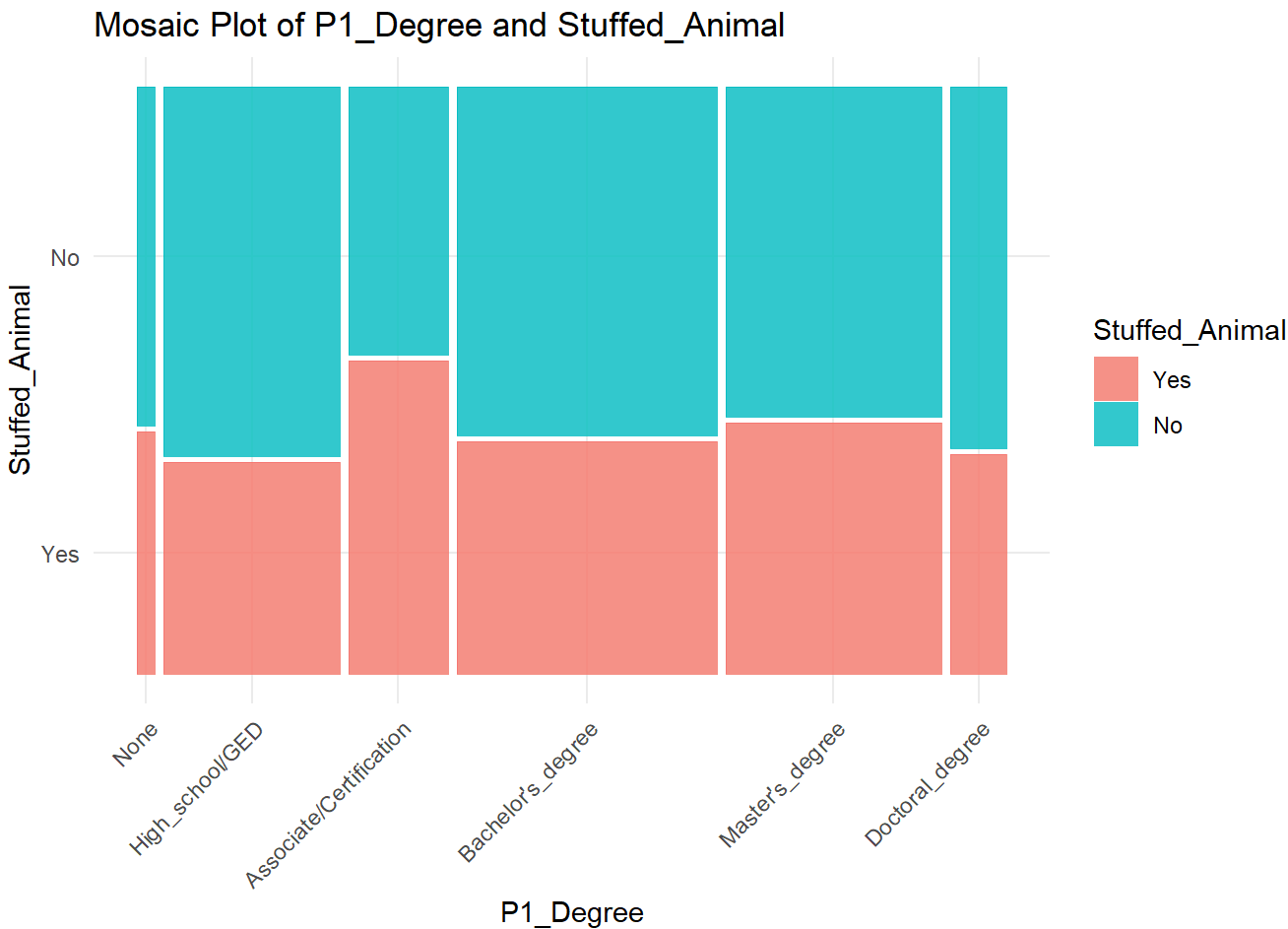
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal and P1_Ethnicity -----
## Bivariate Analysis of Stuffed_Animal and P1_Ethnicity
##
## Frequency Table:
##   Stuffed_Animal Not Hispanic Hispanic
##           Yes           151         74
##           No            200        115
##
## Relative Frequency Table:
##   Stuffed_Animal Not Hispanic Hispanic Total
##           Yes         43.0%    39.2% 41.7%
##           No          57.0%    60.8% 58.3%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

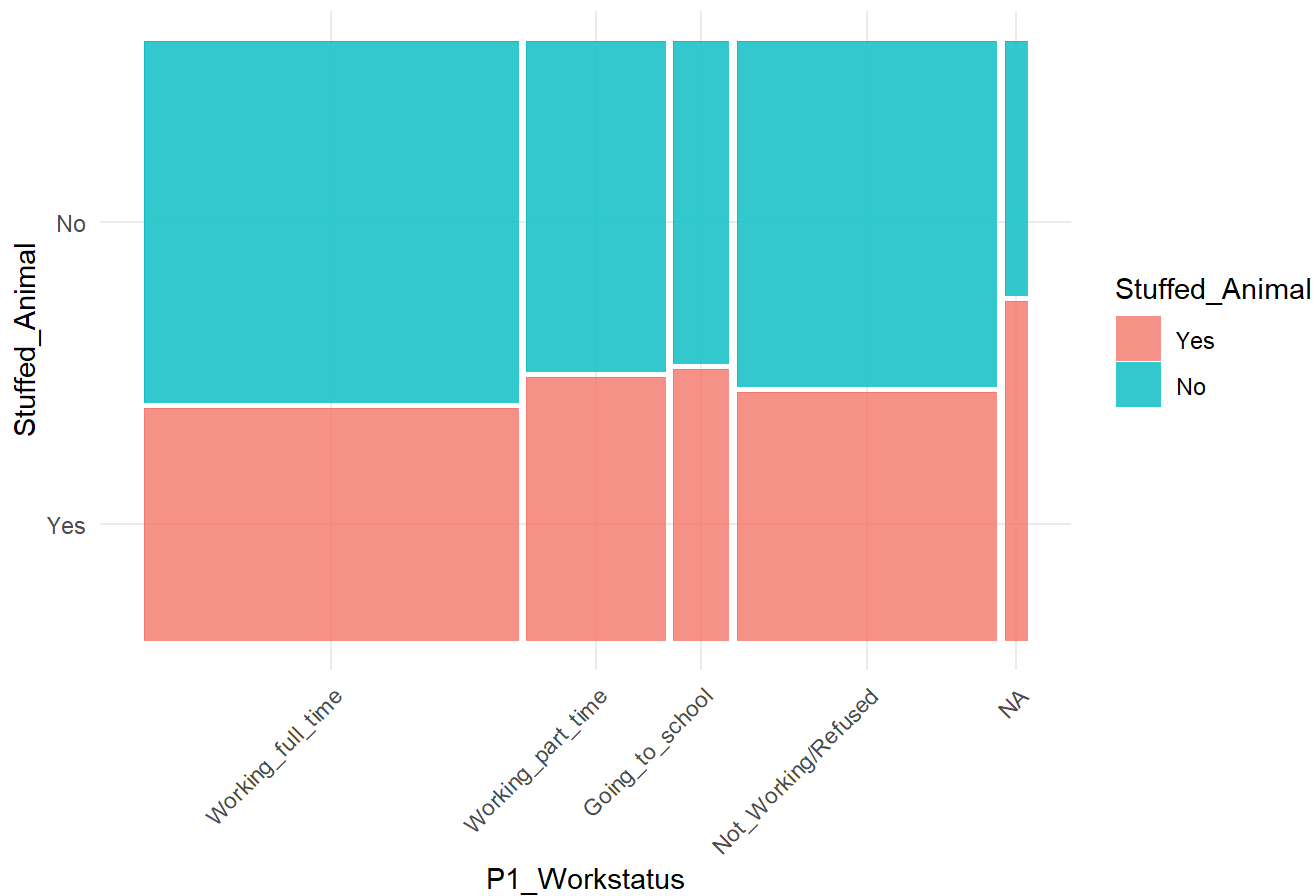
```
##
##
## ----- Analysis for Stuffed_Animal and P1_Degree -----
## Bivariate Analysis of Stuffed_Animal and P1_Degree
##
## Frequency Table:
## Stuffed_Animal None High_school/GED Associate/Certification Bachelor's_degree
##           Yes      5           42           35           68
##           No       7           73           30          102
## Master's_degree Doctoral_degree
##           61          14
##           80          23
##
## Relative Frequency Table:
## Stuffed_Animal  None High_school/GED Associate/Certification Bachelor's_degree
##           Yes 41.7%          36.5%          53.8%          40.0%
##           No 58.3%          63.5%          46.2%          60.0%
## Master's_degree Doctoral_degree Total
##           43.3%          37.8% 41.7%
##           56.7%          62.2% 58.3%
##
## Cramér's V: 0.03792762
```



```
## Saving 7 x 5 in image
```

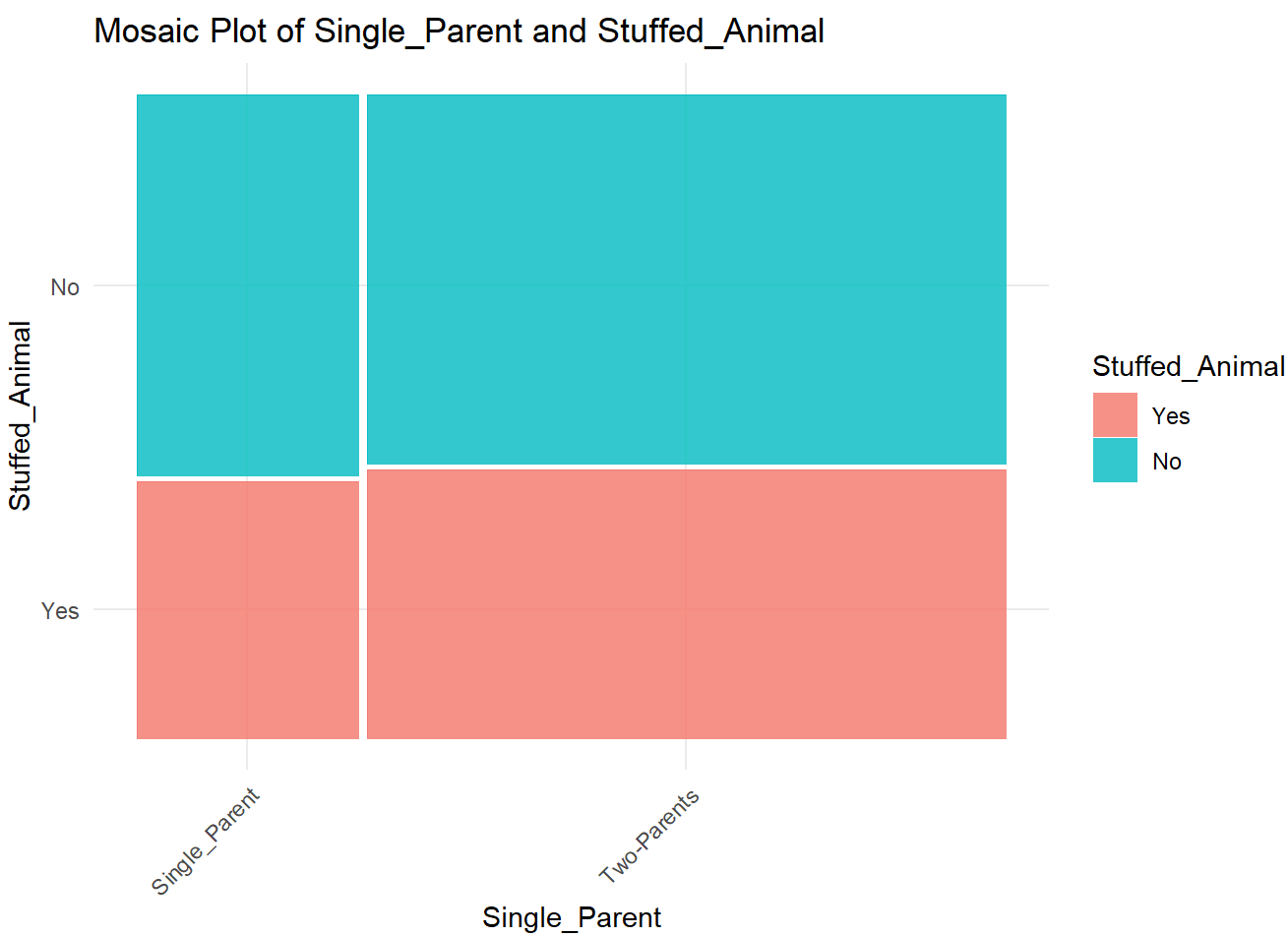
```
##
##
## ----- Analysis for Stuffed_Animal and P1_Workstatus -----
## Bivariate Analysis of Stuffed_Animal and P1_Workstatus
##
## Frequency Table:
##   Stuffed_Animal Working_full_time Working_part_time Going_to_school
##           Yes           93           39           16
##           No           145           49           19
##   Not_Working/Refused NA_
##           69      8
##           96      6
##
## Relative Frequency Table:
##   Stuffed_Animal Working_full_time Working_part_time Going_to_school
##           Yes           39.1%           44.3%           45.7%
##           No           60.9%           55.7%           54.3%
##   Not_Working/Refused   NA_ Total
##           41.8% 57.1% 41.7%
##           58.2% 42.9% 58.3%
##
## Cramér's V:  0
```

Mosaic Plot of P1_Workstatus and Stuffed_Animal



```
## Saving 7 x 5 in image
```

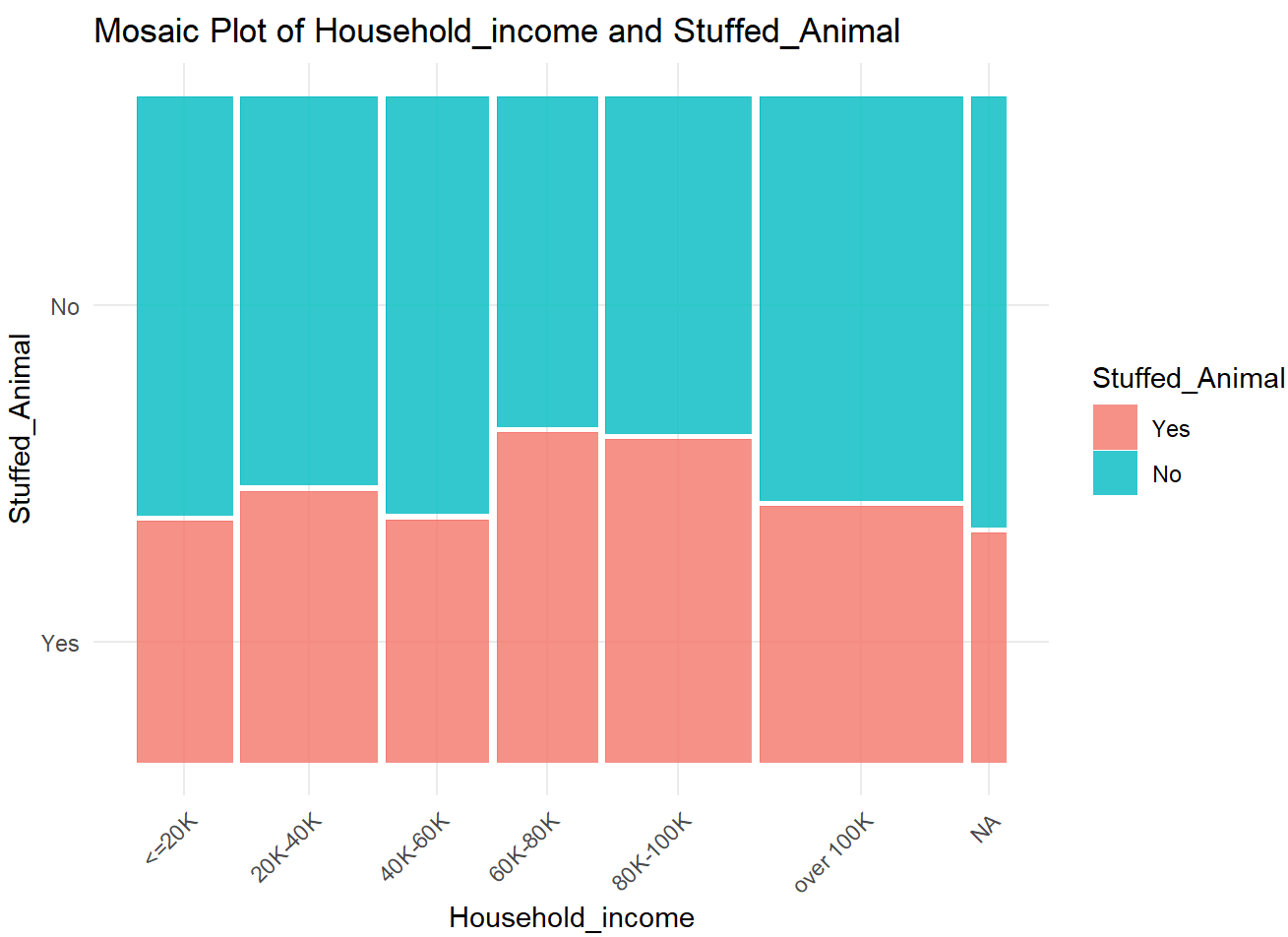
```
##
##
## ----- Analysis for Stuffed_Animal and Single_Parent -----
## Bivariate Analysis of Stuffed_Animal and Single_Parent
##
## Frequency Table:
##   Stuffed_Animal Single_Parent Two-Parents
##           Yes           56           169
##           No            83           232
##
## Relative Frequency Table:
##   Stuffed_Animal Single_Parent Two-Parents Total
##           Yes           40.3%           42.1% 41.7%
##           No            59.7%           57.9% 58.3%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

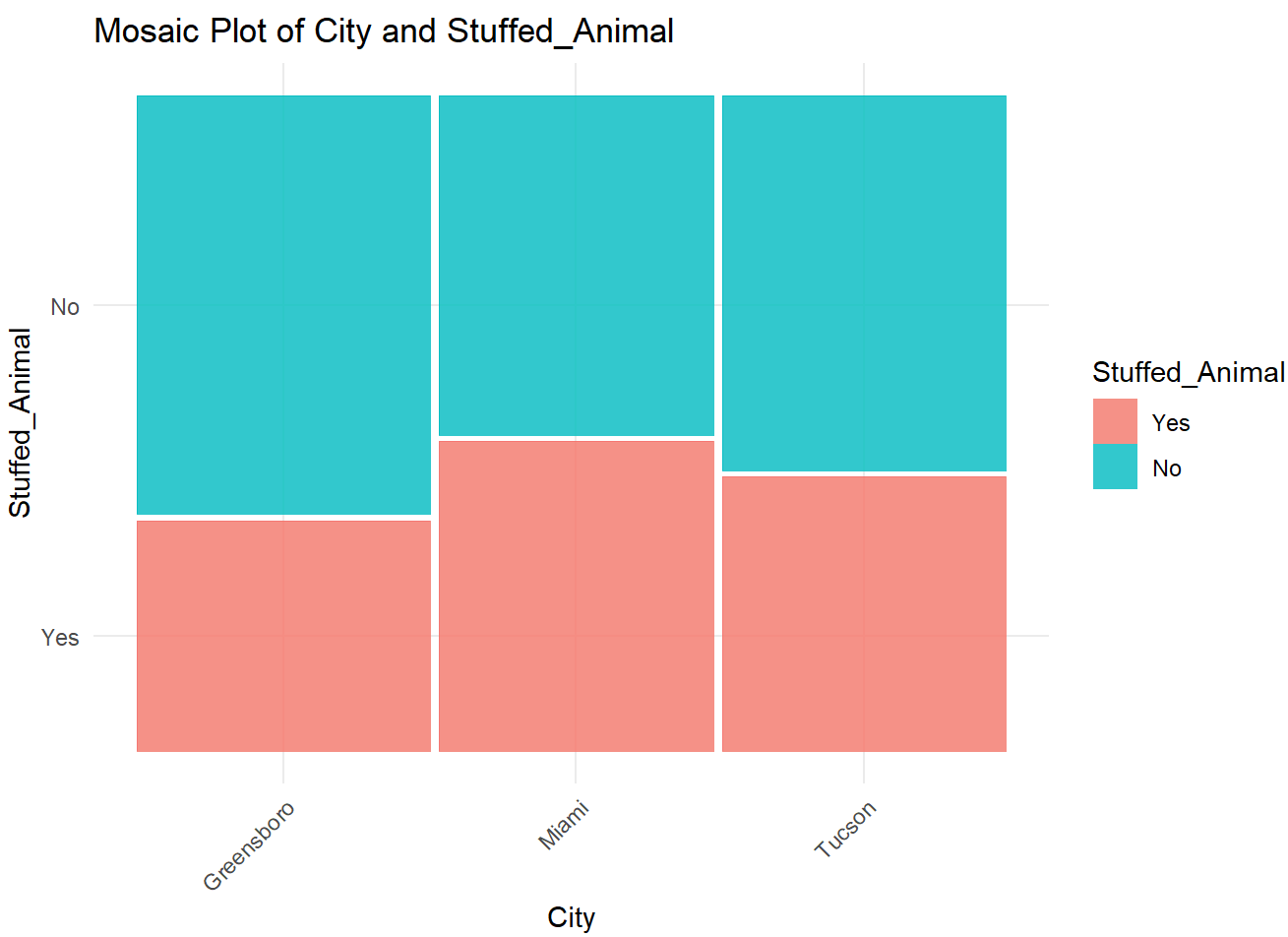


```
##
##
## ----- Analysis for Stuffed_Animal and Household_income -----
## Bivariate Analysis of Stuffed_Animal and Household_income
##
## Frequency Table:
## Stuffed_Animal <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes      23      37      25      33      47      52      8
##           No       40      53      43      33      49      82     15
##
## Relative Frequency Table:
## Stuffed_Animal <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##           Yes 36.5%  41.1%  36.8%  50.0%  49.0%   38.8% 34.8% 41.7%
##           No 63.5%  58.9%  63.2%  50.0%  51.0%   61.2% 65.2% 58.3%
##
## Cramér's V:  0.02143709
```



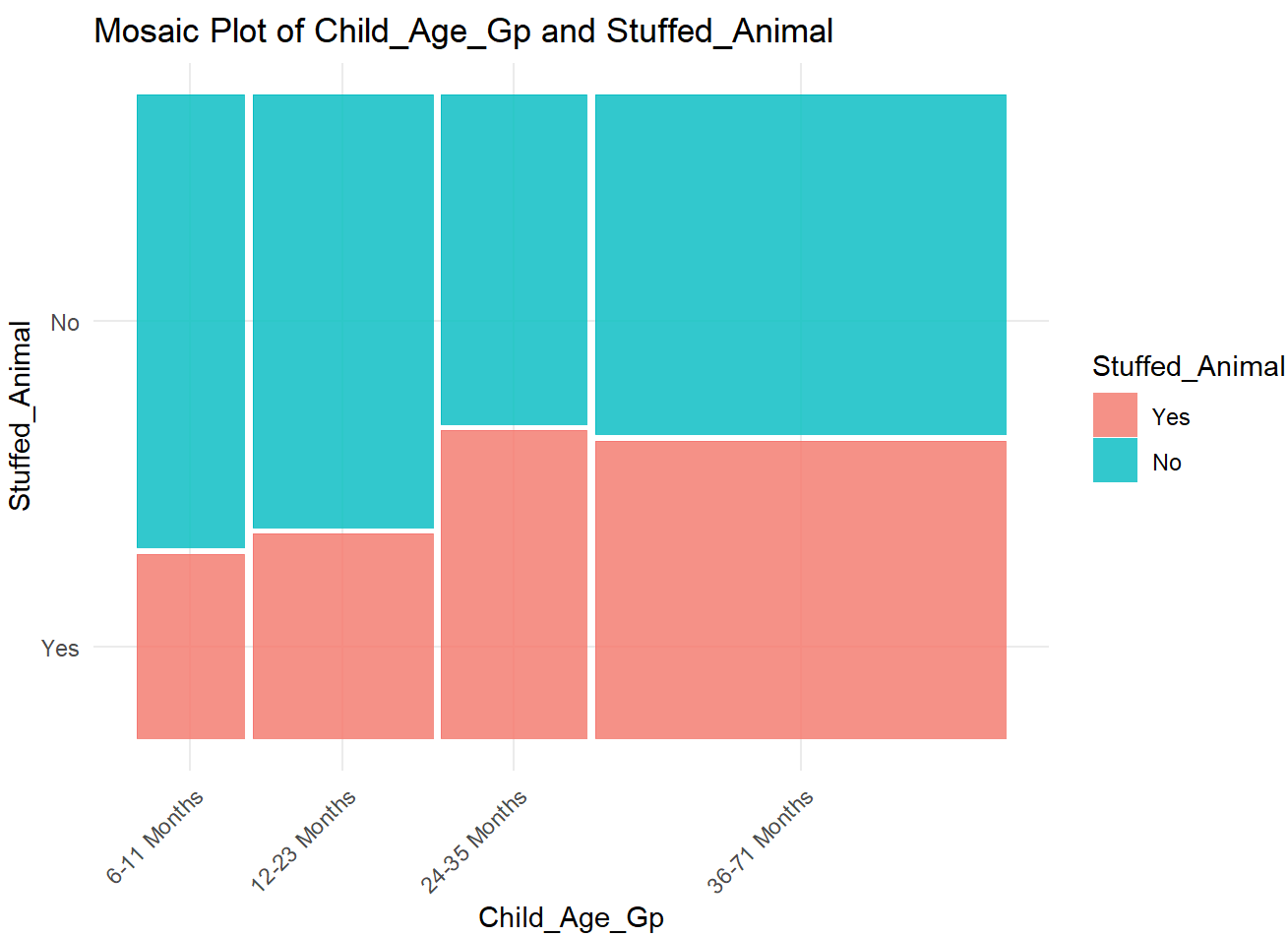
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal and City -----
## Bivariate Analysis of Stuffed_Animal and City
##
## Frequency Table:
##   Stuffed_Animal Greensboro Miami Tucson
##           Yes         66      83      76
##           No         120      91     104
##
## Relative Frequency Table:
##   Stuffed_Animal Greensboro Miami Tucson Total
##           Yes      35.5% 47.7%  42.2% 41.7%
##           No       64.5% 52.3%  57.8% 58.3%
##
## Cramér's V:  0.08117186
```



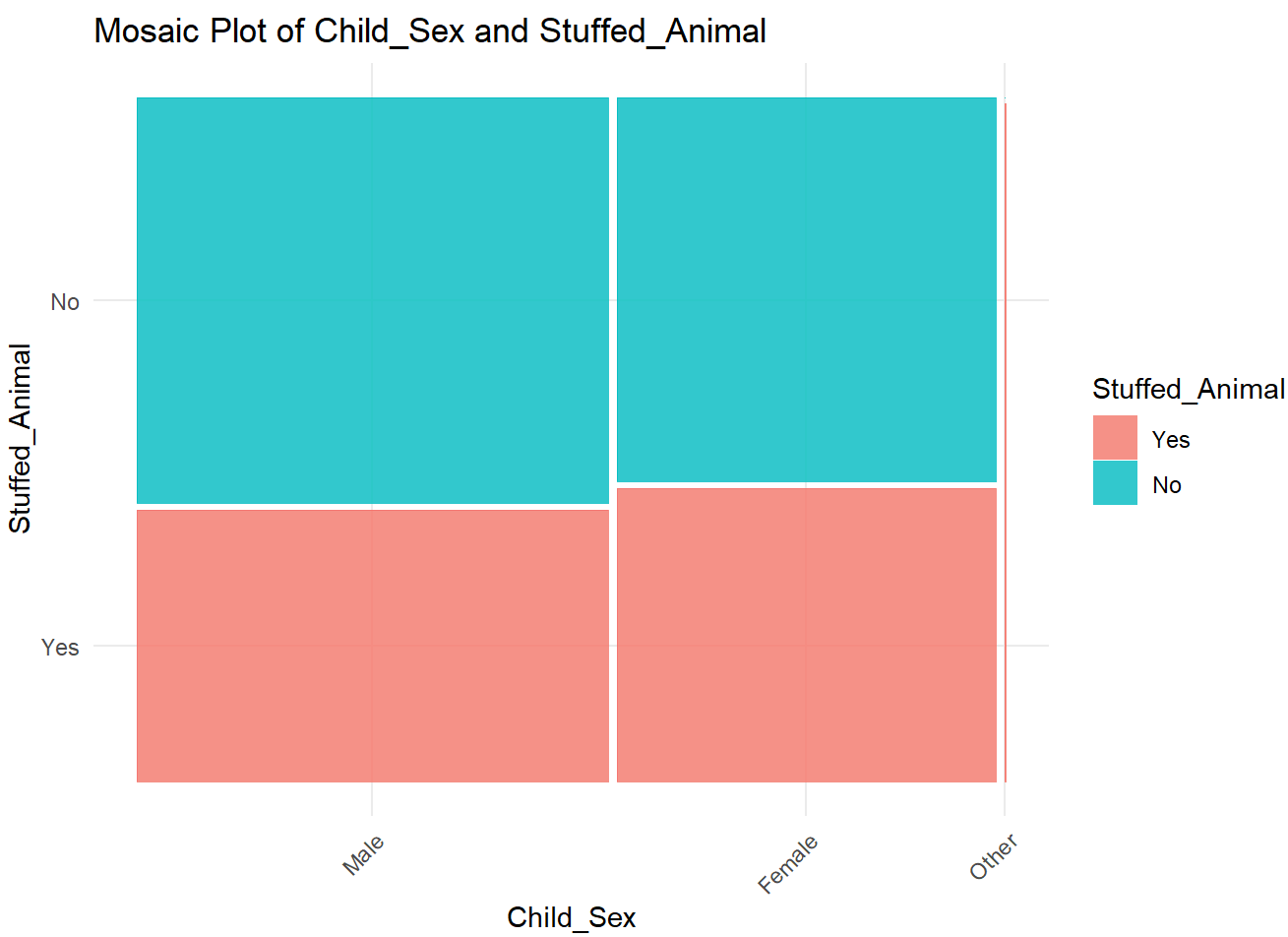
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal and Child_Age_Gp -----
## Bivariate Analysis of Stuffed_Animal and Child_Age_Gp
##
## Frequency Table:
##   Stuffed_Animal 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##               Yes           20           37           45           123
##               No            49           78           48           140
##
## Relative Frequency Table:
##   Stuffed_Animal 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##               Yes           29.0%        32.2%        48.4%        46.8% 41.7%
##               No            71.0%        67.8%        51.6%        53.2% 58.3%
##
## Cramér's V:  0.1386883
```



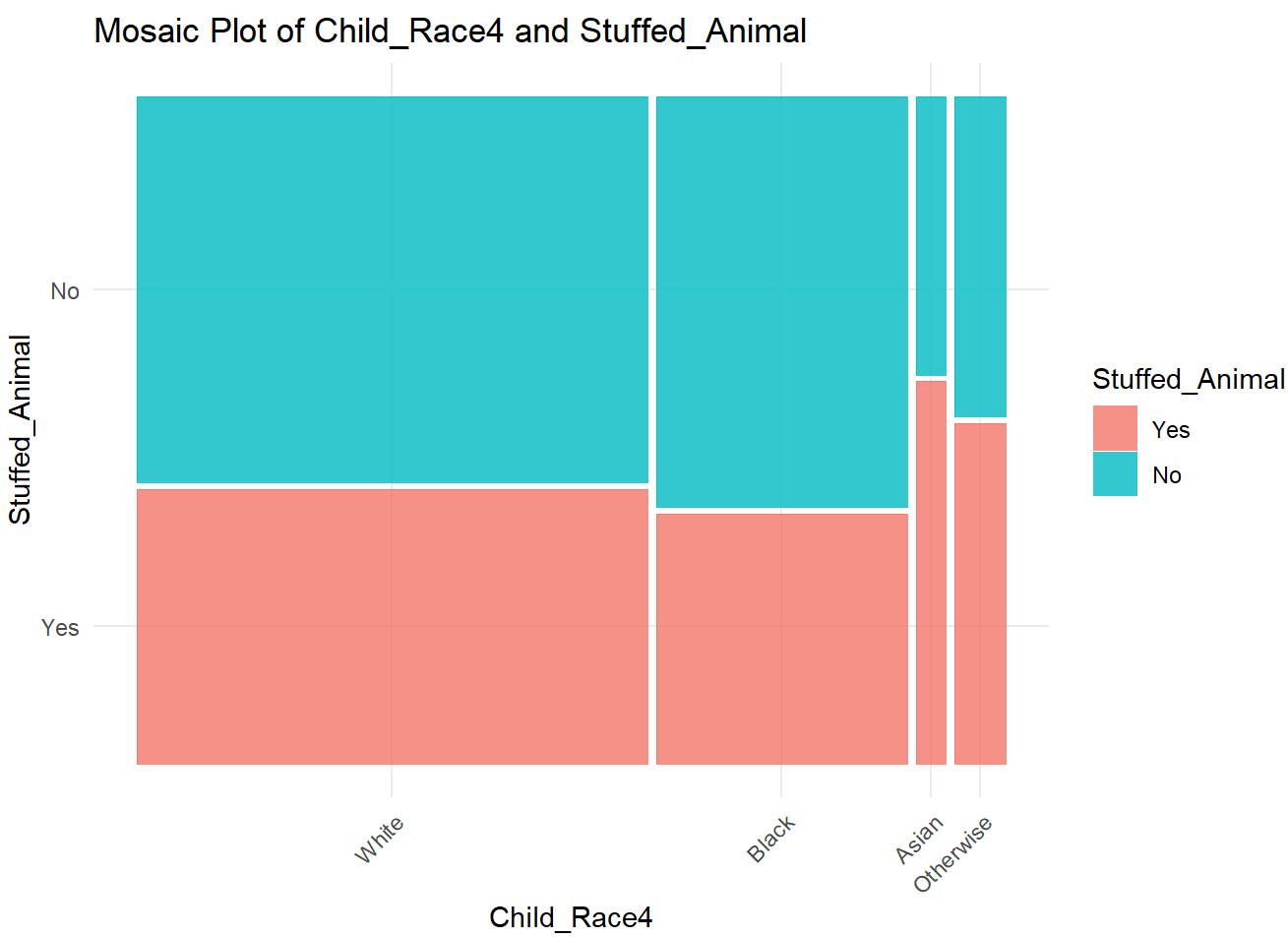
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal and Child_Sex -----
## Bivariate Analysis of Stuffed_Animal and Child_Sex
##
## Frequency Table:
##   Stuffed_Animal Male Female Other
##           Yes  120    104     1
##           No   179    136     0
##
## Relative Frequency Table:
##   Stuffed_Animal Male Female Other Total
##           Yes  40.1%  43.3% 100.0% 41.7%
##           No   59.9%  56.7%   0.0% 58.3%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

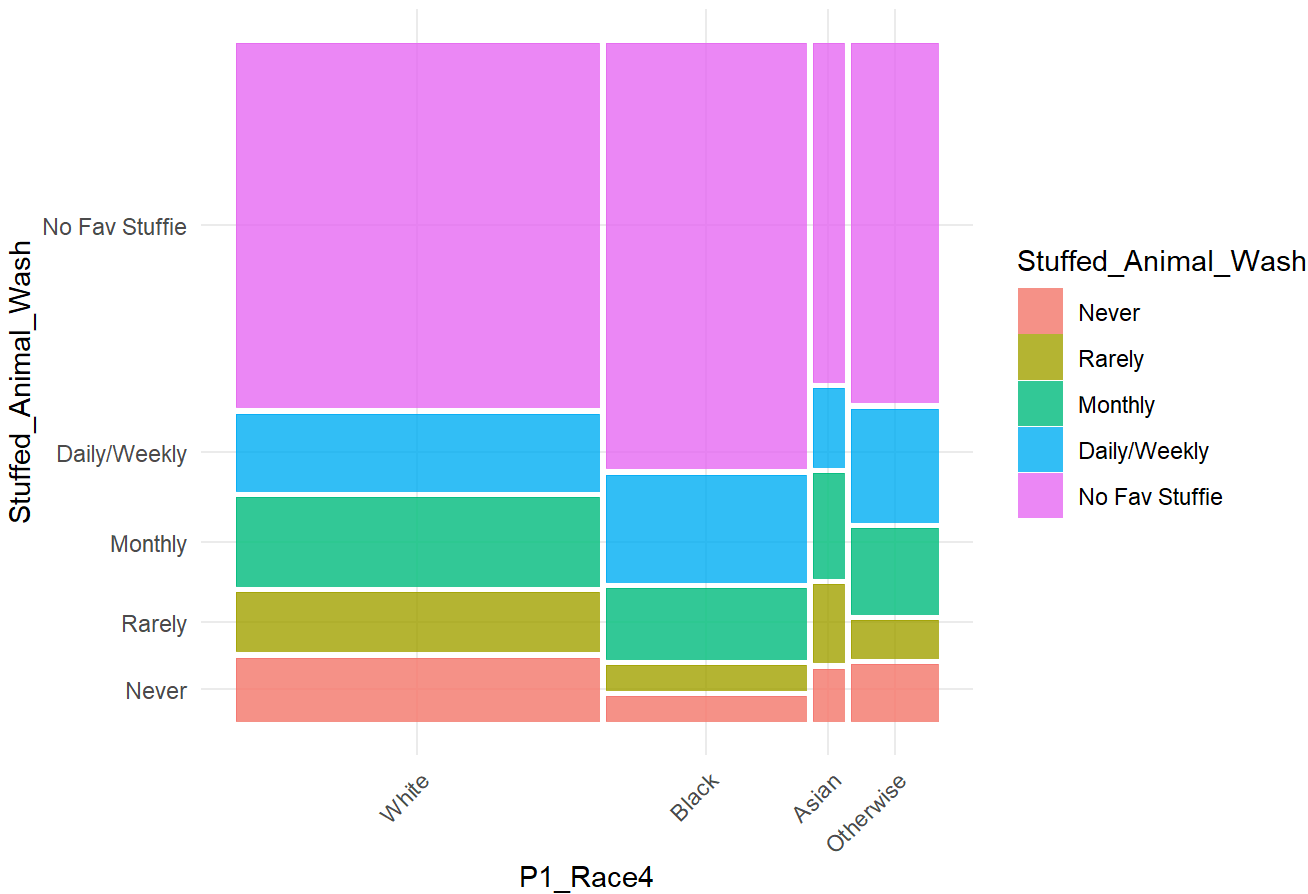
```
##
##
## ----- Analysis for Stuffed_Animal and Child_Race4 -----
## Bivariate Analysis of Stuffed_Animal and Child_Race4
##
## Frequency Table:
##   Stuffed_Animal White Black Asian Otherwise
##           Yes    136    61    11        17
##           No    191   100     8        16
##
## Relative Frequency Table:
##   Stuffed_Animal White Black Asian Otherwise Total
##           Yes  41.6% 37.9% 57.9%    51.5% 41.7%
##           No  58.4% 62.1% 42.1%    48.5% 58.3%
##
## Cramér's V:  0.0494206
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and P1_Race4 -----
## Bivariate Analysis of Stuffed_Animal_Wash and P1_Race4
##
## Frequency Table:
##   Stuffed_Animal_Wash White Black Asian Otherwise
##           Never      28      6      2          6
##           Rarely     26      6      3          4
##           Monthly    39     17      4          9
##           Daily/Weekly 34     26      3         12
##           No Fav Stuffie 161    103     13         38
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash White Black Asian Otherwise Total
##           Never    9.7%  3.8%  8.0%      8.7%  7.8%
##           Rarely    9.0%  3.8% 12.0%      5.8%  7.2%
##           Monthly   13.5% 10.8% 16.0%     13.0% 12.8%
##           Daily/Weekly 11.8% 16.5% 12.0%     17.4% 13.9%
##           No Fav Stuffie 55.9% 65.2% 52.0%     55.1% 58.3%
##
## Cramér's V:  0.03979008
```

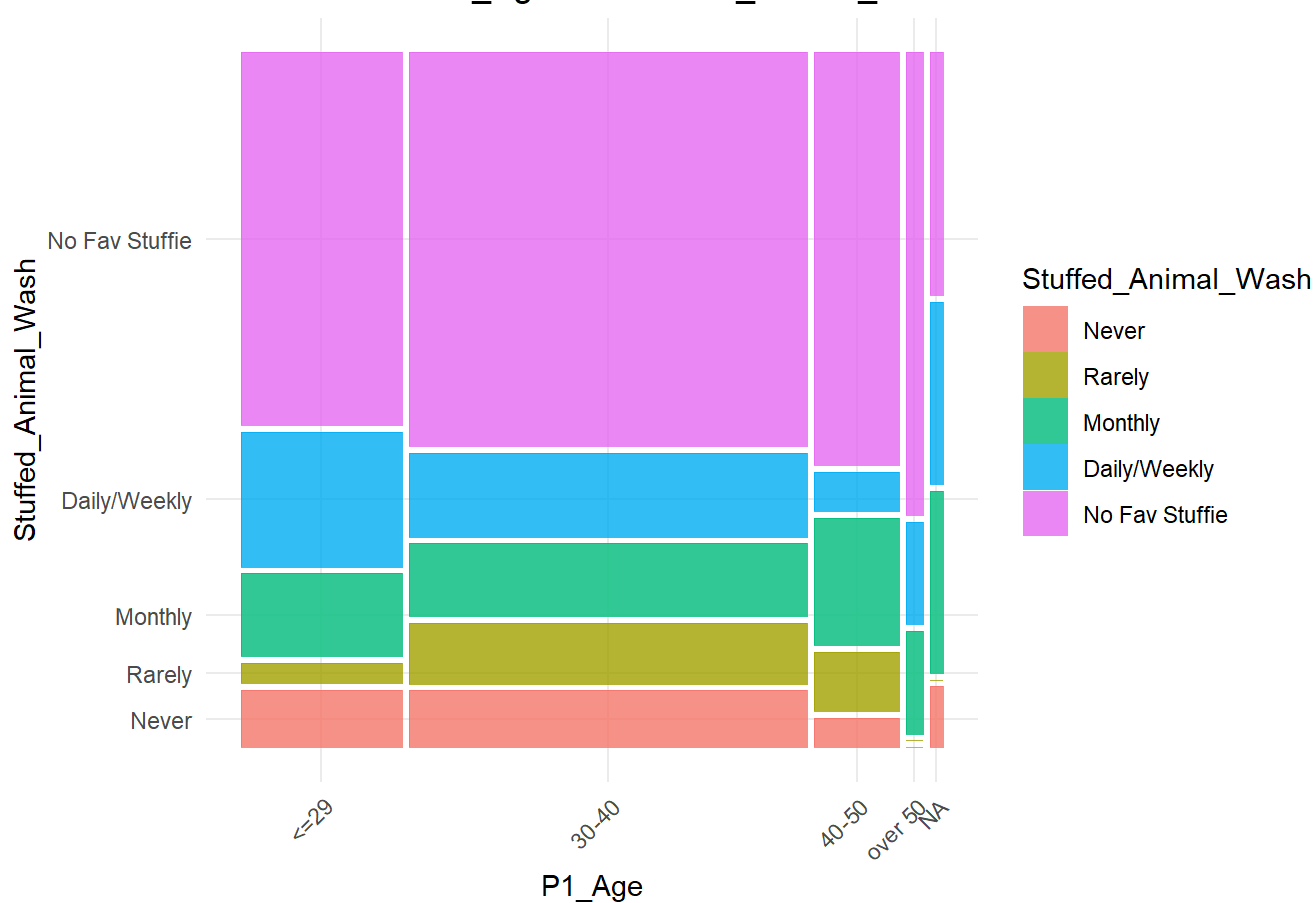
Mosaic Plot of P1_Race4 and Stuffed_Animal_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and P1_Age -----
## Bivariate Analysis of Stuffed_Animal_Wash and P1_Age
##
## Frequency Table:
## Stuffed_Animal_Wash <=29 30-40 40-50 over 50 NA_
##      Never    11    27     3     0    1
##      Rarely     4    29     6     0    0
##      Monthly    16    35    13     2    3
##      Daily/Weekly 26    40     4     2    3
##      No Fav Stuffie 72   188    42     9    4
##
## Relative Frequency Table:
## Stuffed_Animal_Wash <=29 30-40 40-50 over 50 NA_ Total
##      Never    8.5%  8.5%  4.4%    0.0%  9.1%  7.8%
##      Rarely    3.1%  9.1%  8.8%    0.0%  0.0%  7.2%
##      Monthly   12.4% 11.0% 19.1%   15.4% 27.3% 12.8%
##      Daily/Weekly 20.2% 12.5%  5.9%   15.4% 27.3% 13.9%
##      No Fav Stuffie 55.8% 58.9% 61.8%   69.2% 36.4% 58.3%
##
## Cramér's V:  0.06029658
```

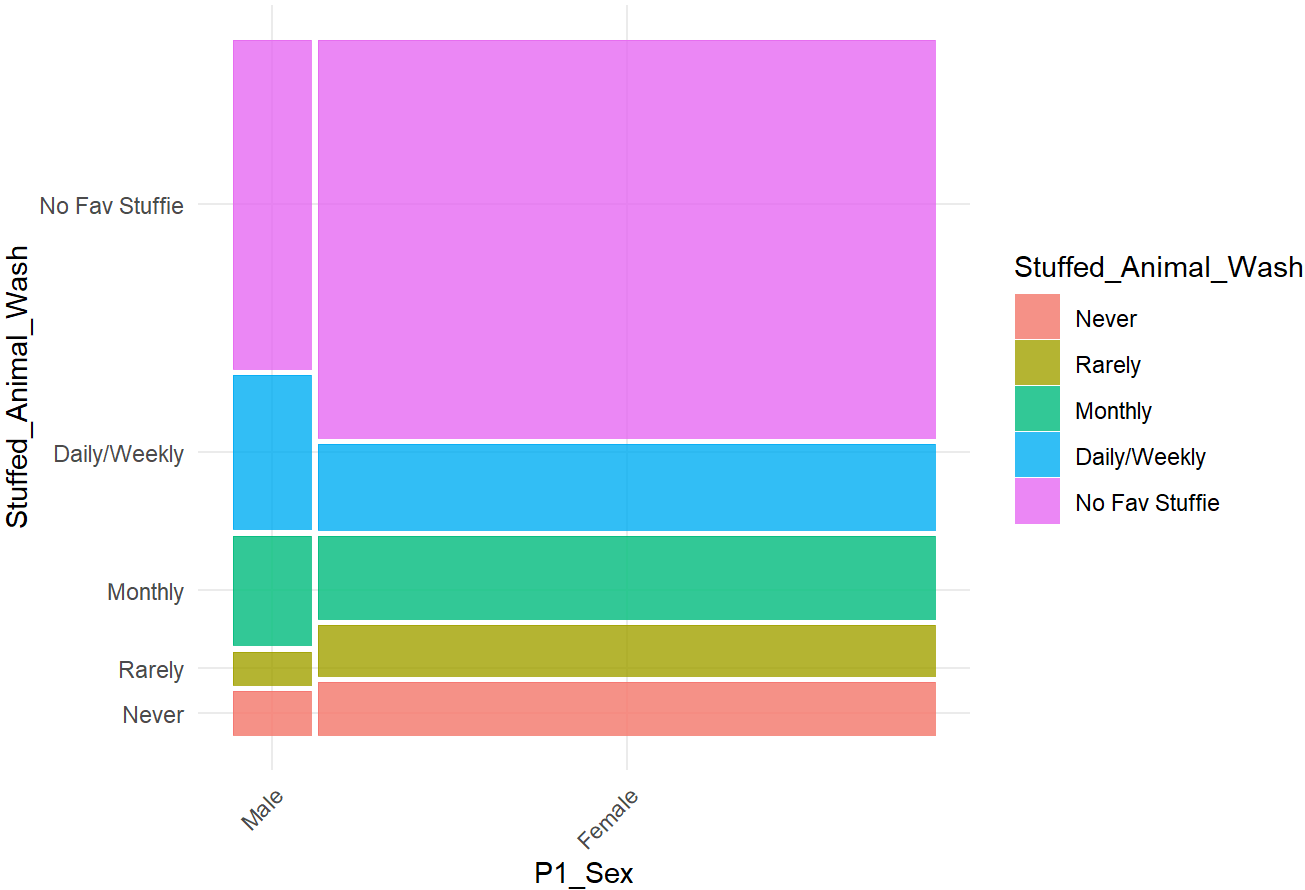
Mosaic Plot of P1_Age and Stuffed_Animal_Wash



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and P1_Sex -----
## Bivariate Analysis of Stuffed_Animal_Wash and P1_Sex
##
## Frequency Table:
##   Stuffed_Animal_Wash   Male Female
##           Never         4      38
##           Rarely        3      36
##           Monthly       10      59
##           Daily/Weekly   14      61
##           No Fav Stuffie 30     285
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash   Male Female Total
##           Never      6.6%   7.9%  7.8%
##           Rarely     4.9%   7.5%  7.2%
##           Monthly    16.4%  12.3% 12.8%
##           Daily/Weekly 23.0% 12.7% 13.9%
##           No Fav Stuffie 49.2% 59.5% 58.3%
##
## Cramér's V:  0.06654714
```

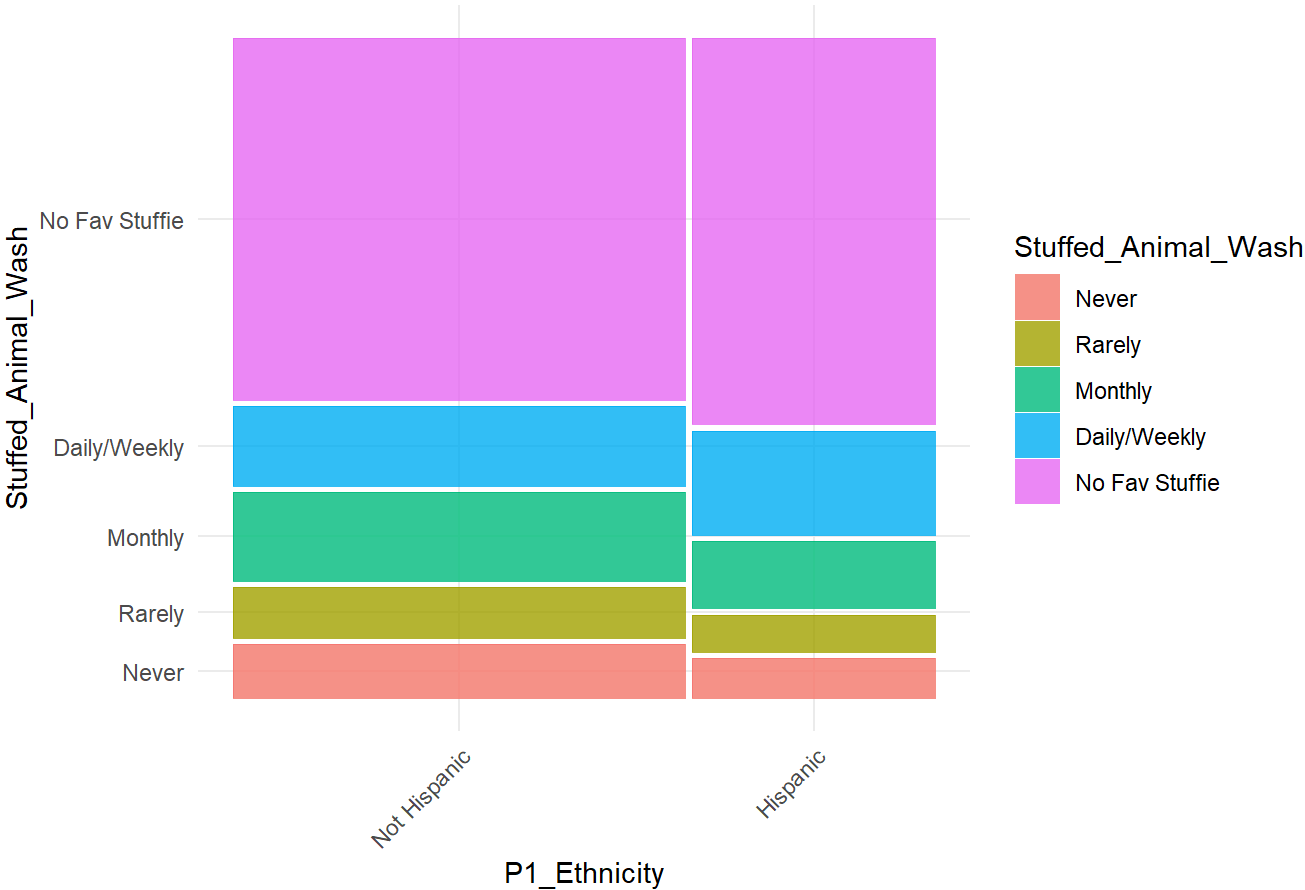
Mosaic Plot of P1_Sex and Stuffed_Animal_Wash



Saving 7 x 5 in image


```
##
##
## ----- Analysis for Stuffed_Animal_Wash and P1_Ethnicity -----
## Bivariate Analysis of Stuffed_Animal_Wash and P1_Ethnicity
##
## Frequency Table:
##   Stuffed_Animal_Wash Not Hispanic Hispanic
##           Never           30           12
##           Rarely          28           11
##           Monthly         49           20
##           Daily/Weekly     44           31
##           No Fav Stuffie   200          115
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash Not Hispanic Hispanic Total
##           Never           8.5%           6.3%  7.8%
##           Rarely           8.0%           5.8%  7.2%
##           Monthly          14.0%          10.6% 12.8%
##           Daily/Weekly     12.5%          16.4% 13.9%
##           No Fav Stuffie   57.0%          60.8% 58.3%
##
## Cramér's V:  0.02284737
```

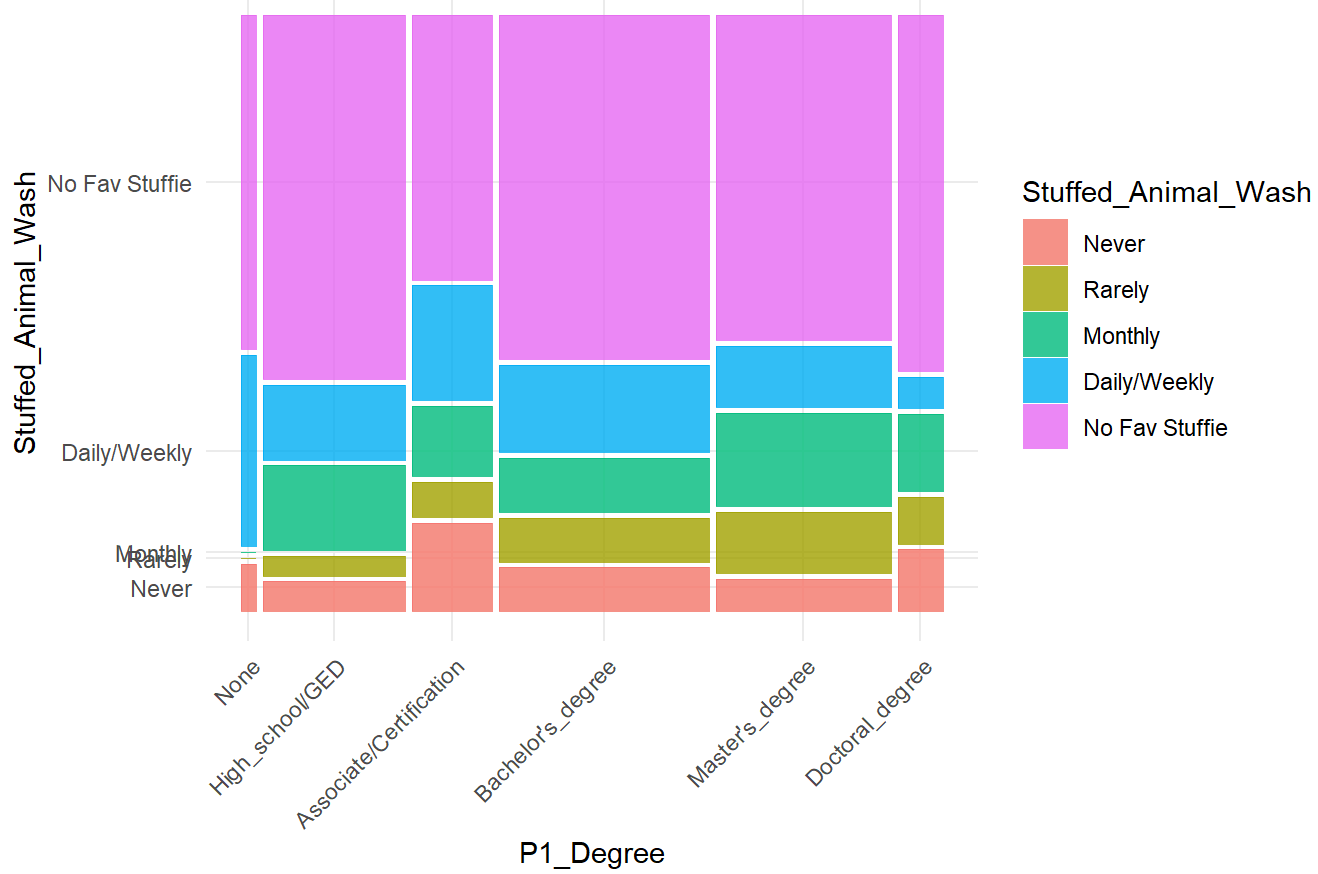
Mosaic Plot of P1_Ethnicity and Stuffed_Animal_Wash



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and P1_Degree -----
## Bivariate Analysis of Stuffed_Animal_Wash and P1_Degree
##
## Frequency Table:
## Stuffed_Animal_Wash None High_school/GED Associate/Certification
##           Never      1           6           10
##           Rarely      0           4            4
##           Monthly      0          17            8
##           Daily/Weekly  4          15           13
##           No Fav Stuffie 7          73           30
## Bachelor's_degree Master's_degree Doctoral_degree
##           13           8            4
##           13          15            3
##           16          23            5
##           26          15            2
##           102         80           23
##
## Relative Frequency Table:
## Stuffed_Animal_Wash None High_school/GED Associate/Certification
##           Never  8.3%          5.2%          15.4%
##           Rarely  0.0%          3.5%           6.2%
##           Monthly 0.0%         14.8%          12.3%
##           Daily/Weekly 33.3%        13.0%          20.0%
##           No Fav Stuffie 58.3%        63.5%          46.2%
## Bachelor's_degree Master's_degree Doctoral_degree Total
##           7.6%          5.7%          10.8%  7.8%
##           7.6%         10.6%           8.1%  7.2%
##           9.4%         16.3%          13.5% 12.8%
##           15.3%        10.6%           5.4% 13.9%
##           60.0%        56.7%          62.2% 58.3%
##
## Cramér's V:  0.06147419
```

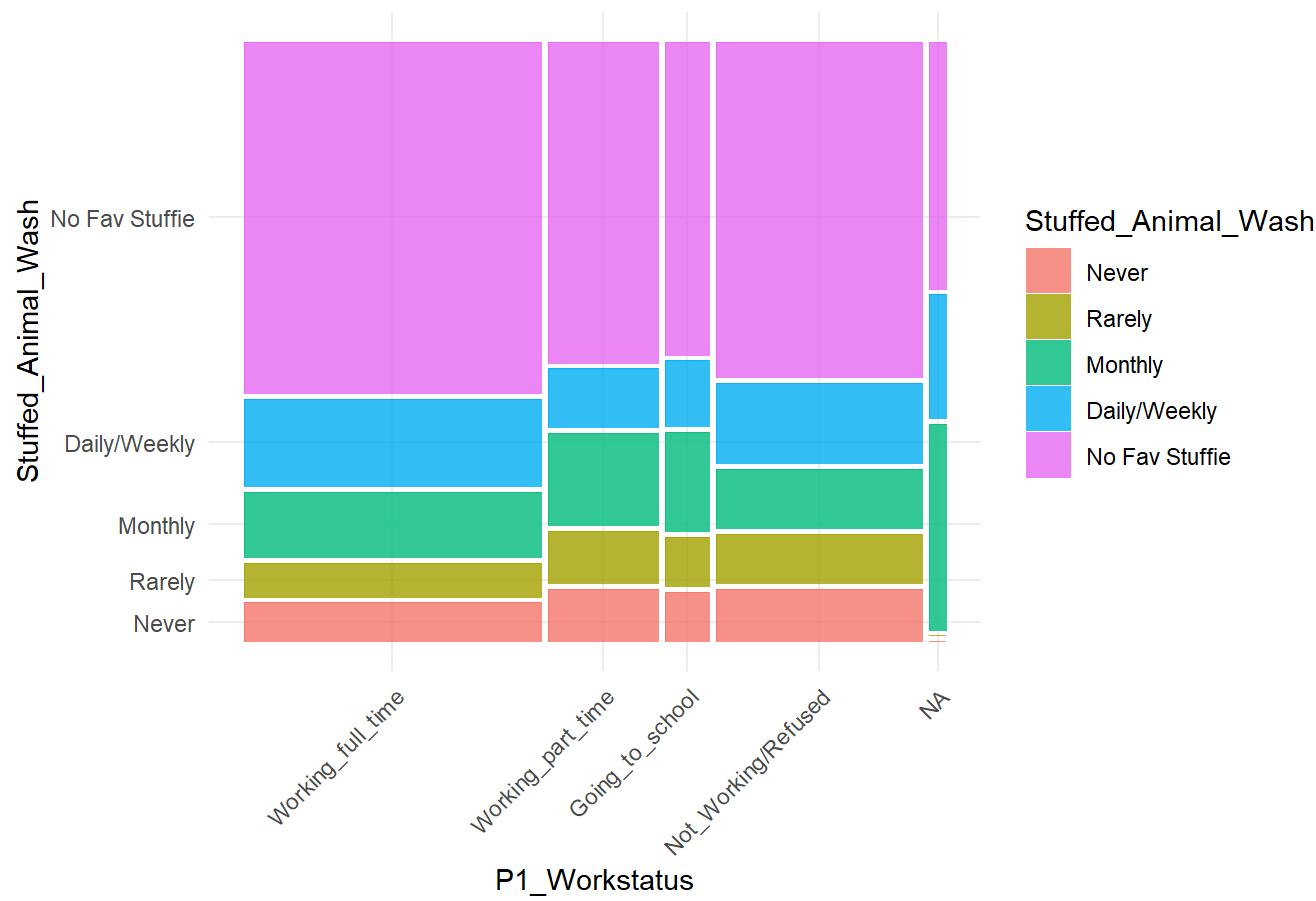
Mosaic Plot of P1_Degree and Stuffed_Animal_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and P1_Workstatus -----
## Bivariate Analysis of Stuffed_Animal_Wash and P1_Workstatus
##
## Frequency Table:
## Stuffed_Animal_Wash Working_full_time Working_part_time Going_to_school
##           Never           16           8           3
##           Rarely          14           8           3
##           Monthly         27          14           6
##           Daily/Weekly     36           9           4
##           No Fav Stuffie    145          49          19
## Not_Working/Refused NA_
##           15    0
##           14    0
##           17    5
##           23    3
##           96    6
##
## Relative Frequency Table:
## Stuffed_Animal_Wash Working_full_time Working_part_time Going_to_school
##           Never           6.7%           9.1%           8.6%
##           Rarely           5.9%           9.1%           8.6%
##           Monthly          11.3%          15.9%          17.1%
##           Daily/Weekly      15.1%          10.2%          11.4%
##           No Fav Stuffie     60.9%          55.7%          54.3%
## Not_Working/Refused  NA_ Total
##           9.1%  0.0%  7.8%
##           8.5%  0.0%  7.2%
##           10.3% 35.7% 12.8%
##           13.9% 21.4% 13.9%
##           58.2% 42.9% 58.3%
##
## Cramér's V:  0
```

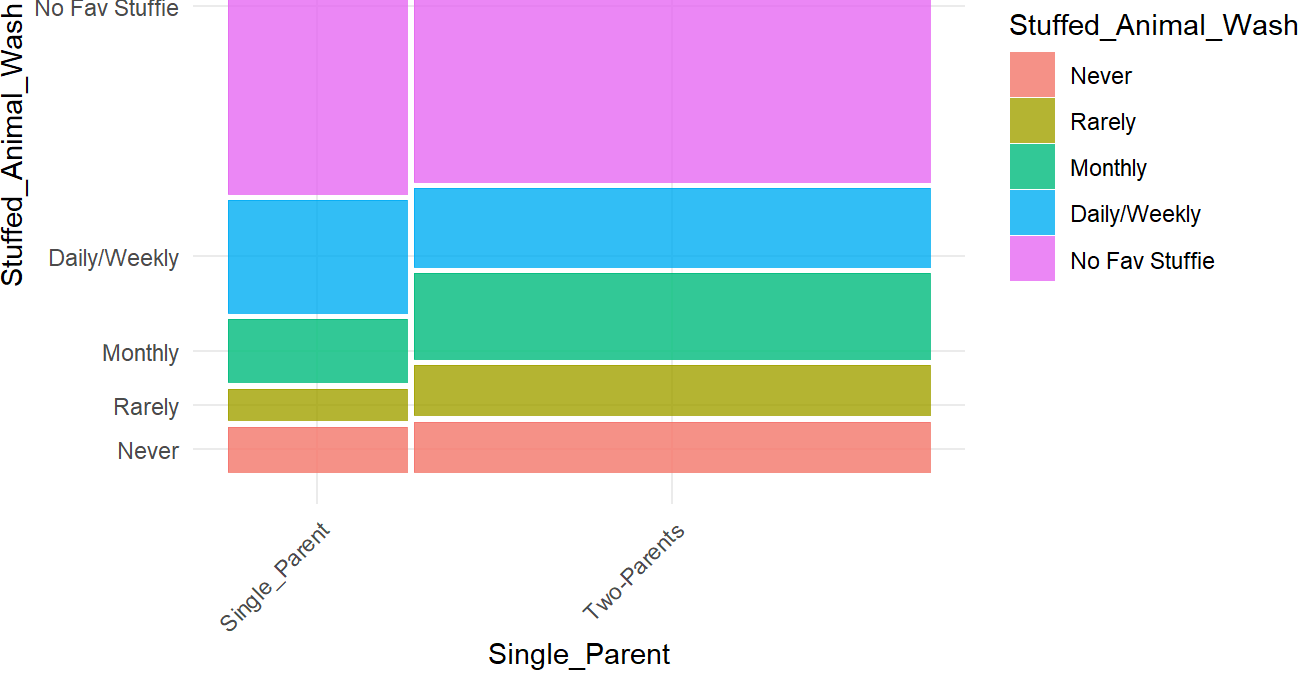
Mosaic Plot of P1_Workstatus and Stuffed_Animal_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and Single_Parent -----
## Bivariate Analysis of Stuffed_Animal_Wash and Single_Parent
##
## Frequency Table:
##   Stuffed_Animal_Wash Single_Parent Two-Parents
##           Never          10           32
##           Rarely           7           32
##           Monthly         14           55
##           Daily/Weekly     25           50
##           No Fav Stuffie    83          232
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash Single_Parent Two-Parents Total
##           Never          7.2%          8.0%  7.8%
##           Rarely          5.0%          8.0%  7.2%
##           Monthly        10.1%         13.7% 12.8%
##           Daily/Weekly    18.0%         12.5% 13.9%
##           No Fav Stuffie  59.7%         57.9% 58.3%
##
## Cramér's V:  0.03626165
```

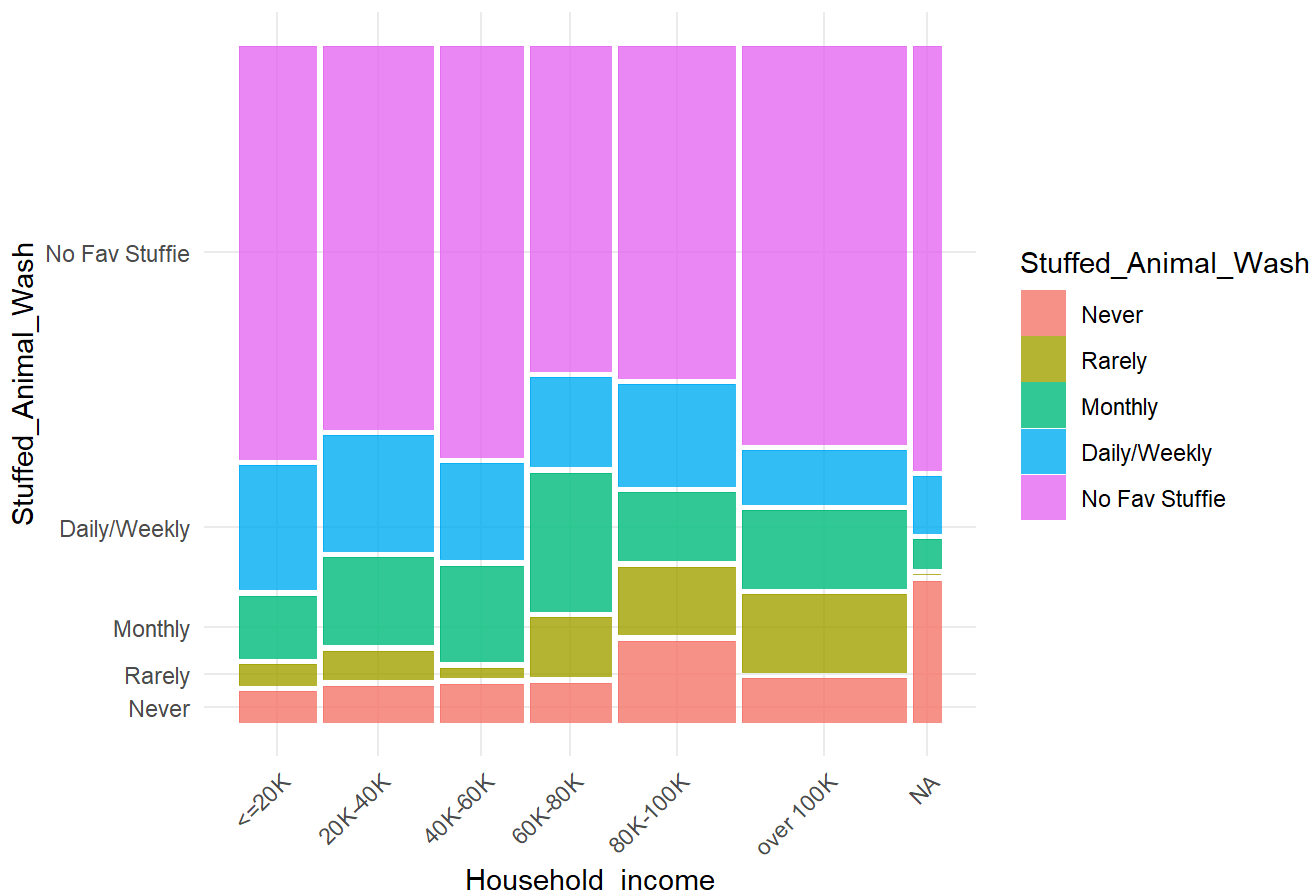
Mosaic Plot of Single_Parent and Stuffed_Animal_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and Household_income -----
## Bivariate Analysis of Stuffed_Animal_Wash and Household_income
##
## Frequency Table:
## Stuffed_Animal_Wash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Never      3      5      4      4      12      9   5
##           Rarely     2      4      1      6      10     16   0
##           Monthly     6     12     10     14     10     16   1
##           Daily/Weekly 12     16     10      9     15     11   2
##           No Fav Stuffie 40     53     43     33     49     82  15
##
## Relative Frequency Table:
## Stuffed_Animal_Wash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_
##           Never  4.8%   5.6%   5.9%   6.1%   12.5%    6.7% 21.7%
##           Rarely  3.2%   4.4%   1.5%   9.1%   10.4%   11.9%  0.0%
##           Monthly  9.5%  13.3%  14.7%  21.2%  10.4%   11.9%  4.3%
##           Daily/Weekly 19.0% 17.8% 14.7% 13.6% 15.6%    8.2%  8.7%
##           No Fav Stuffie 63.5% 58.9% 63.2% 50.0% 51.0%   61.2% 65.2%
## Total
## 7.8%
## 7.2%
## 12.8%
## 13.9%
## 58.3%
##
## Cramér's V:  0.08165921
```

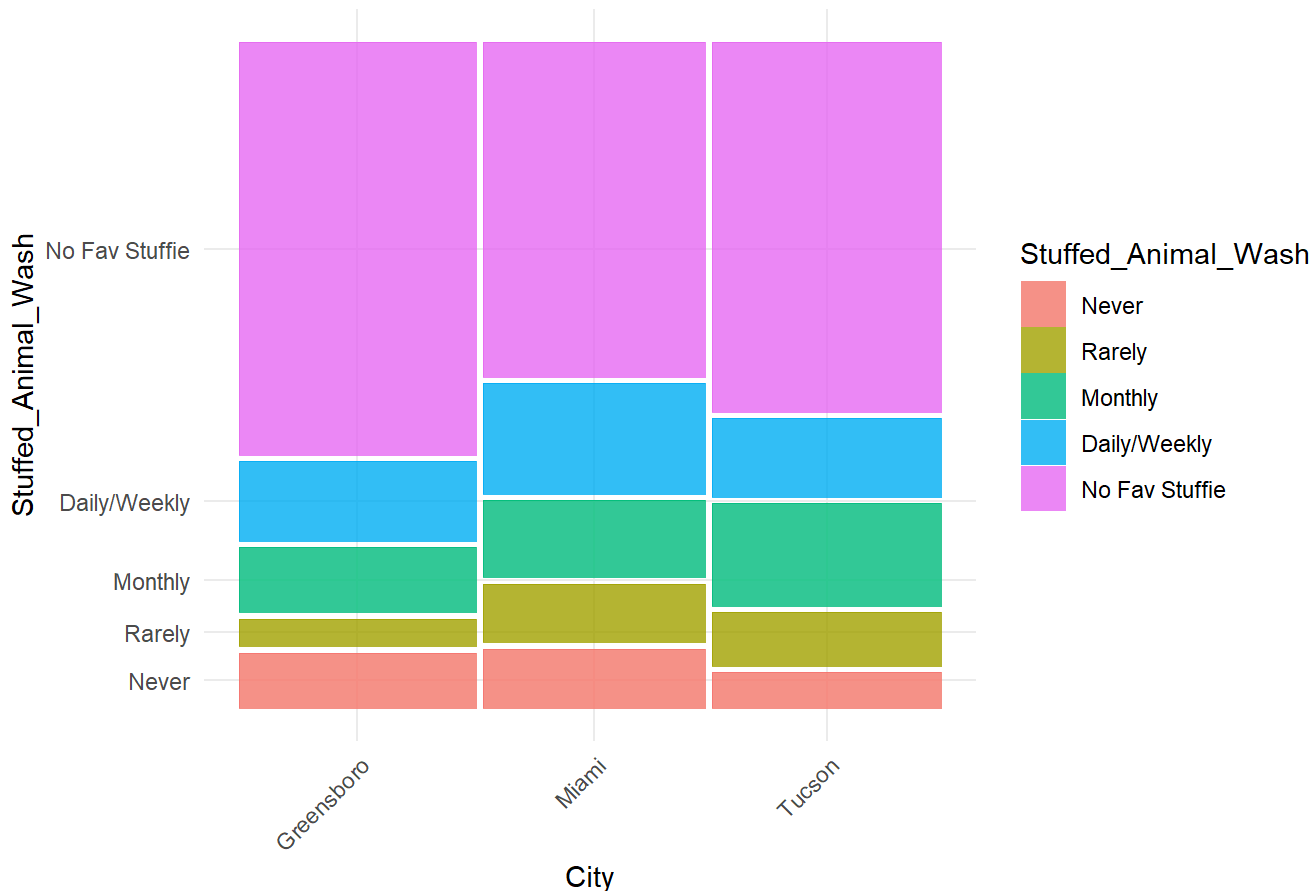
Mosaic Plot of Household_income and Stuffed_Animal_Wash



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and City -----
## Bivariate Analysis of Stuffed_Animal_Wash and City
##
## Frequency Table:
##   Stuffed_Animal_Wash Greensboro Miami Tucson
##           Never          16      16      10
##           Rarely          8      16      15
##           Monthly         19      21      29
##           Daily/Weekly     23      30      22
##           No Fav Stuffie  120      91     104
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash Greensboro Miami Tucson Total
##           Never          8.6%  9.2%   5.6%  7.8%
##           Rarely          4.3%  9.2%   8.3%  7.2%
##           Monthly         10.2% 12.1%  16.1% 12.8%
##           Daily/Weekly     12.4% 17.2%  12.2% 13.9%
##           No Fav Stuffie   64.5% 52.3%  57.8% 58.3%
##
## Cramér's V:  0.06222732
```

Mosaic Plot of City and Stuffed_Animal_Wash

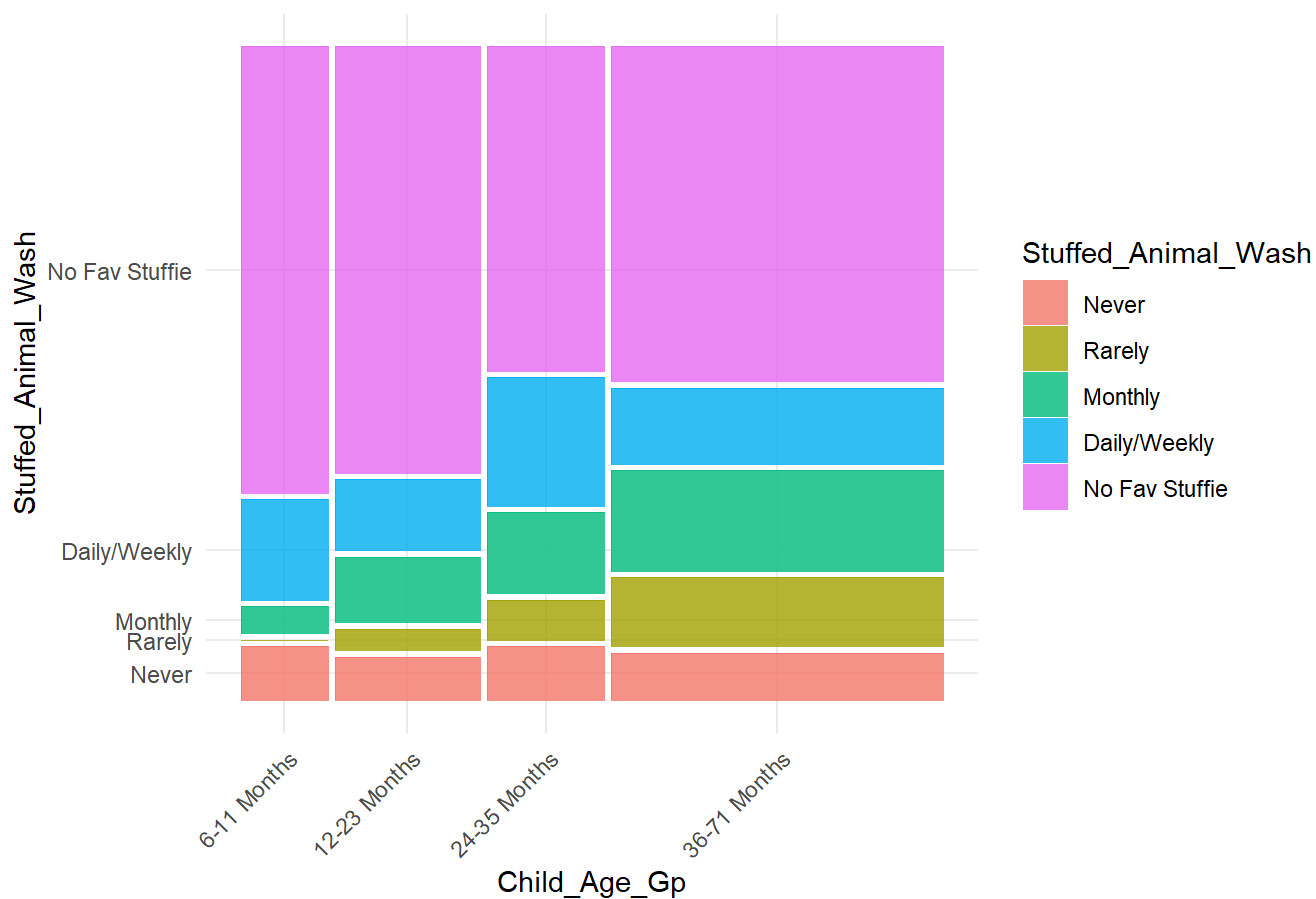


```
## Saving 7 x 5 in image
```



```
##
##
## ----- Analysis for Stuffed_Animal_Wash and Child_Age_Gp -----
## Bivariate Analysis of Stuffed_Animal_Wash and Child_Age_Gp
##
## Frequency Table:
##   Stuffed_Animal_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##           Never           6           8           8           20
##           Rarely          0           4           6           29
##           Monthly         3          12          12          42
##           Daily/Weekly    11          13          19          32
##           No Fav Stuffie  49          78          48          140
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##           Never          8.7%          7.0%          8.6%          7.6%  7.8%
##           Rarely          0.0%          3.5%          6.5%          11.0%  7.2%
##           Monthly         4.3%          10.4%         12.9%          16.0% 12.8%
##           Daily/Weekly    15.9%          11.3%         20.4%          12.2% 13.9%
##           No Fav Stuffie  71.0%          67.8%         51.6%          53.2% 58.3%
##
## Cramér's V:  0.1027146
```

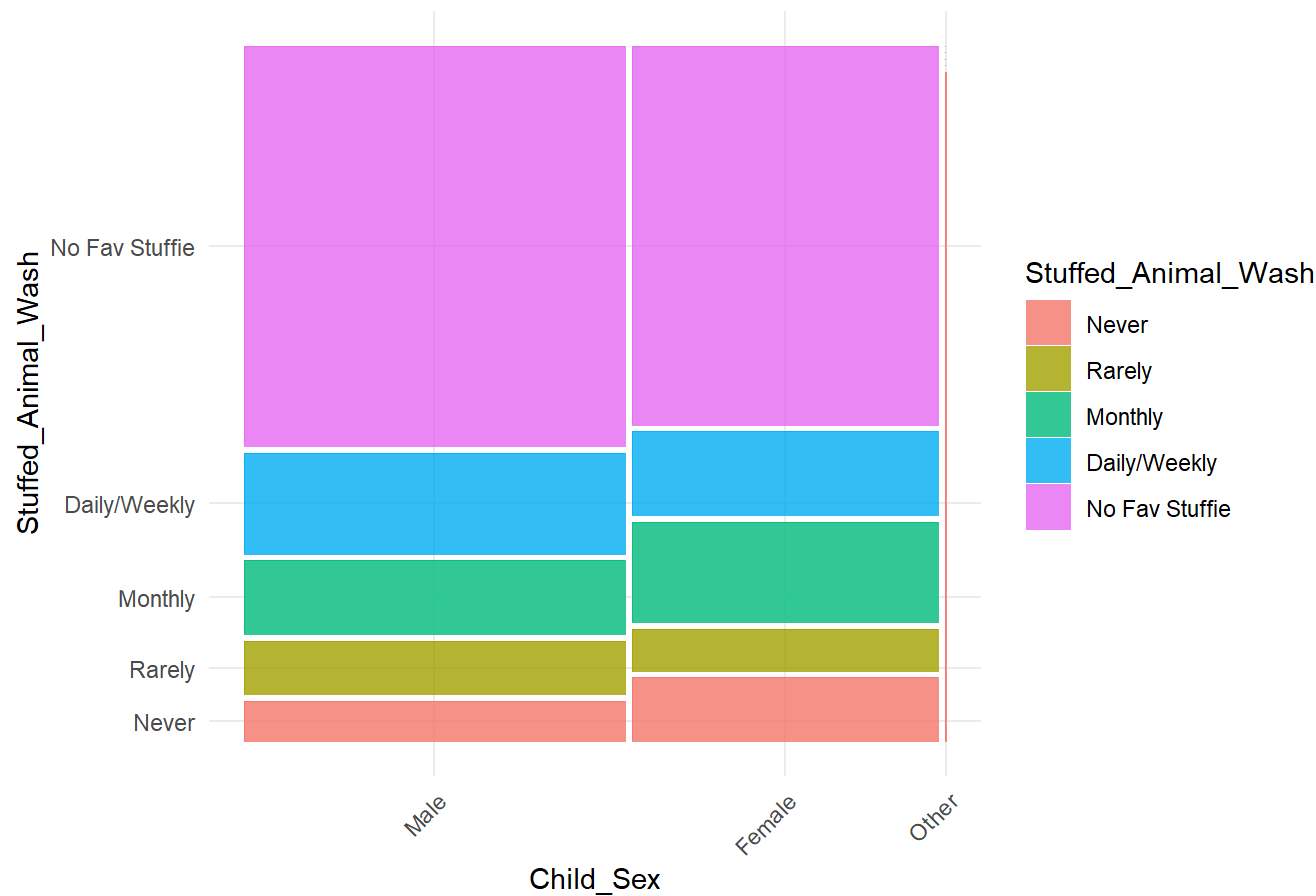
Mosaic Plot of Child_Age_Gp and Stuffed_Animal_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and Child_Sex -----
## Bivariate Analysis of Stuffed_Animal_Wash and Child_Sex
##
## Frequency Table:
##   Stuffed_Animal_Wash Male Female Other
##           Never      18      23      1
##           Rarely     24      15      0
##           Monthly    33      36      0
##           Daily/Weekly 45      30      0
##           No Fav Stuffie 179    136      0
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash Male Female Other Total
##           Never     6.0%    9.6% 100.0%  7.8%
##           Rarely     8.0%    6.2%  0.0%  7.2%
##           Monthly    11.0%   15.0%  0.0% 12.8%
##           Daily/Weekly 15.1%   12.5%  0.0% 13.9%
##           No Fav Stuffie 59.9%   56.7%  0.0% 58.3%
##
## Cramér's V:  0.09203038
```

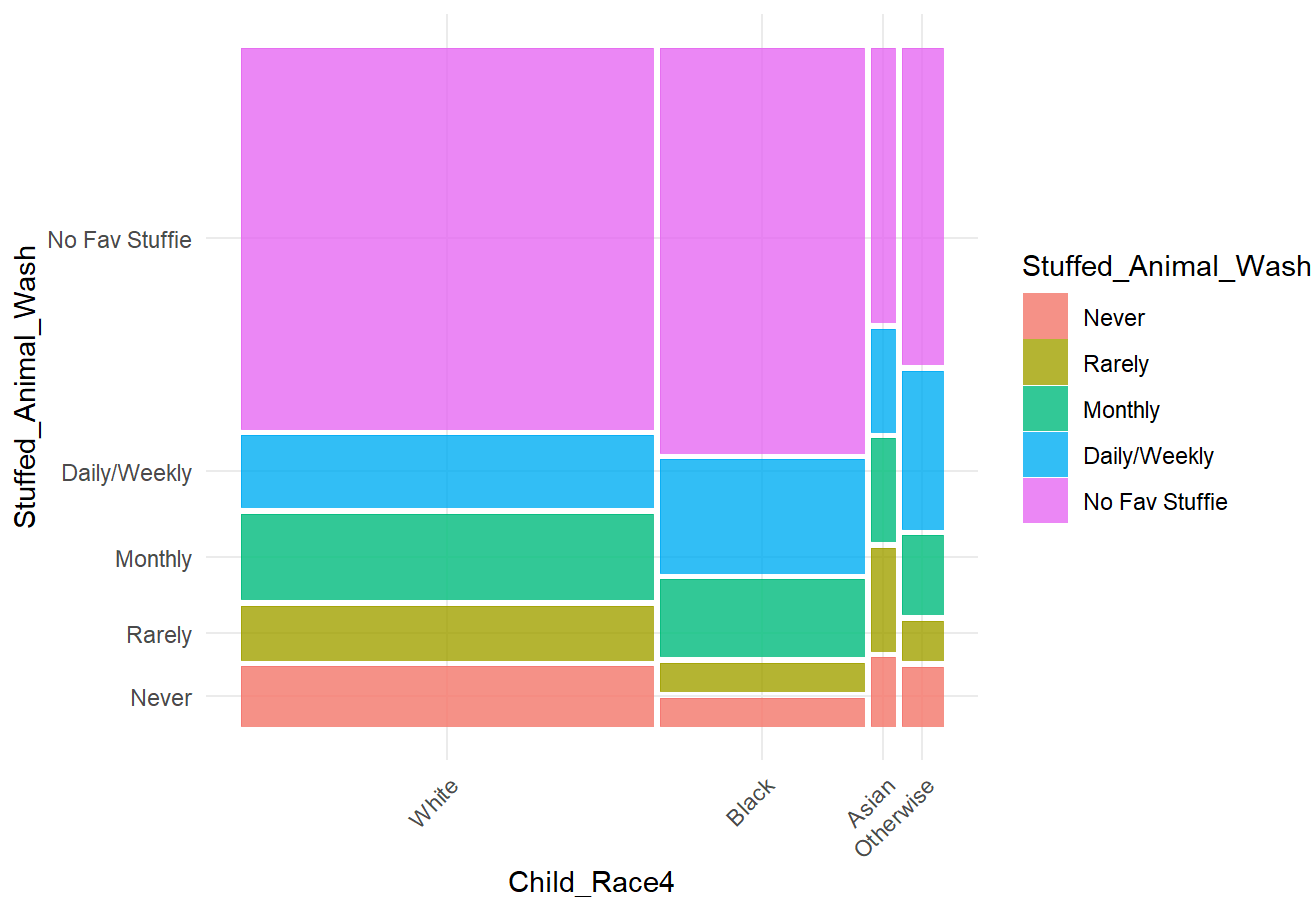
Mosaic Plot of Child_Sex and Stuffed_Animal_Wash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Stuffed_Animal_Wash and Child_Race4 -----
## Bivariate Analysis of Stuffed_Animal_Wash and Child_Race4
##
## Frequency Table:
##   Stuffed_Animal_Wash White Black Asian Otherwise
##           Never      30      7      2         3
##           Rarely     27      7      3         2
##           Monthly    43     19      3         4
##           Daily/Weekly 36     28      3         8
##           No Fav Stuffie 191    100      8        16
##
## Relative Frequency Table:
##   Stuffed_Animal_Wash White Black Asian Otherwise Total
##           Never    9.2%  4.3% 10.5%    9.1%  7.8%
##           Rarely    8.3%  4.3% 15.8%    6.1%  7.2%
##           Monthly   13.1% 11.8% 15.8%   12.1% 12.8%
##           Daily/Weekly 11.0% 17.4% 15.8%   24.2% 13.9%
##           No Fav Stuffie 58.4% 62.1% 42.1%   48.5% 58.3%
##
## Cramér's V:  0.0490952
```

Mosaic Plot of Child_Race4 and Stuffed_Animal_Wash



```
## Saving 7 x 5 in image
```

```
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```

##	var1	var2	cramer_v
## 1	Stuffed_Animal	Child_Age_Gp	0.13868833
## 2	Stuffed_Animal_Wash	Child_Age_Gp	0.10271457
## 3	Stuffed_Animal_Wash	Child_Sex	0.09203038
## 4	Stuffed_Animal_Wash	Household_income	0.08165921
## 5	Stuffed_Animal	City	0.08117186
## 6	Stuffed_Animal_Wash	P1_Sex	0.06654714
## 7	Stuffed_Animal_Wash	City	0.06222732
## 8	Stuffed_Animal_Wash	P1_Degree	0.06147419
## 9	Stuffed_Animal_Wash	P1_Age	0.06029658
## 10	Stuffed_Animal	P1_Race4	0.05213319
## 11	Stuffed_Animal	P1_Sex	0.05038784
## 12	Stuffed_Animal	Child_Race4	0.04942060
## 13	Stuffed_Animal_Wash	Child_Race4	0.04909520
## 14	Stuffed_Animal_Wash	P1_Race4	0.03979008
## 15	Stuffed_Animal	P1_Degree	0.03792762
## 16	Stuffed_Animal_Wash	Single_Parent	0.03626165
## 17	Stuffed_Animal_Wash	P1_Ethnicity	0.02284737
## 18	Stuffed_Animal	Household_income	0.02143709
## 19	Stuffed_Animal	P1_Age	0.00000000
## 20	Stuffed_Animal	P1_Ethnicity	0.00000000
## 21	Stuffed_Animal	P1_Workstatus	0.00000000
## 22	Stuffed_Animal	Single_Parent	0.00000000
## 23	Stuffed_Animal	Child_Sex	0.00000000
## 24	Stuffed_Animal_Wash	P1_Workstatus	0.00000000

Exploring Sucking & Handwashing Variables

```

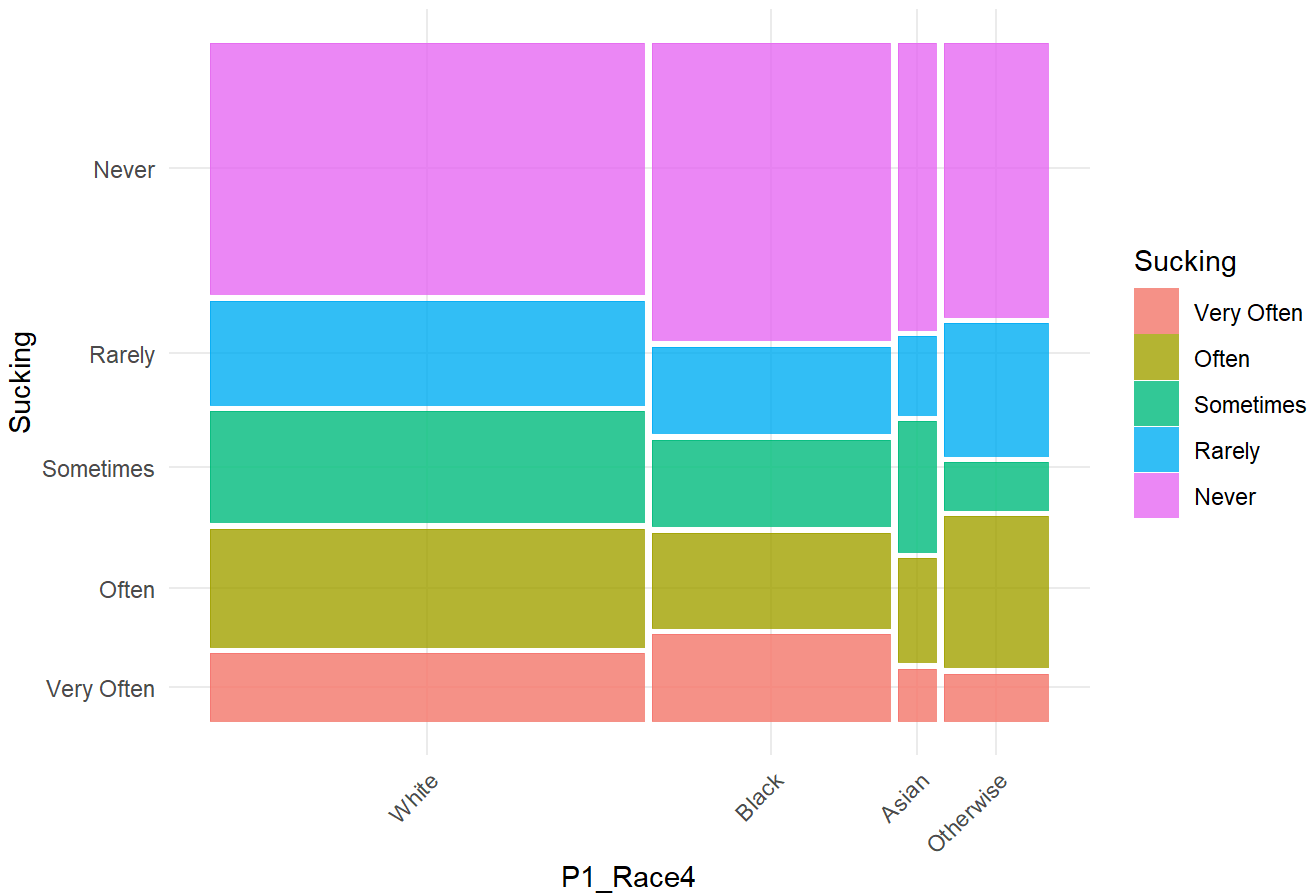
data= data
HabitsVars <- c( "Sucking", "Handwash")
list1= HabitsVars
list2= DemoVars

cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(), stringsAsFactors = FALSE)
results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = "_vs_")]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values,
                              data.frame(var1=var1,var2 = var2, cramer_v = result$cramer_v))
    }
  }
}

```

```
##
##
## ----- Analysis for Sucking and P1_Race4 -----
## Bivariate Analysis of Sucking and P1_Race4
##
## Frequency Table:
##      Sucking White Black Asian Otherwise
## Very Often    30    21     2         5
##      Often    52    23     4        16
## Sometimes    49    21     5         5
##      Rarely   46    21     3        14
##      Never   111    72    11        29
##
## Relative Frequency Table:
##      Sucking White Black Asian Otherwise Total
## Very Often 10.4% 13.3%  8.0%      7.2% 10.7%
##      Often 18.1% 14.6% 16.0%     23.2% 17.6%
## Sometimes 17.0% 13.3% 20.0%      7.2% 14.8%
##      Rarely 16.0% 13.3% 12.0%     20.3% 15.6%
##      Never 38.5% 45.6% 44.0%     42.0% 41.3%
##
## Cramér's V:  0
```

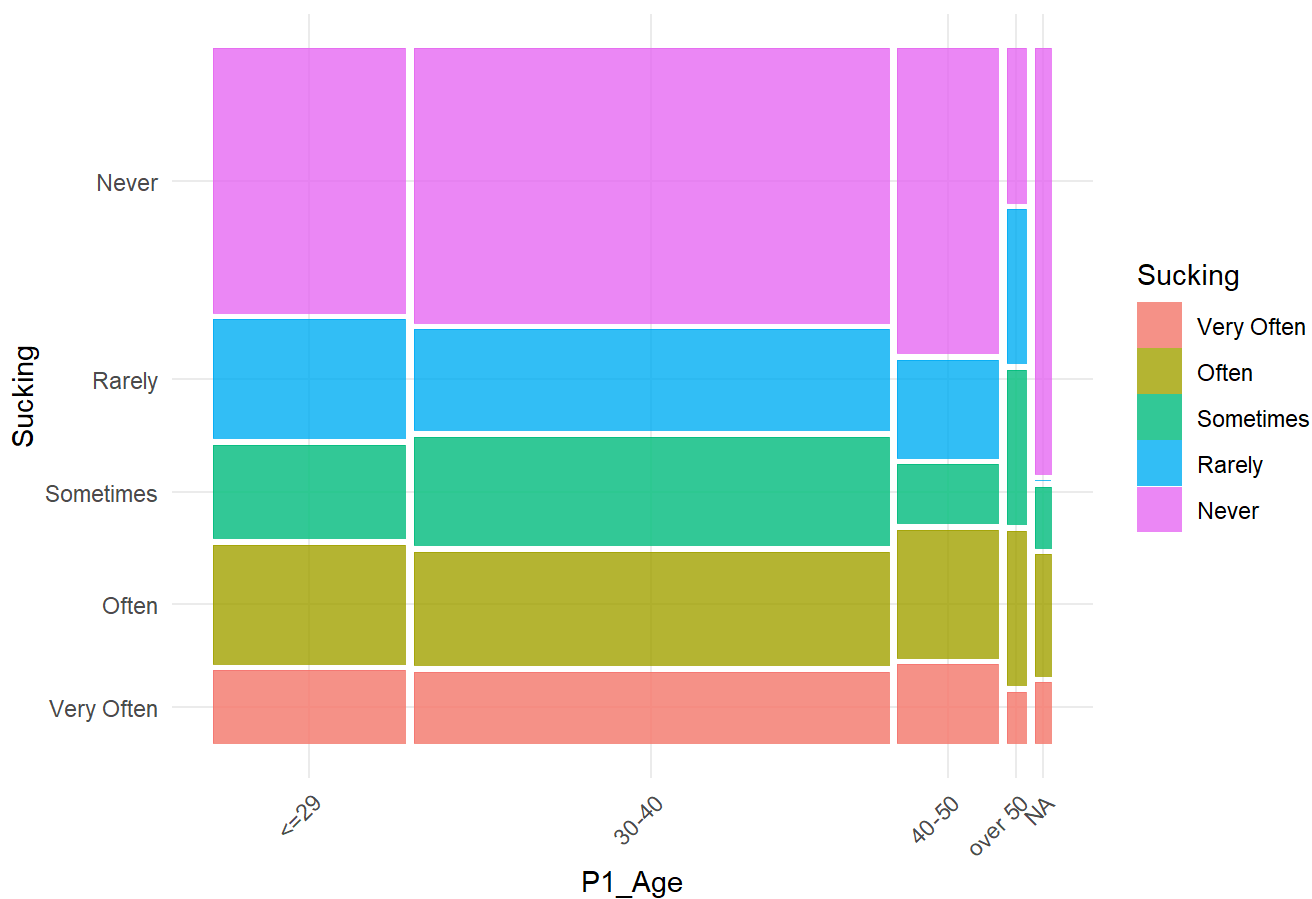
Mosaic Plot of P1_Race4 and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and P1_Age -----
## Bivariate Analysis of Sucking and P1_Age
##
## Frequency Table:
##      Sucking <=29 30-40 40-50 over 50 NA_
## Very Often    14   34    8     1   1
##      Often     23   54   13     3   2
## Sometimes     18   52    6     3   1
##      Rarely    23   48   10     3   0
##      Never     51  131   31     3   7
##
## Relative Frequency Table:
##      Sucking <=29 30-40 40-50 over 50  NA_ Total
## Very Often 10.9% 10.7% 11.8%    7.7%  9.1% 10.7%
##      Often 17.8% 16.9% 19.1%   23.1% 18.2% 17.6%
## Sometimes 14.0% 16.3%  8.8%   23.1%  9.1% 14.8%
##      Rarely 17.8% 15.0% 14.7%   23.1%  0.0% 15.6%
##      Never 39.5% 41.1% 45.6%   23.1% 63.6% 41.3%
##
## Cramér's V:  0
```

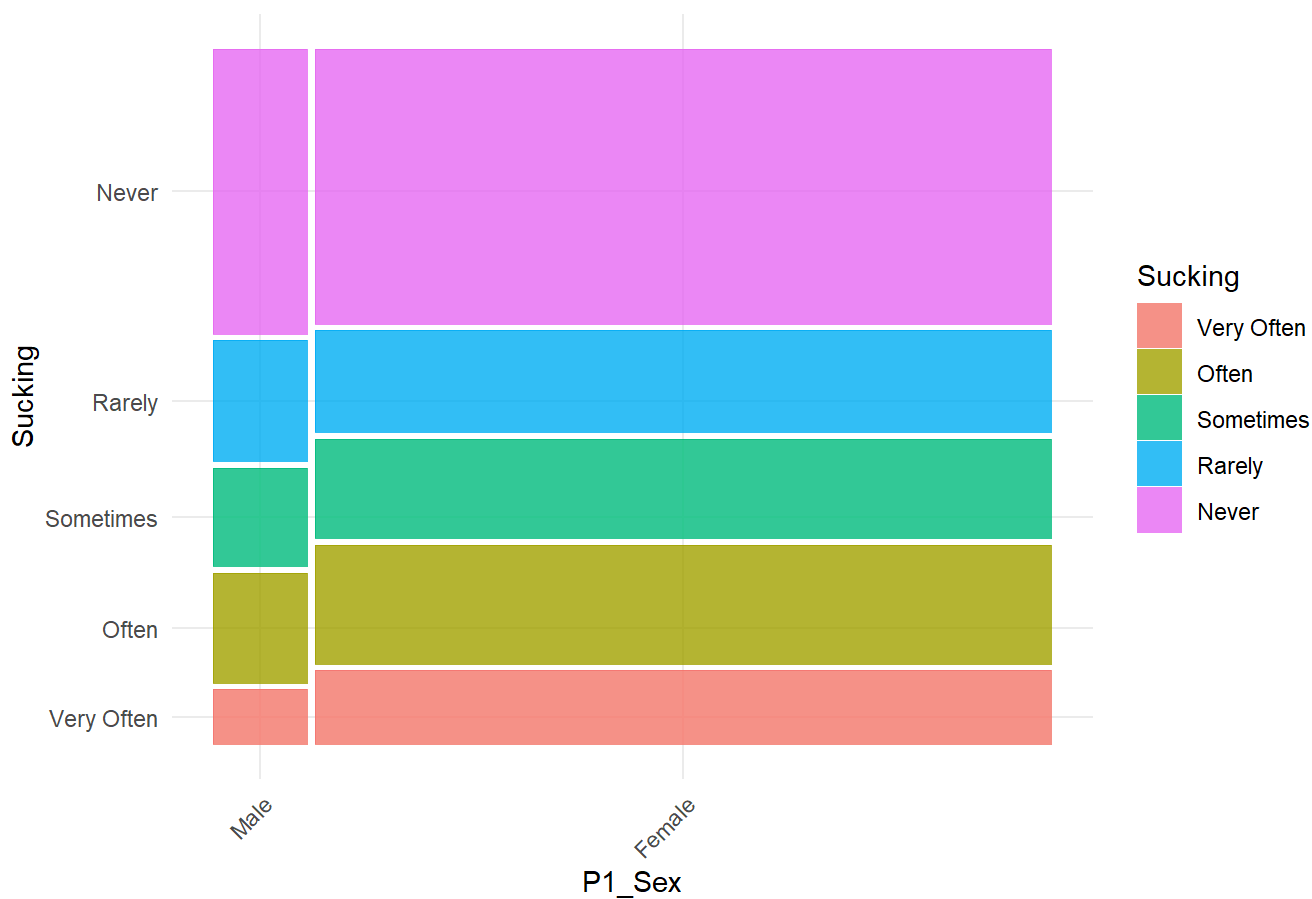
Mosaic Plot of P1_Age and Sucking



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sucking and P1_Sex -----
## Bivariate Analysis of Sucking and P1_Sex
##
## Frequency Table:
##      Sucking  Male Female
## Very Often     5     53
##      Often    10     85
## Sometimes     9     71
##      Rarely   11     73
##      Never    26    197
##
## Relative Frequency Table:
##      Sucking  Male Female Total
## Very Often   8.2%  11.1% 10.7%
##      Often   16.4%  17.7% 17.6%
## Sometimes   14.8%  14.8% 14.8%
##      Rarely  18.0%  15.2% 15.6%
##      Never   42.6%  41.1% 41.3%
##
## Cramér's V:  0
```

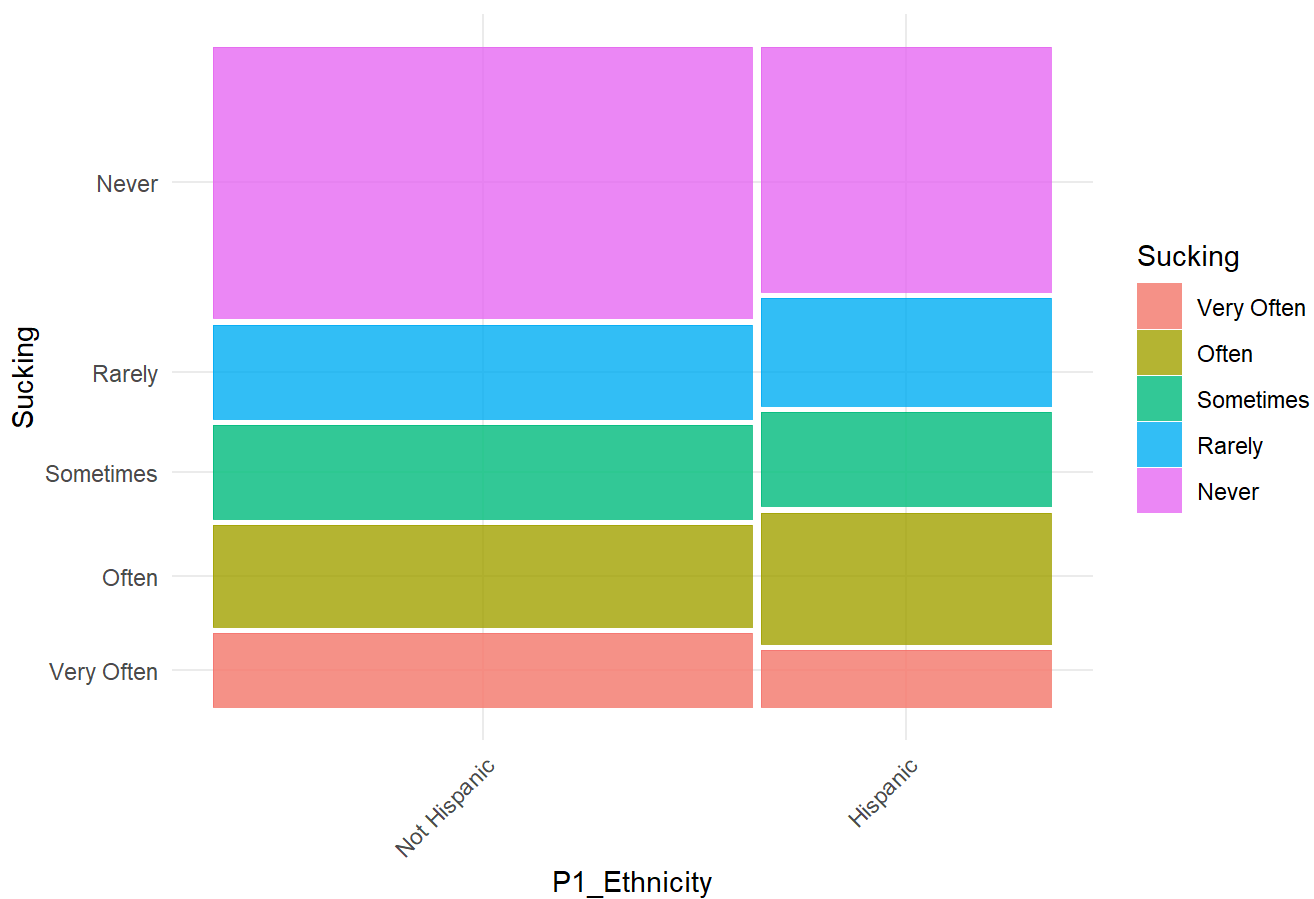
Mosaic Plot of P1_Sex and Sucking



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sucking and P1_Ethnicity -----
## Bivariate Analysis of Sucking and P1_Ethnicity
##
## Frequency Table:
##      Sucking Not Hispanic Hispanic
## Very Often      41      17
##      Often      56      39
##      Sometimes  52      28
##      Rarely     52      32
##      Never     150      73
##
## Relative Frequency Table:
##      Sucking Not Hispanic Hispanic Total
## Very Often    11.7%    9.0% 10.7%
##      Often    16.0%    20.6% 17.6%
##      Sometimes 14.8%    14.8% 14.8%
##      Rarely   14.8%    16.9% 15.6%
##      Never    42.7%    38.6% 41.3%
##
## Cramér's V:  0
```

Mosaic Plot of P1_Ethnicity and Sucking

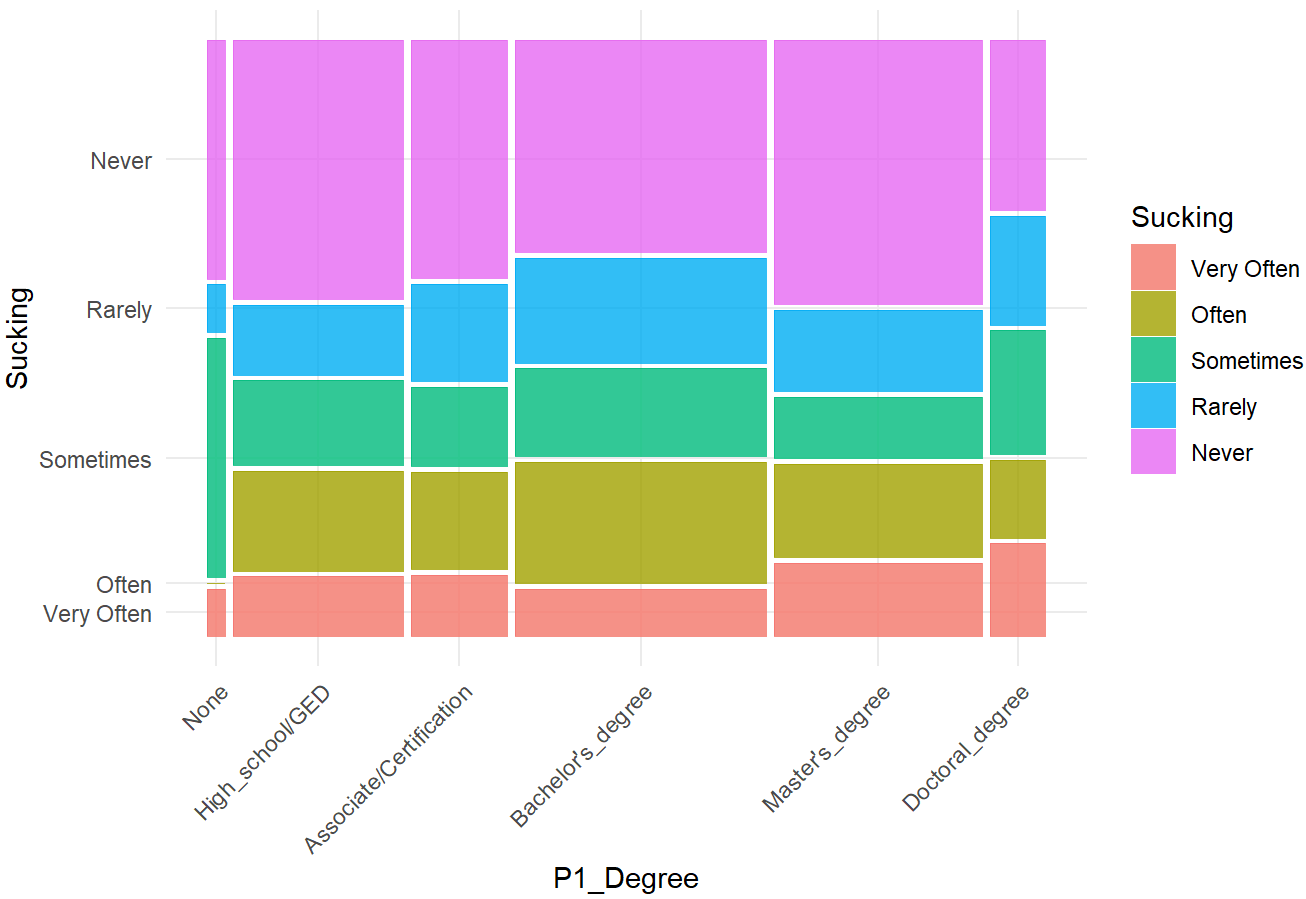


```
## Saving 7 x 5 in image
```



```
##
##
## ----- Analysis for Sucking and P1_Degree -----
## Bivariate Analysis of Sucking and P1_Degree
##
## Frequency Table:
##      Sucking None High_school/GED Associate/Certification Bachelor's_degree
## Very Often    1         12             7             14
##      Often    0         20            11             36
## Sometimes    5         17             9             26
##      Rarely   1         14            11             31
##      Never    5         52            27             63
## Master's_degree Doctoral_degree
##          18          6
##          23          5
##          15          8
##          20          7
##          65         11
##
## Relative Frequency Table:
##      Sucking None High_school/GED Associate/Certification Bachelor's_degree
## Very Often  8.3%        10.4%        10.8%          8.2%
##      Often  0.0%        17.4%        16.9%         21.2%
## Sometimes 41.7%        14.8%        13.8%         15.3%
##      Rarely  8.3%        12.2%        16.9%         18.2%
##      Never 41.7%        45.2%        41.5%         37.1%
## Master's_degree Doctoral_degree Total
##          12.8%        16.2% 10.7%
##          16.3%        13.5% 17.6%
##          10.6%        21.6% 14.8%
##          14.2%        18.9% 15.6%
##          46.1%        29.7% 41.3%
##
## Cramér's V:  0.02032518
```

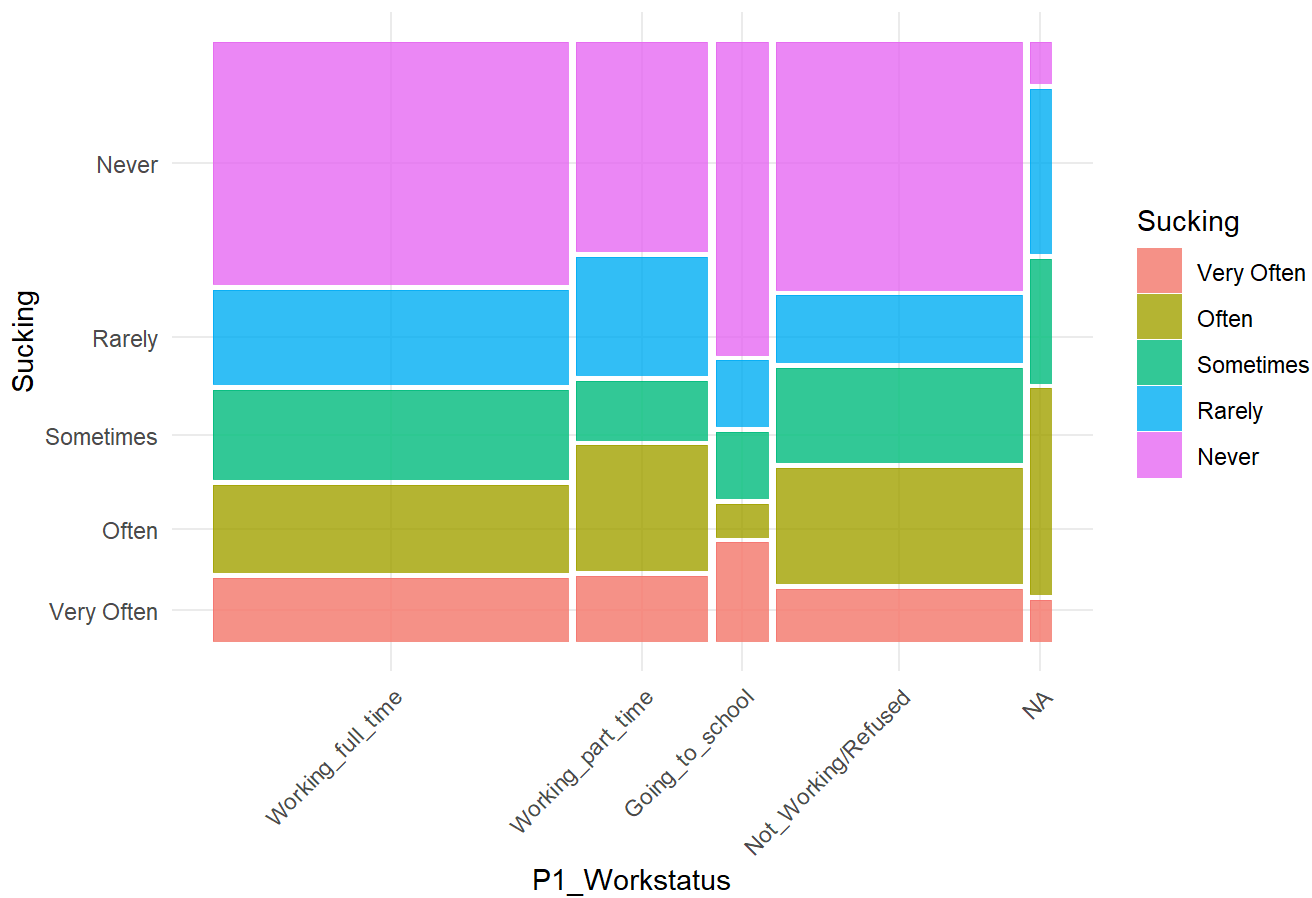
Mosaic Plot of P1_Degree and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and P1_Workstatus -----
## Bivariate Analysis of Sucking and P1_Workstatus
##
## Frequency Table:
##      Sucking Working_full_time Working_part_time Going_to_school
## Very Often           26           10           6
##      Often           36           19           2
##      Sometimes       37           9           4
##      Rarely          39           18           4
##      Never          100           32          19
## Not_Working/Refused NA_
##              15   1
##              33   5
##              27   3
##              19   4
##              71   1
##
## Relative Frequency Table:
##      Sucking Working_full_time Working_part_time Going_to_school
## Very Often          10.9%          11.4%          17.1%
##      Often          15.1%          21.6%           5.7%
##      Sometimes       15.5%          10.2%          11.4%
##      Rarely          16.4%          20.5%          11.4%
##      Never          42.0%          36.4%          54.3%
## Not_Working/Refused  NA_ Total
##              9.1%  7.1% 10.7%
##              20.0% 35.7% 17.6%
##              16.4% 21.4% 14.8%
##              11.5% 28.6% 15.6%
##              43.0%  7.1% 41.3%
##
## Cramér's V:  0.05698498
```

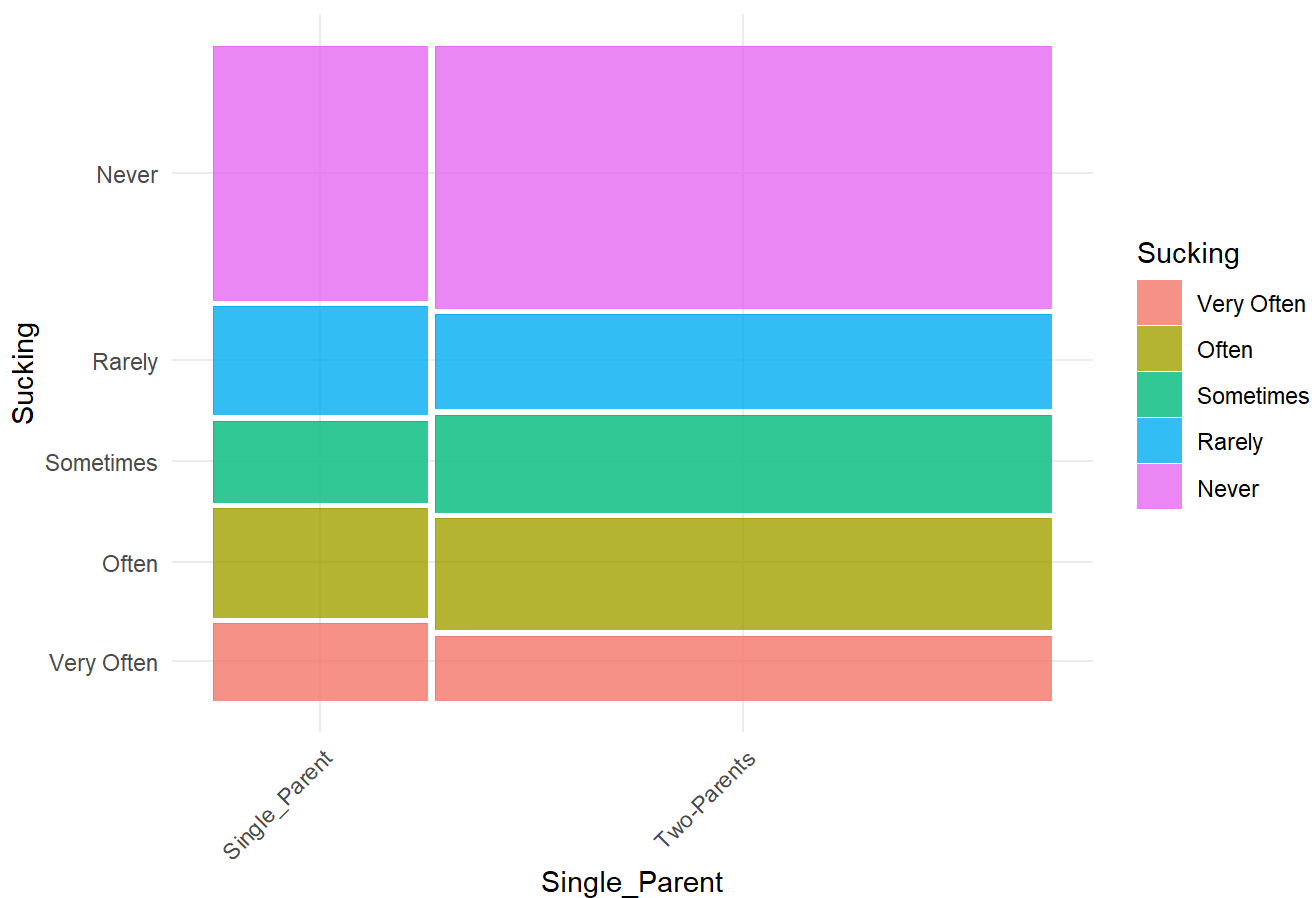
Mosaic Plot of P1_Workstatus and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and Single_Parent -----
## Bivariate Analysis of Sucking and Single_Parent
##
## Frequency Table:
##      Sucking Single_Parent Two-Parents
## Very Often      17         41
##      Often      24         71
## Sometimes      18         62
##      Rarely     24         60
##      Never      56        167
##
## Relative Frequency Table:
##      Sucking Single_Parent Two-Parents Total
## Very Often     12.2%      10.2% 10.7%
##      Often     17.3%      17.7% 17.6%
## Sometimes     12.9%      15.5% 14.8%
##      Rarely     17.3%      15.0% 15.6%
##      Never     40.3%      41.6% 41.3%
##
## Cramér's V:  0
```

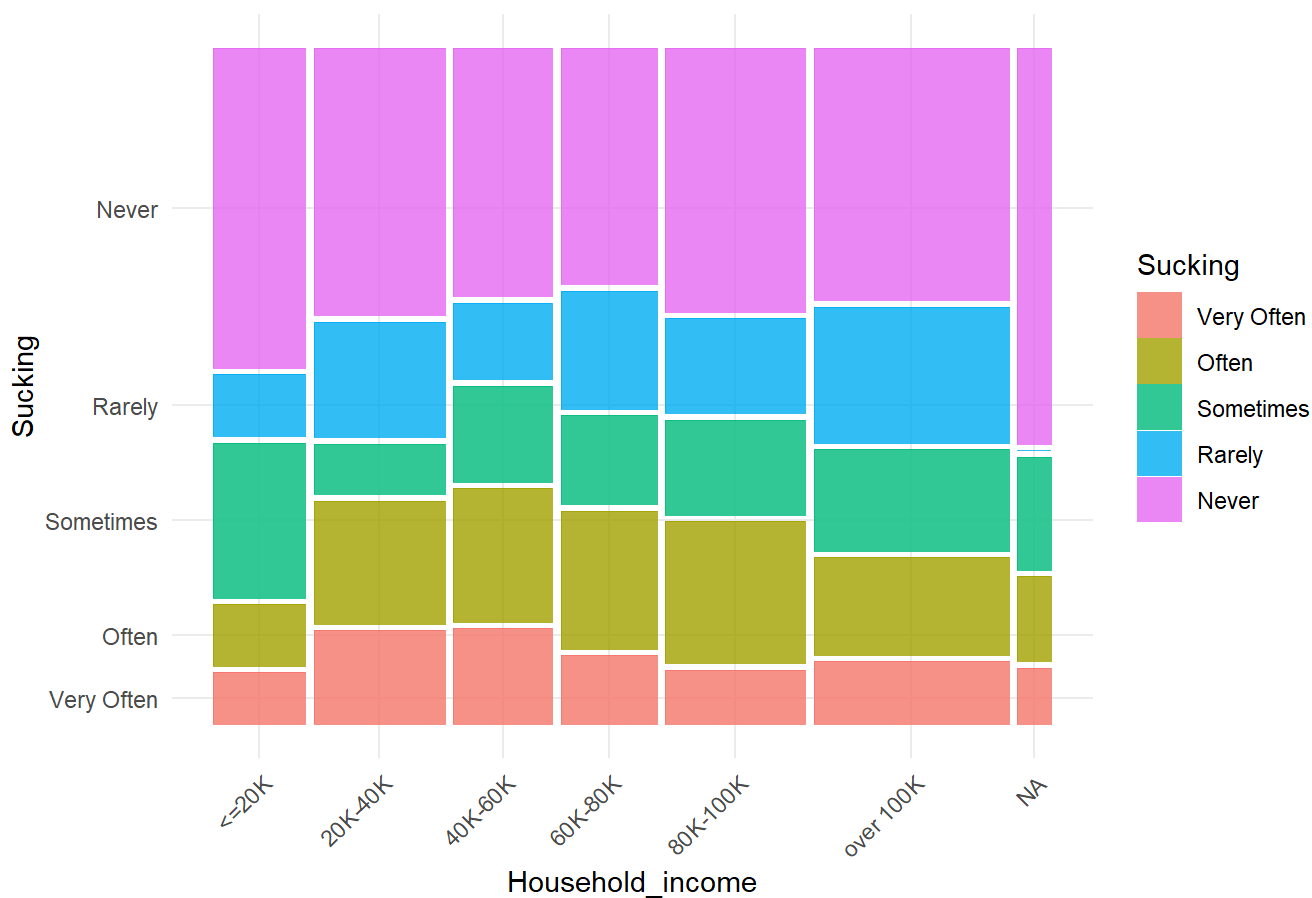
Mosaic Plot of Single_Parent and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and Household_income -----
## Bivariate Analysis of Sucking and Household_income
##
## Frequency Table:
##      Sucking <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
## Very Often      5      13      10       7       8       13   2
##      Often       6      17      14      14      21      20   3
## Sometimes      15       7      10       9      14      21   4
##      Rarely       6      16       8      12      14      28   0
##      Never      31      37      26      24      39      52  14
##
## Relative Frequency Table:
##      Sucking <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_ Total
## Very Often  7.9%  14.4%  14.7%  10.6%   8.3%   9.7%  8.7% 10.7%
##      Often   9.5%  18.9%  20.6%  21.2%  21.9%  14.9% 13.0% 17.6%
## Sometimes  23.8%   7.8%  14.7%  13.6%  14.6%  15.7% 17.4% 14.8%
##      Rarely   9.5%  17.8%  11.8%  18.2%  14.6%  20.9%  0.0% 15.6%
##      Never  49.2%  41.1%  38.2%  36.4%  40.6%  38.8% 60.9% 41.3%
##
## Cramér's V:  0.04139869
```

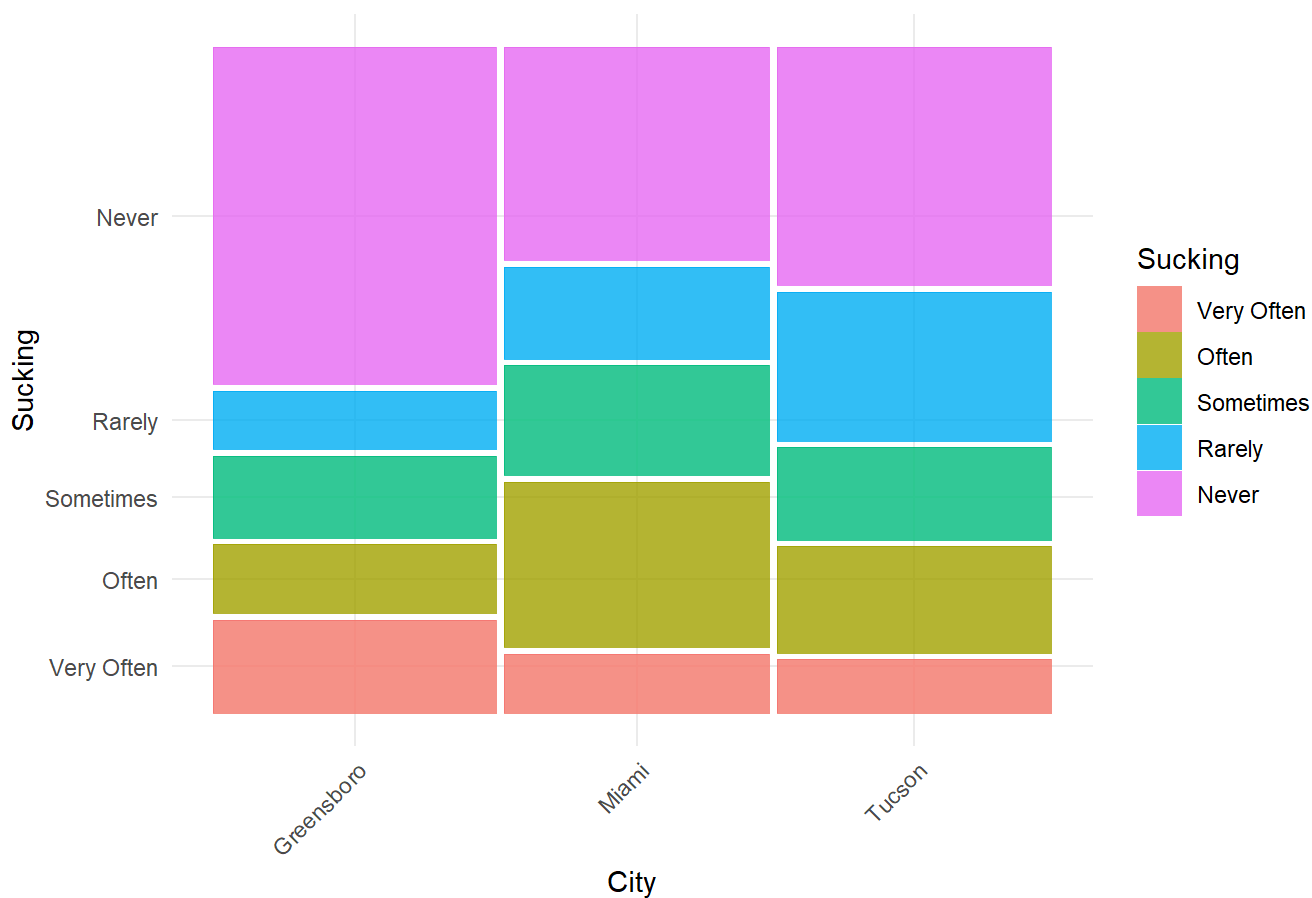
Mosaic Plot of Household_income and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and City -----
## Bivariate Analysis of Sucking and City
##
## Frequency Table:
##      Sucking Greensboro Miami Tucson
## Very Often      27      16      15
##      Often       20      45      30
## Sometimes       24      30      26
##      Rarely      17      25      42
##      Never       98      58      67
##
## Relative Frequency Table:
##      Sucking Greensboro Miami Tucson Total
## Very Often    14.5%  9.2%   8.3% 10.7%
##      Often     10.8% 25.9% 16.7% 17.6%
## Sometimes     12.9% 17.2% 14.4% 14.8%
##      Rarely     9.1% 14.4% 23.3% 15.6%
##      Never     52.7% 33.3% 37.2% 41.3%
##
## Cramér's V:  0.167244
```

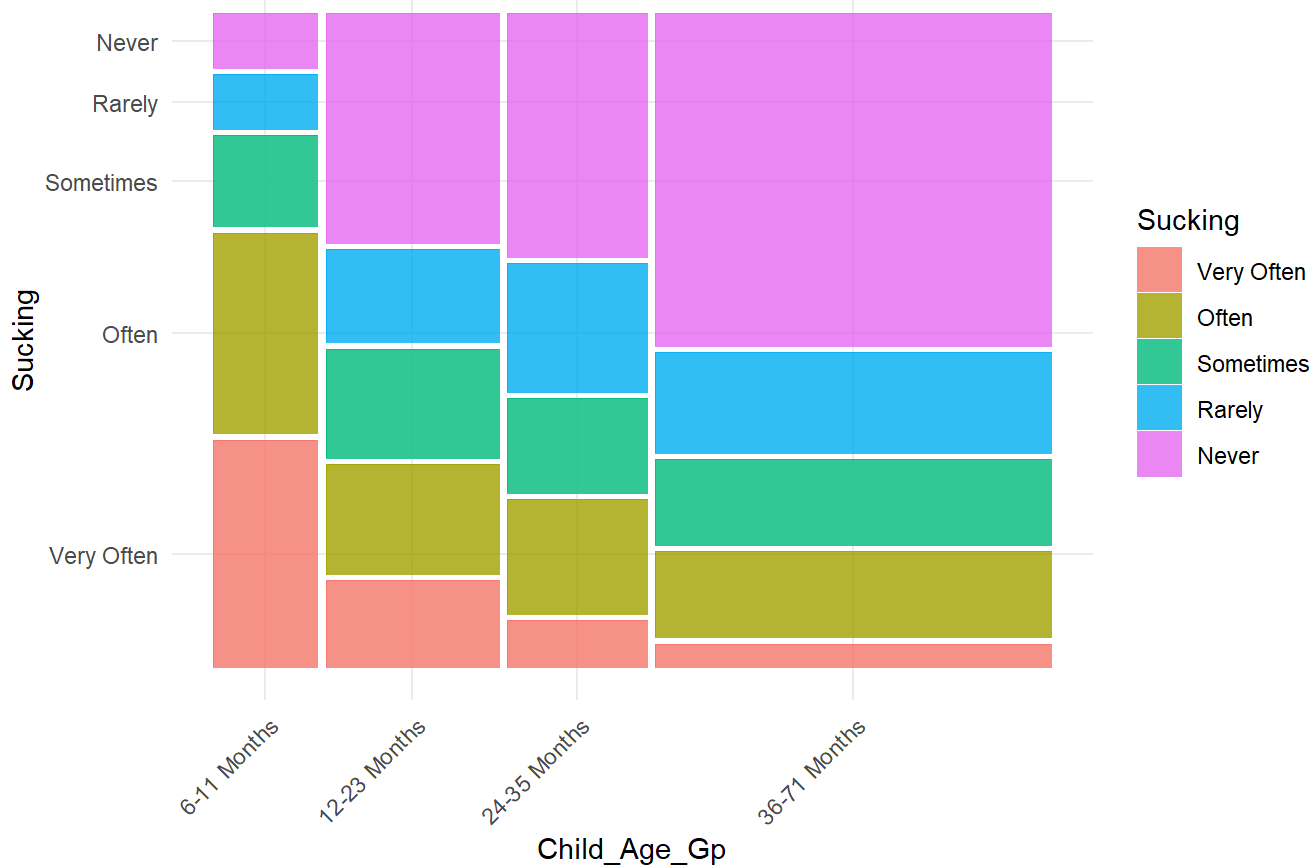
Mosaic Plot of City and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and Child_Age_Gp -----
## Bivariate Analysis of Sucking and Child_Age_Gp
##
## Frequency Table:
##      Sucking 6-11 Months 12-23 Months 24-35 Months 36-71 Months
## Very Often      25         16          7          10
##      Often      22         20         17          36
## Sometimes      10         20         14          36
##      Rarely      6         17         19          42
##      Never       6         42         36         139
##
## Relative Frequency Table:
##      Sucking 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
## Very Often      36.2%      13.9%       7.5%       3.8% 10.7%
##      Often      31.9%      17.4%      18.3%      13.7% 17.6%
## Sometimes      14.5%      17.4%      15.1%      13.7% 14.8%
##      Rarely      8.7%      14.8%      20.4%      16.0% 15.6%
##      Never       8.7%      36.5%      38.7%      52.9% 41.3%
##
## Cramér's V:  0.2299397
```

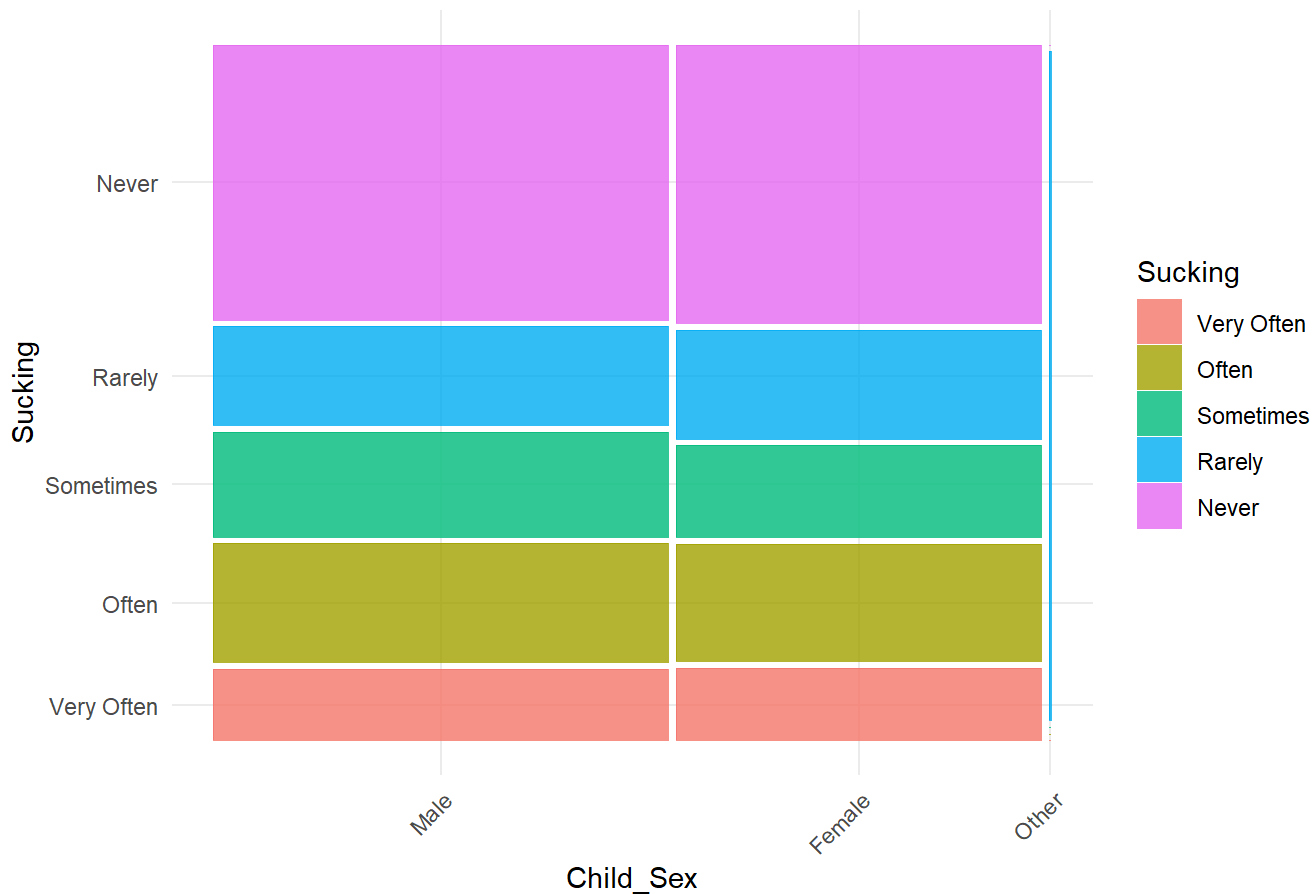
Mosaic Plot of Child_Age_Gp and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and Child_Sex -----
## Bivariate Analysis of Sucking and Child_Sex
##
## Frequency Table:
##      Sucking Male Female Other
## Very Often   32    26    0
##      Often   53    42    0
## Sometimes   47    33    0
##      Rarely  44    39    1
##      Never  123   100    0
##
## Relative Frequency Table:
##      Sucking Male Female Other Total
## Very Often 10.7% 10.8%  0.0% 10.7%
##      Often 17.7% 17.5%  0.0% 17.6%
## Sometimes 15.7% 13.8%  0.0% 14.8%
##      Rarely 14.7% 16.2% 100.0% 15.6%
##      Never 41.1% 41.7%  0.0% 41.3%
##
## Cramér's V:  0
```

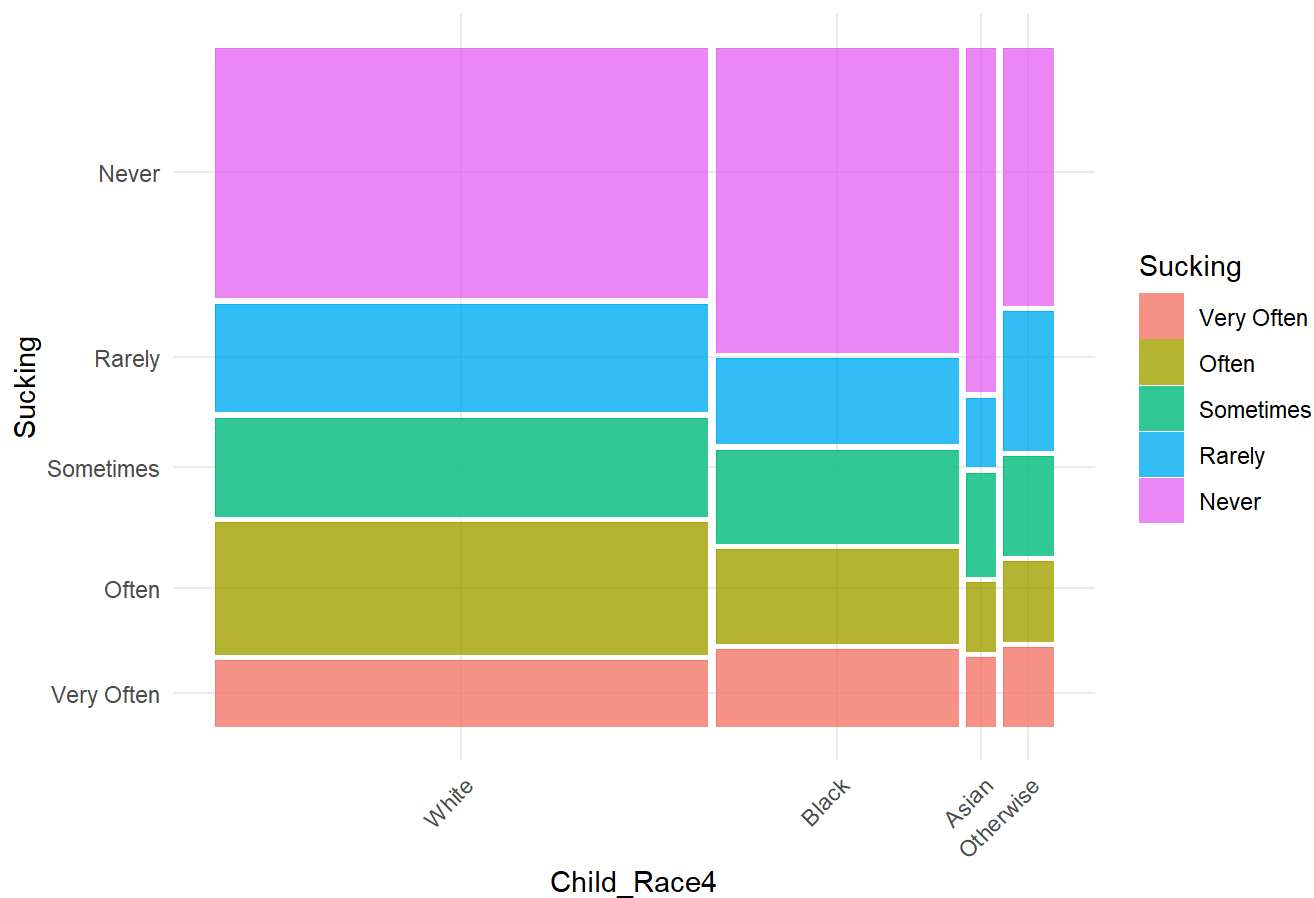

Mosaic Plot of Child_Sex and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sucking and Child_Race4 -----
## Bivariate Analysis of Sucking and Child_Race4
##
## Frequency Table:
##      Sucking White Black Asian Otherwise
## Very Often    33    19     2         4
##      Often    66    23     2         4
## Sometimes    49    23     3         5
##      Rarely   54    21     2         7
##      Never   125    75    10        13
##
## Relative Frequency Table:
##      Sucking White Black Asian Otherwise Total
## Very Often 10.1% 11.8% 10.5%    12.1% 10.7%
##      Often 20.2% 14.3% 10.5%    12.1% 17.6%
## Sometimes 15.0% 14.3% 15.8%    15.2% 14.8%
##      Rarely 16.5% 13.0% 10.5%    21.2% 15.6%
##      Never 38.2% 46.6% 52.6%    39.4% 41.3%
##
## Cramér's V:  0
```

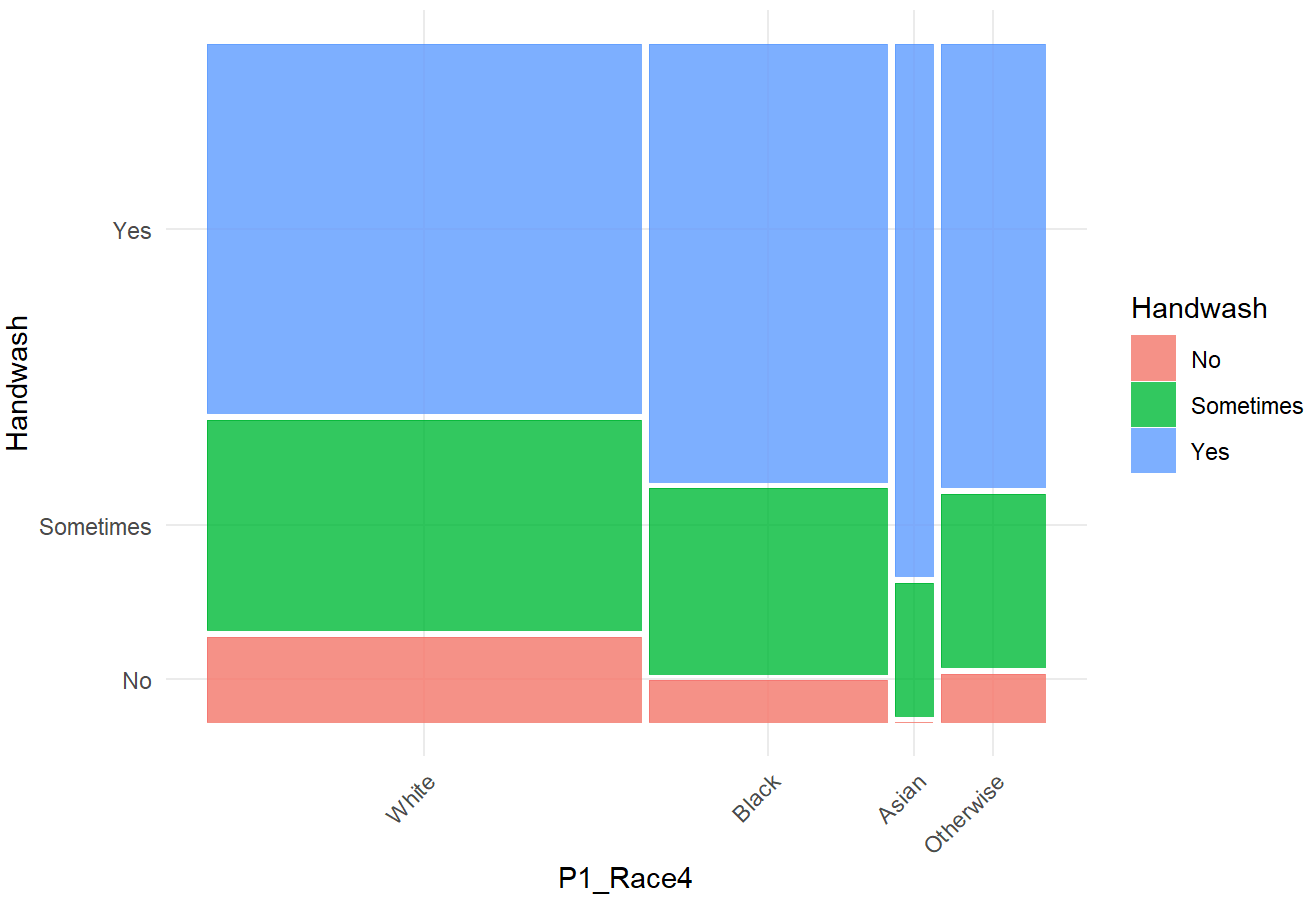
Mosaic Plot of Child_Race4 and Sucking



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and P1_Race4 -----
## Bivariate Analysis of Handwash and P1_Race4
##
## Frequency Table:
##   Handwash White Black Asian Otherwise
##       No    37   10    0      5
##   Sometimes  91   44    5     18
##       Yes  160  104   20     46
##
## Relative Frequency Table:
##   Handwash White Black Asian Otherwise Total
##       No 12.8%  6.3%  0.0%    7.2%  9.6%
##   Sometimes 31.6% 27.8% 20.0%   26.1% 29.3%
##       Yes 55.6% 65.8% 80.0%   66.7% 61.1%
##
## Cramér's V:  0.08145399
```

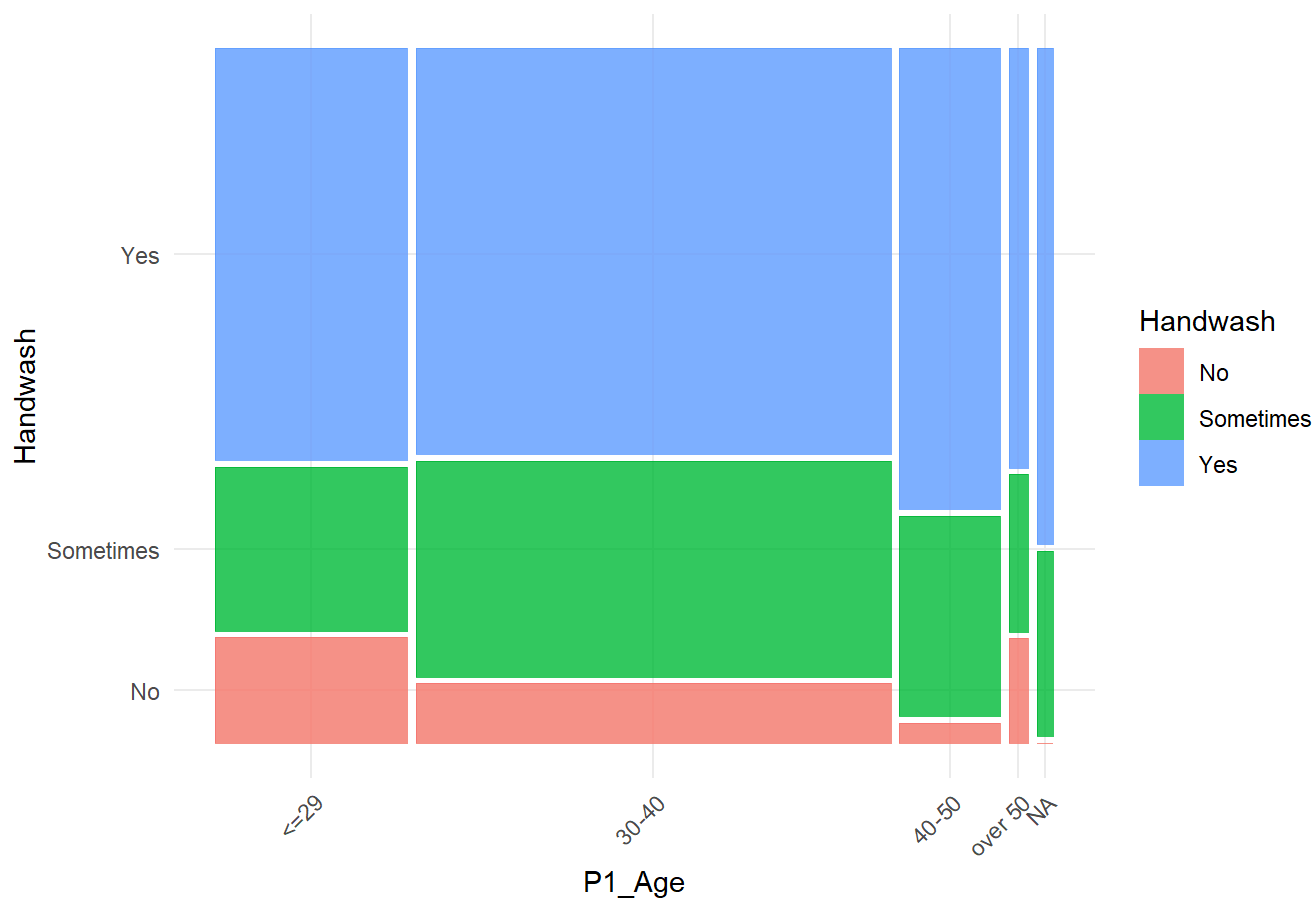
Mosaic Plot of P1_Race4 and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and P1_Age -----
## Bivariate Analysis of Handwash and P1_Age
##
## Frequency Table:
##   Handwash <=29 30-40 40-50 over 50 NA_
##           No    20    28     2      2  0
##   Sometimes  31   101    20     3  3
##           Yes   78   190    46     8  8
##
## Relative Frequency Table:
##   Handwash <=29 30-40 40-50 over 50  NA_ Total
##           No 15.5%  8.8%  2.9%  15.4%  0.0%  9.6%
##   Sometimes 24.0% 31.7% 29.4%  23.1% 27.3% 29.3%
##           Yes 60.5% 59.6% 67.6%  61.5% 72.7% 61.1%
##
## Cramér's V:  0.06384544
```

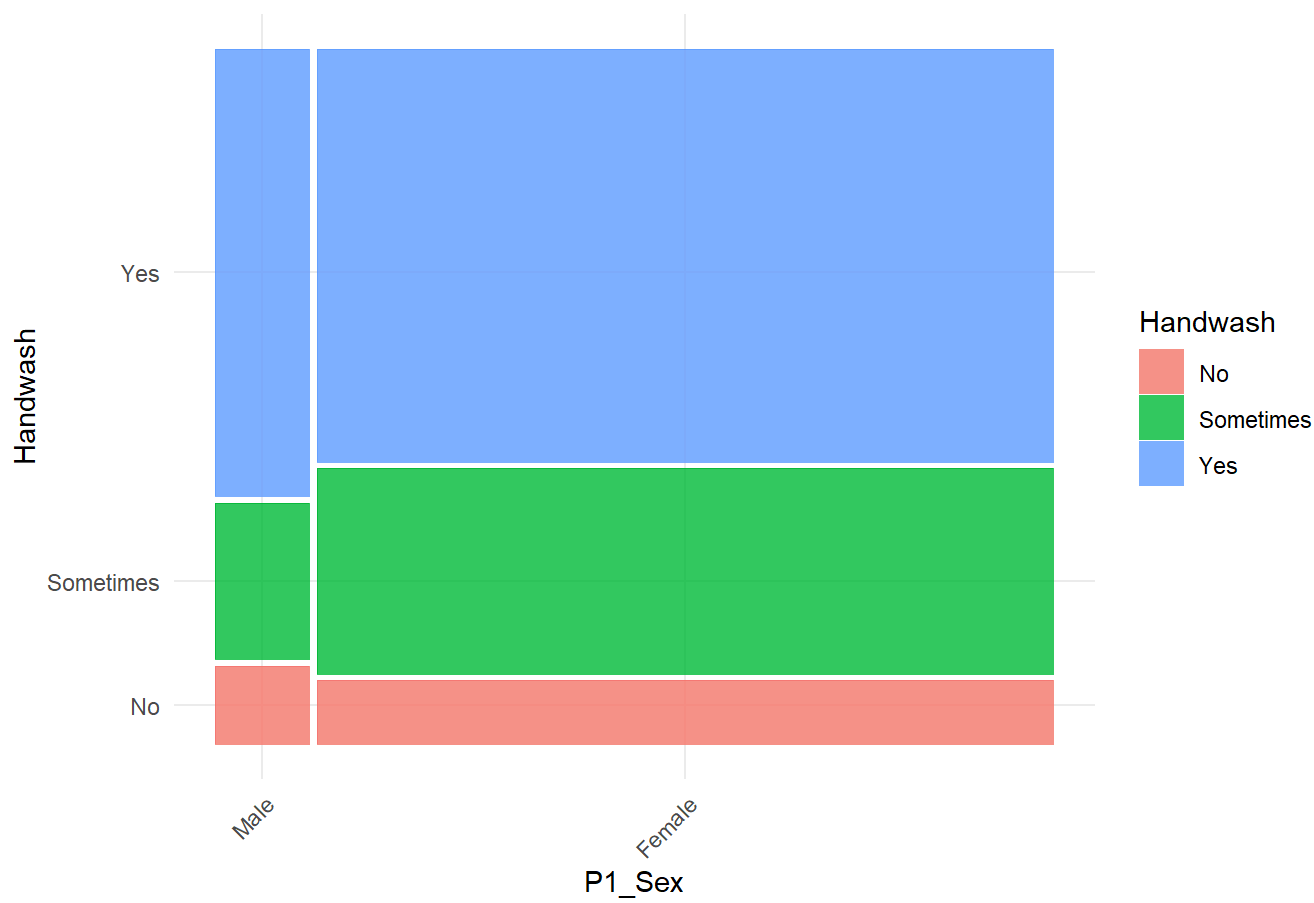
Mosaic Plot of P1_Age and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and P1_Sex -----
## Bivariate Analysis of Handwash and P1_Sex
##
## Frequency Table:
##   Handwash  Male Female
##         No      7     45
##   Sometimes    14    144
##         Yes     40    290
##
## Relative Frequency Table:
##   Handwash  Male Female Total
##         No  11.5%   9.4%  9.6%
##   Sometimes 23.0%  30.1% 29.3%
##         Yes 65.6%  60.5% 61.1%
##
## Cramér's V:  0
```

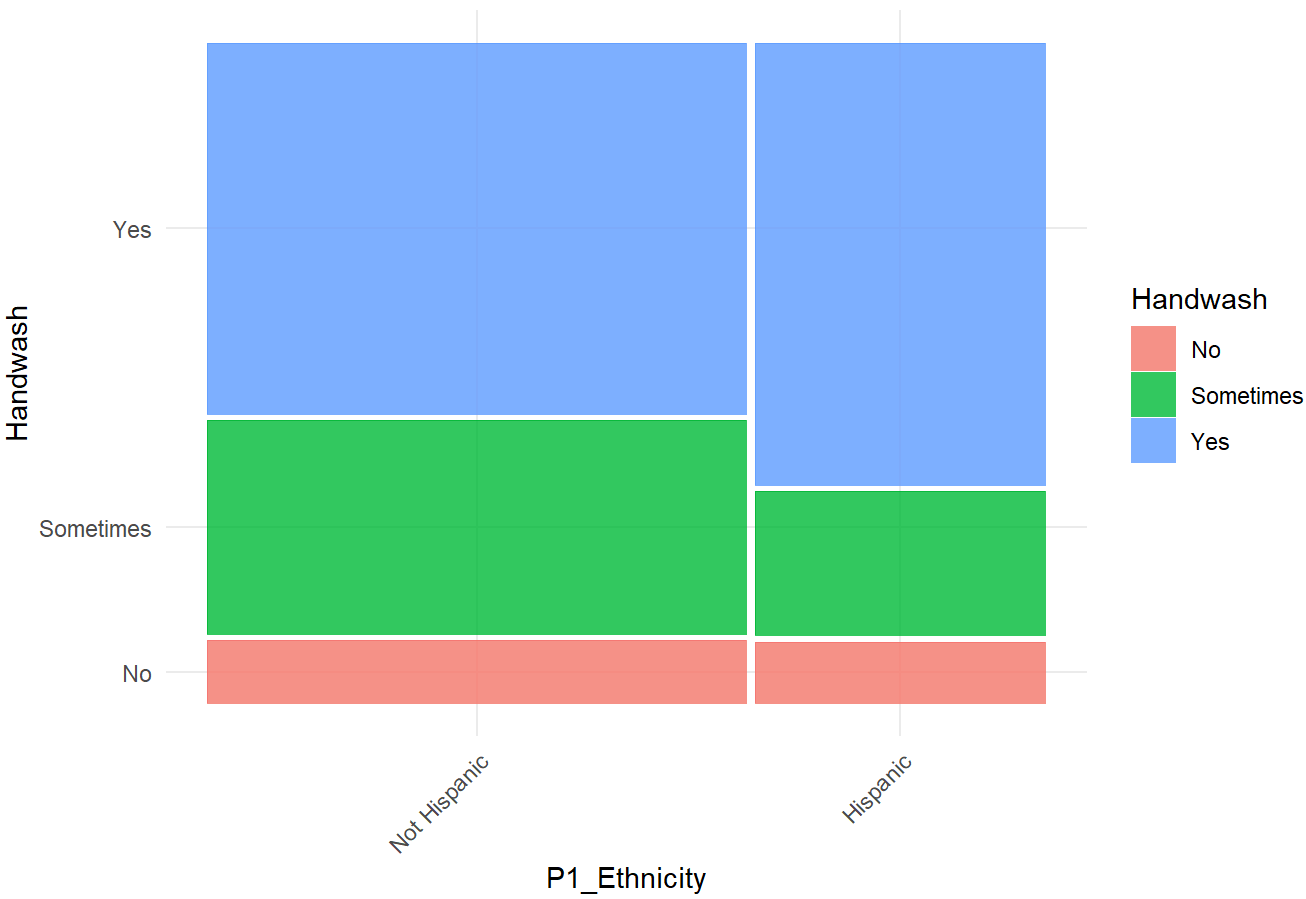
Mosaic Plot of P1_Sex and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and P1_Ethnicity -----
## Bivariate Analysis of Handwash and P1_Ethnicity
##
## Frequency Table:
##   Handwash Not Hispanic Hispanic
##      No          34          18
##   Sometimes      116          42
##      Yes         201         129
##
## Relative Frequency Table:
##   Handwash Not Hispanic Hispanic Total
##      No          9.7%          9.5%  9.6%
##   Sometimes      33.0%         22.2% 29.3%
##      Yes         57.3%         68.3% 61.1%
##
## Cramér's V:  0.09961327
```

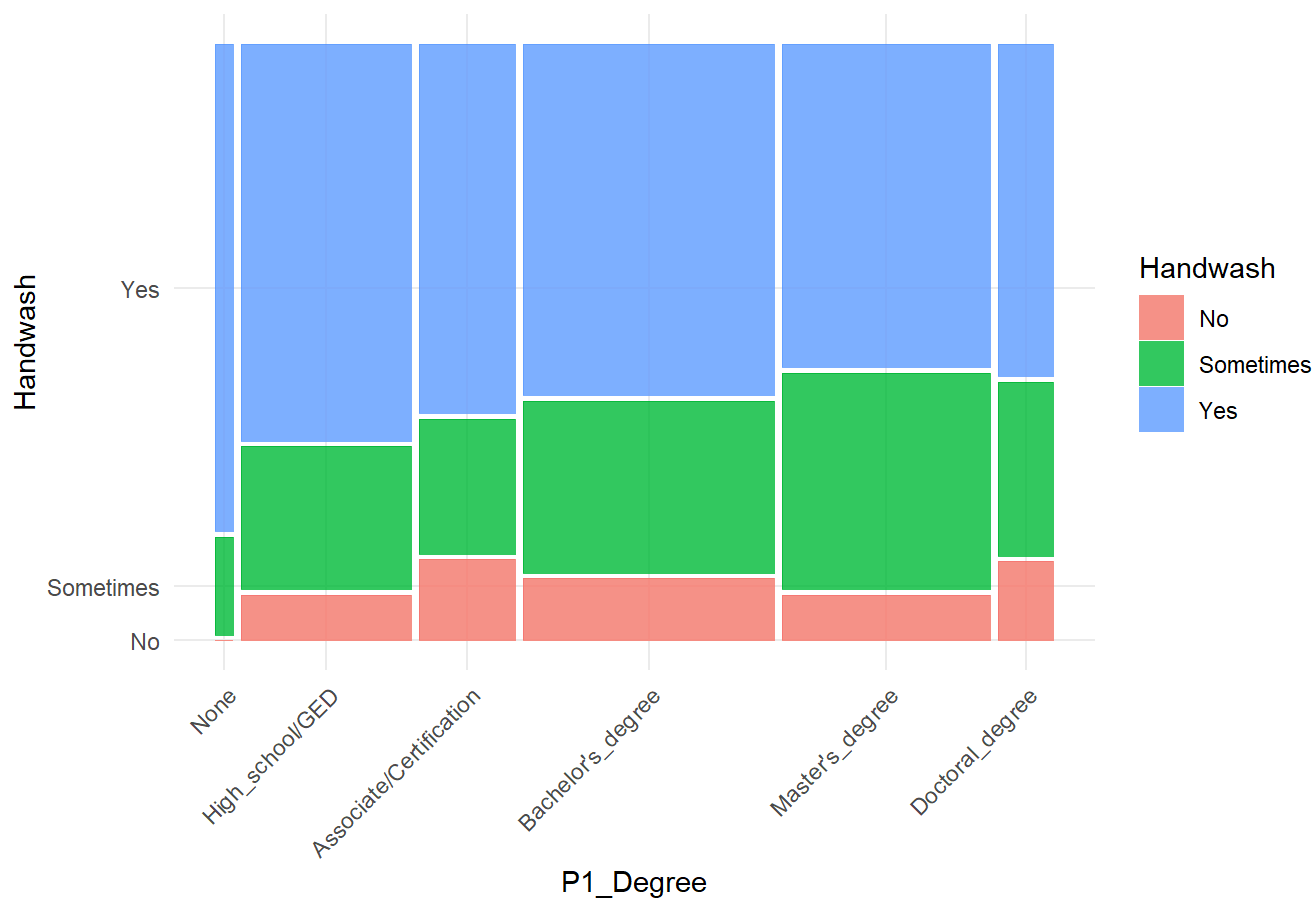
Mosaic Plot of P1_Ethnicity and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and P1_Degree -----
## Bivariate Analysis of Handwash and P1_Degree
##
## Frequency Table:
##   Handwash None High_school/GED Associate/Certification Bachelor's_degree
##       No      0           9              9              18
##   Sometimes  2          28             15             50
##       Yes    10          78             41            102
##   Master's_degree Doctoral_degree
##           11           5
##           52          11
##           78          21
##
## Relative Frequency Table:
##   Handwash None High_school/GED Associate/Certification Bachelor's_degree
##       No  0.0%           7.8%             13.8%           10.6%
##   Sometimes 16.7%          24.3%             23.1%           29.4%
##       Yes  83.3%          67.8%             63.1%           60.0%
##   Master's_degree Doctoral_degree Total
##           7.8%          13.5%  9.6%
##           36.9%          29.7% 29.3%
##           55.3%          56.8% 61.1%
##
## Cramér's V:  0.04299645
```

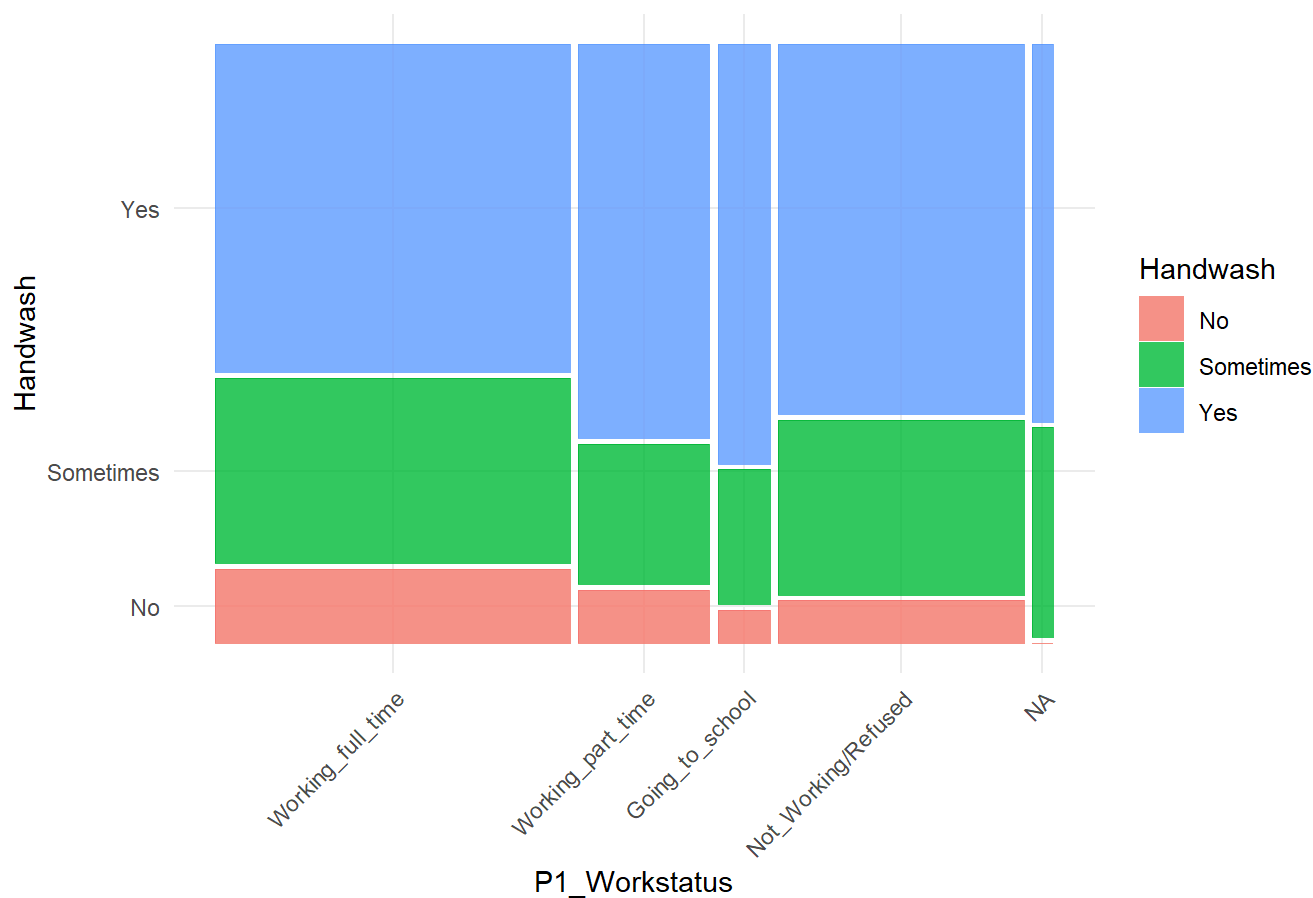
Mosaic Plot of P1_Degree and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and P1_Workstatus -----
## Bivariate Analysis of Handwash and P1_Workstatus
##
## Frequency Table:
##   Handwash Working_full_time Working_part_time Going_to_school
##       No                30                8                2
##   Sometimes              75               21               8
##       Yes               133               59              25
##   Not_Working/Refused NA_
##                   12    0
##                   49    5
##                   104   9
##
## Relative Frequency Table:
##   Handwash Working_full_time Working_part_time Going_to_school
##       No                12.6%                9.1%                5.7%
##   Sometimes              31.5%               23.9%               22.9%
##       Yes               55.9%               67.0%               71.4%
##   Not_Working/Refused   NA_ Total
##                   7.3%  0.0%  9.6%
##                   29.7% 35.7% 29.3%
##                   63.0% 64.3% 61.1%
##
## Cramér's V: 0.03542206
```

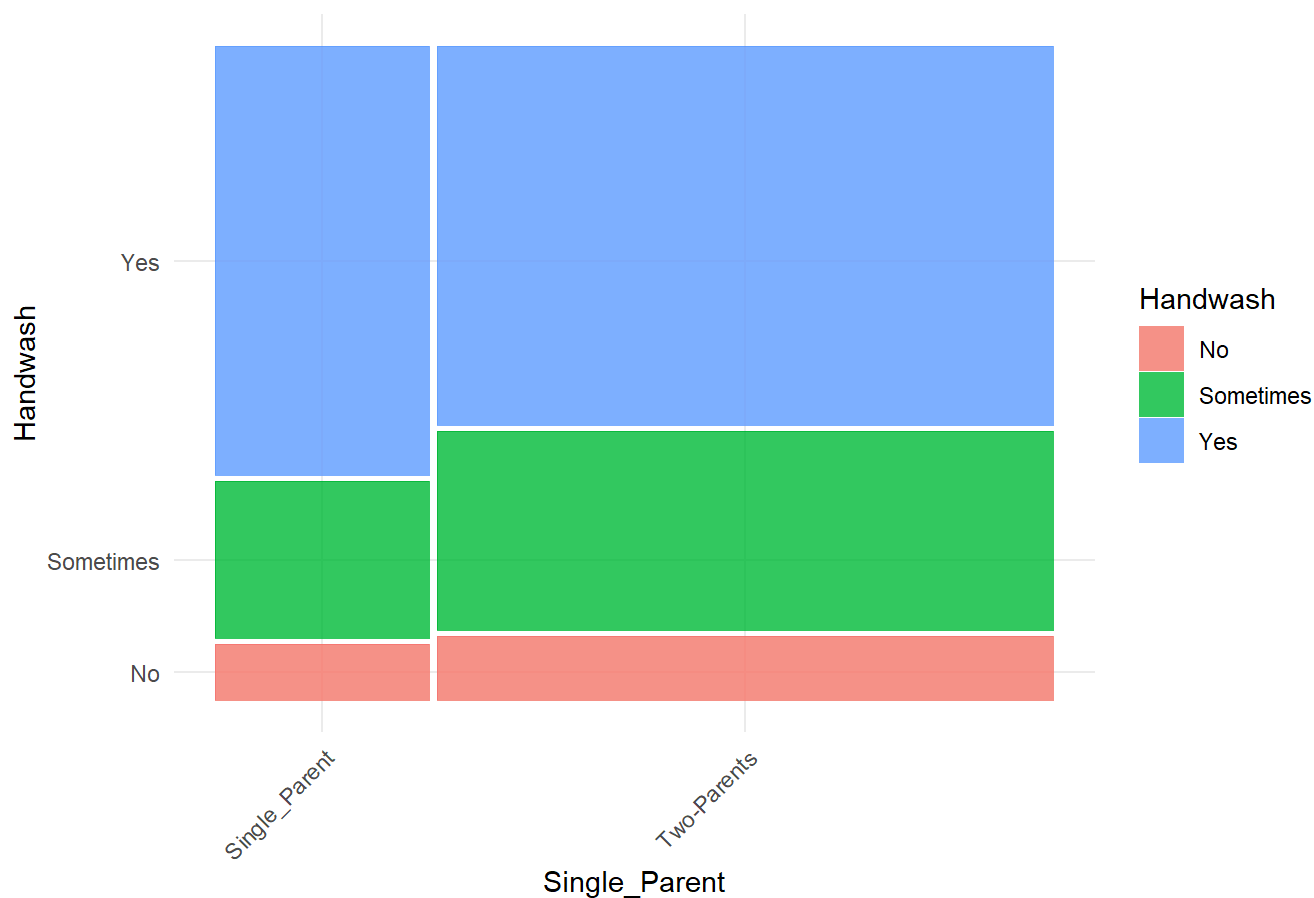
Mosaic Plot of P1_Workstatus and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and Single_Parent -----
## Bivariate Analysis of Handwash and Single_Parent
##
## Frequency Table:
##   Handwash Single_Parent Two-Parents
##       No           12           40
##   Sometimes        34          124
##       Yes           93          237
##
## Relative Frequency Table:
##   Handwash Single_Parent Two-Parents Total
##       No           8.6%          10.0%  9.6%
##   Sometimes        24.5%          30.9% 29.3%
##       Yes          66.9%          59.1% 61.1%
##
## Cramér's V:  0.03581286
```

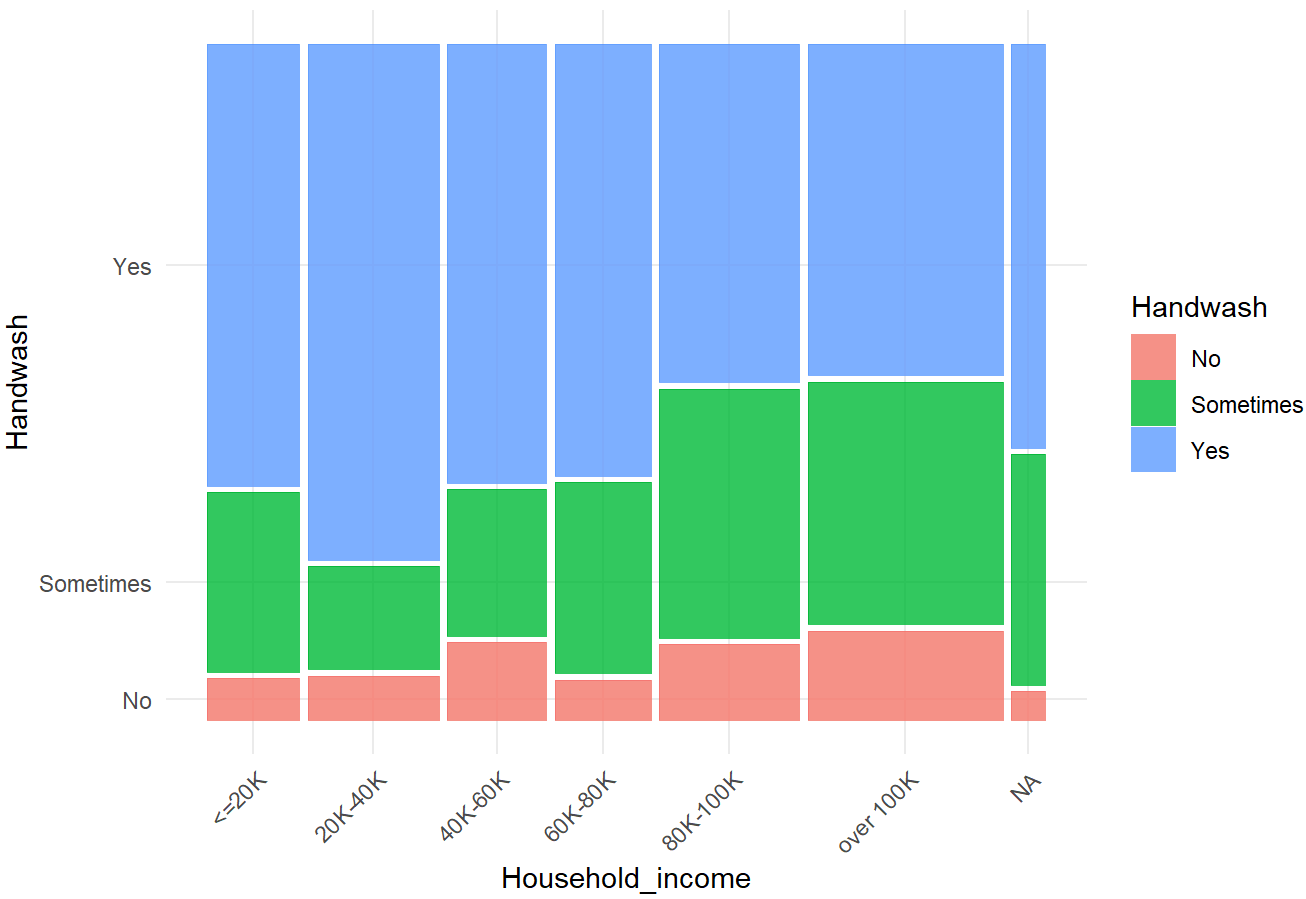

Mosaic Plot of Single_Parent and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and Household_income -----
## Bivariate Analysis of Handwash and Household_income
##
## Frequency Table:
##   Handwash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##         No      4       6       8       4       11      18    1
##   Sometimes  17      14      15      19      36      49    8
##         Yes   42      70      45      43      49      67   14
##
## Relative Frequency Table:
##   Handwash <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##         No   6.3%   6.7%   11.8%   6.1%   11.5%   13.4%  4.3%  9.6%
##   Sometimes 27.0%   15.6%  22.1%  28.8%  37.5%   36.6% 34.8% 29.3%
##         Yes 66.7%   77.8%  66.2%  65.2%  51.0%   50.0% 60.9% 61.1%
##
## Cramér's V:  0.1176455
```

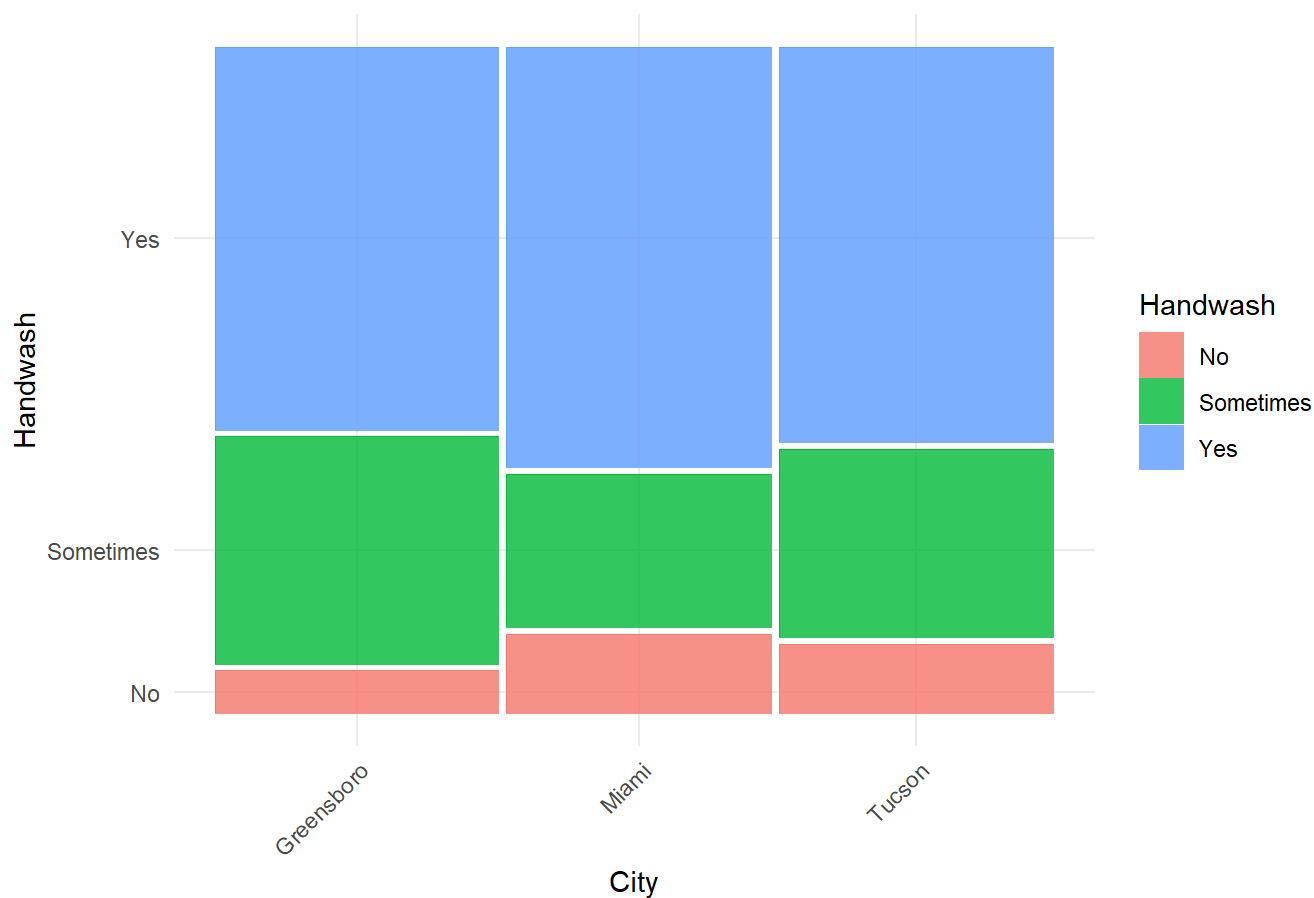
Mosaic Plot of Household_income and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and City -----
## Bivariate Analysis of Handwash and City
##
## Frequency Table:
##   Handwash Greensboro Miami Tucson
##       No           12     21     19
##   Sometimes        65     41     52
##       Yes          109    112    109
##
## Relative Frequency Table:
##   Handwash Greensboro Miami Tucson Total
##       No           6.5% 12.1% 10.6%  9.6%
##   Sometimes       34.9% 23.6% 28.9% 29.3%
##       Yes         58.6% 64.4% 60.6% 61.1%
##
## Cramér's V:  0.05844453
```

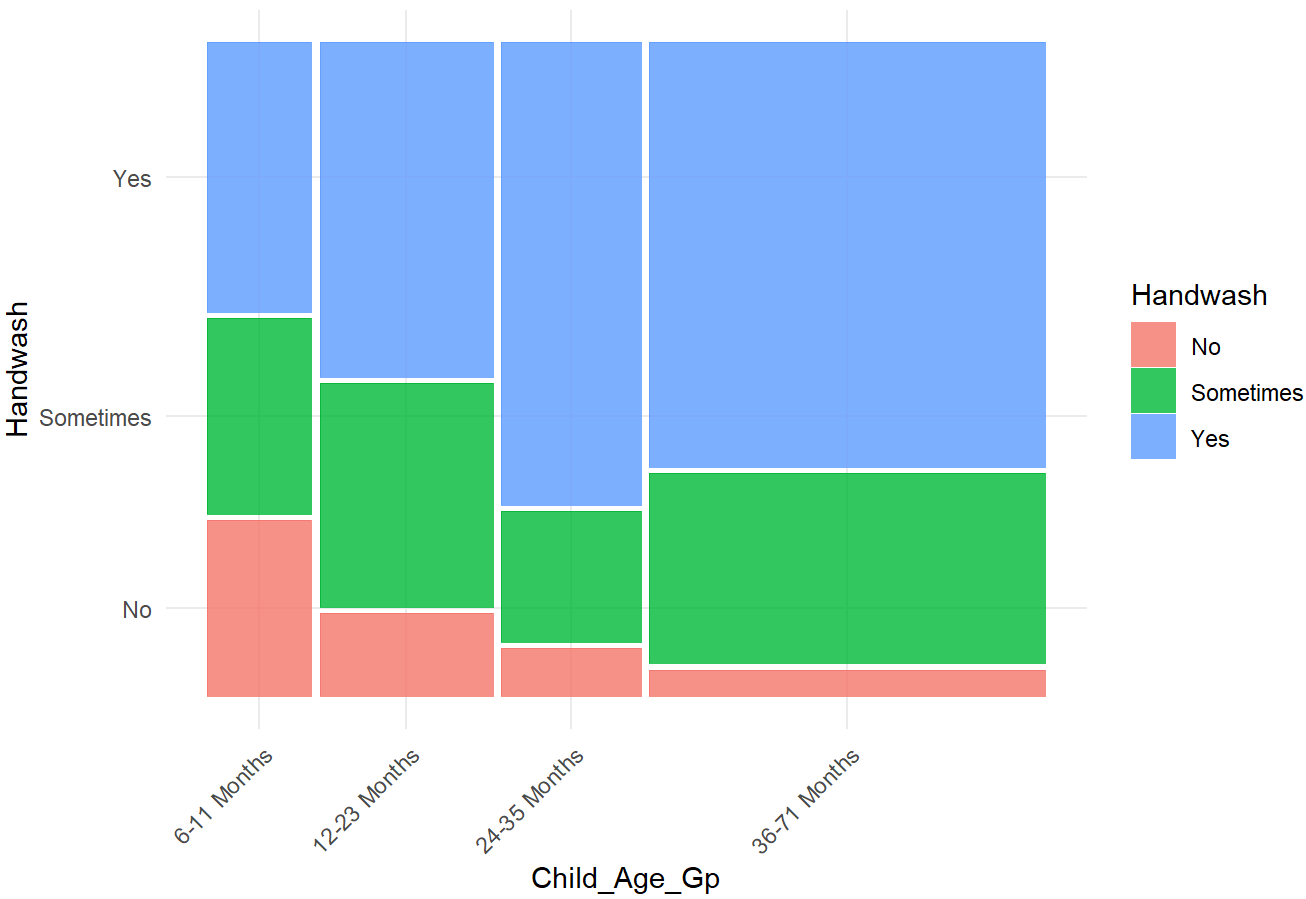
Mosaic Plot of City and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and Child_Age_Gp -----
## Bivariate Analysis of Handwash and Child_Age_Gp
##
## Frequency Table:
##   Handwash 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##         No           19           15           7           11
##   Sometimes           21           40           19          78
##         Yes           29           60           67         174
##
## Relative Frequency Table:
##   Handwash 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##         No           27.5%          13.0%           7.5%           4.2%   9.6%
##   Sometimes           30.4%          34.8%          20.4%          29.7%  29.3%
##         Yes           42.0%          52.2%          72.0%          66.2%  61.1%
##
## Cramér's V:  0.1907255
```

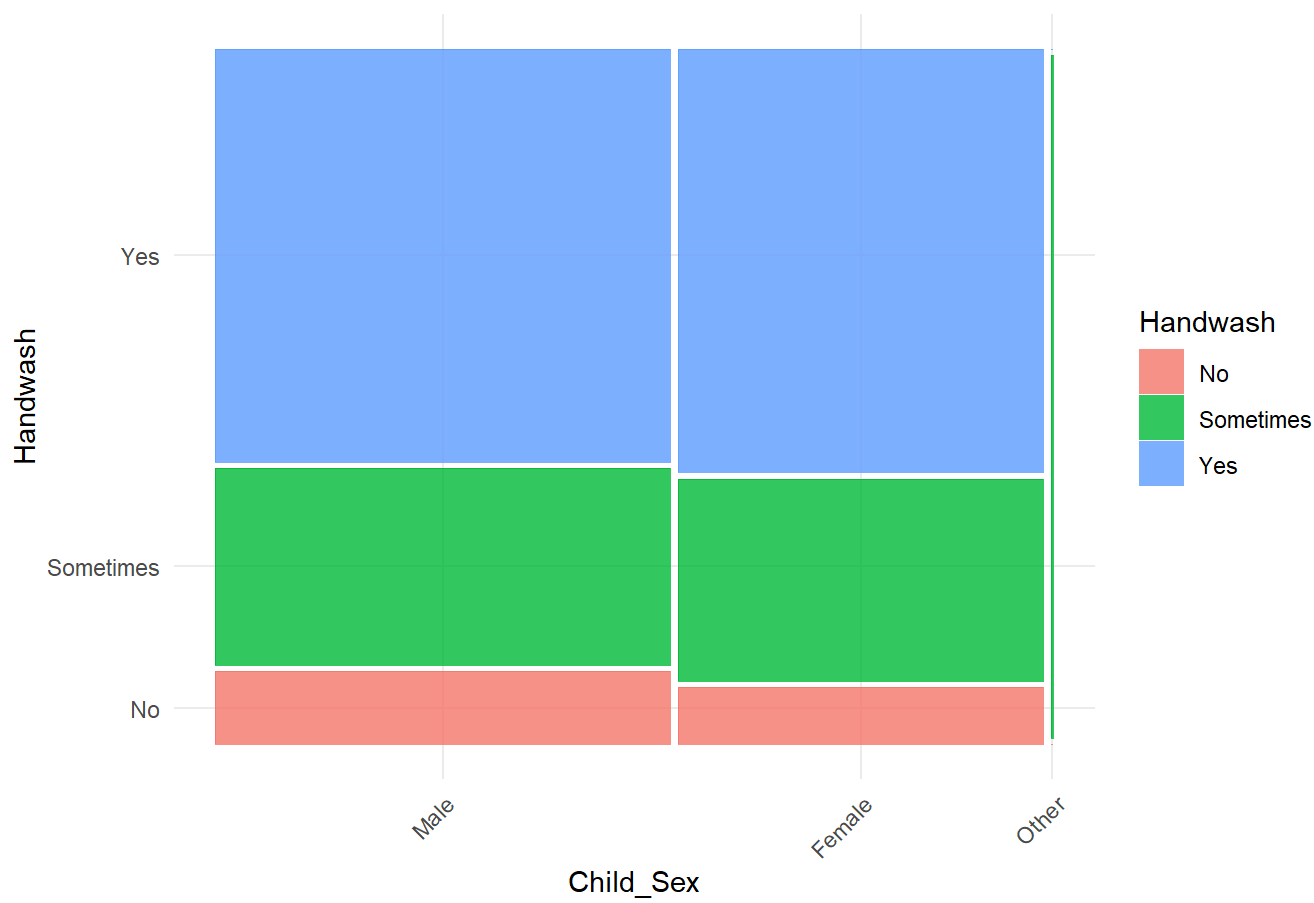
Mosaic Plot of Child_Age_Gp and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and Child_Sex -----
## Bivariate Analysis of Handwash and Child_Sex
##
## Frequency Table:
##   Handwash Male Female Other
##       No    32     20     0
##   Sometimes  86     71     1
##       Yes  181    149     0
##
## Relative Frequency Table:
##   Handwash  Male Female  Other Total
##       No  10.7%   8.3%   0.0%  9.6%
##   Sometimes 28.8%  29.6% 100.0% 29.3%
##       Yes  60.5%  62.1%   0.0% 61.1%
##
## Cramér's V:  0
```

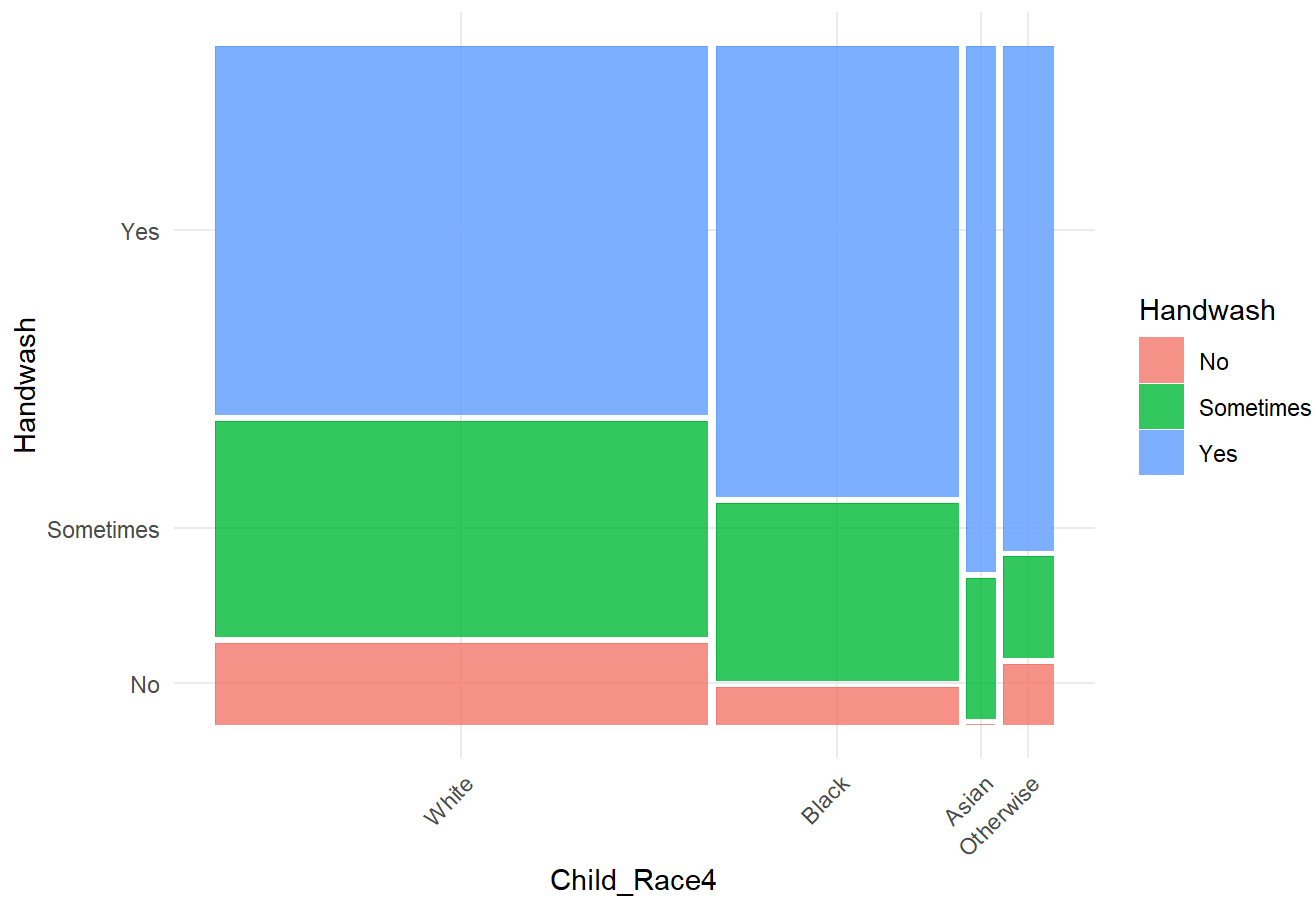
Mosaic Plot of Child_Sex and Handwash



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Handwash and Child_Race4 -----
## Bivariate Analysis of Handwash and Child_Race4
##
## Frequency Table:
##   Handwash White Black Asian Otherwise
##       No    40    9    0        3
##   Sometimes 106   43   4        5
##       Yes   181  109  15       25
##
## Relative Frequency Table:
##   Handwash White Black Asian Otherwise Total
##       No 12.2%  5.6%  0.0%    9.1%  9.6%
##   Sometimes 32.4% 26.7% 21.1%   15.2% 29.3%
##       Yes 55.4% 67.7% 78.9%   75.8% 61.1%
##
## Cramér's V: 0.09677858
```

Mosaic Plot of Child_Race4 and Handwash



```
## Saving 7 x 5 in image
```

```
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```

```
##      var1      var2  cramer_v
## 1  Sucking  Child_Age_Gp 0.22993967
## 2  Handwash  Child_Age_Gp 0.19072550
## 3  Sucking      City 0.16724405
## 4  Handwash Household_income 0.11764552
## 5  Handwash      P1_Ethnicity 0.09961327
## 6  Handwash      Child_Race4 0.09677858
## 7  Handwash      P1_Race4 0.08145399
## 8  Handwash      P1_Age 0.06384544
## 9  Handwash      City 0.05844453
## 10 Sucking      P1_Workstatus 0.05698498
## 11 Handwash      P1_Degree 0.04299645
## 12 Sucking Household_income 0.04139869
## 13 Handwash      Single_Parent 0.03581286
## 14 Handwash      P1_Workstatus 0.03542206
## 15 Sucking      P1_Degree 0.02032518
## 16 Sucking      P1_Race4 0.00000000
## 17 Sucking      P1_Age 0.00000000
## 18 Sucking      P1_Sex 0.00000000
## 19 Sucking      P1_Ethnicity 0.00000000
## 20 Sucking      Single_Parent 0.00000000
## 21 Sucking      Child_Sex 0.00000000
## 22 Sucking      Child_Race4 0.00000000
## 23 Handwash      P1_Sex 0.00000000
## 24 Handwash      Child_Sex 0.00000000
```

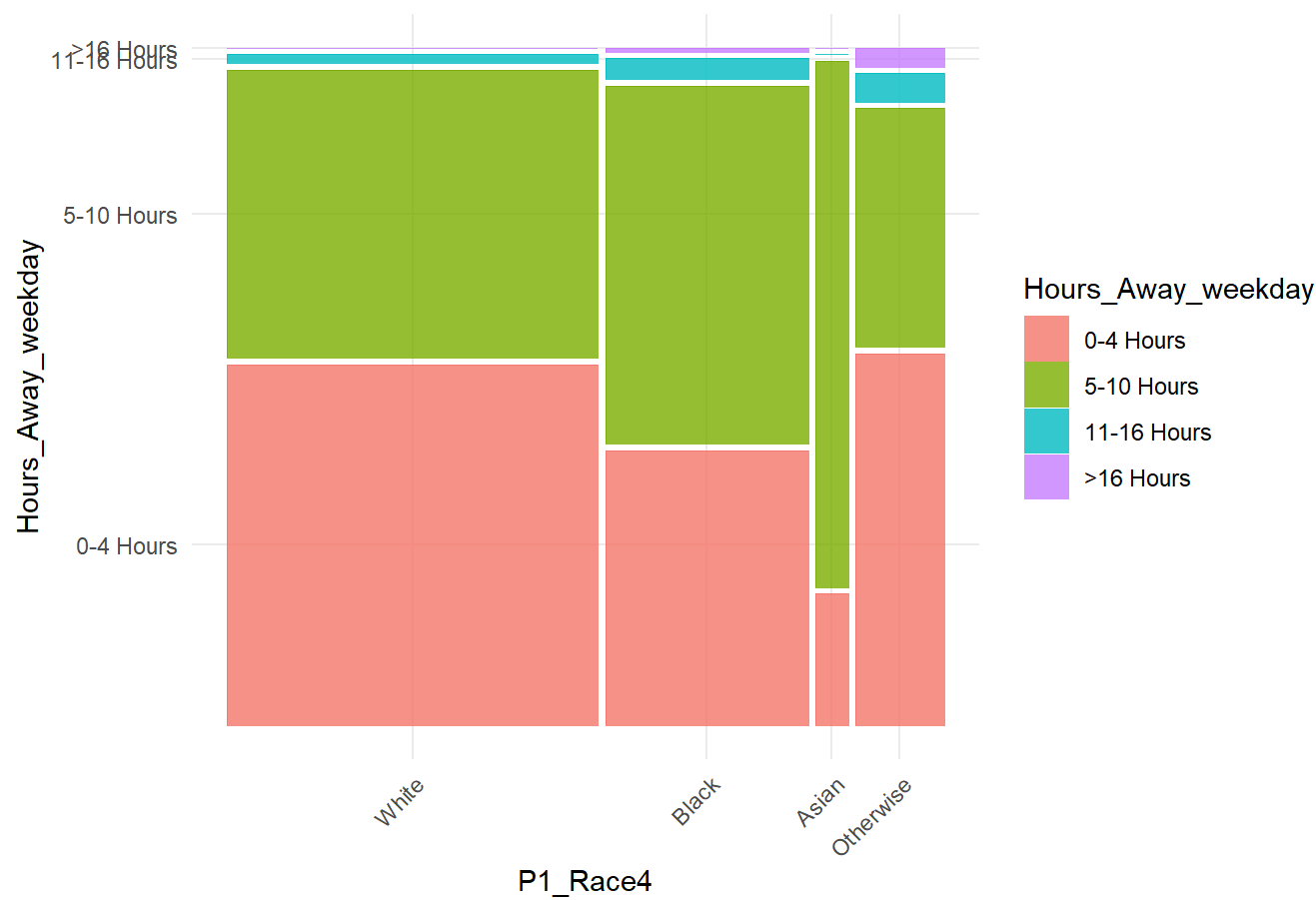
Exploring Hours_Away_var

```
data= ChildrenADF
list1= c("Hours_Away_weekday", "Hours_Outdoors_Home", "Hours_Away_weekend")
list2= DemoVars

cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(), stringsAsFactors = FALSE)
results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = '_vs_')]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values,
                              data.frame(var1 = var1, var2 = var2,
                                          cramer_v = result$cramer_v))
    }
  }
}
```

```
##
##
## ----- Analysis for Hours_Away_weekday and P1_Race4 -----
## Bivariate Analysis of Hours_Away_weekday and P1_Race4
##
## Frequency Table:
##   Hours_Away_weekday White Black Asian Otherwise
##           0-4 Hours   158    66     5        39
##           5-10 Hours   126    86    20        25
##           11-16 Hours     4     5     0         3
##           >16 Hours     0     1     0         2
##
## Relative Frequency Table:
##   Hours_Away_weekday White Black Asian Otherwise Total
##           0-4 Hours  54.9% 41.8% 20.0%    56.5% 49.6%
##           5-10 Hours 43.8% 54.4% 80.0%    36.2% 47.6%
##           11-16 Hours  1.4%  3.2%  0.0%     4.3%  2.2%
##           >16 Hours  0.0%  0.6%  0.0%     2.9%  0.6%
##
## Cramér's V:  0.115571
```

Mosaic Plot of P1_Race4 and Hours_Away_weekday

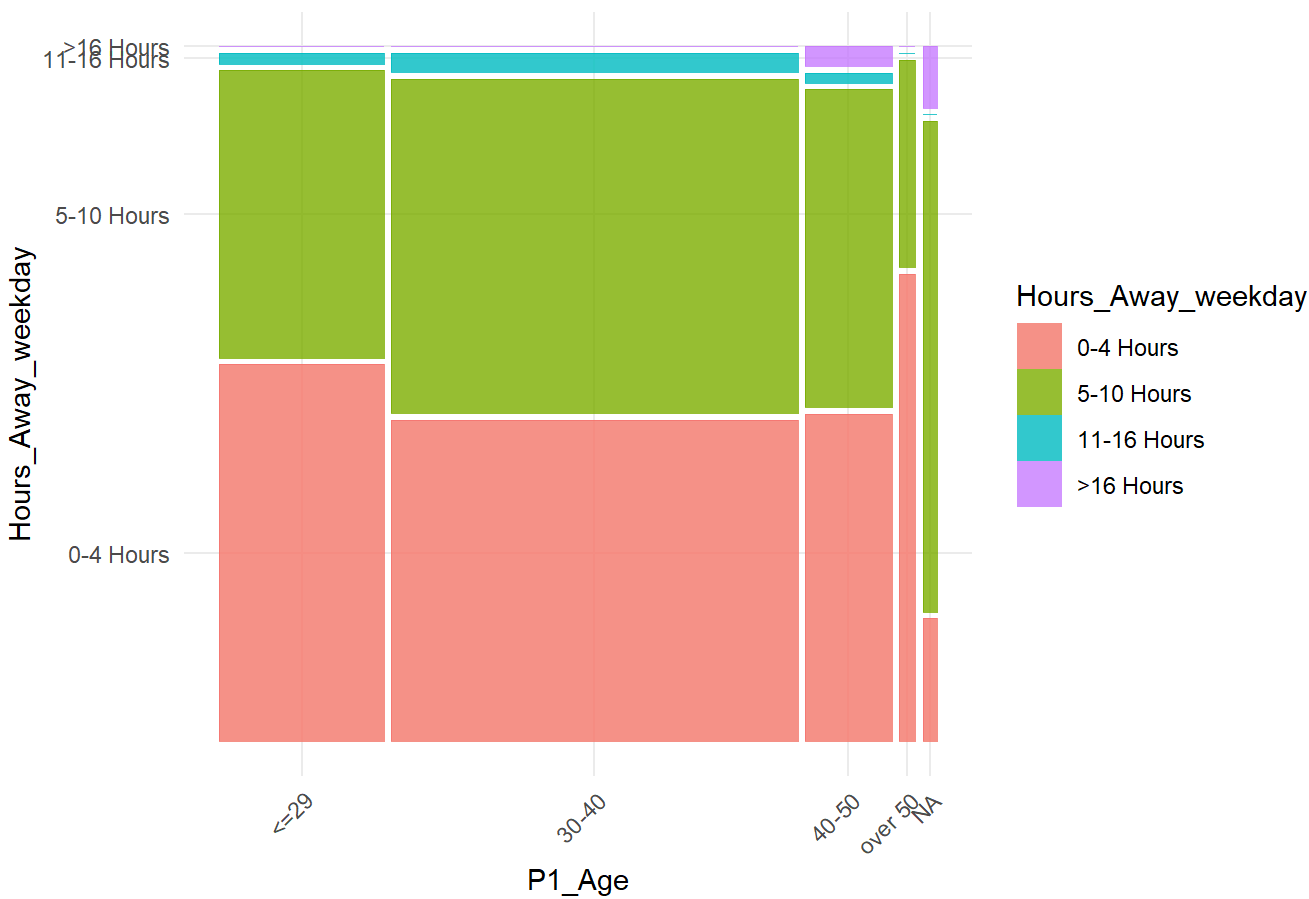


```
## Saving 7 x 5 in image
```



```
##
##
## ----- Analysis for Hours_Away_weekday and P1_Age -----
## Bivariate Analysis of Hours_Away_weekday and P1_Age
##
## Frequency Table:
##   Hours_Away_weekday  <=29  30-40  40-50  over 50  NA_
##           0-4 Hours    72    152    33        9    2
##           5-10 Hours    55    158    32        4    8
##           11-16 Hours     2     9     1         0    0
##           >16 Hours     0     0     2         0    1
##
## Relative Frequency Table:
##   Hours_Away_weekday  <=29  30-40  40-50  over 50   NA_ Total
##           0-4 Hours  55.8%  47.6%  48.5%   69.2% 18.2% 49.6%
##           5-10 Hours  42.6%  49.5%  47.1%   30.8% 72.7% 47.6%
##           11-16 Hours  1.6%   2.8%  1.5%    0.0%  0.0%  2.2%
##           >16 Hours  0.0%   0.0%  2.9%    0.0%  9.1%  0.6%
##
## Cramér's V:  0.1142523
```

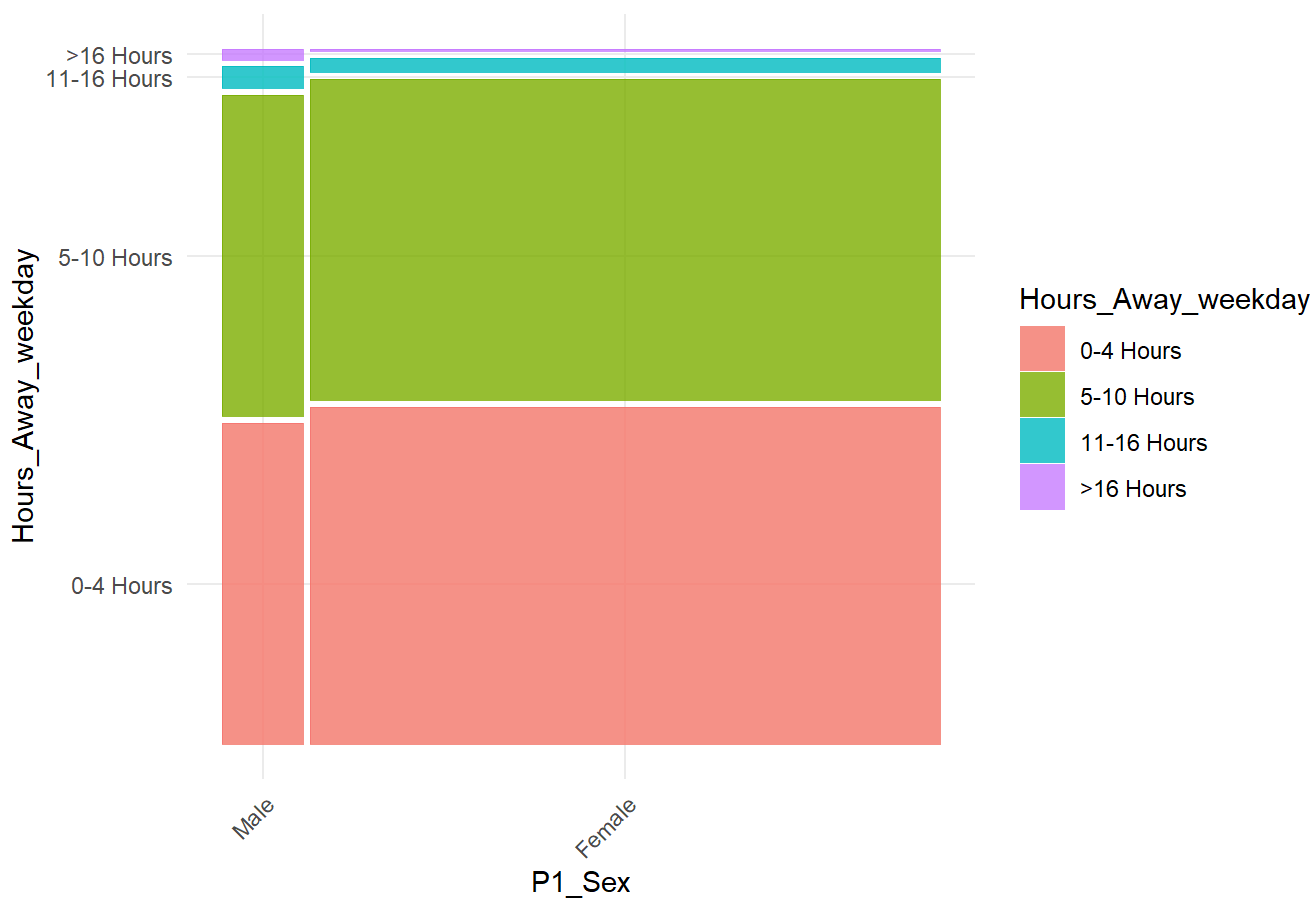
Mosaic Plot of P1_Age and Hours_Away_weekday



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Away_weekday and P1_Sex -----
## Bivariate Analysis of Hours_Away_weekday and P1_Sex
##
## Frequency Table:
##   Hours_Away_weekday   Male  Female
##           0-4 Hours     29    239
##           5-10 Hours     29    228
##          11-16 Hours       2     10
##           >16 Hours       1      2
##
## Relative Frequency Table:
##   Hours_Away_weekday   Male  Female  Total
##           0-4 Hours  47.5%  49.9%  49.6%
##           5-10 Hours  47.5%  47.6%  47.6%
##          11-16 Hours   3.3%   2.1%   2.2%
##           >16 Hours   1.6%   0.4%   0.6%
##
## Cramér's V:  0
```

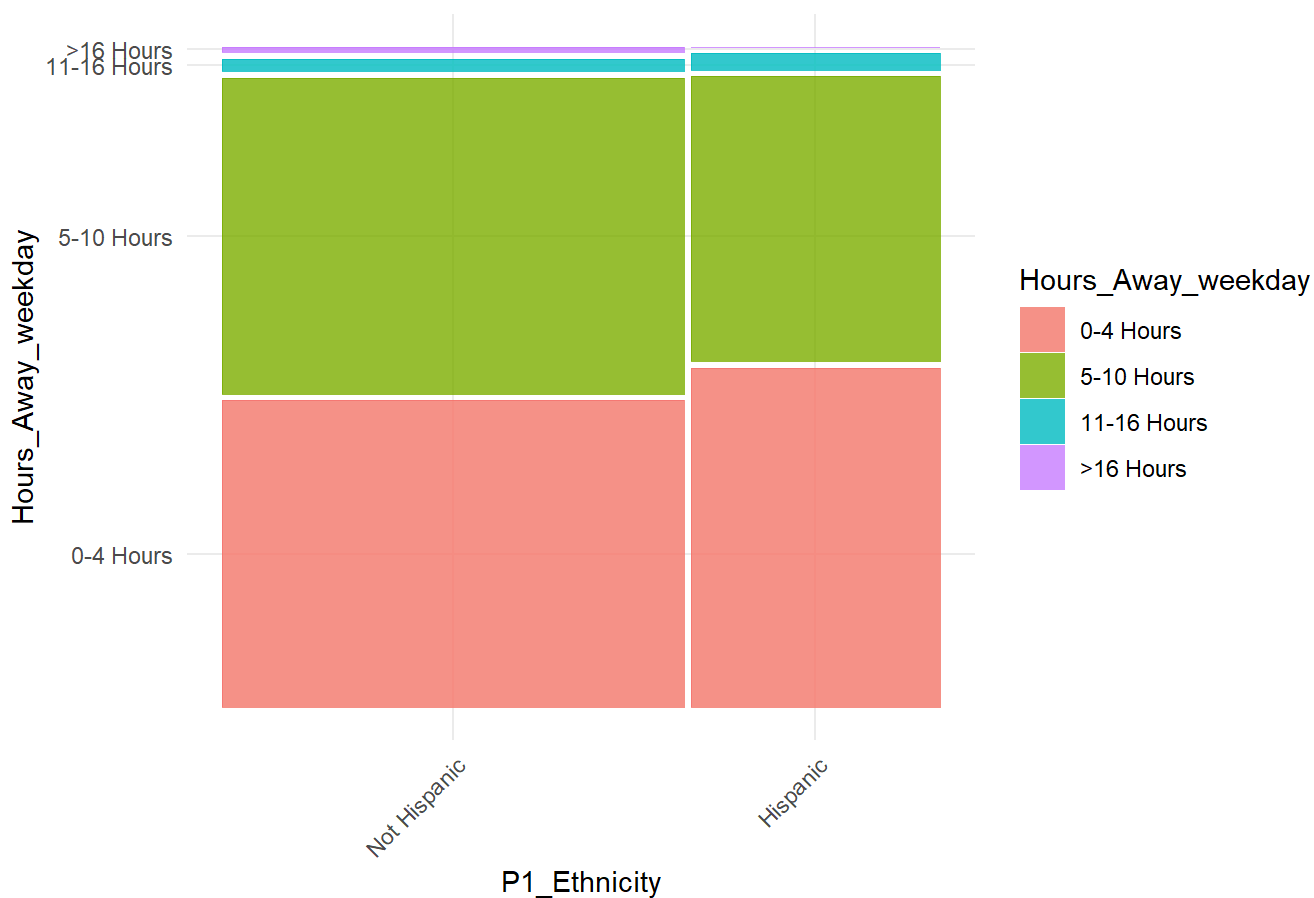
Mosaic Plot of P1_Sex and Hours_Away_weekday



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Away_weekday and P1_Ethnicity -----
## Bivariate Analysis of Hours_Away_weekday and P1_Ethnicity
##
## Frequency Table:
##   Hours_Away_weekday Not Hispanic Hispanic
##           0-4 Hours         168        100
##           5-10 Hours         173         84
##           11-16 Hours          7          5
##           >16 Hours           3           0
##
## Relative Frequency Table:
##   Hours_Away_weekday Not Hispanic Hispanic Total
##           0-4 Hours         47.9%      52.9% 49.6%
##           5-10 Hours         49.3%      44.4% 47.6%
##           11-16 Hours          2.0%       2.6%  2.2%
##           >16 Hours          0.9%       0.0%  0.6%
##
## Cramér's V:  0.0122006
```

Mosaic Plot of P1_Ethnicity and Hours_Away_weekday



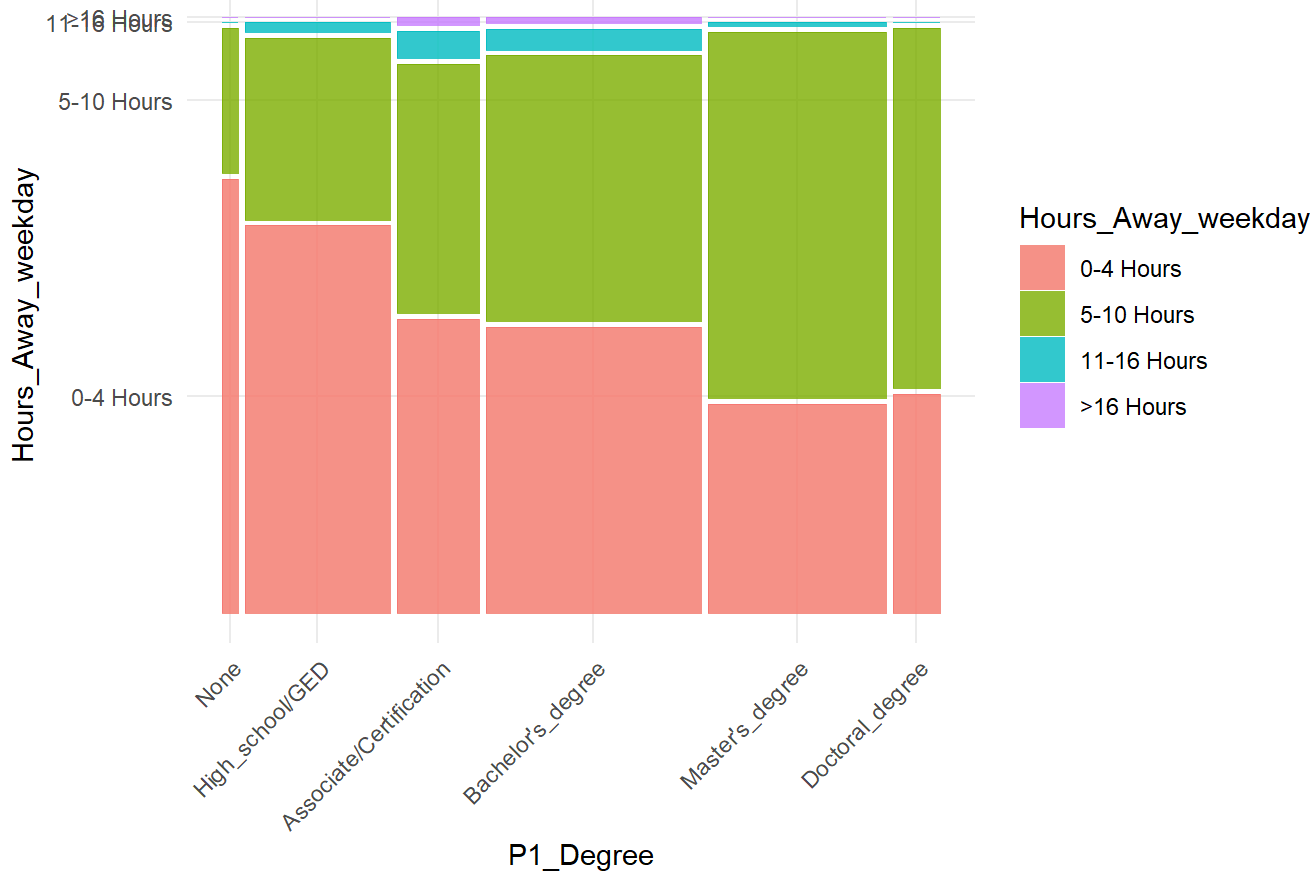
```
## Saving 7 x 5 in image
```

```

##
##
## ----- Analysis for Hours_Away_weekday and P1_Degree -----
## Bivariate Analysis of Hours_Away_weekday and P1_Degree
##
## Frequency Table:
## Hours_Away_weekday None High_school/GED Associate/Certification
##      0-4 Hours      9          77          33
##      5-10 Hours     3          36          28
##     11-16 Hours     0           2           3
##      >16 Hours     0           0           1
## Bachelor's_degree Master's_degree Doctoral_degree
##           84           51           14
##           78           89           23
##            6            1            0
##            2            0            0
##
## Relative Frequency Table:
## Hours_Away_weekday None High_school/GED Associate/Certification
##      0-4 Hours 75.0%          67.0%          50.8%
##      5-10 Hours 25.0%          31.3%          43.1%
##     11-16 Hours  0.0%           1.7%           4.6%
##      >16 Hours  0.0%           0.0%           1.5%
## Bachelor's_degree Master's_degree Doctoral_degree Total
##           49.4%          36.2%          37.8% 49.6%
##           45.9%          63.1%          62.2% 47.6%
##            3.5%           0.7%           0.0%  2.2%
##            1.2%           0.0%           0.0%  0.6%
##
## Cramér's V:  0.1275536

```

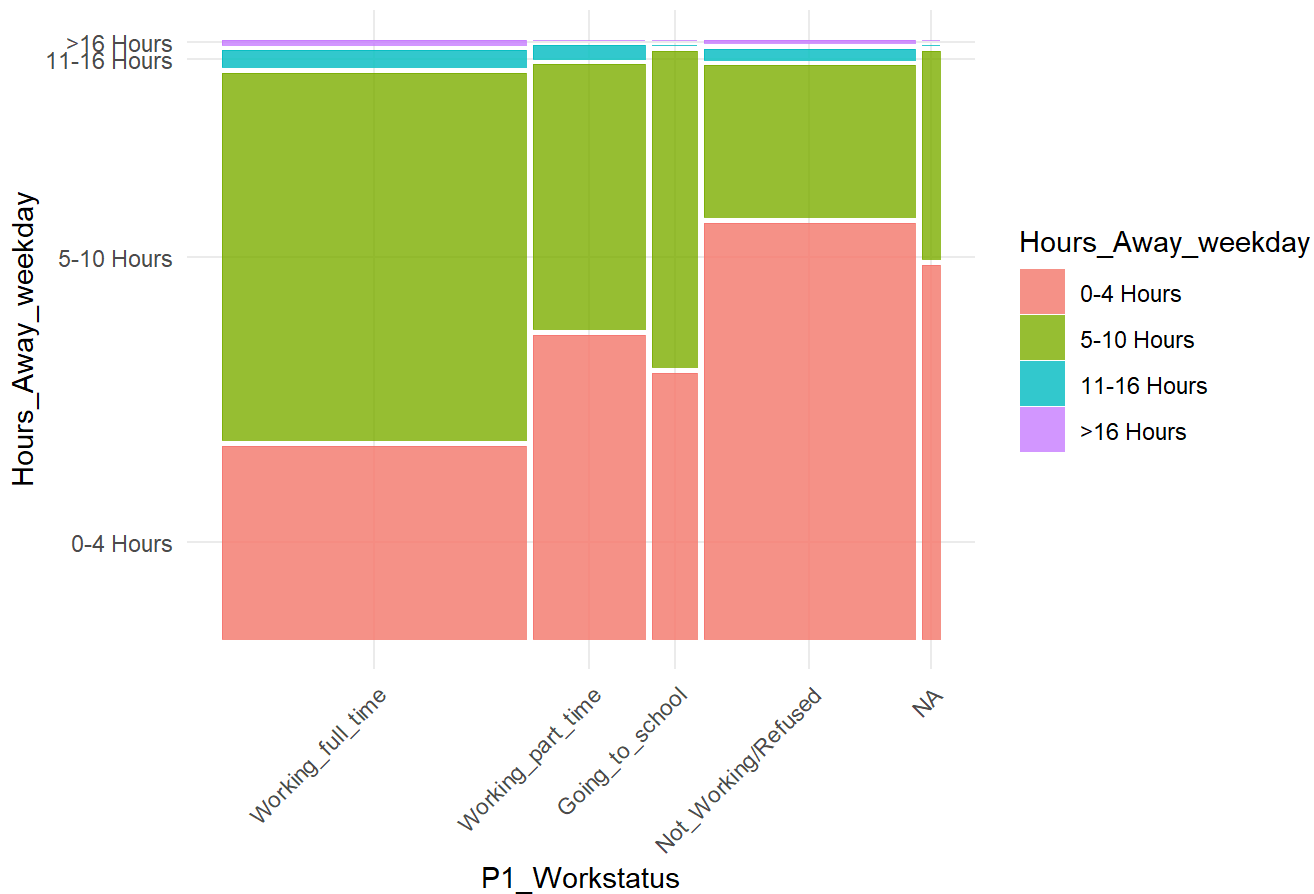
Mosaic Plot of P1_Degree and Hours_Away_weekday



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekday and P1_Workstatus -----
## Bivariate Analysis of Hours_Away_weekday and P1_Workstatus
##
## Frequency Table:
## Hours_Away_weekday Working_full_time Working_part_time Going_to_school
##      0-4 Hours           79           46           16
##      5-10 Hours          150           40           19
##     11-16 Hours           7            2            0
##      >16 Hours           2            0            0
## Not_Working/Refused NA_
##           118    9
##           43    5
##            3    0
##            1    0
##
## Relative Frequency Table:
## Hours_Away_weekday Working_full_time Working_part_time Going_to_school
##      0-4 Hours          33.2%          52.3%          45.7%
##      5-10 Hours          63.0%          45.5%          54.3%
##     11-16 Hours           2.9%           2.3%           0.0%
##      >16 Hours           0.8%           0.0%           0.0%
## Not_Working/Refused  NA_ Total
##           71.5% 64.3% 49.6%
##           26.1% 35.7% 47.6%
##            1.8%  0.0%  2.2%
##            0.6%  0.0%  0.6%
##
## Cramér's V:  0.1750927
```

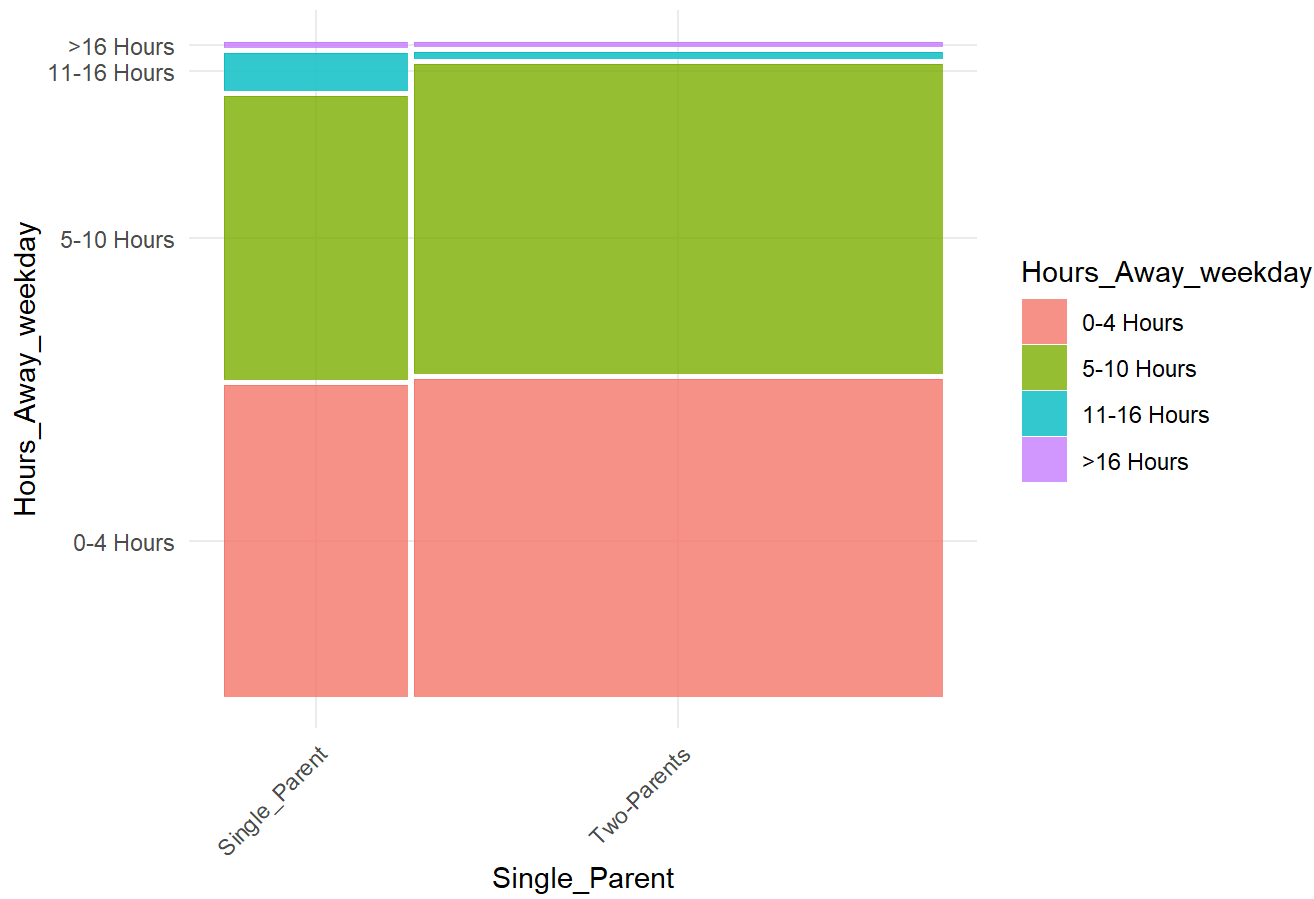
Mosaic Plot of P1_Workstatus and Hours_Away_weekday



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekday and Single_Parent -----
## Bivariate Analysis of Hours_Away_weekday and Single_Parent
##
## Frequency Table:
##   Hours_Away_weekday Single_Parent Two-Parents
##           0-4 Hours           68           200
##           5-10 Hours           62           195
##           11-16 Hours            8            4
##           >16 Hours             1            2
##
## Relative Frequency Table:
##   Hours_Away_weekday Single_Parent Two-Parents Total
##           0-4 Hours           48.9%           49.9% 49.6%
##           5-10 Hours           44.6%           48.6% 47.6%
##           11-16 Hours            5.8%            1.0%  2.2%
##           >16 Hours             0.7%            0.5%  0.6%
##
## Cramér's V:  0.1216006
```

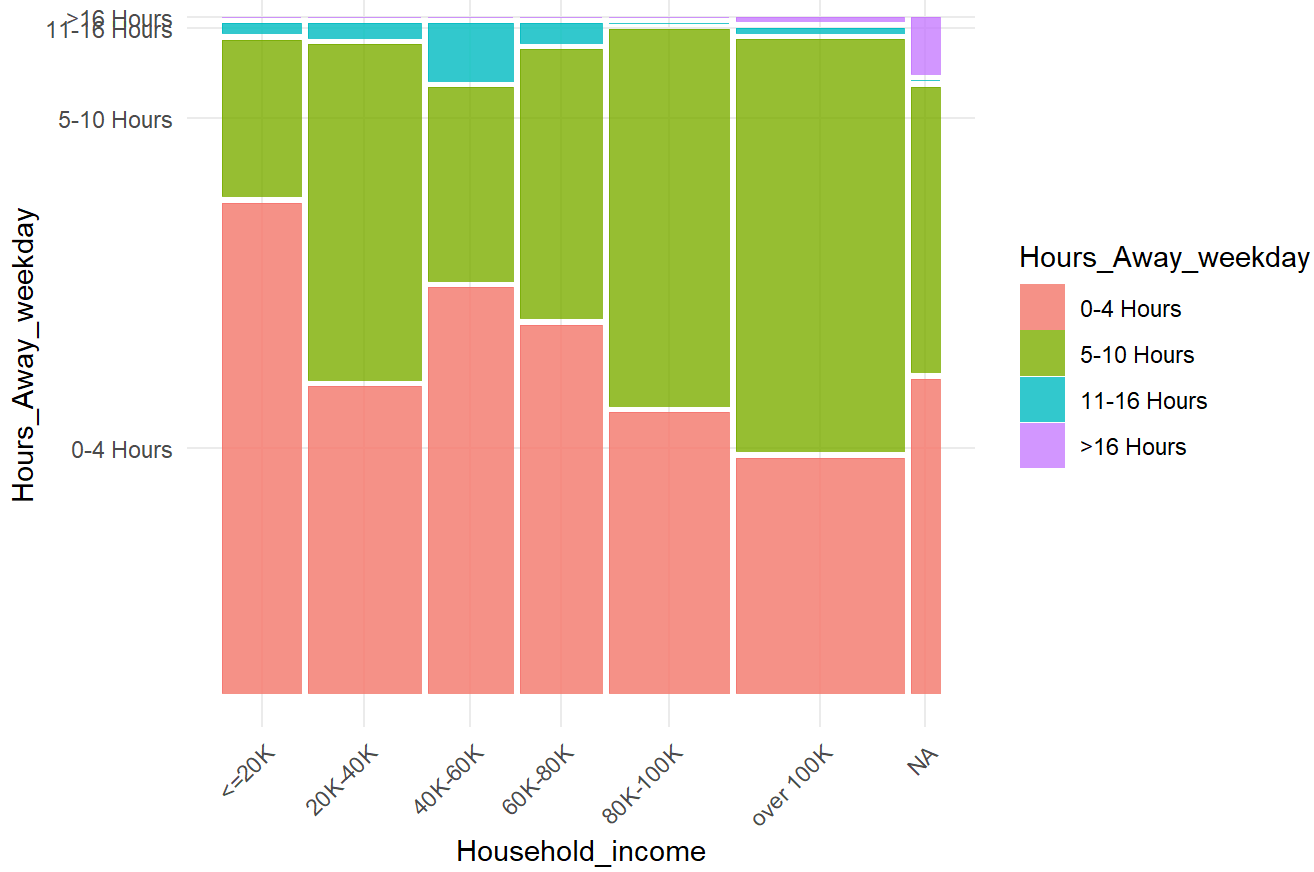
Mosaic Plot of Single_Parent and Hours_Away_weekday



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekday and Household_income -----
## Bivariate Analysis of Hours_Away_weekday and Household_income
##
## Frequency Table:
##   Hours_Away_weekday  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_
##           0-4 Hours      47      42      42      37      41      48    11
##           5-10 Hours     15      46      20      27      55      84    10
##          11-16 Hours       1       2       6       2       0       1     0
##           >16 Hours       0       0       0       0       0       1     2
##
## Relative Frequency Table:
##   Hours_Away_weekday  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_
##           0-4 Hours  74.6%   46.7%   61.8%   56.1%   42.7%   35.8%  47.8%
##           5-10 Hours  23.8%   51.1%   29.4%   40.9%   57.3%   62.7%  43.5%
##          11-16 Hours   1.6%    2.2%    8.8%    3.0%    0.0%    0.7%  0.0%
##           >16 Hours   0.0%    0.0%    0.0%    0.0%    0.0%    0.7%  8.7%
##
## Total
## 49.6%
## 47.6%
## 2.2%
## 0.6%
##
## Cramér's V: 0.2045534
```

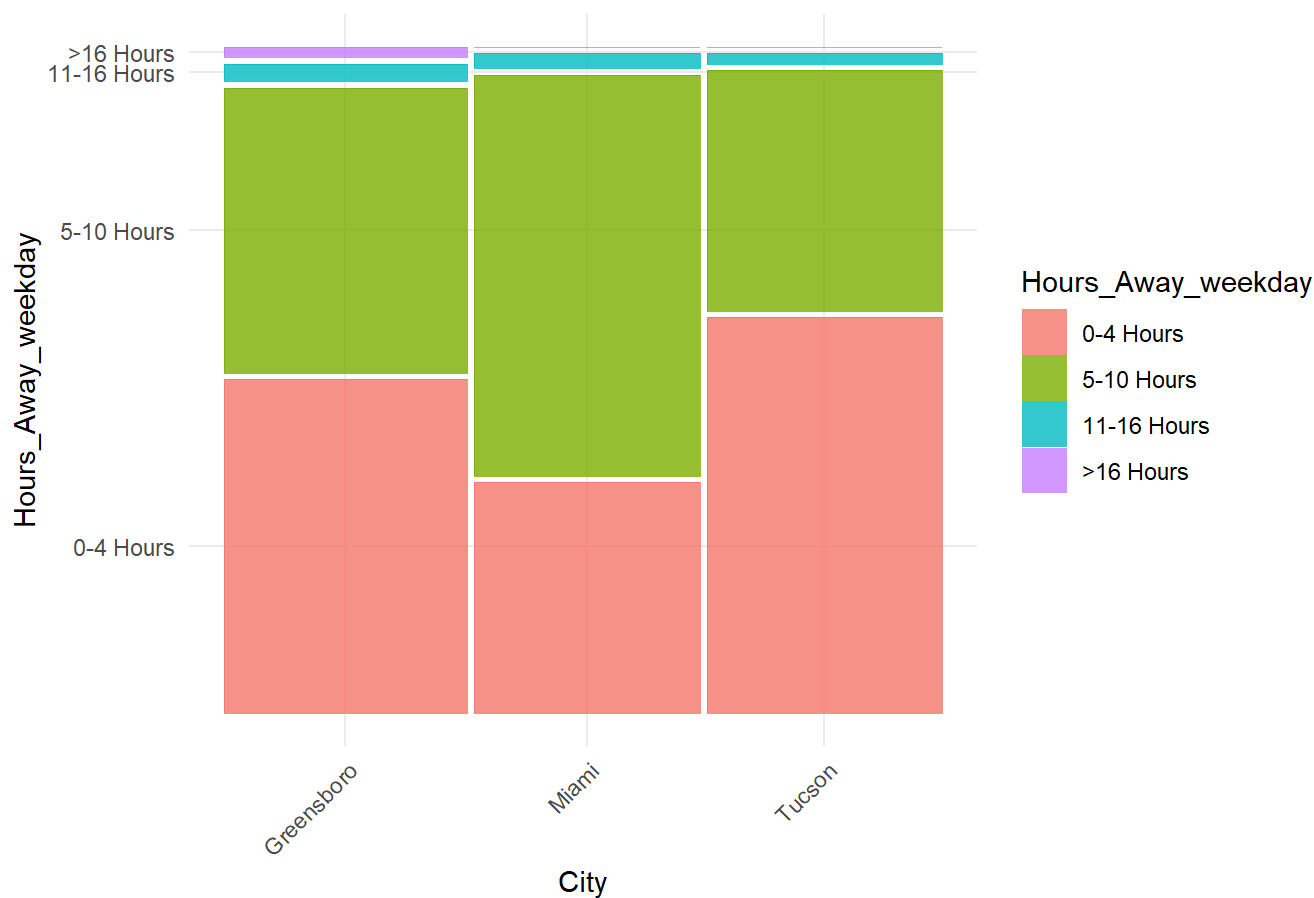

Mosaic Plot of Household_income and Hours_Away_weekday



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekday and City -----
## Bivariate Analysis of Hours_Away_weekday and City
##
## Frequency Table:
##   Hours_Away_weekday Greensboro Miami Tucson
##           0-4 Hours      96     62    110
##           5-10 Hours      82    108     67
##          11-16 Hours       5      4      3
##           >16 Hours       3       0       0
##
## Relative Frequency Table:
##   Hours_Away_weekday Greensboro Miami Tucson Total
##           0-4 Hours    51.6%  35.6%  61.1%  49.6%
##           5-10 Hours    44.1%  62.1%  37.2%  47.6%
##          11-16 Hours     2.7%   2.3%   1.7%   2.2%
##           >16 Hours     1.6%   0.0%   0.0%   0.6%
##
## Cramér's V:  0.1497908
```

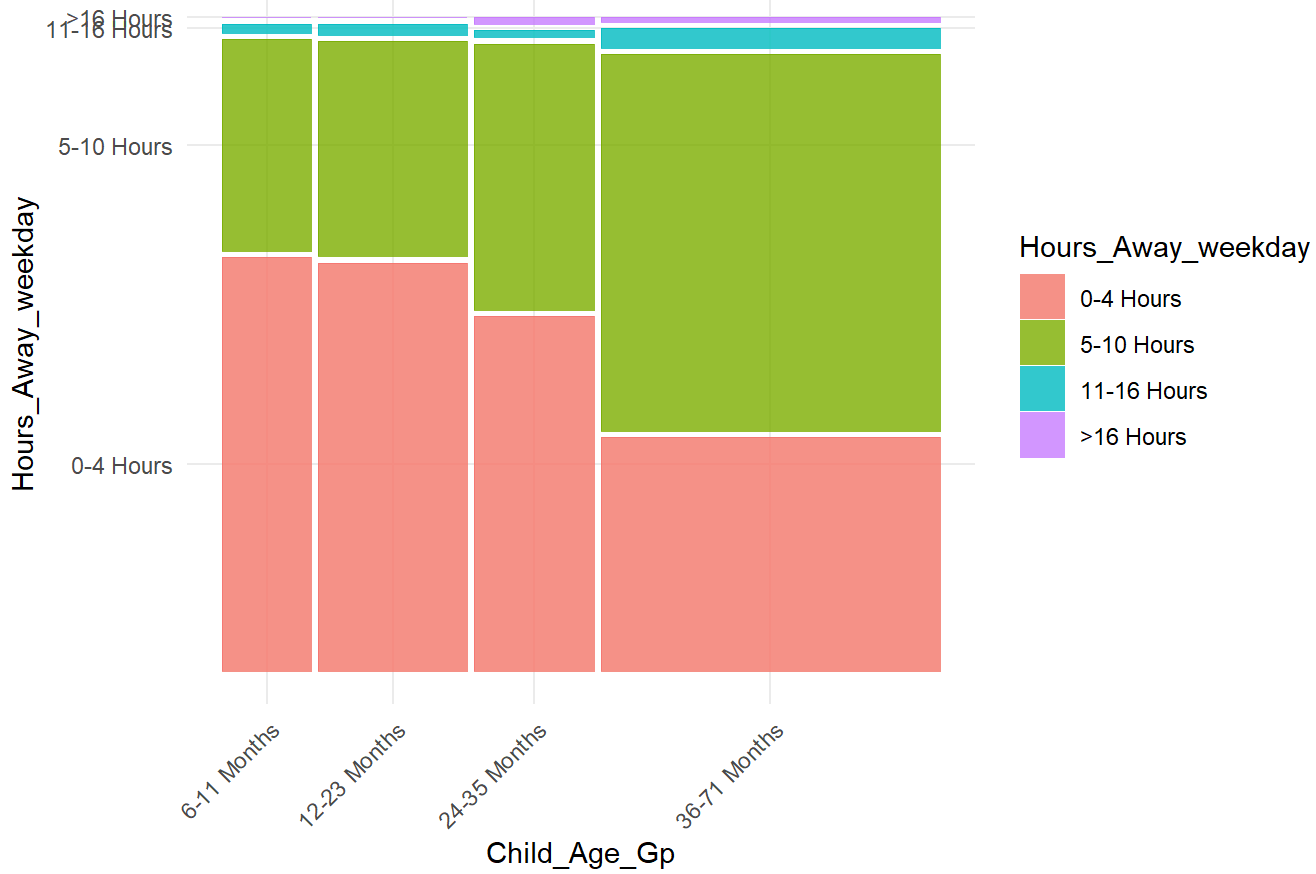
Mosaic Plot of City and Hours_Away_weekday



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekday and Child_Age_Gp -----
## Bivariate Analysis of Hours_Away_weekday and Child_Age_Gp
##
## Frequency Table:
## Hours_Away_weekday 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##      0-4 Hours      45      74      52      97
##      5-10 Hours     23      39      39     156
##     11-16 Hours      1       2       1       8
##      >16 Hours      0       0       1       2
##
## Relative Frequency Table:
## Hours_Away_weekday 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##      0-4 Hours     65.2%     64.3%     55.9%     36.9% 49.6%
##      5-10 Hours     33.3%     33.9%     41.9%     59.3% 47.6%
##     11-16 Hours      1.4%      1.7%      1.1%      3.0%  2.2%
##      >16 Hours      0.0%      0.0%      1.1%      0.8%  0.6%
##
## Cramér's V:  0.1312378
```

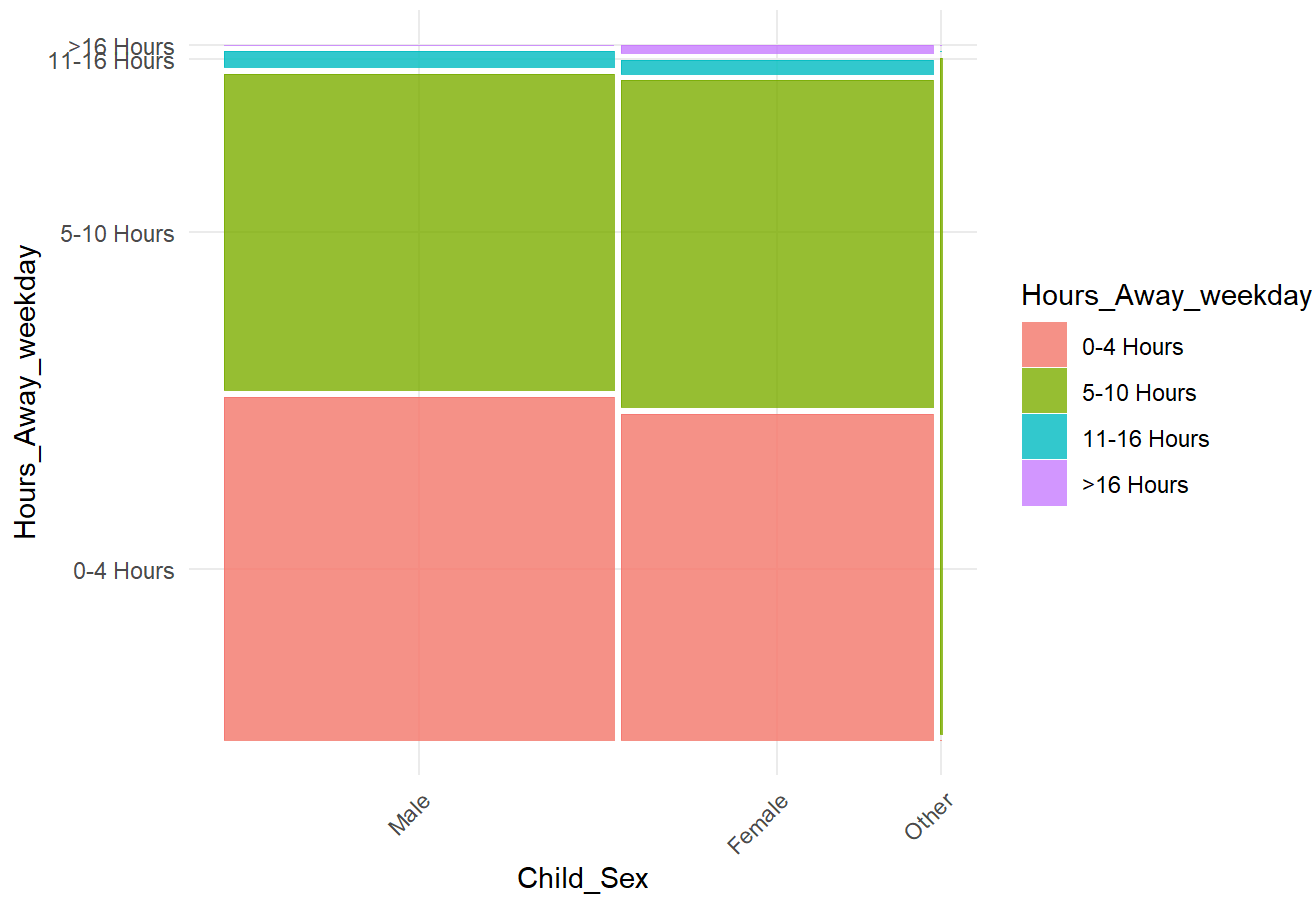
Mosaic Plot of Child_Age_Gp and Hours_Away_weekday



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekday and Child_Sex -----
## Bivariate Analysis of Hours_Away_weekday and Child_Sex
##
## Frequency Table:
##   Hours_Away_weekday Male Female Other
##   0-4 Hours        152    116     0
##   5-10 Hours       140    116     1
##   11-16 Hours        7      5     0
##   >16 Hours         0      3     0
##
## Relative Frequency Table:
##   Hours_Away_weekday Male Female Other Total
##   0-4 Hours        50.8%  48.3%   0.0%  49.6%
##   5-10 Hours       46.8%  48.3% 100.0%  47.6%
##   11-16 Hours       2.3%   2.1%   0.0%   2.2%
##   >16 Hours         0.0%   1.2%   0.0%   0.6%
##
## Cramér's V:  0
```

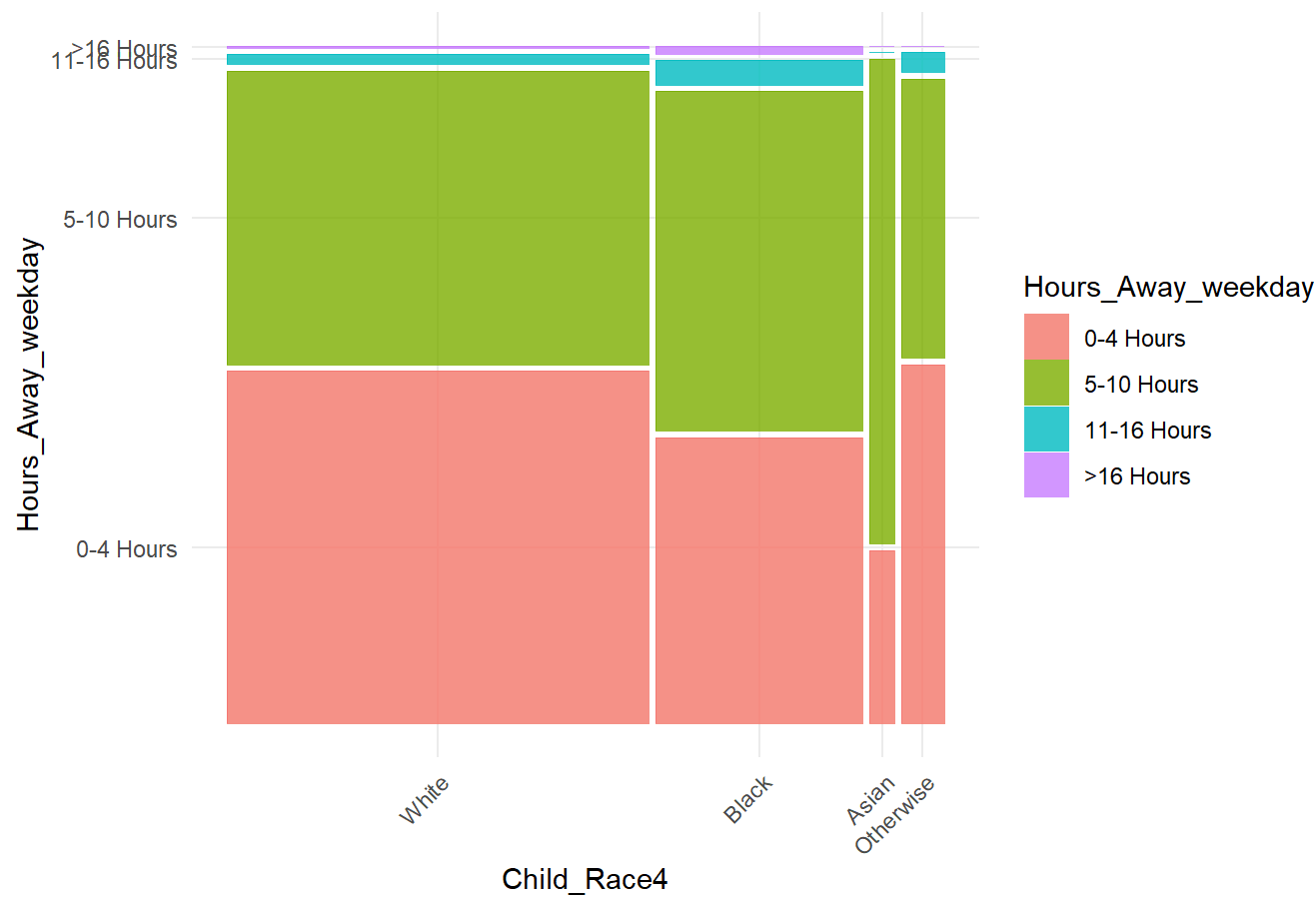
Mosaic Plot of Child_Sex and Hours_Away_weekday



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekday and Child_Race4 -----
## Bivariate Analysis of Hours_Away_weekday and Child_Race4
##
## Frequency Table:
##   Hours_Away_weekday White Black Asian Otherwise
##   0-4 Hours      175    70    5      18
##   5-10 Hours     146    83   14     14
##  11-16 Hours       5     6    0      1
##   >16 Hours       1     2    0      0
##
## Relative Frequency Table:
##   Hours_Away_weekday White Black Asian Otherwise Total
##   0-4 Hours      53.5% 43.5% 26.3%    54.5% 49.6%
##   5-10 Hours     44.6% 51.6% 73.7%    42.4% 47.6%
##  11-16 Hours      1.5%  3.7%  0.0%     3.0%  2.2%
##   >16 Hours       0.3%  1.2%  0.0%     0.0%  0.6%
##
## Cramér's V:  0.05205742
```

Mosaic Plot of Child_Race4 and Hours_Away_weekday



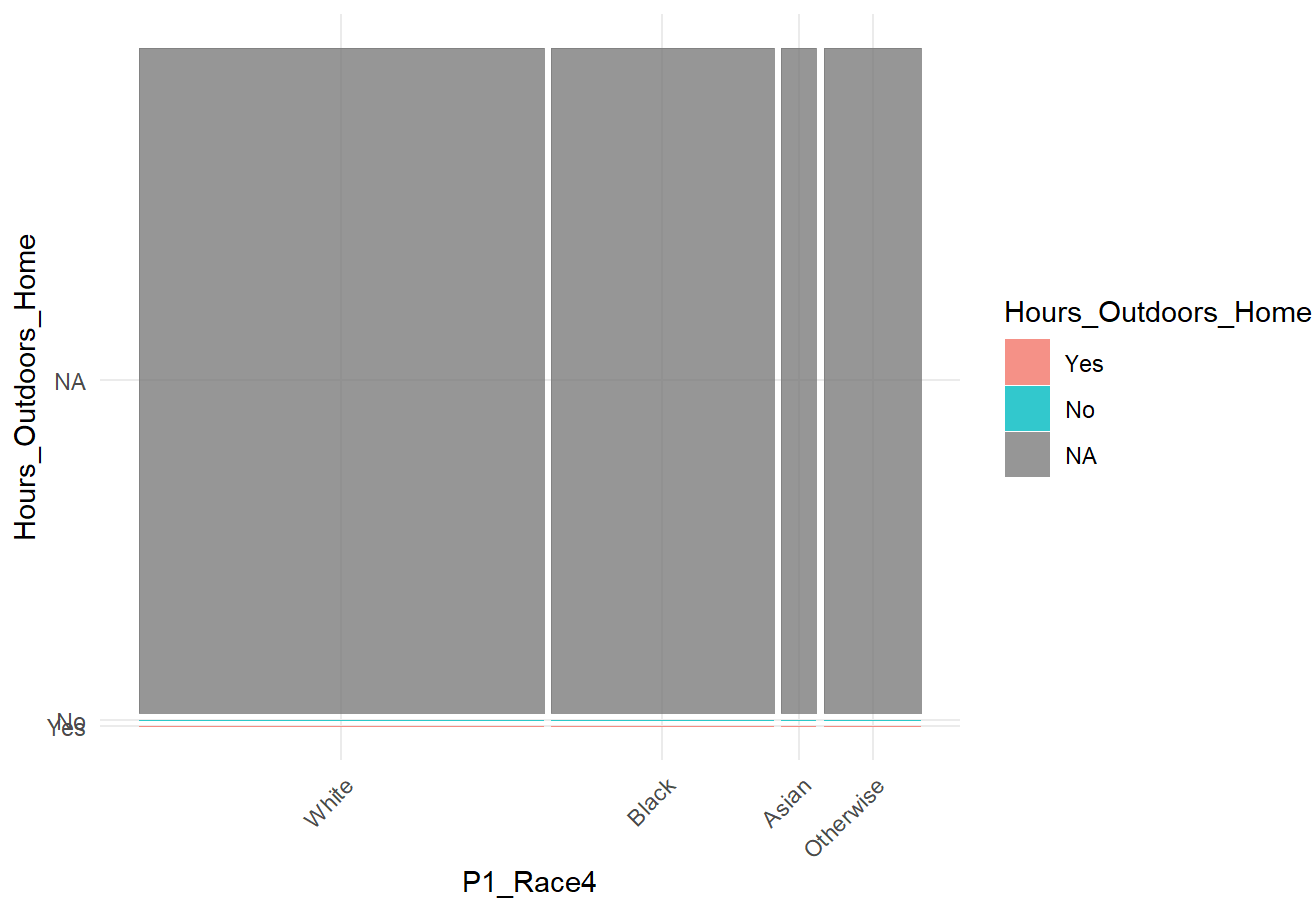
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and P1_Race4 -----
## Bivariate Analysis of Hours_Outdoors_Home and P1_Race4
##
## Frequency Table:
## Hours_Outdoors_Home White Black Asian Otherwise
##           Yes      0      0      0      0
##           No       0      0      0      0
##           <NA>    288    158    25    69
##
## Relative Frequency Table:
## Hours_Outdoors_Home White Black Asian Otherwise Total
##           Yes    0.0%  0.0%  0.0%    0.0%  0.0%
##           No     0.0%  0.0%  0.0%    0.0%  0.0%
##           <NA> 100.0% 100.0% 100.0%  100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of P1_Race4 and Hours_Outdoors_Home



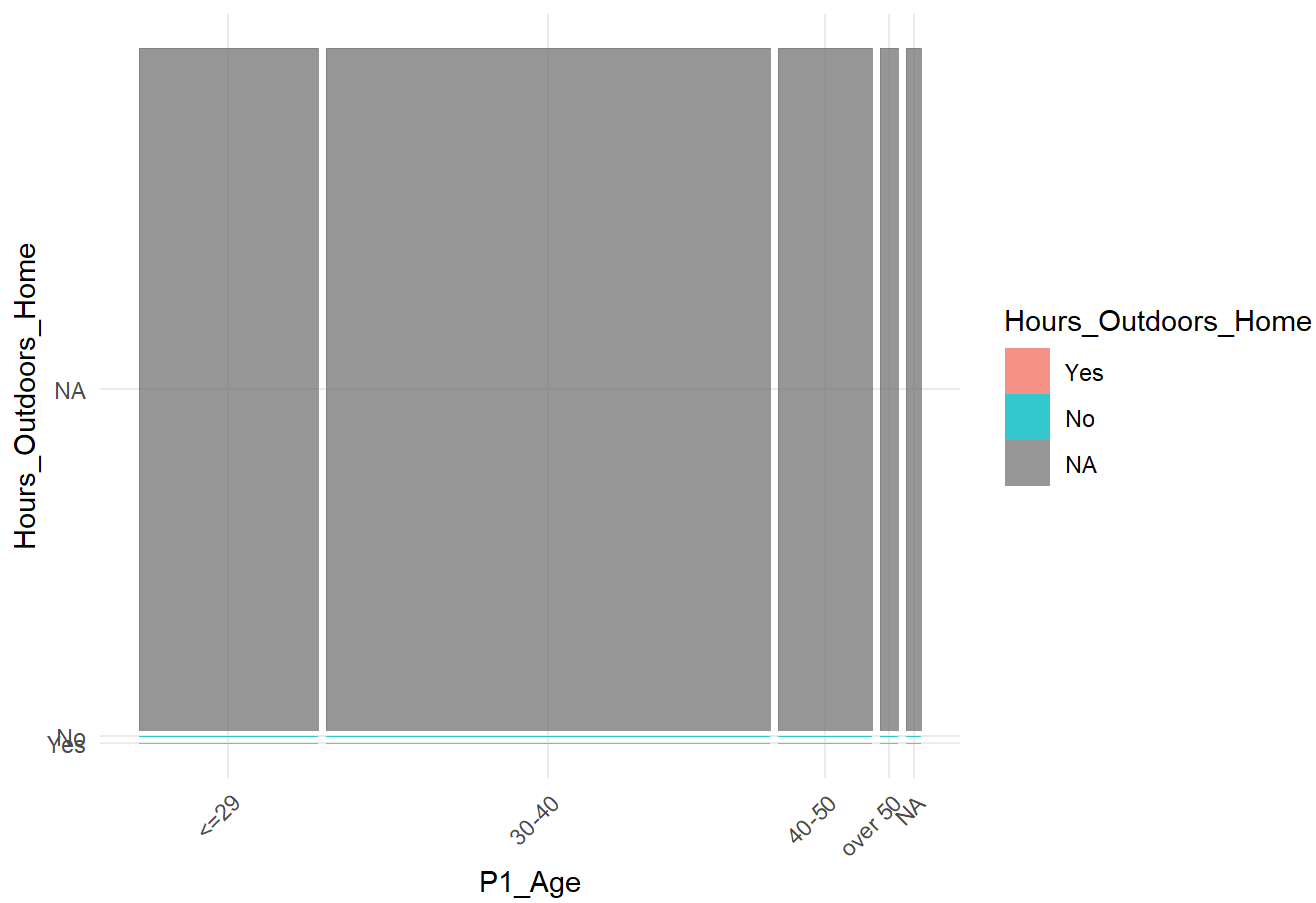
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and P1_Age -----
## Bivariate Analysis of Hours_Outdoors_Home and P1_Age
##
## Frequency Table:
##   Hours_Outdoors_Home  <=29  30-40  40-50  over 50  NA_
##                   Yes      0      0      0      0      0
##                   No       0      0      0      0      0
##                   <NA>  129    319    68     13    11
##
## Relative Frequency Table:
##   Hours_Outdoors_Home  <=29  30-40  40-50  over 50    NA_  Total
##                   Yes    0.0%   0.0%   0.0%   0.0%   0.0%  0.0%
##                   No     0.0%   0.0%   0.0%   0.0%   0.0%  0.0%
##                   <NA> 100.0% 100.0% 100.0% 100.0% 100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of P1_Age and Hours_Outdoors_Home



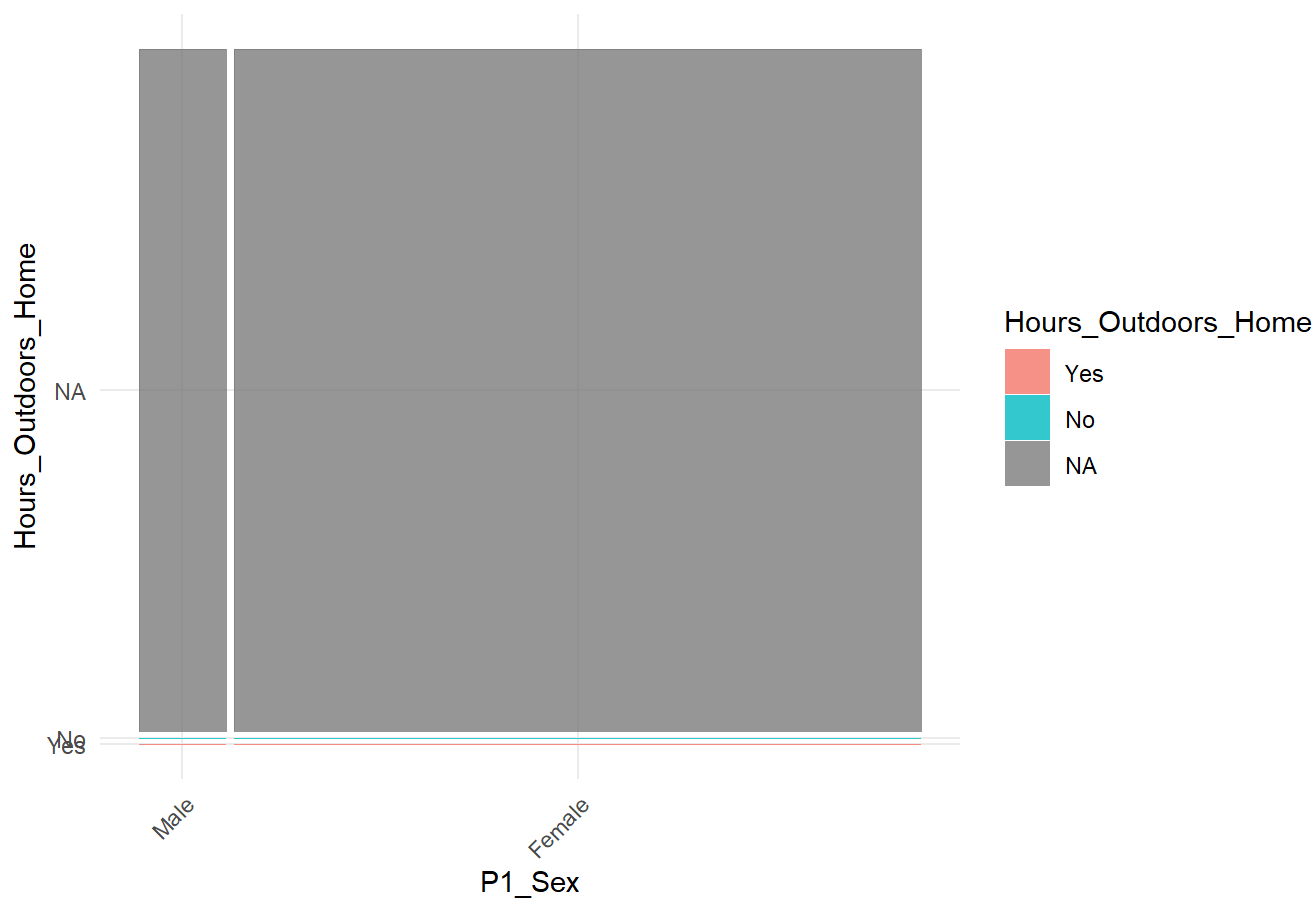
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and P1_Sex -----
## Bivariate Analysis of Hours_Outdoors_Home and P1_Sex
##
## Frequency Table:
##   Hours_Outdoors_Home   Male Female
##           Yes         0         0
##           No          0         0
##           <NA>       61       479
##
## Relative Frequency Table:
##   Hours_Outdoors_Home   Male Female Total
##           Yes        0.0%   0.0%   0.0%
##           No         0.0%   0.0%   0.0%
##           <NA>      100.0% 100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of P1_Sex and Hours_Outdoors_Home



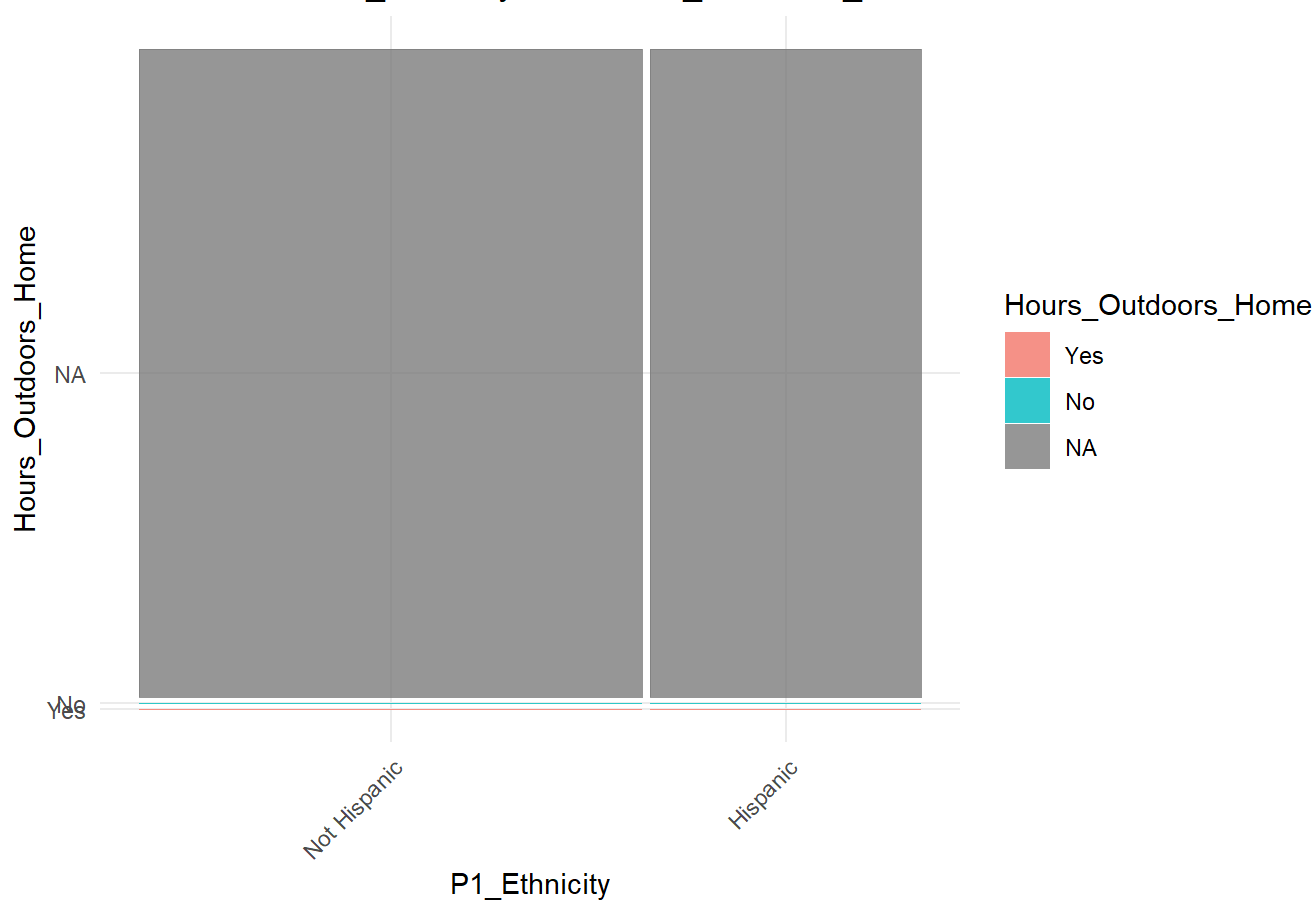
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and P1_Ethnicity -----
## Bivariate Analysis of Hours_Outdoors_Home and P1_Ethnicity
##
## Frequency Table:
##   Hours_Outdoors_Home Not Hispanic Hispanic
##                   Yes           0           0
##                   No            0           0
##                   <NA>        351        189
##
## Relative Frequency Table:
##   Hours_Outdoors_Home Not Hispanic Hispanic Total
##                   Yes           0.0%       0.0%  0.0%
##                   No            0.0%       0.0%  0.0%
##                   <NA>        100.0%    100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```


Mosaic Plot of P1_Ethnicity and Hours_Outdoors_Home



```
## Saving 7 x 5 in image
```

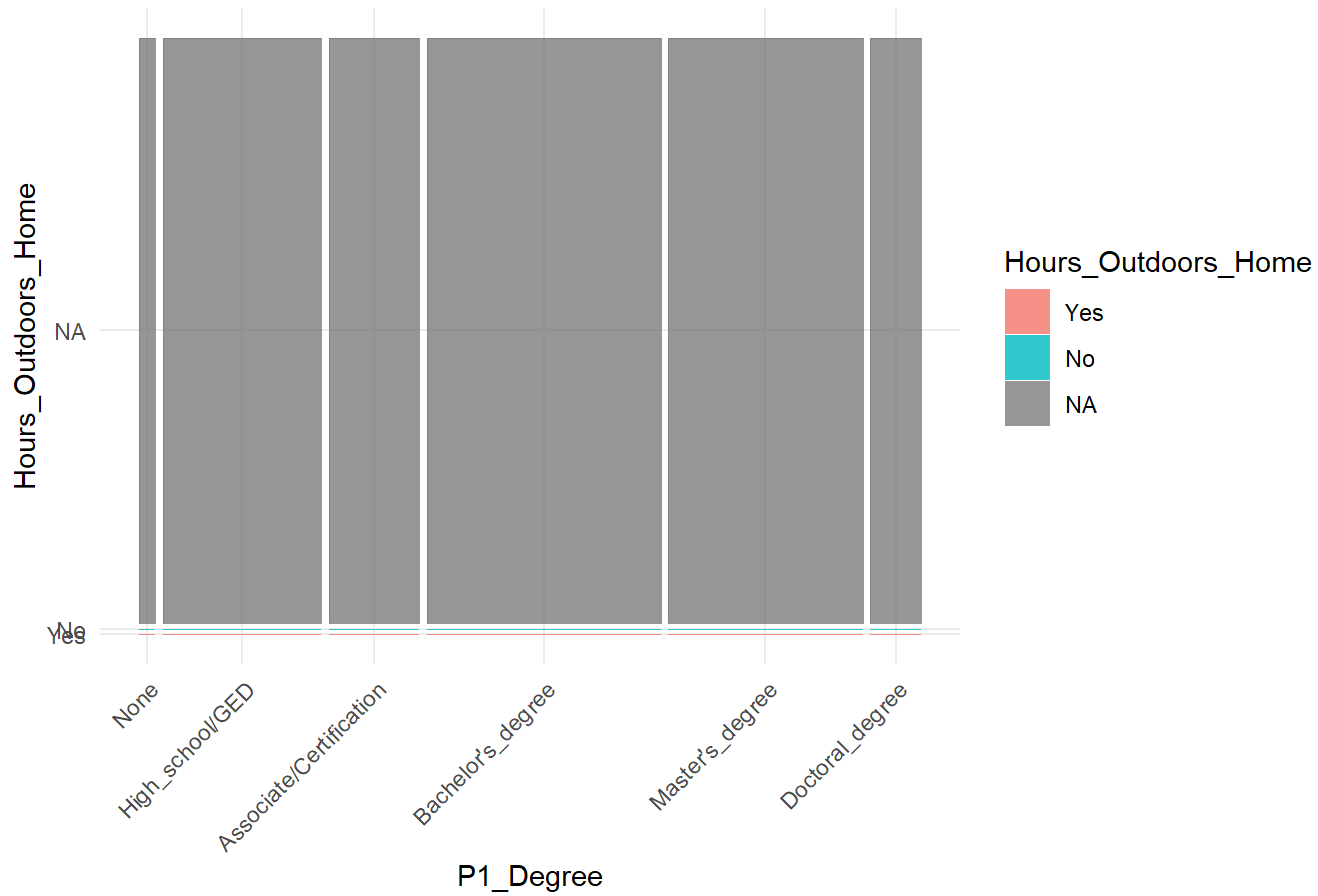
```
##
##
## ----- Analysis for Hours_Outdoors_Home and P1_Degree -----
## Bivariate Analysis of Hours_Outdoors_Home and P1_Degree
##
## Frequency Table:
## Hours_Outdoors_Home None High_school/GED Associate/Certification
##           Yes      0              0              0
##           No       0              0              0
##           <NA>    12             115             65
## Bachelor's_degree Master's_degree Doctoral_degree
##           0          0              0
##           0          0              0
##          170         141             37
##
## Relative Frequency Table:
## Hours_Outdoors_Home None High_school/GED Associate/Certification
##           Yes    0.0%              0.0%              0.0%
##           No     0.0%              0.0%              0.0%
##           <NA> 100.0%             100.0%             100.0%
## Bachelor's_degree Master's_degree Doctoral_degree Total
##           0.0%          0.0%              0.0%  0.0%
##           0.0%          0.0%              0.0%  0.0%
##          100.0%         100.0%             100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

##

Cramér's V: NaN

Mosaic Plot of P1_Degree and Hours_Outdoors_Home



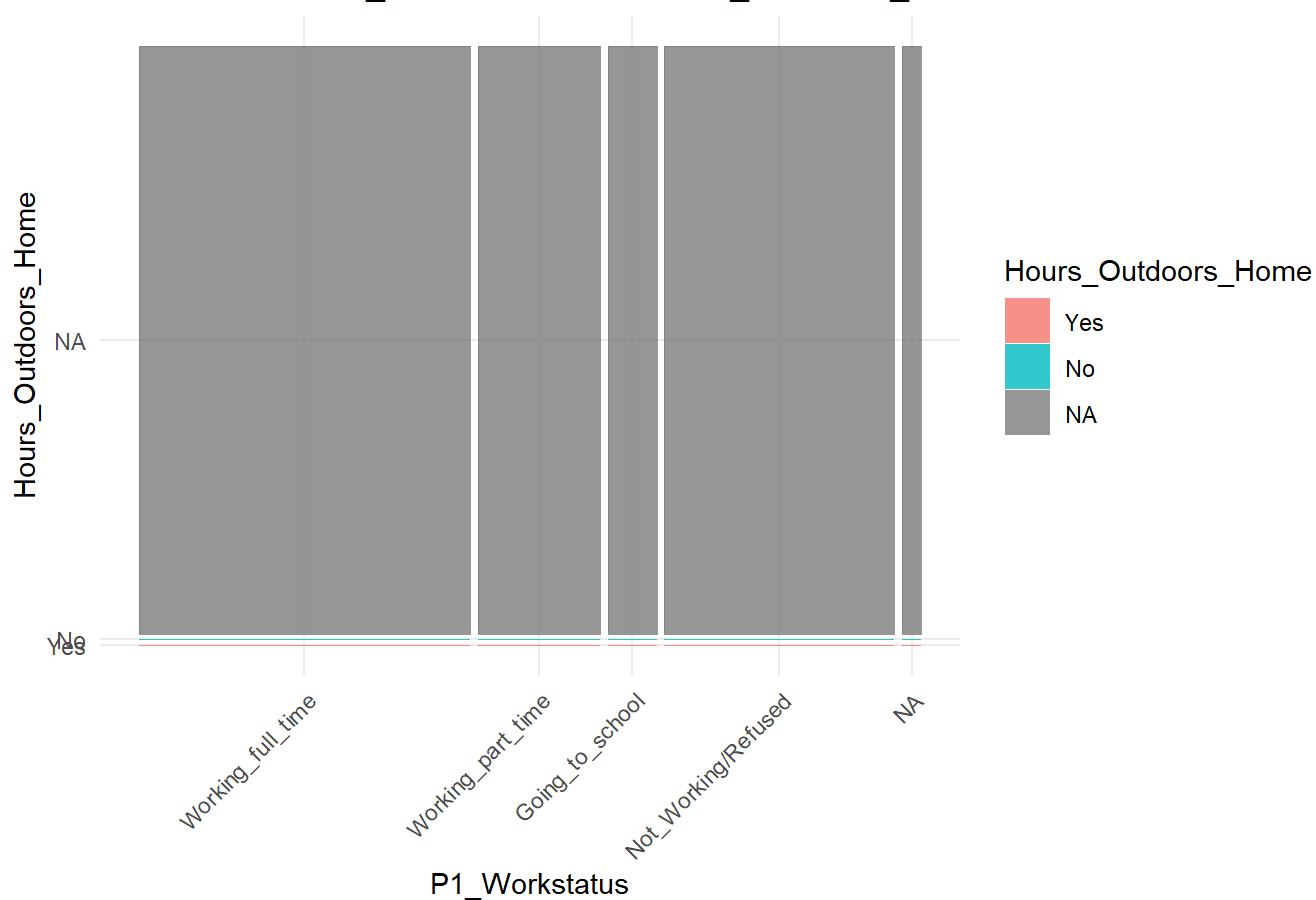
Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Outdoors_Home and P1_Workstatus -----
## Bivariate Analysis of Hours_Outdoors_Home and P1_Workstatus
##
## Frequency Table:
##   Hours_Outdoors_Home Working_full_time Working_part_time Going_to_school
##               Yes                0                0                0
##               No                 0                0                0
##               <NA>             238             88             35
##   Not_Working/Refused NA_
##               0    0
##               0    0
##               165  14
##
## Relative Frequency Table:
##   Hours_Outdoors_Home Working_full_time Working_part_time Going_to_school
##               Yes                0.0%                0.0%                0.0%
##               No                 0.0%                0.0%                0.0%
##               <NA>             100.0%             100.0%             100.0%
##   Not_Working/Refused    NA_    Total
##               0.0%    0.0%    0.0%
##               0.0%    0.0%    0.0%
##               100.0% 100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of P1_Workstatus and Hours_Outdoors_Home



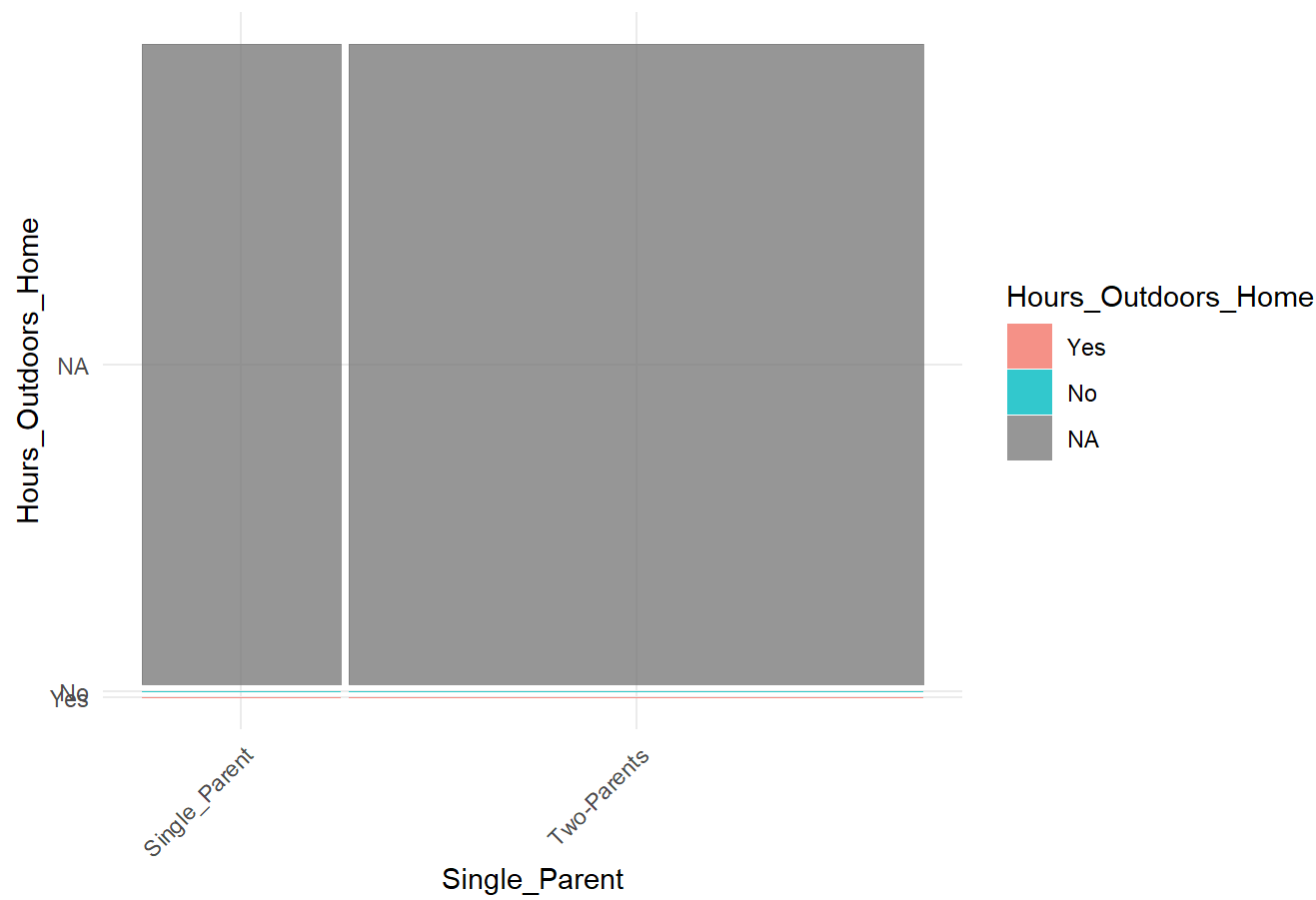
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and Single_Parent -----
## Bivariate Analysis of Hours_Outdoors_Home and Single_Parent
##
## Frequency Table:
##   Hours_Outdoors_Home Single_Parent Two-Parents
##               Yes           0           0
##               No           0           0
##               <NA>        139          401
##
## Relative Frequency Table:
##   Hours_Outdoors_Home Single_Parent Two-Parents  Total
##               Yes           0.0%           0.0%  0.0%
##               No           0.0%           0.0%  0.0%
##               <NA>        100.0%          100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of Single_Parent and Hours_Outdoors_Home



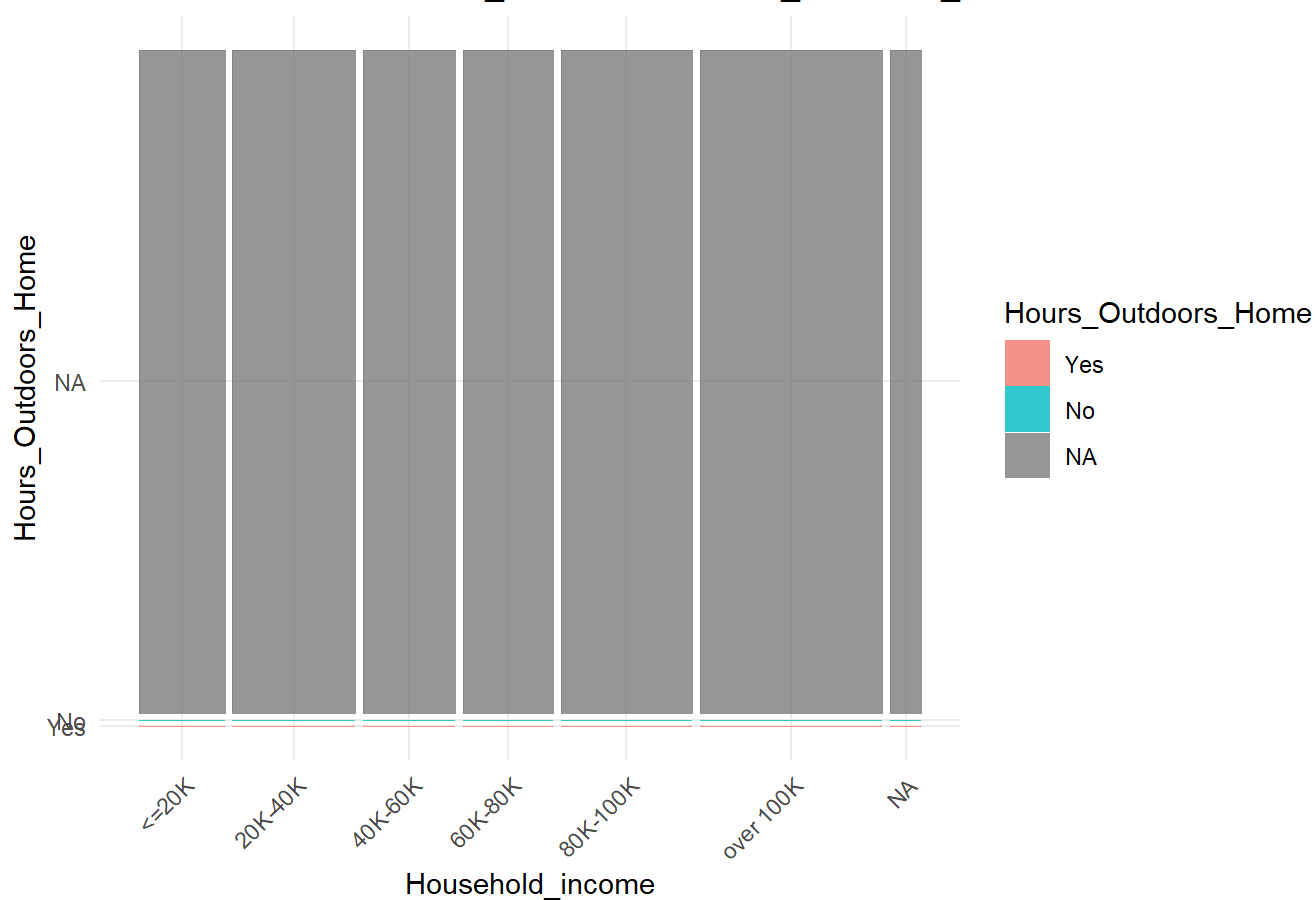
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and Household_income -----
## Bivariate Analysis of Hours_Outdoors_Home and Household_income
##
## Frequency Table:
## Hours_Outdoors_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes      0      0      0      0      0      0  0
##           No       0      0      0      0      0      0  0
##           <NA>    63     90     68     66     96    134  23
##
## Relative Frequency Table:
## Hours_Outdoors_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_
##           Yes    0.0%    0.0%    0.0%    0.0%    0.0%    0.0%  0.0%
##           No     0.0%    0.0%    0.0%    0.0%    0.0%    0.0%  0.0%
##           <NA>  100.0%  100.0%  100.0%  100.0%  100.0%  100.0% 100.0%
##
## Total
##    0.0%
##    0.0%
##   100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of Household_income and Hours_Outdoors_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and City -----
## Bivariate Analysis of Hours_Outdoors_Home and City
##
## Frequency Table:
##   Hours_Outdoors_Home Greensboro Miami Tucson
##                   Yes           0      0      0
##                   No            0      0      0
##                   <NA>        186   174   180
##
## Relative Frequency Table:
##   Hours_Outdoors_Home Greensboro  Miami Tucson  Total
##                   Yes           0.0%  0.0%  0.0%  0.0%
##                   No            0.0%  0.0%  0.0%  0.0%
##                   <NA>        100.0% 100.0% 100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of City and Hours_Outdoors_Home



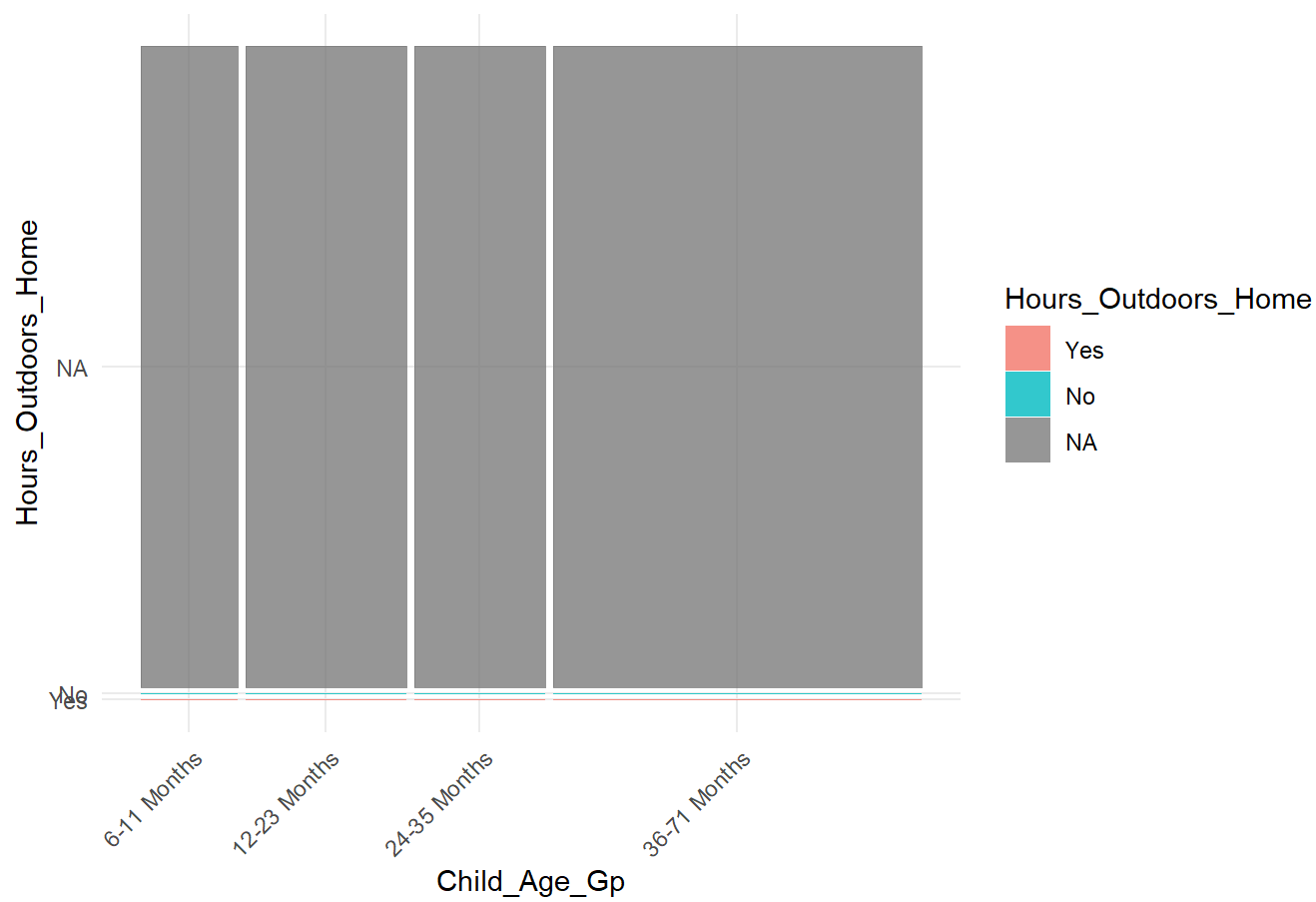
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and Child_Age_Gp -----
## Bivariate Analysis of Hours_Outdoors_Home and Child_Age_Gp
##
## Frequency Table:
##   Hours_Outdoors_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##                   Yes           0           0           0           0
##                   No           0           0           0           0
##                   <NA>        69          115          93          263
##
## Relative Frequency Table:
##   Hours_Outdoors_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months  Total
##                   Yes           0.0%          0.0%          0.0%          0.0%  0.0%
##                   No           0.0%          0.0%          0.0%          0.0%  0.0%
##                   <NA>        100.0%         100.0%         100.0%         100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

Mosaic Plot of Child_Age_Gp and Hours_Outdoors_Home



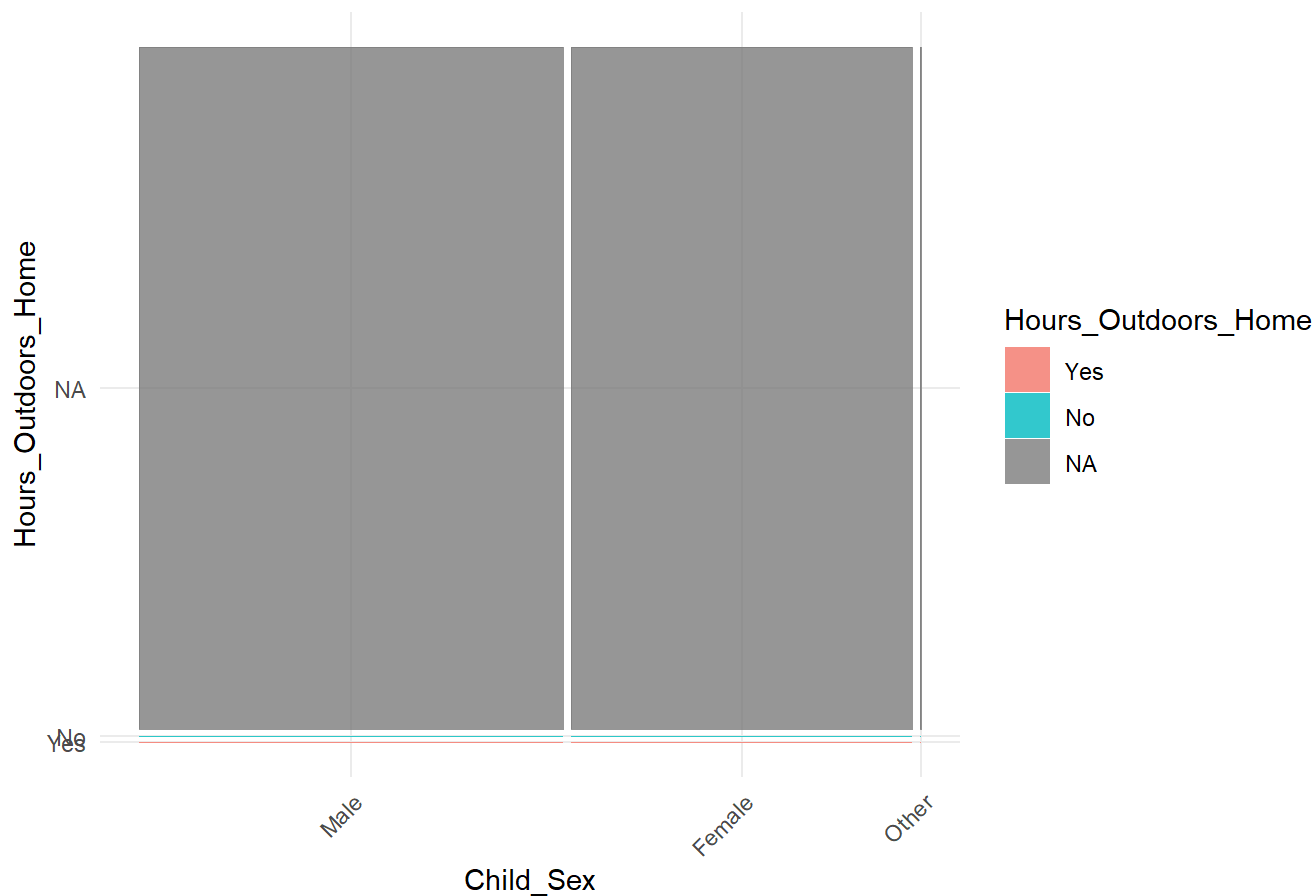
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and Child_Sex -----
## Bivariate Analysis of Hours_Outdoors_Home and Child_Sex
##
## Frequency Table:
##   Hours_Outdoors_Home Male Female Other
##               Yes      0        0      0
##               No       0        0      0
##               <NA>  299    240      1
##
## Relative Frequency Table:
##   Hours_Outdoors_Home   Male Female   Other   Total
##               Yes    0.0%   0.0%   0.0%   0.0%
##               No     0.0%   0.0%   0.0%   0.0%
##               <NA> 100.0% 100.0% 100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```


Mosaic Plot of Child_Sex and Hours_Outdoors_Home



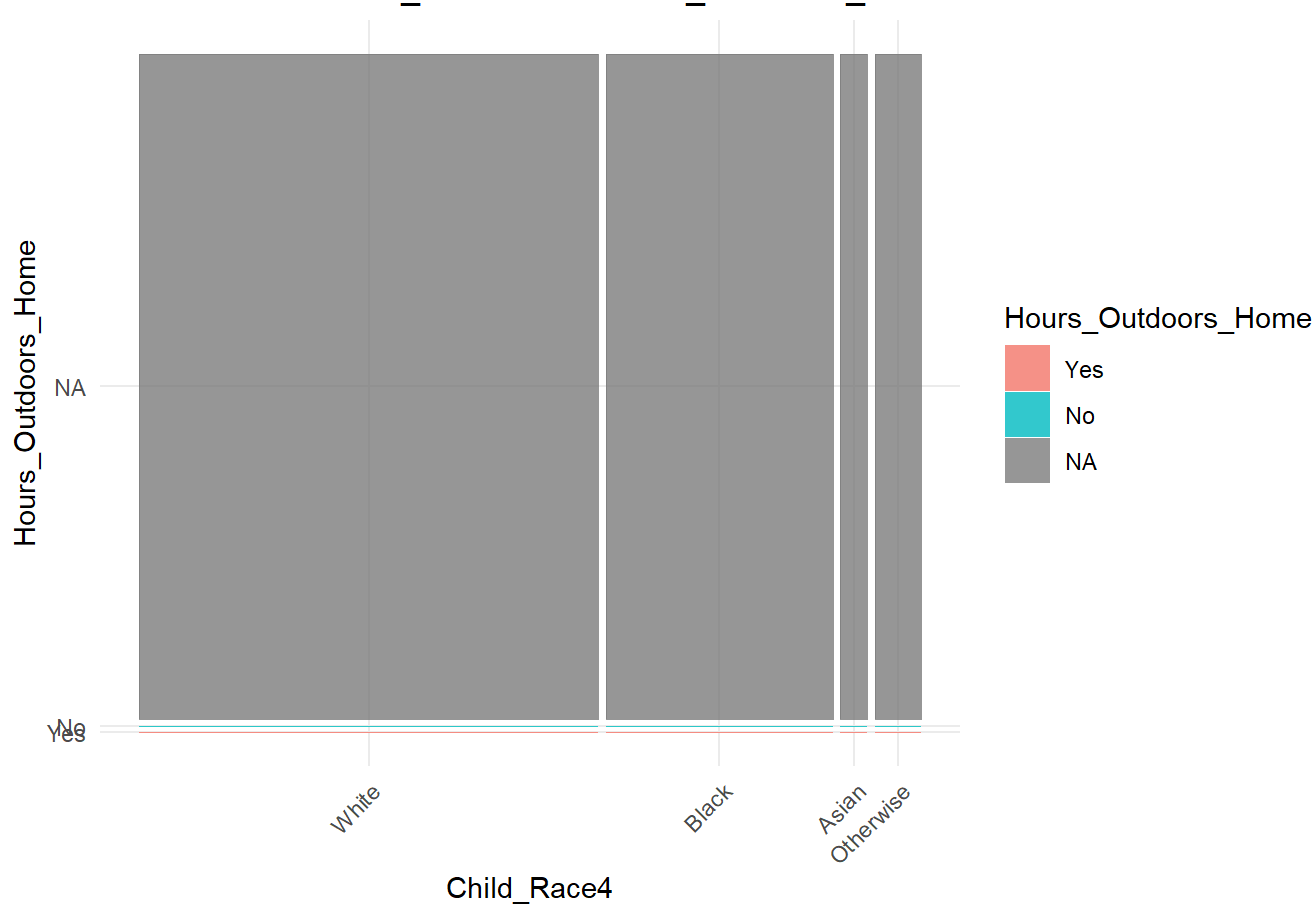
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Outdoors_Home and Child_Race4 -----
## Bivariate Analysis of Hours_Outdoors_Home and Child_Race4
##
## Frequency Table:
##   Hours_Outdoors_Home White Black Asian Otherwise
##           Yes         0      0      0           0
##           No          0      0      0           0
##           <NA>      327    161     19          33
##
## Relative Frequency Table:
##   Hours_Outdoors_Home White Black Asian Otherwise Total
##           Yes         0.0%   0.0%   0.0%         0.0%  0.0%
##           No          0.0%   0.0%   0.0%         0.0%  0.0%
##           <NA>      100.0% 100.0% 100.0%        100.0% 100.0%
```

```
## Error in Fisher's Exact Test: need 2 or more non-zero row marginals
```

```
##
## Cramér's V:  NaN
```

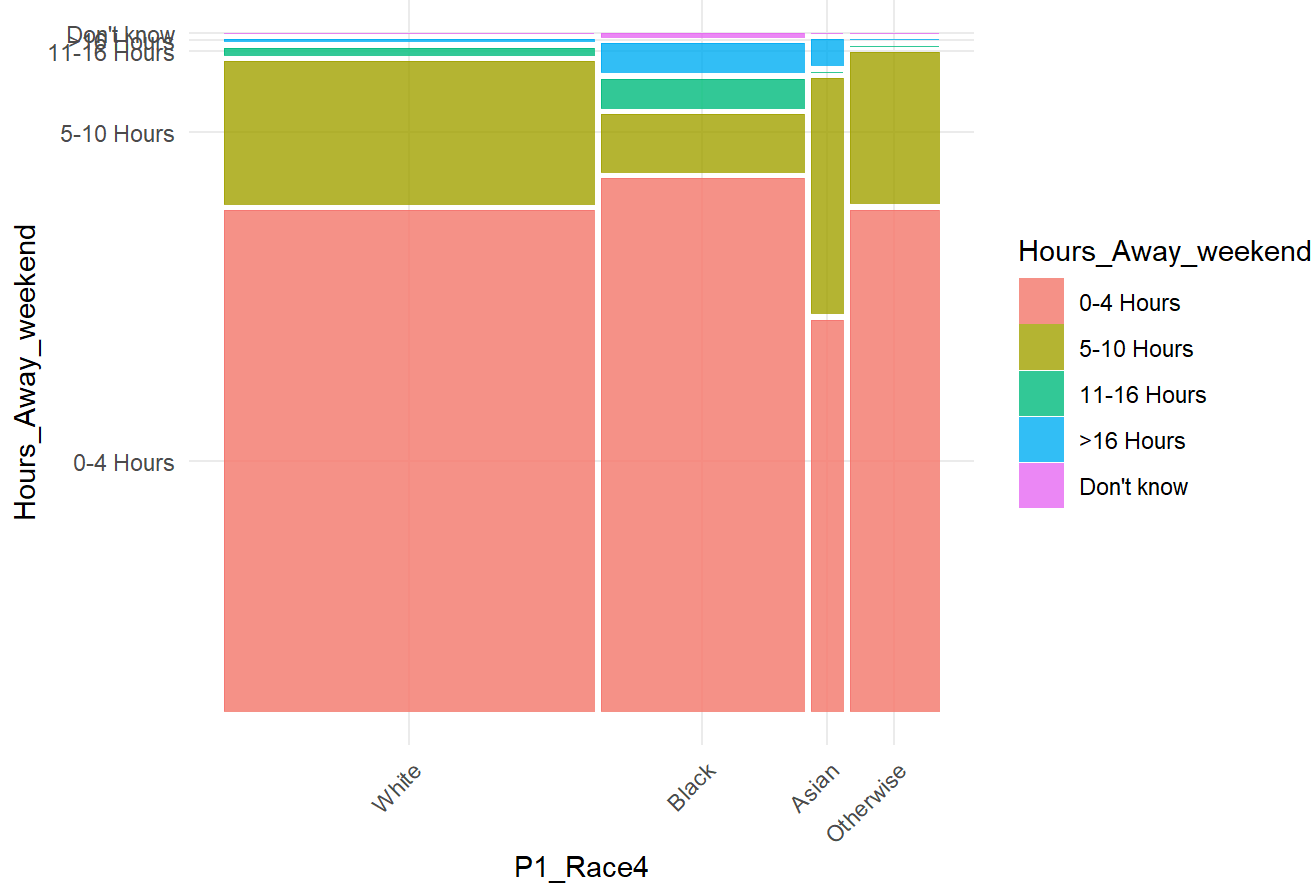
Mosaic Plot of Child_Race4 and Hours_Outdoors_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekend and P1_Race4 -----
## Bivariate Analysis of Hours_Away_weekend and P1_Race4
##
## Frequency Table:
##   Hours_Away_weekend White Black Asian Otherwise
##   0-4 Hours      221   129   15      53
##   5-10 Hours     63    14    9      16
##   11-16 Hours     3     7    0       0
##   >16 Hours      1     7    1       0
##   Don't know      0     1    0       0
##
## Relative Frequency Table:
##   Hours_Away_weekend White Black Asian Otherwise Total
##   0-4 Hours      76.7% 81.6% 60.0%      76.8% 77.4%
##   5-10 Hours     21.9%  8.9% 36.0%     23.2% 18.9%
##   11-16 Hours     1.0%  4.4%  0.0%      0.0%  1.9%
##   >16 Hours      0.3%  4.4%  4.0%      0.0%  1.7%
##   Don't know      0.0%  0.6%  0.0%      0.0%  0.2%
##
## Cramér's V:  0.1288179
```

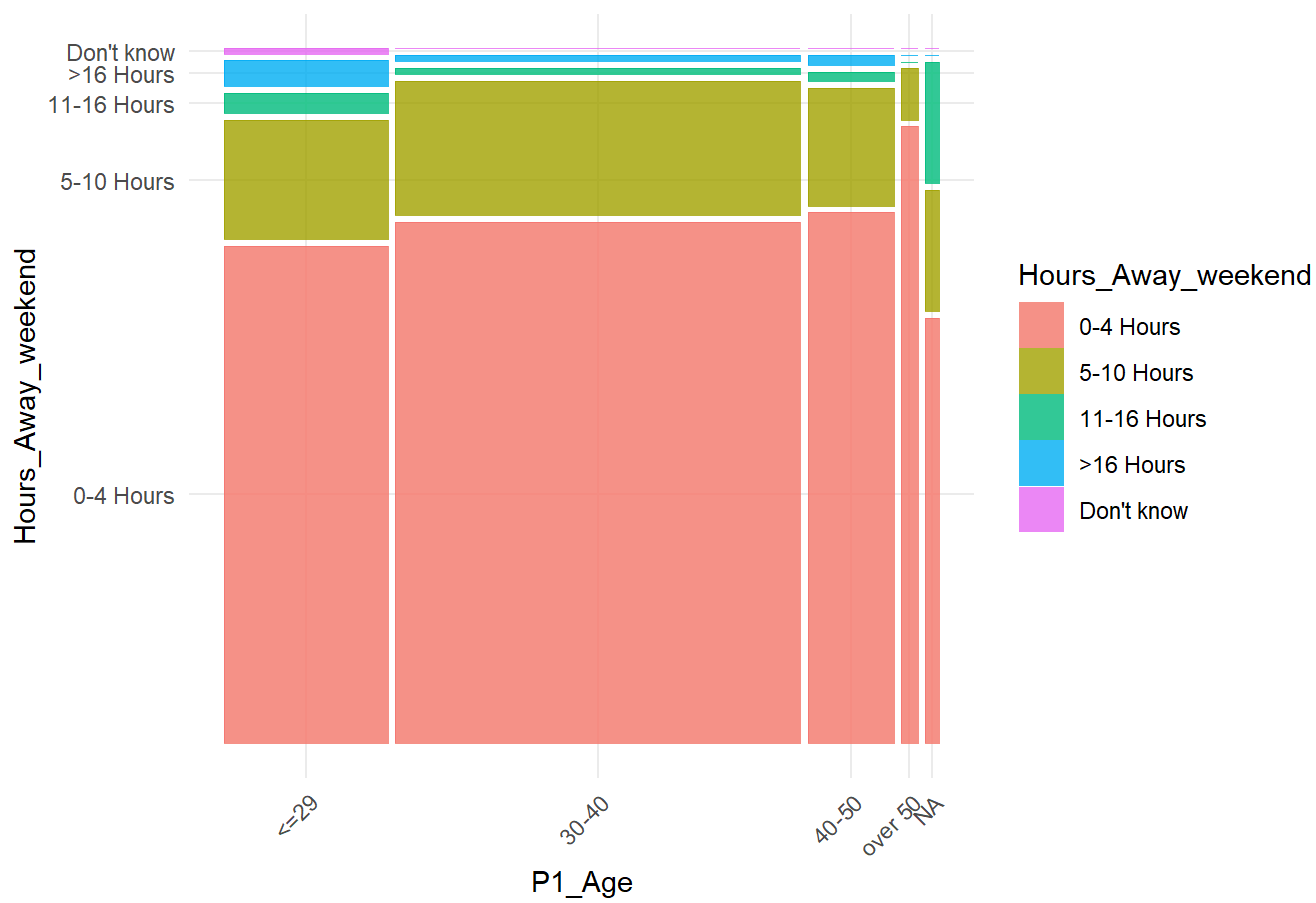
Mosaic Plot of P1_Race4 and Hours_Away_weekend



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekend and P1_Age -----
## Bivariate Analysis of Hours_Away_weekend and P1_Age
##
## Frequency Table:
##   Hours_Away_weekend  <=29  30-40  40-50  over 50  NA_
##           0-4 Hours    96    249    54      12    7
##           5-10 Hours    23     64    12       1    2
##           11-16 Hours     4      3     1        0    2
##           >16 Hours      5      3     1        0    0
##           Don't know      1      0     0        0    0
##
## Relative Frequency Table:
##   Hours_Away_weekend  <=29  30-40  40-50  over 50   NA_ Total
##           0-4 Hours  74.4%  78.1%  79.4%   92.3% 63.6% 77.4%
##           5-10 Hours  17.8%  20.1%  17.6%    7.7% 18.2% 18.9%
##           11-16 Hours   3.1%   0.9%   1.5%    0.0% 18.2%  1.9%
##           >16 Hours    3.9%   0.9%   1.5%    0.0%  0.0%  1.7%
##           Don't know   0.8%   0.0%   0.0%    0.0%  0.0%  0.2%
##
## Cramér's V:  0.07812429
```

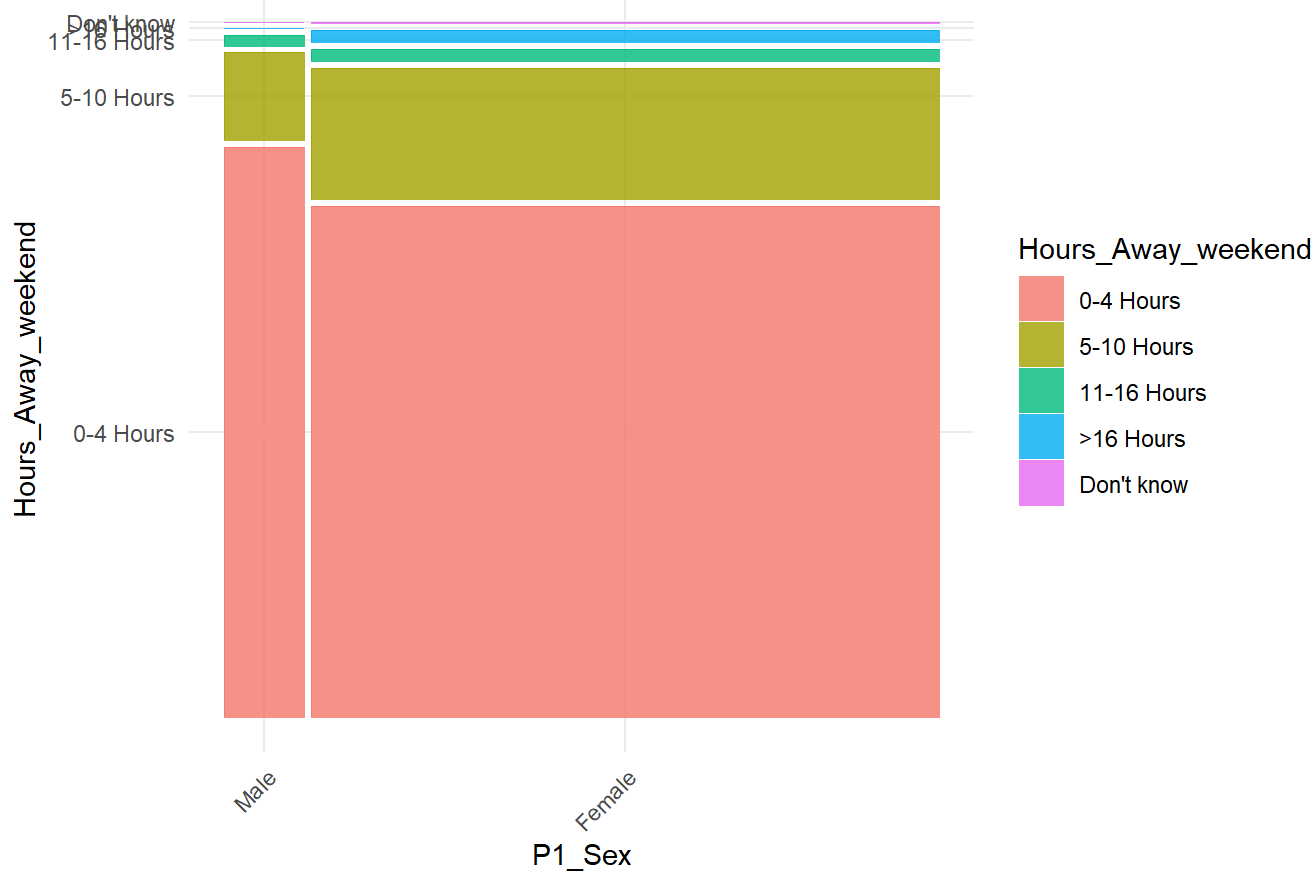
Mosaic Plot of P1_Age and Hours_Away_weekend



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekend and P1_Sex -----
## Bivariate Analysis of Hours_Away_weekend and P1_Sex
##
## Frequency Table:
##   Hours_Away_weekend   Male Female
##   0-4 Hours           52    366
##   5-10 Hours           8     94
##   11-16 Hours          1      9
##   >16 Hours            0      9
##   Don't know           0      1
##
## Relative Frequency Table:
##   Hours_Away_weekend   Male Female Total
##   0-4 Hours           85.2%  76.4%  77.4%
##   5-10 Hours           13.1%  19.6%  18.9%
##   11-16 Hours           1.6%   1.9%   1.9%
##   >16 Hours             0.0%   1.9%   1.7%
##   Don't know            0.0%   0.2%   0.2%
##
## Cramér's V:  0
```

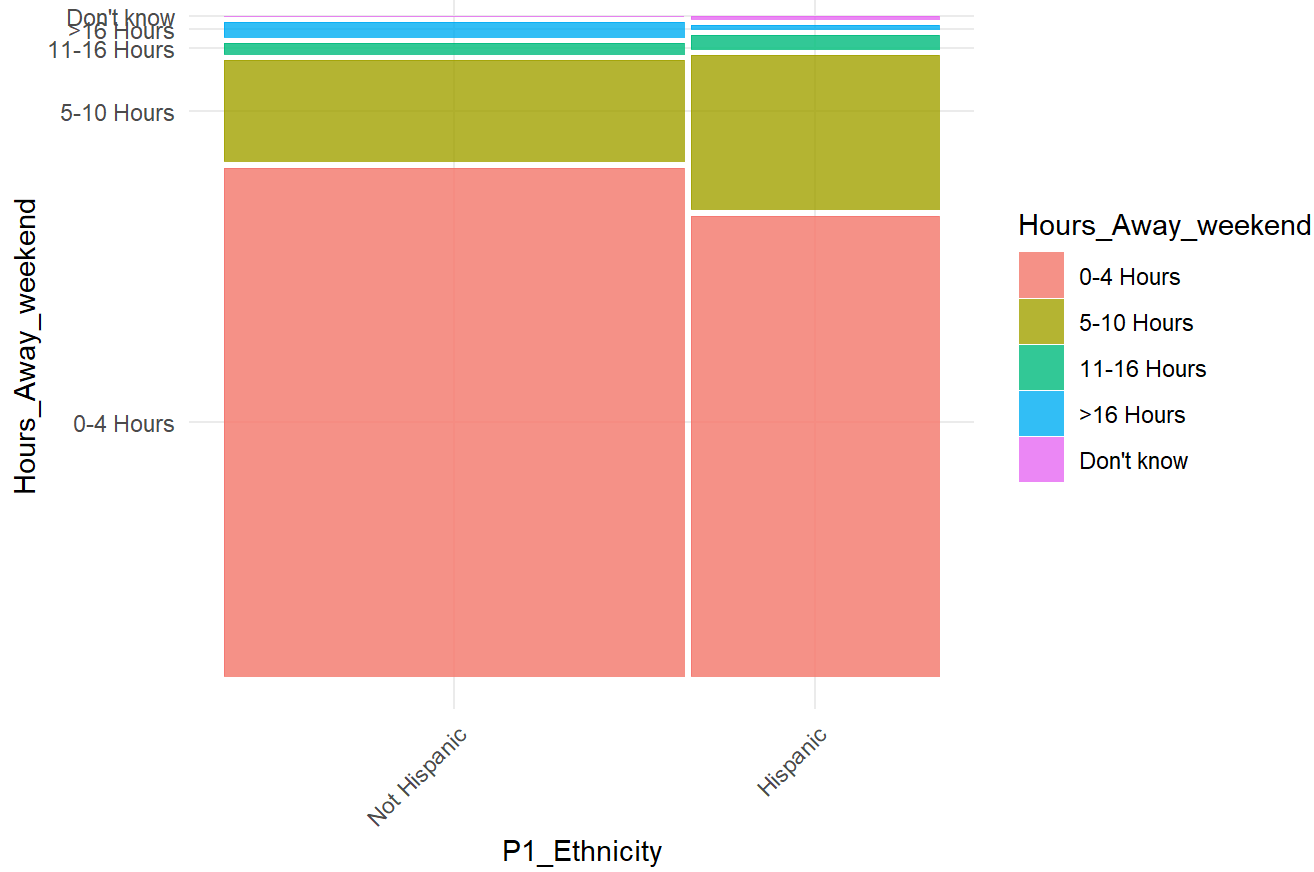
Mosaic Plot of P1_Sex and Hours_Away_weekend



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekend and P1_Ethnicity -----
## Bivariate Analysis of Hours_Away_weekend and P1_Ethnicity
##
## Frequency Table:
##   Hours_Away_weekend Not Hispanic Hispanic
##   0-4 Hours          281          137
##   5-10 Hours          56           46
##   11-16 Hours          6            4
##   >16 Hours           8            1
##   Don't know           0            1
##
## Relative Frequency Table:
##   Hours_Away_weekend Not Hispanic Hispanic Total
##   0-4 Hours          80.1%       72.5% 77.4%
##   5-10 Hours          16.0%       24.3% 18.9%
##   11-16 Hours          1.7%        2.1%  1.9%
##   >16 Hours           2.3%         0.5%  1.7%
##   Don't know           0.0%         0.5%  0.2%
##
## Cramér's V:  0.1028232
```

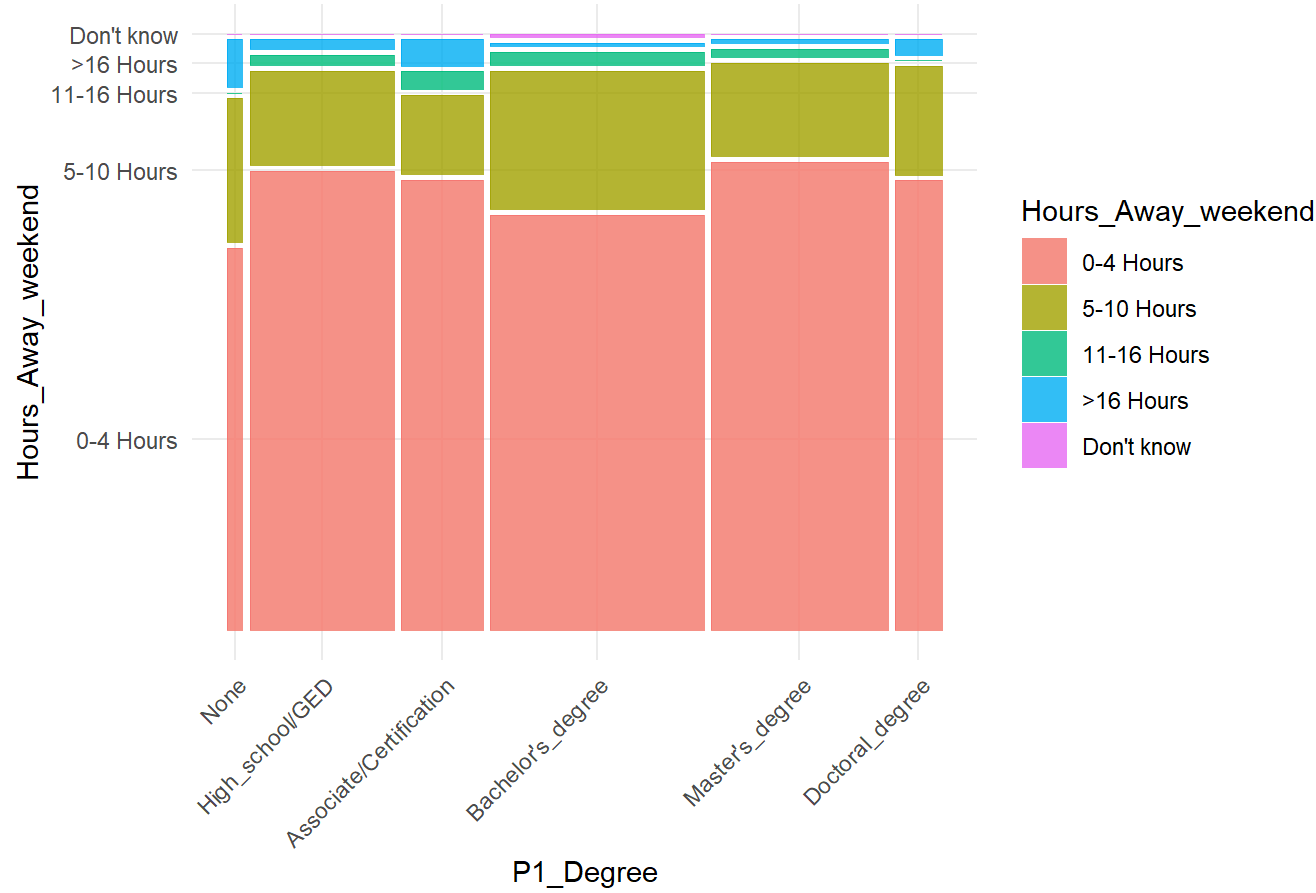
Mosaic Plot of P1_Ethnicity and Hours_Away_weekend



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekend and P1_Degree -----
## Bivariate Analysis of Hours_Away_weekend and P1_Degree
##
## Frequency Table:
## Hours_Away_weekend None High_school/GED Associate/Certification
##      0-4 Hours      8          92          51
##      5-10 Hours     3          19           9
##     11-16 Hours     0           2           2
##      >16 Hours     1           2           3
##     Don't know     0           0           0
## Bachelor's_degree Master's_degree Doctoral_degree
##      123          115          29
##      41           23           7
##       4            2           0
##       1            1           1
##       1            0           0
##
## Relative Frequency Table:
## Hours_Away_weekend None High_school/GED Associate/Certification
##      0-4 Hours 66.7%      80.0%      78.5%
##      5-10 Hours 25.0%      16.5%      13.8%
##     11-16 Hours  0.0%       1.7%       3.1%
##      >16 Hours  8.3%       1.7%       4.6%
##     Don't know  0.0%       0.0%       0.0%
## Bachelor's_degree Master's_degree Doctoral_degree Total
##      72.4%      81.6%      78.4% 77.4%
##      24.1%      16.3%      18.9% 18.9%
##       2.4%       1.4%       0.0%  1.9%
##       0.6%       0.7%       2.7%  1.7%
##       0.6%       0.0%       0.0%  0.2%
##
## Cramér's V:  0
```

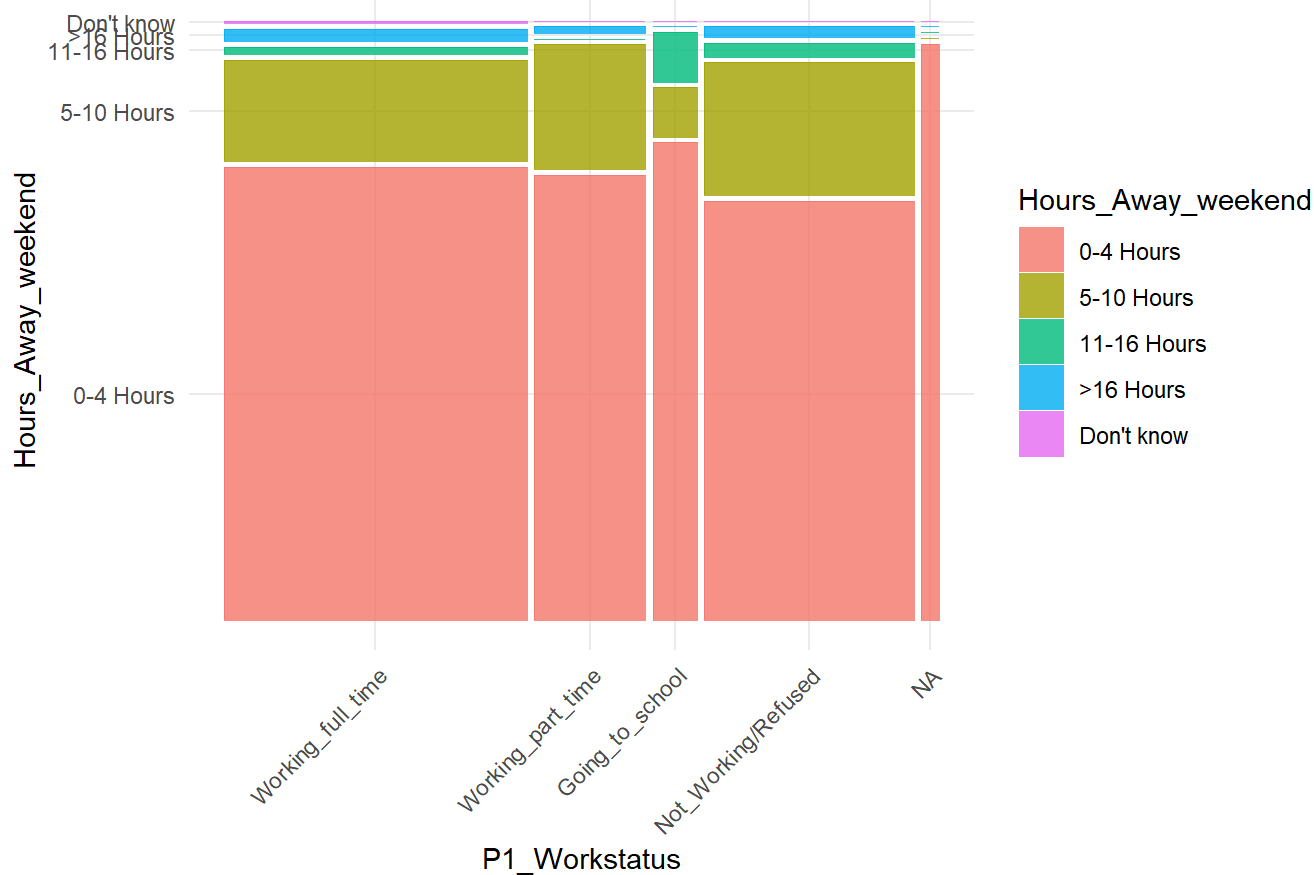
Mosaic Plot of P1_Degree and Hours_Away_weekend



Saving 7 x 5 in image


```
##
##
## ----- Analysis for Hours_Away_weekend and P1_Workstatus -----
## Bivariate Analysis of Hours_Away_weekend and P1_Workstatus
##
## Frequency Table:
## Hours_Away_weekend Working_full_time Working_part_time Going_to_school
##      0-4 Hours      187      68      29
##      5-10 Hours      42      19      3
##     11-16 Hours       3       0      3
##      >16 Hours       5       1      0
##      Don't know       1       0      0
## Not_Working/Refused NA_
##      120  14
##      38   0
##       4   0
##       3   0
##       0   0
##
## Relative Frequency Table:
## Hours_Away_weekend Working_full_time Working_part_time Going_to_school
##      0-4 Hours      78.6%      77.3%      82.9%
##      5-10 Hours      17.6%      21.6%      8.6%
##     11-16 Hours       1.3%       0.0%      8.6%
##      >16 Hours       2.1%       1.1%      0.0%
##      Don't know       0.4%       0.0%      0.0%
## Not_Working/Refused    NA_ Total
##      72.7% 100.0% 77.4%
##      23.0%   0.0% 18.9%
##       2.4%   0.0%  1.9%
##       1.8%   0.0%  1.7%
##       0.0%   0.0%  0.2%
##
## Cramér's V:  0.05235249
```

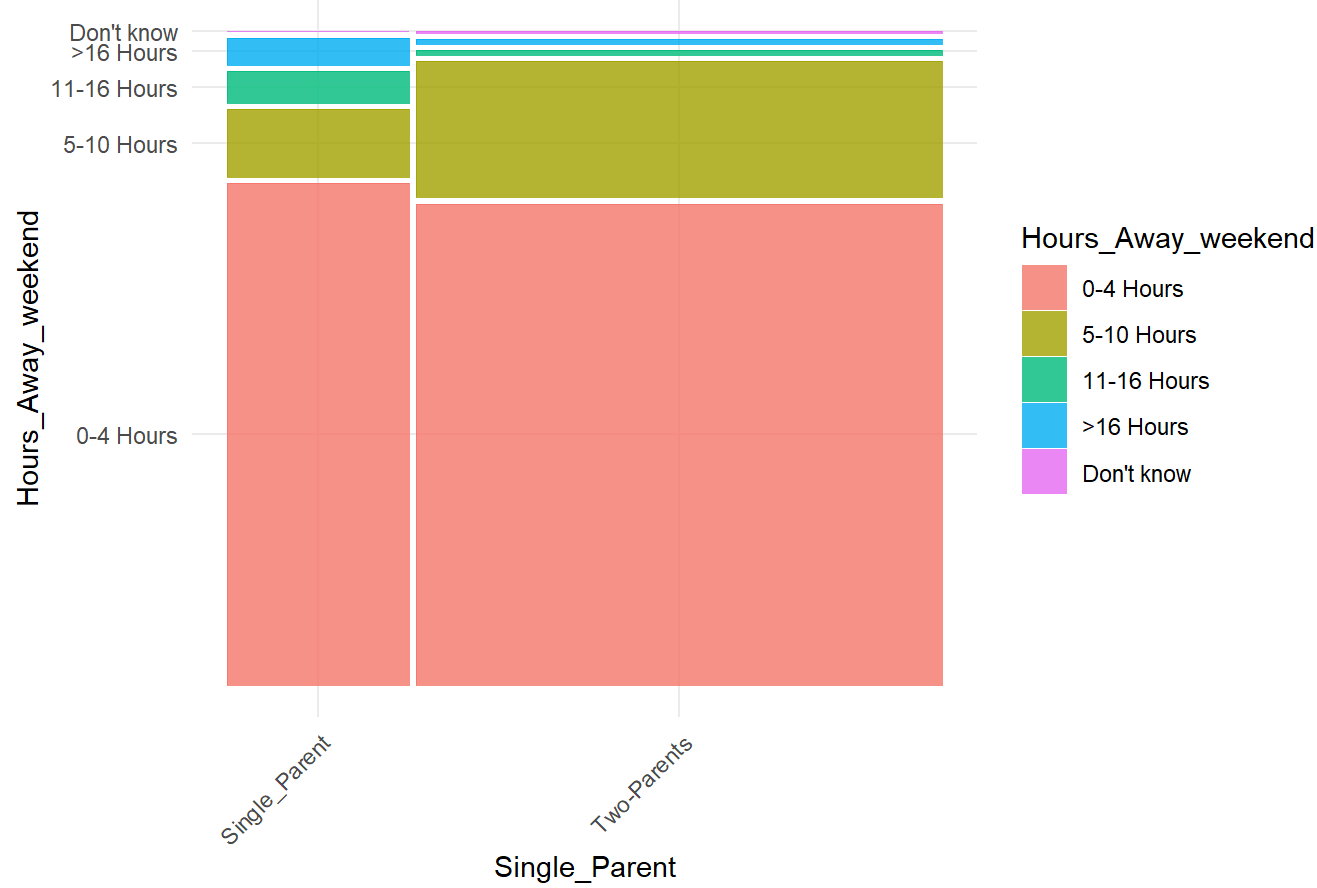
Mosaic Plot of P1_Workstatus and Hours_Away_weekend



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekend and Single_Parent -----
## Bivariate Analysis of Hours_Away_weekend and Single_Parent
##
## Frequency Table:
##   Hours_Away_weekend Single_Parent Two-Parents
##           0-4 Hours           111           307
##           5-10 Hours            15            87
##           11-16 Hours             7             3
##           >16 Hours              6             3
##           Don't know              0             1
##
## Relative Frequency Table:
##   Hours_Away_weekend Single_Parent Two-Parents Total
##           0-4 Hours           79.9%           76.6% 77.4%
##           5-10 Hours           10.8%           21.7% 18.9%
##           11-16 Hours            5.0%            0.7%  1.9%
##           >16 Hours             4.3%            0.7%  1.7%
##           Don't know            0.0%            0.2%  0.2%
##
## Cramér's V:  0.1979307
```

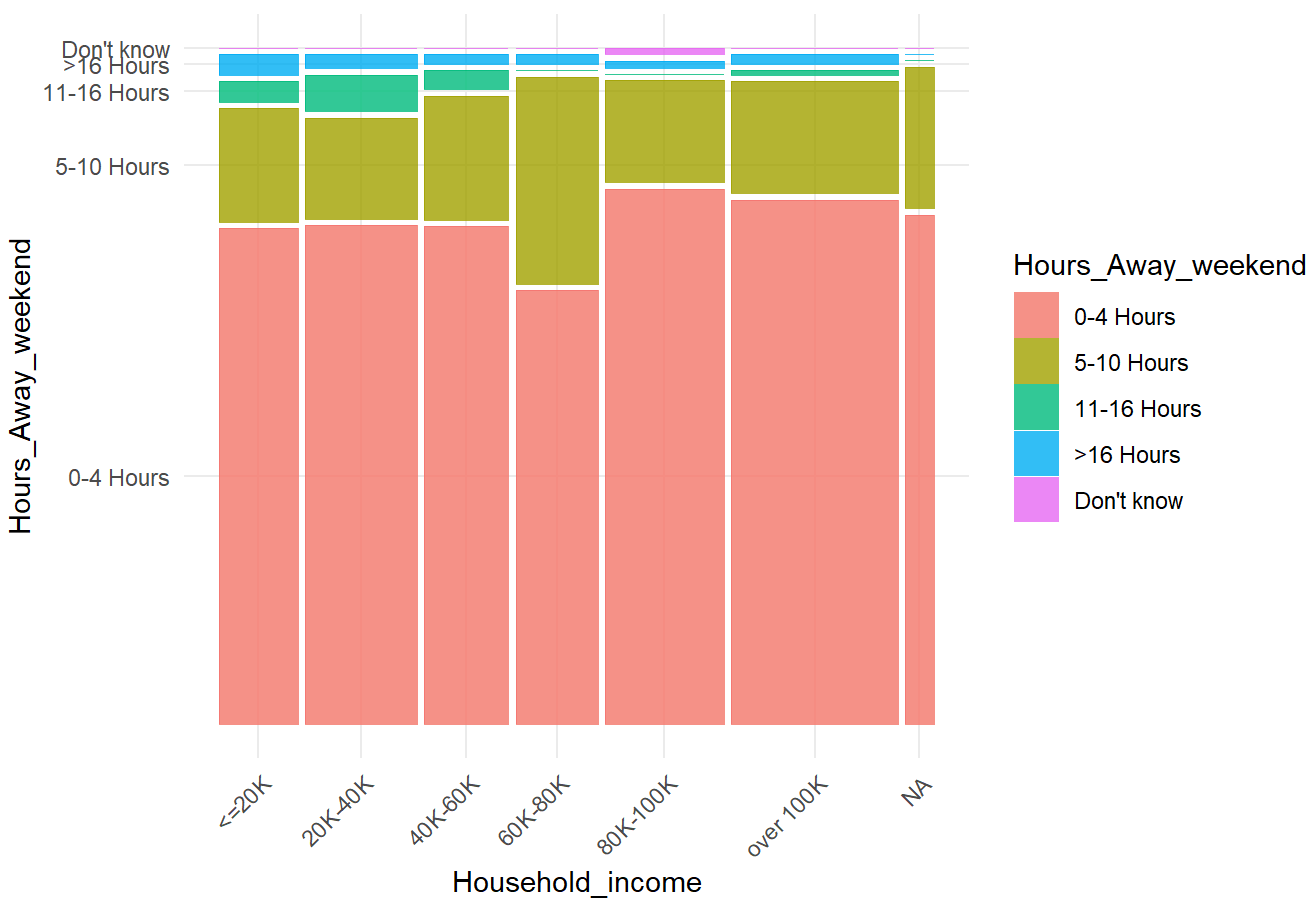
Mosaic Plot of Single_Parent and Hours_Away_weekend



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Hours_Away_weekend and Household_income -----
## Bivariate Analysis of Hours_Away_weekend and Household_income
##
## Frequency Table:
##   Hours_Away_weekend <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           0-4 Hours    48      69      52      44      79      108  18
##           5-10 Hours   11      14      13      21      15      23   5
##          11-16 Hours    2       5       2       0       0       1   0
##           >16 Hours    2       2       1       1       1       2   0
##           Don't know    0       0       0       0       1       0   0
##
## Relative Frequency Table:
##   Hours_Away_weekend <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_
##           0-4 Hours 76.2%  76.7%  76.5%  66.7%   82.3%   80.6% 78.3%
##           5-10 Hours 17.5%  15.6%  19.1%  31.8%  15.6%   17.2% 21.7%
##          11-16 Hours  3.2%   5.6%   2.9%   0.0%   0.0%    0.7% 0.0%
##           >16 Hours  3.2%   2.2%   1.5%   1.5%   1.0%    1.5% 0.0%
##           Don't know  0.0%   0.0%   0.0%   0.0%   1.0%    0.0% 0.0%
##
## Total
## 77.4%
## 18.9%
##  1.9%
##  1.7%
##  0.2%
##
## Cramér's V:  0.03770377
```

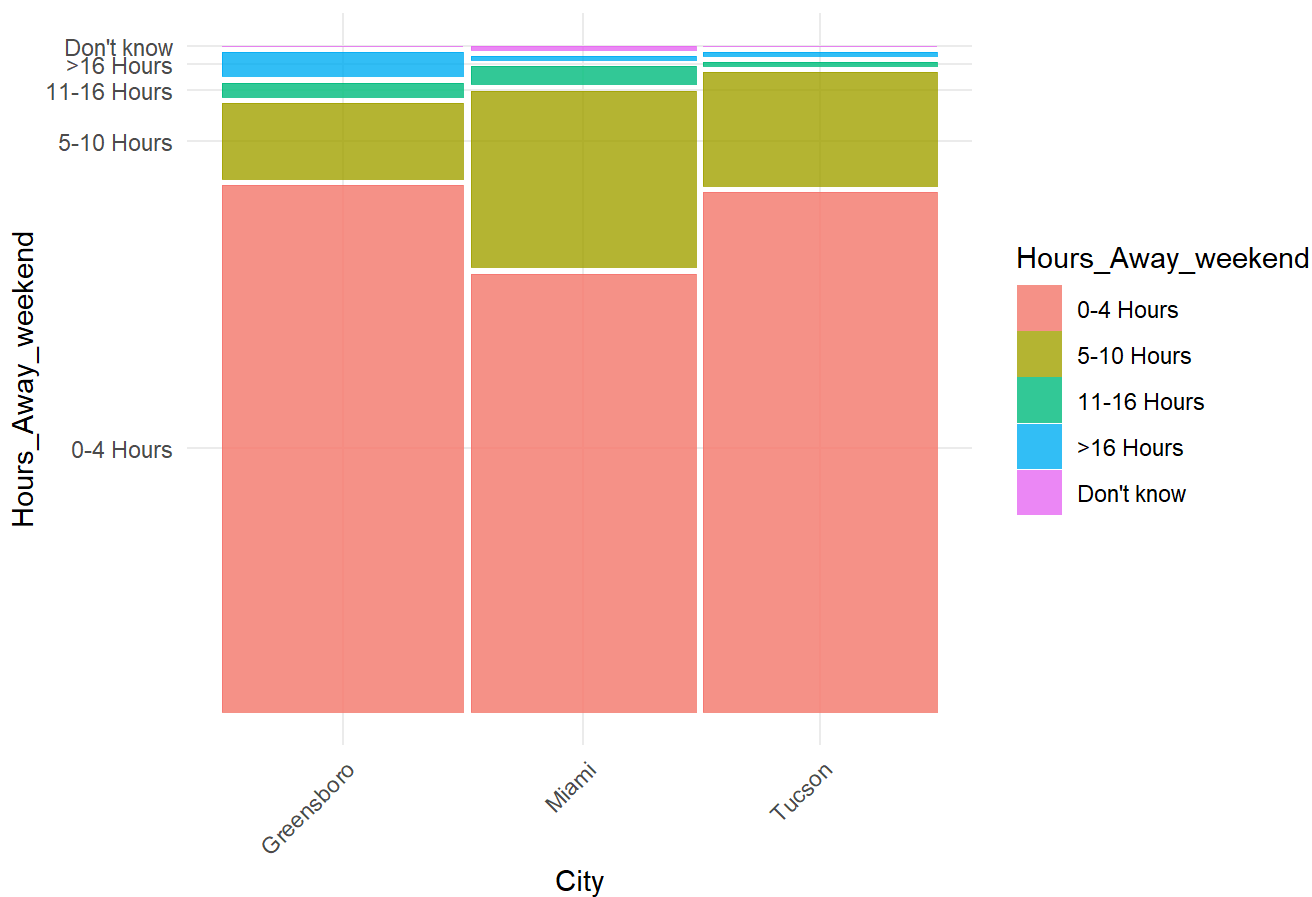
Mosaic Plot of Household_income and Hours_Away_weekend



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Away_weekend and City -----
## Bivariate Analysis of Hours_Away_weekend and City
##
## Frequency Table:
##   Hours_Away_weekend Greensboro Miami Tucson
##           0-4 Hours      153    119    146
##           5-10 Hours       22     48     32
##          11-16 Hours        4      5      1
##           >16 Hours        7      1      1
##           Don't know        0      1      0
##
## Relative Frequency Table:
##   Hours_Away_weekend Greensboro Miami Tucson Total
##           0-4 Hours    82.3% 68.4% 81.1% 77.4%
##           5-10 Hours   11.8% 27.6% 17.8% 18.9%
##          11-16 Hours    2.2%  2.9%  0.6%  1.9%
##           >16 Hours    3.8%  0.6%  0.6%  1.7%
##           Don't know    0.0%  0.6%  0.0%  0.2%
##
## Cramér's V:  0.1328278
```

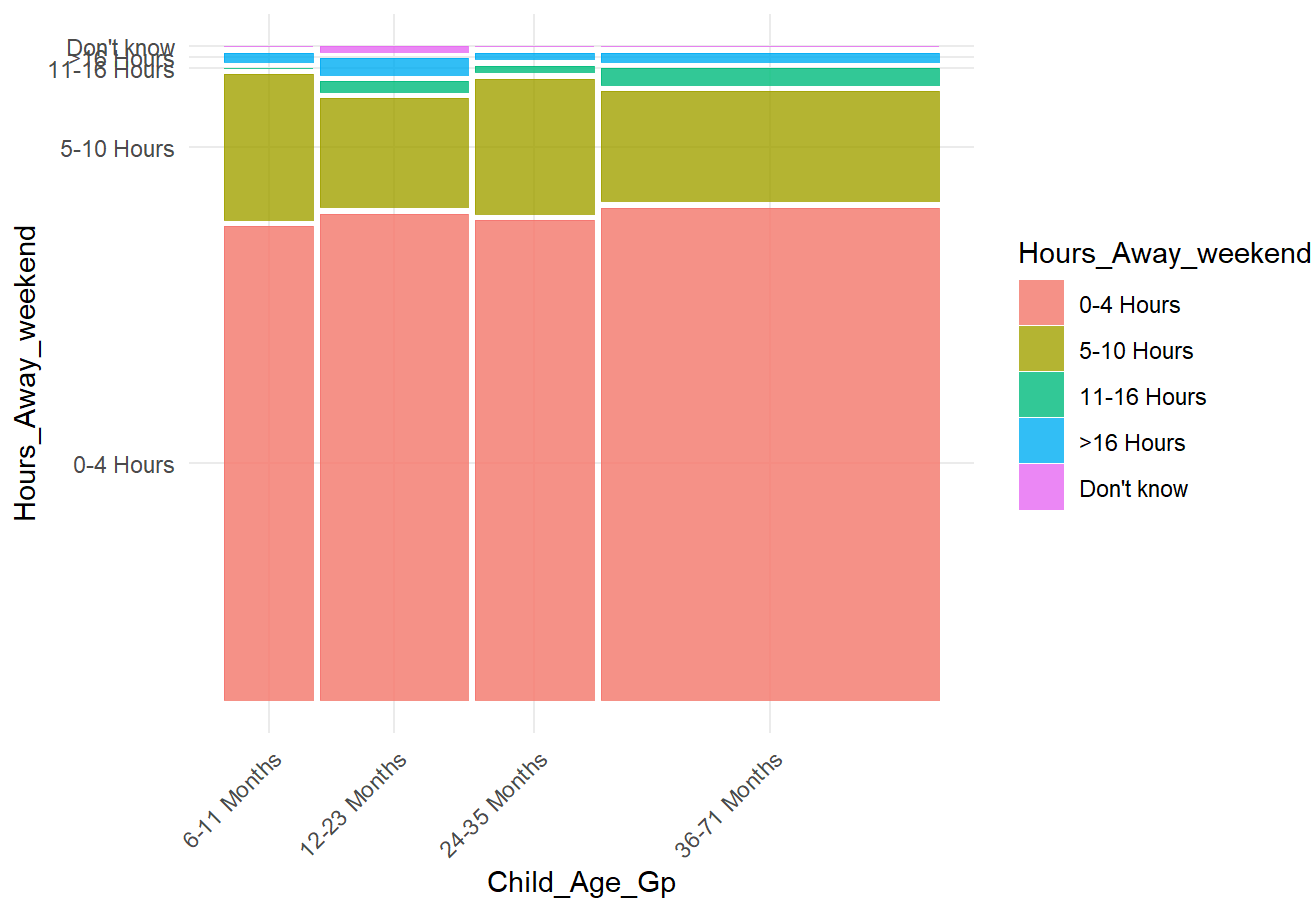
Mosaic Plot of City and Hours_Away_weekend



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Away_weekend and Child_Age_Gp -----
## Bivariate Analysis of Hours_Away_weekend and Child_Age_Gp
##
## Frequency Table:
##   Hours_Away_weekend 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##           0-4 Hours           52           89           71           206
##           5-10 Hours           16           20           20           46
##           11-16 Hours           0            2            1            7
##           >16 Hours            1            3            1            4
##           Don't know            0            1            0            0
##
## Relative Frequency Table:
##   Hours_Away_weekend 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##           0-4 Hours       75.4%       77.4%       76.3%       78.3% 77.4%
##           5-10 Hours       23.2%       17.4%       21.5%       17.5% 18.9%
##           11-16 Hours        0.0%        1.7%        1.1%        2.7%  1.9%
##           >16 Hours         1.4%        2.6%        1.1%        1.5%  1.7%
##           Don't know         0.0%        0.9%        0.0%        0.0%  0.2%
##
## Cramér's V:  0
```

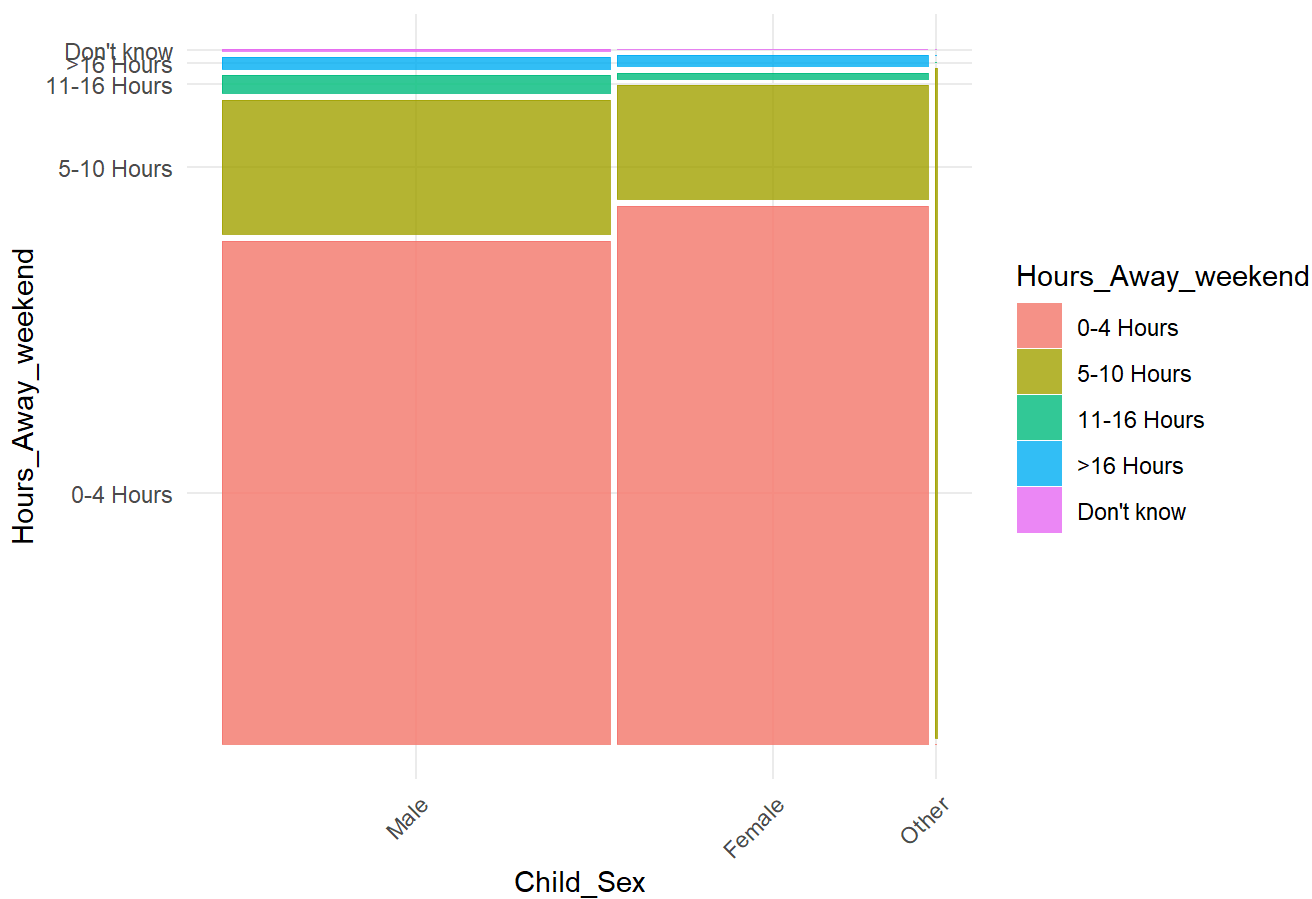
Mosaic Plot of Child_Age_Gp and Hours_Away_weekend



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Away_weekend and Child_Sex -----
## Bivariate Analysis of Hours_Away_weekend and Child_Sex
##
## Frequency Table:
##   Hours_Away_weekend Male Female Other
##           0-4 Hours  225    193    0
##           5-10 Hours   60    41    1
##          11-16 Hours    8     2    0
##           >16 Hours    5     4    0
##          Don't know    1     0    0
##
## Relative Frequency Table:
##   Hours_Away_weekend  Male Female  Other Total
##           0-4 Hours 75.3%  80.4%   0.0% 77.4%
##           5-10 Hours 20.1%  17.1% 100.0% 18.9%
##          11-16 Hours  2.7%   0.8%   0.0%  1.9%
##           >16 Hours  1.7%   1.7%   0.0%  1.7%
##          Don't know  0.3%   0.0%   0.0%  0.2%
##
## Cramér's V:  0.02396181
```

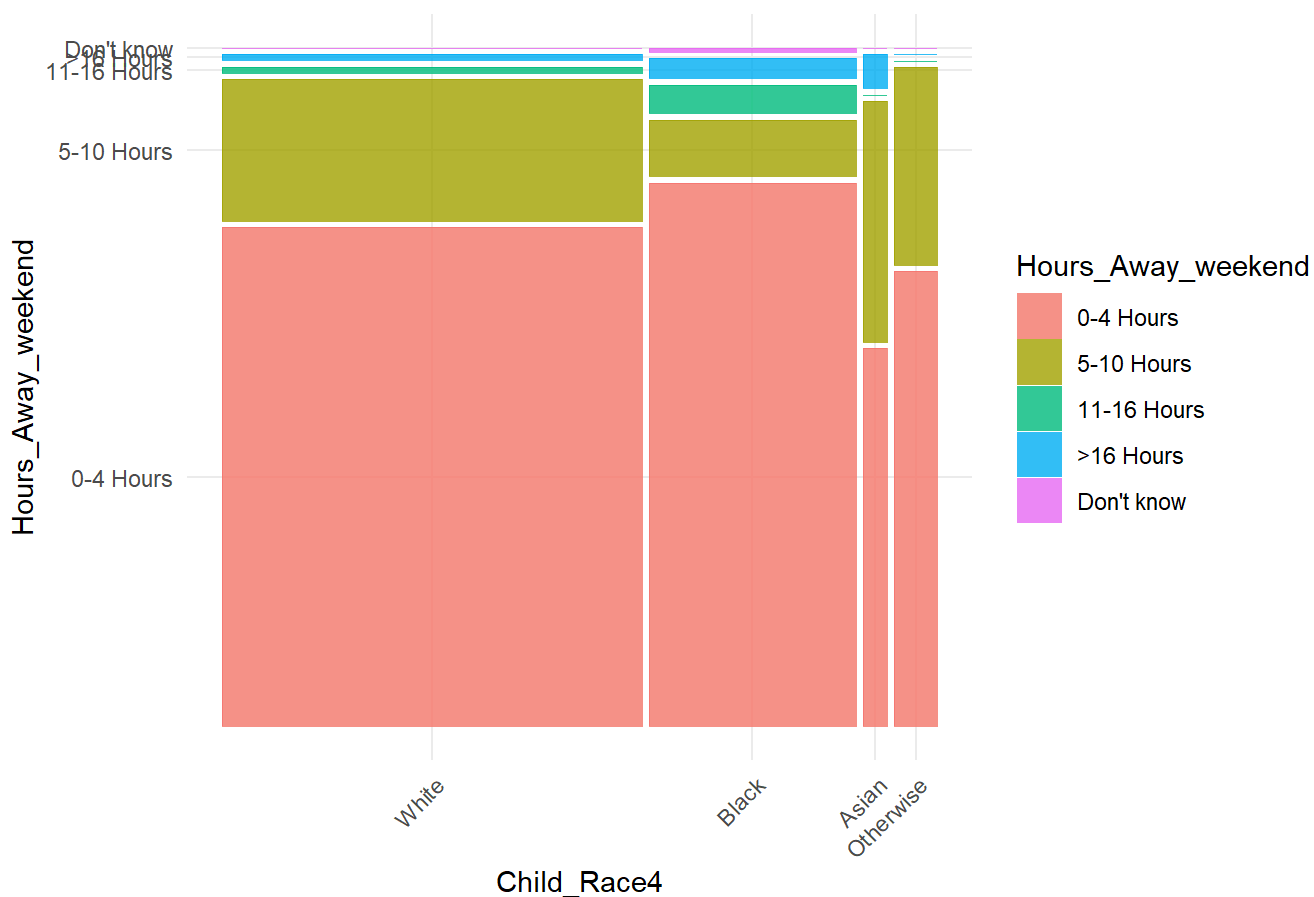
Mosaic Plot of Child_Sex and Hours_Away_weekend



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Hours_Away_weekend and Child_Race4 -----
## Bivariate Analysis of Hours_Away_weekend and Child_Race4
##
## Frequency Table:
##   Hours_Away_weekend White Black Asian Otherwise
##           0-4 Hours   250   134    11      23
##           5-10 Hours    71    14     7     10
##          11-16 Hours     3     7     0      0
##           >16 Hours     3     5     1      0
##           Don't know     0     1     0      0
##
## Relative Frequency Table:
##   Hours_Away_weekend White Black Asian Otherwise Total
##           0-4 Hours 76.5% 83.2% 57.9%   69.7% 77.4%
##           5-10 Hours 21.7%  8.7% 36.8%   30.3% 18.9%
##          11-16 Hours  0.9%  4.3%  0.0%    0.0%  1.9%
##           >16 Hours  0.9%  3.1%  5.3%    0.0%  1.7%
##           Don't know  0.0%  0.6%  0.0%    0.0%  0.2%
##
## Cramér's V:  0.11435
```

Mosaic Plot of Child_Race4 and Hours_Away_weekend



```
## Saving 7 x 5 in image
```

```
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```


##	var1	var2	cramer_v
## 1	Hours_Away_weekday	Household_income	0.20455345
## 2	Hours_Away_weekend	Single_Parent	0.19793067
## 3	Hours_Away_weekday	P1_Workstatus	0.17509273
## 4	Hours_Away_weekday	City	0.14979076
## 5	Hours_Away_weekend	City	0.13282783
## 6	Hours_Away_weekday	Child_Age_Gp	0.13123777
## 7	Hours_Away_weekend	P1_Race4	0.12881795
## 8	Hours_Away_weekday	P1_Degree	0.12755359
## 9	Hours_Away_weekday	Single_Parent	0.12160055
## 10	Hours_Away_weekday	P1_Race4	0.11557102
## 11	Hours_Away_weekend	Child_Race4	0.11434999
## 12	Hours_Away_weekday	P1_Age	0.11425232
## 13	Hours_Away_weekend	P1_Ethnicity	0.10282317
## 14	Hours_Away_weekend	P1_Age	0.07812429
## 15	Hours_Away_weekend	P1_Workstatus	0.05235249
## 16	Hours_Away_weekday	Child_Race4	0.05205742
## 17	Hours_Away_weekend	Household_income	0.03770377
## 18	Hours_Away_weekend	Child_Sex	0.02396181
## 19	Hours_Away_weekday	P1_Ethnicity	0.01220060
## 20	Hours_Away_weekday	P1_Sex	0.00000000
## 21	Hours_Away_weekday	Child_Sex	0.00000000
## 22	Hours_Away_weekend	P1_Sex	0.00000000
## 23	Hours_Away_weekend	P1_Degree	0.00000000
## 24	Hours_Away_weekend	Child_Age_Gp	0.00000000
## 25	Hours_Outdoors_Home	P1_Race4	NaN
## 26	Hours_Outdoors_Home	P1_Age	NaN
## 27	Hours_Outdoors_Home	P1_Sex	NaN
## 28	Hours_Outdoors_Home	P1_Ethnicity	NaN
## 29	Hours_Outdoors_Home	P1_Degree	NaN
## 30	Hours_Outdoors_Home	P1_Workstatus	NaN
## 31	Hours_Outdoors_Home	Single_Parent	NaN
## 32	Hours_Outdoors_Home	Household_income	NaN
## 33	Hours_Outdoors_Home	City	NaN
## 34	Hours_Outdoors_Home	Child_Age_Gp	NaN
## 35	Hours_Outdoors_Home	Child_Sex	NaN
## 36	Hours_Outdoors_Home	Child_Race4	NaN

Exploring Playoutside Variables

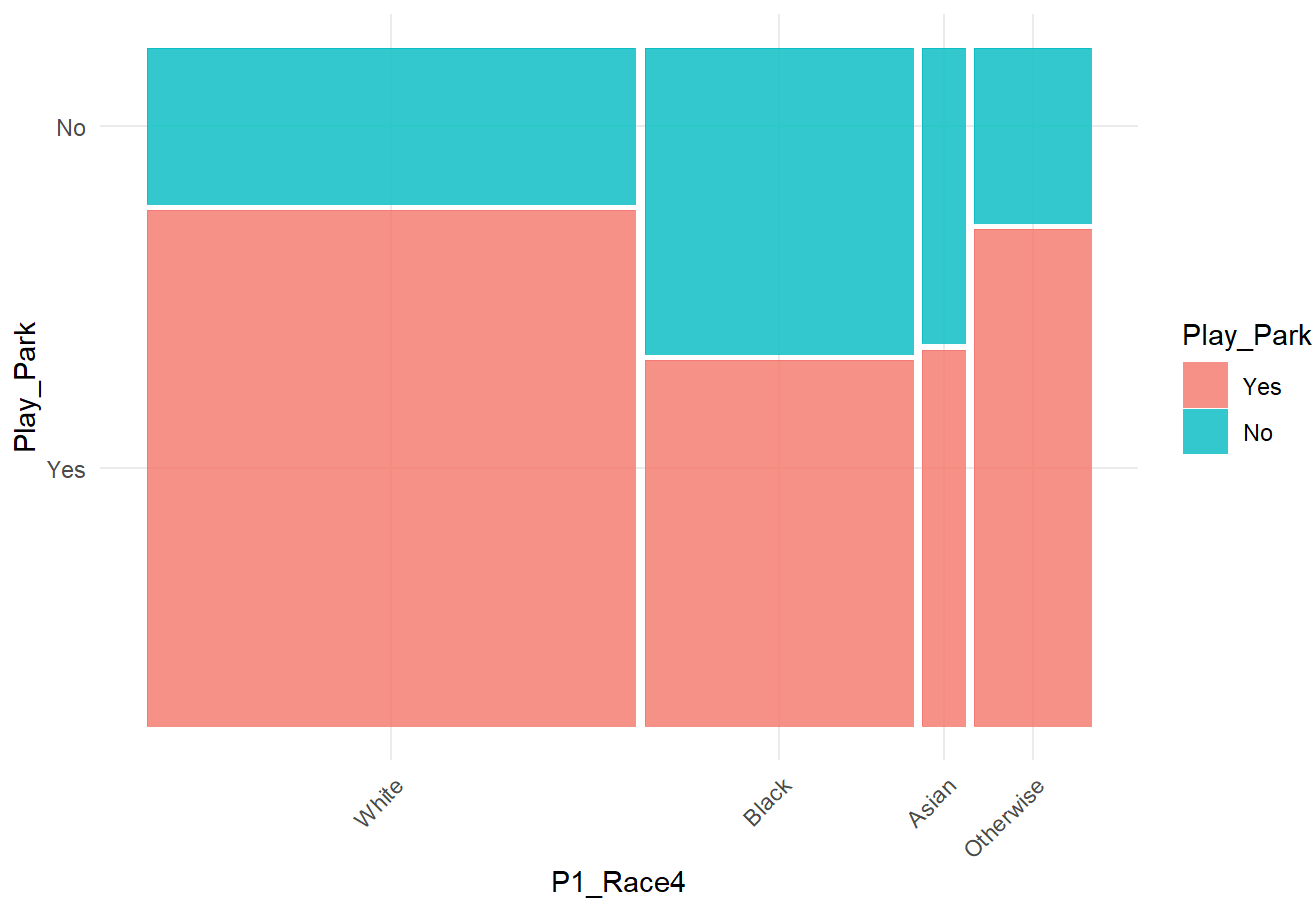
```
## Redo "Hours_Outdoors_Home",
PlayVars <- c("Play_Park", "Play_Daycare", "Play_Babysitter_Home", "Play_Relative_Home", "Play_
Other")

list1= PlayVars
list2= DemoVars

cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(), stringsAsFactors = FALSE)
results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = "_vs_")]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values,
                              data.frame(var1=var1, var2 = var2,
                                          cramer_v = result$cramer_v))
    }
  }
}
```

```
##
##
## ----- Analysis for Play_Park and P1_Race4 -----
## Bivariate Analysis of Play_Park and P1_Race4
##
## Frequency Table:
##   Play_Park White Black Asian Otherwise
##         Yes   221    86    14        51
##         No    67    72    11        18
##
## Relative Frequency Table:
##   Play_Park White Black Asian Otherwise Total
##         Yes 76.7% 54.4% 56.0%    73.9% 68.9%
##         No 23.3% 45.6% 44.0%    26.1% 31.1%
##
## Cramér's V: 0.2084981
```

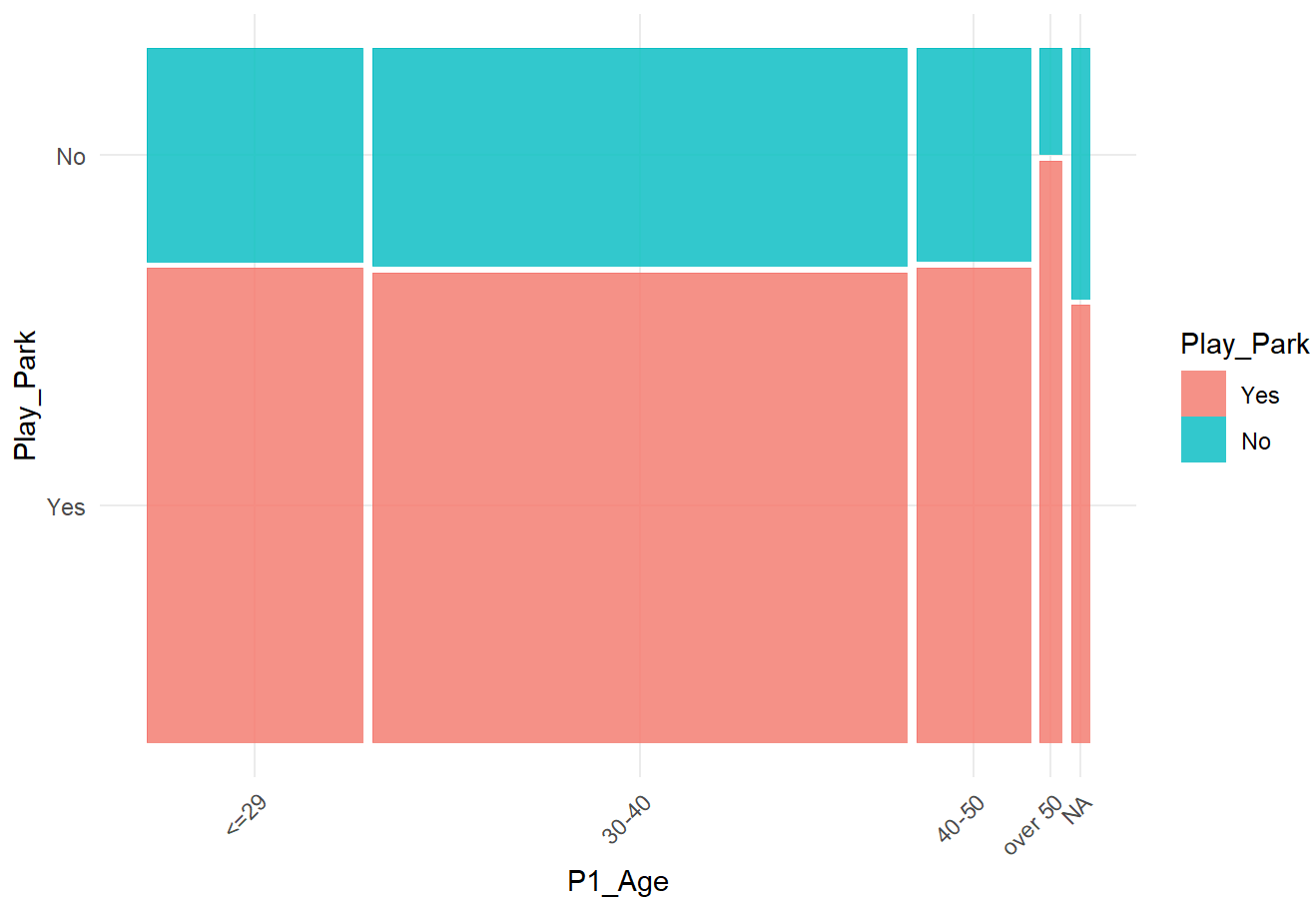
Mosaic Plot of P1_Race4 and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and P1_Age -----
## Bivariate Analysis of Play_Park and P1_Age
##
## Frequency Table:
## Play_Park <=29 30-40 40-50 over 50 NA_
##      Yes    89   218    47    11    7
##      No     40   101    21     2    4
##
## Relative Frequency Table:
## Play_Park <=29 30-40 40-50 over 50  NA_ Total
##      Yes 69.0% 68.3% 69.1%   84.6% 63.6% 68.9%
##      No 31.0% 31.7% 30.9%   15.4% 36.4% 31.1%
##
## Cramér's V:  0
```

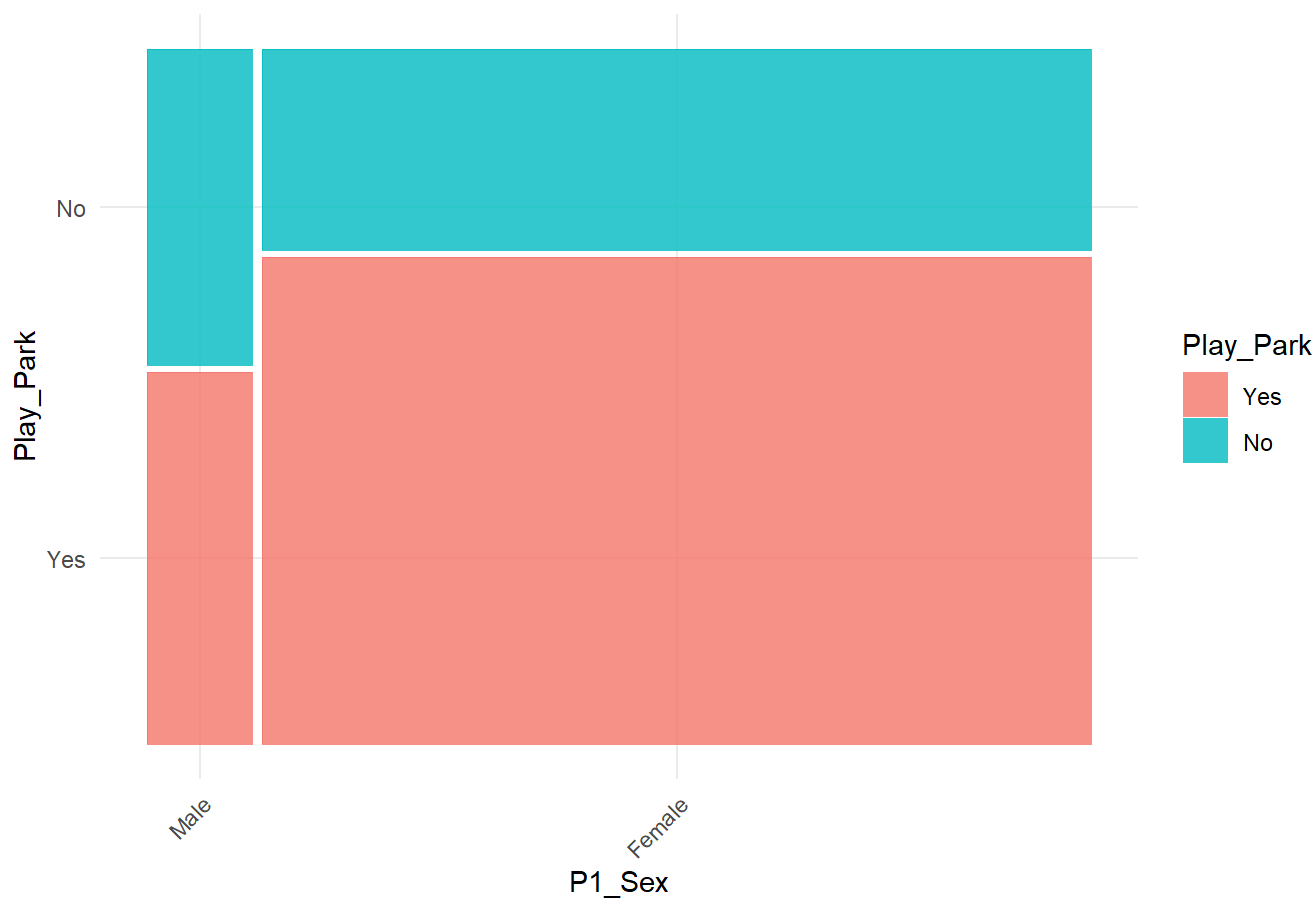
Mosaic Plot of P1_Age and Play_Park



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Park and P1_Sex -----
## Bivariate Analysis of Play_Park and P1_Sex
##
## Frequency Table:
##   Play_Park   Male Female
##         Yes     33    339
##         No      28    140
##
## Relative Frequency Table:
##   Play_Park   Male Female Total
##         Yes  54.1%  70.8% 68.9%
##         No   45.9%  29.2% 31.1%
##
## Cramér's V:  0.10566
```

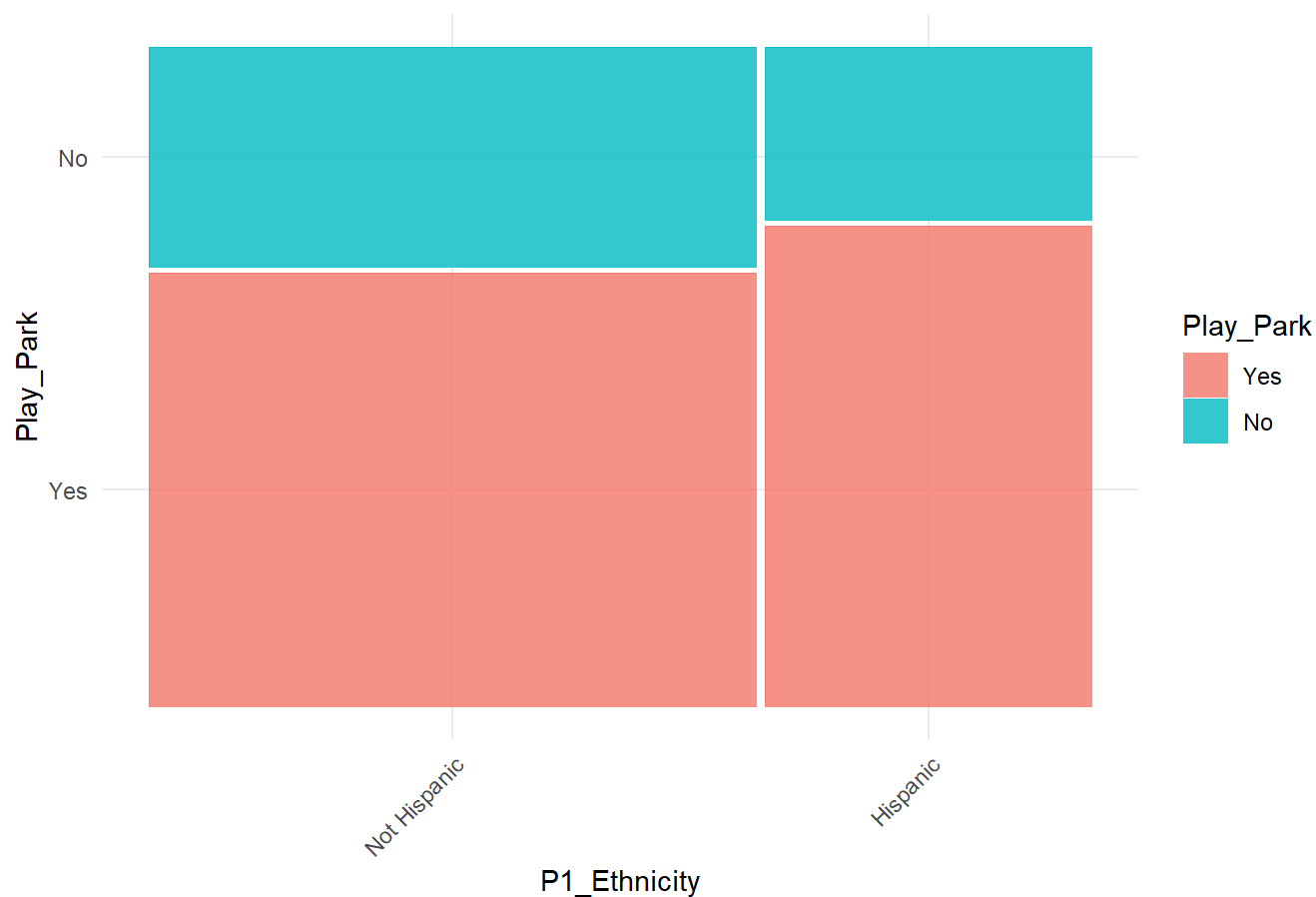
Mosaic Plot of P1_Sex and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and P1_Ethnicity -----
## Bivariate Analysis of Play_Park and P1_Ethnicity
##
## Frequency Table:
##   Play_Park Not Hispanic Hispanic
##         Yes         233         139
##         No          118          50
##
## Relative Frequency Table:
##   Play_Park Not Hispanic Hispanic Total
##         Yes         66.4%       73.5% 68.9%
##         No          33.6%       26.5% 31.1%
##
## Cramér's V:  0.05998388
```

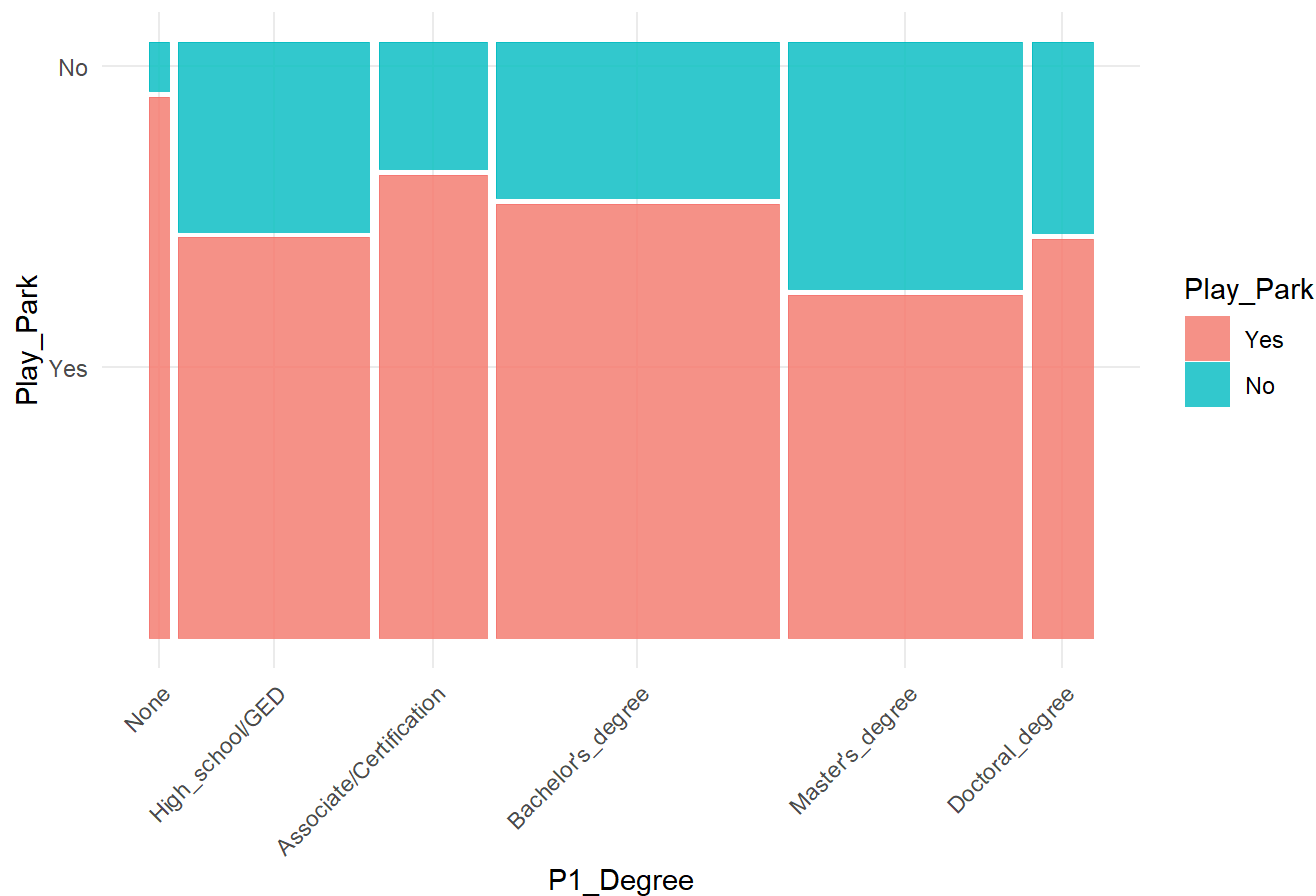
Mosaic Plot of P1_Ethnicity and Play_Park



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Park and P1_Degree -----
## Bivariate Analysis of Play_Park and P1_Degree
##
## Frequency Table:
## Play_Park None High_school/GED Associate/Certification Bachelor's_degree
## Yes 11 78 51 125
## No 1 37 14 45
## Master's_degree Doctoral_degree
## 82 25
## 59 12
##
## Relative Frequency Table:
## Play_Park None High_school/GED Associate/Certification Bachelor's_degree
## Yes 91.7% 67.8% 78.5% 73.5%
## No 8.3% 32.2% 21.5% 26.5%
## Master's_degree Doctoral_degree Total
## 58.2% 67.6% 68.9%
## 41.8% 32.4% 31.1%
##
## Cramér's V: 0.1365645
```

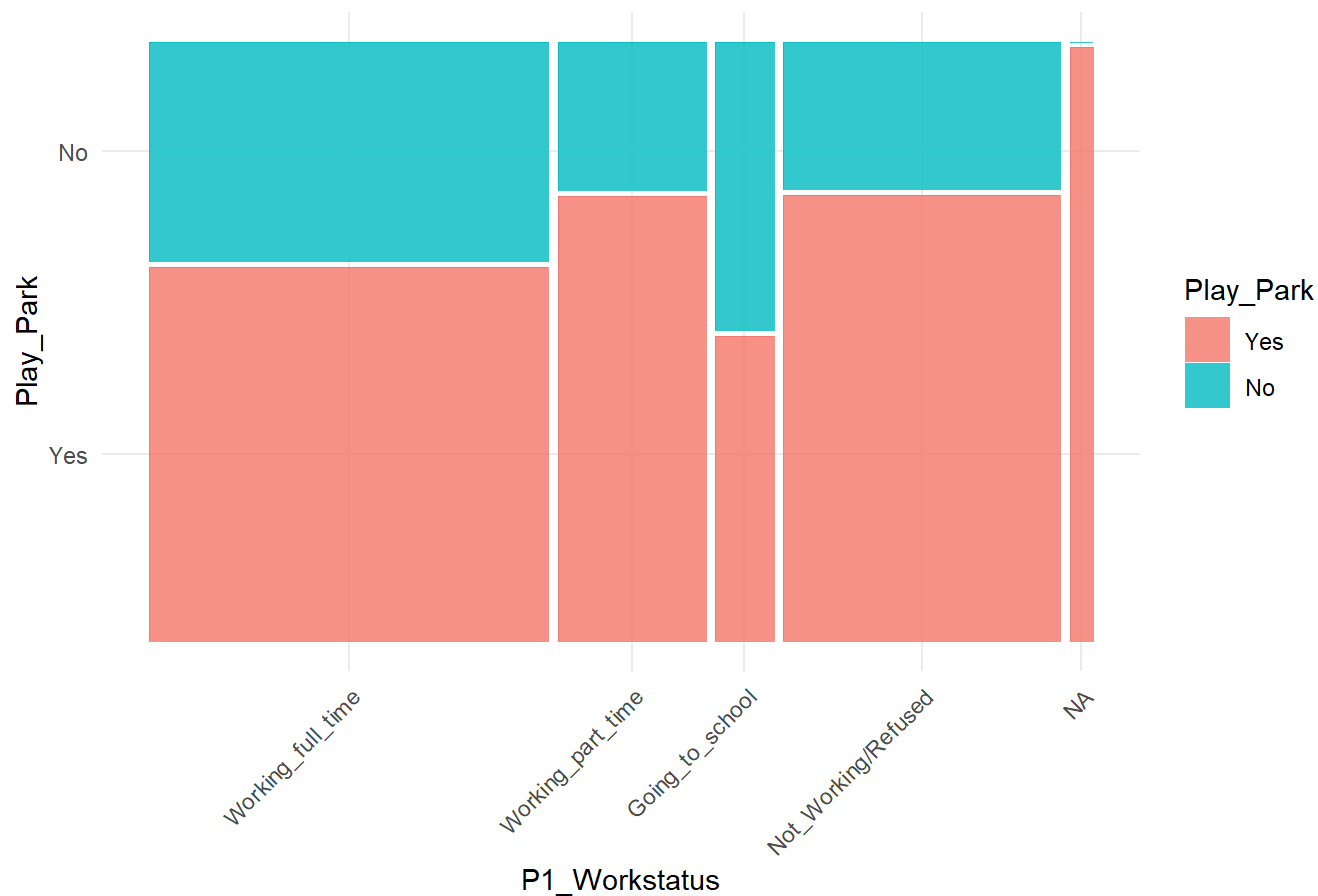
Mosaic Plot of P1_Degree and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and P1_Workstatus -----
## Bivariate Analysis of Play_Park and P1_Workstatus
##
## Frequency Table:
## Play_Park Working_full_time Working_part_time Going_to_school
##      Yes           150           66           18
##      No            88           22           17
## Not_Working/Refused NA_
##                124  14
##                41   0
##
## Relative Frequency Table:
## Play_Park Working_full_time Working_part_time Going_to_school
##      Yes           63.0%           75.0%           51.4%
##      No            37.0%           25.0%           48.6%
## Not_Working/Refused  NA_ Total
##                75.2% 100.0% 68.9%
##                24.8%   0.0% 31.1%
##
## Cramér's V:  0.1704782
```

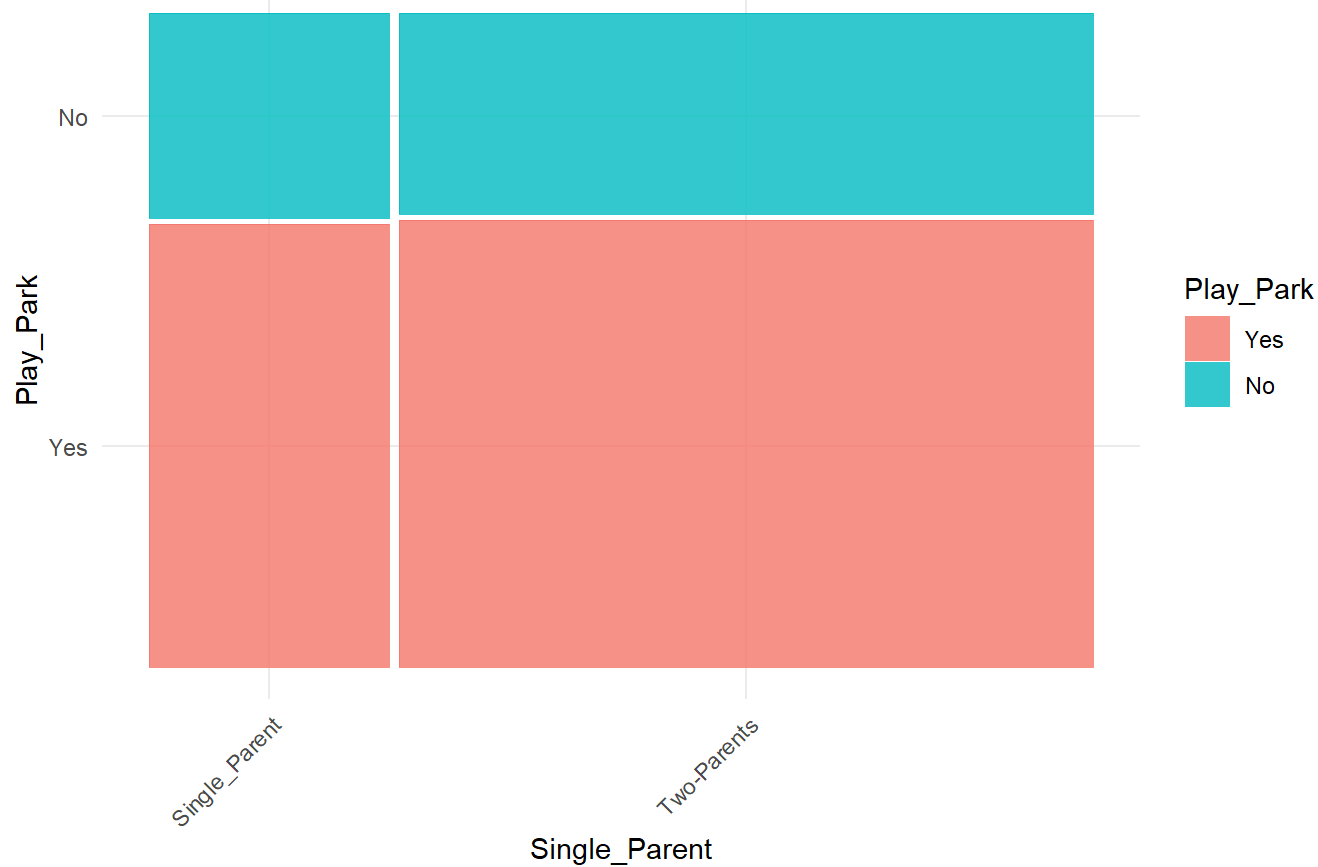
Mosaic Plot of P1_Workstatus and Play_Park



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Park and Single_Parent -----
## Bivariate Analysis of Play_Park and Single_Parent
##
## Frequency Table:
##   Play_Park Single_Parent Two-Parents
##         Yes          95          277
##         No           44          124
##
## Relative Frequency Table:
##   Play_Park Single_Parent Two-Parents Total
##         Yes          68.3%          69.1% 68.9%
##         No           31.7%          30.9% 31.1%
##
## Cramér's V:  0
```

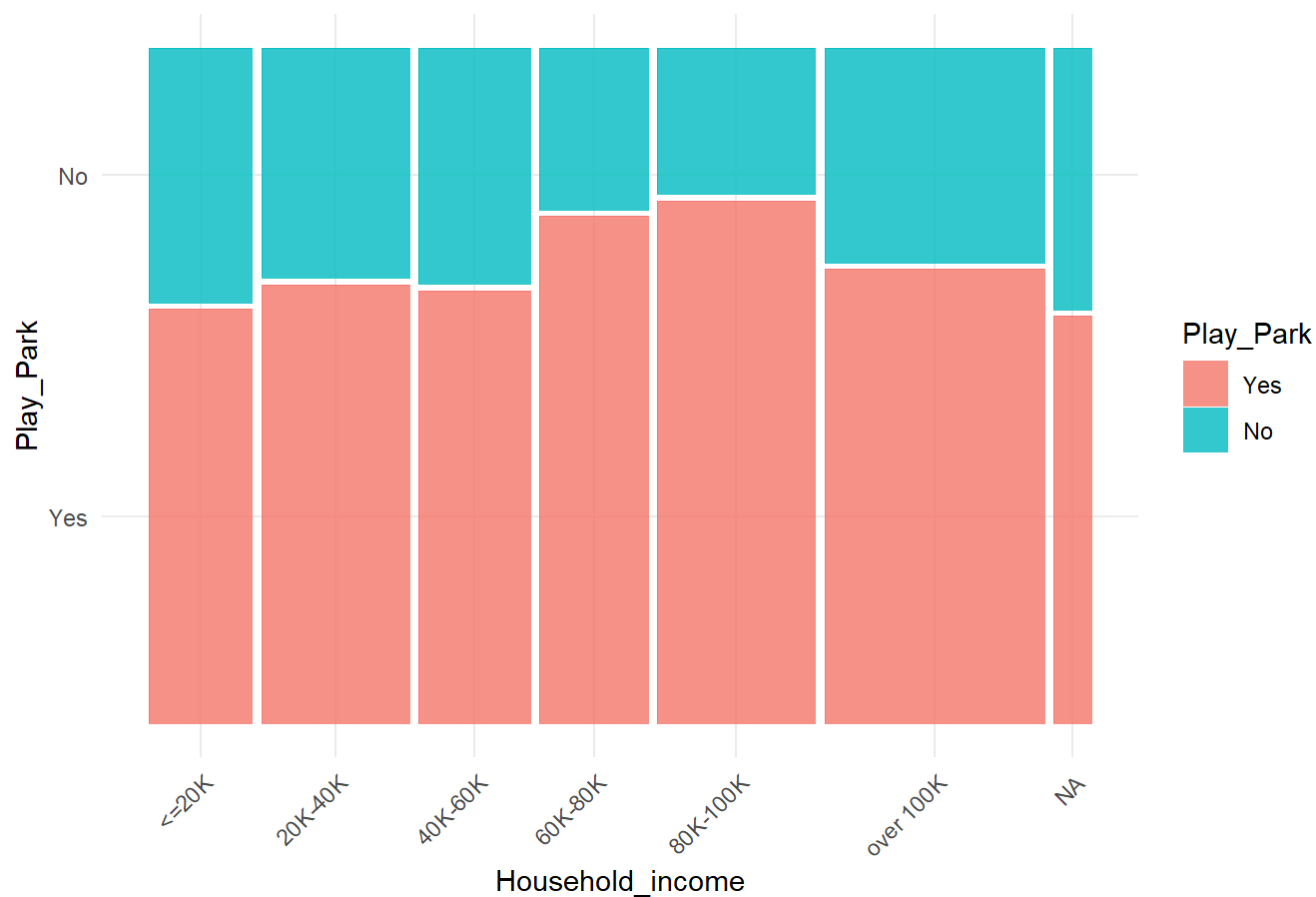

Mosaic Plot of Single_Parent and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and Household_income -----
## Bivariate Analysis of Play_Park and Household_income
##
## Frequency Table:
## Play_Park <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##      Yes      39      59      44      50      75      91  14
##      No       24      31      24      16      21      43   9
##
## Relative Frequency Table:
## Play_Park <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_ Total
##      Yes 61.9%  65.6%  64.7%  75.8%  78.1%    67.9% 60.9% 68.9%
##      No 38.1%  34.4%  35.3%  24.2%  21.9%    32.1% 39.1% 31.1%
##
## Cramér's V: 0.06767276
```

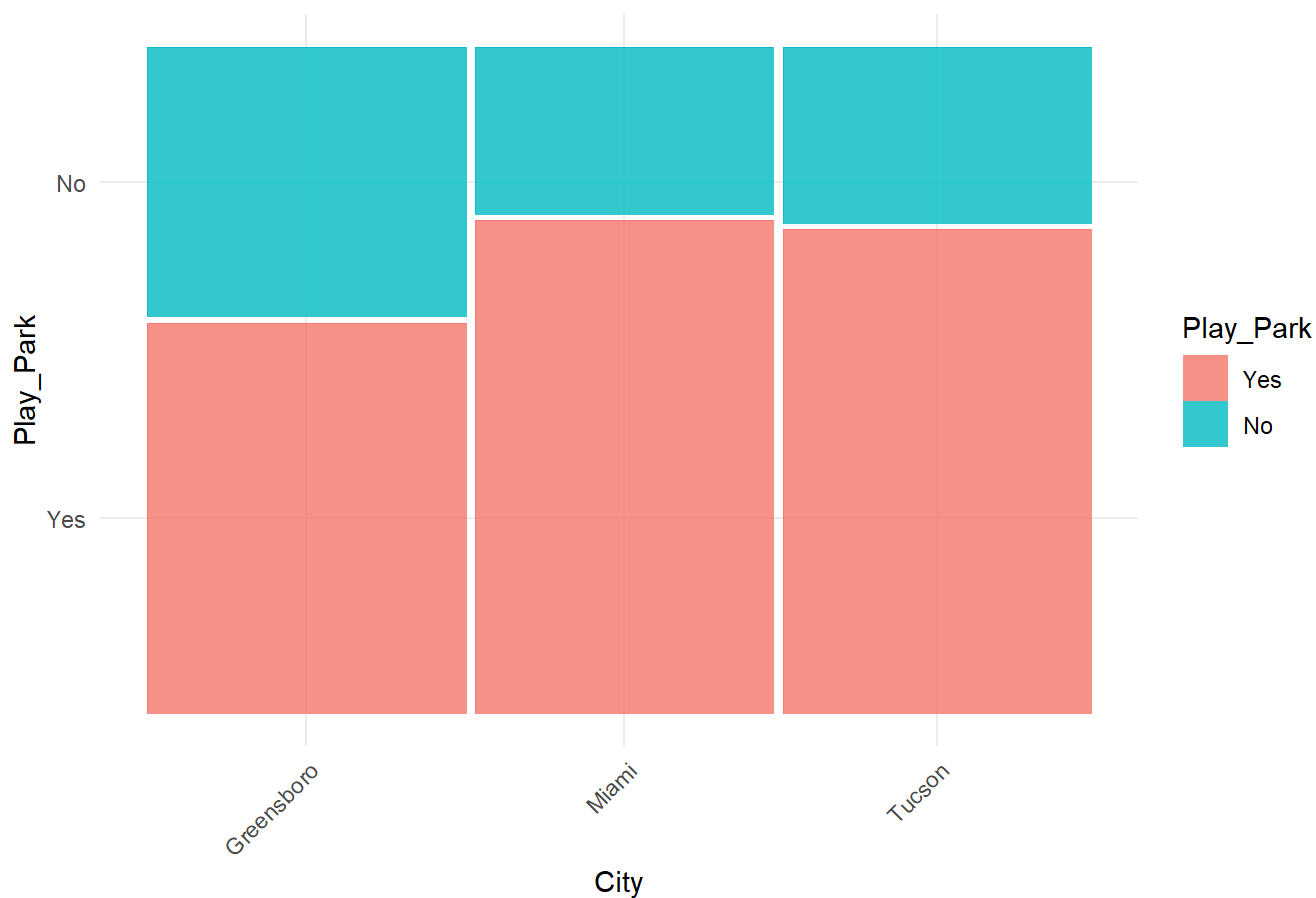
Mosaic Plot of Household_income and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and City -----
## Bivariate Analysis of Play_Park and City
##
## Frequency Table:
## Play_Park Greensboro Miami Tucson
##      Yes      110    130    132
##      No       76     44     48
##
## Relative Frequency Table:
## Play_Park Greensboro Miami Tucson Total
##      Yes     59.1% 74.7% 73.3% 68.9%
##      No      40.9% 25.3% 26.7% 31.1%
##
## Cramér's V: 0.1406149
```

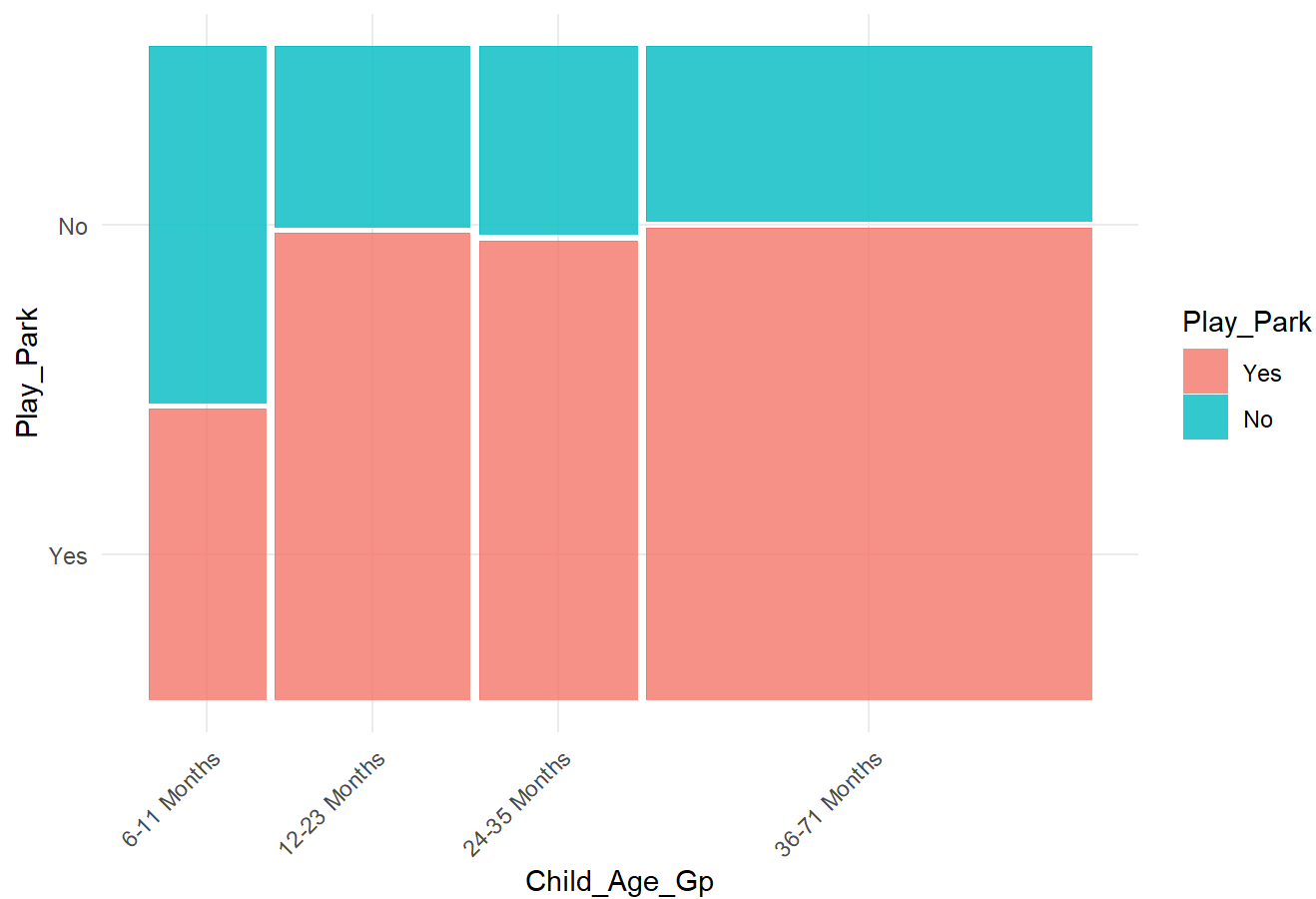
Mosaic Plot of City and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and Child_Age_Gp -----
## Bivariate Analysis of Play_Park and Child_Age_Gp
##
## Frequency Table:
## Play_Park 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##      Yes      31      83      66      192
##      No       38      32      27      71
##
## Relative Frequency Table:
## Play_Park 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##      Yes      44.9%      72.2%      71.0%      73.0% 68.9%
##      No       55.1%      27.8%      29.0%      27.0% 31.1%
##
## Cramér's V:  0.1843777
```

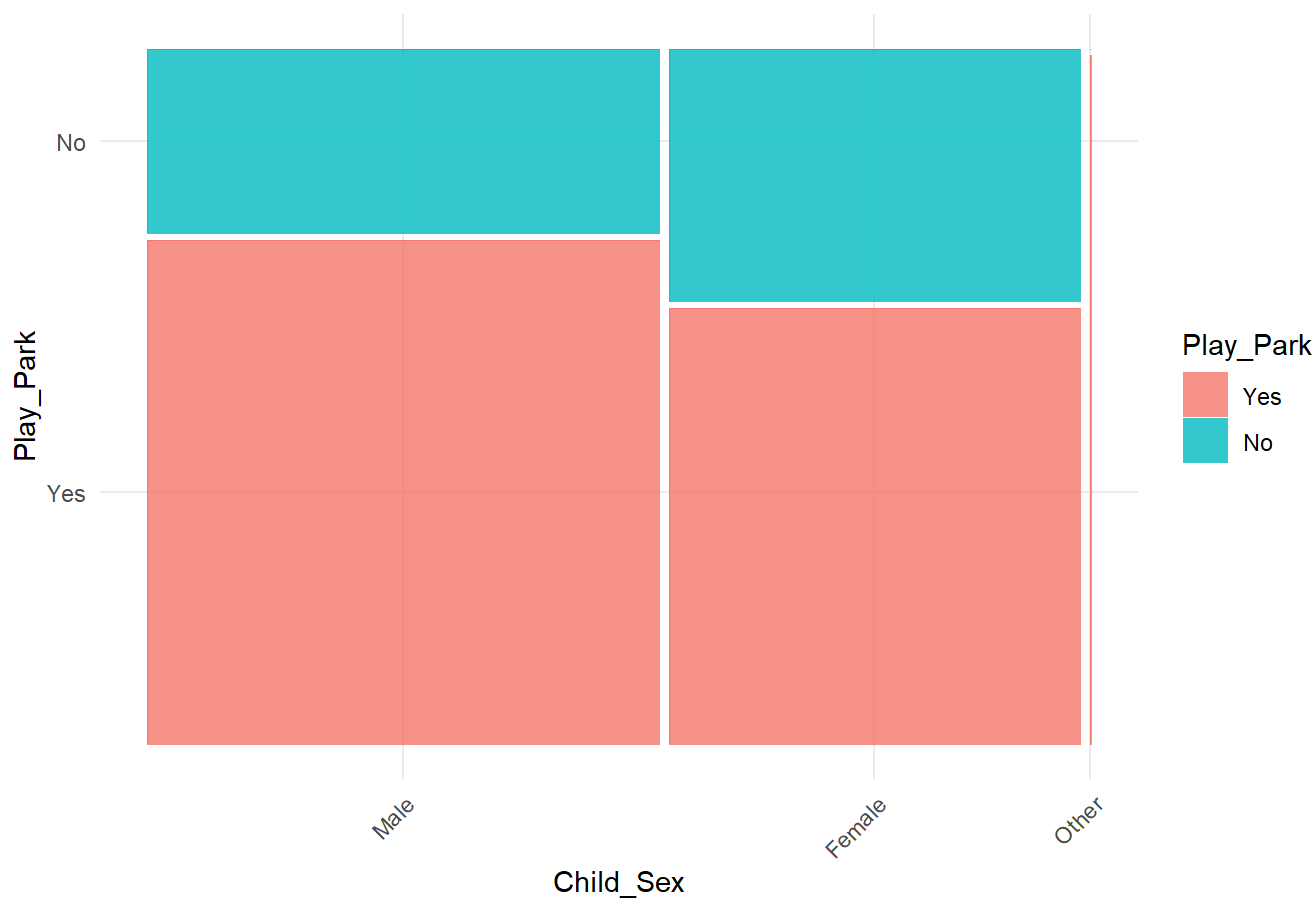
Mosaic Plot of Child_Age_Gp and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and Child_Sex -----
## Bivariate Analysis of Play_Park and Child_Sex
##
## Frequency Table:
##   Play_Park Male Female Other
##         Yes   219    152     1
##         No    80     88     0
##
## Relative Frequency Table:
##   Play_Park  Male Female  Other Total
##         Yes 73.2%  63.3% 100.0% 68.9%
##         No 26.8%  36.7%   0.0% 31.1%
##
## Cramér's V: 0.0918821
```

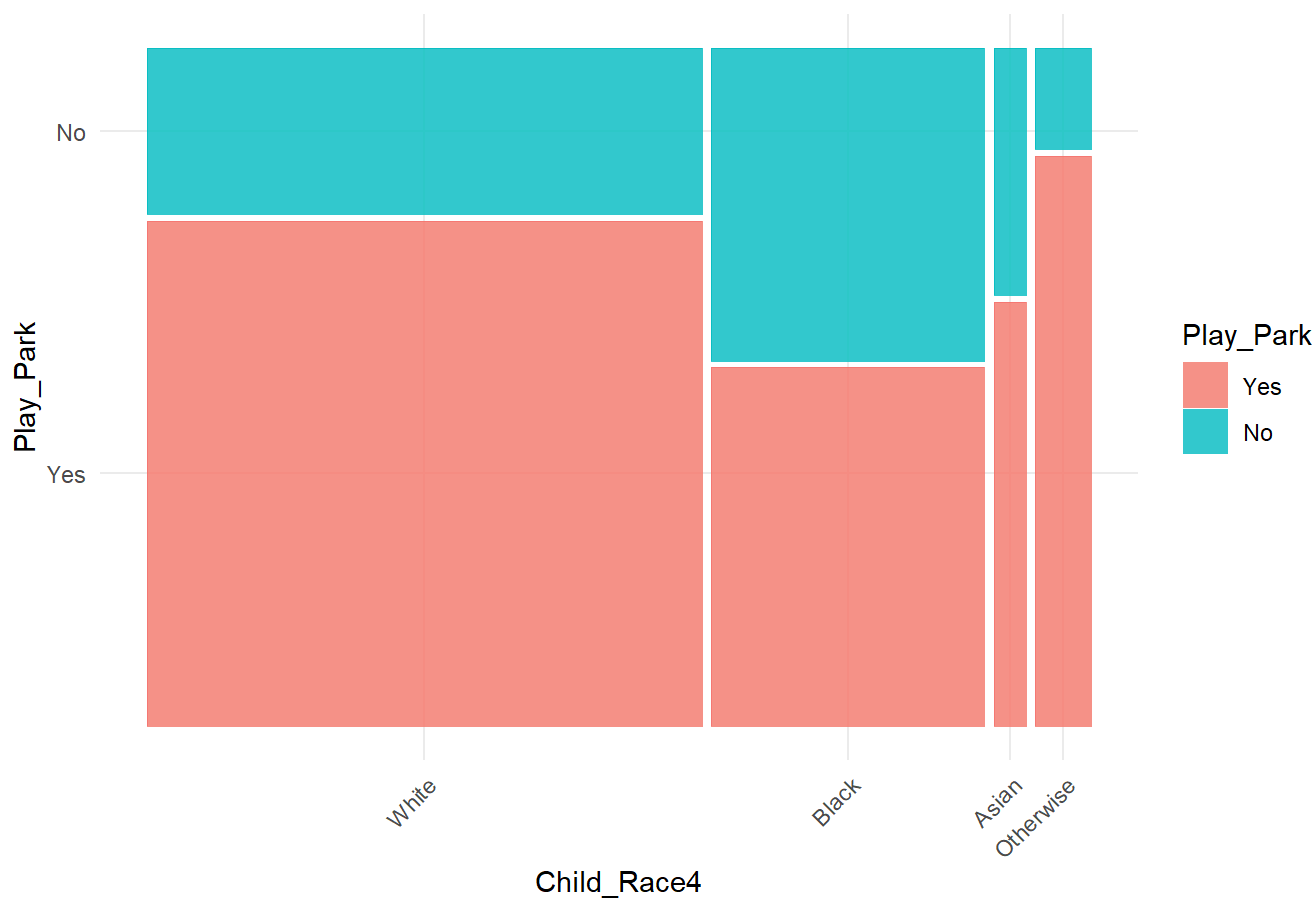
Mosaic Plot of Child_Sex and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and Child_Race4 -----
## Bivariate Analysis of Play_Park and Child_Race4
##
## Frequency Table:
##   Play_Park White Black Asian Otherwise
##         Yes   246    86    12        28
##         No    81    75     7         5
##
## Relative Frequency Table:
##   Play_Park White Black Asian Otherwise Total
##         Yes 75.2% 53.4% 63.2%    84.8% 68.9%
##         No 24.8% 46.6% 36.8%    15.2% 31.1%
##
## Cramér's V: 0.2167636
```

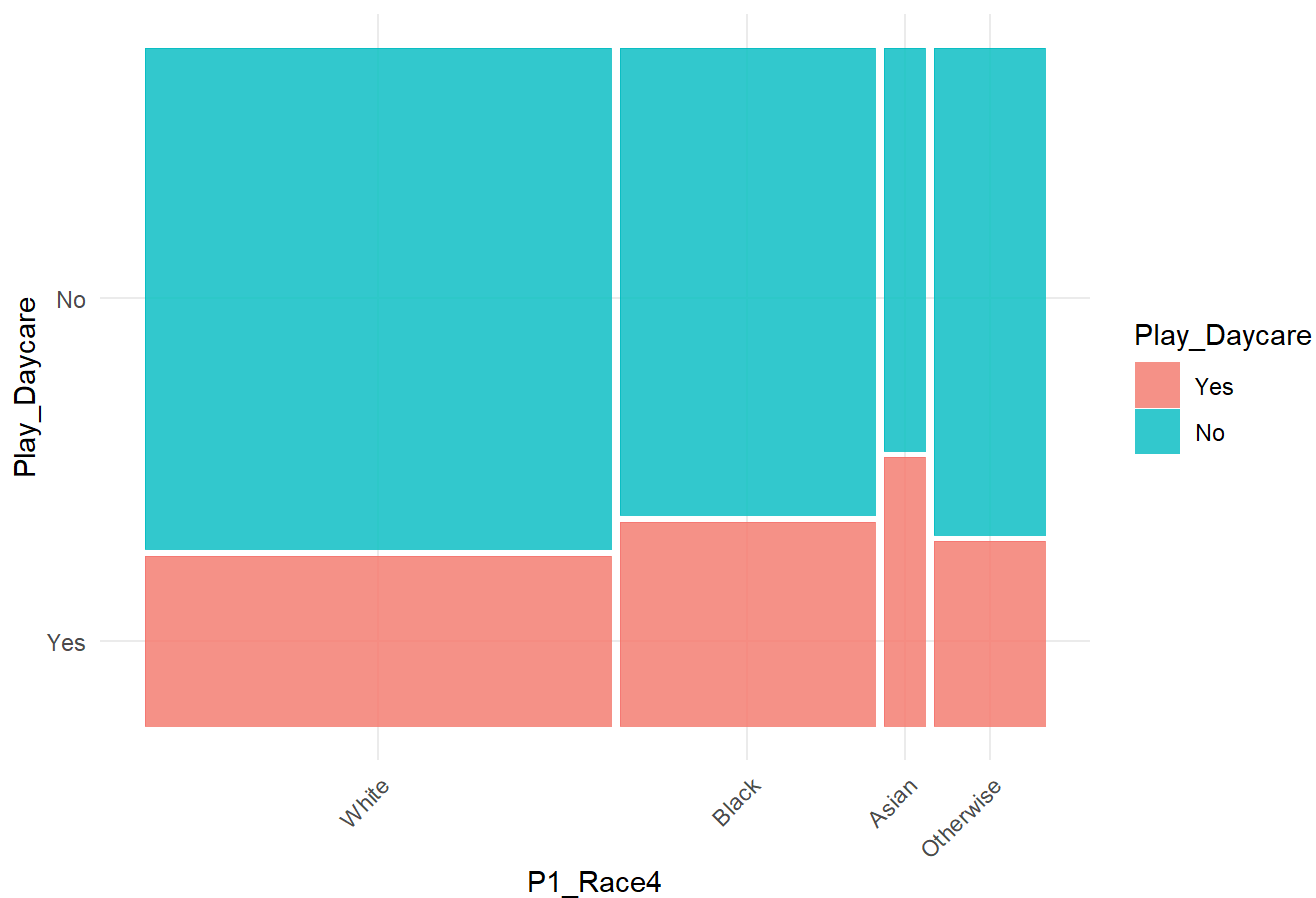
Mosaic Plot of Child_Race4 and Play_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and P1_Race4 -----
## Bivariate Analysis of Play_Daycare and P1_Race4
##
## Frequency Table:
##   Play_Daycare White Black Asian Otherwise
##           Yes    73    48    10         19
##           No   215   110    15         50
##
## Relative Frequency Table:
##   Play_Daycare White Black Asian Otherwise Total
##           Yes 25.3% 30.4% 40.0%      27.5% 27.8%
##           No 74.7% 69.6% 60.0%      72.5% 72.2%
##
## Cramér's V: 0.02106887
```

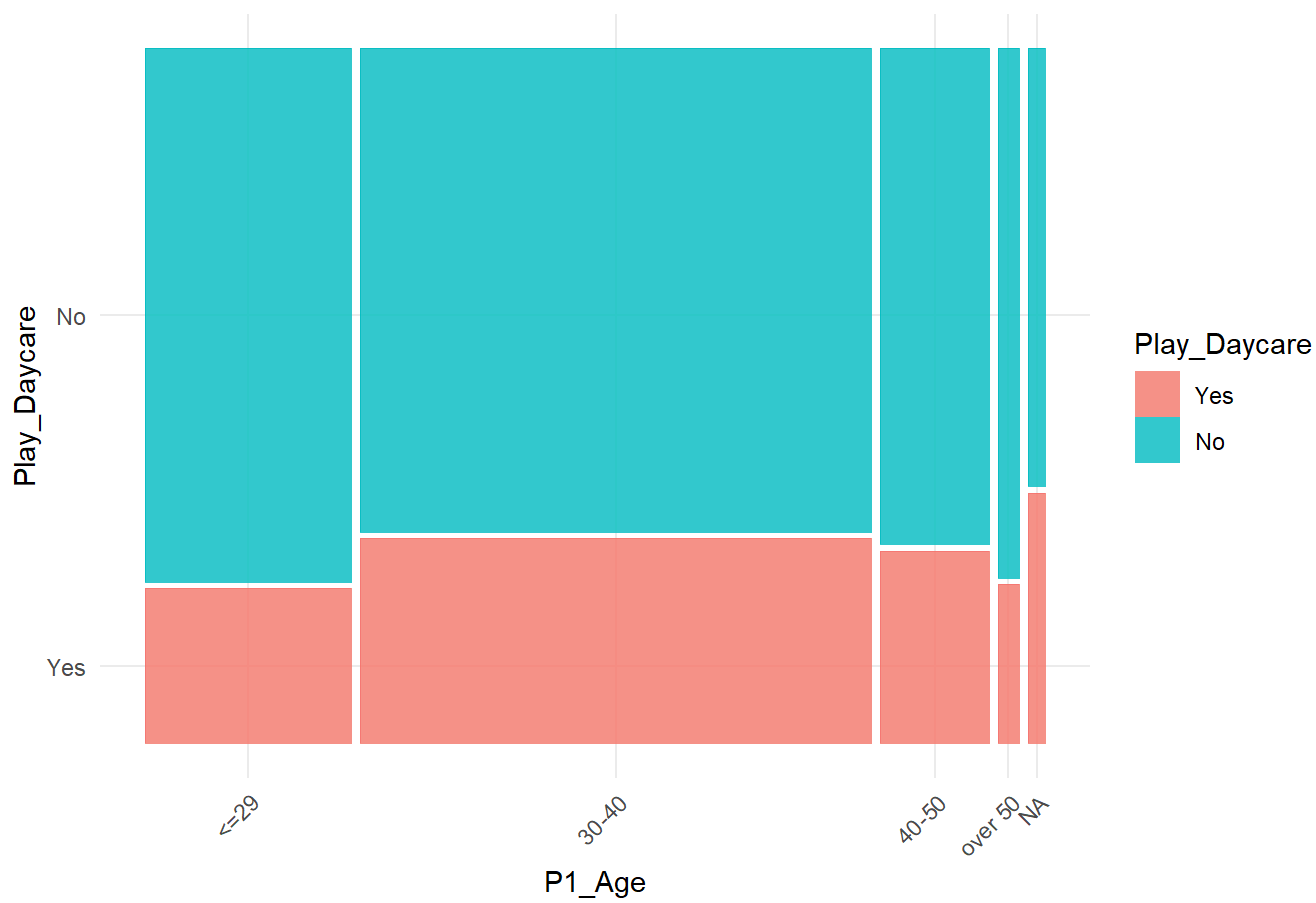
Mosaic Plot of P1_Race4 and Play_Daycare



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Daycare and P1_Age -----
## Bivariate Analysis of Play_Daycare and P1_Age
##
## Frequency Table:
## Play_Daycare <=29 30-40 40-50 over 50 NA_
##           Yes   29   95   19     3   4
##           No  100  224   49    10   7
##
## Relative Frequency Table:
## Play_Daycare <=29 30-40 40-50 over 50  NA_ Total
##           Yes 22.5% 29.8% 27.9%  23.1% 36.4% 27.8%
##           No 77.5% 70.2% 72.1%  76.9% 63.6% 72.2%
##
## Cramér's V:  0
```

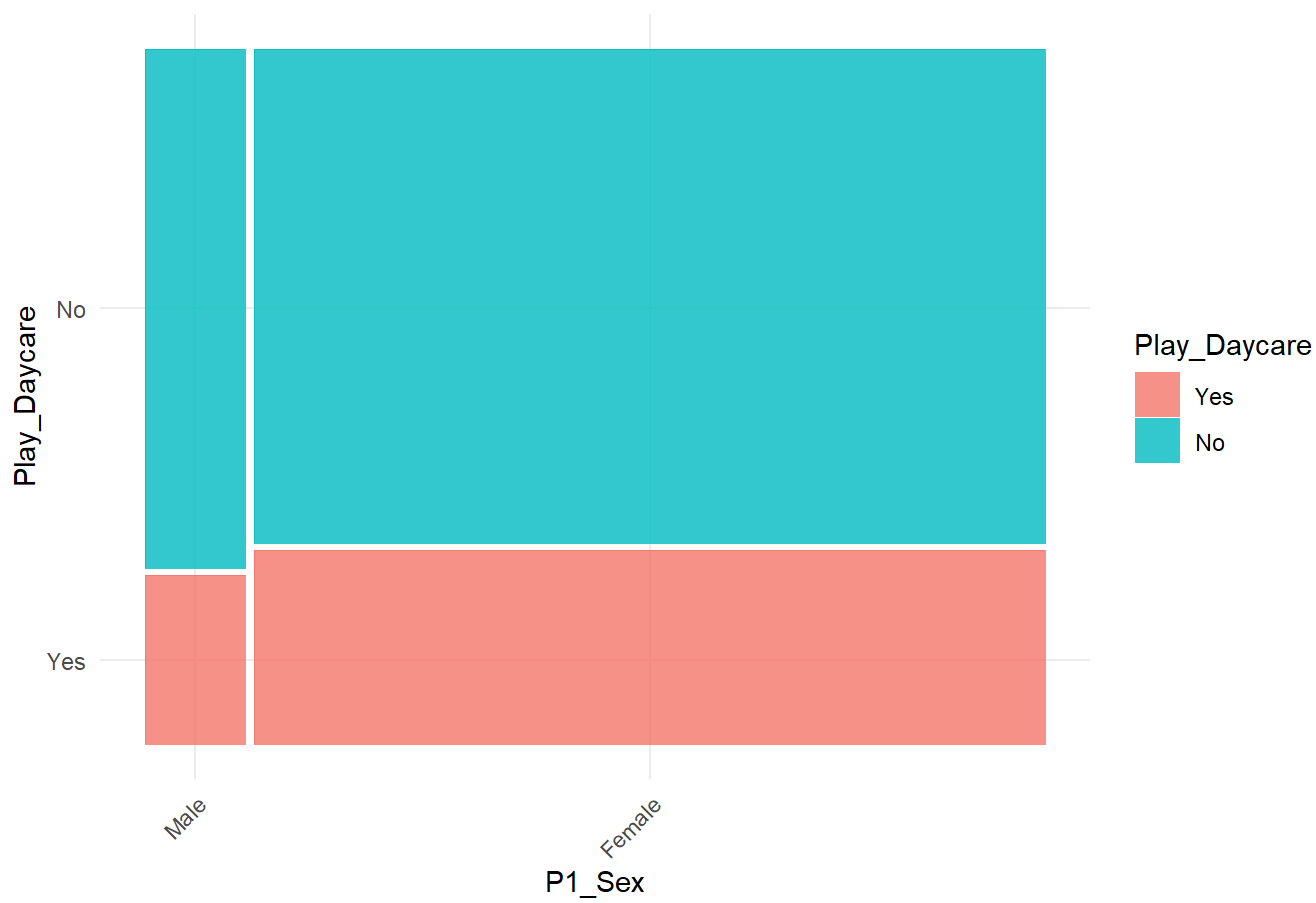
Mosaic Plot of P1_Age and Play_Daycare



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Daycare and P1_Sex -----
## Bivariate Analysis of Play_Daycare and P1_Sex
##
## Frequency Table:
## Play_Daycare  Male Female
##           Yes    15    135
##           No     46    344
##
## Relative Frequency Table:
## Play_Daycare  Male Female Total
##           Yes 24.6% 28.2% 27.8%
##           No 75.4% 71.8% 72.2%
##
## Cramér's V:  0
```

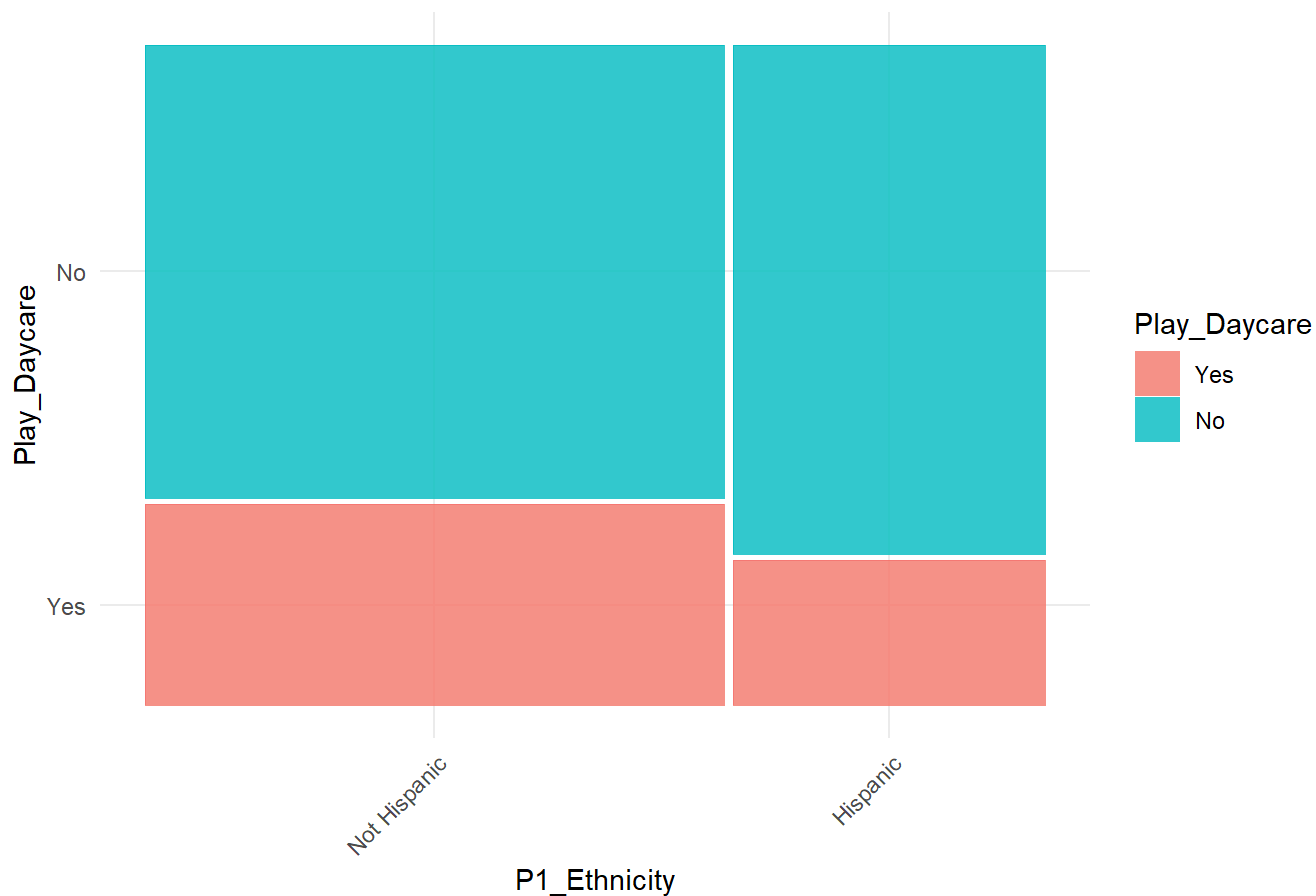

Mosaic Plot of P1_Sex and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and P1_Ethnicity -----
## Bivariate Analysis of Play_Daycare and P1_Ethnicity
##
## Frequency Table:
##   Play_Daycare Not Hispanic Hispanic
##           Yes           108         42
##           No            243        147
##
## Relative Frequency Table:
##   Play_Daycare Not Hispanic Hispanic Total
##           Yes           30.8%      22.2% 27.8%
##           No            69.2%      77.8% 72.2%
##
## Cramér's V:  0.08025388
```

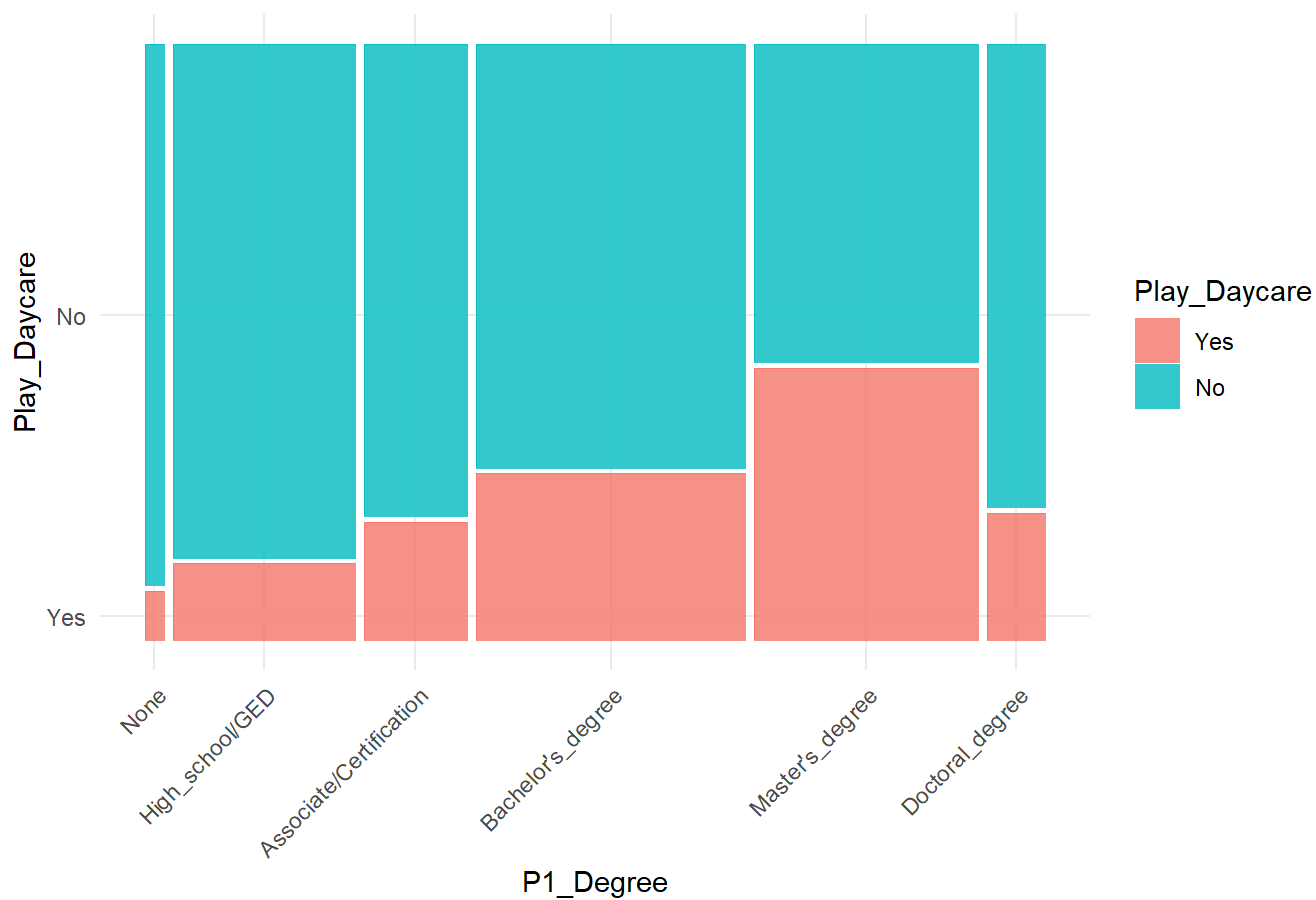
Mosaic Plot of P1_Ethnicity and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and P1_Degree -----
## Bivariate Analysis of Play_Daycare and P1_Degree
##
## Frequency Table:
## Play_Daycare None High_school/GED Associate/Certification Bachelor's_degree
##      Yes      1          15          13          48
##      No     11         100          52         122
## Master's_degree Doctoral_degree
##      65          8
##      76         29
##
## Relative Frequency Table:
## Play_Daycare None High_school/GED Associate/Certification Bachelor's_degree
##      Yes  8.3%      13.0%      20.0%      28.2%
##      No  91.7%      87.0%      80.0%      71.8%
## Master's_degree Doctoral_degree Total
##      46.1%      21.6%  27.8%
##      53.9%      78.4%  72.2%
##
## Cramér's V:  0.2583176
```

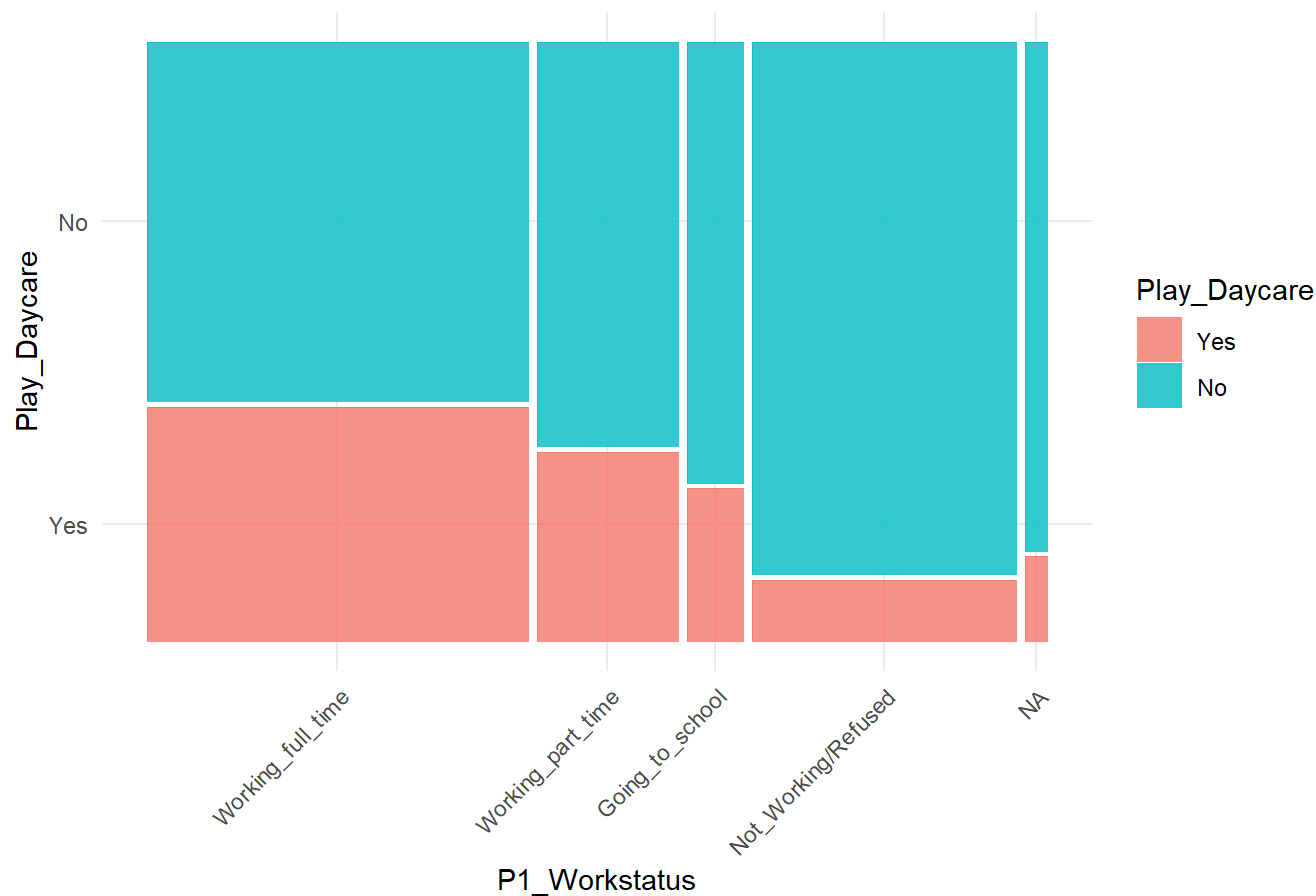
Mosaic Plot of P1_Degree and Play_Daycare



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Daycare and P1_Workstatus -----
## Bivariate Analysis of Play_Daycare and P1_Workstatus
##
## Frequency Table:
##   Play_Daycare Working_full_time Working_part_time Going_to_school
##           Yes           94           28           9
##           No           144           60           26
##   Not_Working/Refused NA_
##               17    2
##               148   12
##
## Relative Frequency Table:
##   Play_Daycare Working_full_time Working_part_time Going_to_school
##           Yes           39.5%           31.8%           25.7%
##           No           60.5%           68.2%           74.3%
##   Not_Working/Refused   NA_ Total
##               10.3% 14.3% 27.8%
##               89.7% 85.7% 72.2%
##
## Cramér's V:  0.2705677
```

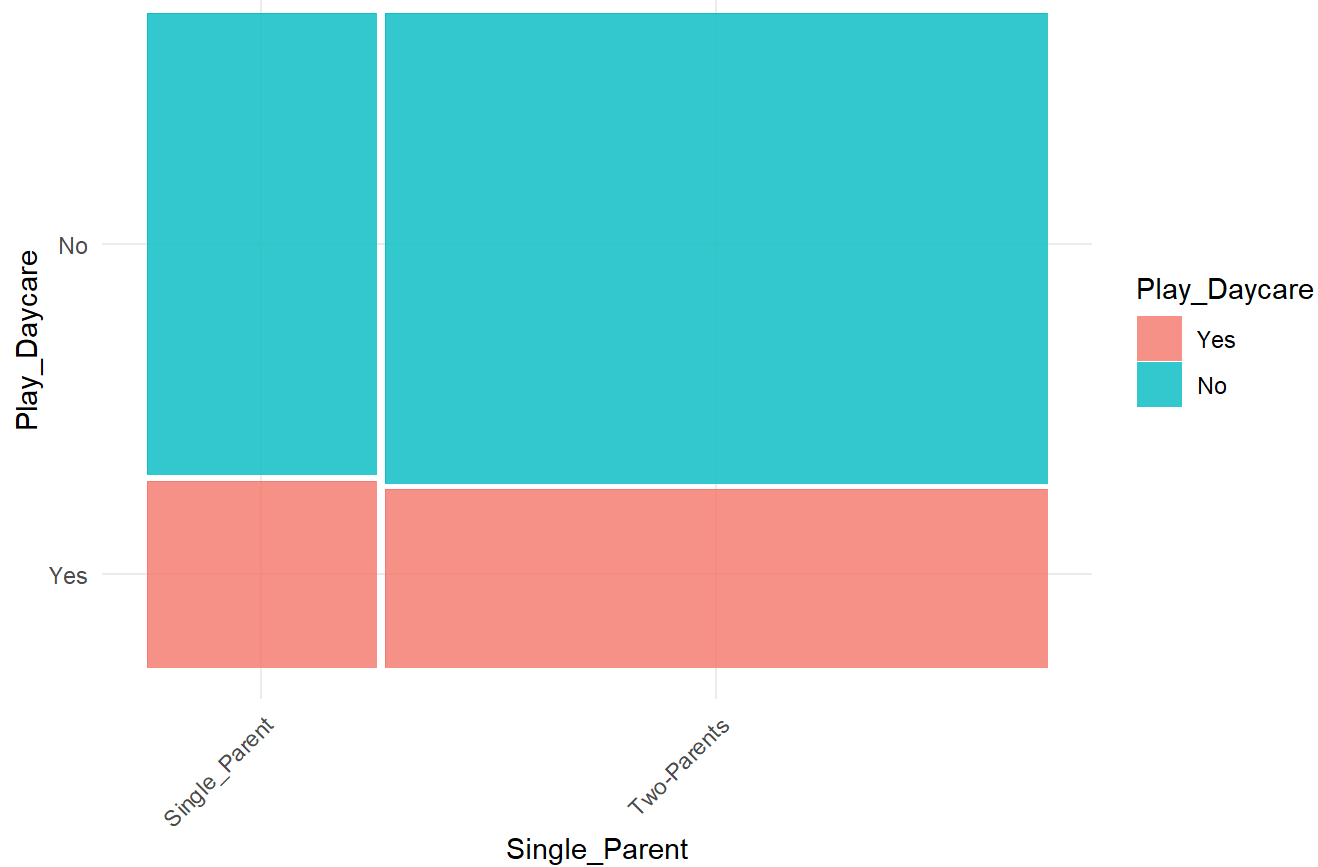
Mosaic Plot of P1_Workstatus and Play_Daycare



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Daycare and Single_Parent -----
## Bivariate Analysis of Play_Daycare and Single_Parent
##
## Frequency Table:
##   Play_Daycare Single_Parent Two-Parents
##           Yes           40           110
##           No            99           291
##
## Relative Frequency Table:
##   Play_Daycare Single_Parent Two-Parents Total
##           Yes           28.8%           27.4% 27.8%
##           No            71.2%           72.6% 72.2%
##
## Cramér's V:  0
```

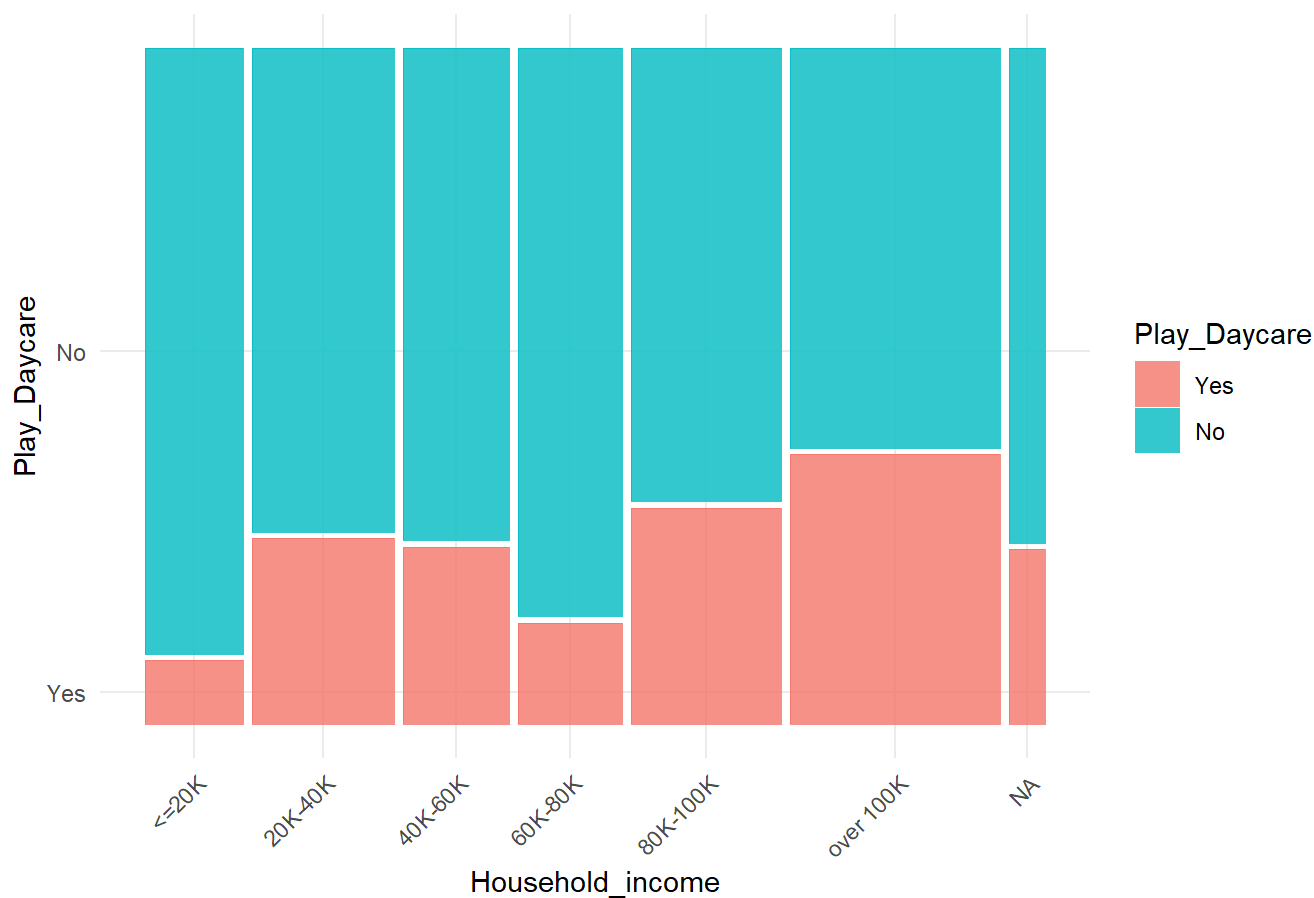
Mosaic Plot of Single_Parent and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and Household_income -----
## Bivariate Analysis of Play_Daycare and Household_income
##
## Frequency Table:
## Play_Daycare <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes      6      25      18      10      31      54      6
##           No      57      65      50      56      65      80     17
##
## Relative Frequency Table:
## Play_Daycare <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##           Yes  9.5%  27.8%  26.5%  15.2%  32.3%  40.3% 26.1% 27.8%
##           No  90.5%  72.2%  73.5%  84.8%  67.7%  59.7% 73.9% 72.2%
##
## Cramér's V:  0.1984836
```

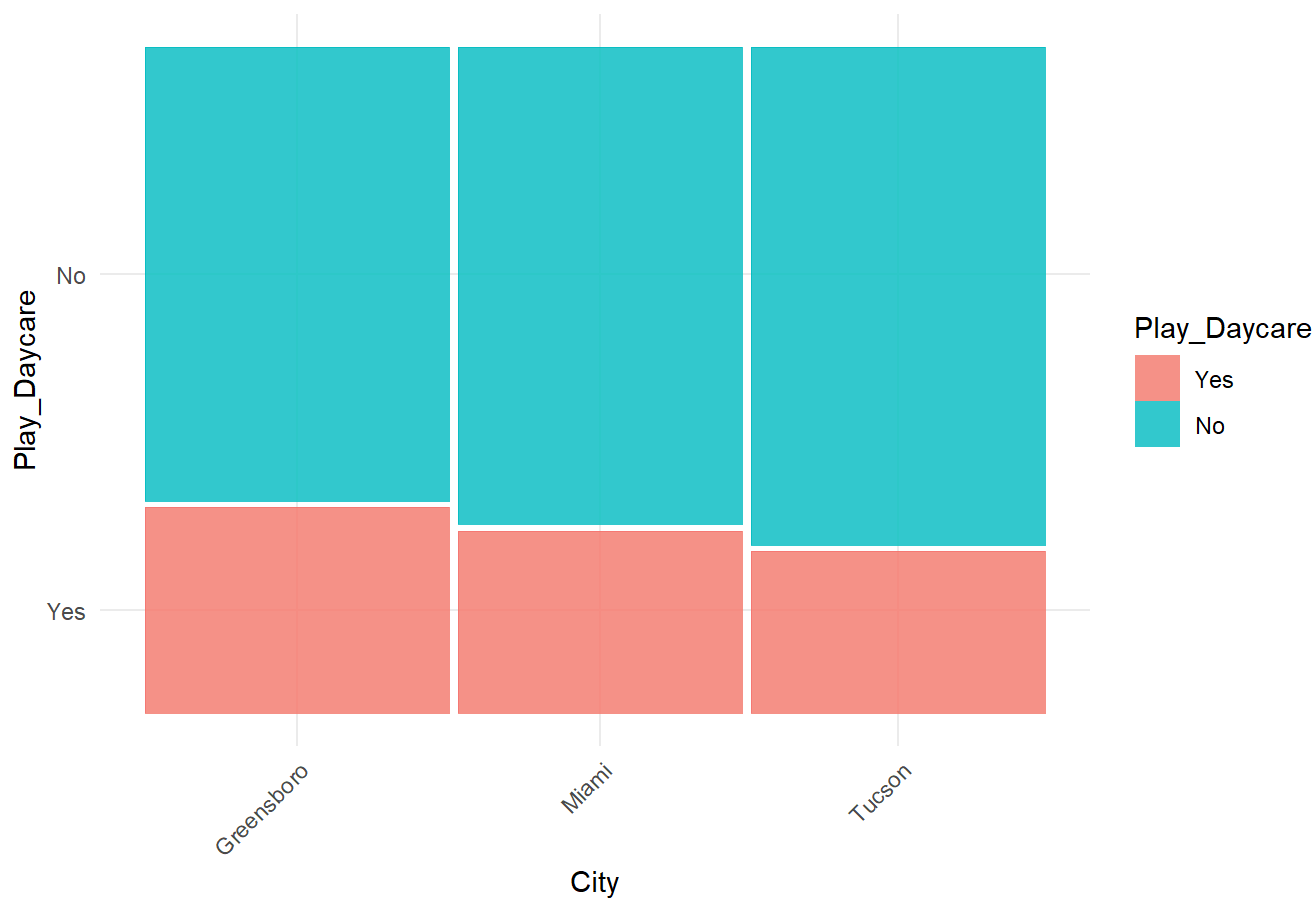
Mosaic Plot of Household_income and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and City -----
## Bivariate Analysis of Play_Daycare and City
##
## Frequency Table:
## Play_Daycare Greensboro Miami Tucson
##           Yes      58      48      44
##           No      128     126     136
##
## Relative Frequency Table:
## Play_Daycare Greensboro Miami Tucson Total
##           Yes     31.2% 27.6% 24.4% 27.8%
##           No     68.8% 72.4% 75.6% 72.2%
##
## Cramér's V:  0.01150411
```

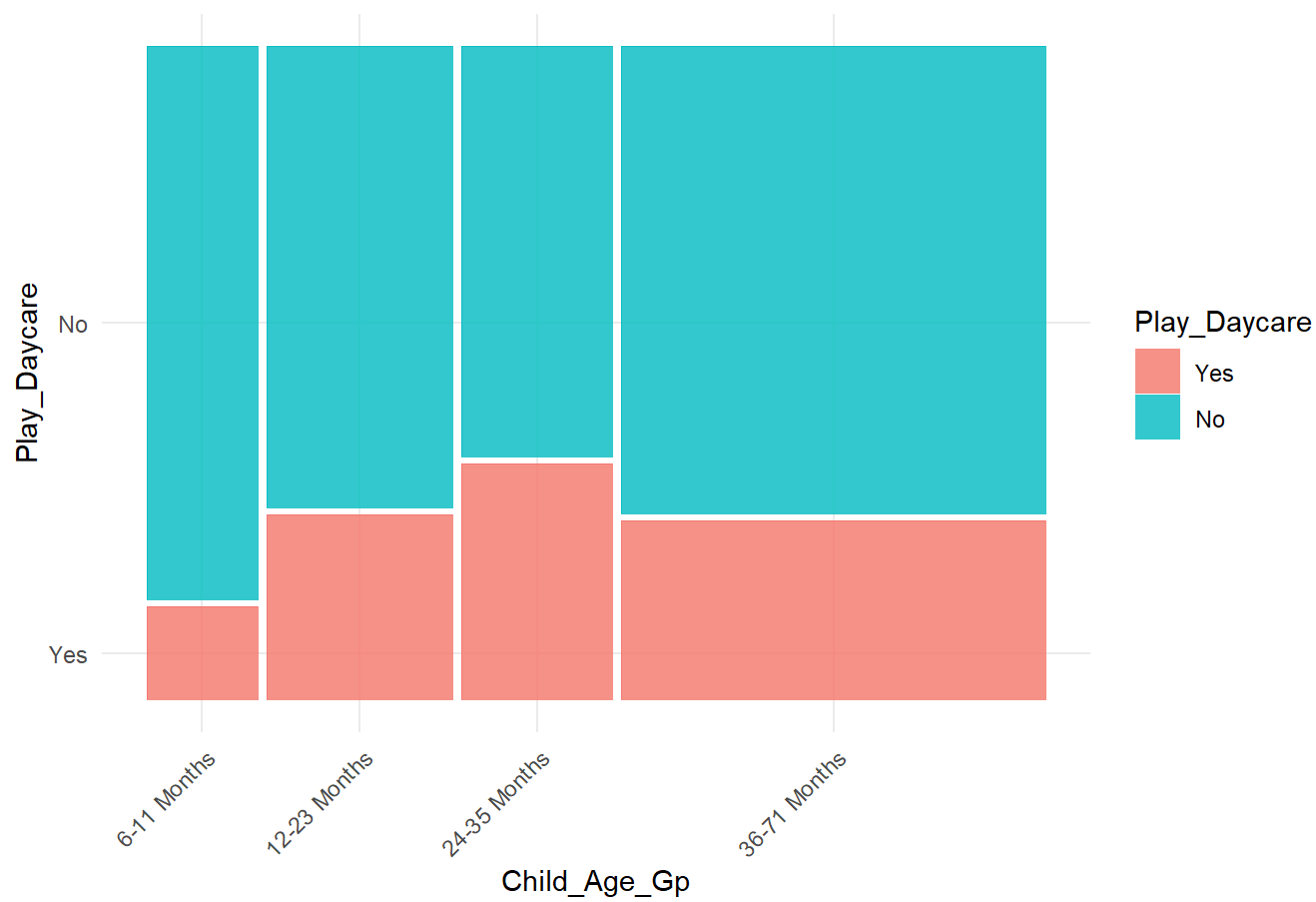
Mosaic Plot of City and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and Child_Age_Gp -----
## Bivariate Analysis of Play_Daycare and Child_Age_Gp
##
## Frequency Table:
## Play_Daycare 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##           Yes           10           33           34           73
##           No            59           82           59          190
##
## Relative Frequency Table:
## Play_Daycare 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##           Yes          14.5%          28.7%          36.6%          27.8% 27.8%
##           No            85.5%          71.3%          63.4%          72.2% 72.2%
##
## Cramér's V:  0.1113897
```

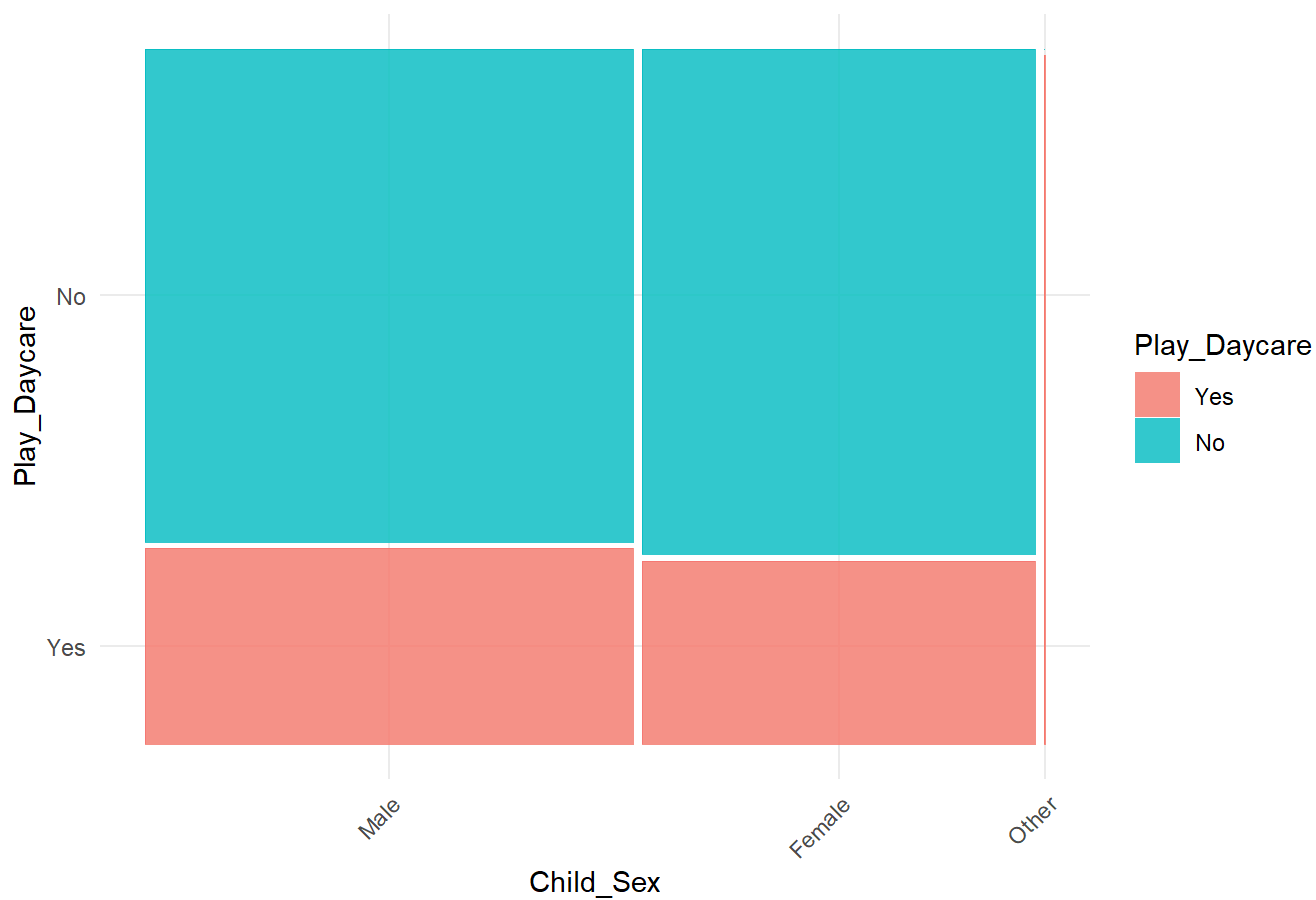
Mosaic Plot of Child_Age_Gp and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and Child_Sex -----
## Bivariate Analysis of Play_Daycare and Child_Sex
##
## Frequency Table:
##   Play_Daycare Male Female Other
##           Yes    85     64     1
##           No   214    176     0
##
## Relative Frequency Table:
##   Play_Daycare  Male Female  Other Total
##           Yes 28.4% 26.7% 100.0% 27.8%
##           No 71.6% 73.3%   0.0% 72.2%
##
## Cramér's V: 0.03869425
```

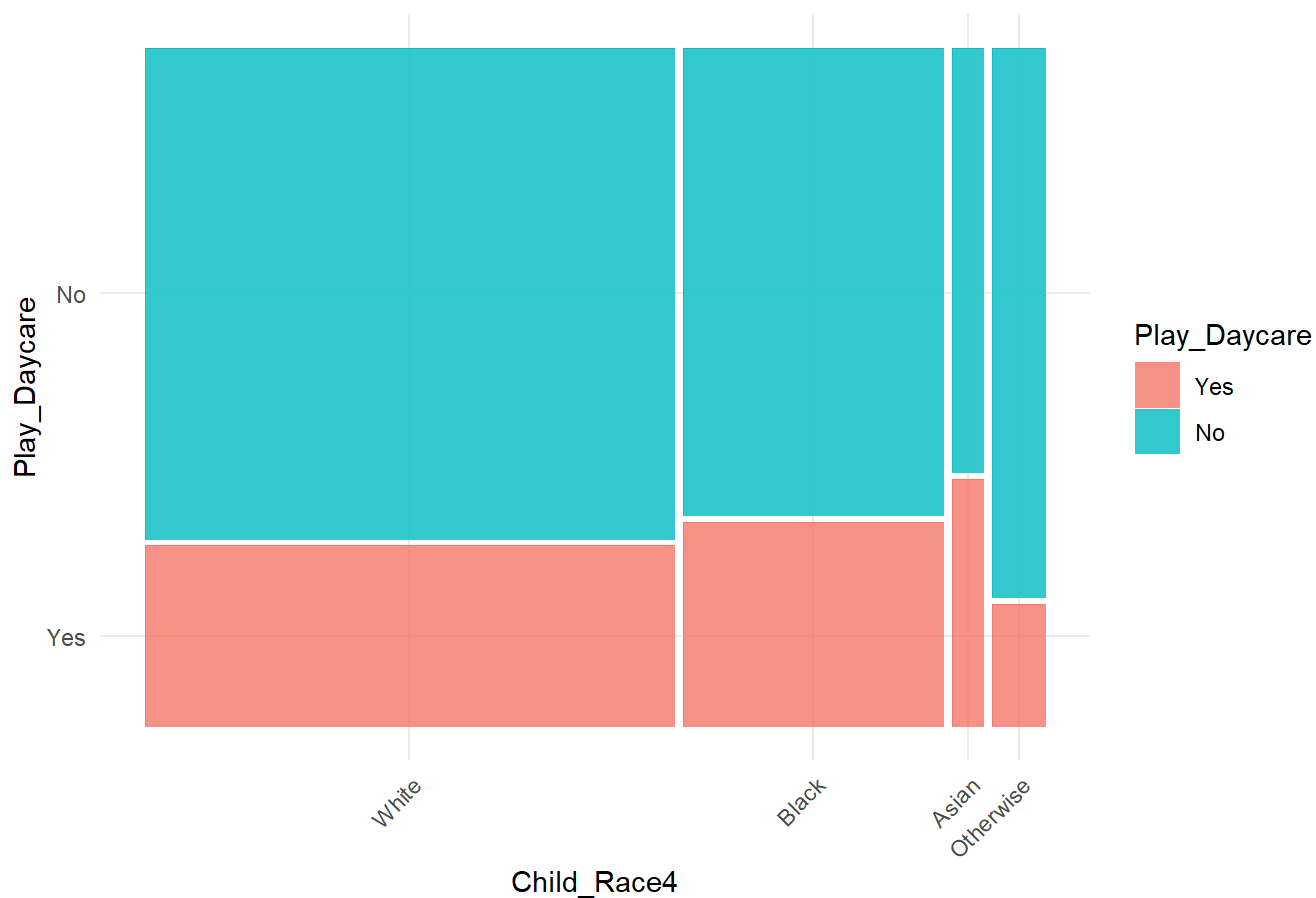

Mosaic Plot of Child_Sex and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and Child_Race4 -----
## Bivariate Analysis of Play_Daycare and Child_Race4
##
## Frequency Table:
##   Play_Daycare White Black Asian Otherwise
##           Yes    88    49     7         6
##           No   239   112    12        27
##
## Relative Frequency Table:
##   Play_Daycare White Black Asian Otherwise Total
##           Yes 26.9% 30.4% 36.8%    18.2% 27.8%
##           No 73.1% 69.6% 63.2%    81.8% 72.2%
##
## Cramér's V:  0
```

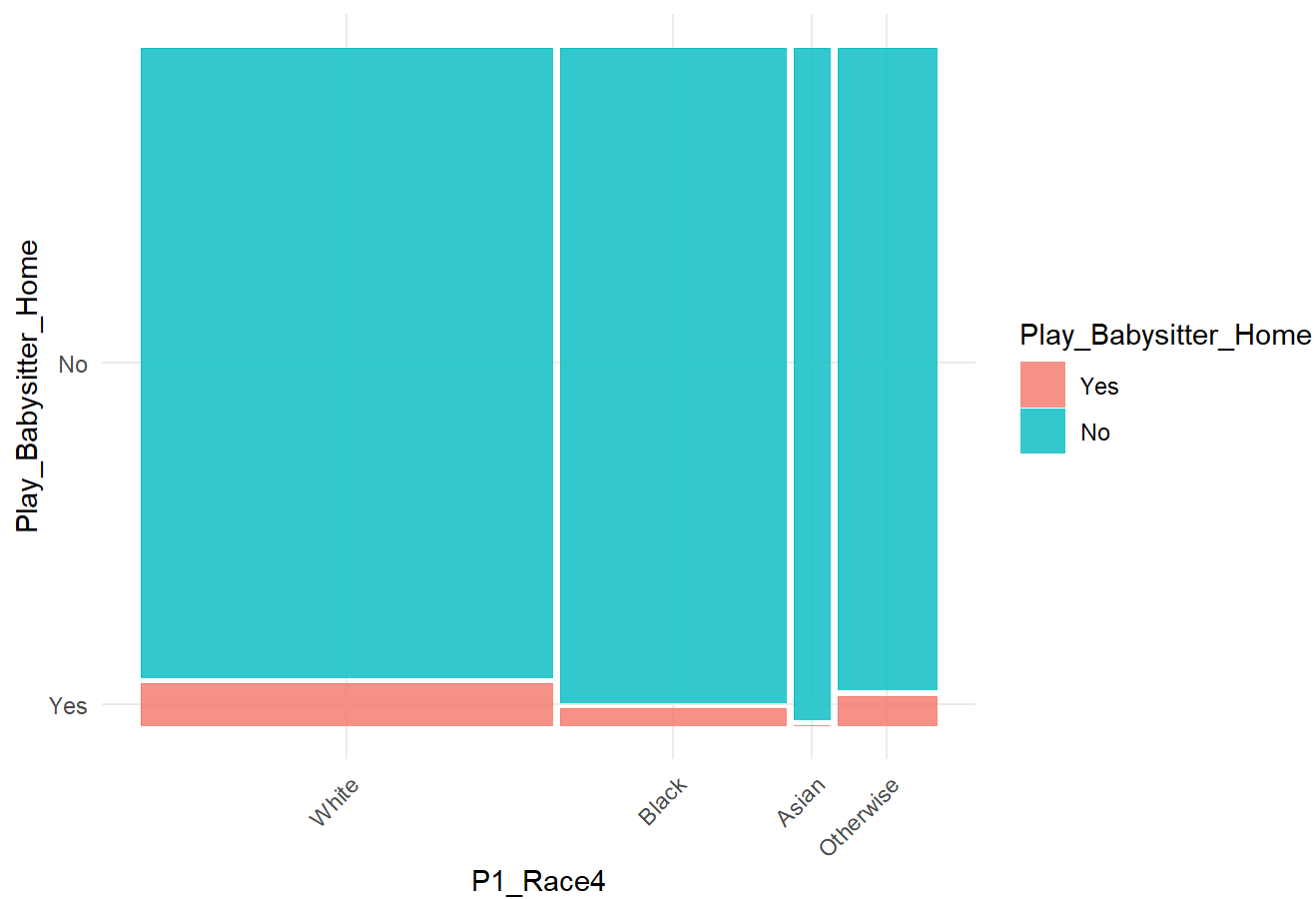
Mosaic Plot of Child_Race4 and Play_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and P1_Race4 -----
## Bivariate Analysis of Play_Babysitter_Home and P1_Race4
##
## Frequency Table:
## Play_Babysitter_Home White Black Asian Otherwise
##           Yes      18      4      0      3
##           No     270    154     25     66
##
## Relative Frequency Table:
## Play_Babysitter_Home White Black Asian Otherwise Total
##           Yes   6.2%   2.5%   0.0%    4.3%   4.6%
##           No  93.8%  97.5% 100.0%   95.7%  95.4%
##
## Cramér's V:  0.05289644
```

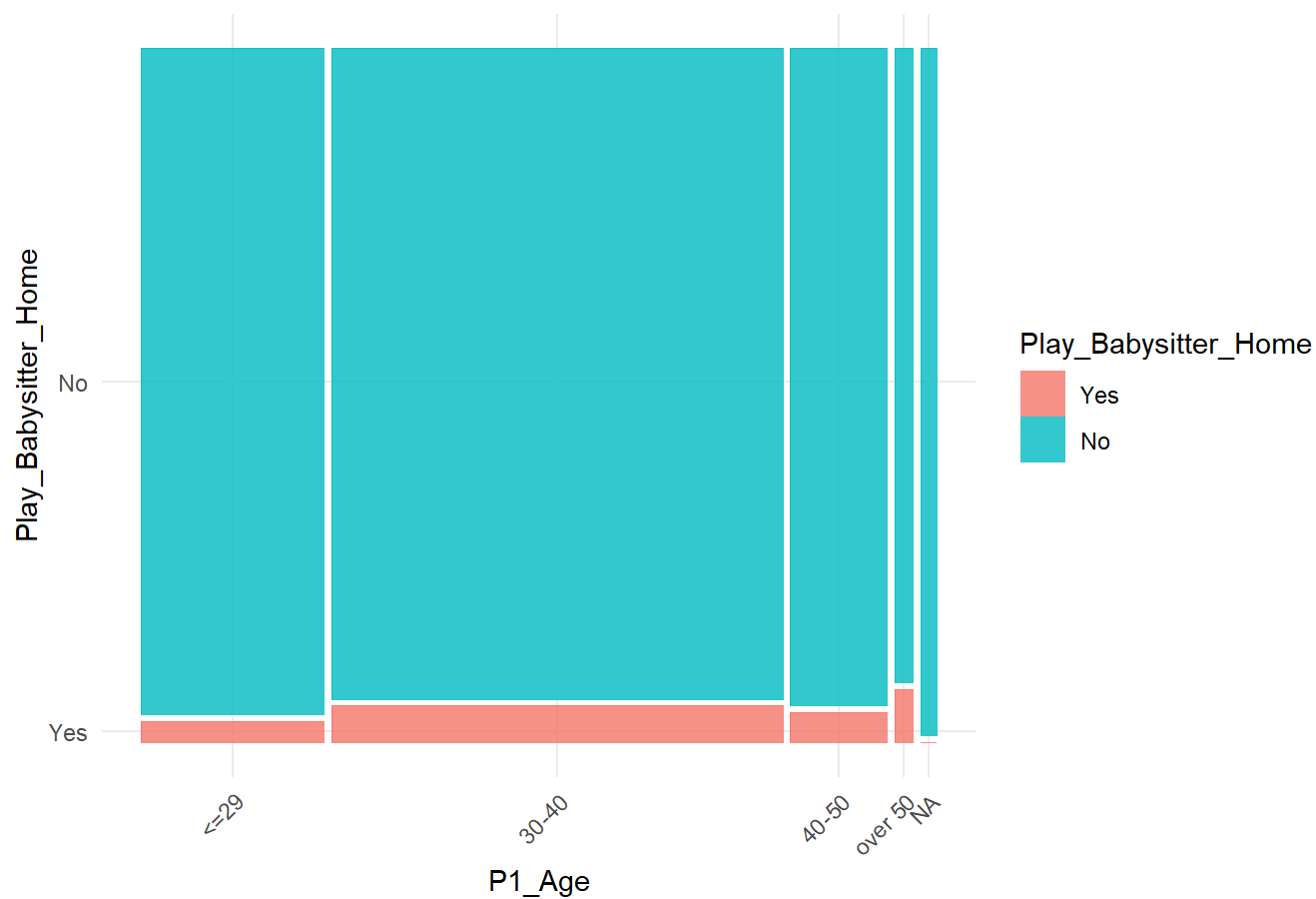
Mosaic Plot of P1_Race4 and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and P1_Age -----
## Bivariate Analysis of Play_Babysitter_Home and P1_Age
##
## Frequency Table:
## Play_Babysitter_Home <=29 30-40 40-50 over 50 NA_
##           Yes      4      17      3      1      0
##           No     125     302     65     12     11
##
## Relative Frequency Table:
## Play_Babysitter_Home <=29 30-40 40-50 over 50    NA_ Total
##           Yes    3.1%  5.3%  4.4%    7.7%   0.0%  4.6%
##           No   96.9%  94.7%  95.6%   92.3% 100.0% 95.4%
##
## Cramér's V:  0
```

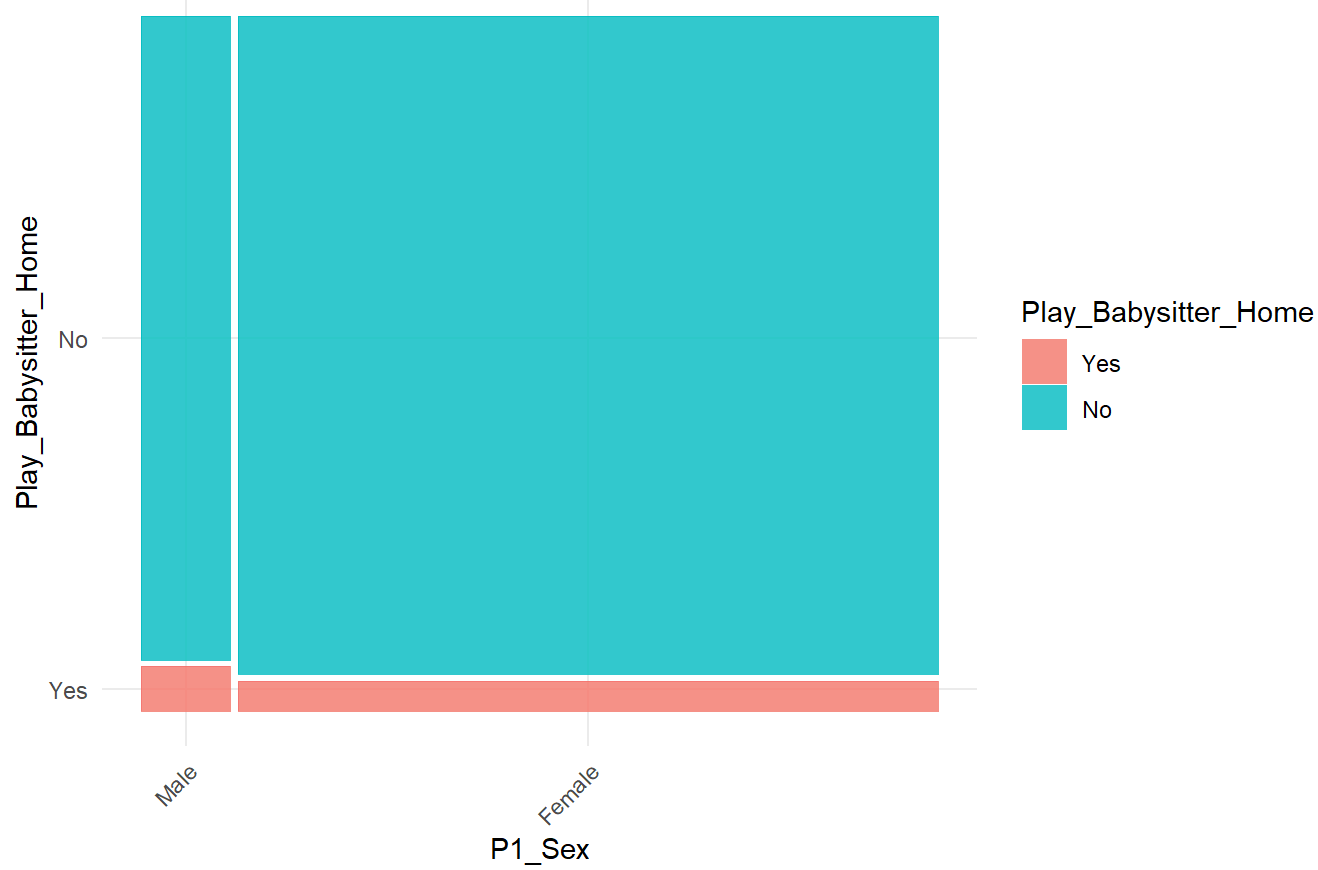
Mosaic Plot of P1_Age and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and P1_Sex -----
## Bivariate Analysis of Play_Babysitter_Home and P1_Sex
##
## Frequency Table:
##   Play_Babysitter_Home   Male Female
##               Yes         4      21
##               No        57     458
##
## Relative Frequency Table:
##   Play_Babysitter_Home   Male Female Total
##               Yes    6.6%   4.4%  4.6%
##               No   93.4%  95.6% 95.4%
##
## Cramér's V:  0
```

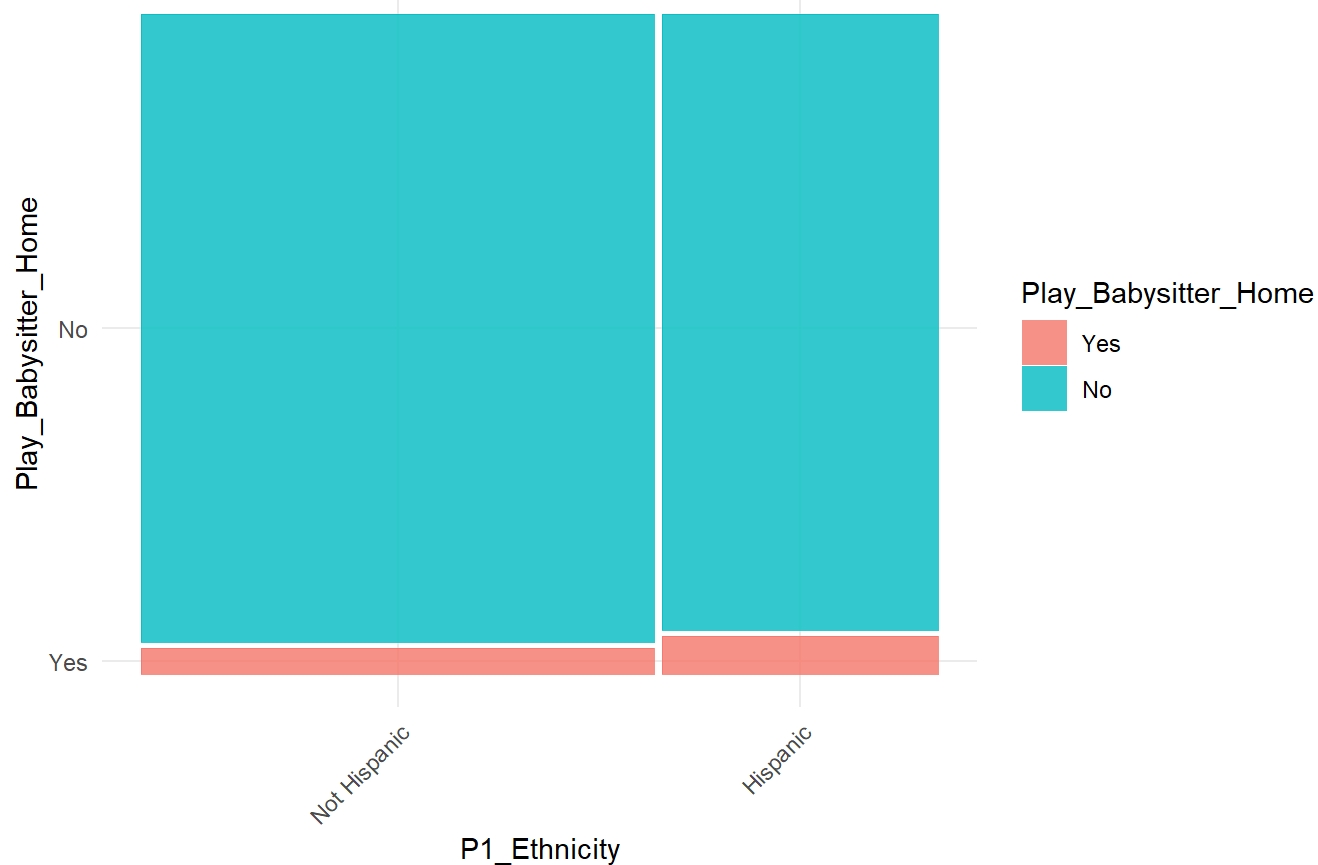
Mosaic Plot of P1_Sex and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and P1_Ethnicity -----
## Bivariate Analysis of Play_Babysitter_Home and P1_Ethnicity
##
## Frequency Table:
##   Play_Babysitter_Home Not Hispanic Hispanic
##               Yes           14           11
##               No           337          178
##
## Relative Frequency Table:
##   Play_Babysitter_Home Not Hispanic Hispanic Total
##               Yes           4.0%           5.8%  4.6%
##               No           96.0%          94.2% 95.4%
##
## Cramér's V:  0
```

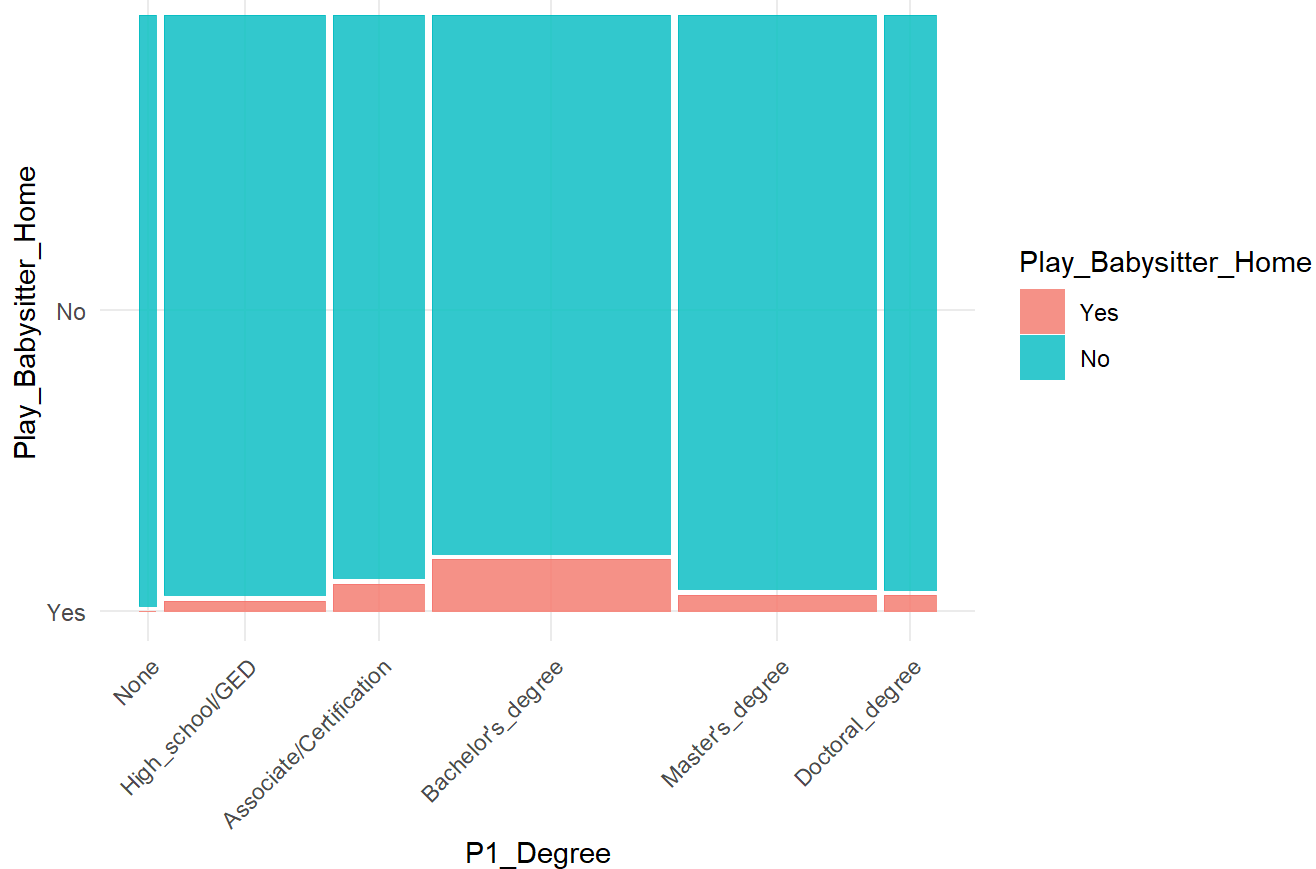
Mosaic Plot of P1_Ethnicity and Play_Babysitter_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Babysitter_Home and P1_Degree -----
## Bivariate Analysis of Play_Babysitter_Home and P1_Degree
##
## Frequency Table:
##   Play_Babysitter_Home  None High_school/GED Associate/Certification
##               Yes      0              2              3
##               No      12             113             62
##   Bachelor's_degree Master's_degree Doctoral_degree
##               15              4              1
##               155             137             36
##
## Relative Frequency Table:
##   Play_Babysitter_Home  None High_school/GED Associate/Certification
##               Yes      0.0%              1.7%              4.6%
##               No     100.0%              98.3%              95.4%
##   Bachelor's_degree Master's_degree Doctoral_degree Total
##               8.8%              2.8%              2.7%  4.6%
##               91.2%              97.2%              97.3% 95.4%
##
## Cramér's V:  0.1042605
```

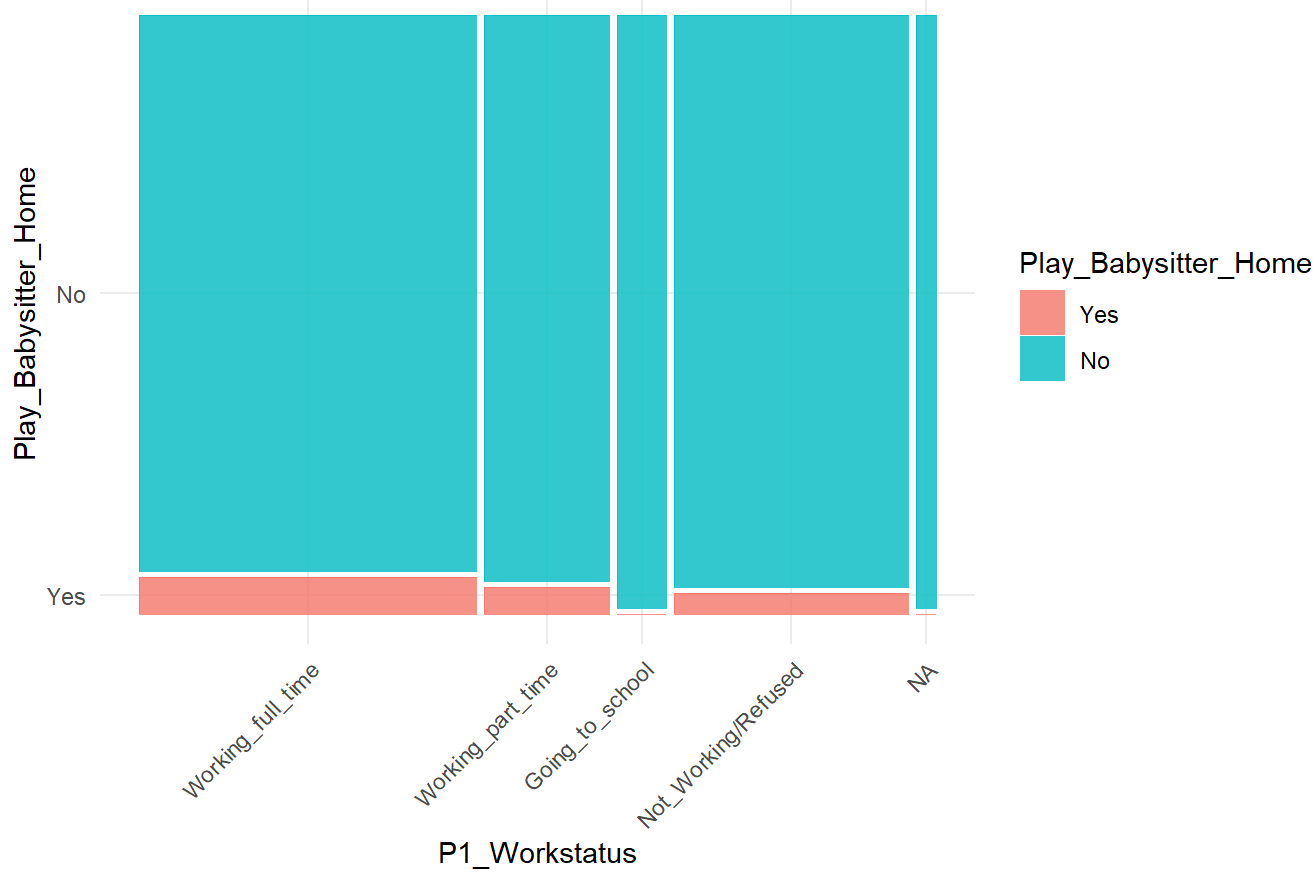
Mosaic Plot of P1_Degree and Play_Babysitter_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Babysitter_Home and P1_Workstatus -----
## Bivariate Analysis of Play_Babysitter_Home and P1_Workstatus
##
## Frequency Table:
## Play_Babysitter_Home Working_full_time Working_part_time Going_to_school
## Yes 15 4 0
## No 223 84 35
## Not_Working/Refused NA_
## 6 0
## 159 14
##
## Relative Frequency Table:
## Play_Babysitter_Home Working_full_time Working_part_time Going_to_school
## Yes 6.3% 4.5% 0.0%
## No 93.7% 95.5% 100.0%
## Not_Working/Refused NA_ Total
## 3.6% 0.0% 4.6%
## 96.4% 100.0% 95.4%
##
## Cramér's V: 0.02152977
```

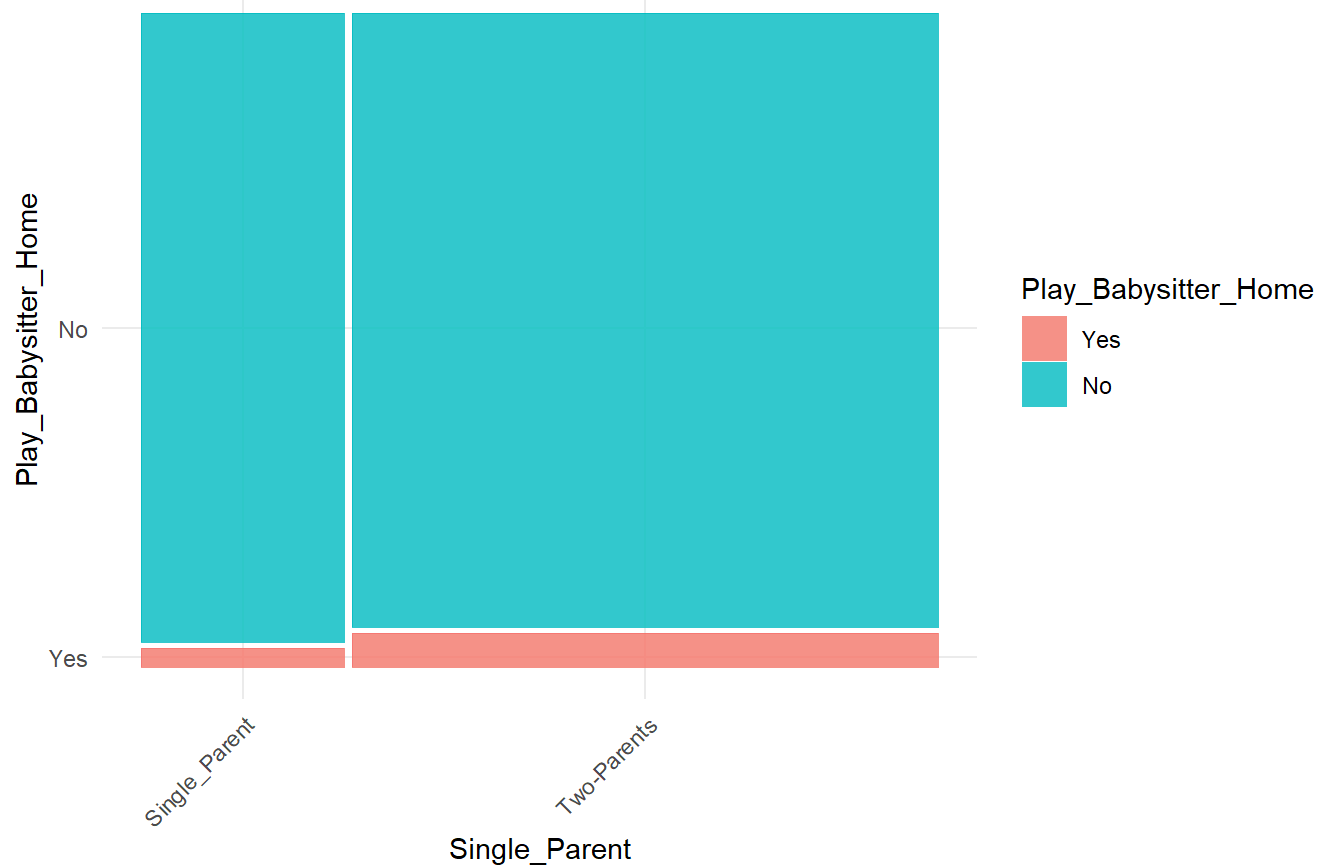
Mosaic Plot of P1_Workstatus and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and Single_Parent -----
## Bivariate Analysis of Play_Babysitter_Home and Single_Parent
##
## Frequency Table:
##   Play_Babysitter_Home Single_Parent Two-Parents
##               Yes           4           21
##               No          135          380
##
## Relative Frequency Table:
##   Play_Babysitter_Home Single_Parent Two-Parents Total
##               Yes           2.9%           5.2%  4.6%
##               No          97.1%          94.8% 95.4%
##
## Cramér's V:  0.02356573
```

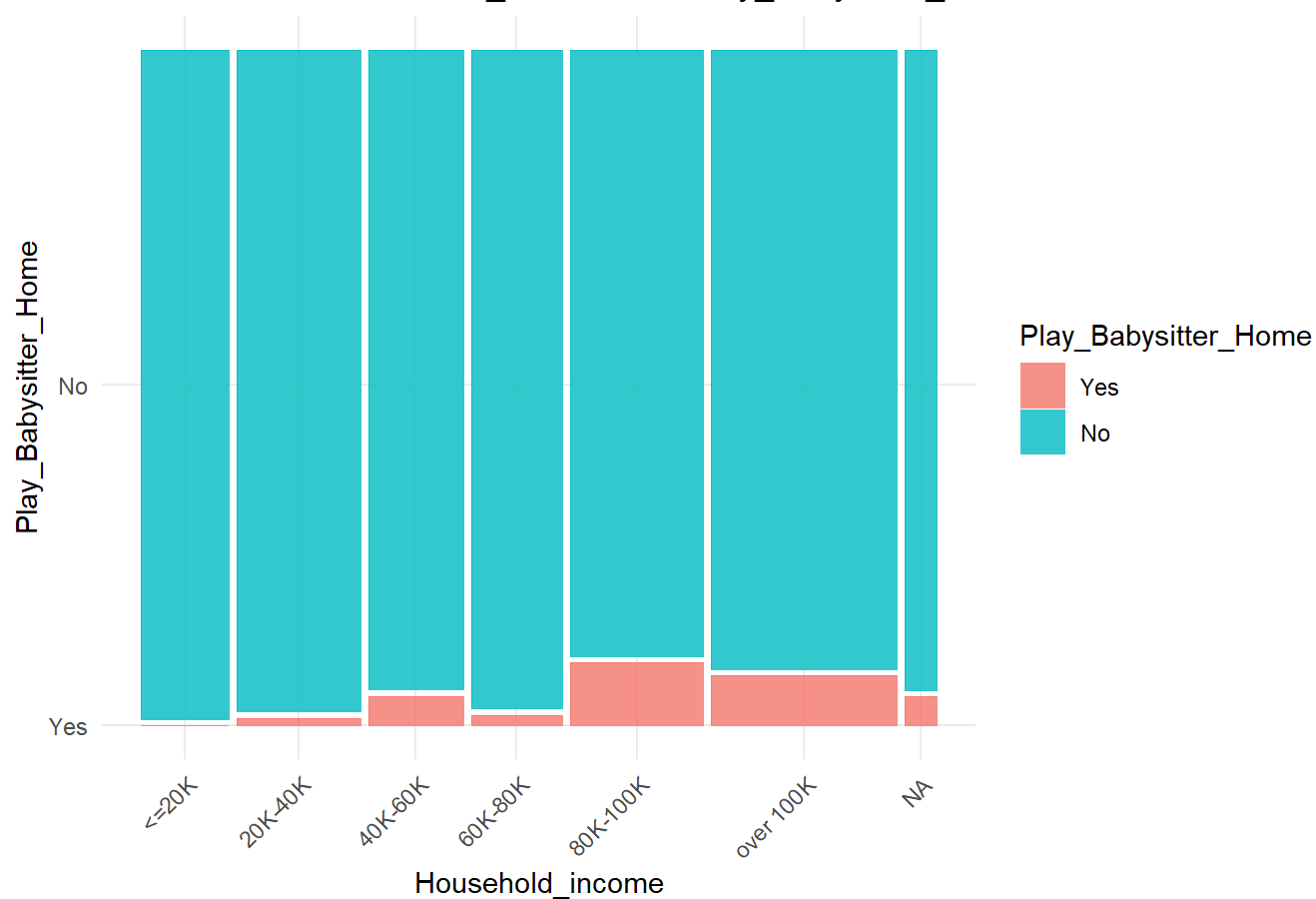

Mosaic Plot of Single_Parent and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and Household_income -----
## Bivariate Analysis of Play_Babysitter_Home and Household_income
##
## Frequency Table:
## Play_Babysitter_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes      0      1      3      1      9      10      1
##           No     63     89     65     65     87     124     22
##
## Relative Frequency Table:
## Play_Babysitter_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes    0.0%    1.1%    4.4%    1.5%    9.4%    7.5%  4.3%
##           No  100.0%   98.9%   95.6%   98.5%   90.6%   92.5% 95.7%
##
## Total
##    4.6%
##   95.4%
##
## Cramér's V:  0.1245701
```

Mosaic Plot of Household_income and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and City -----
## Bivariate Analysis of Play_Babysitter_Home and City
##
## Frequency Table:
## Play_Babysitter_Home Greensboro Miami Tucson
##           Yes           5           9          11
##           No          181          165          169
##
## Relative Frequency Table:
## Play_Babysitter_Home Greensboro Miami Tucson Total
##           Yes          2.7%   5.2%   6.1%   4.6%
##           No          97.3%  94.8%  93.9%  95.4%
##
## Cramér's V:  0.03322494
```

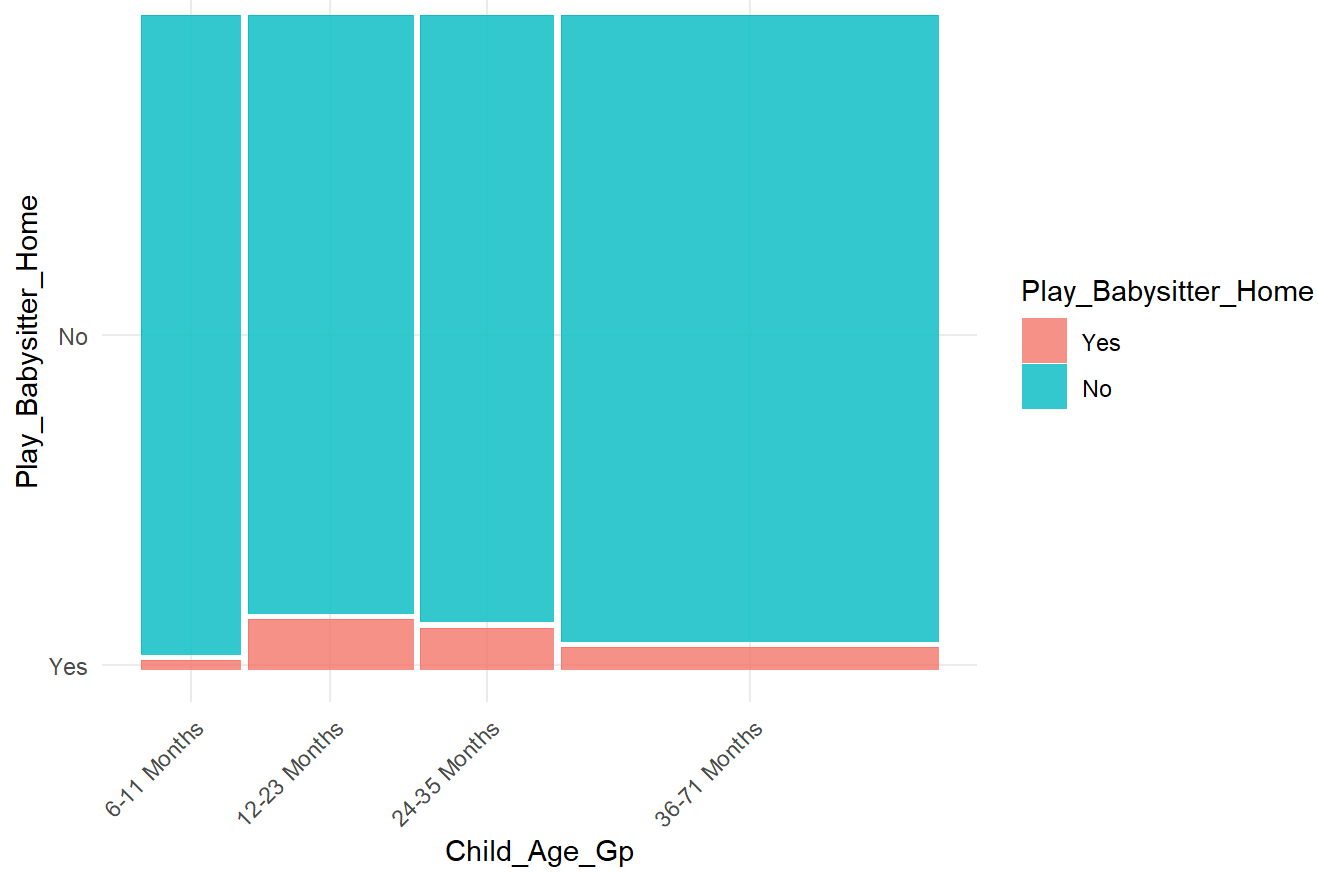
Mosaic Plot of City and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and Child_Age_Gp -----
## Bivariate Analysis of Play_Babysitter_Home and Child_Age_Gp
##
## Frequency Table:
##   Play_Babysitter_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##                        Yes          1           9           6           9
##                        No          68          106          87          254
##
## Relative Frequency Table:
##   Play_Babysitter_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##                        Yes          1.4%         7.8%         6.5%         3.4%  4.6%
##                        No          98.6%         92.2%         93.5%         96.6% 95.4%
##
## Cramér's V:  0.07212812
```

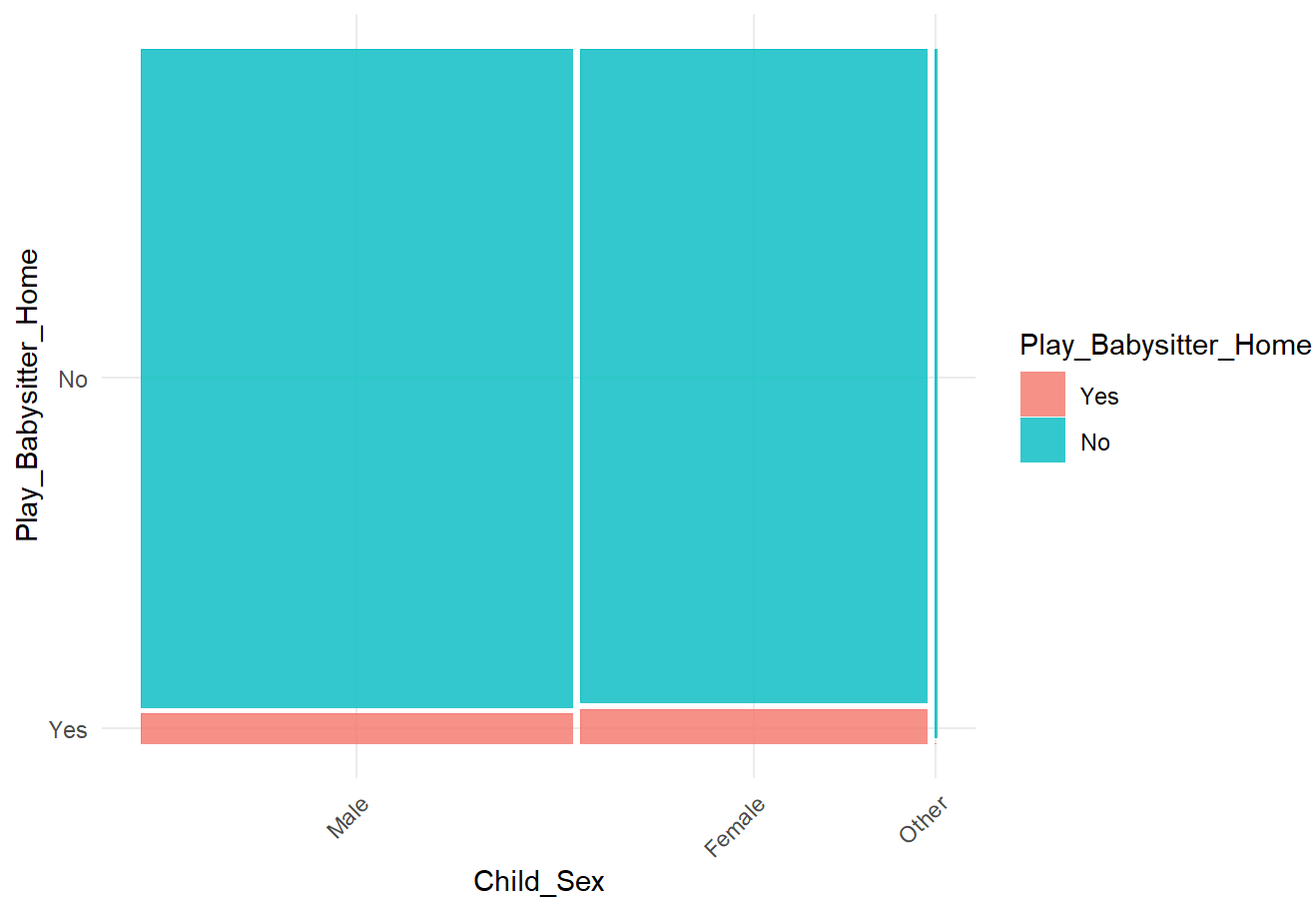
Mosaic Plot of Child_Age_Gp and Play_Babysitter_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Babysitter_Home and Child_Sex -----
## Bivariate Analysis of Play_Babysitter_Home and Child_Sex
##
## Frequency Table:
## Play_Babysitter_Home Male Female Other
##           Yes      13      12      0
##           No     286     228      1
##
## Relative Frequency Table:
## Play_Babysitter_Home  Male Female  Other Total
##           Yes    4.3%   5.0%   0.0%  4.6%
##           No   95.7%  95.0% 100.0% 95.4%
##
## Cramér's V:  0
```

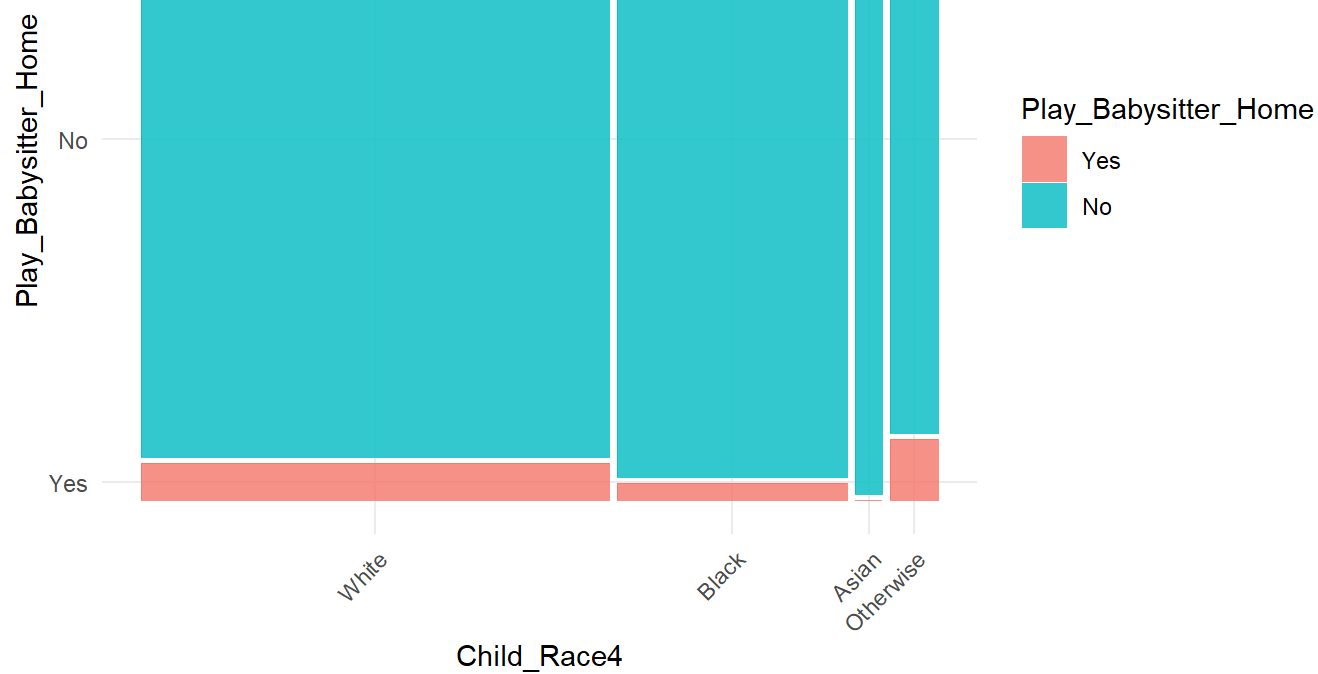
Mosaic Plot of Child_Sex and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Babysitter_Home and Child_Race4 -----
## Bivariate Analysis of Play_Babysitter_Home and Child_Race4
##
## Frequency Table:
##   Play_Babysitter_Home White Black Asian Otherwise
##               Yes      18     4     0         3
##               No     309    157    19        30
##
## Relative Frequency Table:
##   Play_Babysitter_Home White Black  Asian Otherwise Total
##               Yes    5.5%  2.5%   0.0%     9.1%  4.6%
##               No   94.5% 97.5% 100.0%    90.9% 95.4%
##
## Cramér's V:  0.05531608
```

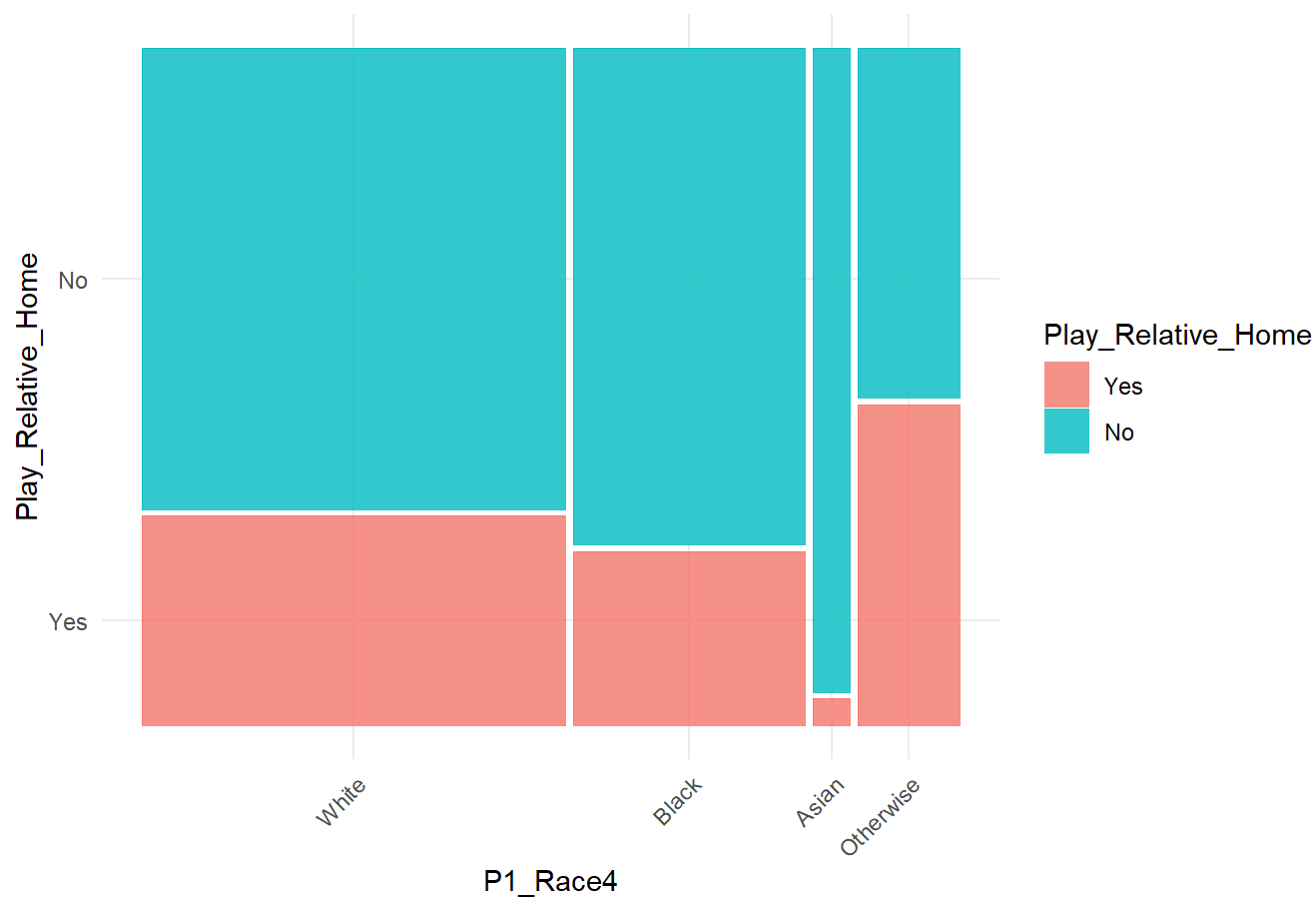
Mosaic Plot of Child_Race4 and Play_Babysitter_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and P1_Race4 -----
## Bivariate Analysis of Play_Relative_Home and P1_Race4
##
## Frequency Table:
## Play_Relative_Home White Black Asian Otherwise
##           Yes      90      41       1        33
##           No     198     117      24        36
##
## Relative Frequency Table:
## Play_Relative_Home White Black Asian Otherwise Total
##           Yes 31.2% 25.9%  4.0%      47.8% 30.6%
##           No 68.8% 74.1% 96.0%      52.2% 69.4%
##
## Cramér's V:  0.1757434
```

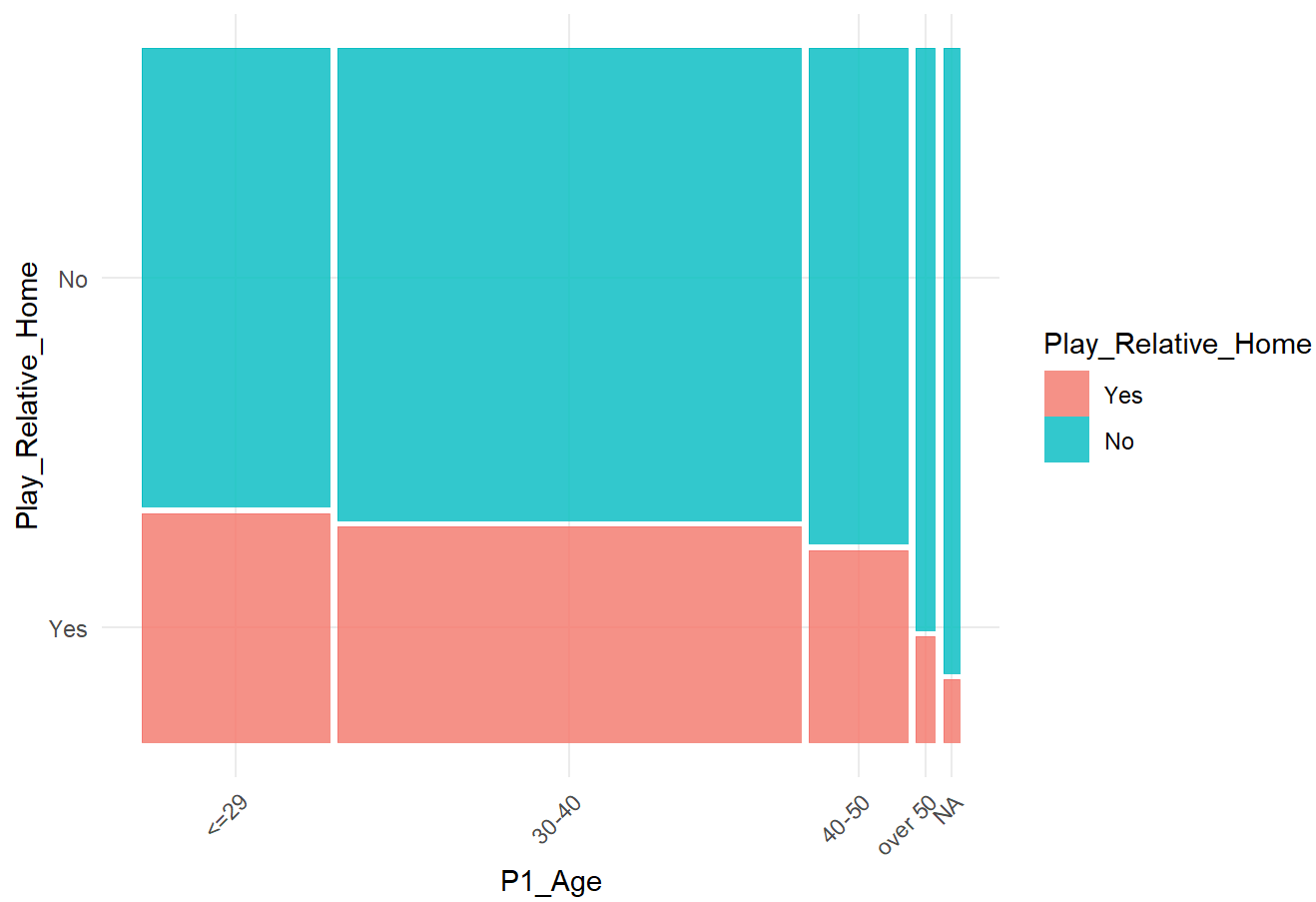
Mosaic Plot of P1_Race4 and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and P1_Age -----
## Bivariate Analysis of Play_Relative_Home and P1_Age
##
## Frequency Table:
## Play_Relative_Home <=29 30-40 40-50 over 50 NA_
##           Yes    43    100    19      2    1
##           No     86    219    49     11   10
##
## Relative Frequency Table:
## Play_Relative_Home <=29 30-40 40-50 over 50  NA_ Total
##           Yes 33.3% 31.3% 27.9%  15.4%  9.1% 30.6%
##           No 66.7% 68.7% 72.1%  84.6% 90.9% 69.4%
##
## Cramér's V: 0.03262214
```

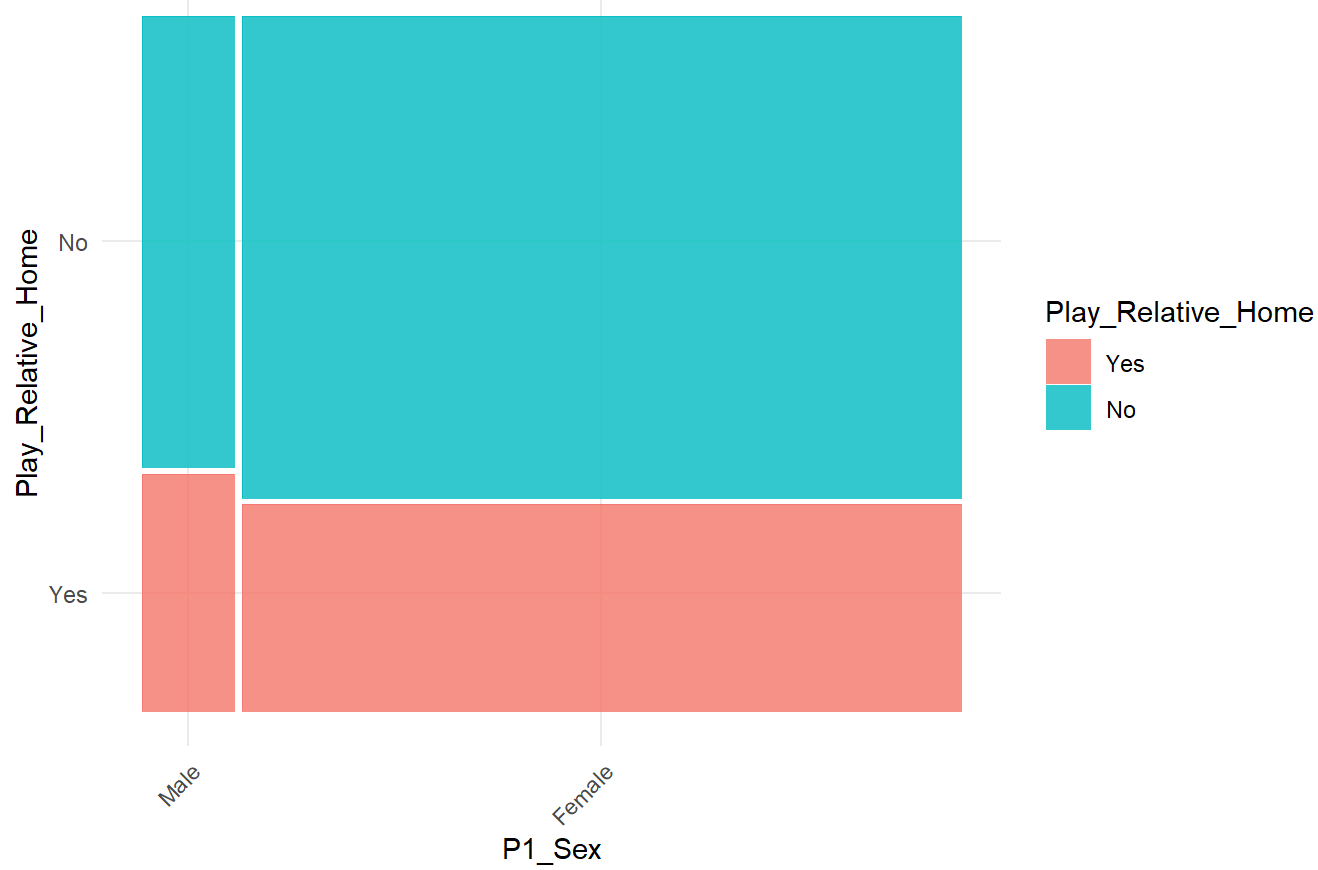
Mosaic Plot of P1_Age and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and P1_Sex -----
## Bivariate Analysis of Play_Relative_Home and P1_Sex
##
## Frequency Table:
##   Play_Relative_Home   Male Female
##           Yes         21    144
##           No          40    335
##
## Relative Frequency Table:
##   Play_Relative_Home   Male Female Total
##           Yes      34.4%  30.1% 30.6%
##           No      65.6%  69.9% 69.4%
##
##
## Cramér's V:  0
```

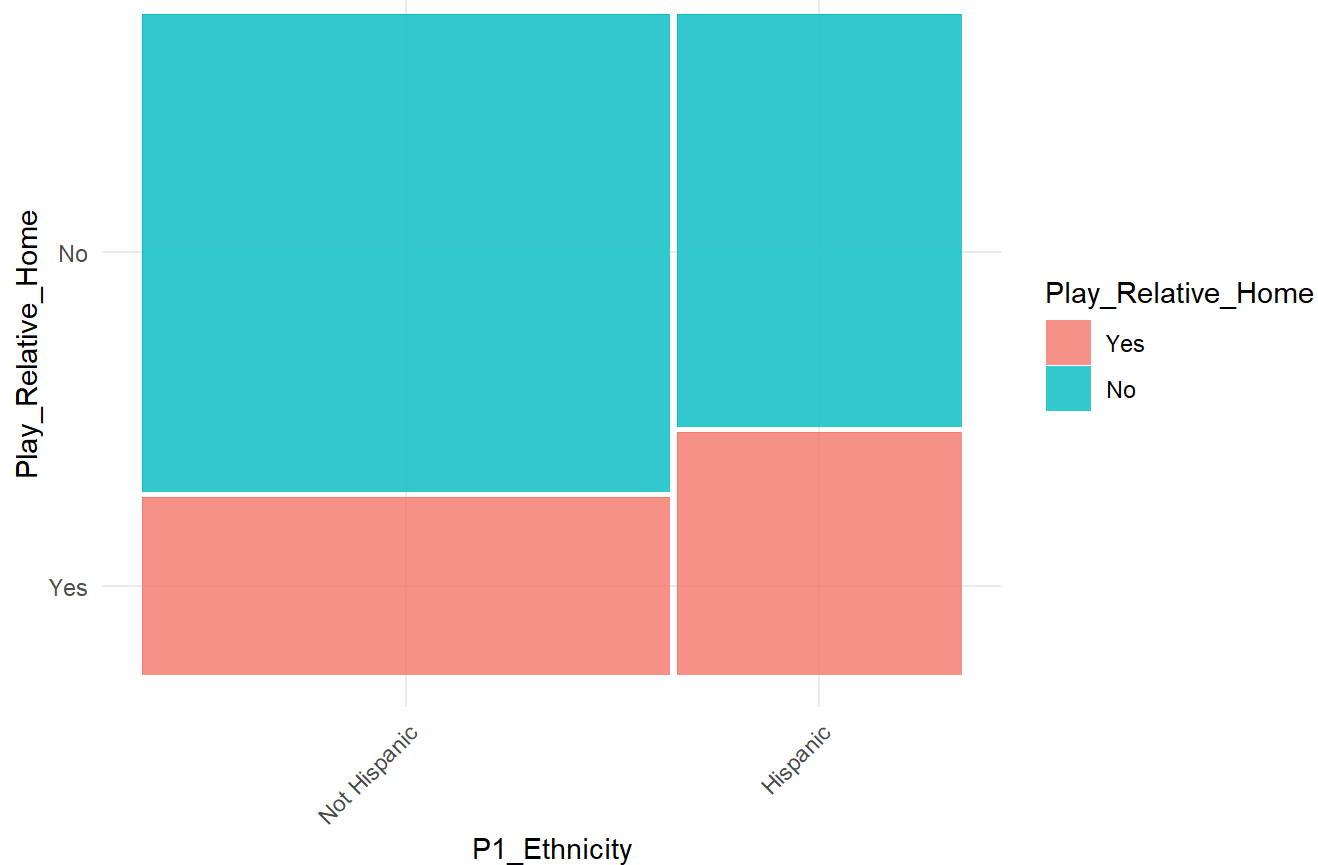

Mosaic Plot of P1_Sex and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and P1_Ethnicity -----
## Bivariate Analysis of Play_Relative_Home and P1_Ethnicity
##
## Frequency Table:
##   Play_Relative_Home Not Hispanic Hispanic
##               Yes           95           70
##               No           256          119
##
## Relative Frequency Table:
##   Play_Relative_Home Not Hispanic Hispanic Total
##               Yes      27.1%    37.0% 30.6%
##               No      72.9%    63.0% 69.4%
##
## Cramér's V:  0.093923
```

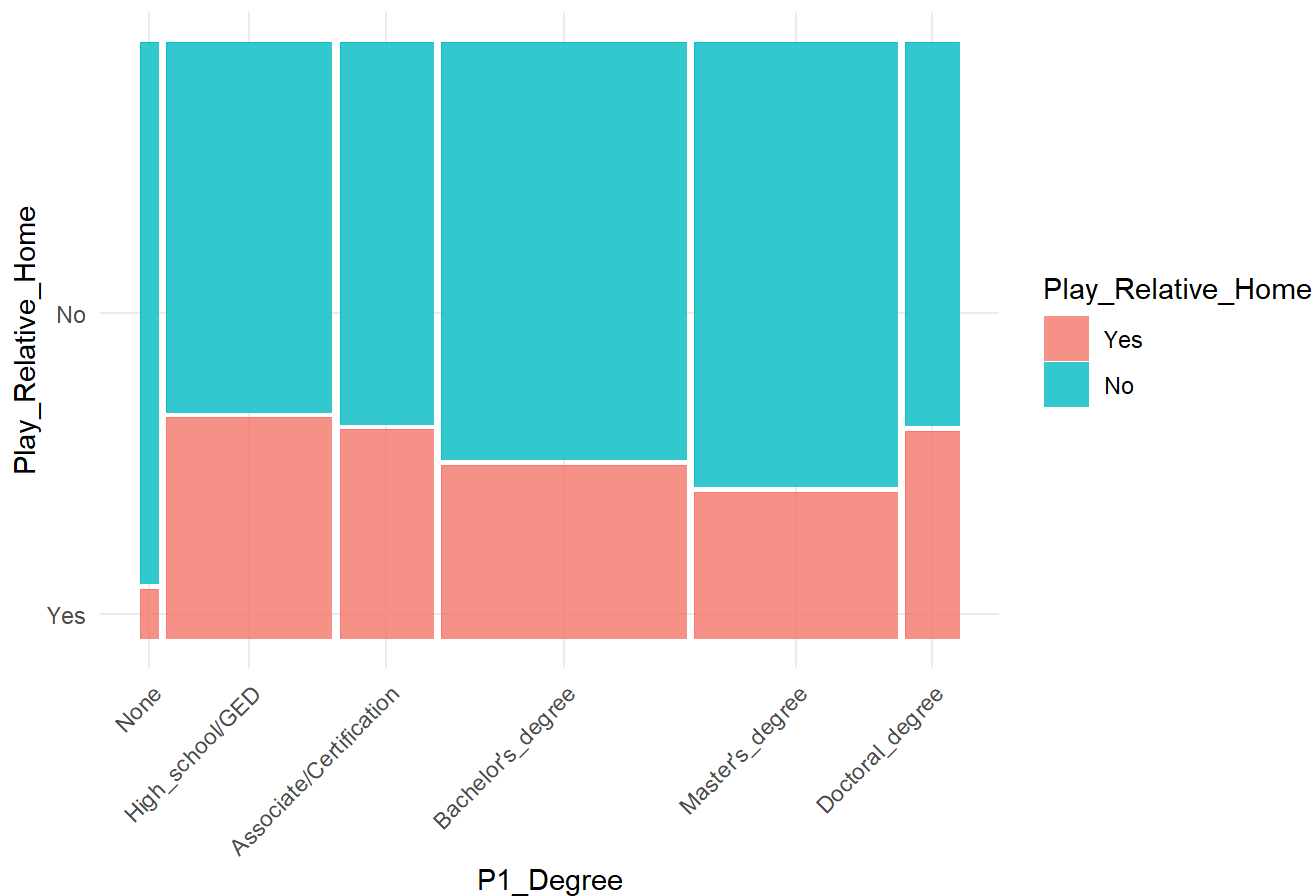
Mosaic Plot of P1_Ethnicity and Play_Relative_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Relative_Home and P1_Degree -----
## Bivariate Analysis of Play_Relative_Home and P1_Degree
##
## Frequency Table:
## Play_Relative_Home None High_school/GED Associate/Certification
##           Yes      1           43           23
##           No      11           72           42
## Bachelor's_degree Master's_degree Doctoral_degree
##           50           35           13
##           120          106           24
##
## Relative Frequency Table:
## Play_Relative_Home None High_school/GED Associate/Certification
##           Yes  8.3%           37.4%           35.4%
##           No  91.7%           62.6%           64.6%
## Bachelor's_degree Master's_degree Doctoral_degree Total
##           29.4%           24.8%           35.1% 30.6%
##           70.6%           75.2%           64.9% 69.4%
##
## Cramér's V:  0.08268109
```

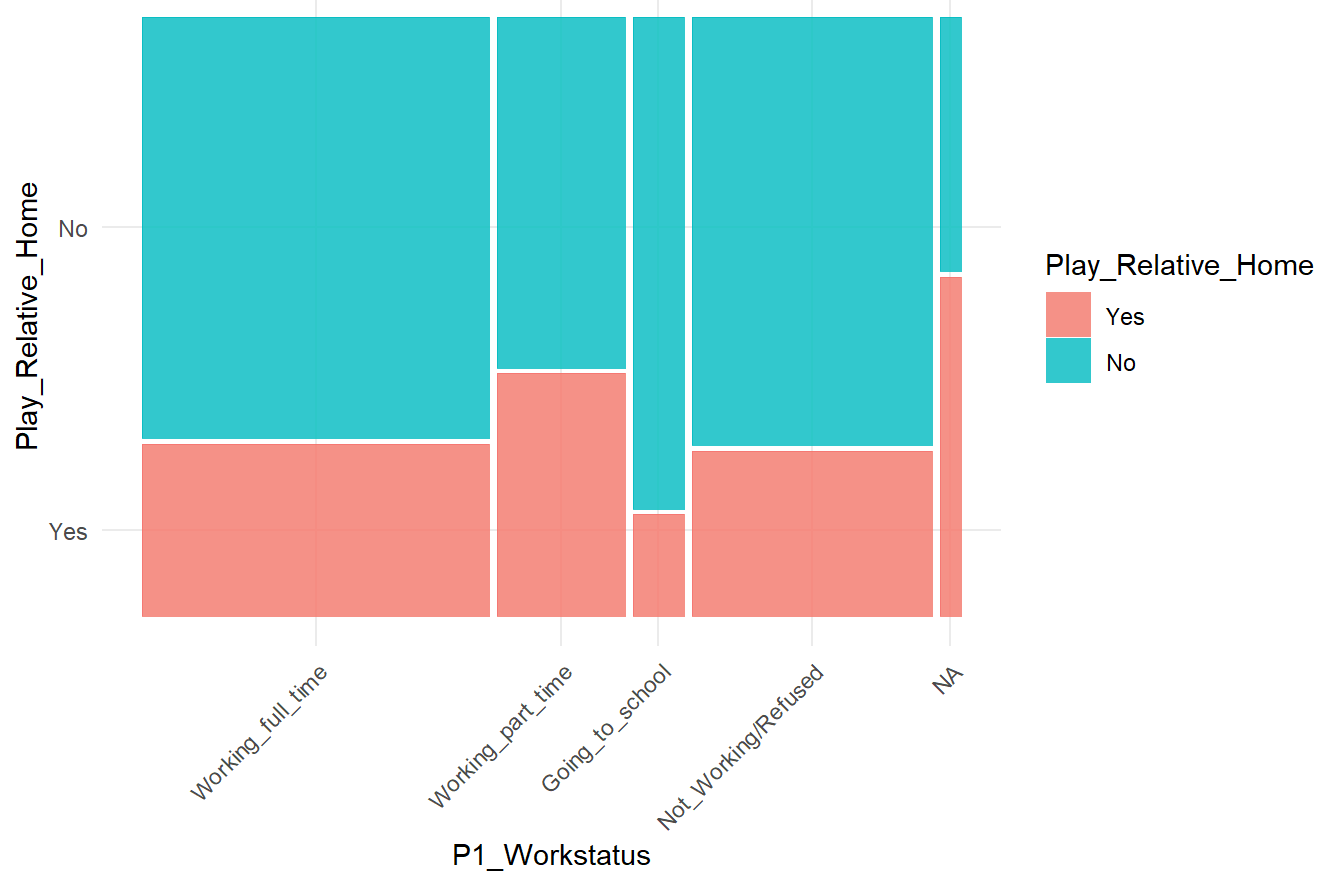
Mosaic Plot of P1_Degree and Play_Relative_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Relative_Home and P1_Workstatus -----
## Bivariate Analysis of Play_Relative_Home and P1_Workstatus
##
## Frequency Table:
## Play_Relative_Home Working_full_time Working_part_time Going_to_school
##           Yes           69           36           6
##           No           169           52           29
## Not_Working/Refused NA_
##           46      8
##           119     6
##
## Relative Frequency Table:
## Play_Relative_Home Working_full_time Working_part_time Going_to_school
##           Yes           29.0%           40.9%           17.1%
##           No           71.0%           59.1%           82.9%
## Not_Working/Refused  NA_ Total
##           27.9% 57.1% 30.6%
##           72.1% 42.9% 69.4%
##
## Cramér's V:  0.1285065
```

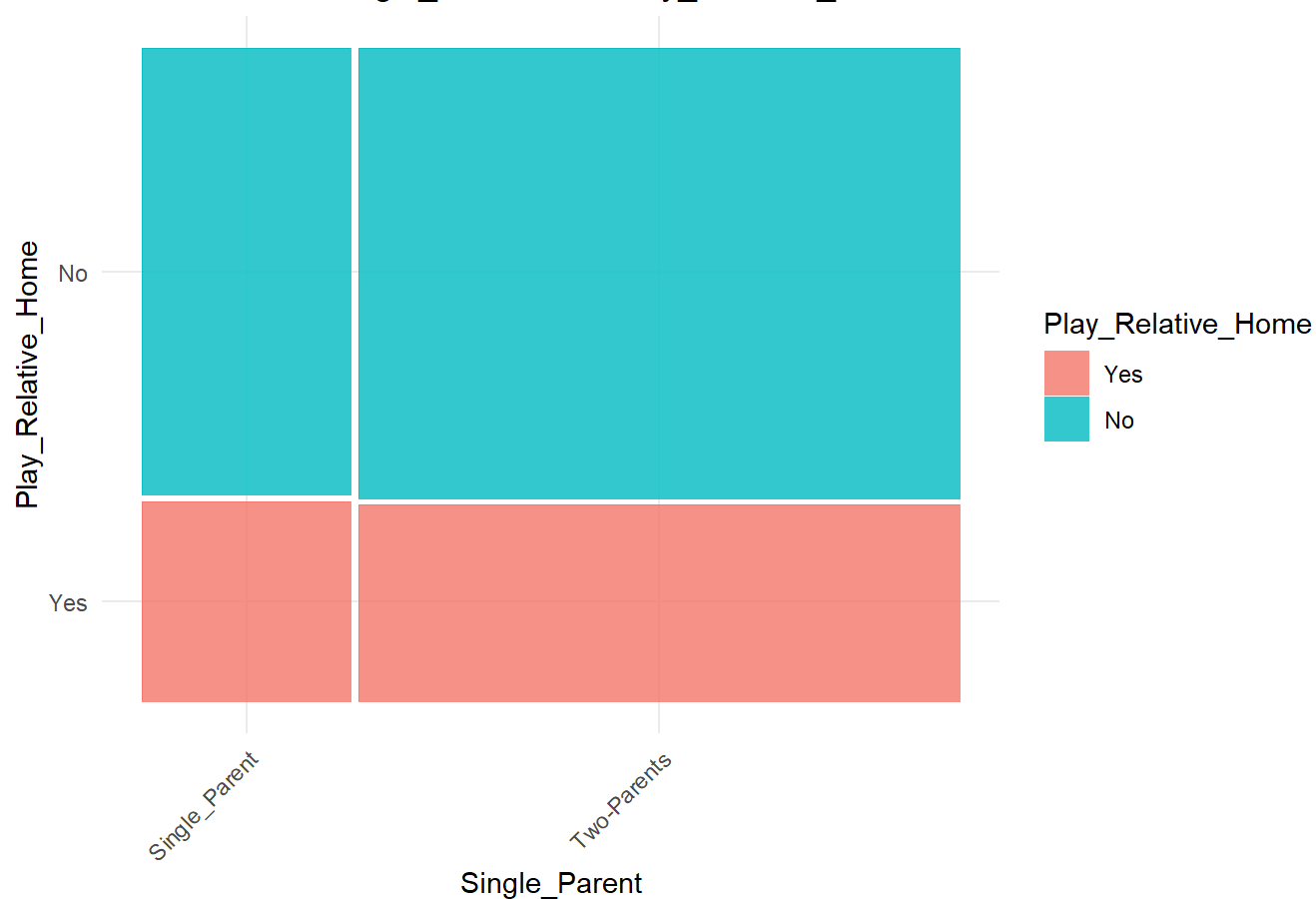
Mosaic Plot of P1_Workstatus and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and Single_Parent -----
## Bivariate Analysis of Play_Relative_Home and Single_Parent
##
## Frequency Table:
##   Play_Relative_Home Single_Parent Two-Parents
##               Yes           43           122
##               No            96           279
##
## Relative Frequency Table:
##   Play_Relative_Home Single_Parent Two-Parents Total
##               Yes           30.9%           30.4% 30.6%
##               No            69.1%           69.6% 69.4%
##
## Cramér's V:  0
```

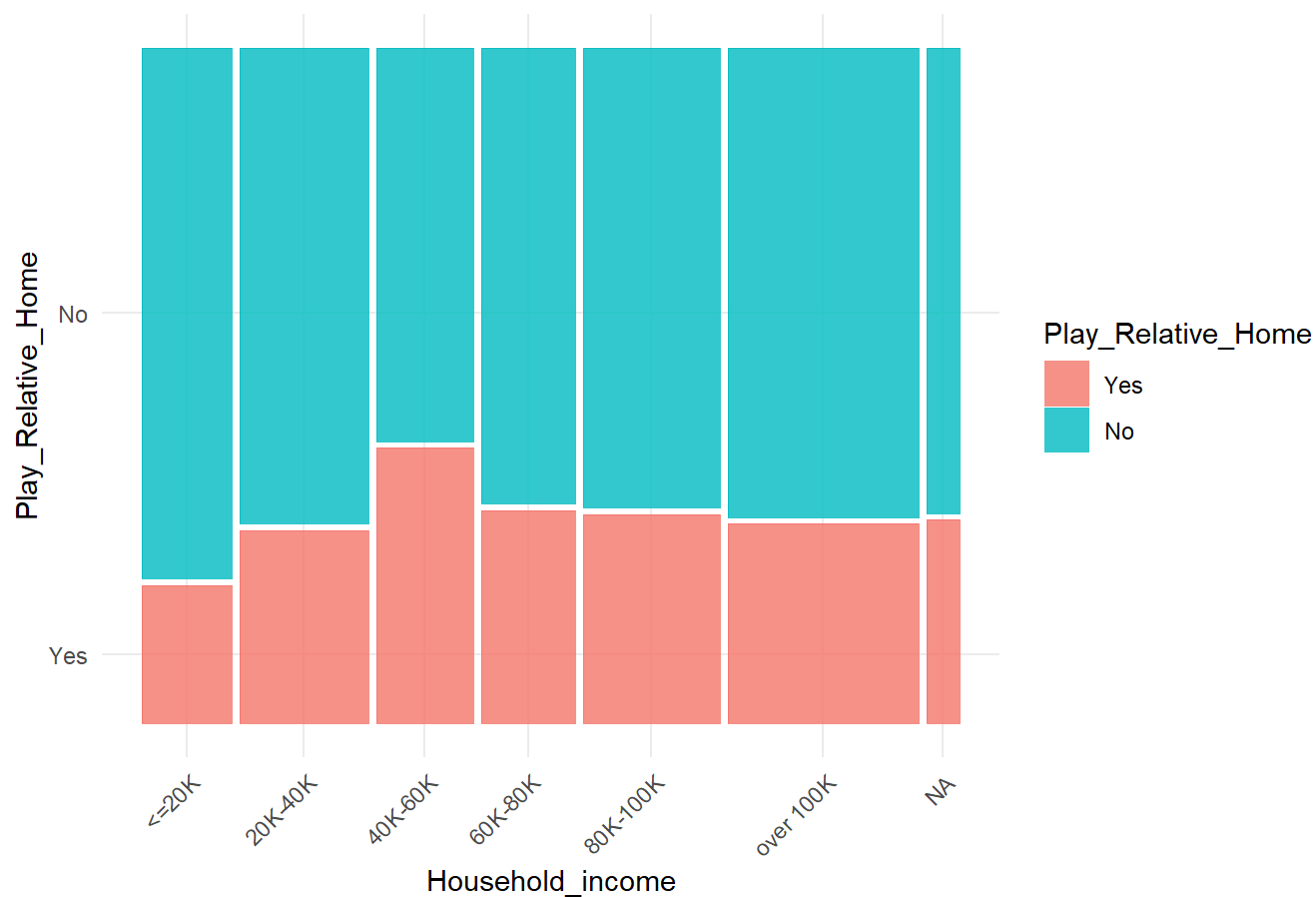
Mosaic Plot of Single_Parent and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and Household_income -----
## Bivariate Analysis of Play_Relative_Home and Household_income
##
## Frequency Table:
## Play_Relative_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes      13      26      28      21      30      40      7
##           No       50      64      40      45      66      94     16
##
## Relative Frequency Table:
## Play_Relative_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_
##           Yes 20.6%   28.9%   41.2%   31.8%   31.2%   29.9% 30.4%
##           No 79.4%   71.1%   58.8%   68.2%   68.8%   70.1% 69.6%
## Total
## 30.6%
## 69.4%
##
## Cramér's V: 0.03721954
```

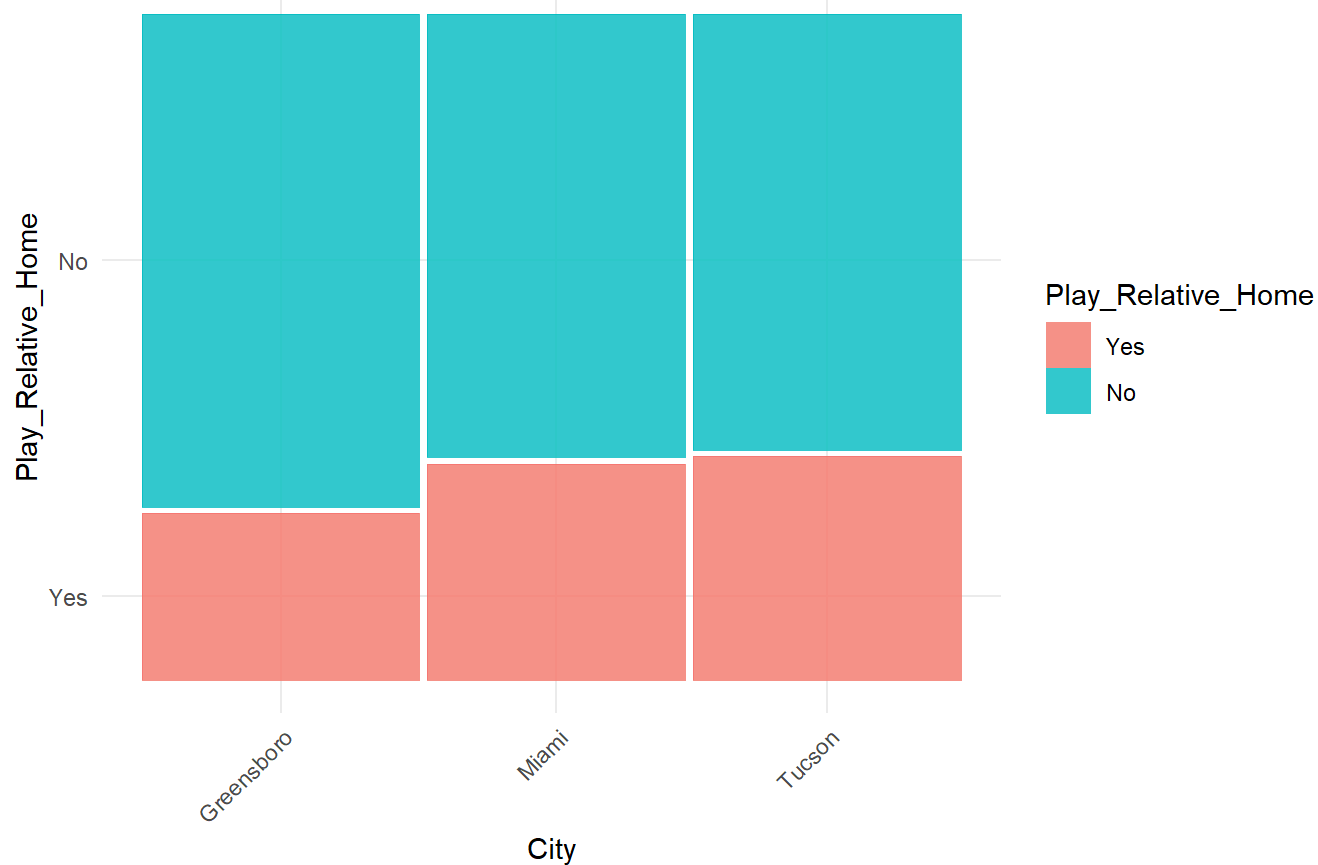
Mosaic Plot of Household_income and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and City -----
## Bivariate Analysis of Play_Relative_Home and City
##
## Frequency Table:
## Play_Relative_Home Greensboro Miami Tucson
##           Yes         47      57      61
##           No         139     117     119
##
## Relative Frequency Table:
## Play_Relative_Home Greensboro Miami Tucson Total
##           Yes      25.3% 32.8% 33.9% 30.6%
##           No      74.7% 67.2% 66.1% 69.4%
##
## Cramér's V: 0.05757637
```

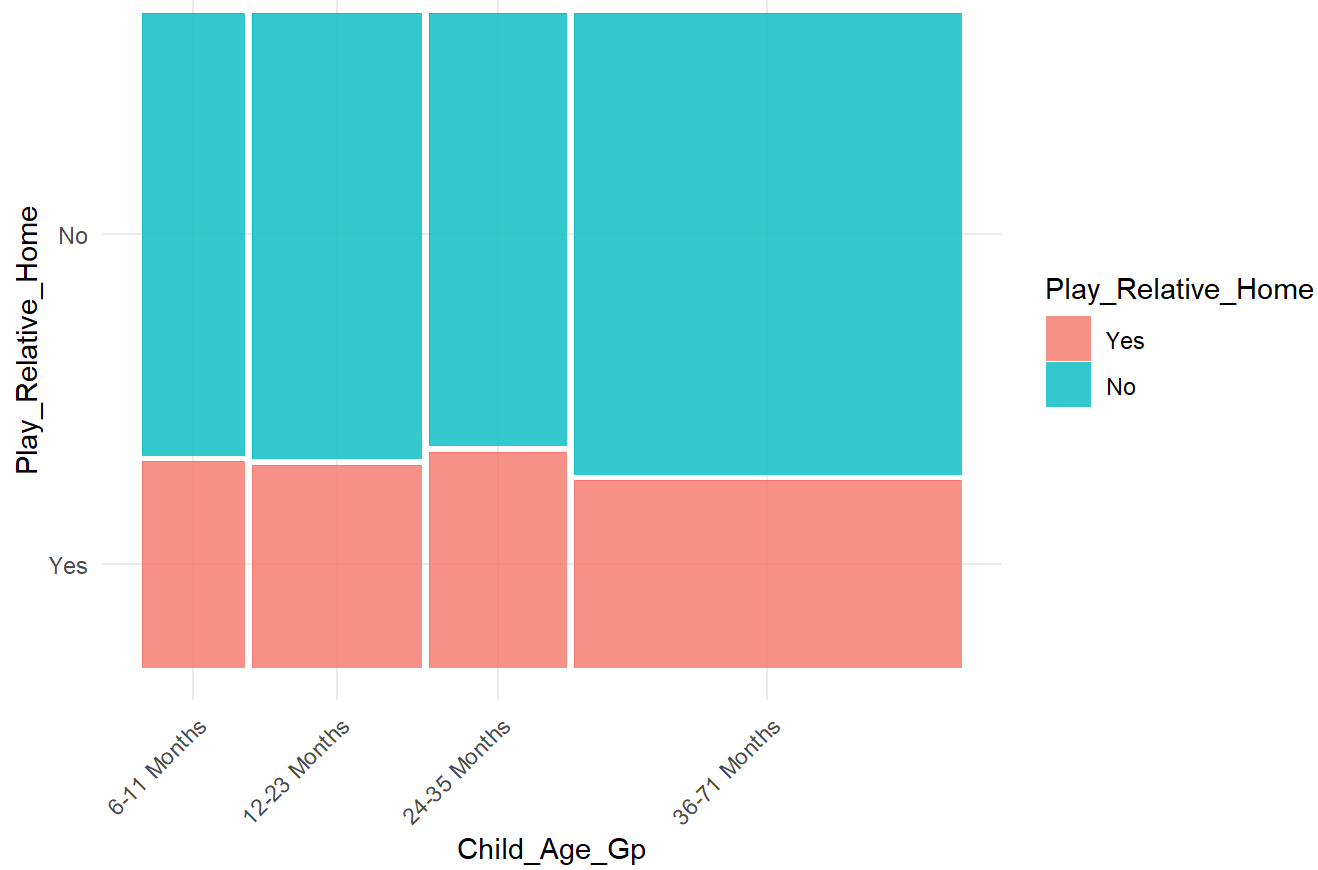
Mosaic Plot of City and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and Child_Age_Gp -----
## Bivariate Analysis of Play_Relative_Home and Child_Age_Gp
##
## Frequency Table:
## Play_Relative_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##                      Yes          22          36          31          76
##                      No           47          79          62         187
##
## Relative Frequency Table:
## Play_Relative_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##                      Yes        31.9%        31.3%        33.3%        28.9% 30.6%
##                      No         68.1%        68.7%        66.7%        71.1% 69.4%
##
## Cramér's V:  0
```

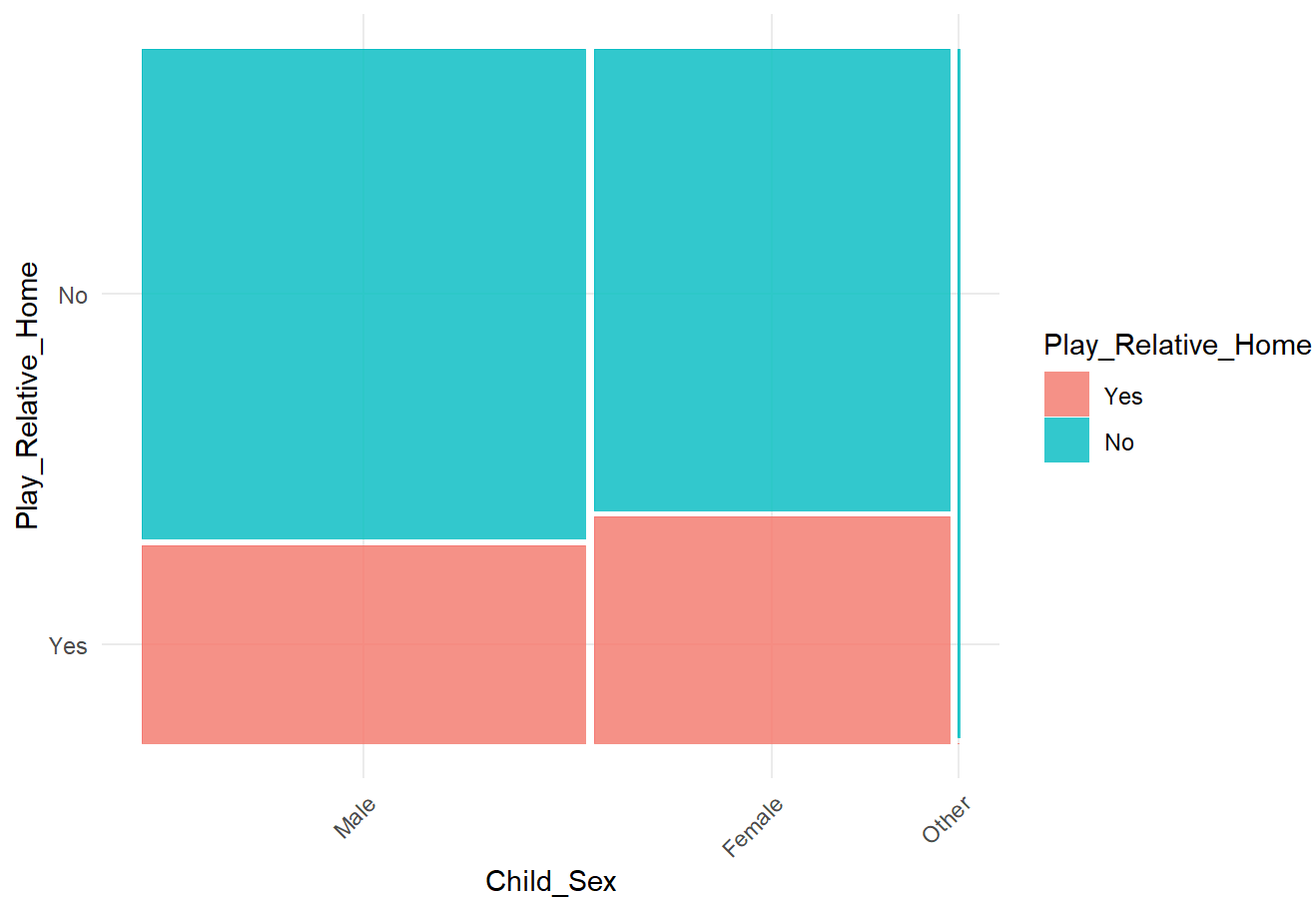
Mosaic Plot of Child_Age_Gp and Play_Relative_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Relative_Home and Child_Sex -----
## Bivariate Analysis of Play_Relative_Home and Child_Sex
##
## Frequency Table:
## Play_Relative_Home Male Female Other
##           Yes      86      79      0
##           No      213     161      1
##
## Relative Frequency Table:
## Play_Relative_Home  Male Female  Other Total
##           Yes 28.8%  32.9%   0.0% 30.6%
##           No 71.2%  67.1% 100.0% 69.4%
##
## Cramér's V:  0
```

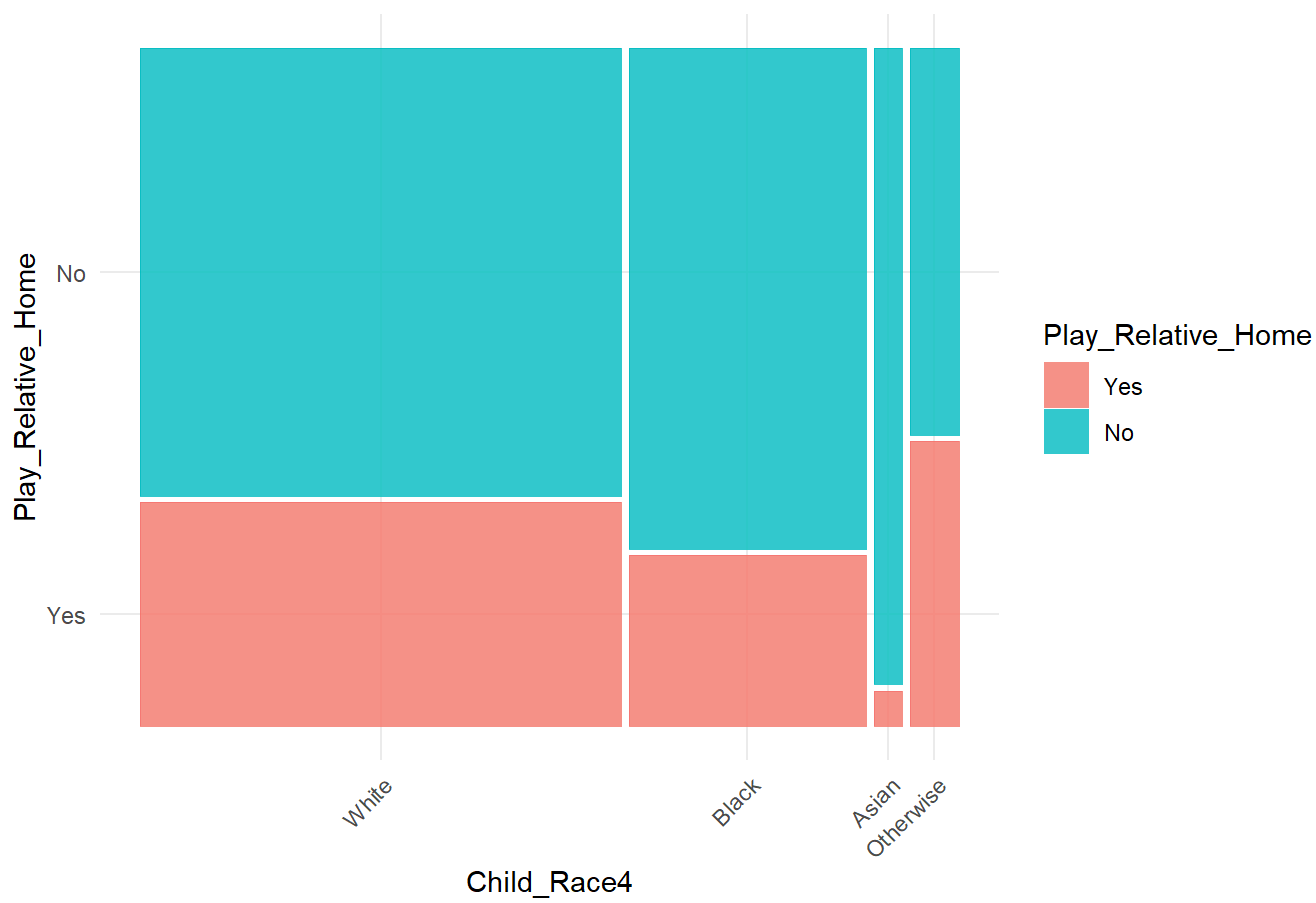

Mosaic Plot of Child_Sex and Play_Relative_Home



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Relative_Home and Child_Race4 -----
## Bivariate Analysis of Play_Relative_Home and Child_Race4
##
## Frequency Table:
##   Play_Relative_Home White Black Asian Otherwise
##               Yes    109    41    1      14
##               No     218   120   18      19
##
## Relative Frequency Table:
##   Play_Relative_Home White Black Asian Otherwise Total
##               Yes 33.3% 25.5%  5.3%    42.4% 30.6%
##               No 66.7% 74.5% 94.7%    57.6% 69.4%
##
## Cramér's V:  0.1223448
```

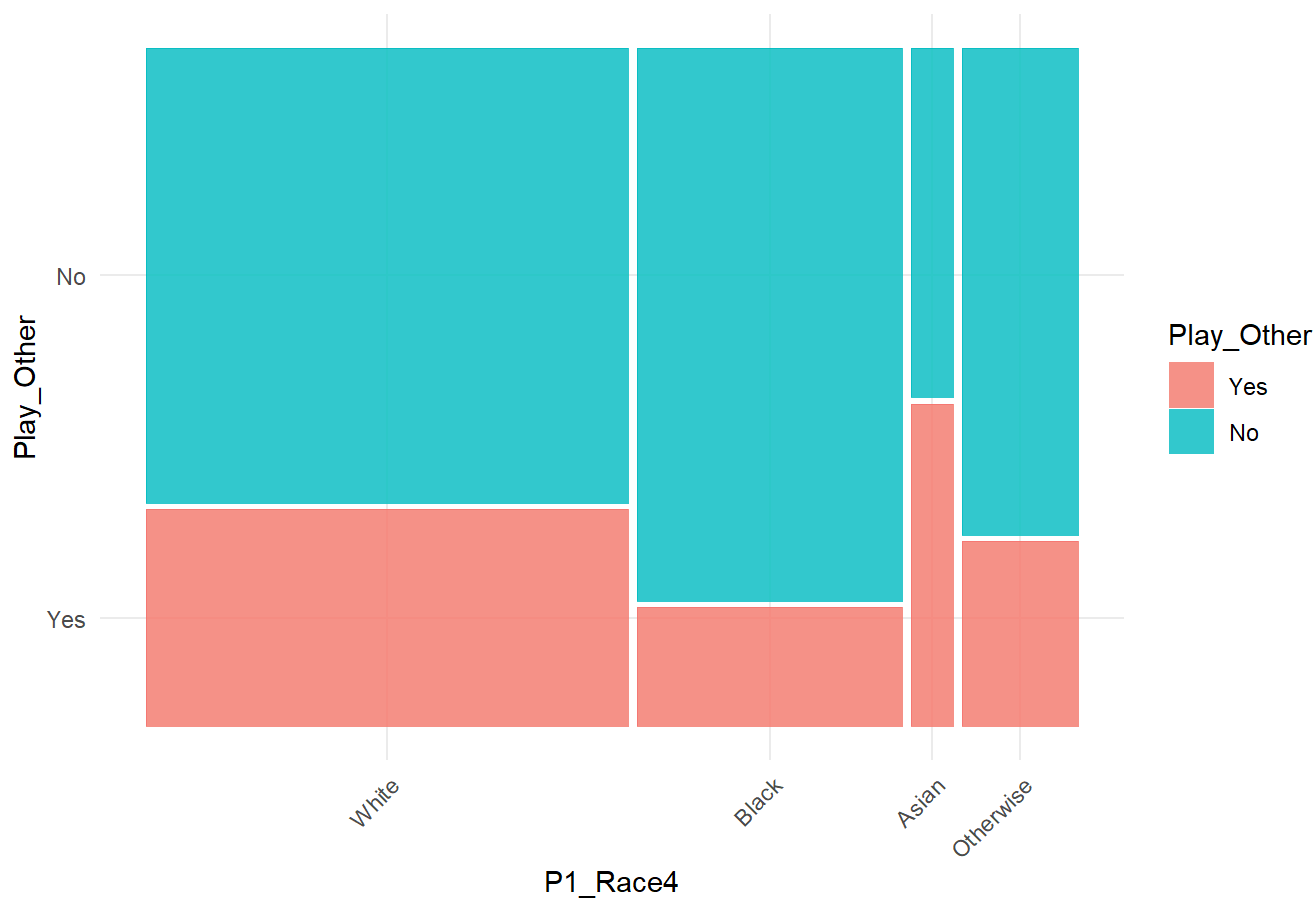
Mosaic Plot of Child_Race4 and Play_Relative_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Other and P1_Race4 -----
## Bivariate Analysis of Play_Other and P1_Race4
##
## Frequency Table:
## Play_Other White Black Asian Otherwise
##      Yes    93    28    12    19
##      No   195   130    13    50
##
## Relative Frequency Table:
## Play_Other White Black Asian Otherwise Total
##      Yes 32.3% 17.7% 48.0%    27.5% 28.1%
##      No 67.7% 82.3% 52.0%    72.5% 71.9%
##
## Cramér's V: 0.1542014
```

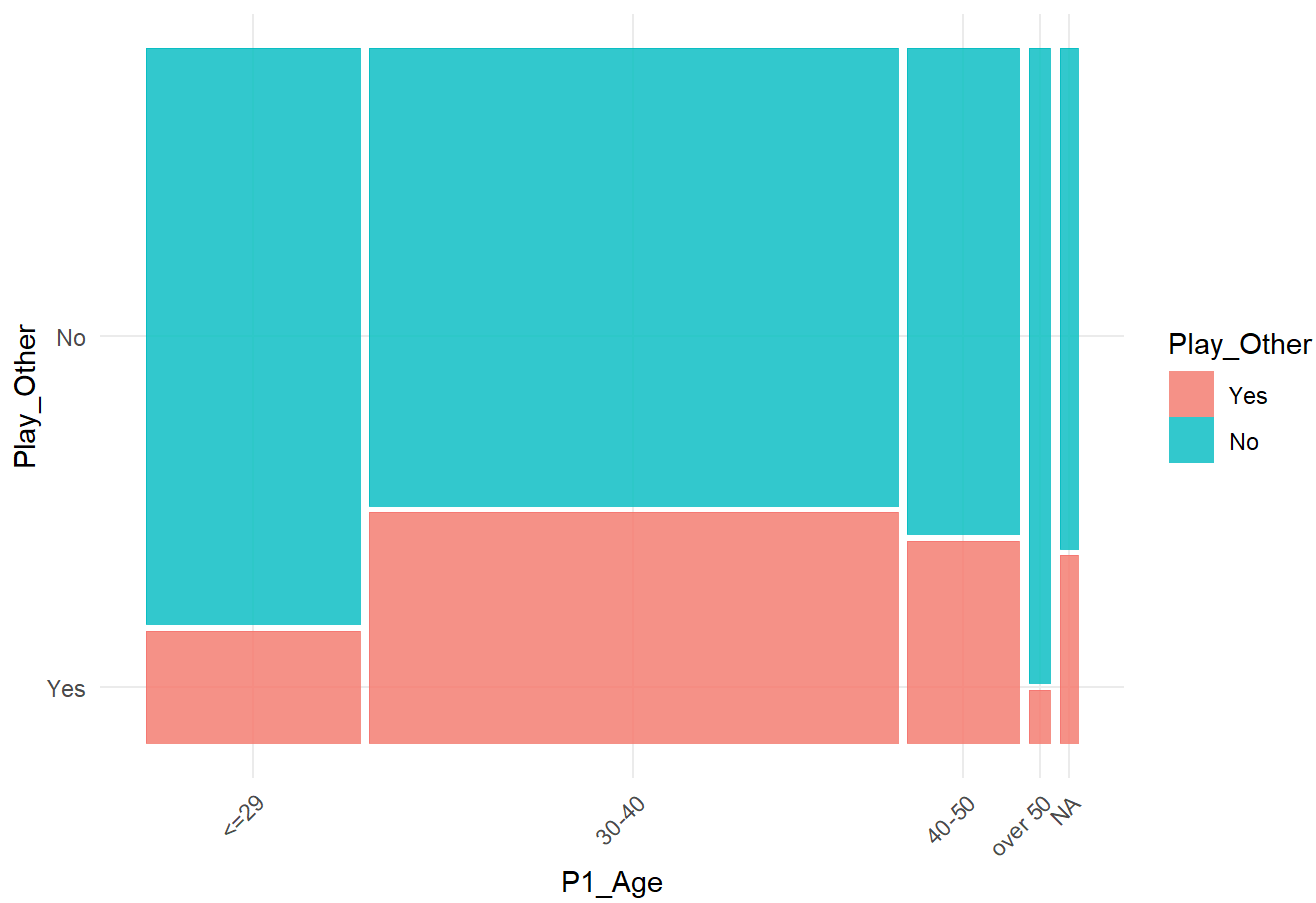
Mosaic Plot of P1_Race4 and Play_Other



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Other and P1_Age -----
## Bivariate Analysis of Play_Other and P1_Age
##
## Frequency Table:
## Play_Other <=29 30-40 40-50 over 50 NA_
##      Yes    21   107    20     1    3
##      No   108   212    48    12    8
##
## Relative Frequency Table:
## Play_Other  <=29 30-40 40-50 over 50  NA_ Total
##      Yes 16.3% 33.5% 29.4%   7.7% 27.3% 28.1%
##      No 83.7% 66.5% 70.6%  92.3% 72.7% 71.9%
##
## Cramér's V:  0.151154
```

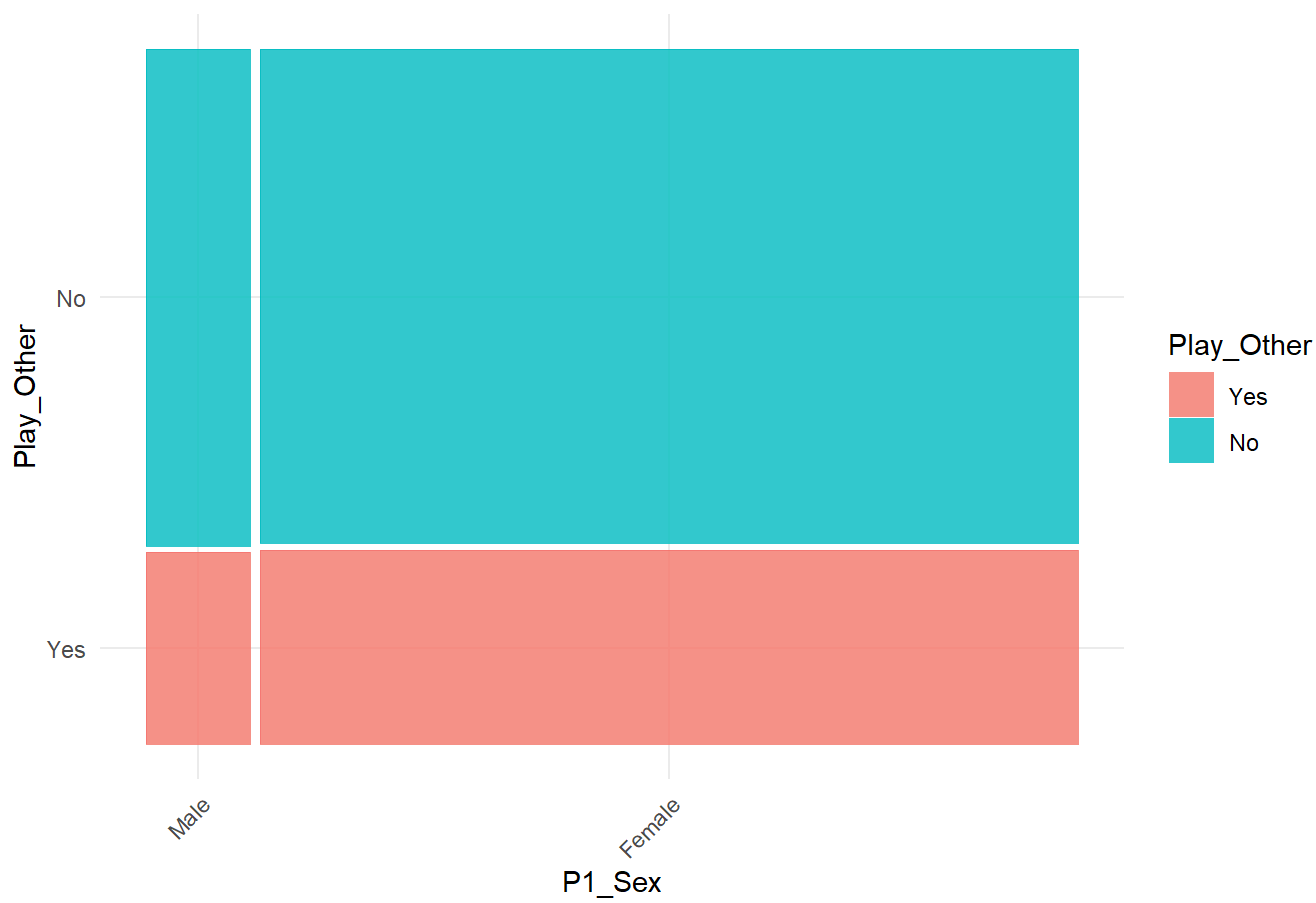
Mosaic Plot of P1_Age and Play_Other



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Other and P1_Sex -----
## Bivariate Analysis of Play_Other and P1_Sex
##
## Frequency Table:
##   Play_Other  Male Female
##           Yes    17    135
##           No     44    344
##
## Relative Frequency Table:
##   Play_Other  Male Female Total
##           Yes 27.9% 28.2% 28.1%
##           No 72.1% 71.8% 71.9%
##
## Cramér's V:  0
```

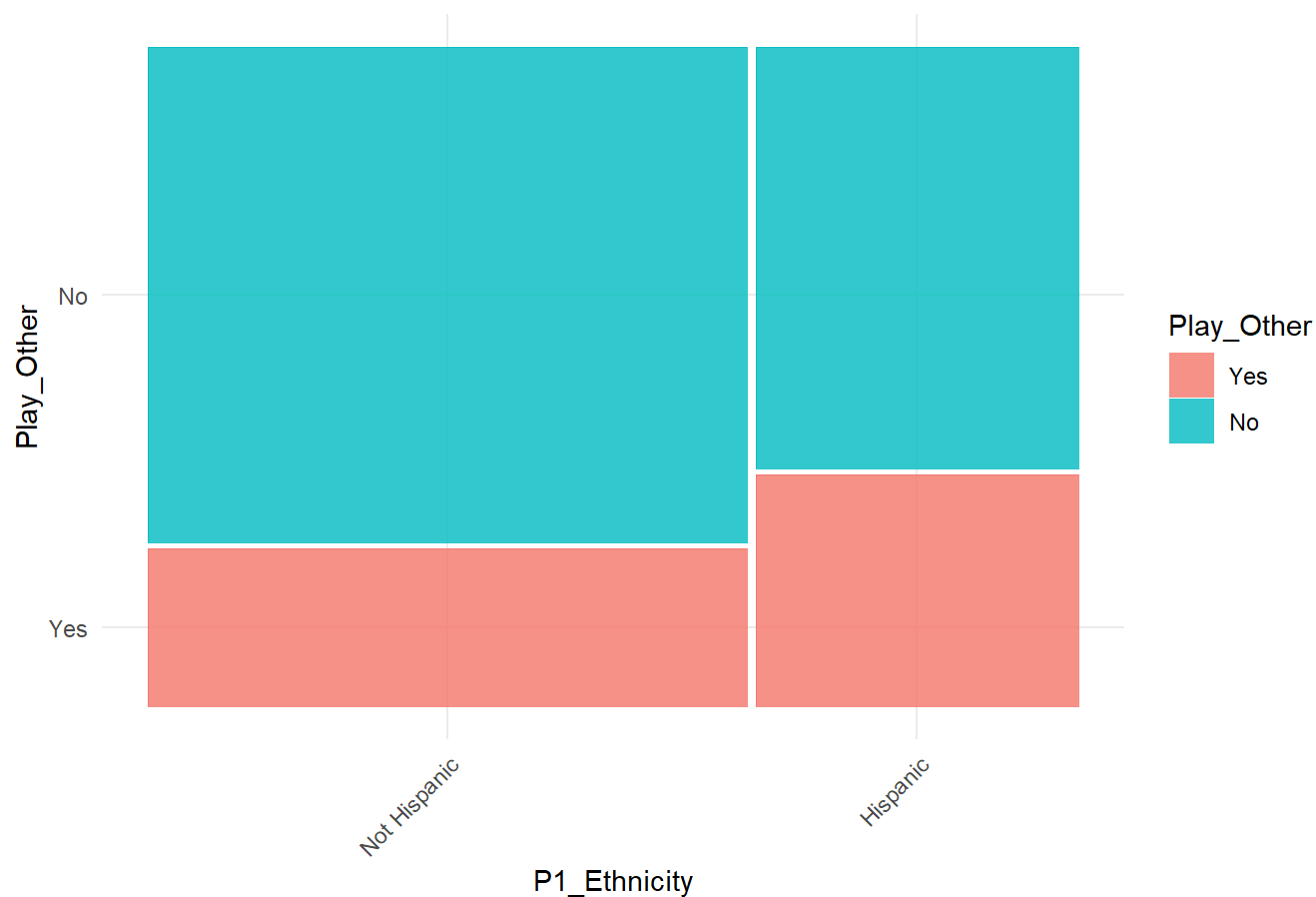
Mosaic Plot of P1_Sex and Play_Other



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Other and P1_Ethnicity -----
## Bivariate Analysis of Play_Other and P1_Ethnicity
##
## Frequency Table:
##   Play_Other Not Hispanic Hispanic
##           Yes           85         67
##           No           266        122
##
## Relative Frequency Table:
##   Play_Other Not Hispanic Hispanic Total
##           Yes        24.2%    35.4% 28.1%
##           No         75.8%    64.6% 71.9%
##
## Cramér's V:  0.1111825
```

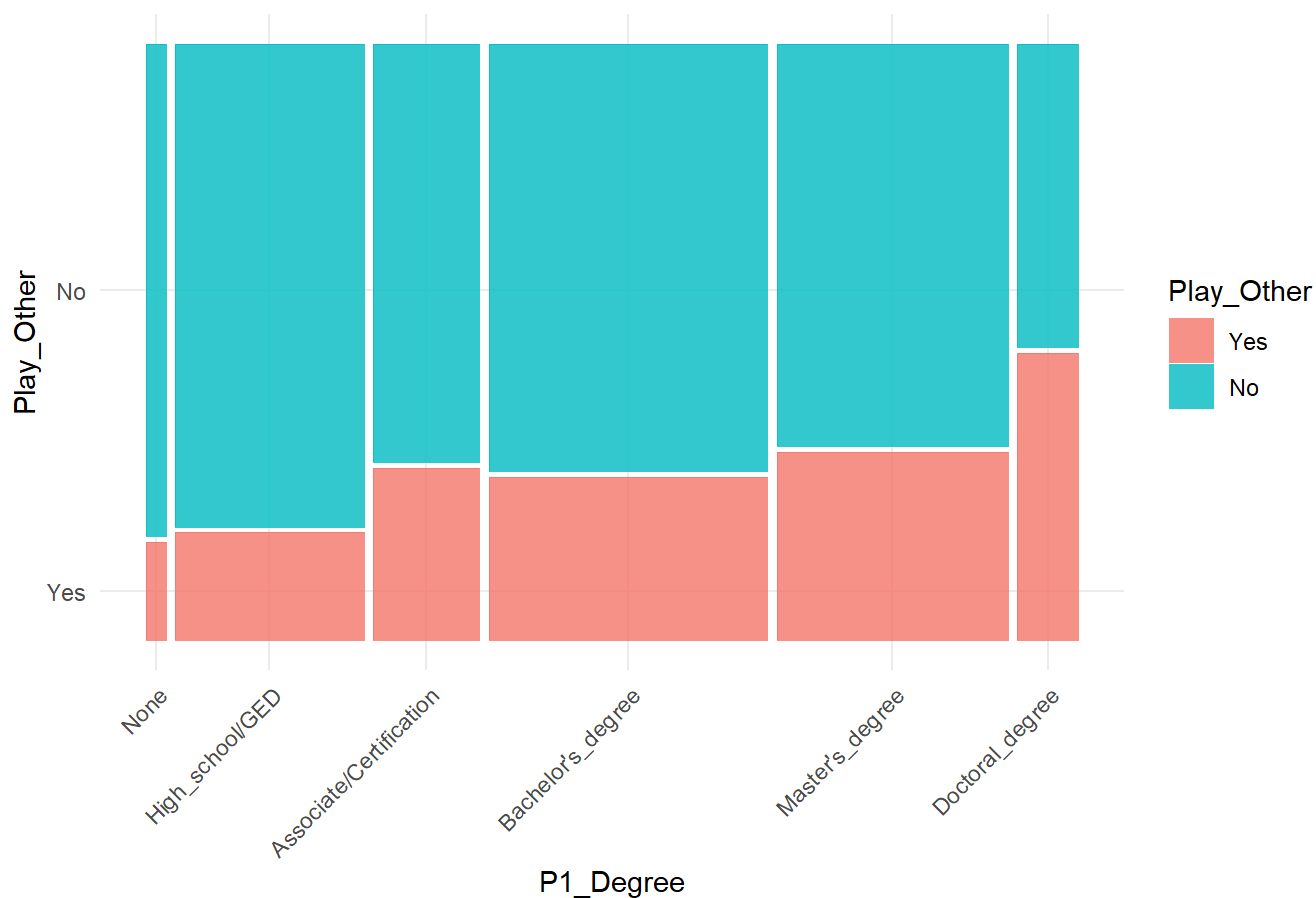
Mosaic Plot of P1_Ethnicity and Play_Other



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Other and P1_Degree -----
## Bivariate Analysis of Play_Other and P1_Degree
##
## Frequency Table:
## Play_Other None High_school/GED Associate/Certification Bachelor's_degree
##      Yes    2          21          19          47
##      No    10          94          46          123
## Master's_degree Doctoral_degree
##           45          18
##           96          19
##
## Relative Frequency Table:
## Play_Other None High_school/GED Associate/Certification Bachelor's_degree
##      Yes 16.7%      18.3%      29.2%      27.6%
##      No 83.3%      81.7%      70.8%      72.4%
## Master's_degree Doctoral_degree Total
##           31.9%      48.6% 28.1%
##           68.1%      51.4% 71.9%
##
## Cramér's V: 0.1366708
```

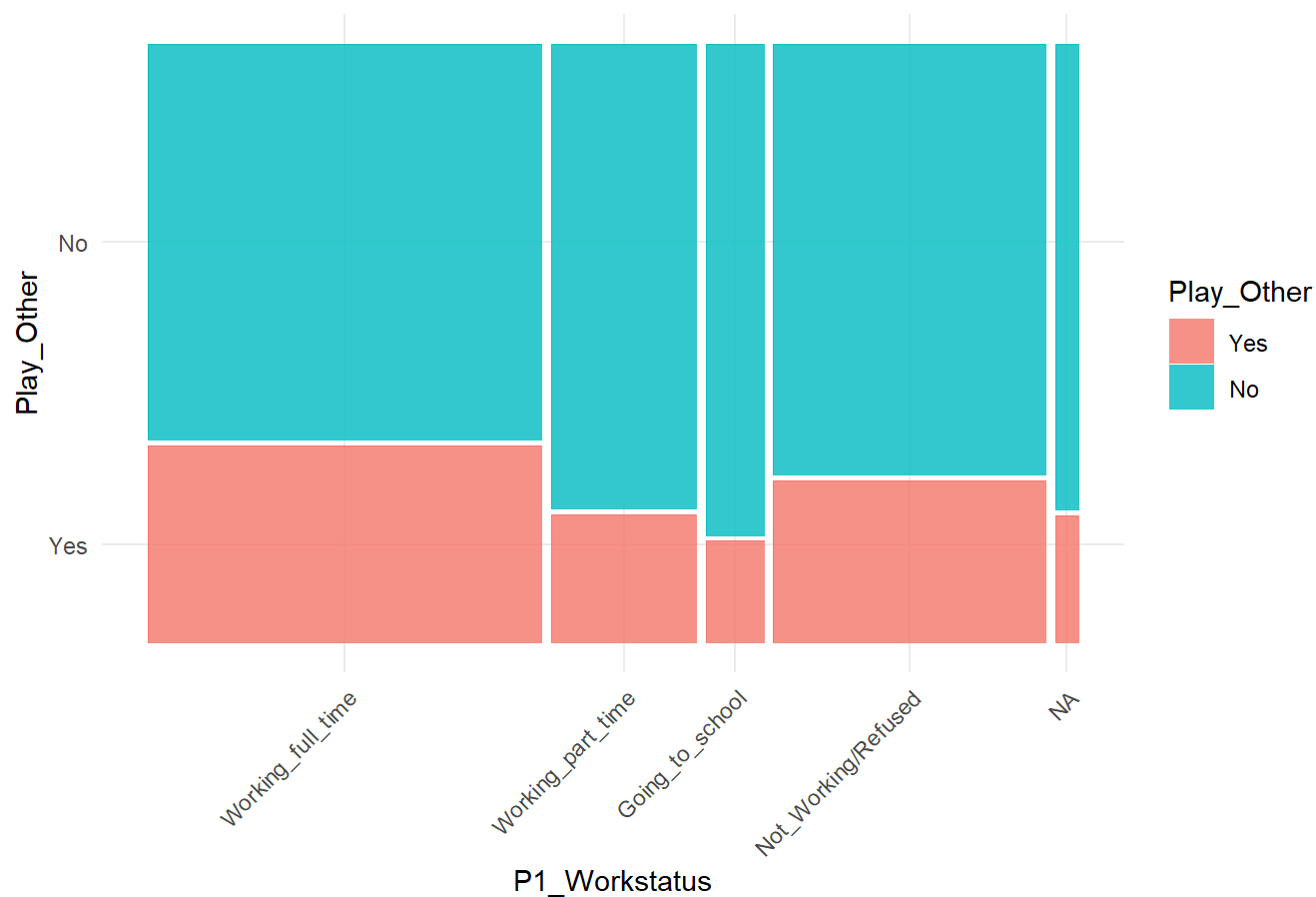
Mosaic Plot of P1_Degree and Play_Other



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Other and P1_Workstatus -----
## Bivariate Analysis of Play_Other and P1_Workstatus
##
## Frequency Table:
## Play_Other Working_full_time Working_part_time Going_to_school
##      Yes           79           19           6
##      No          159           69          29
## Not_Working/Refused NA_
##           45    3
##          120   11
##
## Relative Frequency Table:
## Play_Other Working_full_time Working_part_time Going_to_school
##      Yes           33.2%           21.6%           17.1%
##      No           66.8%           78.4%           82.9%
## Not_Working/Refused  NA_ Total
##           27.3% 21.4% 28.1%
##           72.7% 78.6% 71.9%
##
## Cramér's V:  0.07859766
```

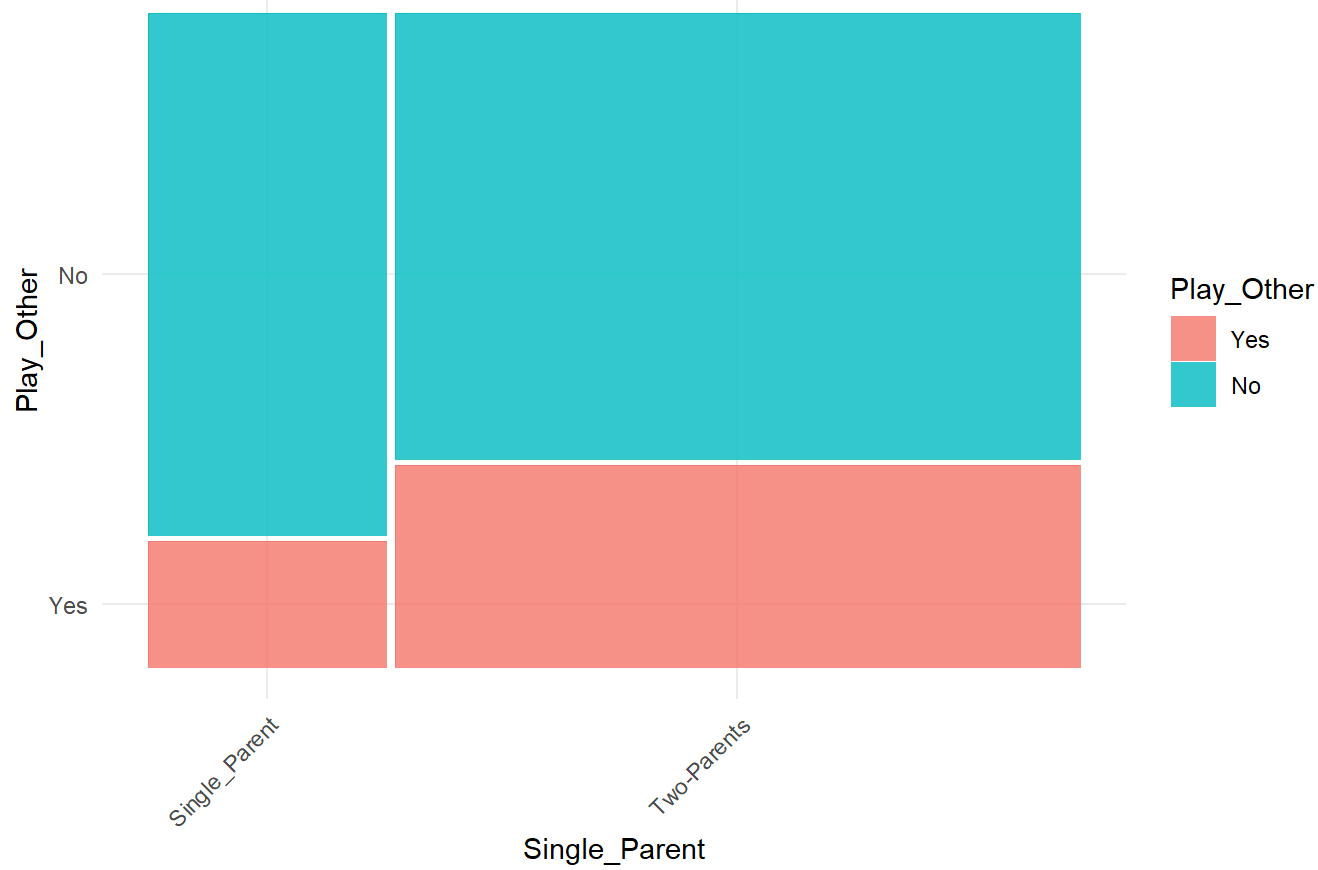
Mosaic Plot of P1_Workstatus and Play_Other



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Play_Other and Single_Parent -----
## Bivariate Analysis of Play_Other and Single_Parent
##
## Frequency Table:
##   Play_Other Single_Parent Two-Parents
##         Yes          27          125
##         No           112          276
##
## Relative Frequency Table:
##   Play_Other Single_Parent Two-Parents Total
##         Yes          19.4%          31.2% 28.1%
##         No           80.6%          68.8% 71.9%
##
## Cramér's V:  0.105871
```

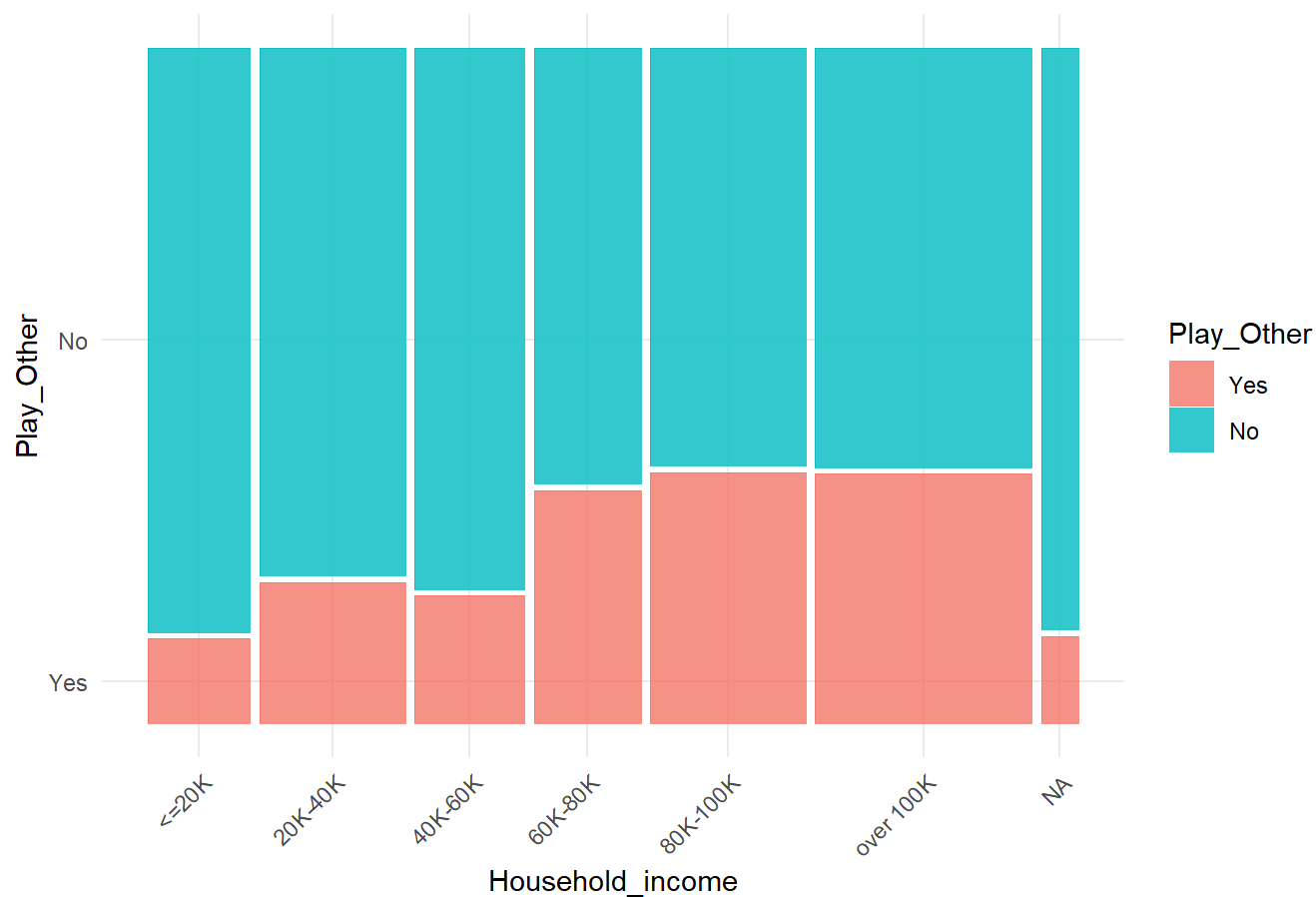

Mosaic Plot of Single_Parent and Play_Other



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Other and Household_income -----
## Bivariate Analysis of Play_Other and Household_income
##
## Frequency Table:
## Play_Other <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##      Yes      8      19      13      23      36      50      3
##      No      55      71      55      43      60      84     20
##
## Relative Frequency Table:
## Play_Other <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##      Yes 12.7%  21.1%  19.1%  34.8%  37.5%  37.3% 13.0% 28.1%
##      No 87.3%  78.9%  80.9%  65.2%  62.5%  62.7% 87.0% 71.9%
##
## Cramér's V: 0.1933305
```

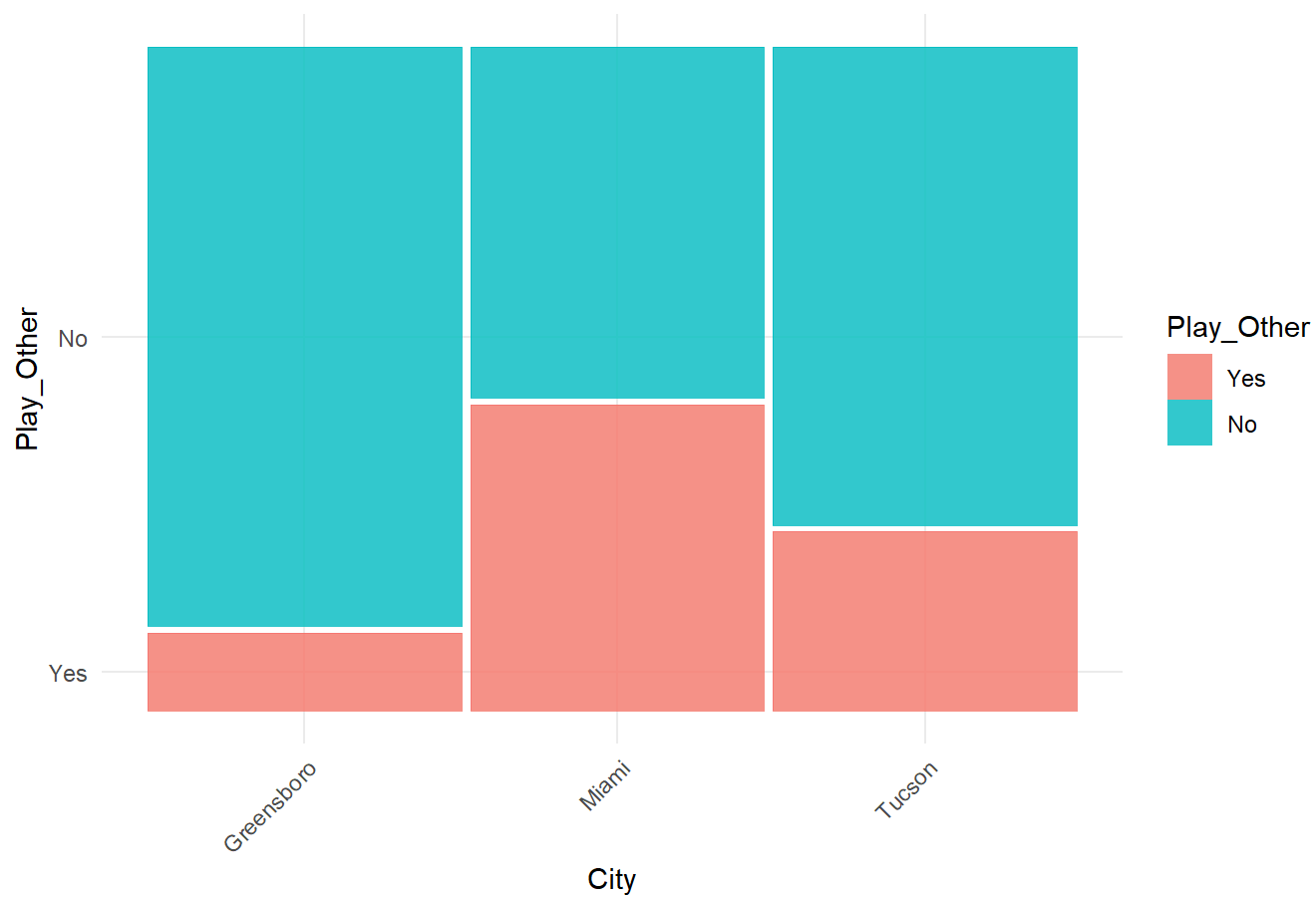
Mosaic Plot of Household_income and Play_Other



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Other and City -----
## Bivariate Analysis of Play_Other and City
##
## Frequency Table:
## Play_Other Greensboro Miami Tucson
##      Yes      22      81      49
##      No      164      93     131
##
## Relative Frequency Table:
## Play_Other Greensboro Miami Tucson Total
##      Yes     11.8% 46.6% 27.2% 28.1%
##      No     88.2% 53.4% 72.8% 71.9%
##
## Cramér's V: 0.3097251
```

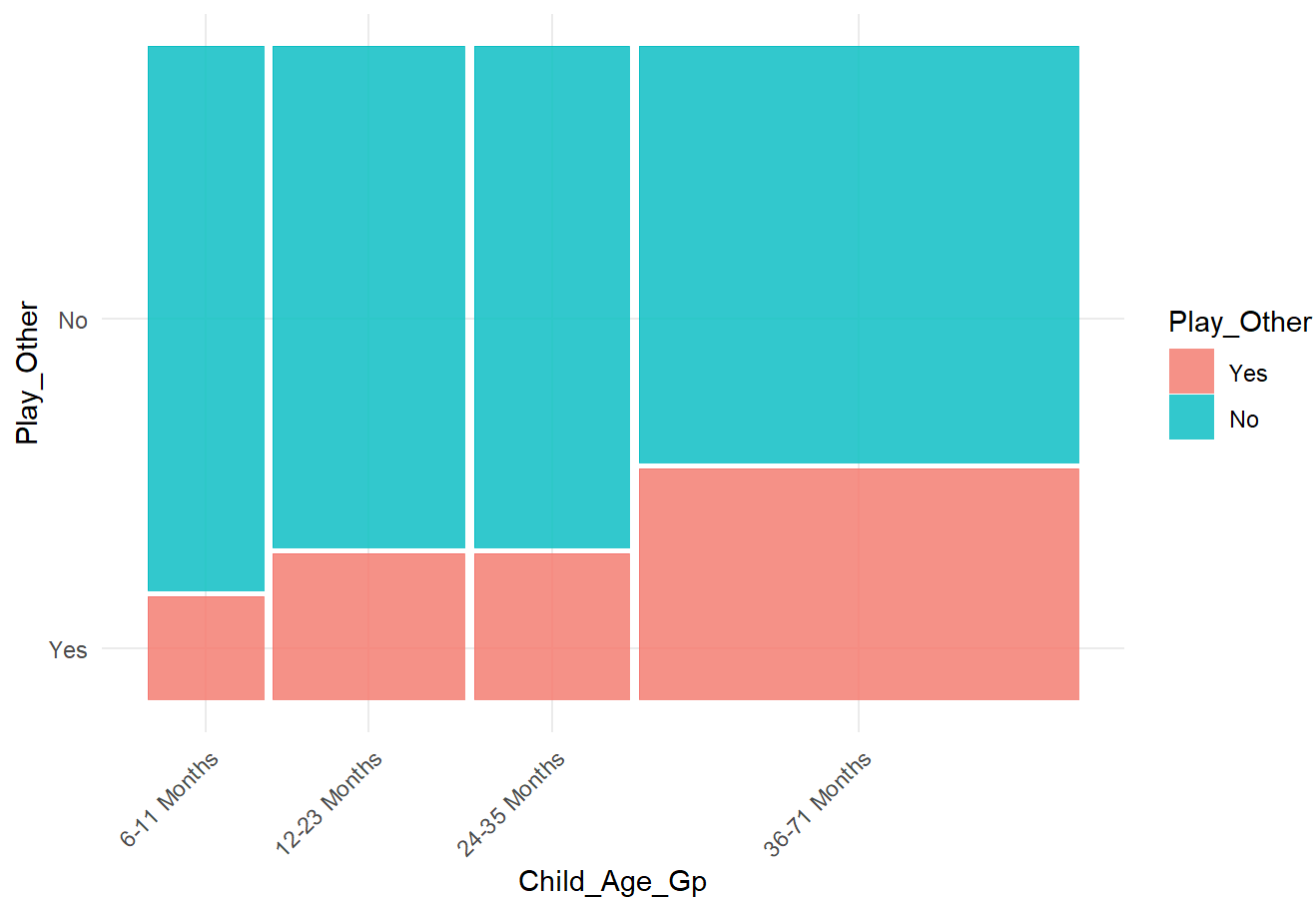
Mosaic Plot of City and Play_Other



```
## Saving 7 x 5 in image
```

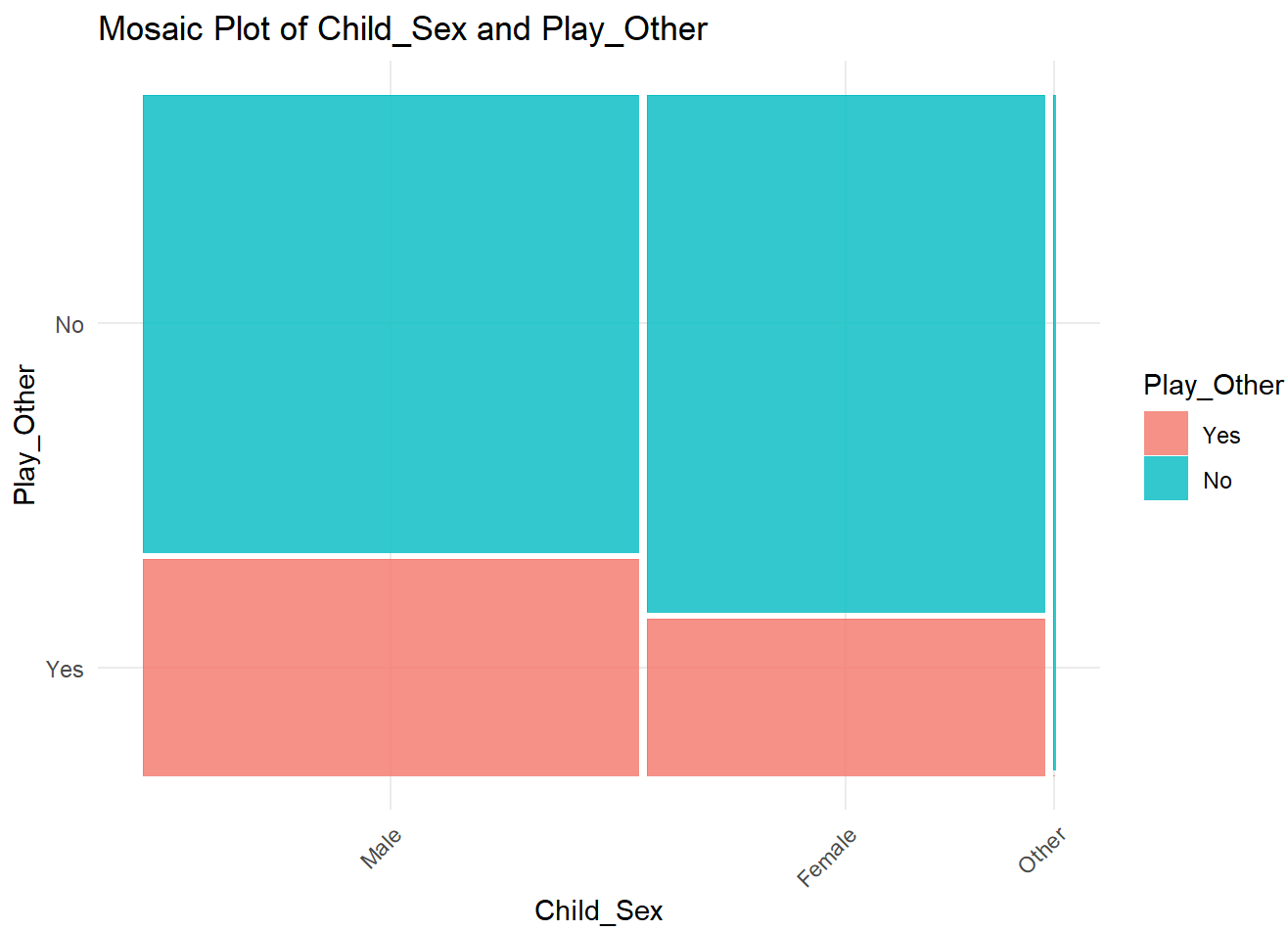
```
##
##
## ----- Analysis for Play_Other and Child_Age_Gp -----
## Bivariate Analysis of Play_Other and Child_Age_Gp
##
## Frequency Table:
## Play_Other 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##      Yes      11      26      21      94
##      No      58      89      72     169
##
## Relative Frequency Table:
## Play_Other 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##      Yes     15.9%     22.6%     22.6%     35.7% 28.1%
##      No     84.1%     77.4%     77.4%     64.3% 71.9%
##
## Cramér's V:  0.153773
```

Mosaic Plot of Child_Age_Gp and Play_Other



```
## Saving 7 x 5 in image
```

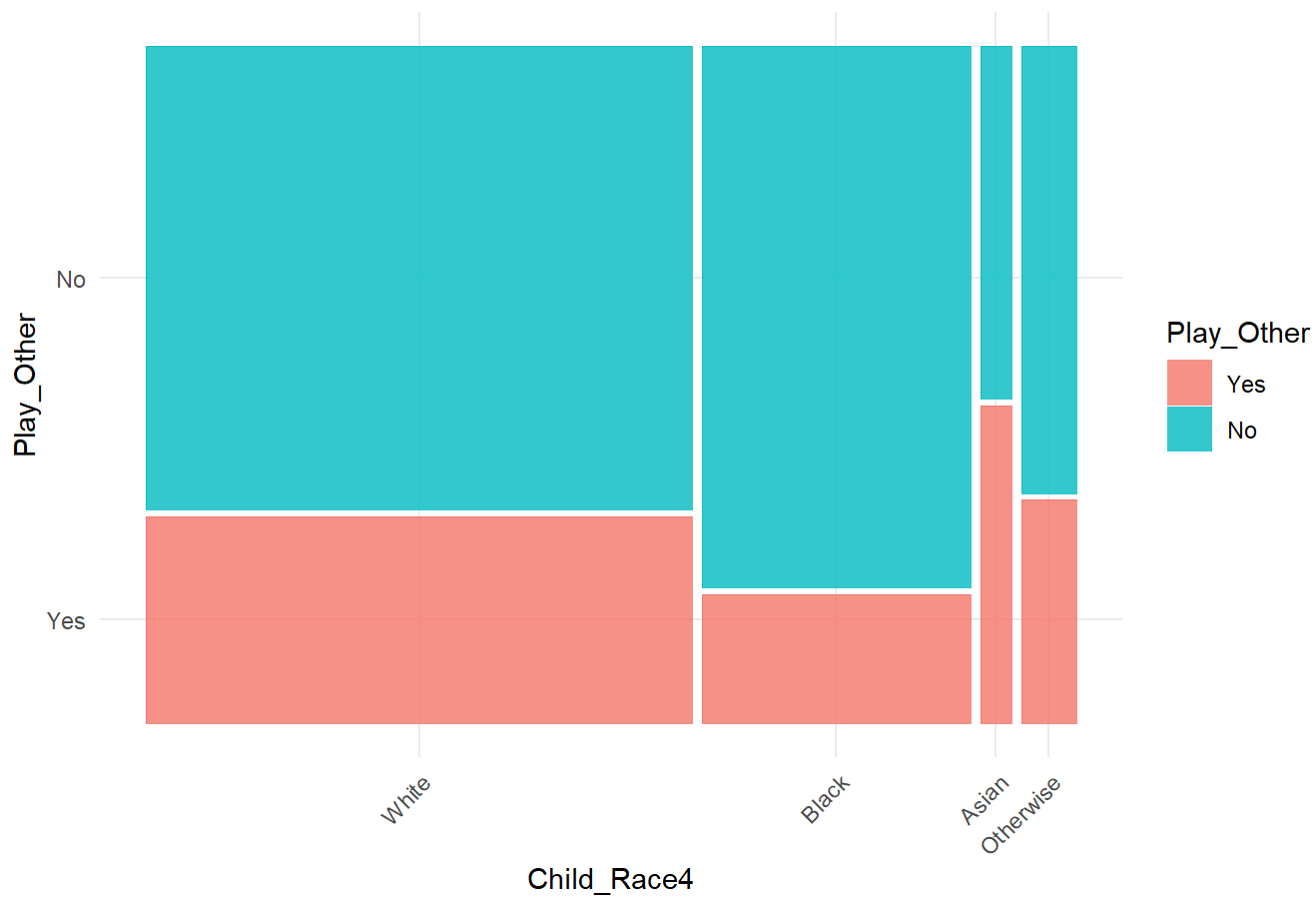
```
##
##
## ----- Analysis for Play_Other and Child_Sex -----
## Bivariate Analysis of Play_Other and Child_Sex
##
## Frequency Table:
##   Play_Other Male Female Other
##         Yes   96    56    0
##         No  203   184    1
##
## Relative Frequency Table:
##   Play_Other  Male Female  Other Total
##         Yes 32.1% 23.3%  0.0% 28.1%
##         No 67.9% 76.7% 100.0% 71.9%
##
##
## Cramér's V: 0.08007428
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Other and Child_Race4 -----
## Bivariate Analysis of Play_Other and Child_Race4
##
## Frequency Table:
## Play_Other White Black Asian Otherwise
##      Yes   101    31    9     11
##      No   226   130   10     22
##
## Relative Frequency Table:
## Play_Other White Black Asian Otherwise Total
##      Yes 30.9% 19.3% 47.4%    33.3% 28.1%
##      No 69.1% 80.7% 52.6%    66.7% 71.9%
##
## Cramér's V: 0.1249307
```

Mosaic Plot of Child_Race4 and Play_Other



```
## Saving 7 x 5 in image
```

```
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```

##	var1	var2	cramer_v
## 1	Play_Other	City	0.30972512
## 2	Play_Daycare	P1_Workstatus	0.27056771
## 3	Play_Daycare	P1_Degree	0.25831756
## 4	Play_Park	Child_Race4	0.21676357
## 5	Play_Park	P1_Race4	0.20849808
## 6	Play_Daycare	Household_income	0.19848363
## 7	Play_Other	Household_income	0.19333055
## 8	Play_Park	Child_Age_Gp	0.18437771
## 9	Play_Relative_Home	P1_Race4	0.17574345
## 10	Play_Park	P1_Workstatus	0.17047818
## 11	Play_Other	P1_Race4	0.15420140
## 12	Play_Other	Child_Age_Gp	0.15377303
## 13	Play_Other	P1_Age	0.15115401
## 14	Play_Park	City	0.14061486
## 15	Play_Other	P1_Degree	0.13667081
## 16	Play_Park	P1_Degree	0.13656449
## 17	Play_Relative_Home	P1_Workstatus	0.12850652
## 18	Play_Other	Child_Race4	0.12493070
## 19	Play_Babysitter_Home	Household_income	0.12457008
## 20	Play_Relative_Home	Child_Race4	0.12234483
## 21	Play_Daycare	Child_Age_Gp	0.11138972
## 22	Play_Other	P1_Ethnicity	0.11118245
## 23	Play_Other	Single_Parent	0.10587100
## 24	Play_Park	P1_Sex	0.10566001
## 25	Play_Babysitter_Home	P1_Degree	0.10426048
## 26	Play_Relative_Home	P1_Ethnicity	0.09392300
## 27	Play_Park	Child_Sex	0.09188210
## 28	Play_Relative_Home	P1_Degree	0.08268109
## 29	Play_Daycare	P1_Ethnicity	0.08025388
## 30	Play_Other	Child_Sex	0.08007428
## 31	Play_Other	P1_Workstatus	0.07859766
## 32	Play_Babysitter_Home	Child_Age_Gp	0.07212812
## 33	Play_Park	Household_income	0.06767276
## 34	Play_Park	P1_Ethnicity	0.05998388
## 35	Play_Relative_Home	City	0.05757637
## 36	Play_Babysitter_Home	Child_Race4	0.05531608
## 37	Play_Babysitter_Home	P1_Race4	0.05289644
## 38	Play_Daycare	Child_Sex	0.03869425
## 39	Play_Relative_Home	Household_income	0.03721954
## 40	Play_Babysitter_Home	City	0.03322494
## 41	Play_Relative_Home	P1_Age	0.03262214
## 42	Play_Babysitter_Home	Single_Parent	0.02356573
## 43	Play_Babysitter_Home	P1_Workstatus	0.02152977
## 44	Play_Daycare	P1_Race4	0.02106887
## 45	Play_Daycare	City	0.01150411
## 46	Play_Park	P1_Age	0.00000000
## 47	Play_Park	Single_Parent	0.00000000
## 48	Play_Daycare	P1_Age	0.00000000
## 49	Play_Daycare	P1_Sex	0.00000000
## 50	Play_Daycare	Single_Parent	0.00000000
## 51	Play_Daycare	Child_Race4	0.00000000
## 52	Play_Babysitter_Home	P1_Age	0.00000000
## 53	Play_Babysitter_Home	P1_Sex	0.00000000
## 54	Play_Babysitter_Home	P1_Ethnicity	0.00000000
## 55	Play_Babysitter_Home	Child_Sex	0.00000000

```
## 56   Play_Relative_Home      P1_Sex 0.00000000
## 57   Play_Relative_Home      Single_Parent 0.00000000
## 58   Play_Relative_Home      Child_Age_Gp 0.00000000
## 59   Play_Relative_Home      Child_Sex 0.00000000
## 60           Play_Other      P1_Sex 0.00000000
```

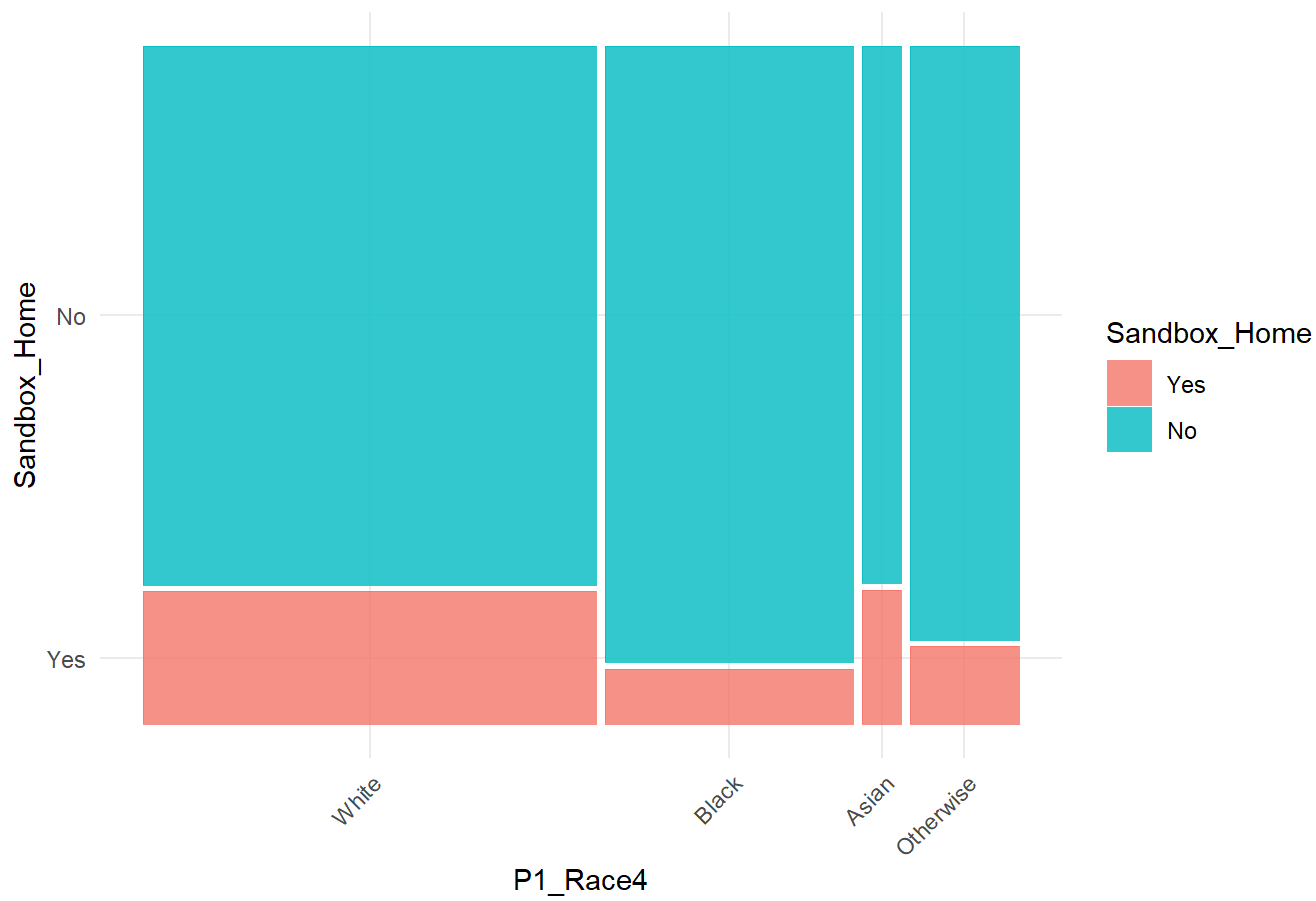
Exploring Sandbox Variables

```
SandboxVars <- c( "Sandbox_Home", "Sandbox_Park", "Sandbox_Daycare",
  "Sandbox_Babysitter", "Sandbox_Relative", "Sandbox_Other")
list1= SandboxVars
list2= DemoVars

cramer_v_values <- data.frame(var2 = character(), cramer_v = numeric(), stringsAsFactors = TRUE)
results <- list()
for (var1 in list1) {
  for (var2 in list2) {
    cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")
    result <- Analyze_Bivarite(data, var1, var2)
    results[[paste(var1, var2, sep = "_vs_")]] <- result
    if (!is.null(result$cramer_v)) {
      cramer_v_values <- rbind(cramer_v_values,
        data.frame(var1 = var1, var2 = var2, cramer_v = result$cramer_v))
    }
  }
}
```

```
##
##
## ----- Analysis for Sandbox_Home and P1_Race4 -----
## Bivariate Analysis of Sandbox_Home and P1_Race4
##
## Frequency Table:
##   Sandbox_Home White Black Asian Otherwise
##           Yes    57    13     5         8
##           No   231   145    20        61
##
## Relative Frequency Table:
##   Sandbox_Home White Black Asian Otherwise Total
##           Yes 19.8%  8.2% 20.0%    11.6% 15.4%
##           No 80.2% 91.8% 80.0%    88.4% 84.6%
##
## Cramér's V:  0.1269549
```

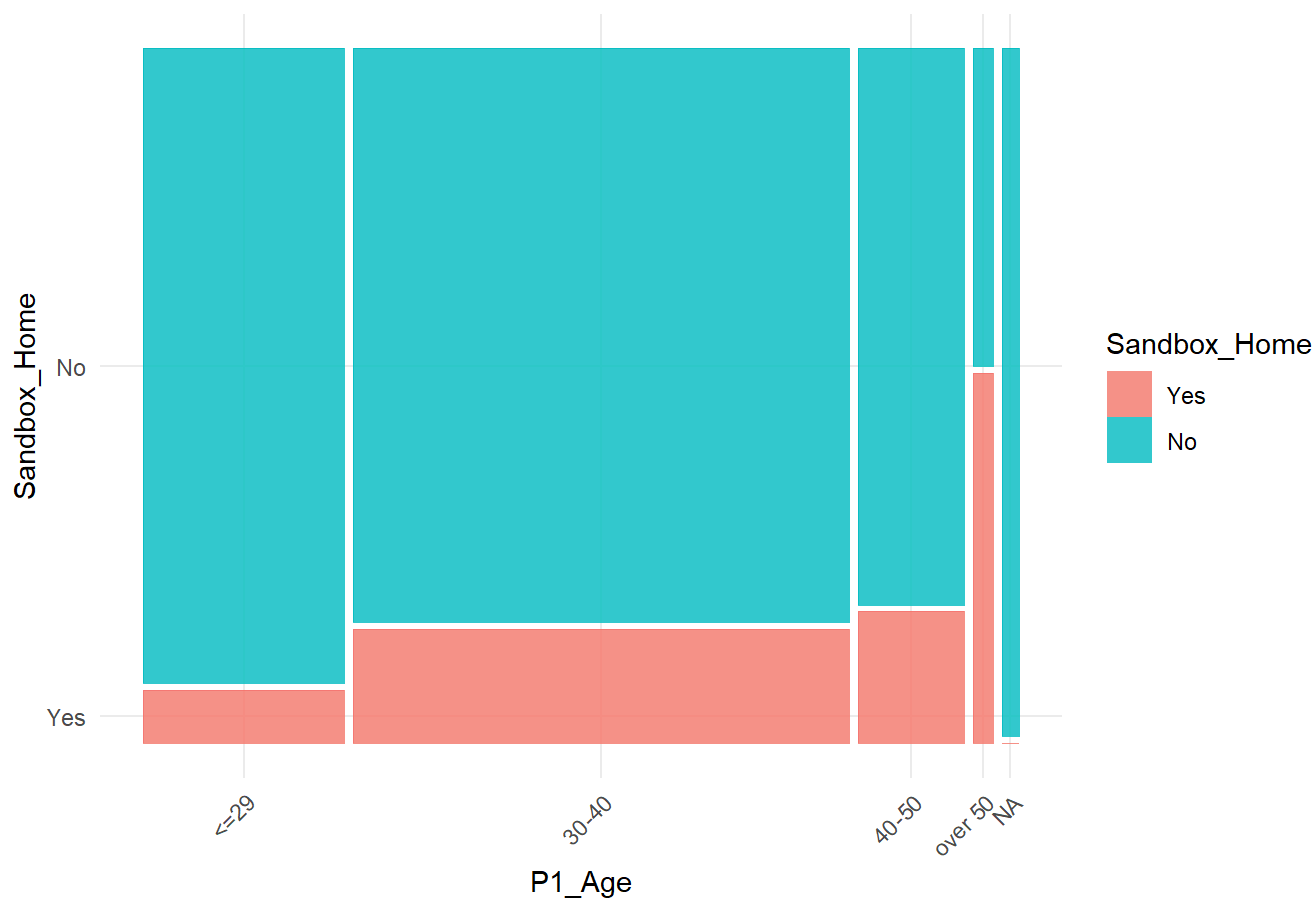

Mosaic Plot of P1_Race4 and Sandbox_Home



```
## Saving 7 x 5 in image
```

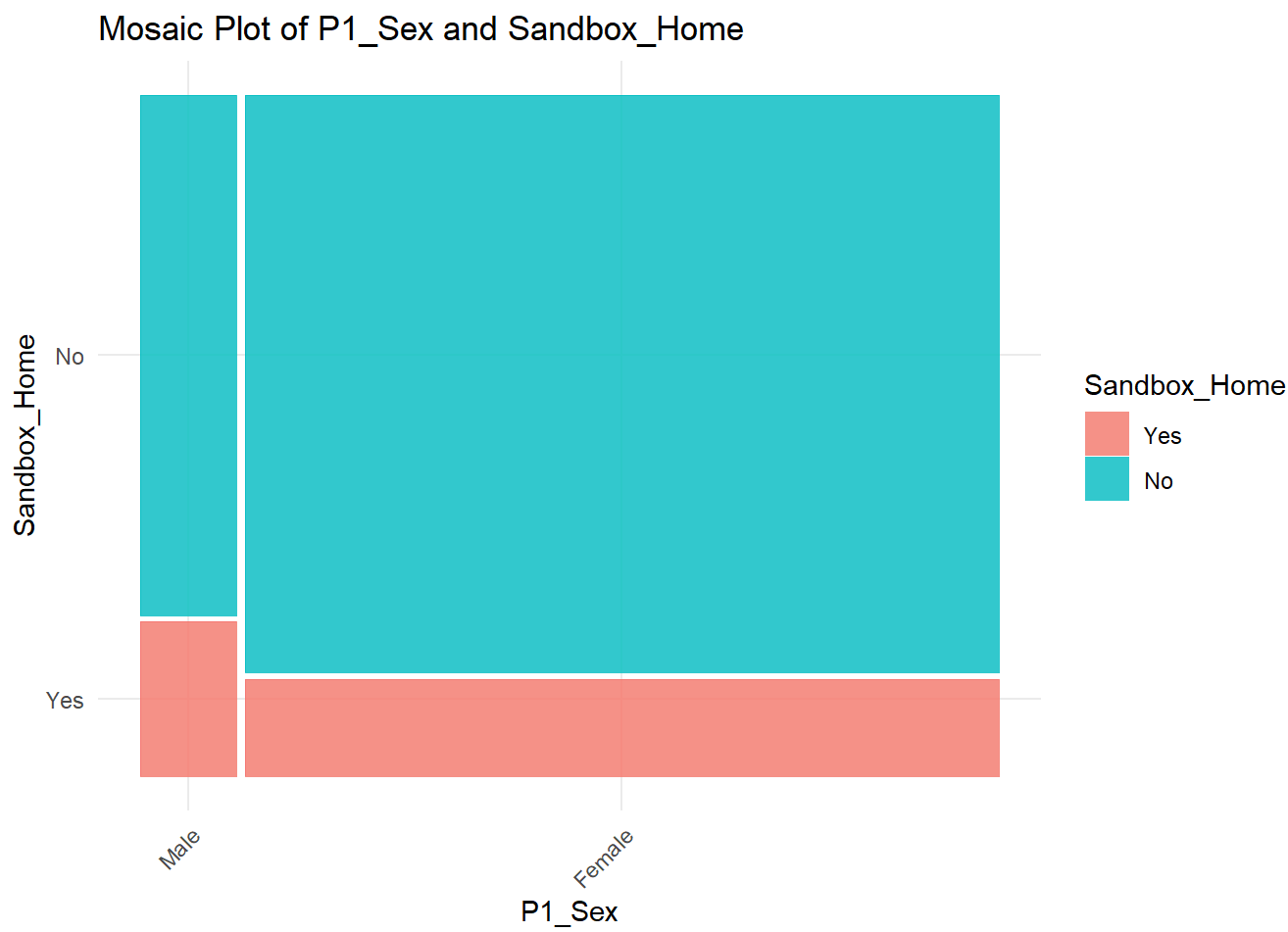
```
##
##
## ----- Analysis for Sandbox_Home and P1_Age -----
## Bivariate Analysis of Sandbox_Home and P1_Age
##
## Frequency Table:
##   Sandbox_Home  <=29 30-40 40-50 over 50 NA_
##           Yes    10    53    13     7    0
##           No   119   266    55     6   11
##
## Relative Frequency Table:
##   Sandbox_Home  <=29 30-40 40-50 over 50   NA_ Total
##           Yes   7.8% 16.6% 19.1%  53.8%   0.0% 15.4%
##           No  92.2% 83.4% 80.9%  46.2% 100.0% 84.6%
##
## Cramér's V:  0.1909592
```

Mosaic Plot of P1_Age and Sandbox_Home



```
## Saving 7 x 5 in image
```

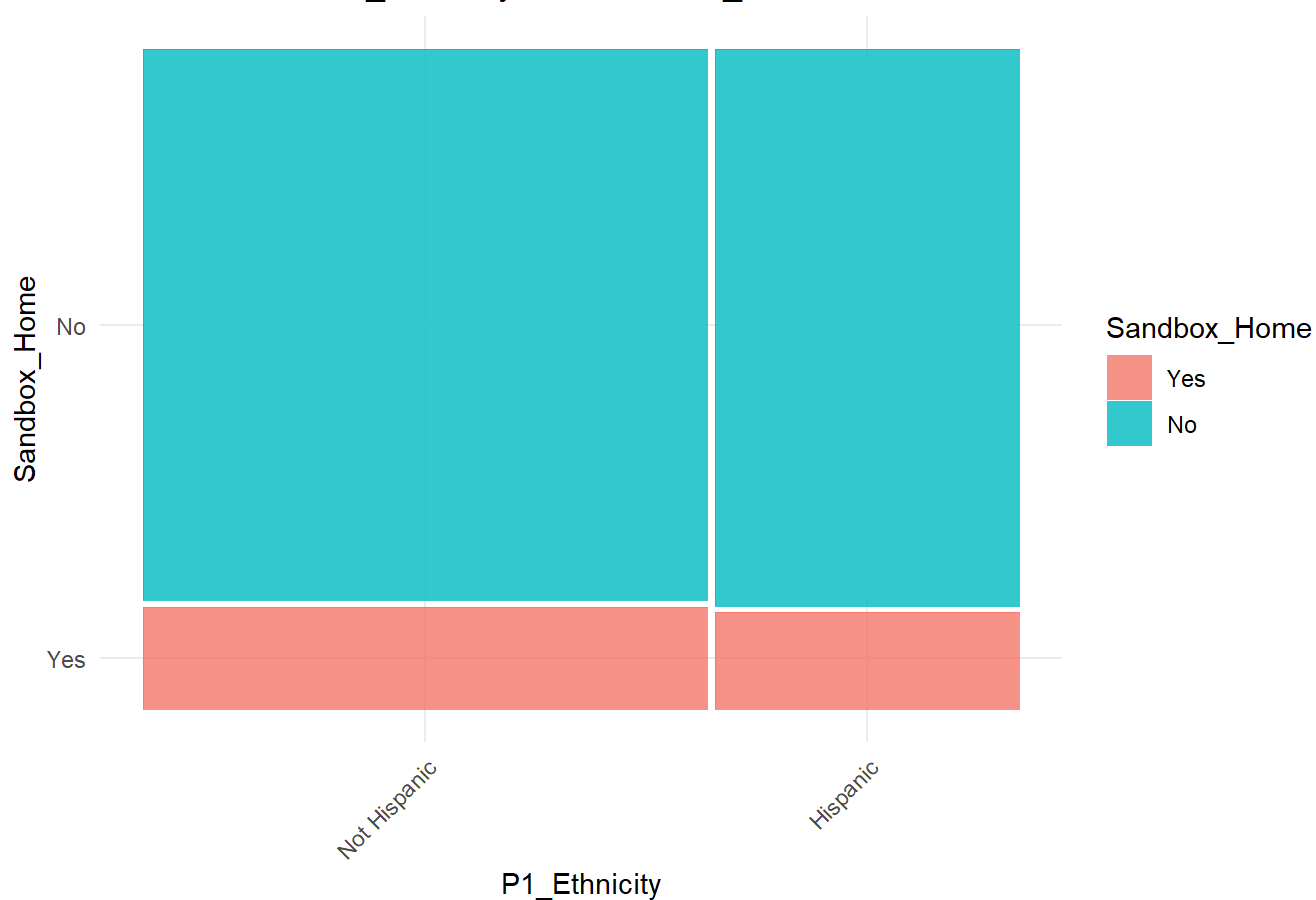
```
##
##
## ----- Analysis for Sandbox_Home and P1_Sex -----
## Bivariate Analysis of Sandbox_Home and P1_Sex
##
## Frequency Table:
##   Sandbox_Home  Male Female
##           Yes    14     69
##           No     47    410
##
## Relative Frequency Table:
##   Sandbox_Home  Male Female Total
##           Yes 23.0% 14.4% 15.4%
##           No 77.0% 85.6% 84.6%
##
##
## Cramér's V: 0.06146082
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Home and P1_Ethnicity -----
## Bivariate Analysis of Sandbox_Home and P1_Ethnicity
##
## Frequency Table:
##   Sandbox_Home Not Hispanic Hispanic
##           Yes           55         28
##           No           296        161
##
## Relative Frequency Table:
##   Sandbox_Home Not Hispanic Hispanic Total
##           Yes       15.7%    14.8% 15.4%
##           No       84.3%    85.2% 84.6%
##
## Cramér's V:  0
```

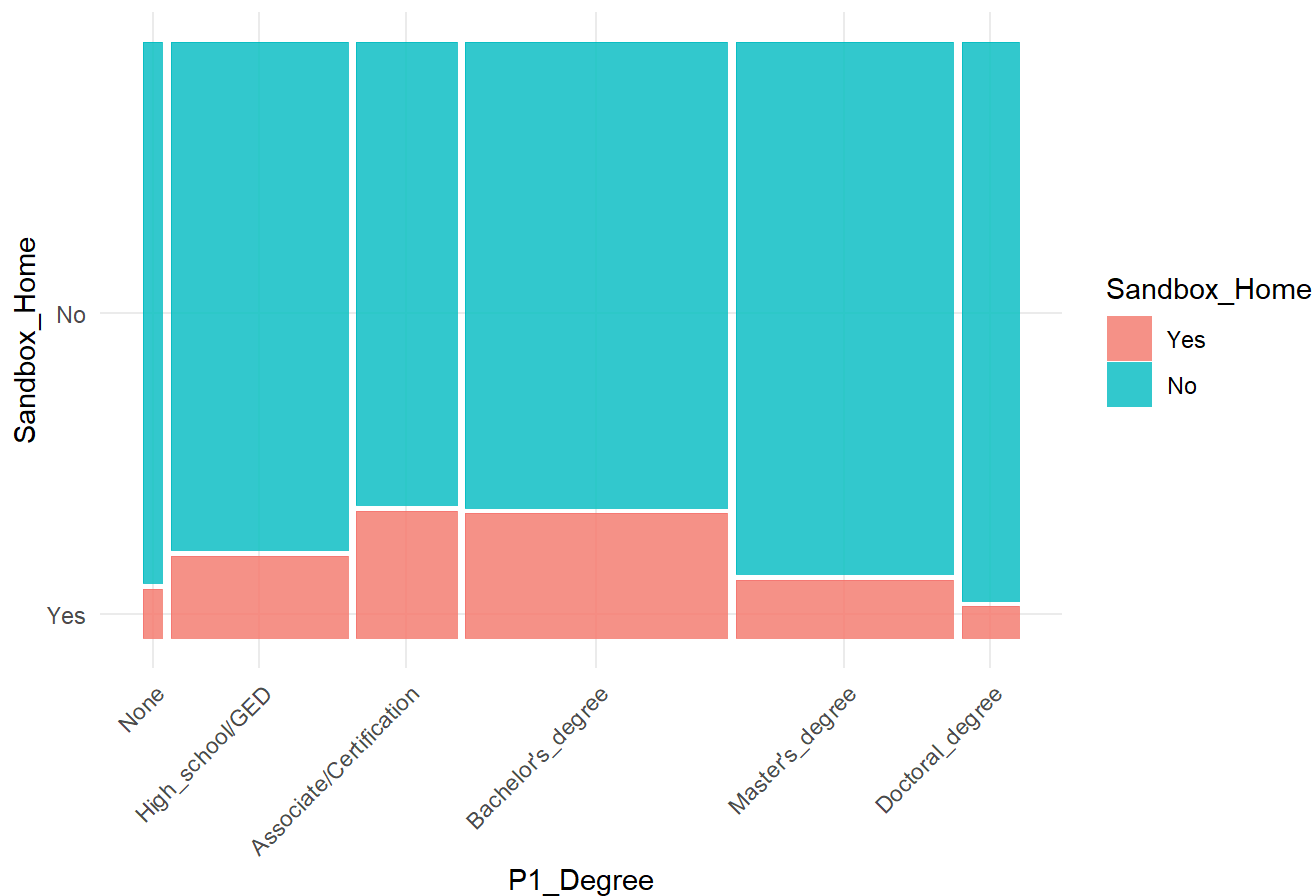
Mosaic Plot of P1_Ethnicity and Sandbox_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Home and P1_Degree -----
## Bivariate Analysis of Sandbox_Home and P1_Degree
##
## Frequency Table:
##   Sandbox_Home None High_school/GED Associate/Certification Bachelor's_degree
##           Yes      1             16                   14             36
##           No      11             99                   51             134
##   Master's_degree Doctoral_degree
##           14             2
##           127            35
##
## Relative Frequency Table:
##   Sandbox_Home None High_school/GED Associate/Certification Bachelor's_degree
##           Yes  8.3%             13.9%                   21.5%             21.2%
##           No  91.7%             86.1%                   78.5%             78.8%
##   Master's_degree Doctoral_degree Total
##           9.9%             5.4% 15.4%
##           90.1%            94.6% 84.6%
##
## Cramér's V:  0.1216461
```

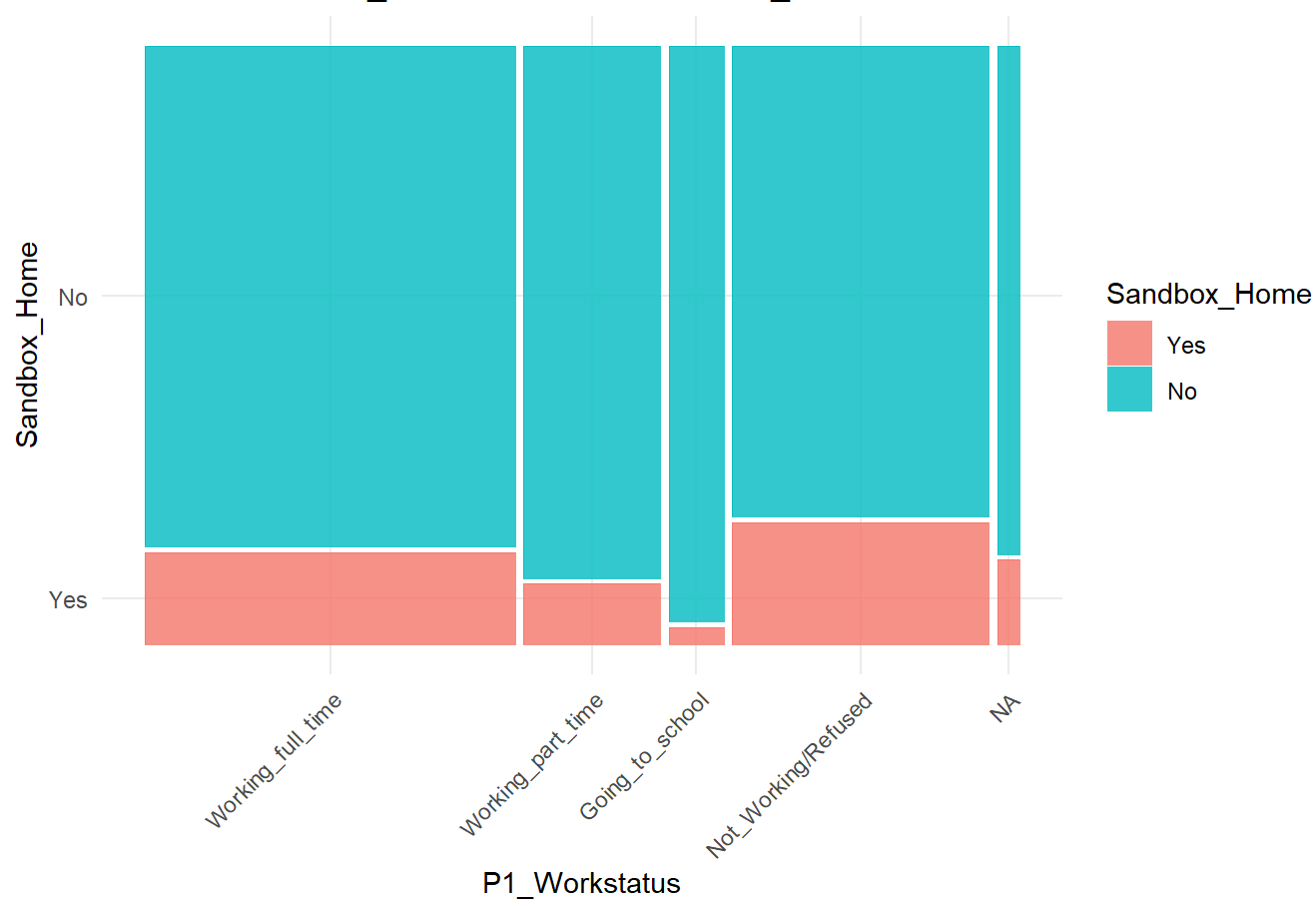
Mosaic Plot of P1_Degree and Sandbox_Home



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Home and P1_Workstatus -----
## Bivariate Analysis of Sandbox_Home and P1_Workstatus
##
## Frequency Table:
##   Sandbox_Home Working_full_time Working_part_time Going_to_school
##           Yes           37           9           1
##           No           201           79          34
##   Not_Working/Refused NA_
##           34    2
##           131  12
##
## Relative Frequency Table:
##   Sandbox_Home Working_full_time Working_part_time Going_to_school
##           Yes           15.5%           10.2%           2.9%
##           No           84.5%           89.8%           97.1%
##   Not_Working/Refused   NA_ Total
##           20.6% 14.3% 15.4%
##           79.4% 85.7% 84.6%
##
## Cramér's V:  0.1009292
```

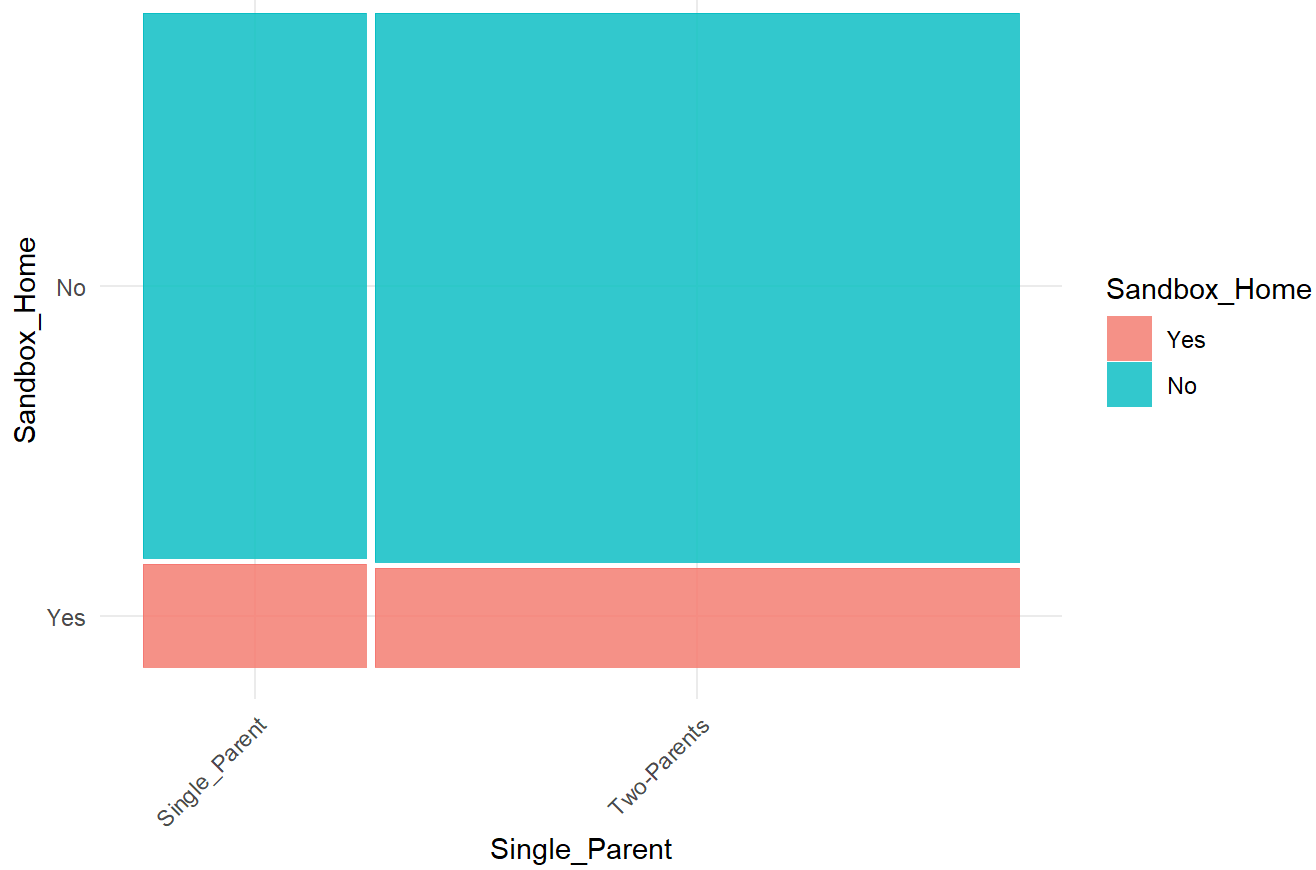
Mosaic Plot of P1_Workstatus and Sandbox_Home



Saving 7 x 5 in image

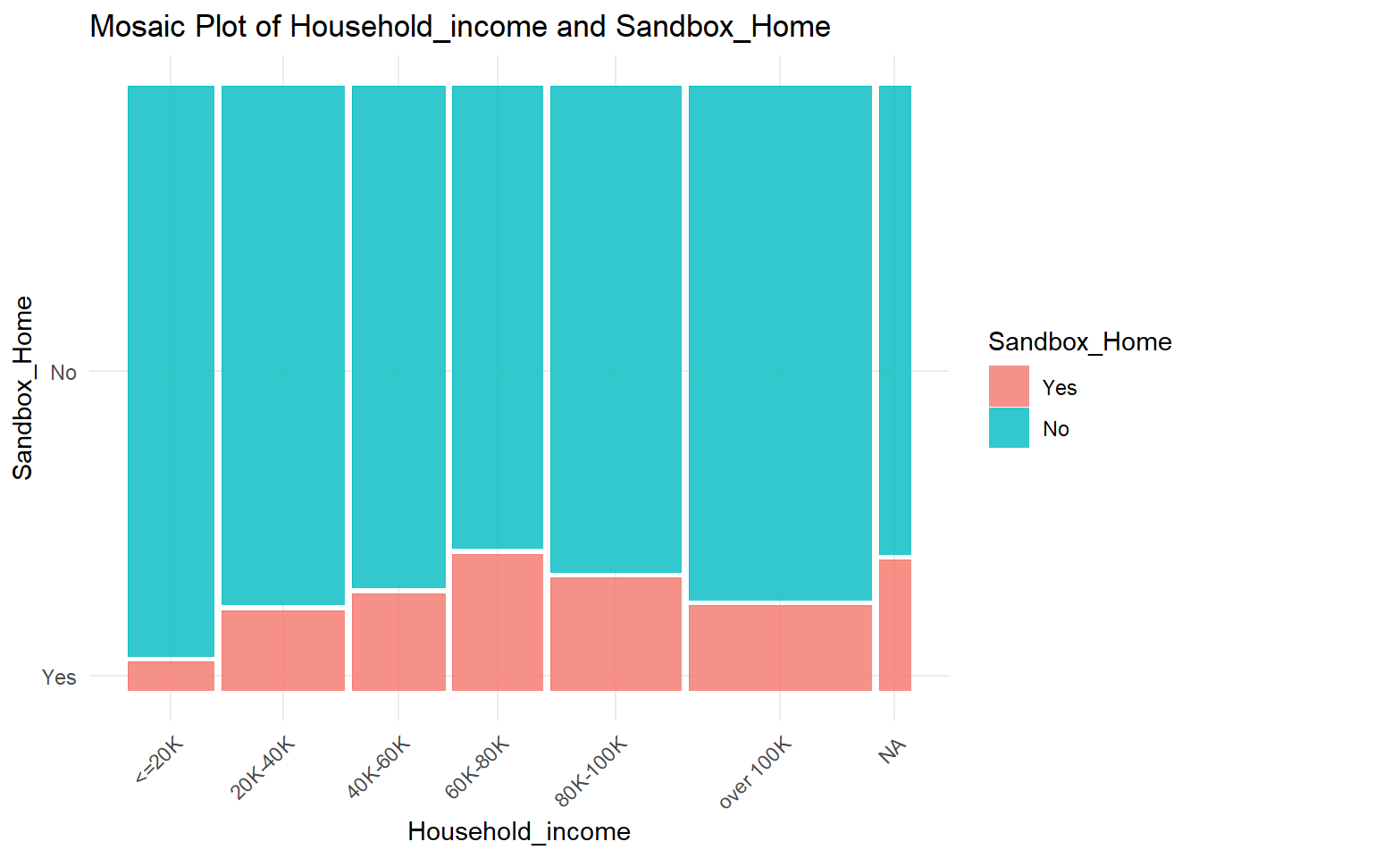
```
##
##
## ----- Analysis for Sandbox_Home and Single_Parent -----
## Bivariate Analysis of Sandbox_Home and Single_Parent
##
## Frequency Table:
##   Sandbox_Home Single_Parent Two-Parents
##           Yes           22           61
##           No            117          340
##
## Relative Frequency Table:
##   Sandbox_Home Single_Parent Two-Parents Total
##           Yes           15.8%          15.2% 15.4%
##           No            84.2%          84.8% 84.6%
##
## Cramér's V:  0
```

Mosaic Plot of Single_Parent and Sandbox_Home



```
## Saving 7 x 5 in image
```

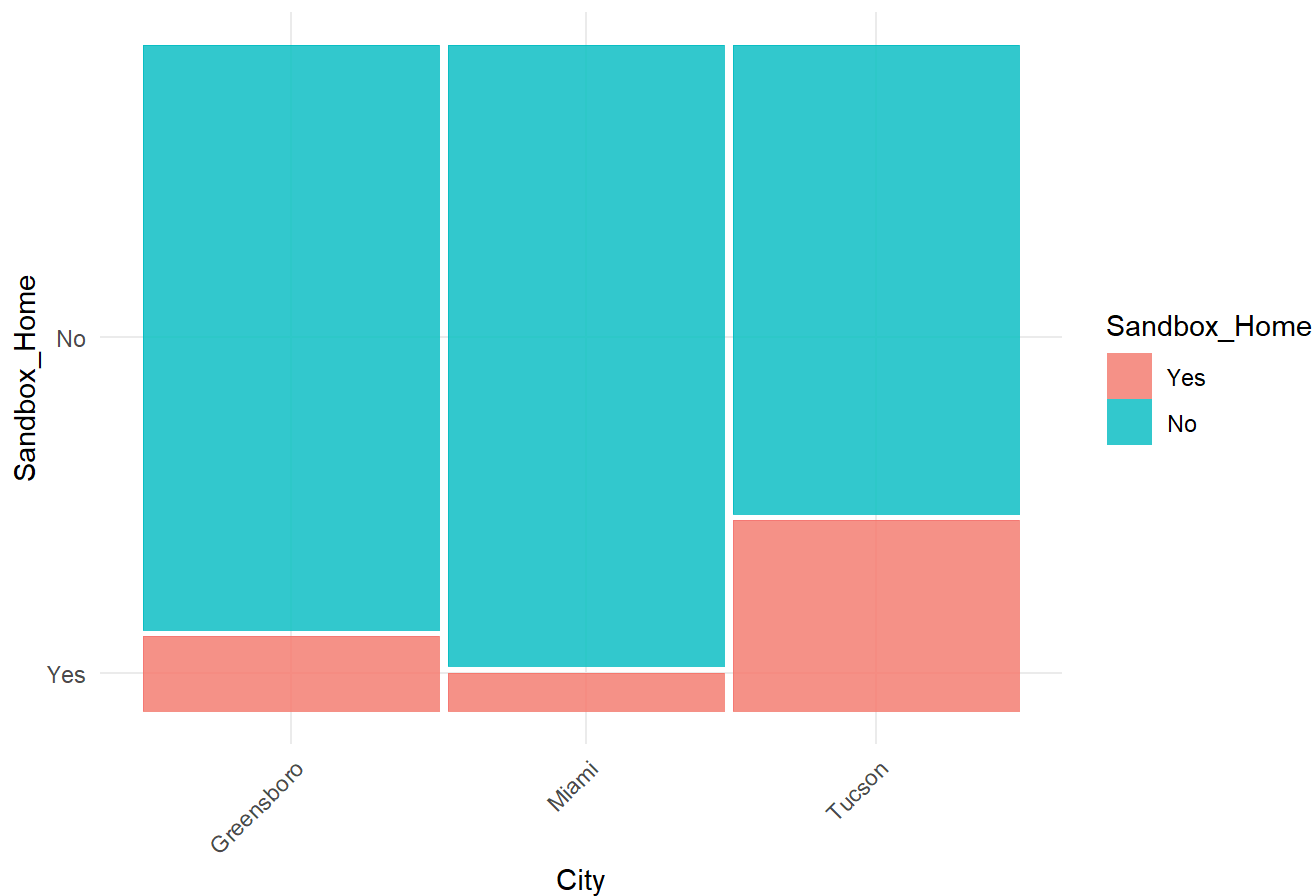
```
##
##
## ----- Analysis for Sandbox_Home and Household_income -----
## Bivariate Analysis of Sandbox_Home and Household_income
##
## Frequency Table:
##   Sandbox_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##             Yes      3      12      11      15      18      19    5
##             No      60      78      57      51      78     115   18
##
## Relative Frequency Table:
##   Sandbox_Home <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_ Total
##             Yes  4.8%  13.3%  16.2%  22.7%  18.8%    14.2% 21.7% 15.4%
##             No  95.2%  86.7%  83.8%  77.3%  81.2%    85.8% 78.3% 84.6%
##
## Cramér's V:  0.08840976
```



```
## Saving 7 x 5 in image

##
## ----- Analysis for Sandbox_Home and City -----
## Bivariate Analysis of Sandbox_Home and City
##
## Frequency Table:
##   Sandbox_Home Greensboro Miami Tucson
##           Yes      21      10      52
##           No      165     164     128
##
## Relative Frequency Table:
##   Sandbox_Home Greensboro Miami Tucson Total
##           Yes     11.3%   5.7%  28.9% 15.4%
##           No     88.7%  94.3%  71.1% 84.6%
##
## Cramér's V:  0.2657047
```

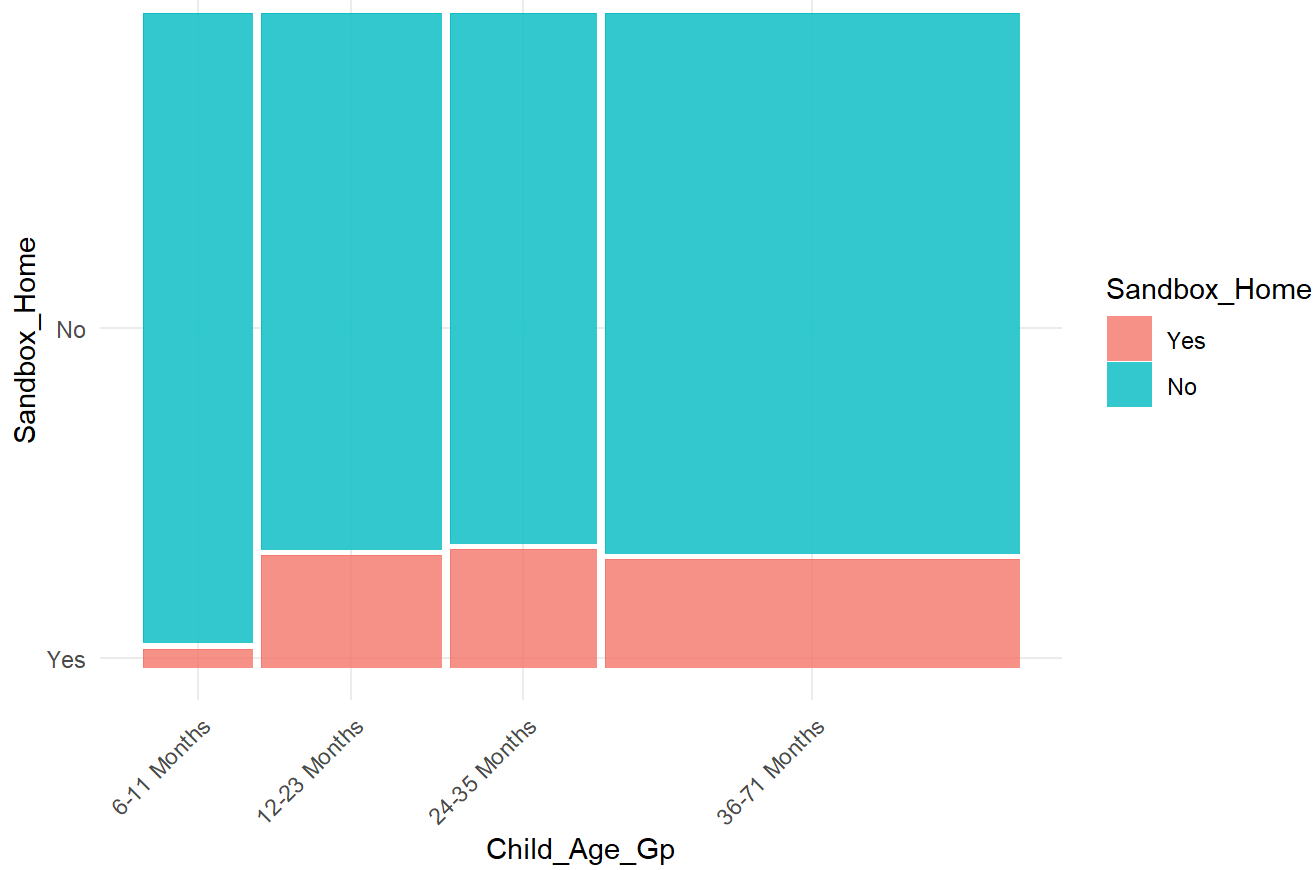

Mosaic Plot of City and Sandbox_Home



```
## Saving 7 x 5 in image
```

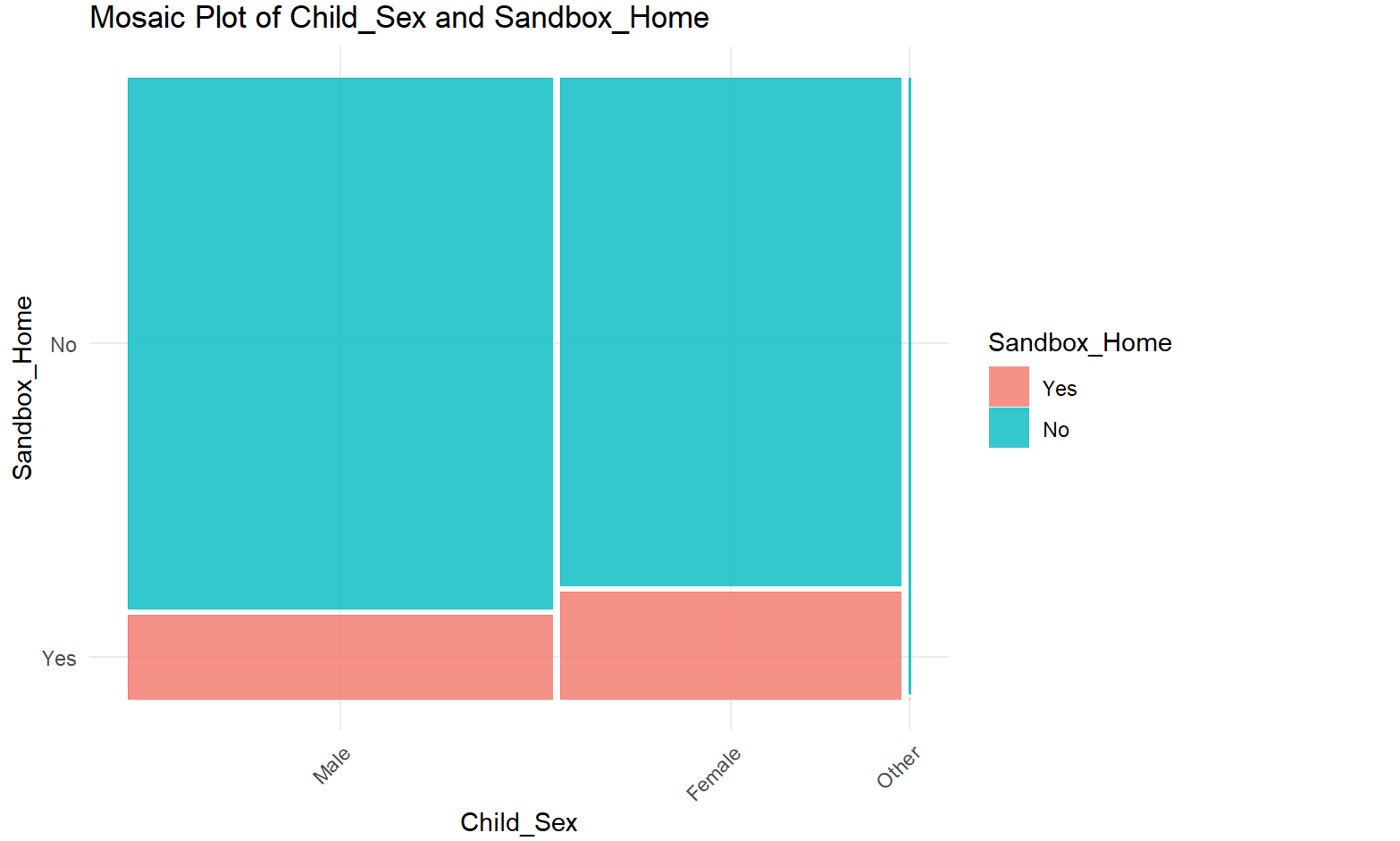
```
##
##
## ----- Analysis for Sandbox_Home and Child_Age_Gp -----
## Bivariate Analysis of Sandbox_Home and Child_Age_Gp
##
## Frequency Table:
##   Sandbox_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##           Yes           2           20           17           44
##           No            67           95           76          219
##
## Relative Frequency Table:
##   Sandbox_Home 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##           Yes           2.9%          17.4%          18.3%          16.7% 15.4%
##           No            97.1%          82.6%          81.7%          83.3% 84.6%
##
## Cramér's V:  0.1105335
```

Mosaic Plot of Child_Age_Gp and Sandbox_Home



```
## Saving 7 x 5 in image
```

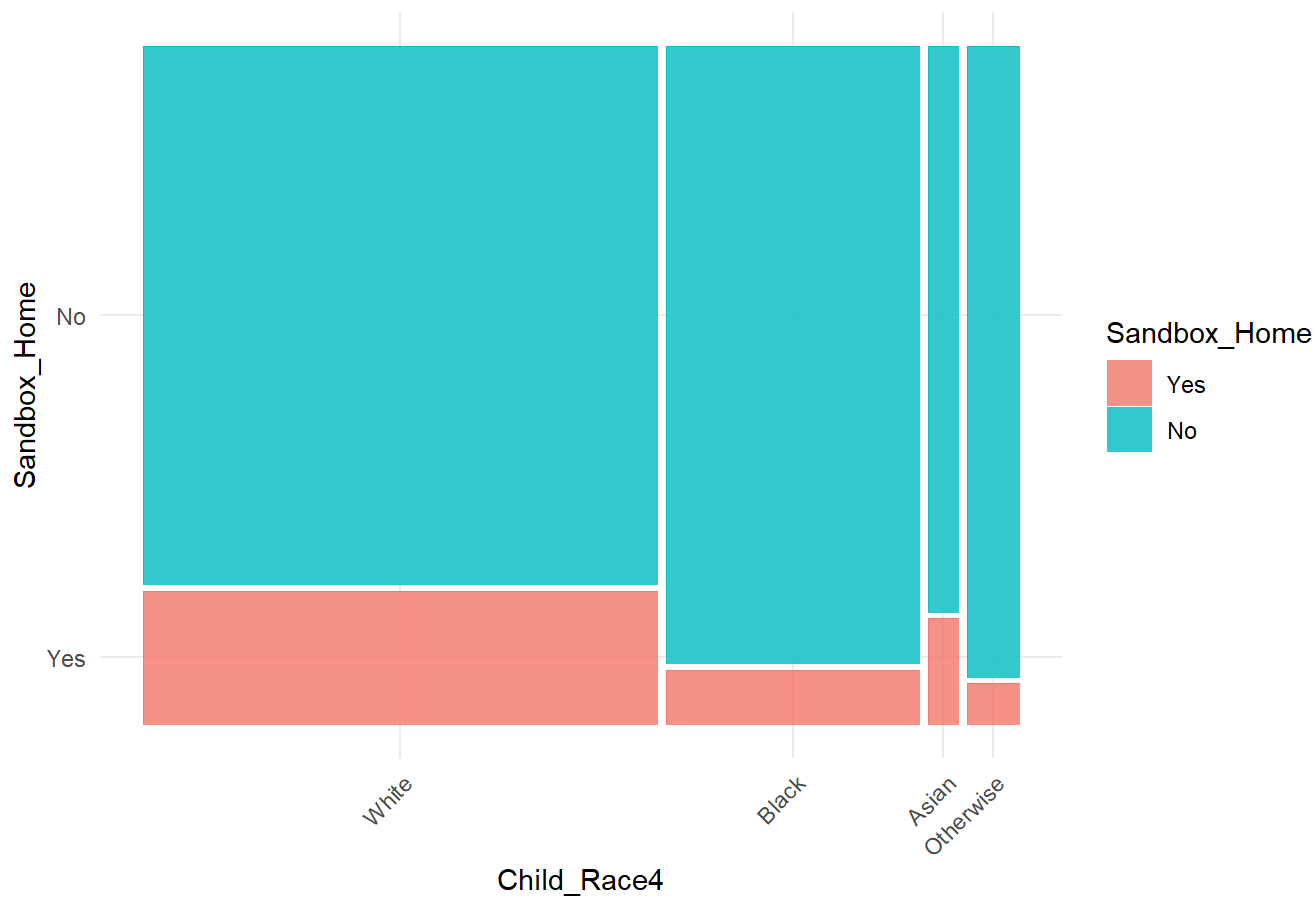
```
##
##
## ----- Analysis for Sandbox_Home and Child_Sex -----
## Bivariate Analysis of Sandbox_Home and Child_Sex
##
## Frequency Table:
##   Sandbox_Home Male Female Other
##           Yes   41    42    0
##           No  258   198    1
##
## Relative Frequency Table:
##   Sandbox_Home  Male Female  Other Total
##           Yes 13.7%  17.5%   0.0% 15.4%
##           No 86.3%  82.5% 100.0% 84.6%
##
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Home and Child_Race4 -----
## Bivariate Analysis of Sandbox_Home and Child_Race4
##
## Frequency Table:
##   Sandbox_Home White Black Asian Otherwise
##           Yes    65    13     3         2
##           No   262   148    16        31
##
## Relative Frequency Table:
##   Sandbox_Home White Black Asian Otherwise Total
##           Yes 19.9%  8.1% 15.8%      6.1% 15.4%
##           No 80.1% 91.9% 84.2%     93.9% 84.6%
##
## Cramér's V:  0.1421487
```

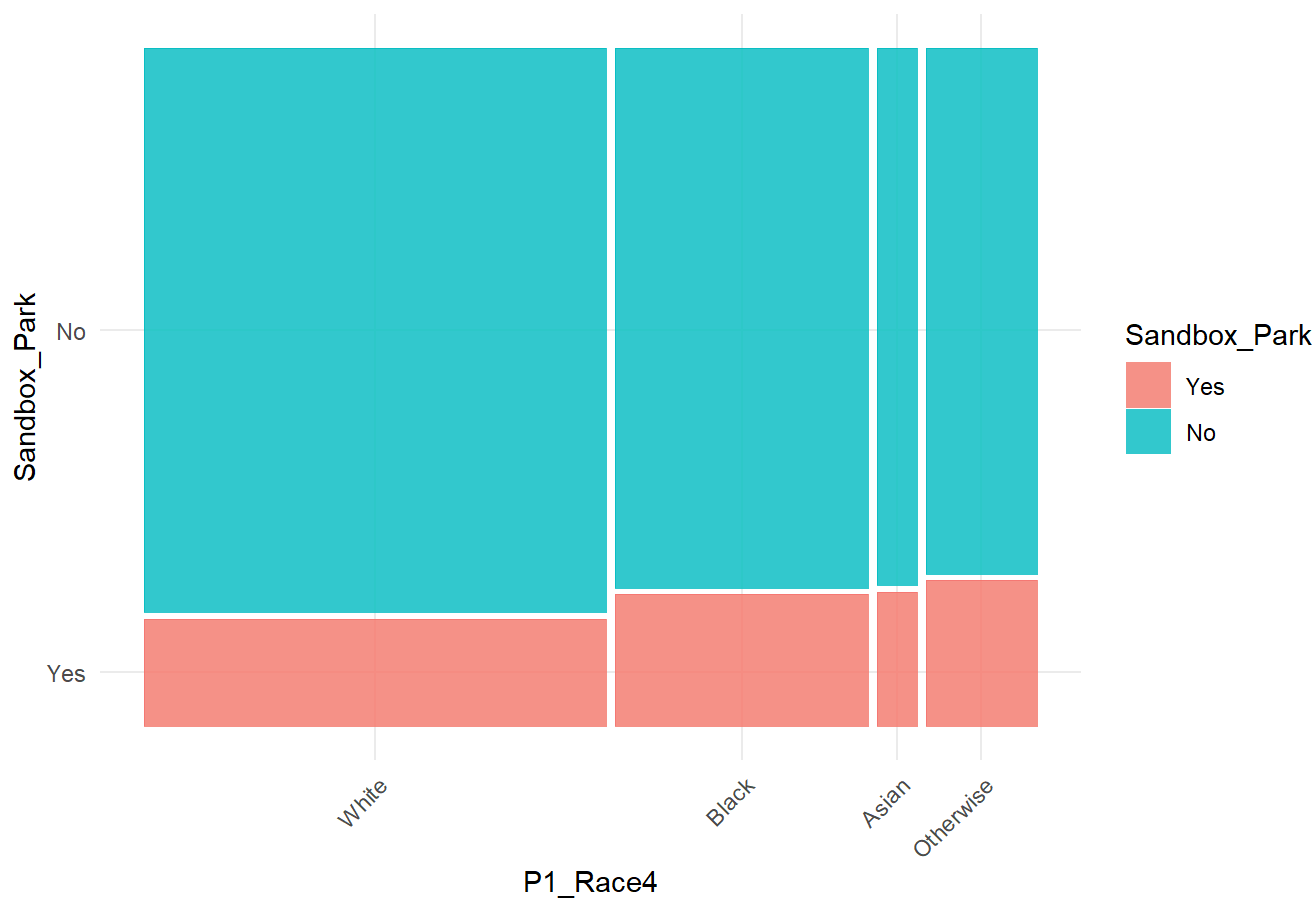
Mosaic Plot of Child_Race4 and Sandbox_Home



```
## Saving 7 x 5 in image
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```
##
##
## ----- Analysis for Sandbox_Park and P1_Race4 -----
## Bivariate Analysis of Sandbox_Park and P1_Race4
##
## Frequency Table:
##   Sandbox_Park White Black Asian Otherwise
##           Yes    46    31    5         15
##           No   242   127   20         54
##
## Relative Frequency Table:
##   Sandbox_Park White Black Asian Otherwise Total
##           Yes 16.0% 19.6% 20.0%      21.7% 18.0%
##           No 84.0% 80.4% 80.0%      78.3% 82.0%
##
##
## Cramér's V:  0
```

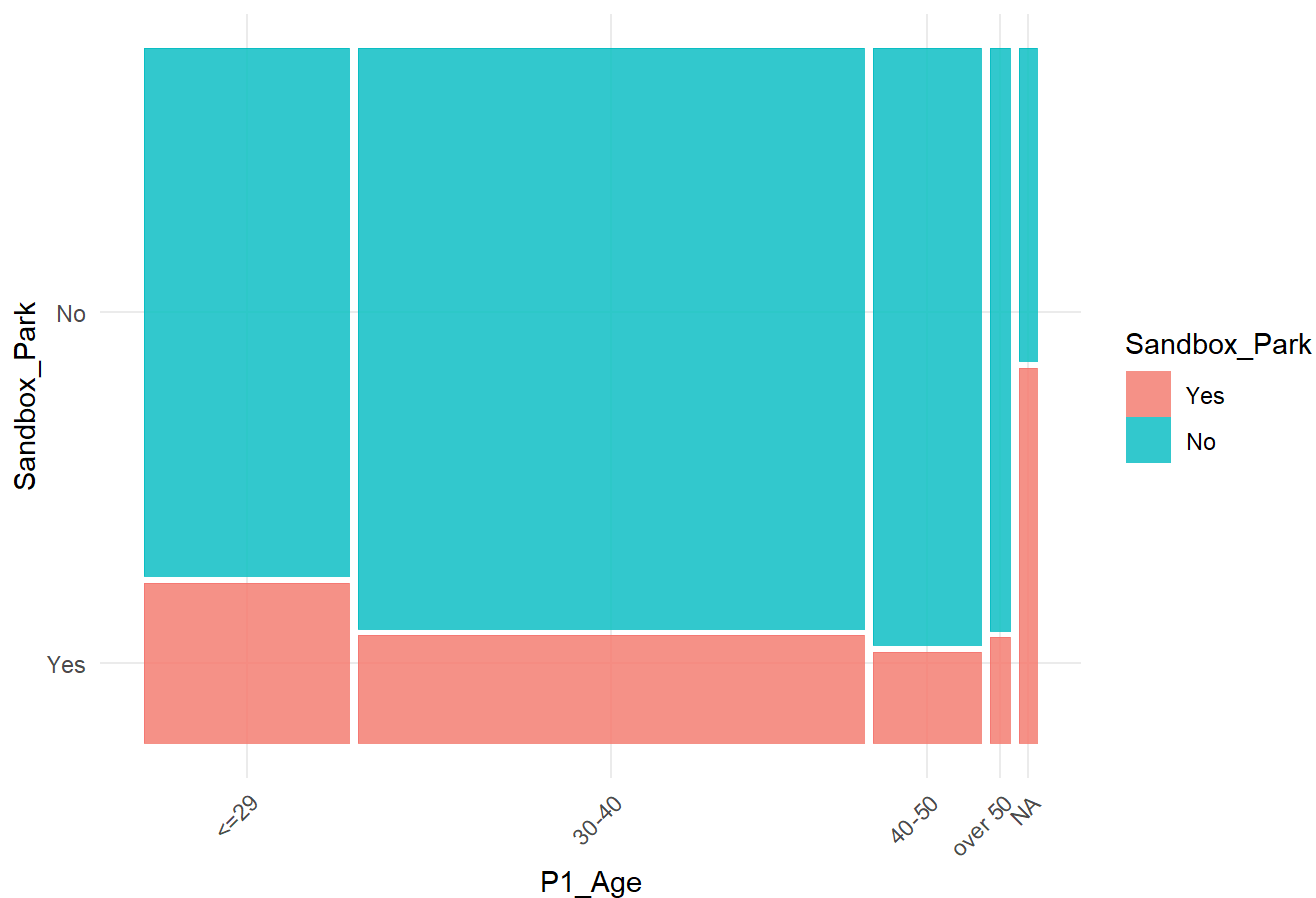
Mosaic Plot of P1_Race4 and Sandbox_Park



```
## Saving 7 x 5 in image
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```
##
##
## ----- Analysis for Sandbox_Park and P1_Age -----
## Bivariate Analysis of Sandbox_Park and P1_Age
##
## Frequency Table:
##   Sandbox_Park  <=29  30-40  40-50  over 50  NA_
##           Yes    30    50    9        2    6
##           No    99   269   59       11    5
##
## Relative Frequency Table:
##   Sandbox_Park  <=29  30-40  40-50  over 50   NA_ Total
##           Yes  23.3%  15.7%  13.2%   15.4%  54.5%  18.0%
##           No  76.7%  84.3%  86.8%   84.6%  45.5%  82.0%
##
## Cramér's V:  0.1406244
```

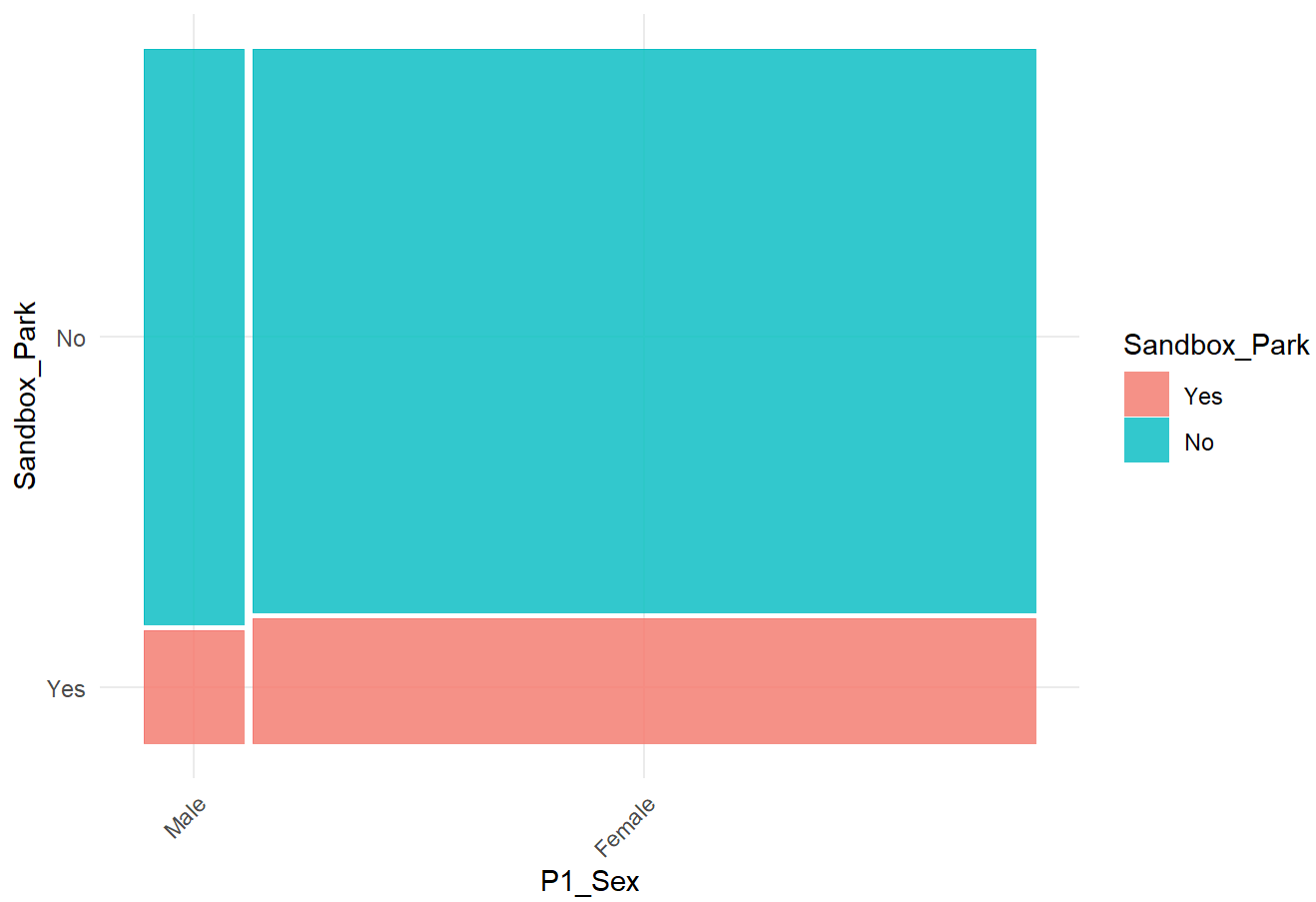
Mosaic Plot of P1_Age and Sandbox_Park



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Park and P1_Sex -----
## Bivariate Analysis of Sandbox_Park and P1_Sex
##
## Frequency Table:
##   Sandbox_Park  Male Female
##           Yes    10     87
##           No     51    392
##
## Relative Frequency Table:
##   Sandbox_Park  Male Female Total
##           Yes 16.4%  18.2% 18.0%
##           No 83.6%  81.8% 82.0%
##
## Cramér's V:  0
```

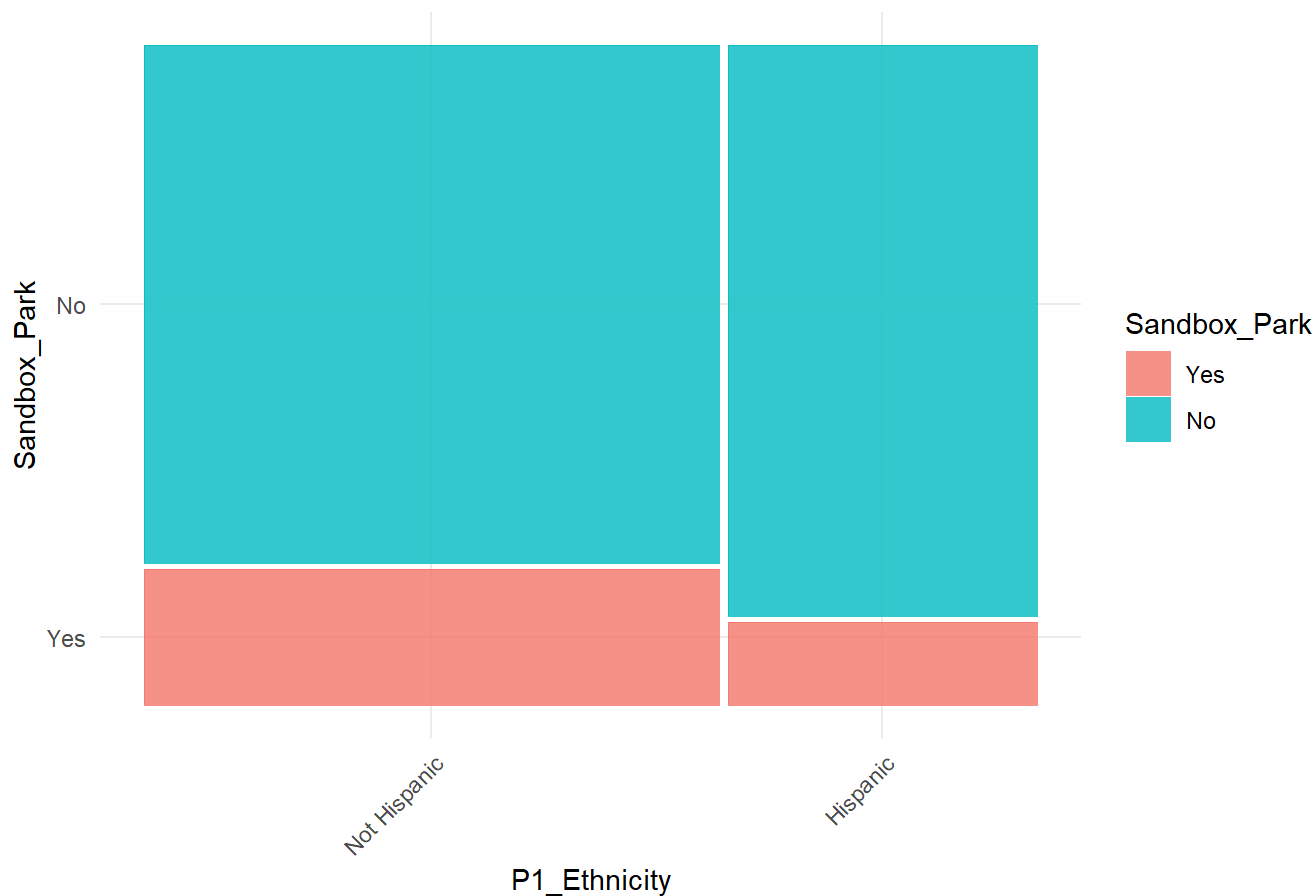
Mosaic Plot of P1_Sex and Sandbox_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Park and P1_Ethnicity -----
## Bivariate Analysis of Sandbox_Park and P1_Ethnicity
##
## Frequency Table:
##   Sandbox_Park Not Hispanic Hispanic
##           Yes           73          24
##           No            278         165
##
## Relative Frequency Table:
##   Sandbox_Park Not Hispanic Hispanic Total
##           Yes          20.8%    12.7% 18.0%
##           No           79.2%    87.3% 82.0%
##
## Cramér's V:  0.09103453
```

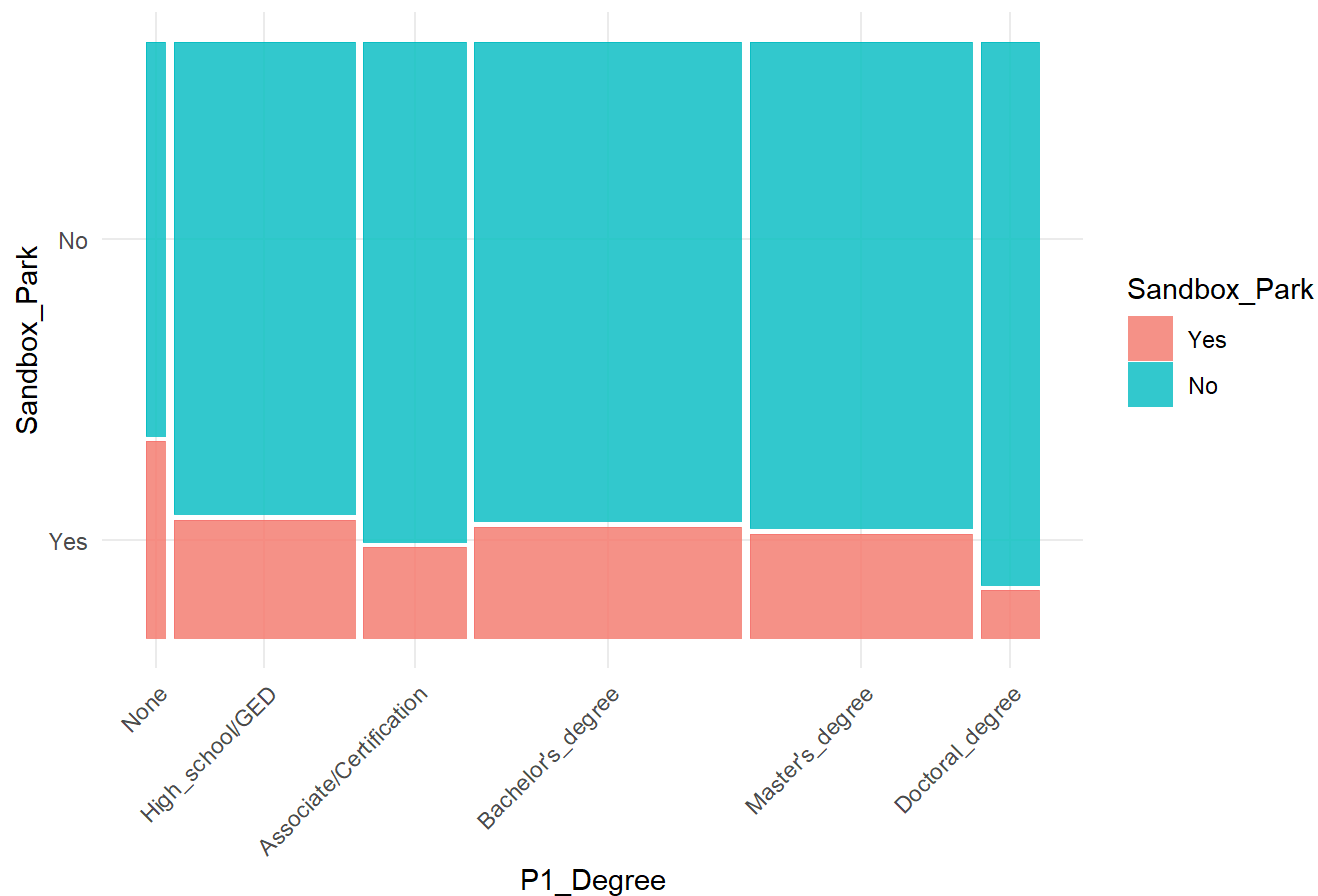
Mosaic Plot of P1_Ethnicity and Sandbox_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Park and P1_Degree -----
## Bivariate Analysis of Sandbox_Park and P1_Degree
##
## Frequency Table:
##   Sandbox_Park None High_school/GED Associate/Certification Bachelor's_degree
##           Yes    4           23           10           32
##           No     8           92           55           138
##   Master's_degree Doctoral_degree
##           25           3
##           116          34
##
## Relative Frequency Table:
##   Sandbox_Park None High_school/GED Associate/Certification Bachelor's_degree
##           Yes 33.3%           20.0%           15.4%           18.8%
##           No 66.7%           80.0%           84.6%           81.2%
##   Master's_degree Doctoral_degree Total
##           17.7%           8.1% 18.0%
##           82.3%           91.9% 82.0%
##
## Cramér's V: 0.01060705
```


Mosaic Plot of P1_Degree and Sandbox_Park



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Park and P1_Workstatus -----
## Bivariate Analysis of Sandbox_Park and P1_Workstatus
##
## Frequency Table:
##   Sandbox_Park Working_full_time Working_part_time Going_to_school
##           Yes                28                20                8
##           No                 210                68               27
##   Not_Working/Refused NA_
##               40      1
##               125    13
##
## Relative Frequency Table:
##   Sandbox_Park Working_full_time Working_part_time Going_to_school
##           Yes                11.8%                22.7%                22.9%
##           No                 88.2%                77.3%                77.1%
##   Not_Working/Refused  NA_ Total
##               24.2%  7.1% 18.0%
##               75.8% 92.9% 82.0%
##
## Cramér's V:  0.1337982
```

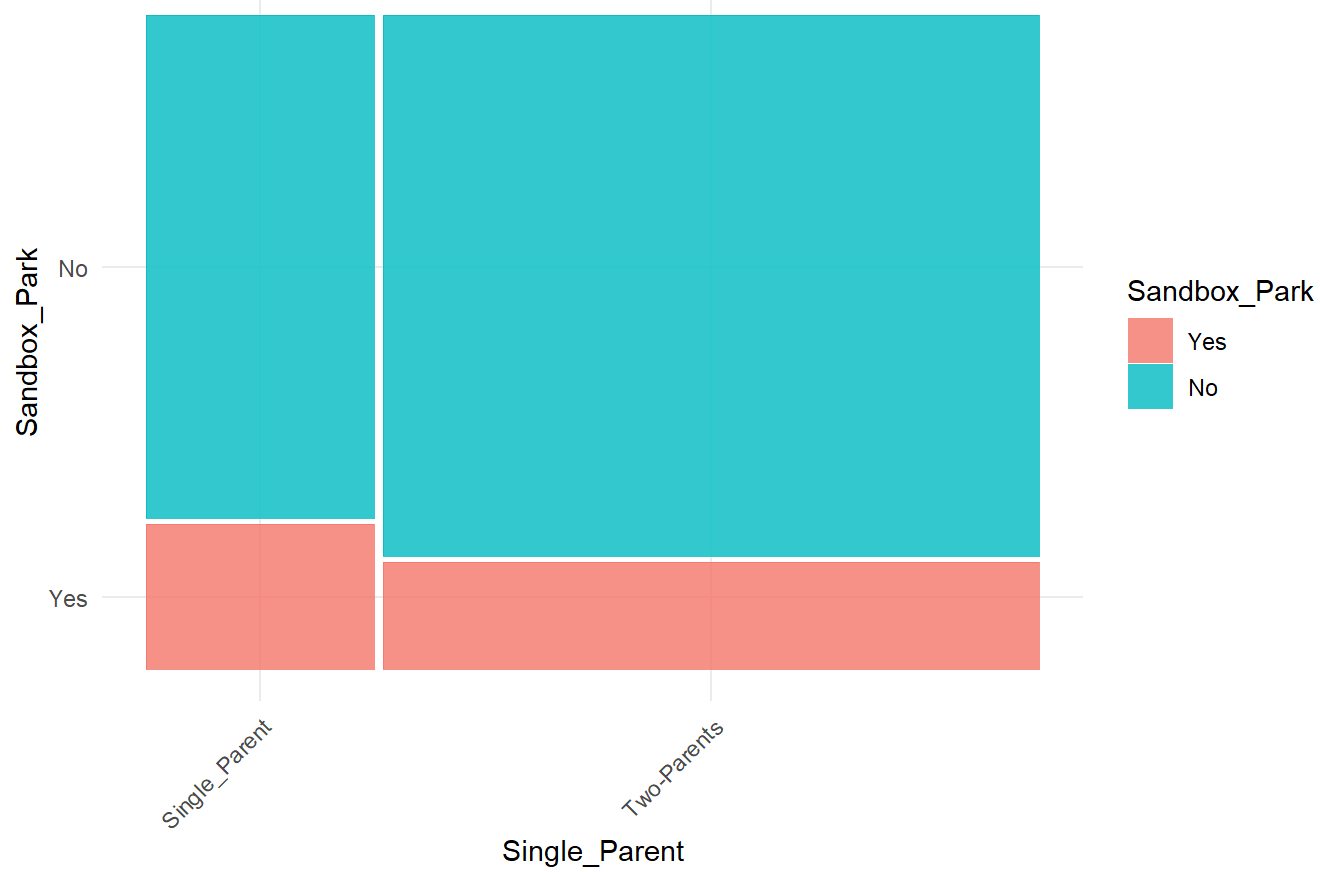
Mosaic Plot of P1_Workstatus and Sandbox_Park



```
## Saving 7 x 5 in image
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```
##
##
## ----- Analysis for Sandbox_Park and Single_Parent -----
## Bivariate Analysis of Sandbox_Park and Single_Parent
##
## Frequency Table:
##   Sandbox_Park Single_Parent Two-Parents
##           Yes           31           66
##           No            108          335
##
## Relative Frequency Table:
##   Sandbox_Park Single_Parent Two-Parents Total
##           Yes           22.3%          16.5% 18.0%
##           No            77.7%          83.5% 82.0%
##
## Cramér's V:  0.05077841
```

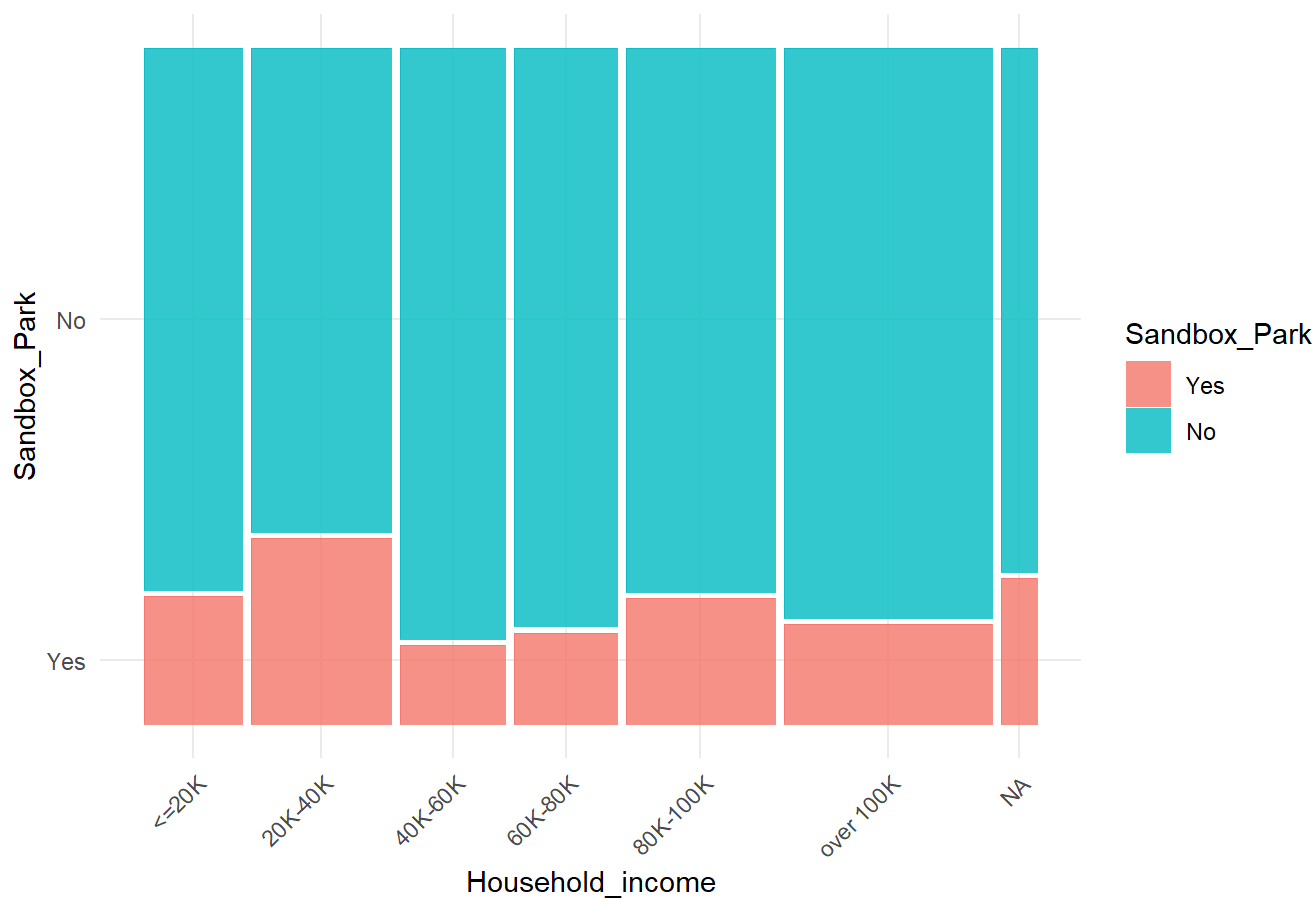
Mosaic Plot of Single_Parent and Sandbox_Park



```
## Saving 7 x 5 in image
```

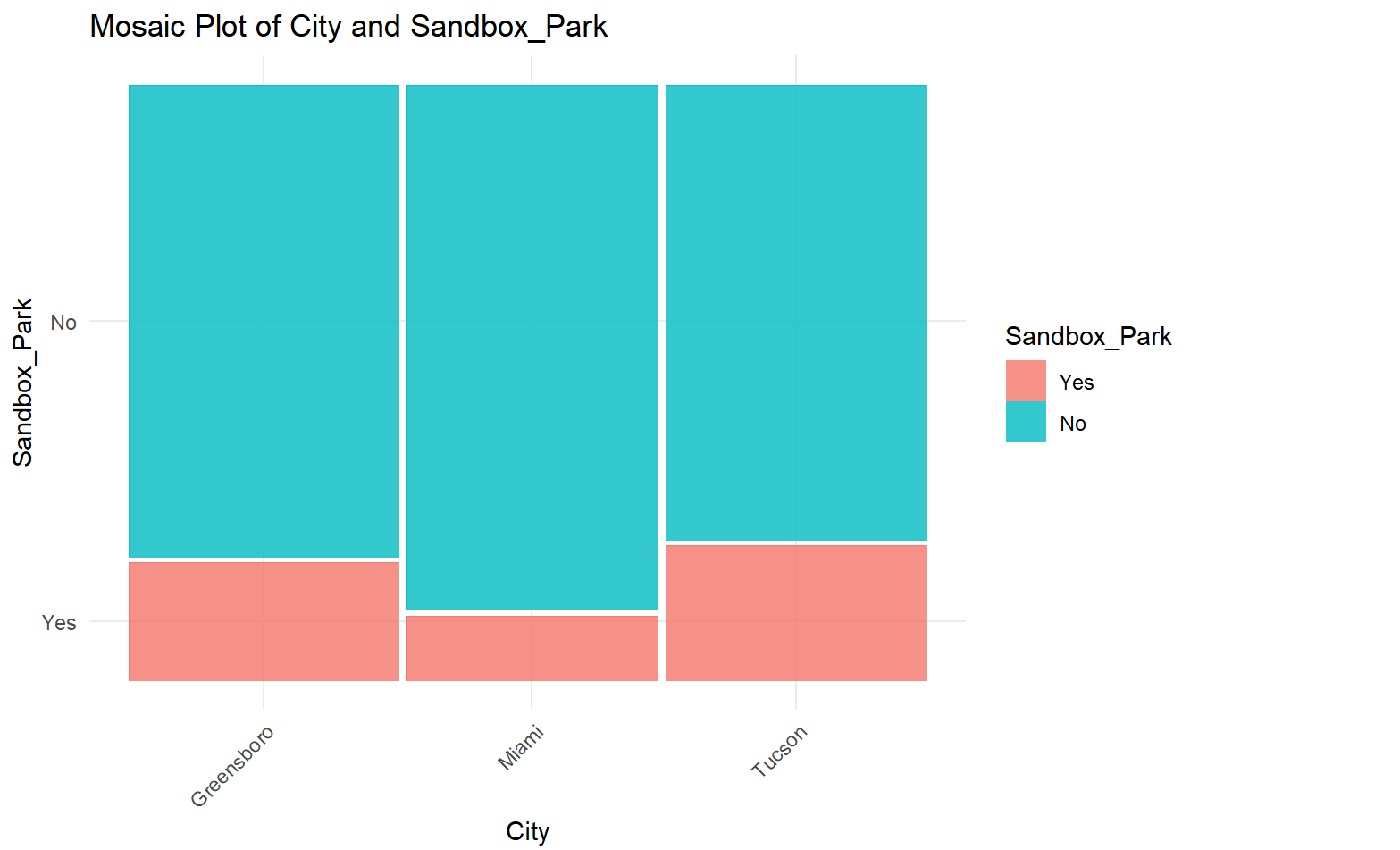
```
##
##
## ----- Analysis for Sandbox_Park and Household_income -----
## Bivariate Analysis of Sandbox_Park and Household_income
##
## Frequency Table:
##   Sandbox_Park <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes      12      25       8       9       18       20   5
##           No       51      65      60      57      78      114  18
##
## Relative Frequency Table:
##   Sandbox_Park <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_ Total
##           Yes 19.0%  27.8%  11.8%  13.6%  18.8%   14.9% 21.7% 18.0%
##           No 81.0%  72.2%  88.2%  86.4%  81.2%   85.1% 78.3% 82.0%
##
## Cramér's V:  0.08212824
```

Mosaic Plot of Household_income and Sandbox_Park



```
## Saving 7 x 5 in image
```

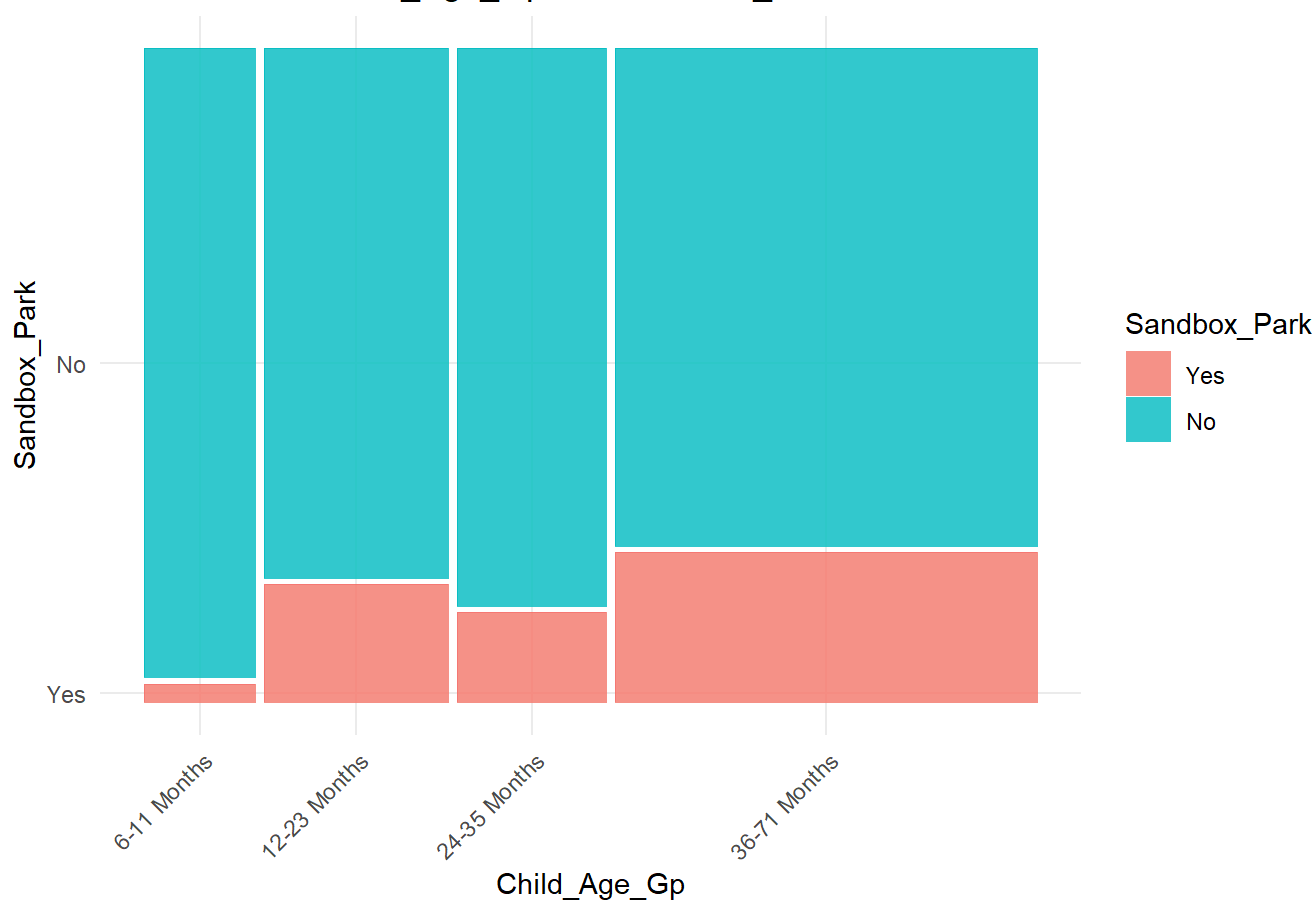
```
##
##
## ----- Analysis for Sandbox_Park and City -----
## Bivariate Analysis of Sandbox_Park and City
##
## Frequency Table:
##   Sandbox_Park Greensboro Miami Tucson
##           Yes      37      19      41
##           No      149     155     139
##
## Relative Frequency Table:
##   Sandbox_Park Greensboro Miami Tucson Total
##           Yes      19.9% 10.9% 22.8% 18.0%
##           No      80.1% 89.1% 77.2% 82.0%
##
## Cramér's V:  0.1152204
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Park and Child_Age_Gp -----
## Bivariate Analysis of Sandbox_Park and Child_Age_Gp
##
## Frequency Table:
##   Sandbox_Park 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##           Yes           2           21           13           61
##           No           67           94           80          202
##
## Relative Frequency Table:
##   Sandbox_Park 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##           Yes           2.9%          18.3%          14.0%          23.2% 18.0%
##           No           97.1%          81.7%          86.0%          76.8% 82.0%
##
## Cramér's V:  0.158335
```

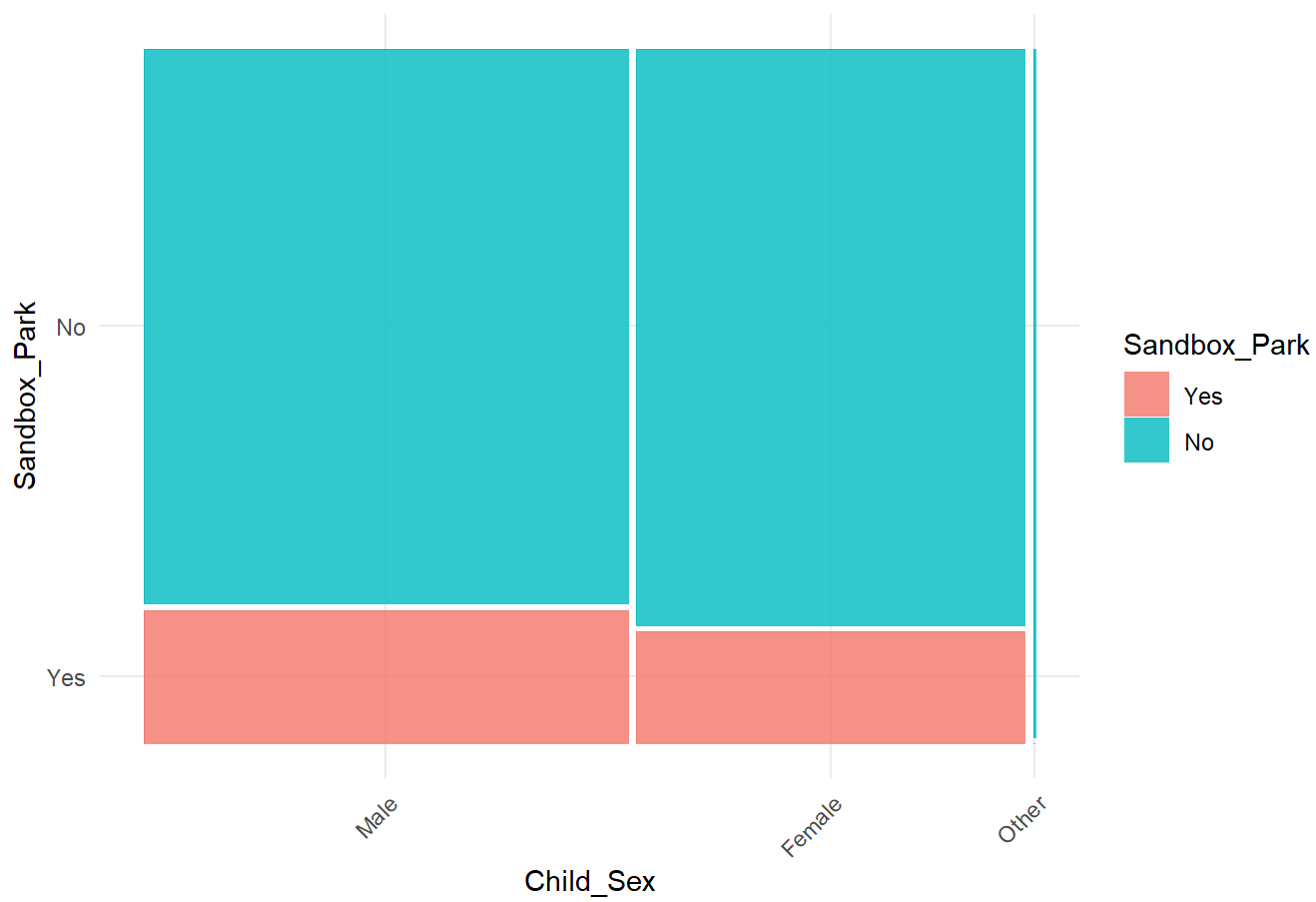
Mosaic Plot of Child_Age_Gp and Sandbox_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Park and Child_Sex -----
## Bivariate Analysis of Sandbox_Park and Child_Sex
##
## Frequency Table:
##   Sandbox_Park Male Female Other
##           Yes   58     39     0
##           No  241    201     1
##
## Relative Frequency Table:
##   Sandbox_Park Male Female Other Total
##           Yes 19.4% 16.2%  0.0% 18.0%
##           No 80.6% 83.8% 100.0% 82.0%
##
##
## Cramér's V:  0
```

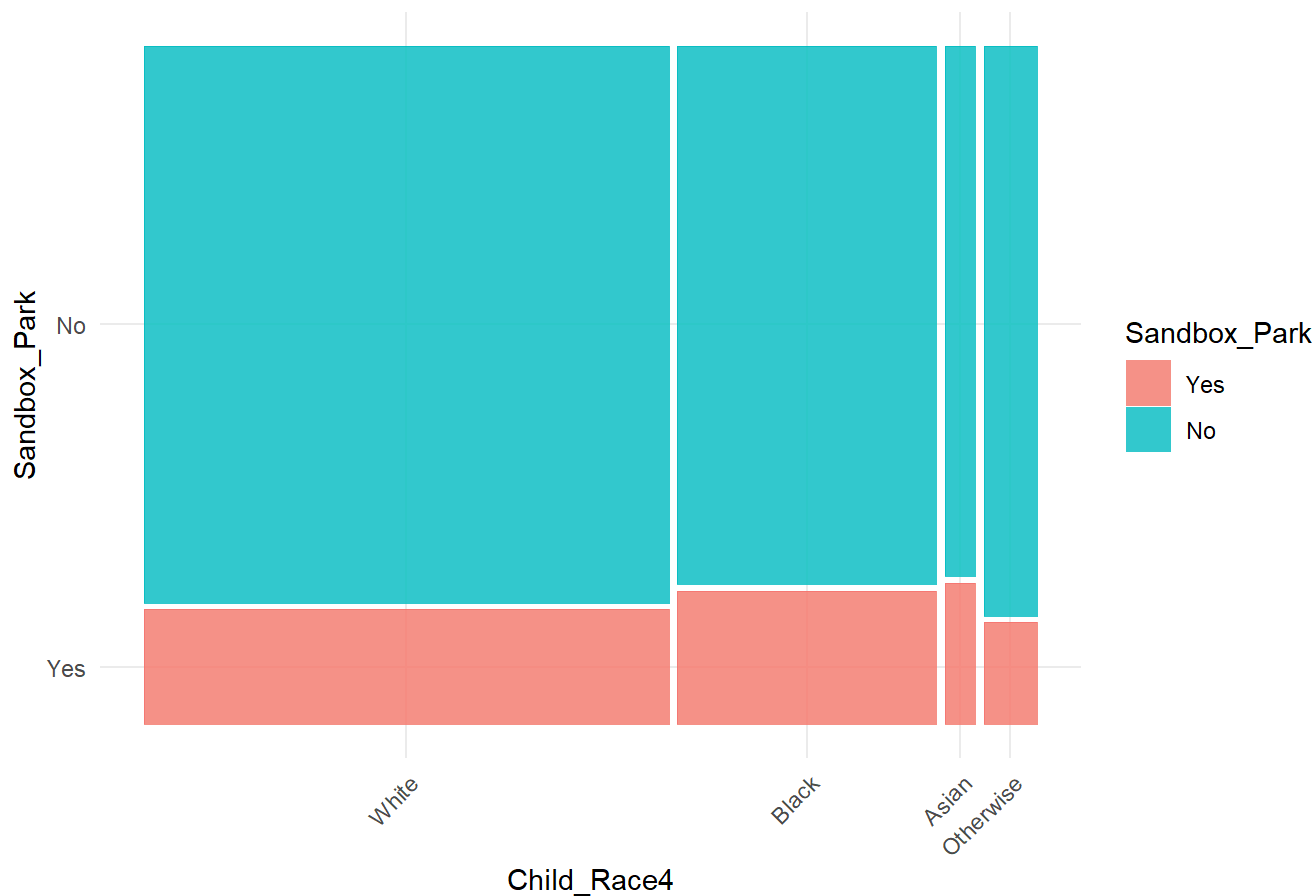
Mosaic Plot of Child_Sex and Sandbox_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Park and Child_Race4 -----
## Bivariate Analysis of Sandbox_Park and Child_Race4
##
## Frequency Table:
##   Sandbox_Park White Black Asian Otherwise
##           Yes    56    32    4         5
##           No   271   129   15        28
##
## Relative Frequency Table:
##   Sandbox_Park White Black Asian Otherwise Total
##           Yes 17.1% 19.9% 21.1%    15.2% 18.0%
##           No 82.9% 80.1% 78.9%    84.8% 82.0%
##
## Cramér's V:  0
```

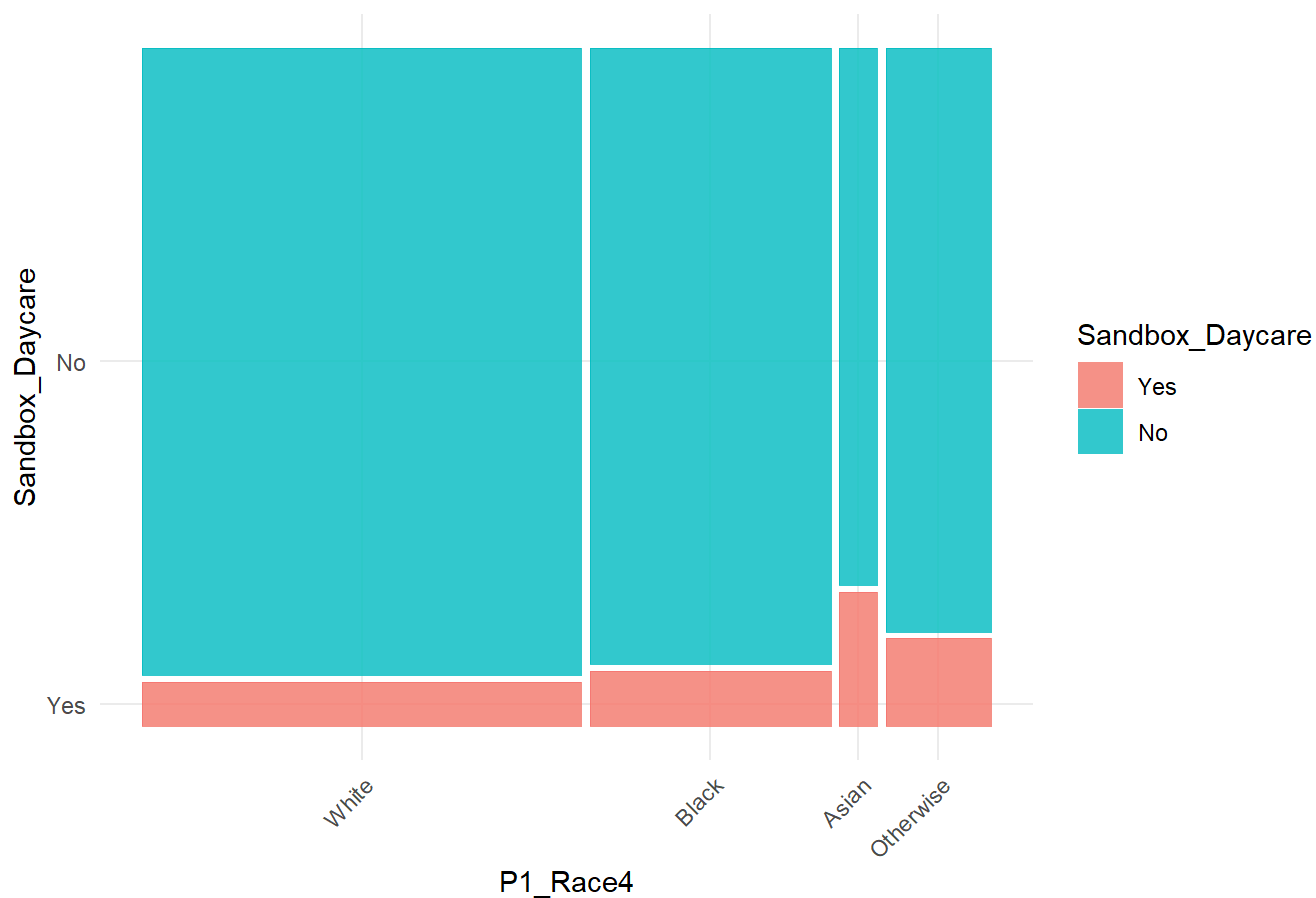
Mosaic Plot of Child_Race4 and Sandbox_Park



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and P1_Race4 -----
## Bivariate Analysis of Sandbox_Daycare and P1_Race4
##
## Frequency Table:
##   Sandbox_Daycare White Black Asian Otherwise
##           Yes      19    13     5         9
##           No     269   145    20        60
##
## Relative Frequency Table:
##   Sandbox_Daycare White Black Asian Otherwise Total
##           Yes   6.6%  8.2% 20.0%    13.0%  8.5%
##           No  93.4% 91.8% 80.0%    87.0% 91.5%
##
## Cramér's V:  0.09053272
```

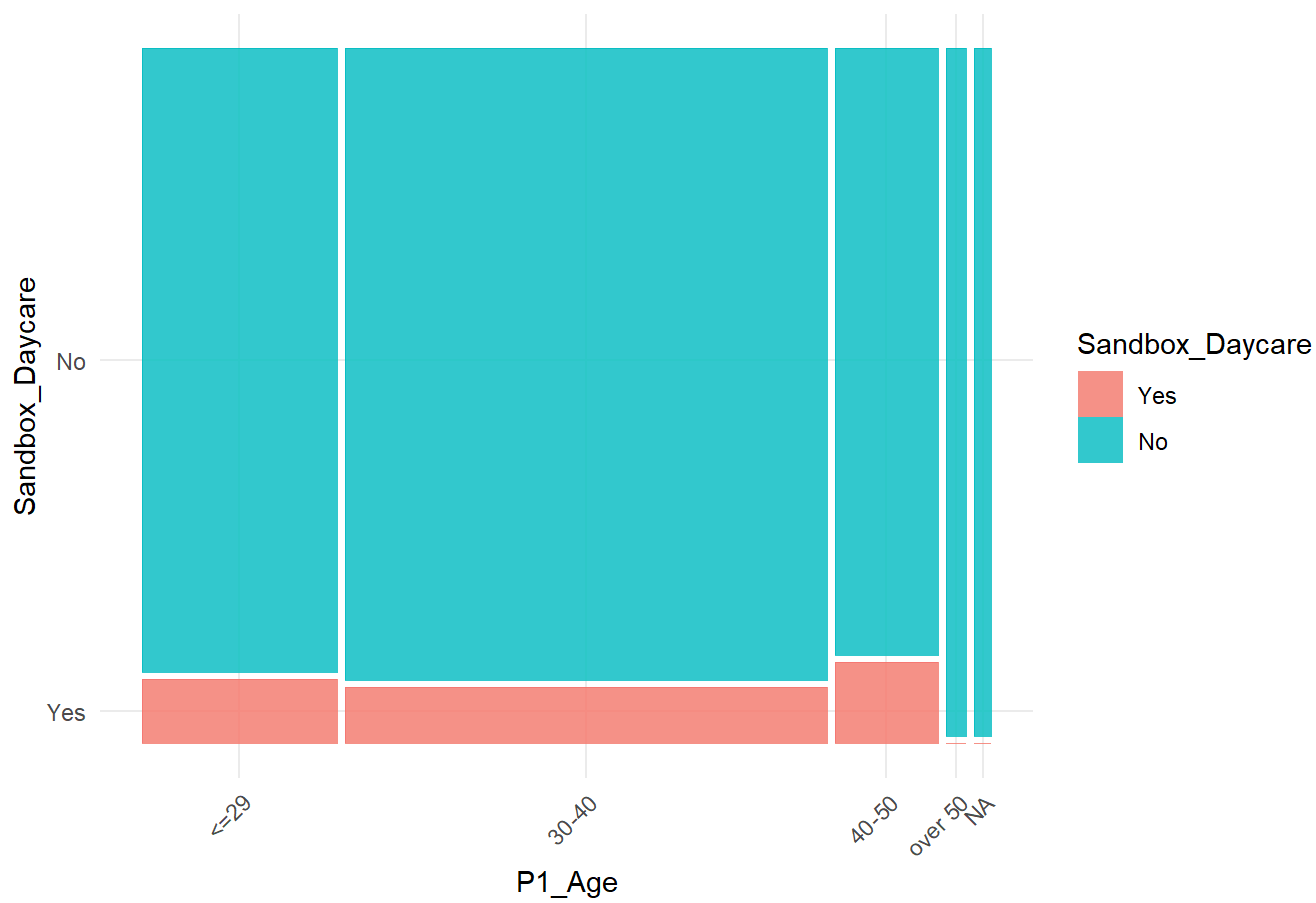

Mosaic Plot of P1_Race4 and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and P1_Age -----
## Bivariate Analysis of Sandbox_Daycare and P1_Age
##
## Frequency Table:
##   Sandbox_Daycare  <=29  30-40  40-50  over 50  NA_
##               Yes    12    26    8      0    0
##               No   117   293   60     13   11
##
## Relative Frequency Table:
##   Sandbox_Daycare  <=29  30-40  40-50  over 50    NA_ Total
##               Yes  9.3%  8.2% 11.8%    0.0%  0.0%  8.5%
##               No 90.7% 91.8% 88.2%   100.0% 100.0% 91.5%
##
## Cramér's V:  0
```

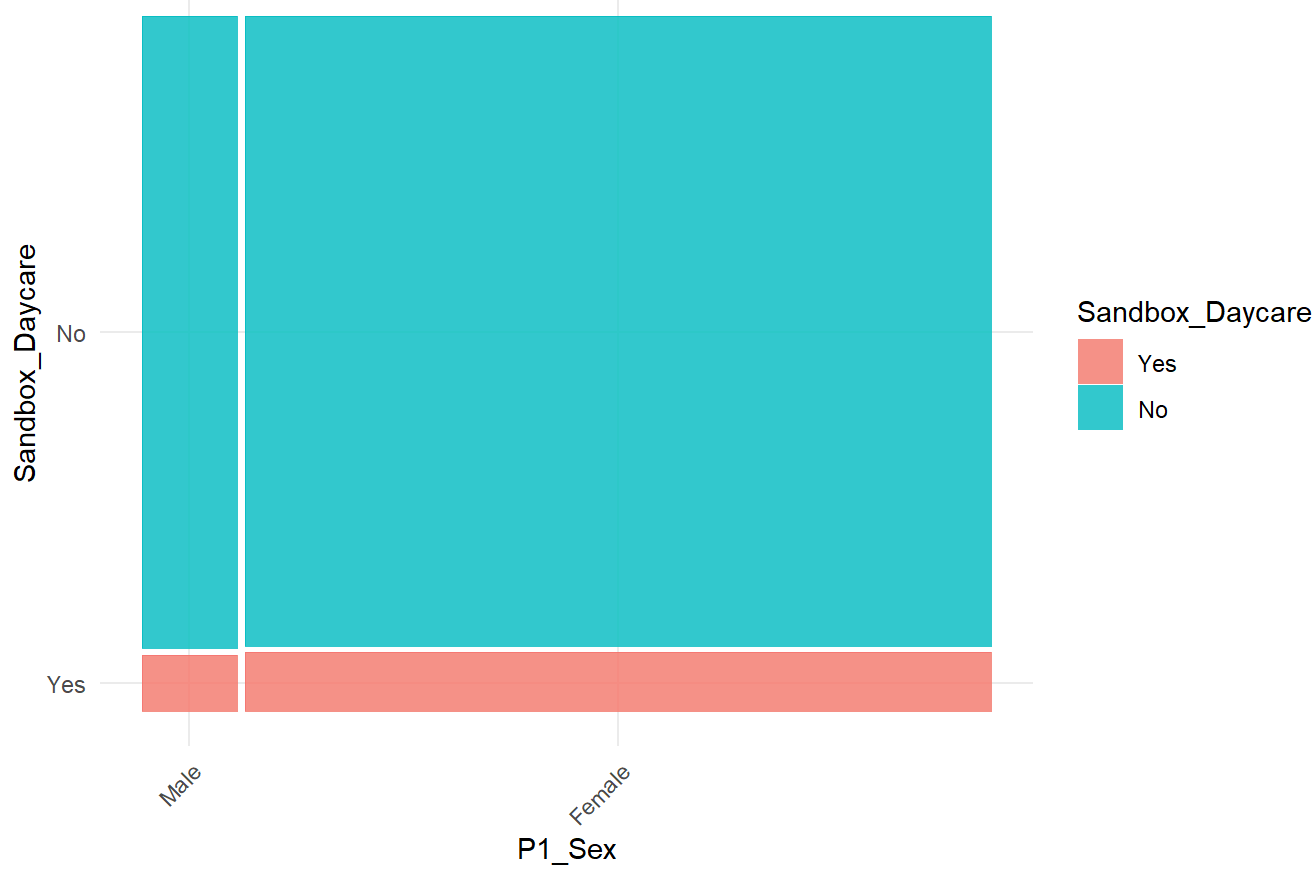
Mosaic Plot of P1_Age and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and P1_Sex -----
## Bivariate Analysis of Sandbox_Daycare and P1_Sex
##
## Frequency Table:
##   Sandbox_Daycare  Male  Female
##           Yes         5      41
##           No        56     438
##
## Relative Frequency Table:
##   Sandbox_Daycare  Male  Female  Total
##           Yes    8.2%   8.6%   8.5%
##           No   91.8%  91.4%  91.5%
##
##
## Cramér's V:  0
```

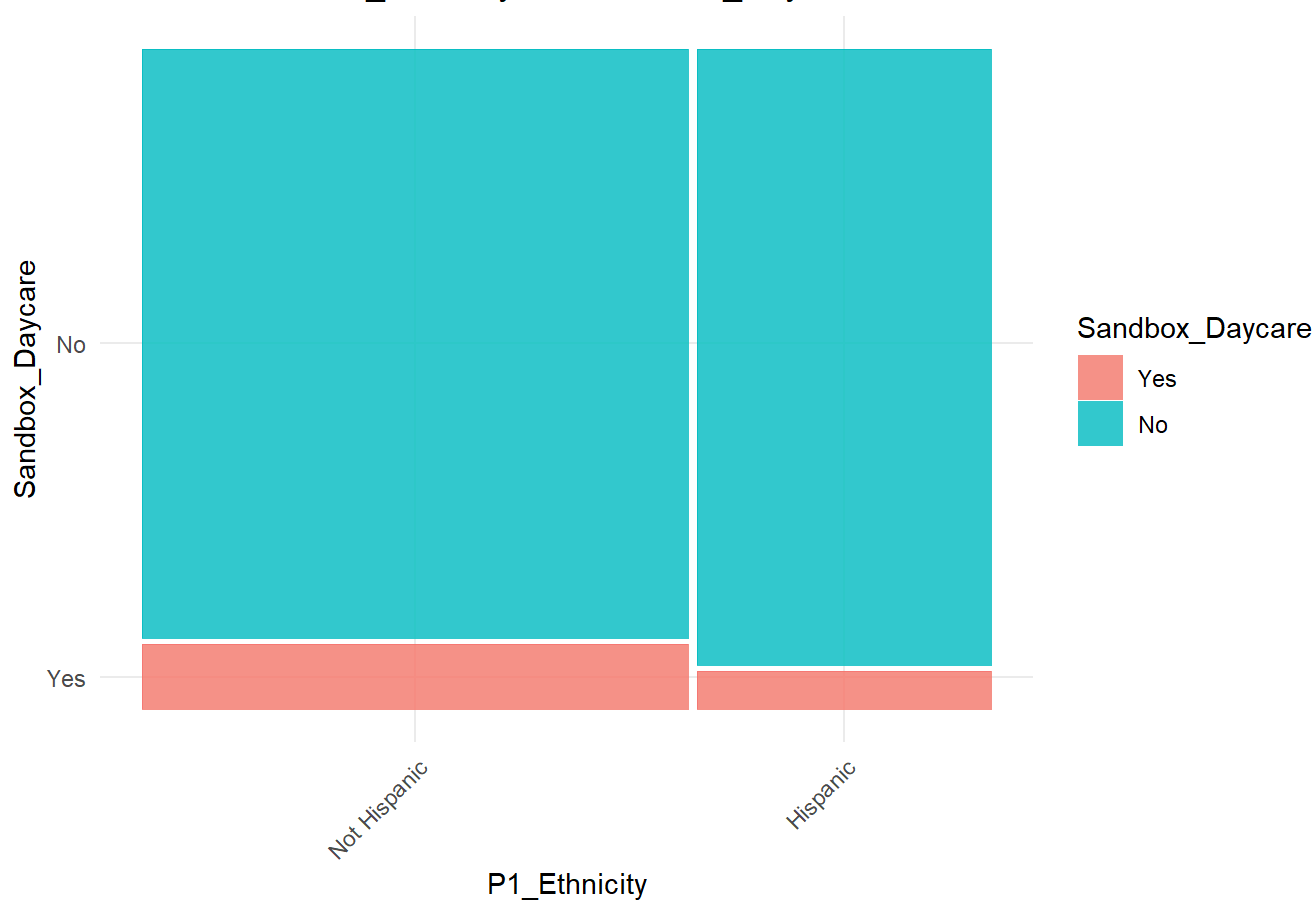
Mosaic Plot of P1_Sex and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and P1_Ethnicity -----
## Bivariate Analysis of Sandbox_Daycare and P1_Ethnicity
##
## Frequency Table:
##   Sandbox_Daycare Not Hispanic Hispanic
##           Yes           35           11
##           No           316          178
##
## Relative Frequency Table:
##   Sandbox_Daycare Not Hispanic Hispanic Total
##           Yes          10.0%    5.8%  8.5%
##           No           90.0%   94.2% 91.5%
##
## Cramér's V:  0.05640785
```

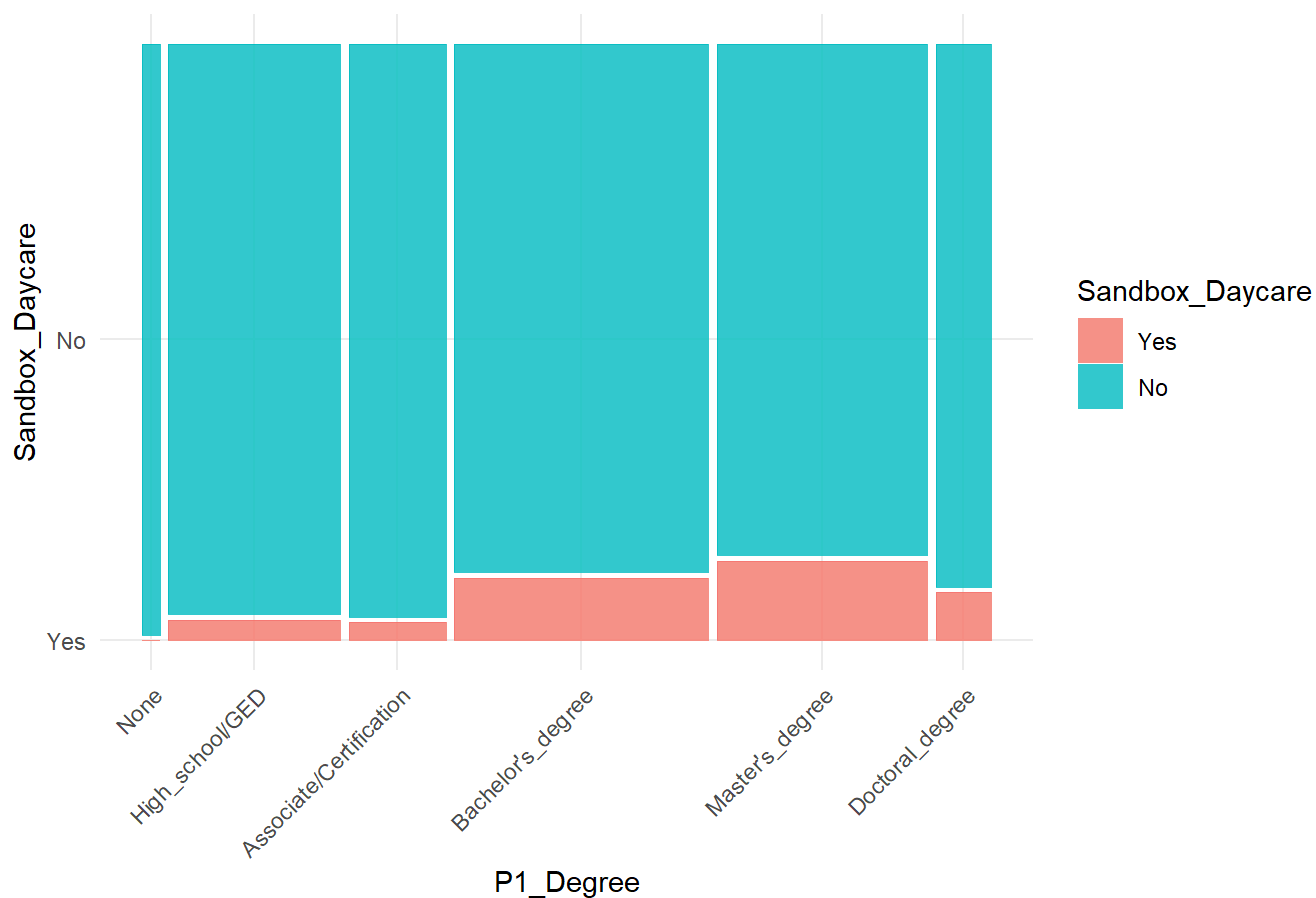
Mosaic Plot of P1_Ethnicity and Sandbox_Daycare



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Daycare and P1_Degree -----
## Bivariate Analysis of Sandbox_Daycare and P1_Degree
##
## Frequency Table:
##   Sandbox_Daycare  None  High_school/GED  Associate/Certification  Bachelor's_degree
##           Yes      0           4           2           18
##           No     12          111          63          152
##   Master's_degree  Doctoral_degree
##           19           3
##           122          34
##
## Relative Frequency Table:
##   Sandbox_Daycare  None  High_school/GED  Associate/Certification
##           Yes    0.0%           3.5%           3.1%
##           No  100.0%          96.5%          96.9%
##   Bachelor's_degree  Master's_degree  Doctoral_degree  Total
##           10.6%           13.5%           8.1%  8.5%
##           89.4%           86.5%          91.9% 91.5%
##
## Cramér's V:  0.1196368
```

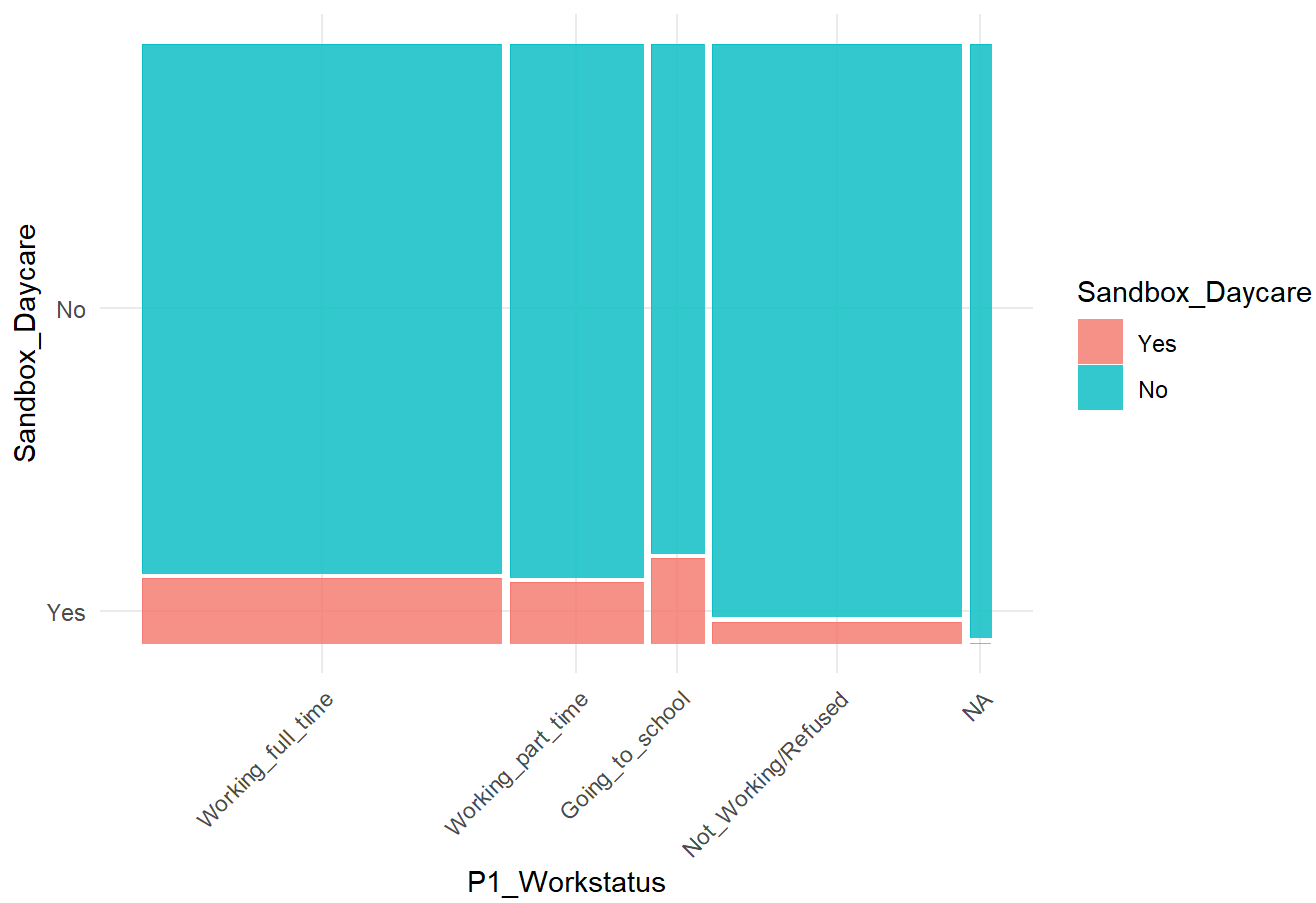
Mosaic Plot of P1_Degree and Sandbox_Daycare



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Daycare and P1_Workstatus -----
## Bivariate Analysis of Sandbox_Daycare and P1_Workstatus
##
## Frequency Table:
##   Sandbox_Daycare Working_full_time Working_part_time Going_to_school
##               Yes                26                9                5
##               No                 212               79               30
##   Not_Working/Refused NA_
##               6      0
##               159  14
##
## Relative Frequency Table:
##   Sandbox_Daycare Working_full_time Working_part_time Going_to_school
##               Yes                10.9%                10.2%                14.3%
##               No                 89.1%                 89.8%                 85.7%
##   Not_Working/Refused    NA_ Total
##               3.6%    0.0%  8.5%
##               96.4% 100.0% 91.5%
##
## Cramér's V:  0.1049273
```

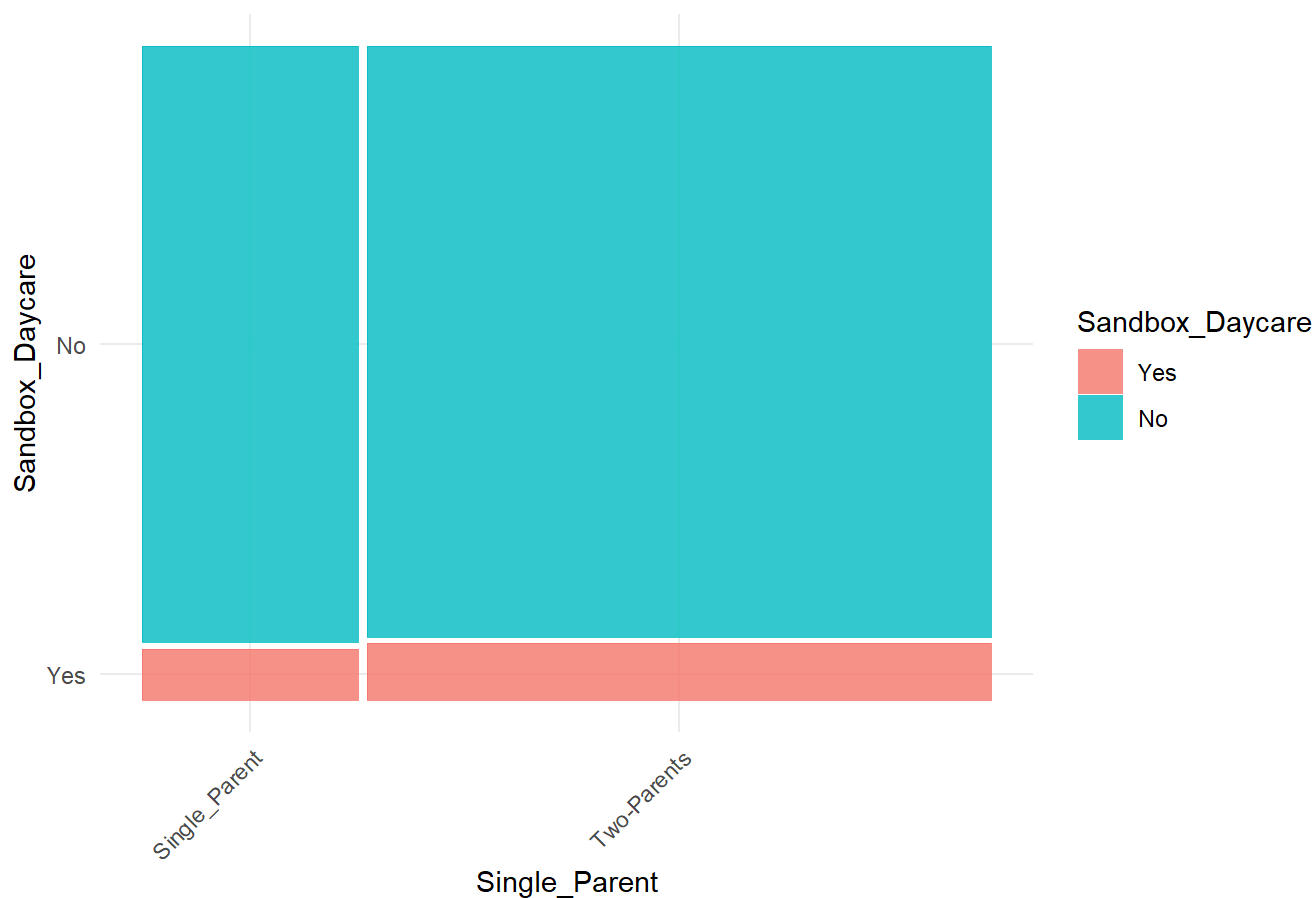
Mosaic Plot of P1_Workstatus and Sandbox_Daycare



```
## Saving 7 x 5 in image
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```
##
##
## ----- Analysis for Sandbox_Daycare and Single_Parent -----
## Bivariate Analysis of Sandbox_Daycare and Single_Parent
##
## Frequency Table:
##   Sandbox_Daycare Single_Parent Two-Parents
##           Yes           11           35
##           No            128          366
##
## Relative Frequency Table:
##   Sandbox_Daycare Single_Parent Two-Parents Total
##           Yes           7.9%           8.7%  8.5%
##           No            92.1%          91.3% 91.5%
##
## Cramér's V:  0
```

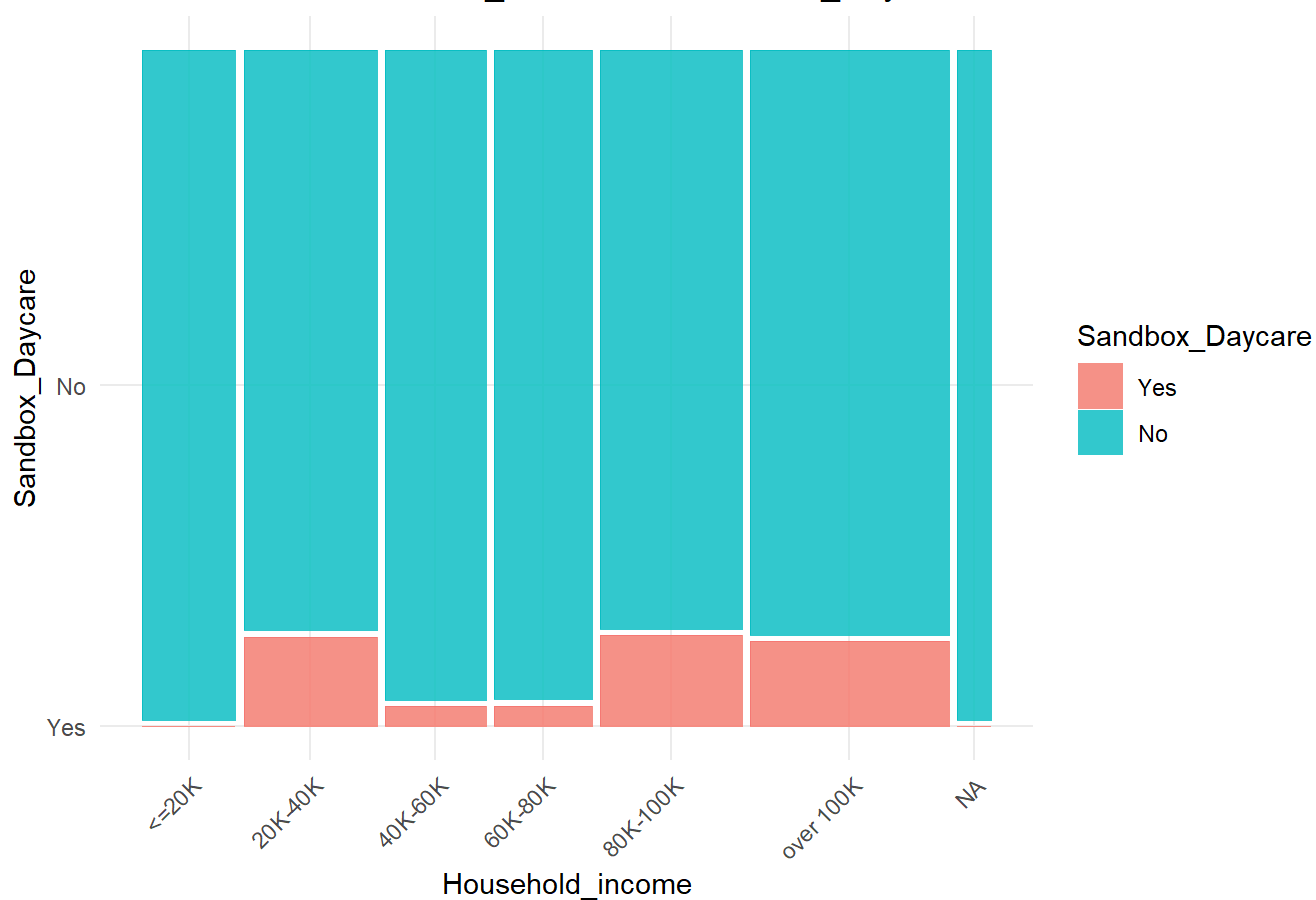
Mosaic Plot of Single_Parent and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and Household_income -----
## Bivariate Analysis of Sandbox_Daycare and Household_income
##
## Frequency Table:
##   Sandbox_Daycare  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_
##                   Yes      0       12       2       2       13       17      0
##                   No      63       78       66       64       83      117     23
##
## Relative Frequency Table:
##   Sandbox_Daycare  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_ Total
##                   Yes    0.0%   13.3%    2.9%    3.0%   13.5%   12.7%  0.0%  8.5%
##                   No  100.0%   86.7%   97.1%   97.0%   86.5%   87.3% 100.0% 91.5%
##
## Cramér's V:  0.1724821
```

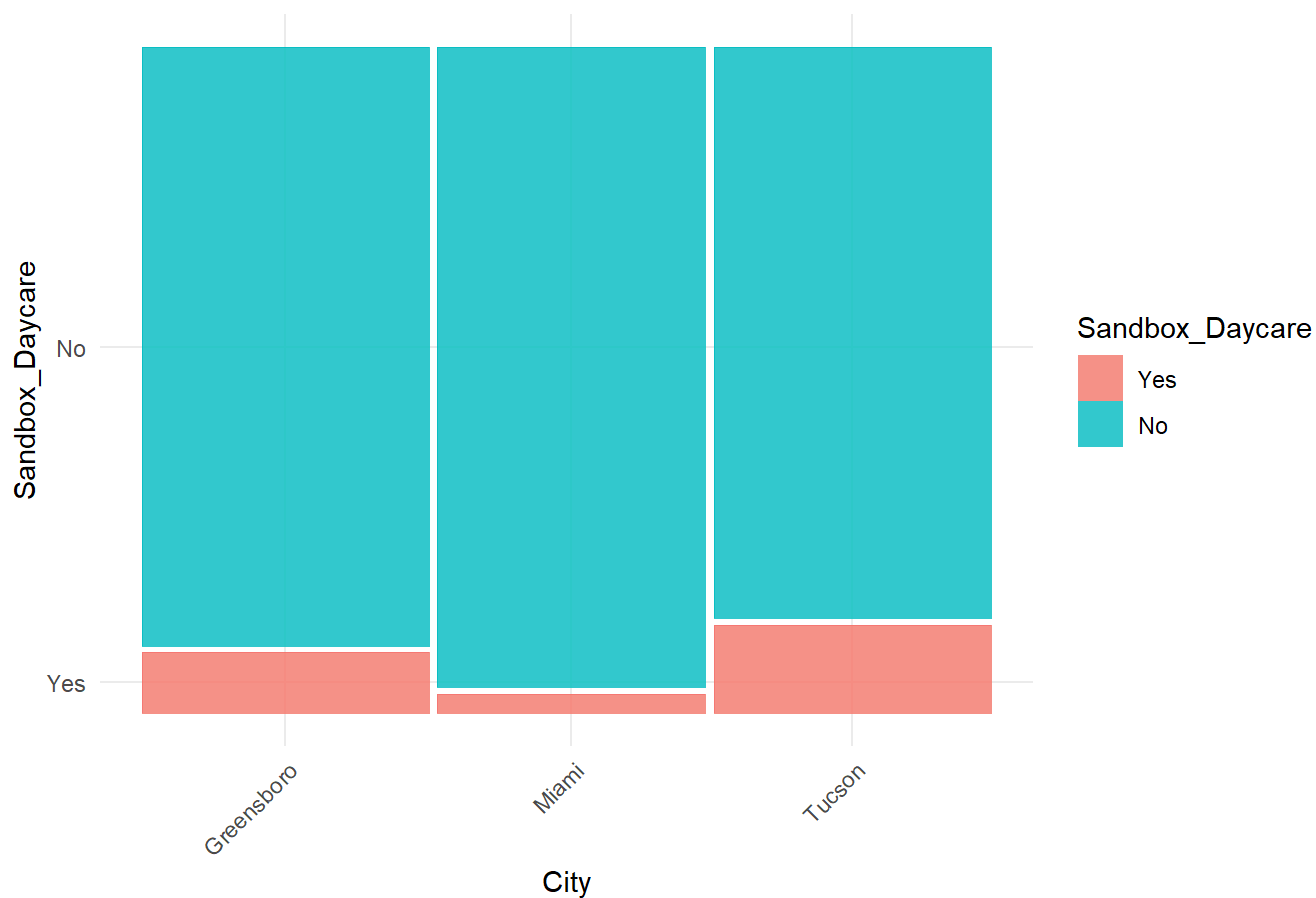
Mosaic Plot of Household_income and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and City -----
## Bivariate Analysis of Sandbox_Daycare and City
##
## Frequency Table:
##   Sandbox_Daycare Greensboro Miami Tucson
##           Yes         17      5      24
##           No        169    169    156
##
## Relative Frequency Table:
##   Sandbox_Daycare Greensboro Miami Tucson Total
##           Yes      9.1%  2.9%  13.3%  8.5%
##           No      90.9% 97.1%  86.7% 91.5%
##
## Cramér's V: 0.1399583
```

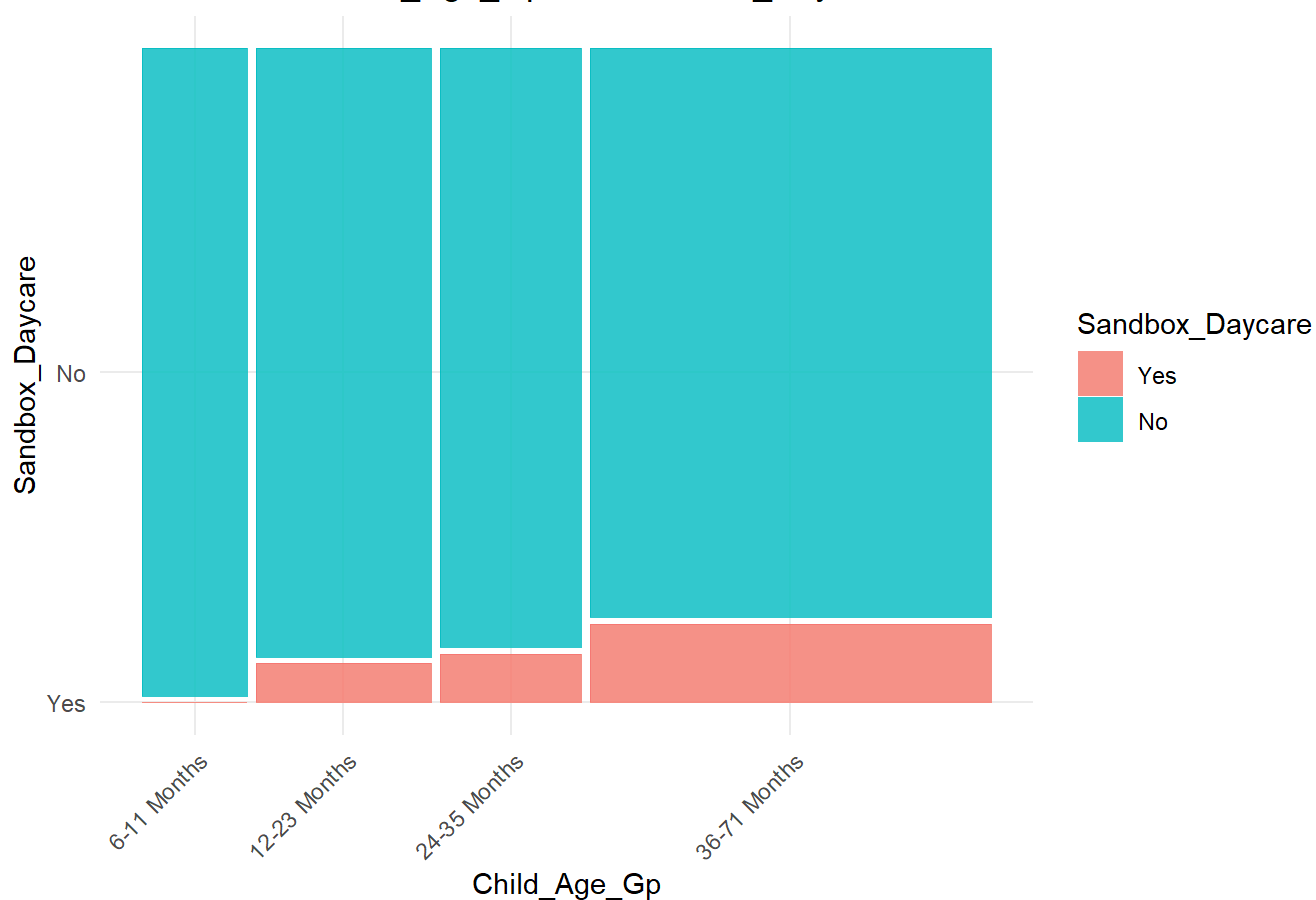

Mosaic Plot of City and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and Child_Age_Gp -----
## Bivariate Analysis of Sandbox_Daycare and Child_Age_Gp
##
## Frequency Table:
##   Sandbox_Daycare 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##               Yes           0           7           7           32
##               No          69          108          86          231
##
## Relative Frequency Table:
##   Sandbox_Daycare 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##               Yes           0.0%           6.1%           7.5%          12.2%  8.5%
##               No          100.0%          93.9%          92.5%          87.8% 91.5%
##
## Cramér's V:  0.1285181
```

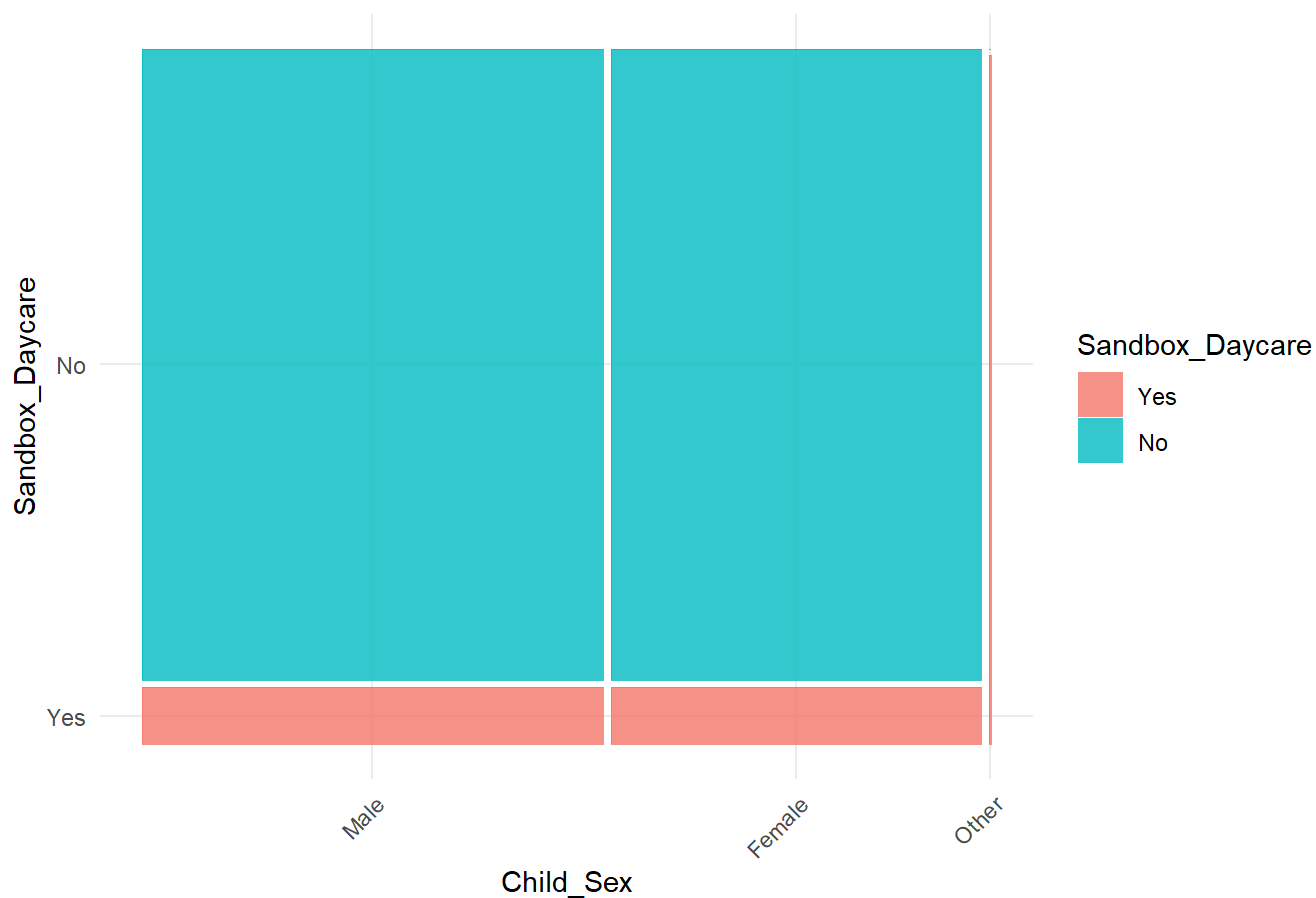
Mosaic Plot of Child_Age_Gp and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and Child_Sex -----
## Bivariate Analysis of Sandbox_Daycare and Child_Sex
##
## Frequency Table:
##   Sandbox_Daycare Male Female Other
##             Yes    25     20     1
##             No   274    220     0
##
## Relative Frequency Table:
##   Sandbox_Daycare Male Female Other Total
##             Yes  8.4%   8.3% 100.0%  8.5%
##             No  91.6%  91.7%   0.0%  91.5%
##
## Cramér's V:  0.1274519
```

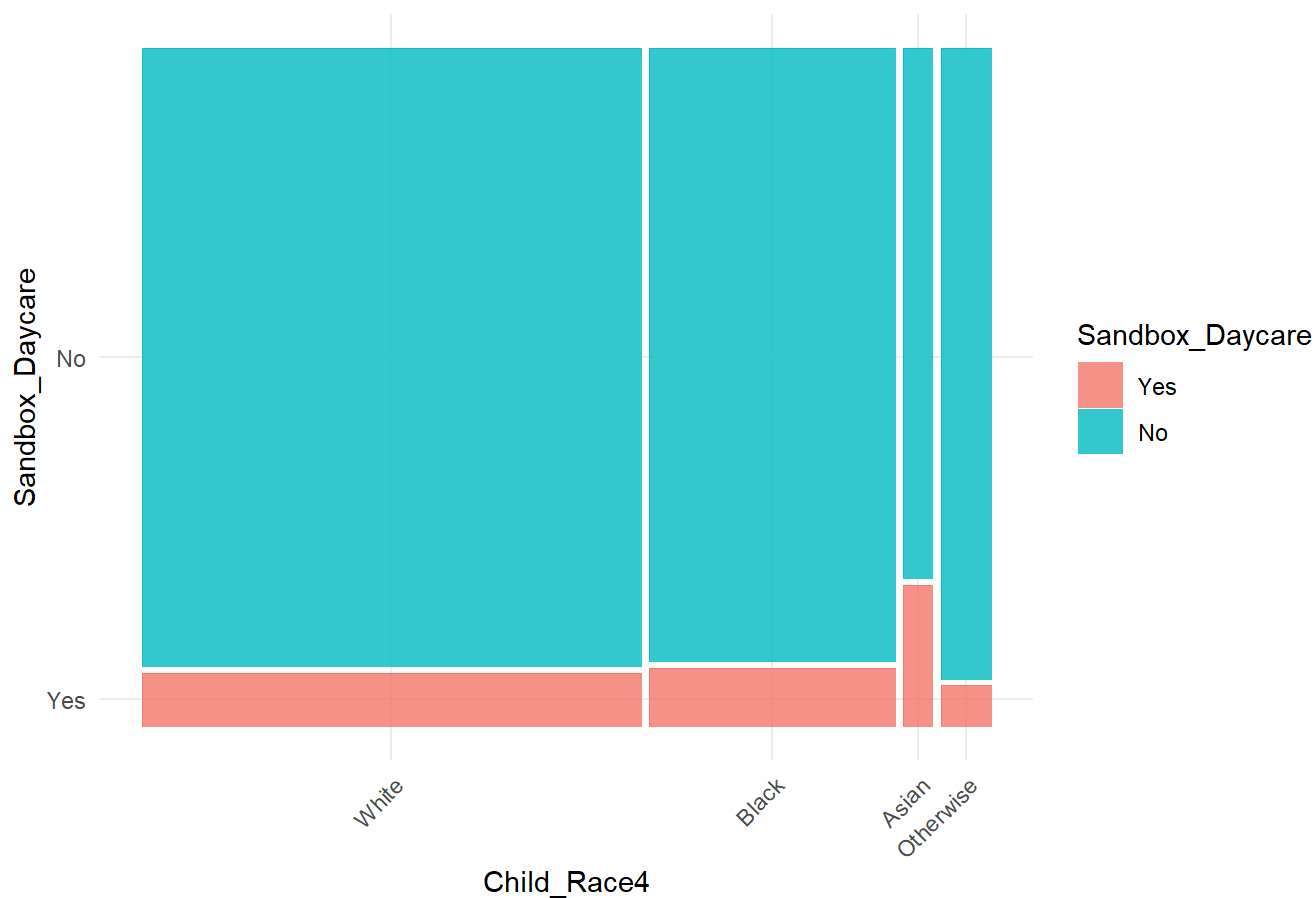
Mosaic Plot of Child_Sex and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Daycare and Child_Race4 -----
## Bivariate Analysis of Sandbox_Daycare and Child_Race4
##
## Frequency Table:
##   Sandbox_Daycare White Black Asian Otherwise
##           Yes      26    14     4         2
##           No     301   147    15        31
##
## Relative Frequency Table:
##   Sandbox_Daycare White Black Asian Otherwise Total
##           Yes   8.0%  8.7% 21.1%      6.1%  8.5%
##           No  92.0% 91.3% 78.9%     93.9% 91.5%
##
## Cramér's V:  0.04761977
```

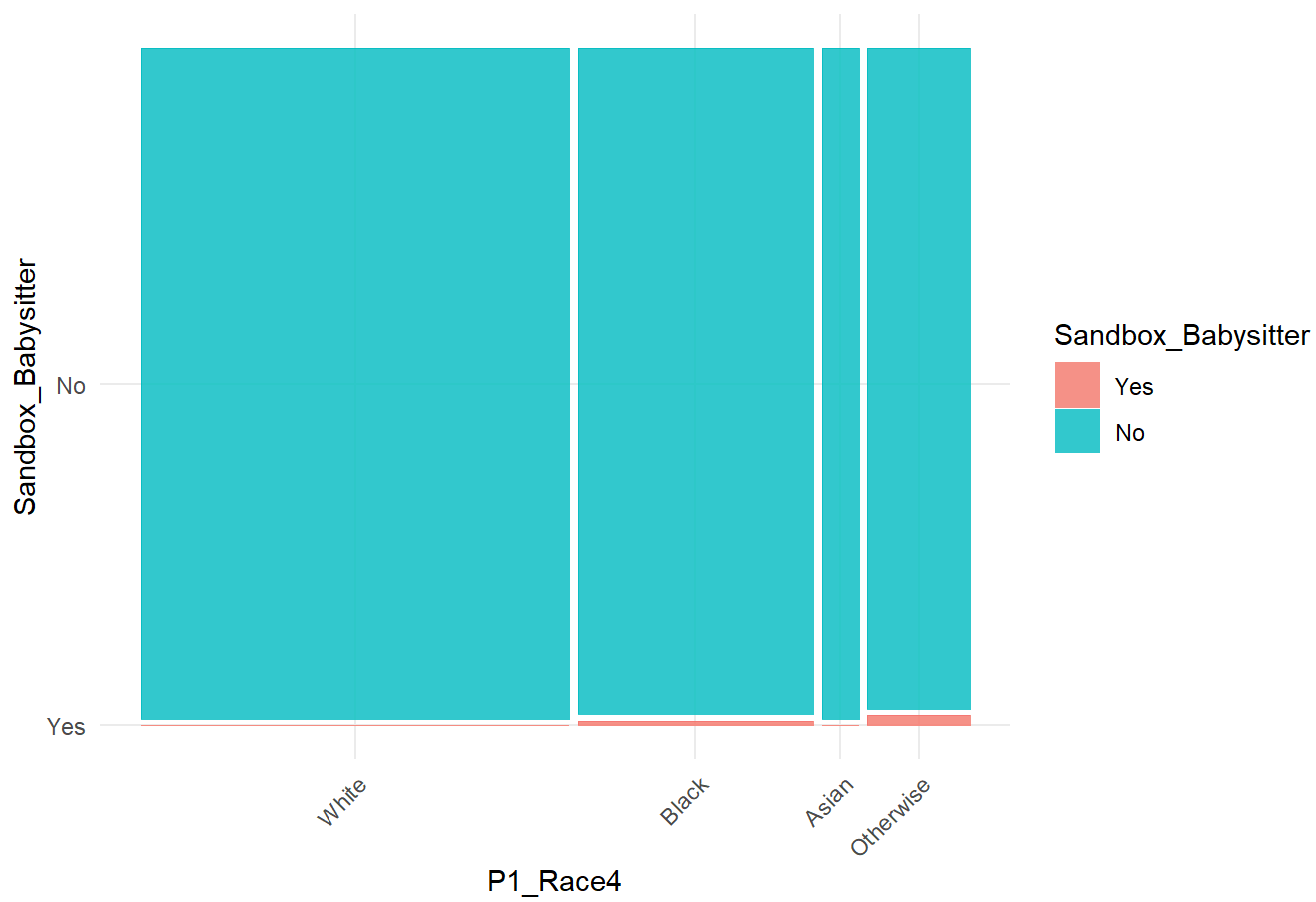
Mosaic Plot of Child_Race4 and Sandbox_Daycare



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and P1_Race4 -----
## Bivariate Analysis of Sandbox_Babysitter and P1_Race4
##
## Frequency Table:
##   Sandbox_Babysitter White Black Asian Otherwise
##           Yes        0      1      0          1
##           No       288    157    25         68
##
## Relative Frequency Table:
##   Sandbox_Babysitter  White Black  Asian Otherwise Total
##           Yes    0.0%  0.6%  0.0%    1.4%  0.4%
##           No  100.0% 99.4% 100.0%   98.6% 99.6%
##
##
## Cramér's V:  0.03418288
```

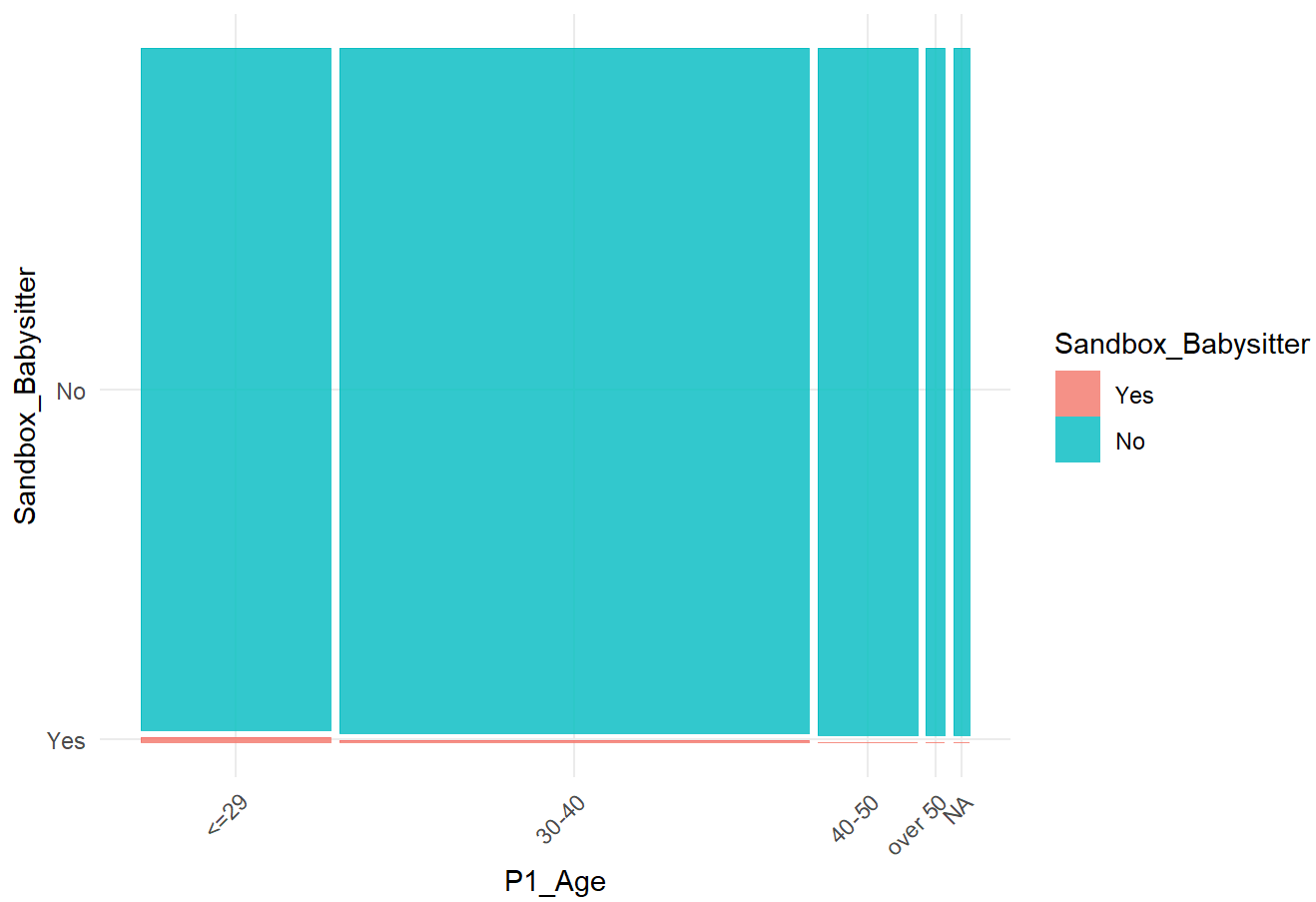
Mosaic Plot of P1_Race4 and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and P1_Age -----
## Bivariate Analysis of Sandbox_Babysitter and P1_Age
##
## Frequency Table:
##   Sandbox_Babysitter  <=29  30-40  40-50  over 50  NA_
##                   Yes      1      1      0      0      0
##                   No    128    318    68     13    11
##
## Relative Frequency Table:
##   Sandbox_Babysitter  <=29  30-40  40-50  over 50    NA_ Total
##                   Yes   0.8%  0.3%   0.0%   0.0%   0.0%  0.4%
##                   No  99.2% 99.7% 100.0% 100.0% 100.0% 99.6%
##
## Cramér's V:  0
```

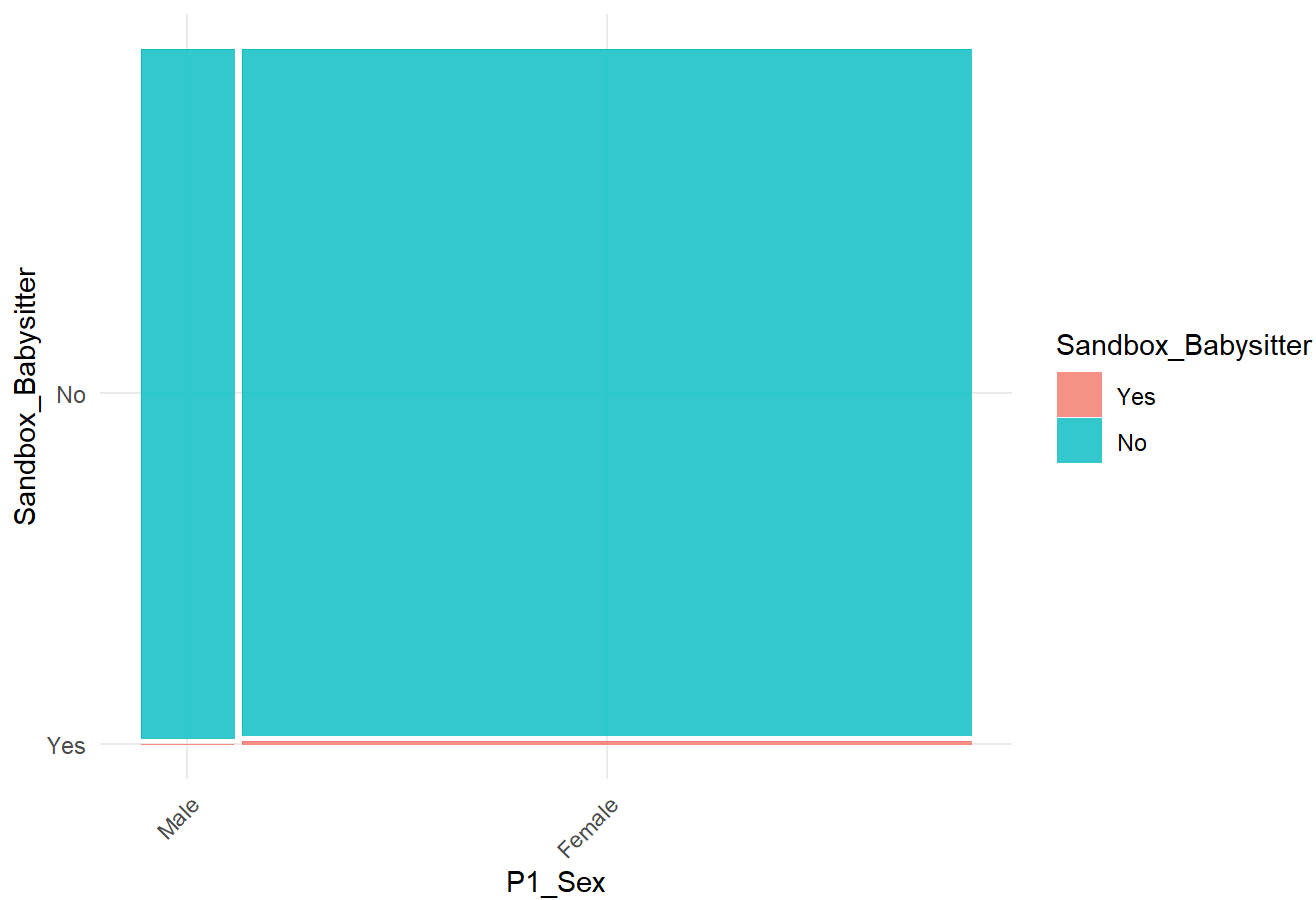
Mosaic Plot of P1_Age and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and P1_Sex -----
## Bivariate Analysis of Sandbox_Babysitter and P1_Sex
##
## Frequency Table:
##   Sandbox_Babysitter   Male Female
##           Yes         0         2
##           No        61       477
##
## Relative Frequency Table:
##   Sandbox_Babysitter   Male Female Total
##           Yes       0.0%   0.4%  0.4%
##           No      100.0%  99.6% 99.6%
##
##
## Cramér's V:  0
```

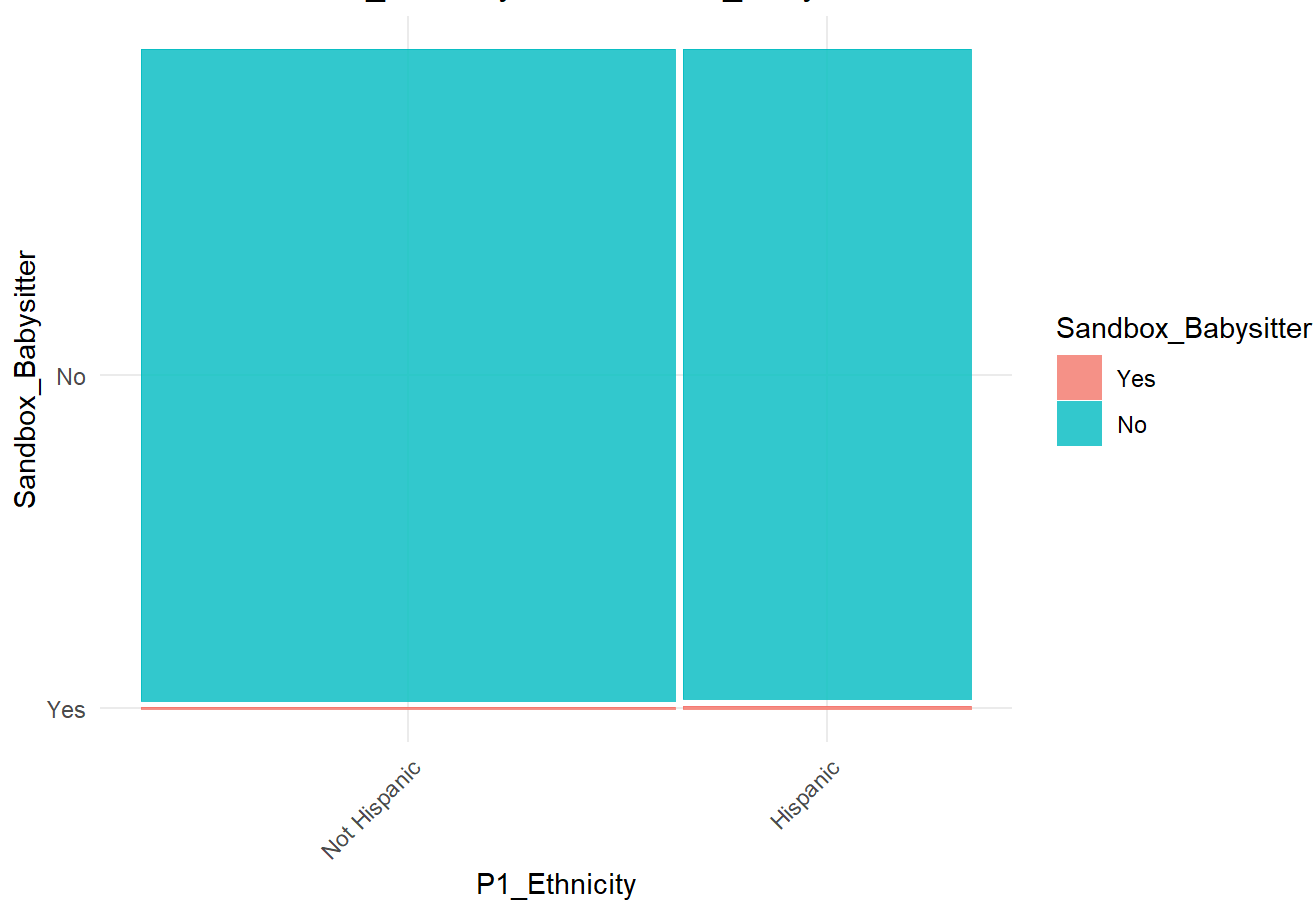
Mosaic Plot of P1_Sex and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and P1_Ethnicity -----
## Bivariate Analysis of Sandbox_Babysitter and P1_Ethnicity
##
## Frequency Table:
##   Sandbox_Babysitter Not Hispanic Hispanic
##               Yes           1           1
##               No          350          188
##
## Relative Frequency Table:
##   Sandbox_Babysitter Not Hispanic Hispanic Total
##               Yes           0.3%           0.5%  0.4%
##               No          99.7%          99.5% 99.6%
##
## Cramér's V:  0
```

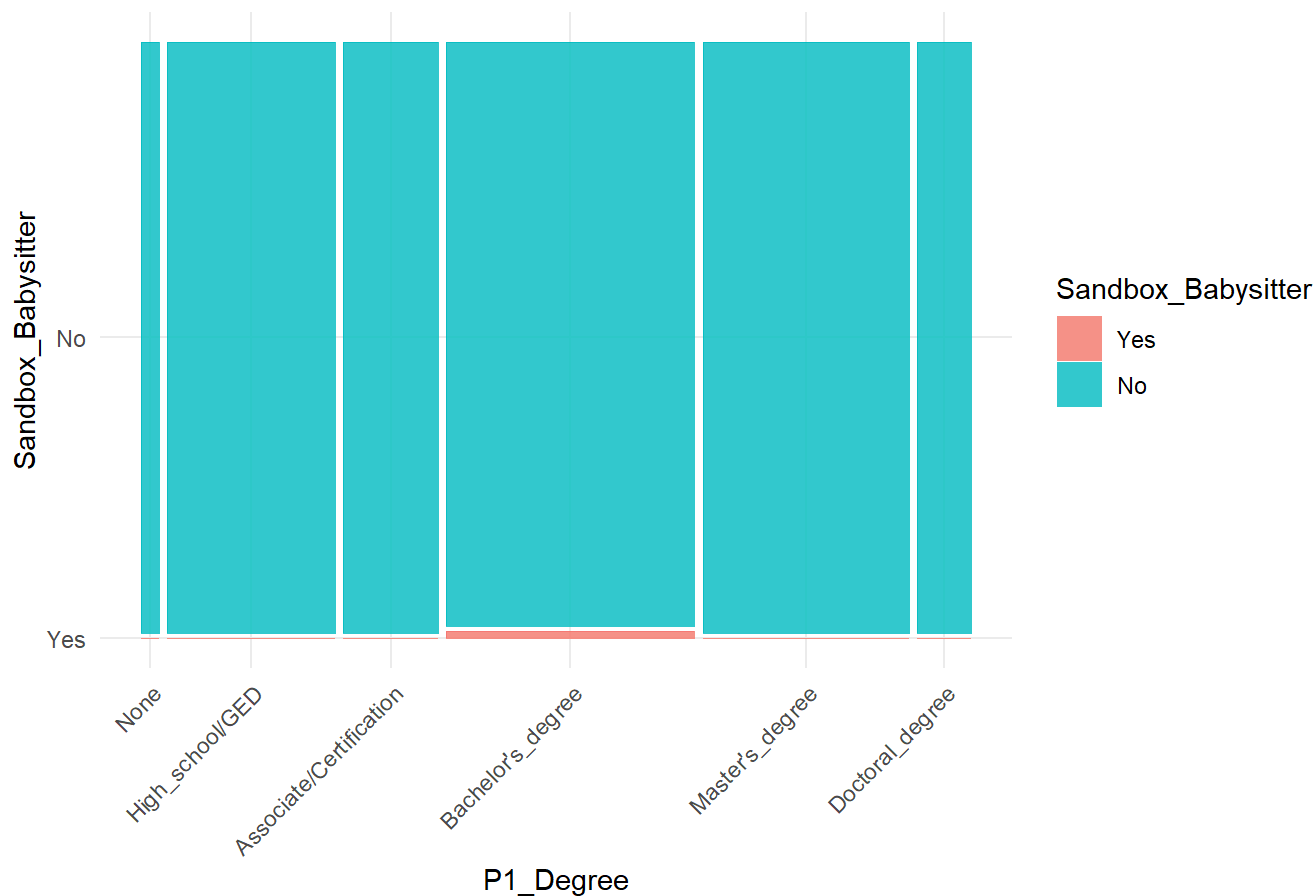
Mosaic Plot of P1_Ethnicity and Sandbox_Babysitter



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Babysitter and P1_Degree -----
## Bivariate Analysis of Sandbox_Babysitter and P1_Degree
##
## Frequency Table:
##   Sandbox_Babysitter None High_school/GED Associate/Certification
##               Yes      0              0              0
##               No      12             115             65
##   Bachelor's_degree Master's_degree Doctoral_degree
##               2              0              0
##             168             141             37
##
## Relative Frequency Table:
##   Sandbox_Babysitter   None High_school/GED Associate/Certification
##               Yes    0.0%              0.0%              0.0%
##               No  100.0%             100.0%             100.0%
##   Bachelor's_degree Master's_degree Doctoral_degree Total
##               1.2%              0.0%              0.0%  0.4%
##               98.8%             100.0%             100.0% 99.6%
##
## Cramér's V:  0
```

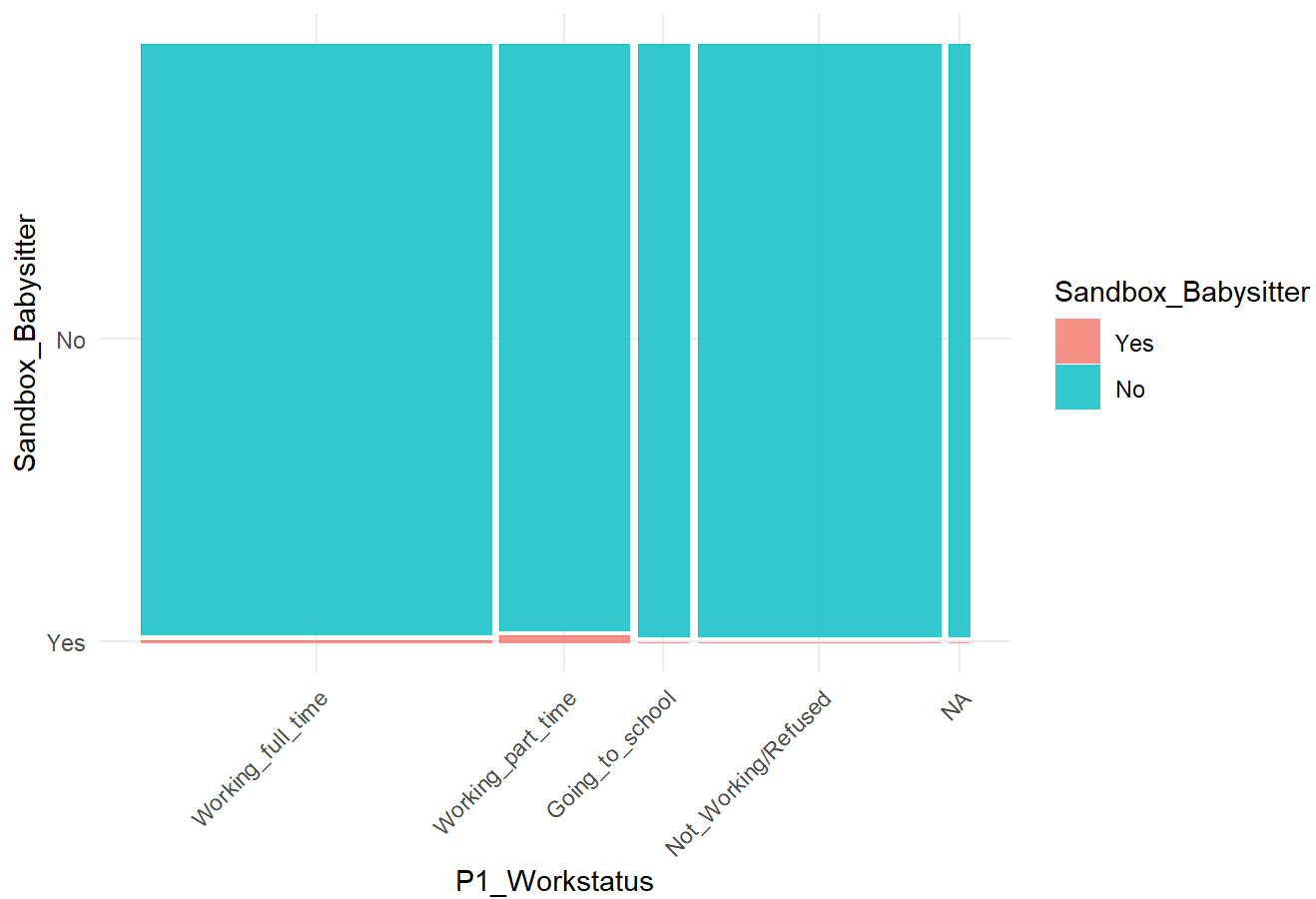

Mosaic Plot of P1_Degree and Sandbox_Babysitter



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Babysitter and P1_Workstatus -----
## Bivariate Analysis of Sandbox_Babysitter and P1_Workstatus
##
## Frequency Table:
##   Sandbox_Babysitter Working_full_time Working_part_time Going_to_school
##               Yes                1                1                0
##               No               237               87               35
##   Not_Working/Refused NA_
##               0    0
##               165  14
##
## Relative Frequency Table:
##   Sandbox_Babysitter Working_full_time Working_part_time Going_to_school
##               Yes                0.4%                1.1%                0.0%
##               No               99.6%               98.9%               100.0%
##   Not_Working/Refused   NA_ Total
##               0.0%    0.0%  0.4%
##               100.0% 100.0% 99.6%
##
## Cramér's V:  0
```

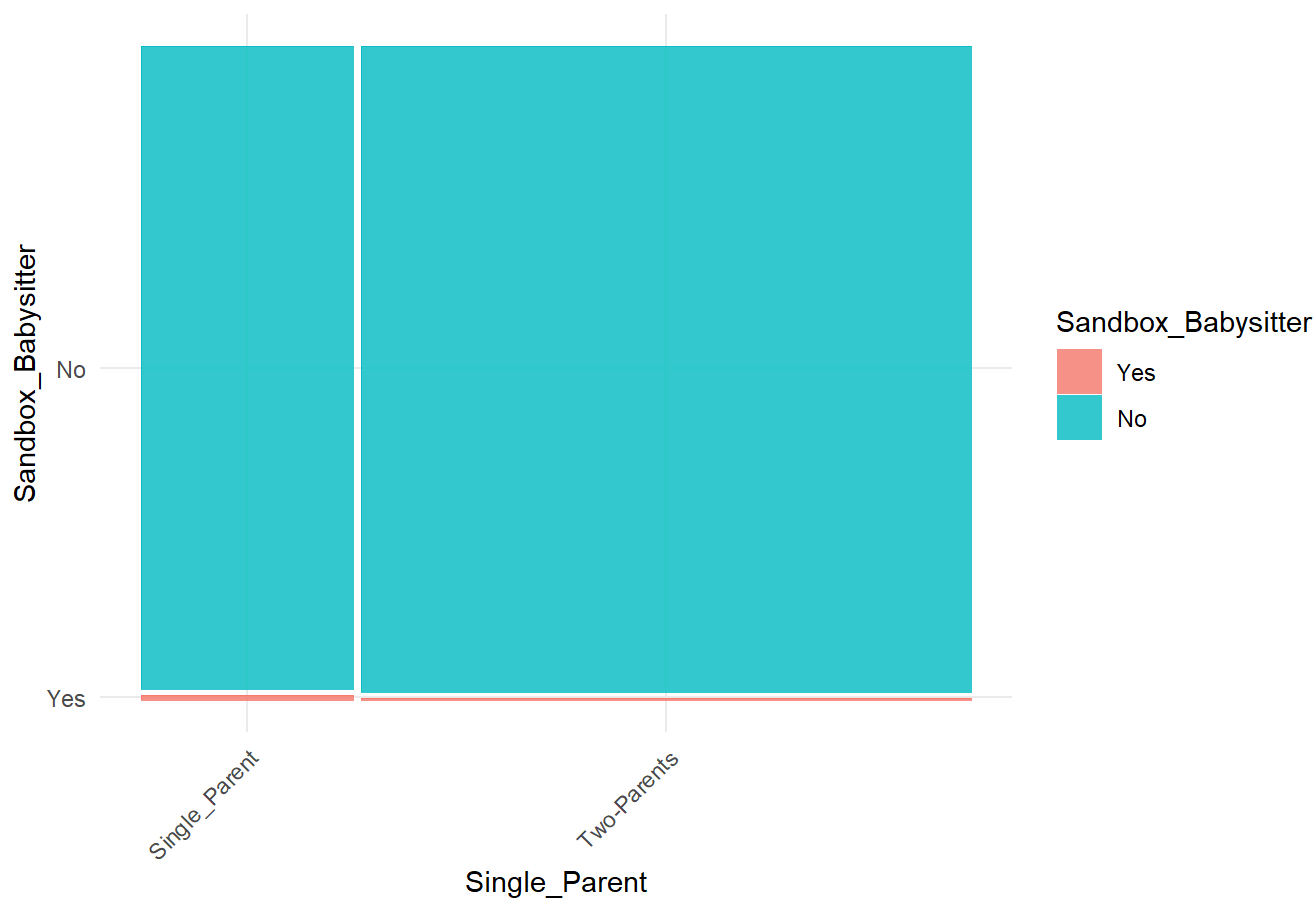
Mosaic Plot of P1_Workstatus and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and Single_Parent -----
## Bivariate Analysis of Sandbox_Babysitter and Single_Parent
##
## Frequency Table:
##   Sandbox_Babysitter Single_Parent Two-Parents
##           Yes           1             1
##           No          138          400
##
## Relative Frequency Table:
##   Sandbox_Babysitter Single_Parent Two-Parents Total
##           Yes           0.7%          0.2%  0.4%
##           No          99.3%          99.8% 99.6%
##
## Cramér's V:  0
```

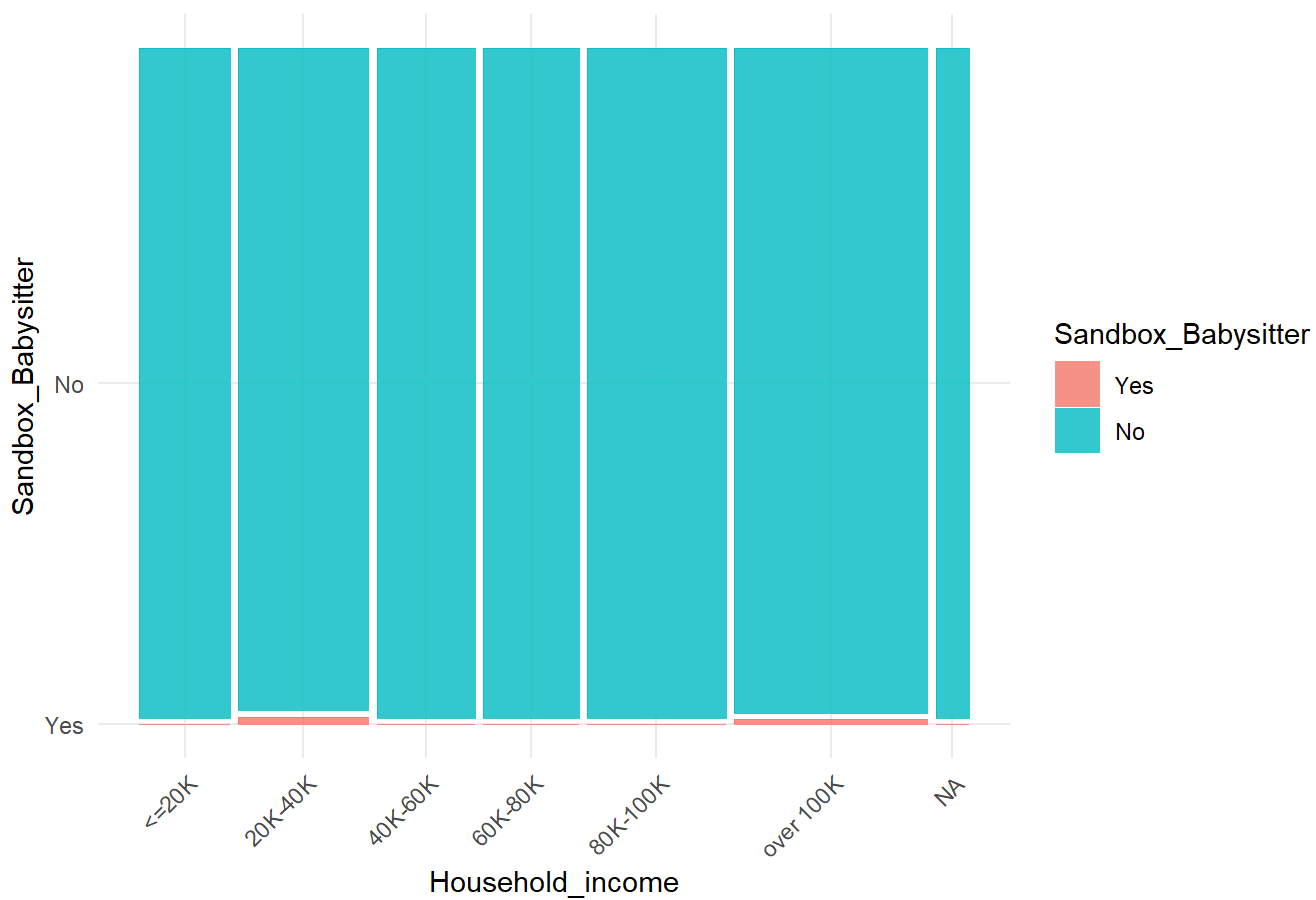
Mosaic Plot of Single_Parent and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and Household_income -----
## Bivariate Analysis of Sandbox_Babysitter and Household_income
##
## Frequency Table:
##   Sandbox_Babysitter  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_
##                   Yes      0      1      0      0      0      1      0
##                   No     63     89     68     66     96    133    23
##
## Relative Frequency Table:
##   Sandbox_Babysitter  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_
##                   Yes    0.0%    1.1%    0.0%    0.0%    0.0%    0.7%    0.0%
##                   No  100.0%   98.9%  100.0%  100.0%  100.0%   99.3%  100.0%
##
## Total
##   0.4%
##  99.6%
##
## Cramér's V:  0
```

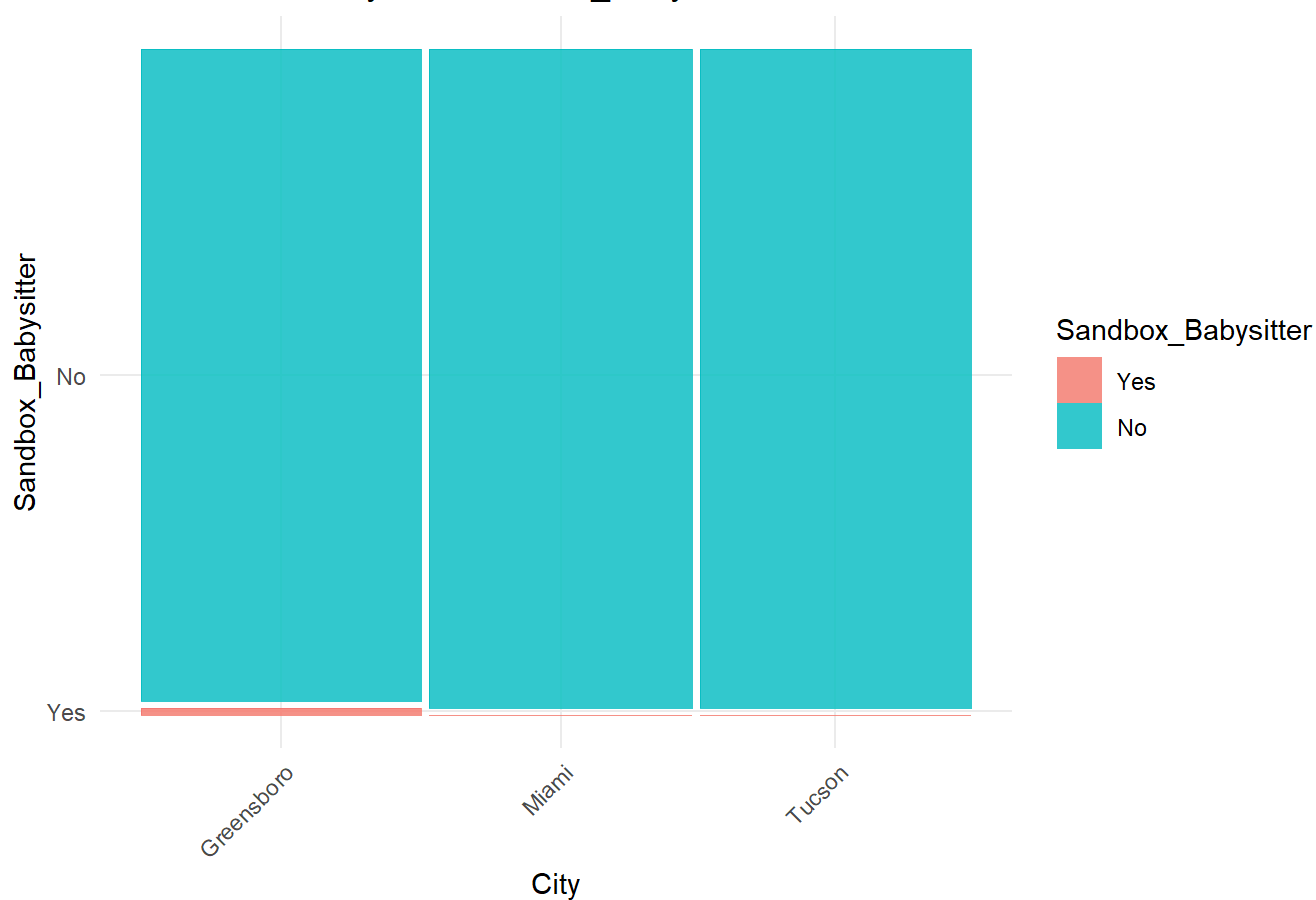
Mosaic Plot of Household_income and Sandbox_Babysitter



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Babysitter and City -----
## Bivariate Analysis of Sandbox_Babysitter and City
##
## Frequency Table:
##   Sandbox_Babysitter Greensboro Miami Tucson
##                   Yes           2      0      0
##                   No          184   174   180
##
## Relative Frequency Table:
##   Sandbox_Babysitter Greensboro  Miami Tucson Total
##                   Yes         1.1%   0.0%   0.0%  0.4%
##                   No         98.9% 100.0% 100.0% 99.6%
##
## Cramér's V:  0.05805917
```

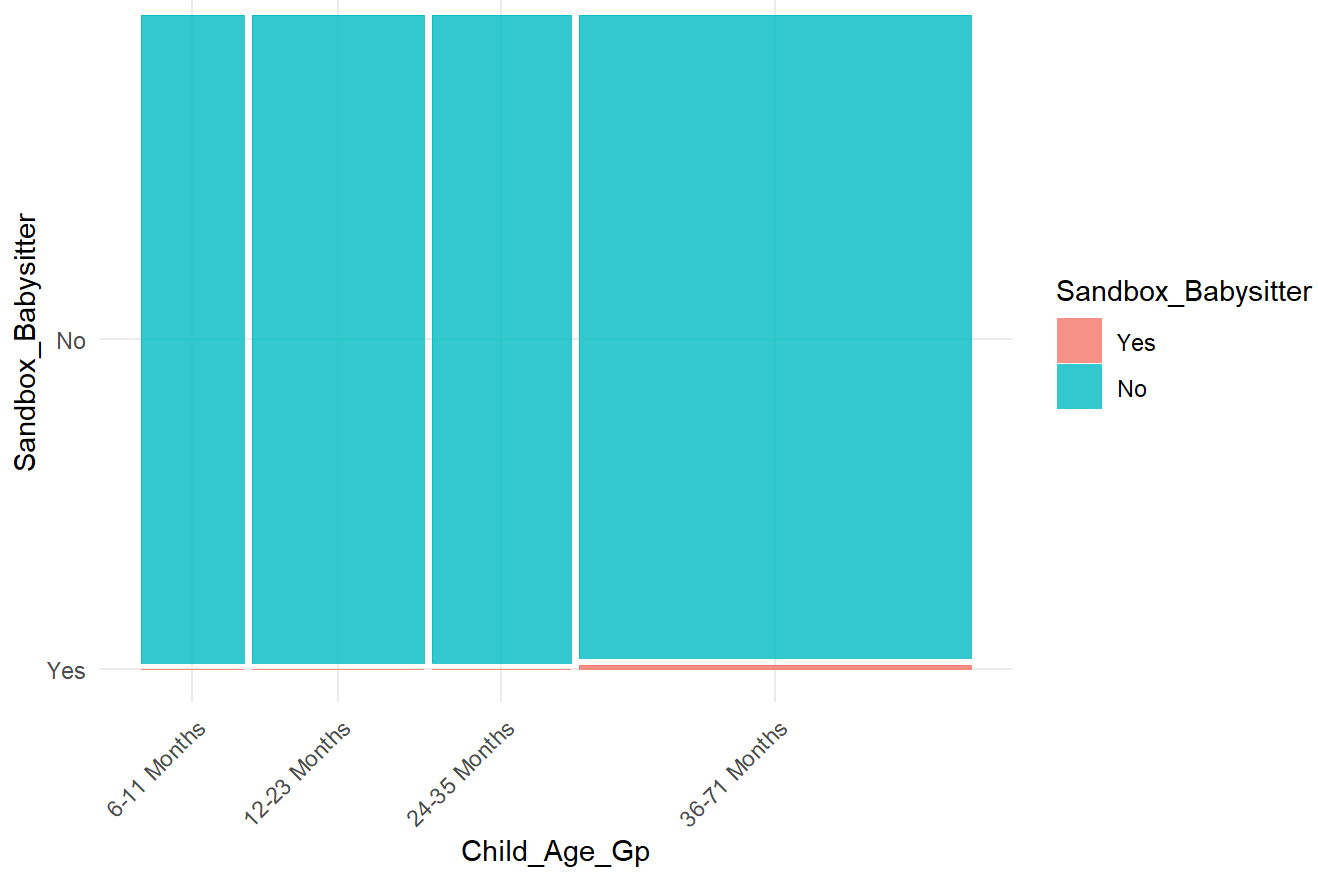
Mosaic Plot of City and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and Child_Age_Gp -----
## Bivariate Analysis of Sandbox_Babysitter and Child_Age_Gp
##
## Frequency Table:
##   Sandbox_Babysitter 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##                   Yes           0           0           0           2
##                   No           69          115          93          261
##
## Relative Frequency Table:
##   Sandbox_Babysitter 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##                   Yes           0.0%          0.0%          0.0%          0.8%  0.4%
##                   No          100.0%         100.0%         100.0%         99.2% 99.6%
##
## Cramér's V:  0
```

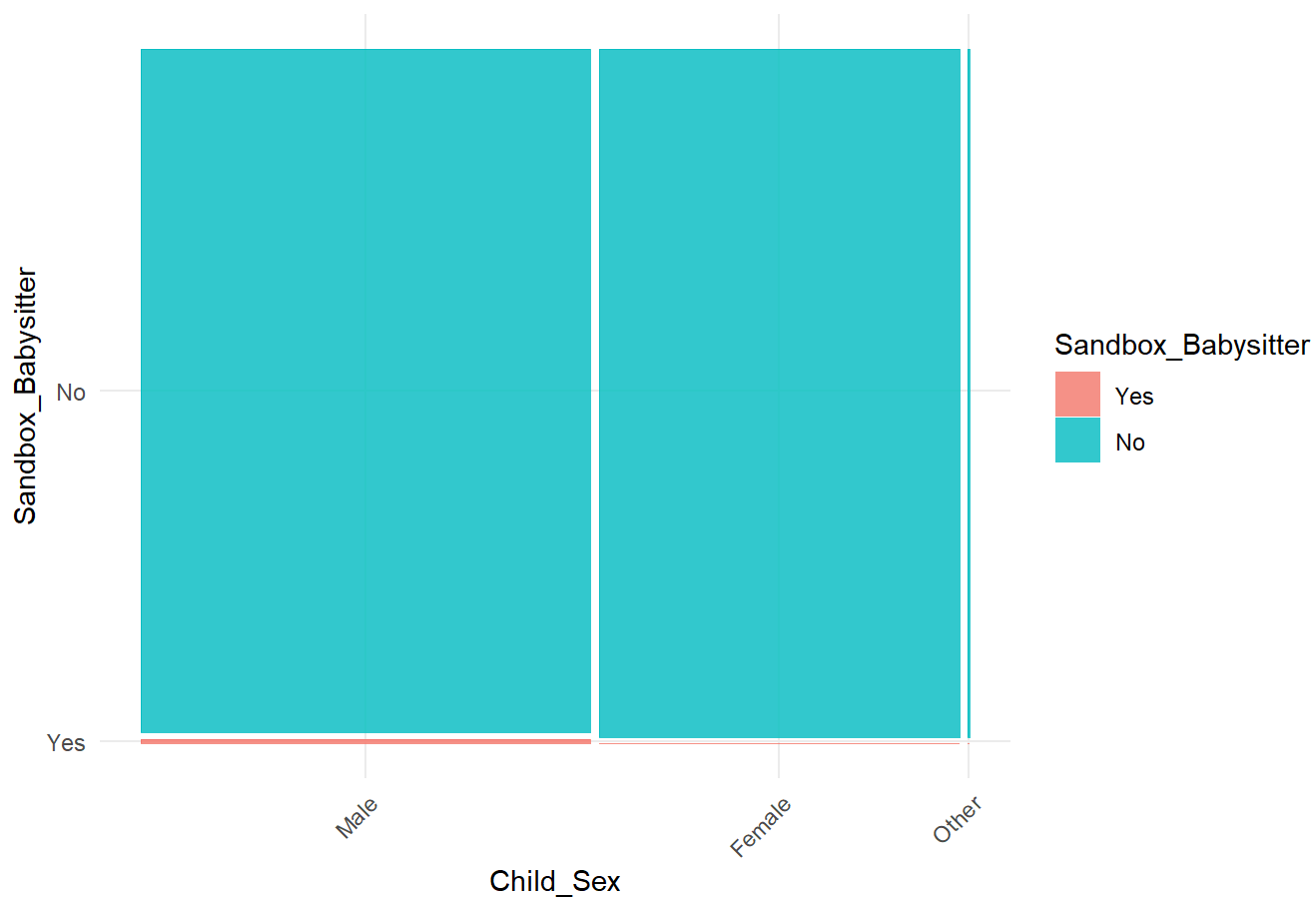
Mosaic Plot of Child_Age_Gp and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Babysitter and Child_Sex -----
## Bivariate Analysis of Sandbox_Babysitter and Child_Sex
##
## Frequency Table:
##   Sandbox_Babysitter Male Female Other
##               Yes      2        0      0
##               No    297    240      1
##
## Relative Frequency Table:
##   Sandbox_Babysitter  Male Female  Other Total
##               Yes    0.7%   0.0%   0.0%  0.4%
##               No   99.3% 100.0% 100.0% 99.6%
##
## Cramér's V:  0
```

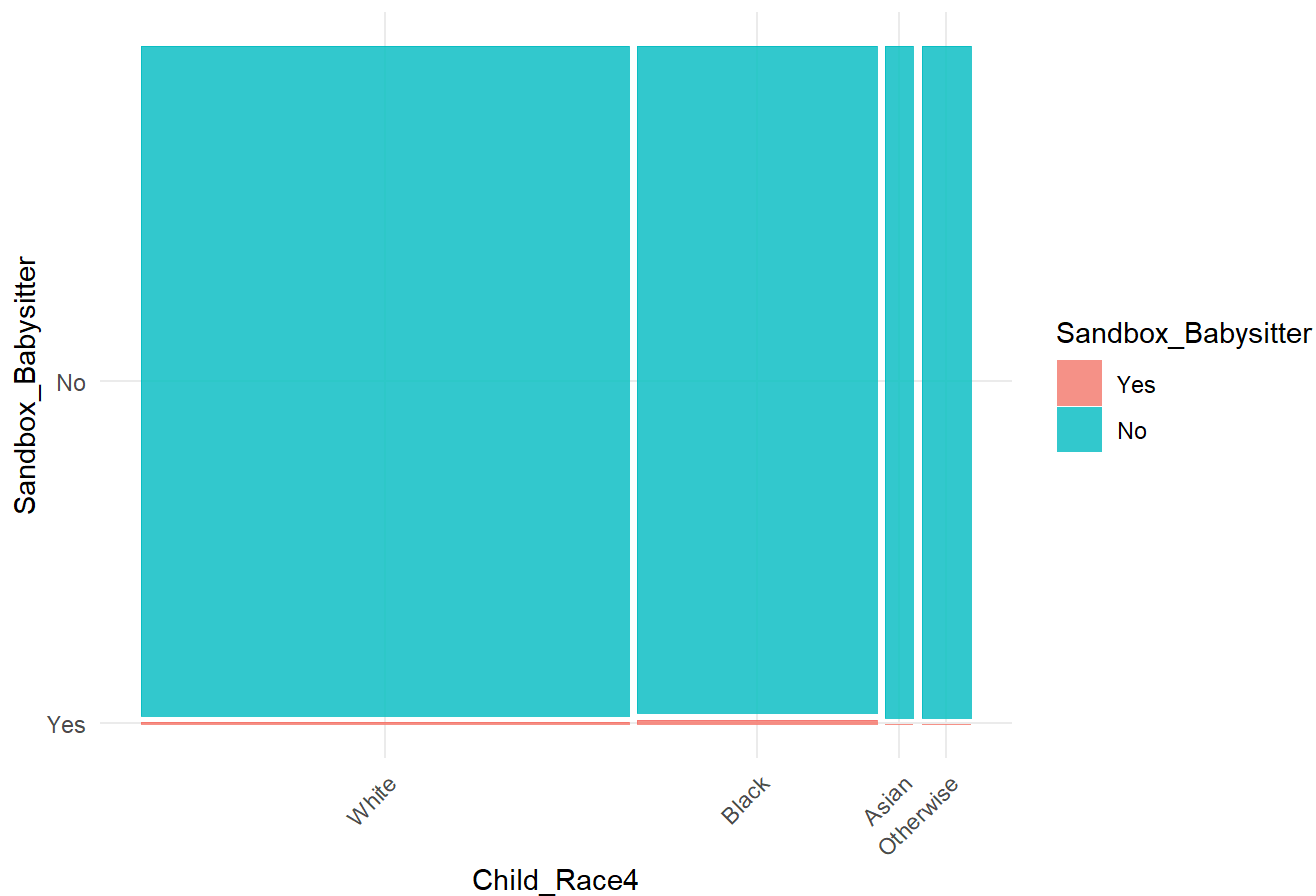
Mosaic Plot of Child_Sex and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

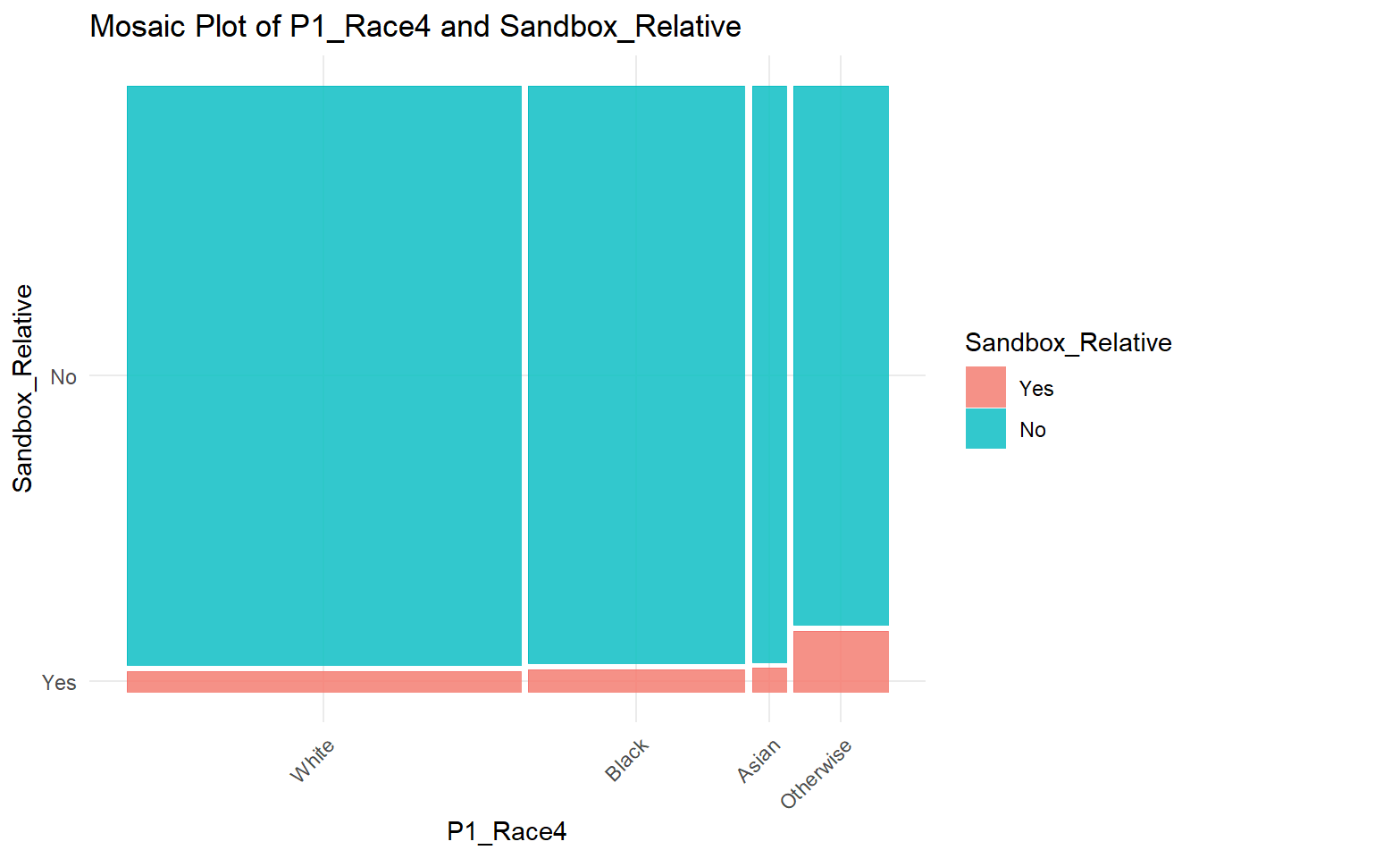
```
##
##
## ----- Analysis for Sandbox_Babysitter and Child_Race4 -----
## Bivariate Analysis of Sandbox_Babysitter and Child_Race4
##
## Frequency Table:
##   Sandbox_Babysitter White Black Asian Otherwise
##               Yes      1      1      0          0
##               No    326   160    19         33
##
## Relative Frequency Table:
##   Sandbox_Babysitter White Black  Asian Otherwise Total
##               Yes   0.3%  0.6%   0.0%      0.0%  0.4%
##               No  99.7% 99.4% 100.0%    100.0% 99.6%
##
## Cramér's V:  0
```

Mosaic Plot of Child_Race4 and Sandbox_Babysitter



```
## Saving 7 x 5 in image
```

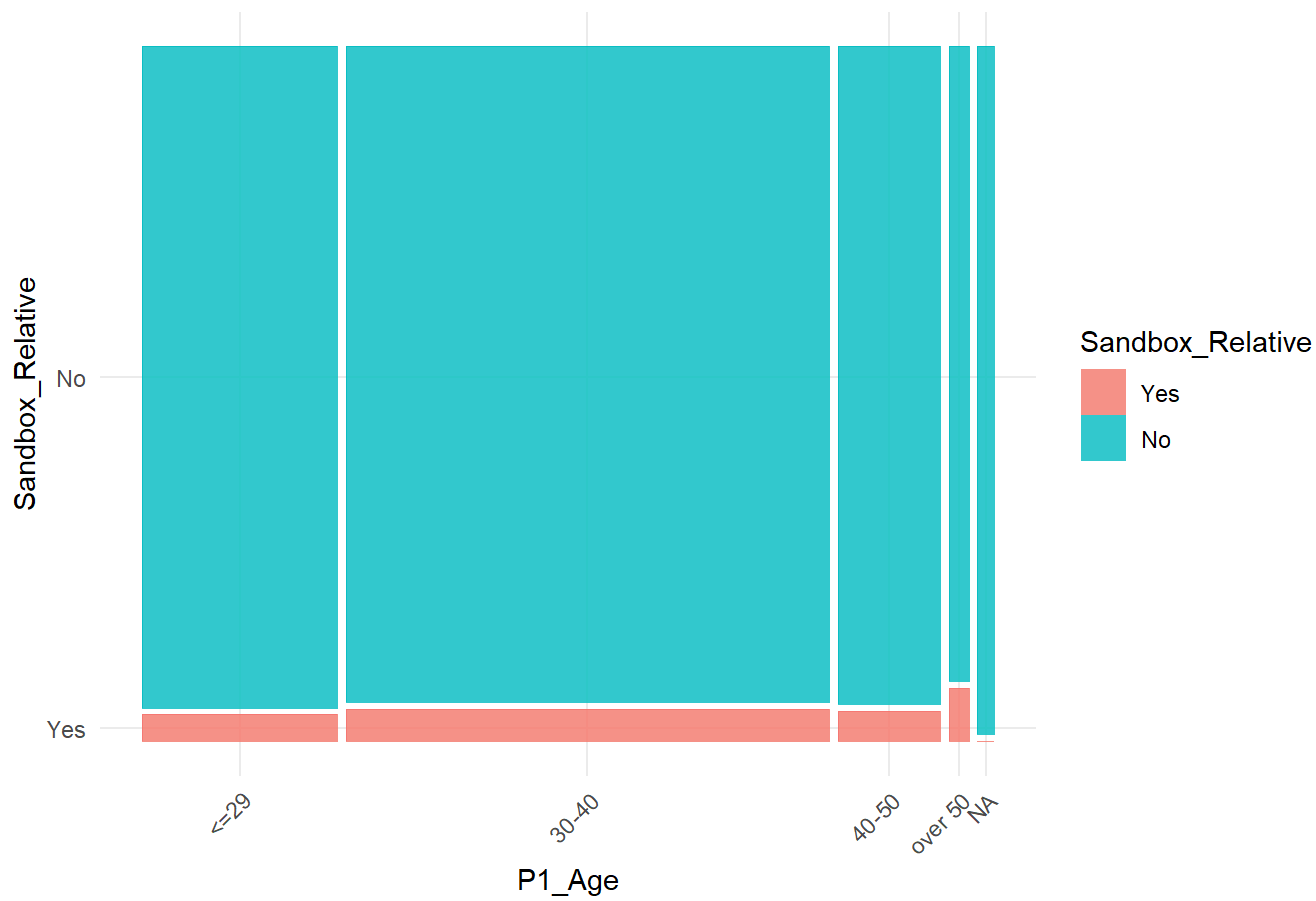
```
##
##
## ----- Analysis for Sandbox_Relative and P1_Race4 -----
## Bivariate Analysis of Sandbox_Relative and P1_Race4
##
## Frequency Table:
##   Sandbox_Relative White Black Asian Otherwise
##           Yes      10      6      1          7
##           No     278    152     24         62
##
## Relative Frequency Table:
##   Sandbox_Relative White Black Asian Otherwise Total
##           Yes   3.5%   3.8%  4.0%      10.1%  4.4%
##           No  96.5%  96.2% 96.0%      89.9% 95.6%
##
## Cramér's V:  0.07562187
```

```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Relative and P1_Age -----
## Bivariate Analysis of Sandbox_Relative and P1_Age
##
## Frequency Table:
##   Sandbox_Relative  <=29  30-40  40-50  over 50  NA_
##                   Yes    5    15    3      1    0
##                   No   124   304   65     12   11
##
## Relative Frequency Table:
##   Sandbox_Relative  <=29  30-40  40-50  over 50    NA_ Total
##                   Yes  3.9%  4.7%  4.4%   7.7%   0.0%  4.4%
##                   No 96.1% 95.3% 95.6%  92.3% 100.0% 95.6%
##
## Cramér's V:  0
```

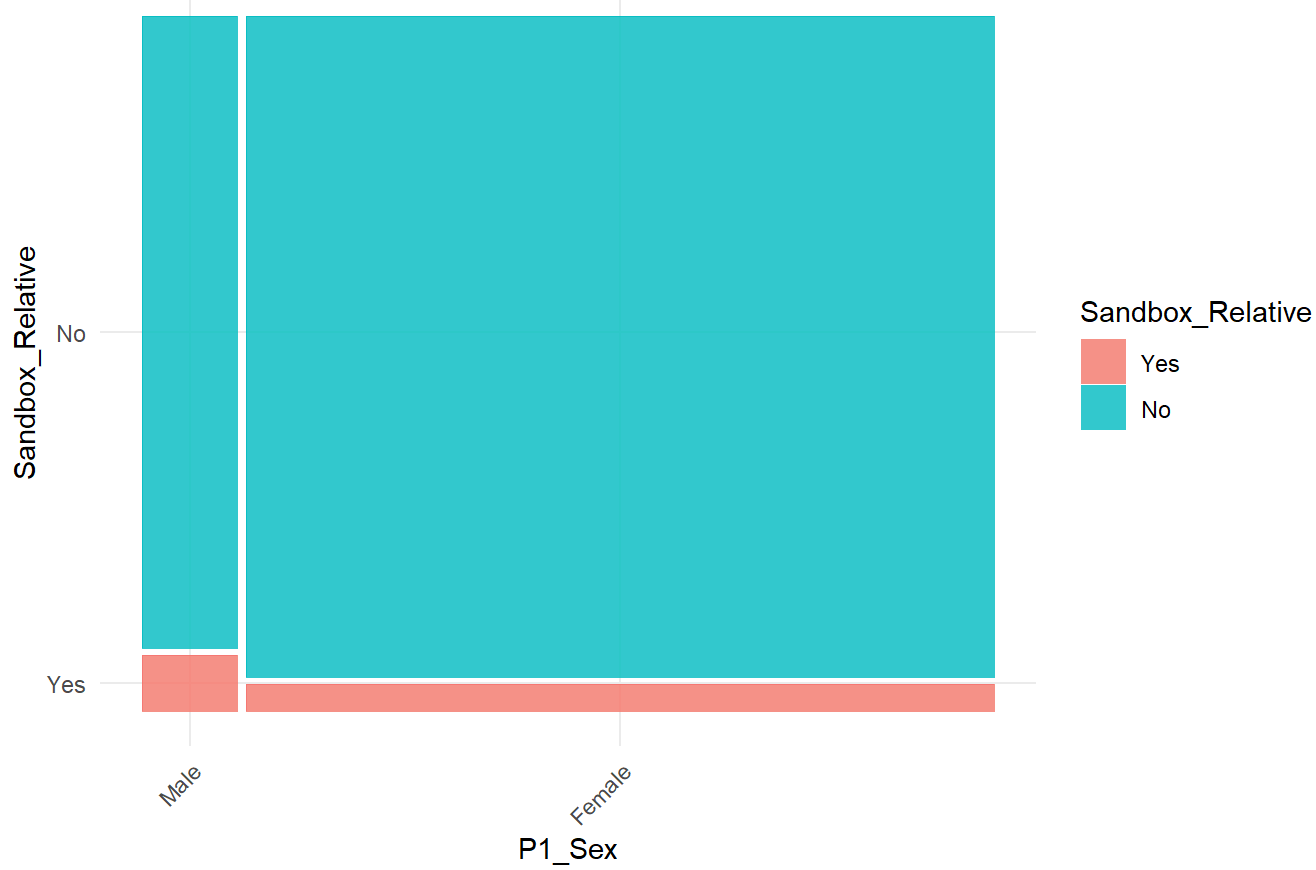
Mosaic Plot of P1_Age and Sandbox_Relative



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Relative and P1_Sex -----
## Bivariate Analysis of Sandbox_Relative and P1_Sex
##
## Frequency Table:
##   Sandbox_Relative   Male   Female
##           Yes         5       19
##           No        56      460
##
## Relative Frequency Table:
##   Sandbox_Relative   Male   Female   Total
##           Yes      8.2%    4.0%    4.4%
##           No     91.8%    96.0%   95.6%
##
## Cramér's V:  0.0486933
```

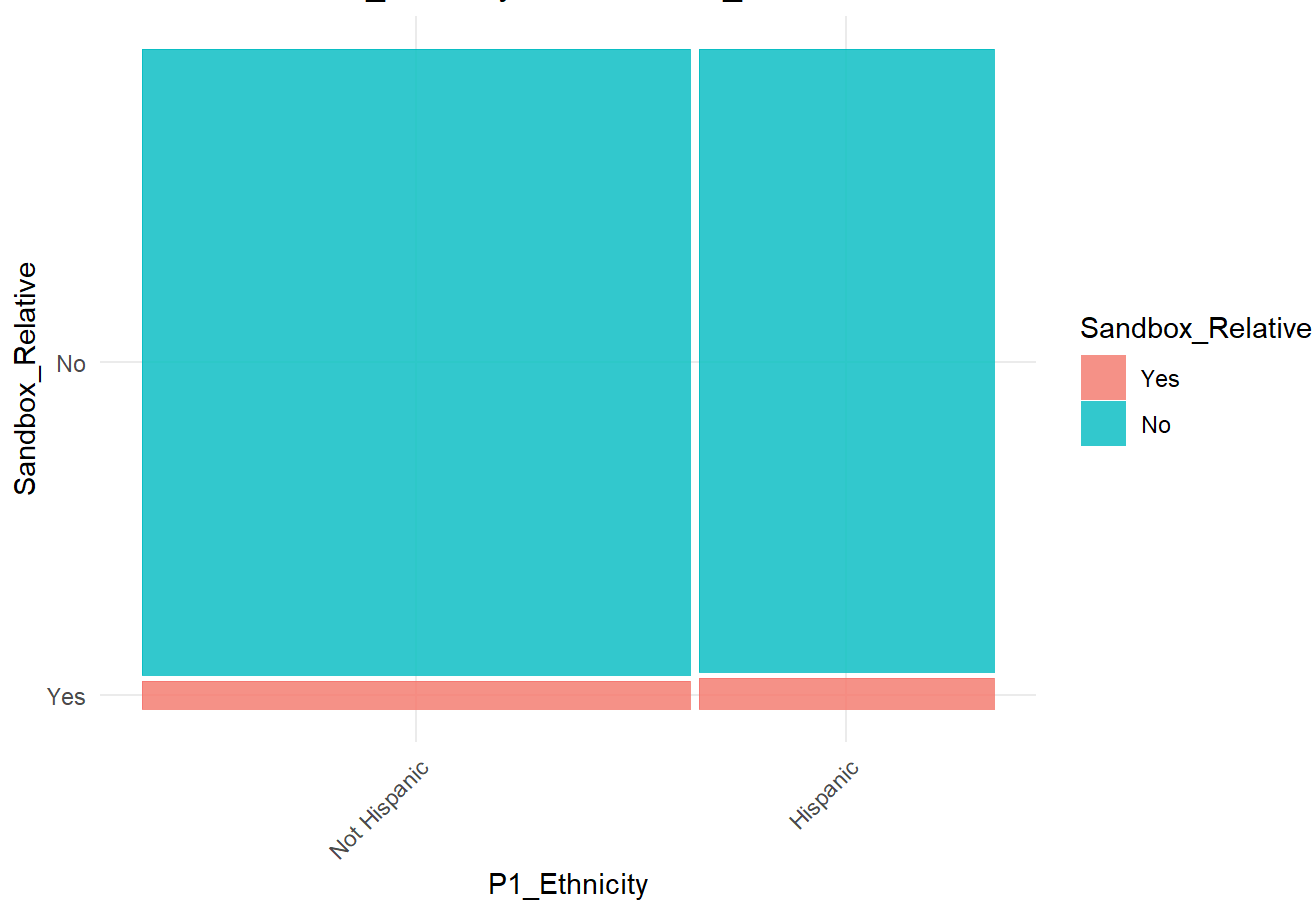
Mosaic Plot of P1_Sex and Sandbox_Relative



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Relative and P1_Ethnicity -----
## Bivariate Analysis of Sandbox_Relative and P1_Ethnicity
##
## Frequency Table:
##   Sandbox_Relative Not Hispanic Hispanic
##               Yes           15           9
##               No          336          180
##
## Relative Frequency Table:
##   Sandbox_Relative Not Hispanic Hispanic Total
##               Yes           4.3%          4.8%  4.4%
##               No          95.7%          95.2% 95.6%
##
## Cramér's V:  0
```

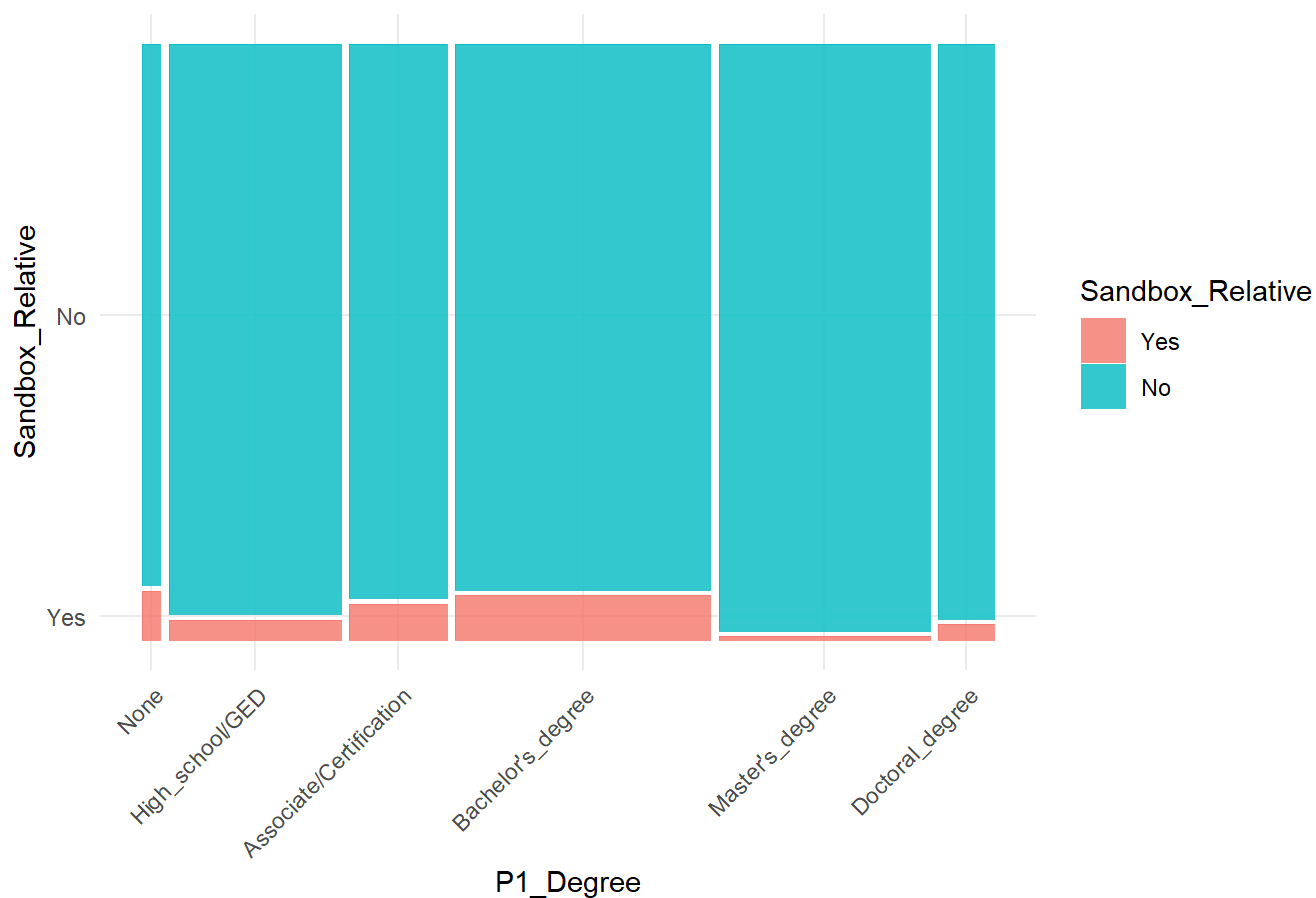
Mosaic Plot of P1_Ethnicity and Sandbox_Relative



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Relative and P1_Degree -----
## Bivariate Analysis of Sandbox_Relative and P1_Degree
##
## Frequency Table:
##   Sandbox_Relative  None  High_school/GED  Associate/Certification
##           Yes      1           4           4
##           No     11          111          61
##   Bachelor's_degree  Master's_degree  Doctoral_degree
##           13           1           1
##           157          140          36
##
## Relative Frequency Table:
##   Sandbox_Relative  None  High_school/GED  Associate/Certification
##           Yes   8.3%       3.5%           6.2%
##           No  91.7%       96.5%          93.8%
##   Bachelor's_degree  Master's_degree  Doctoral_degree  Total
##           7.6%         0.7%         2.7%   4.4%
##           92.4%        99.3%        97.3%  95.6%
##
## Cramér's V:  0.09746474
```

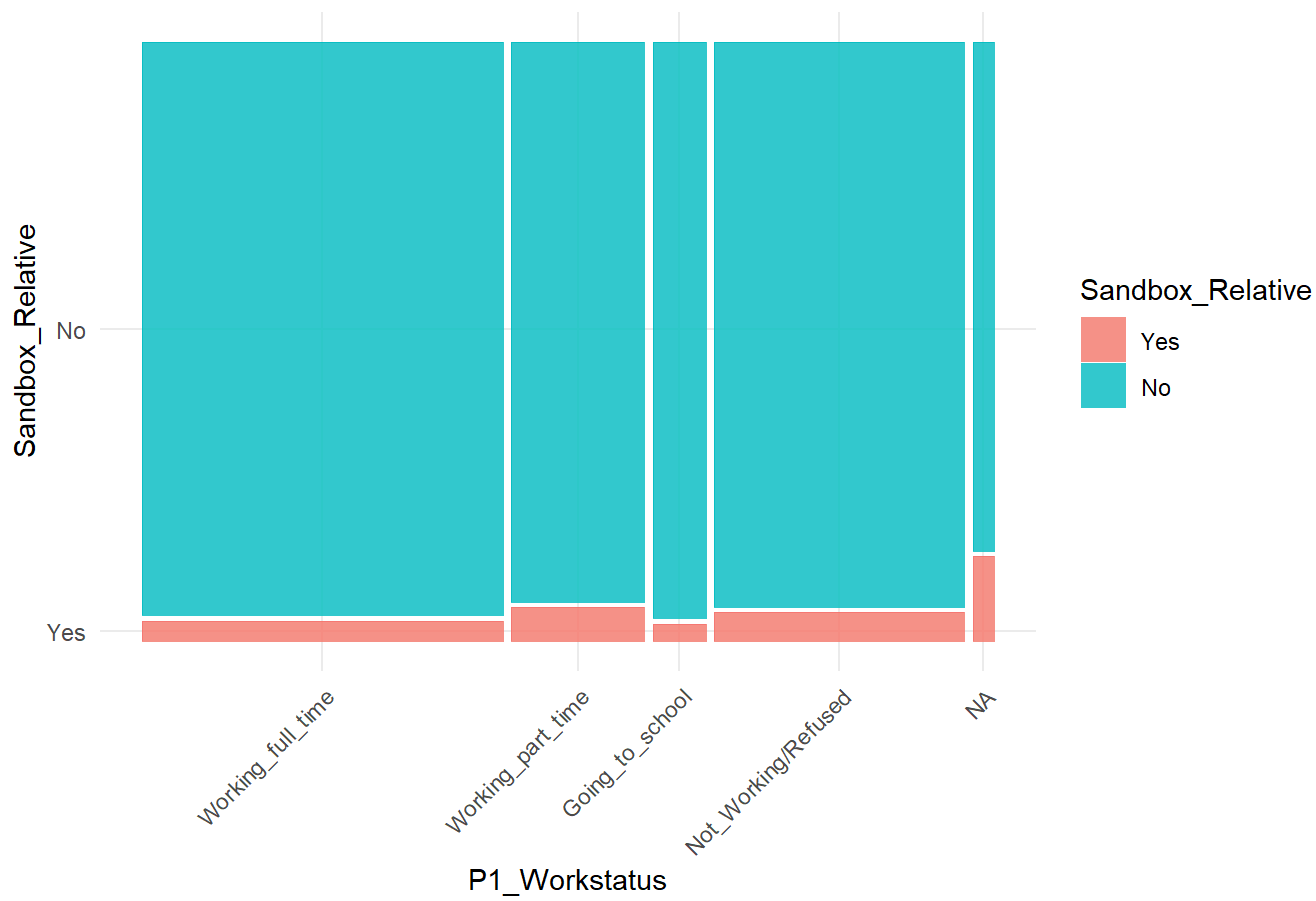
Mosaic Plot of P1_Degree and Sandbox_Relative



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Relative and P1_Workstatus -----
## Bivariate Analysis of Sandbox_Relative and P1_Workstatus
##
## Frequency Table:
##   Sandbox_Relative Working_full_time Working_part_time Going_to_school
##               Yes                8                5                1
##               No               230               83               34
##   Not_Working/Refused NA_
##               8      2
##               157   12
##
## Relative Frequency Table:
##   Sandbox_Relative Working_full_time Working_part_time Going_to_school
##               Yes                3.4%                5.7%                2.9%
##               No               96.6%               94.3%               97.1%
##   Not_Working/Refused   NA_ Total
##               4.8% 14.3%  4.4%
##               95.2% 85.7% 95.6%
##
## Cramér's V:  0.02827807
```

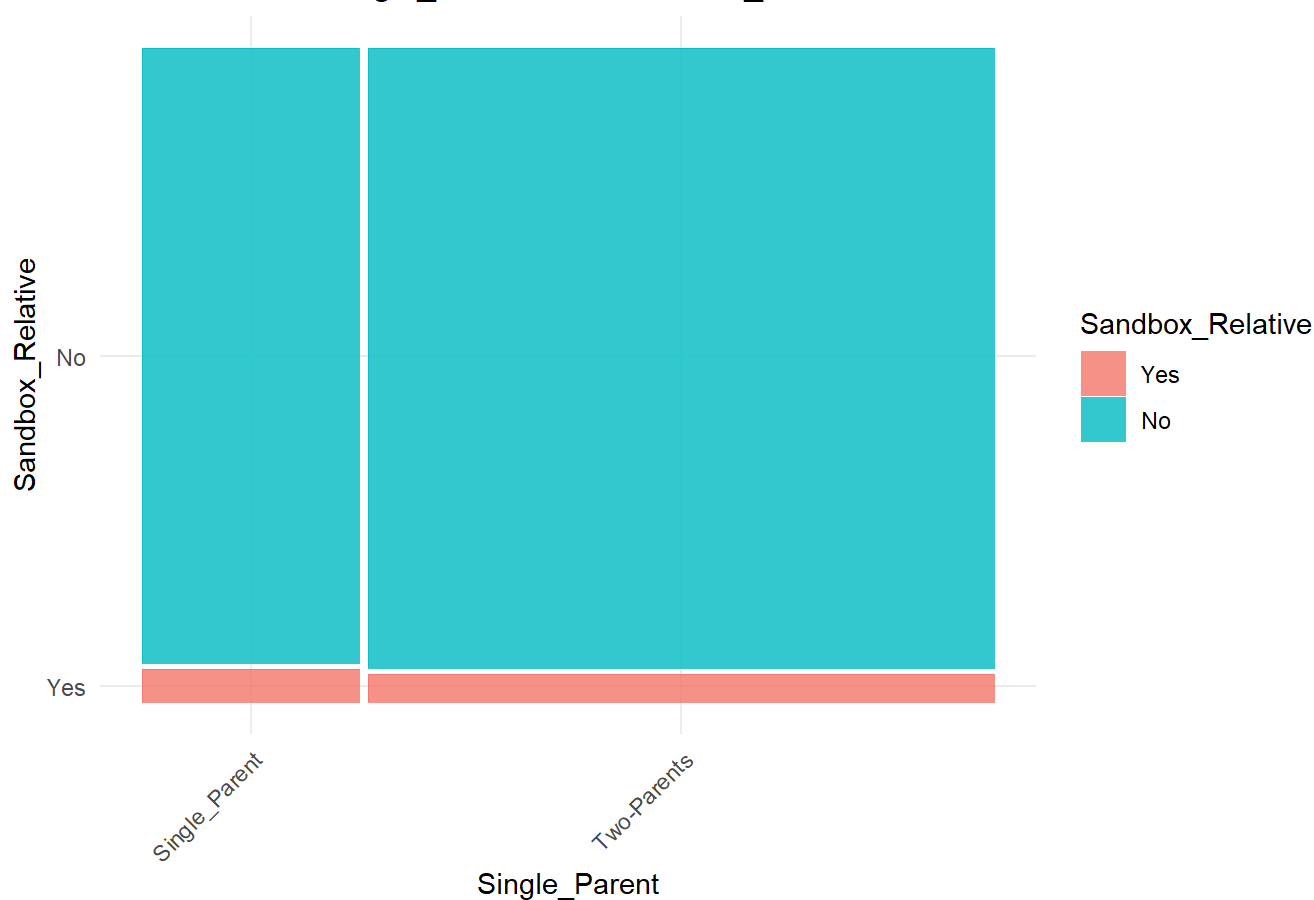
Mosaic Plot of P1_Workstatus and Sandbox_Relative



Saving 7 x 5 in image

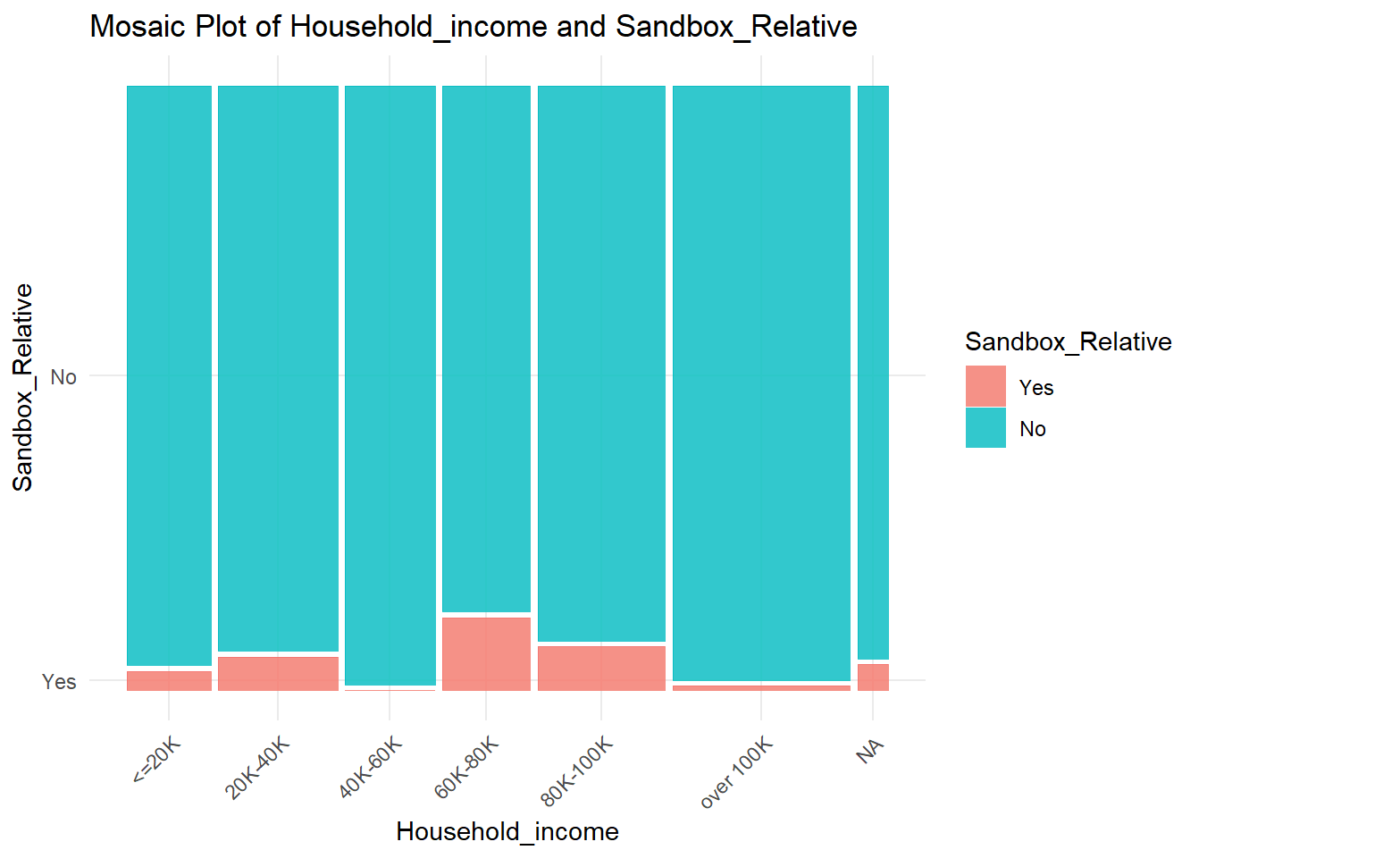
```
##
##
## ----- Analysis for Sandbox_Relative and Single_Parent -----
## Bivariate Analysis of Sandbox_Relative and Single_Parent
##
## Frequency Table:
##   Sandbox_Relative Single_Parent Two-Parents
##           Yes           7           17
##           No          132          384
##
## Relative Frequency Table:
##   Sandbox_Relative Single_Parent Two-Parents Total
##           Yes           5.0%         4.2%  4.4%
##           No          95.0%         95.8% 95.6%
##
## Cramér's V:  0
```

Mosaic Plot of Single_Parent and Sandbox_Relative



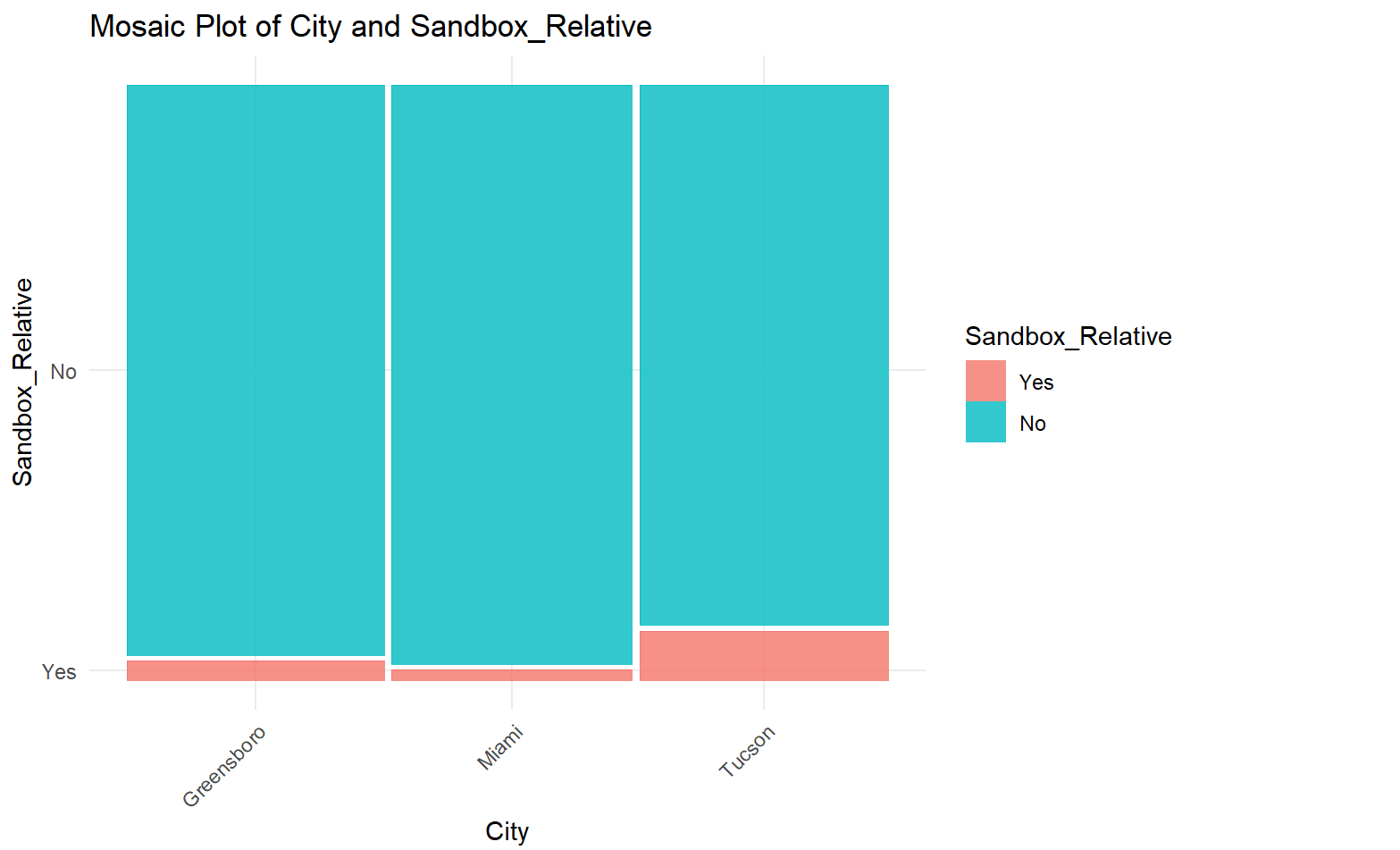
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Relative and Household_income -----
## Bivariate Analysis of Sandbox_Relative and Household_income
##
## Frequency Table:
##   Sandbox_Relative  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_
##                   Yes      2        5        0        8        7        1      1
##                   No      61       85       68       58       89      133    22
##
## Relative Frequency Table:
##   Sandbox_Relative  <=20K  20K-40K  40K-60K  60K-80K  80K-100K  over 100K  NA_ Total
##                   Yes  3.2%   5.6%   0.0%   12.1%   7.3%   0.7%  4.3%  4.4%
##                   No 96.8%  94.4% 100.0%  87.9%  92.7%  99.3% 95.7% 95.6%
##
## Cramér's V:  0.155059
```



```
## Saving 7 x 5 in image

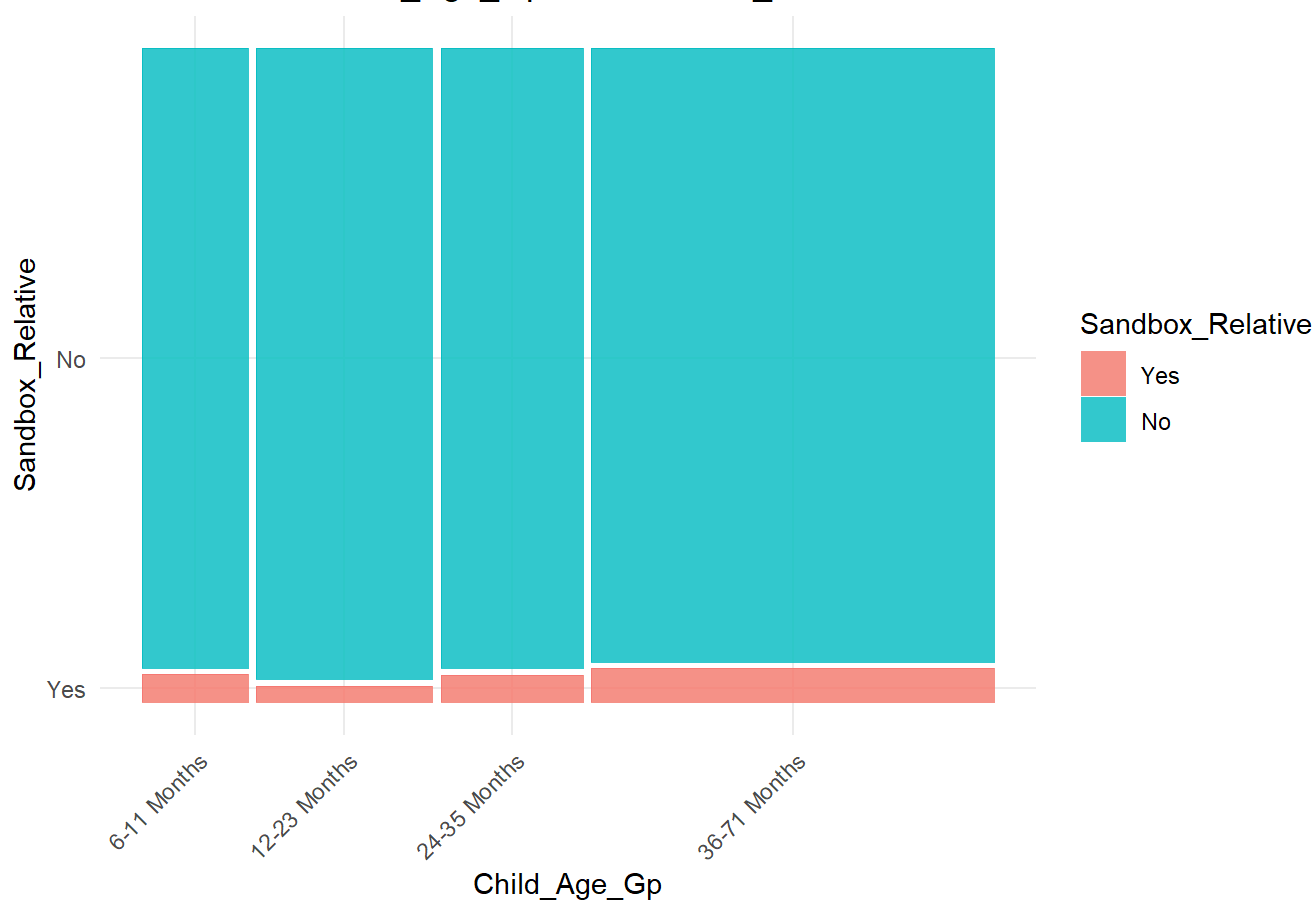
##
## ----- Analysis for Sandbox_Relative and City -----
## Bivariate Analysis of Sandbox_Relative and City
##
## Frequency Table:
##   Sandbox_Relative Greensboro Miami Tucson
##               Yes           6      3     15
##               No          180    171    165
##
## Relative Frequency Table:
##   Sandbox_Relative Greensboro Miami Tucson Total
##               Yes      3.2%  1.7%   8.3%  4.4%
##               No      96.8% 98.3%  91.7% 95.6%
##
## Cramér's V:  0.1225009
```

```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Relative and Child_Age_Gp -----
## Bivariate Analysis of Sandbox_Relative and Child_Age_Gp
##
## Frequency Table:
##   Sandbox_Relative 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##               Yes           3           3           4           14
##               No           66          112          89          249
##
## Relative Frequency Table:
##   Sandbox_Relative 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##               Yes      4.3%      2.6%      4.3%      5.3%  4.4%
##               No      95.7%      97.4%      95.7%      94.7% 95.6%
##
## Cramér's V:  0
```

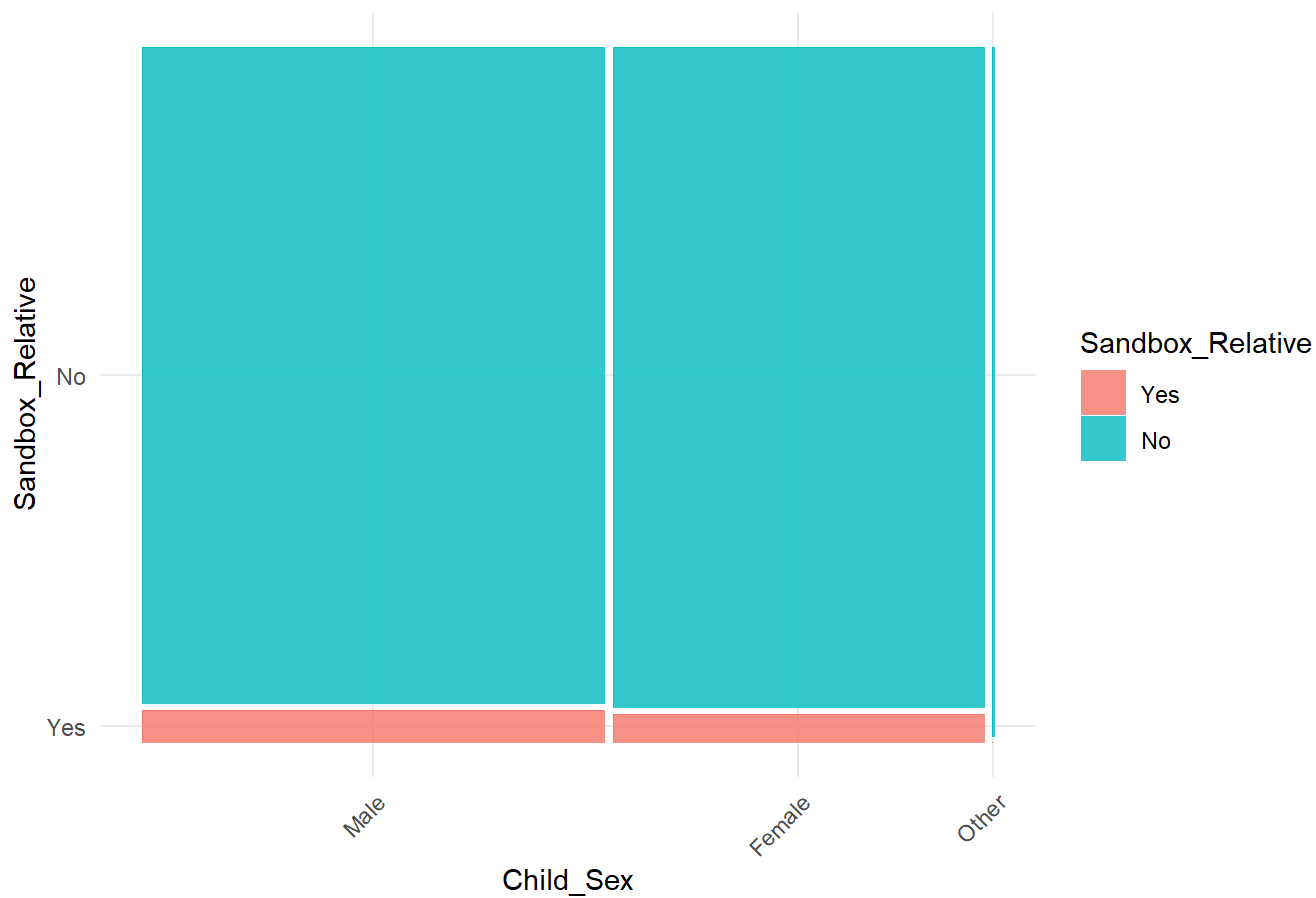
Mosaic Plot of Child_Age_Gp and Sandbox_Relative



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Relative and Child_Sex -----
## Bivariate Analysis of Sandbox_Relative and Child_Sex
##
## Frequency Table:
##   Sandbox_Relative Male Female Other
##           Yes      14      10      0
##           No     285     230      1
##
## Relative Frequency Table:
##   Sandbox_Relative Male Female Other Total
##           Yes    4.7%    4.2%    0.0%  4.4%
##           No   95.3%   95.8% 100.0% 95.6%
##
##
## Cramér's V:  0
```

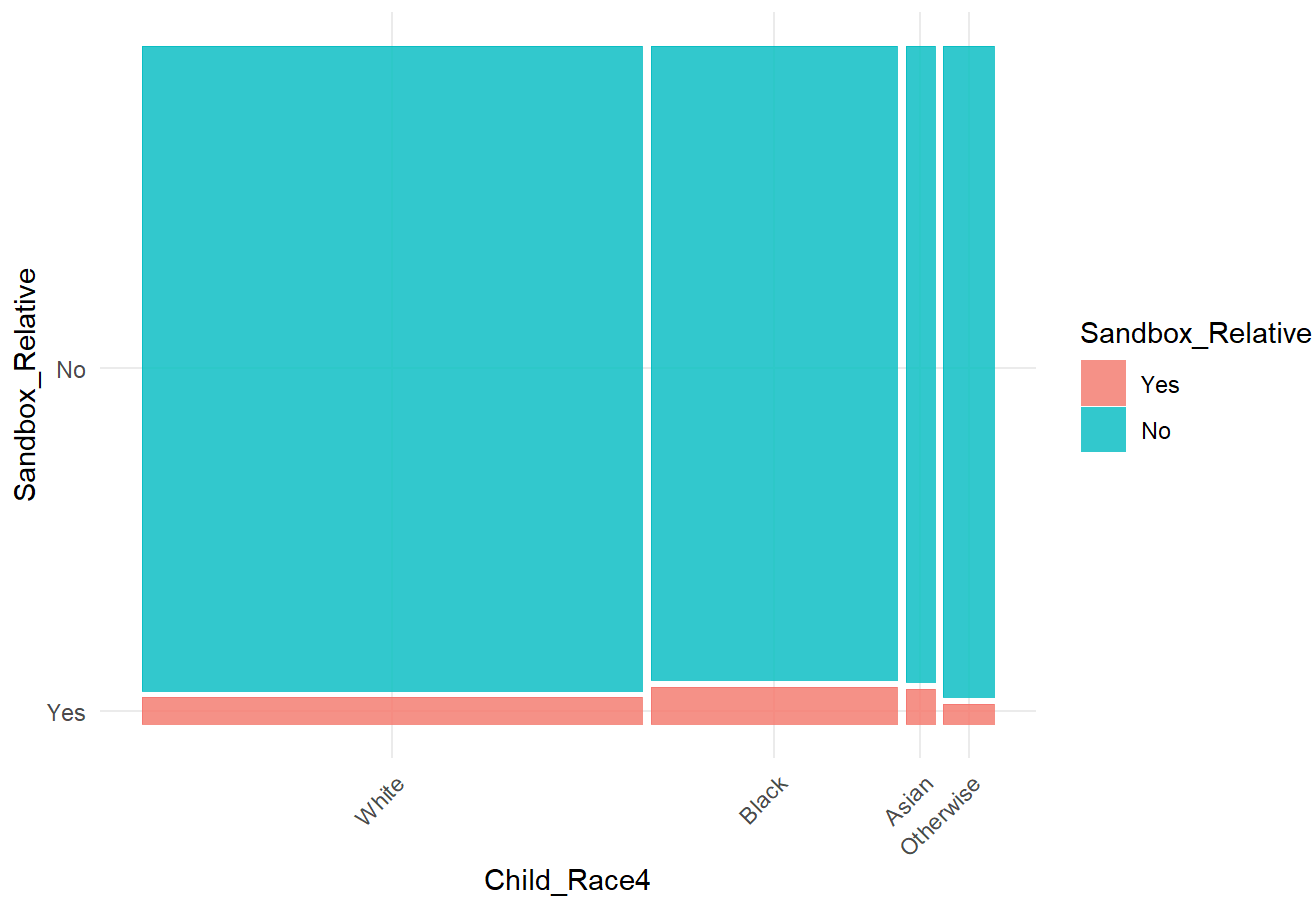
Mosaic Plot of Child_Sex and Sandbox_Relative



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Relative and Child_Race4 -----
## Bivariate Analysis of Sandbox_Relative and Child_Race4
##
## Frequency Table:
##   Sandbox_Relative White Black Asian Otherwise
##           Yes      13      9      1          1
##           No     314    152     18         32
##
## Relative Frequency Table:
##   Sandbox_Relative White Black Asian Otherwise Total
##           Yes    4.0%   5.6%   5.3%      3.0%   4.4%
##           No   96.0%  94.4%  94.7%     97.0%  95.6%
##
## Cramér's V:  0
```

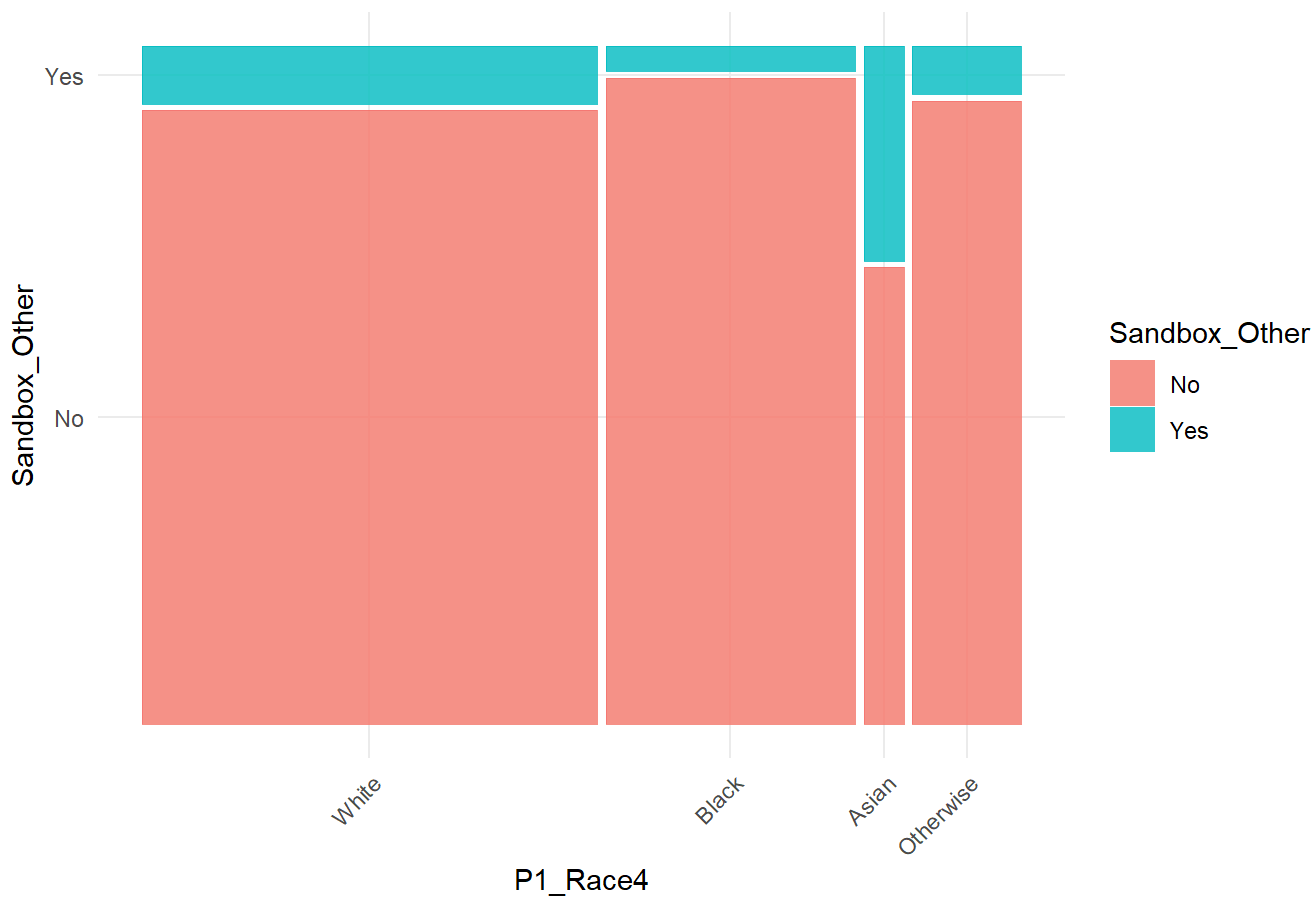
Mosaic Plot of Child_Race4 and Sandbox_Relative



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Other and P1_Race4 -----
## Bivariate Analysis of Sandbox_Other and P1_Race4
##
## Frequency Table:
##   Sandbox_Other White Black Asian Otherwise
##           No    263   152   17      64
##           Yes    25    6    8      5
##
## Relative Frequency Table:
##   Sandbox_Other White Black Asian Otherwise Total
##           No  91.3% 96.2% 68.0%    92.8% 91.9%
##           Yes   8.7%  3.8% 32.0%     7.2%  8.1%
##
## Cramér's V:  0.1934848
```

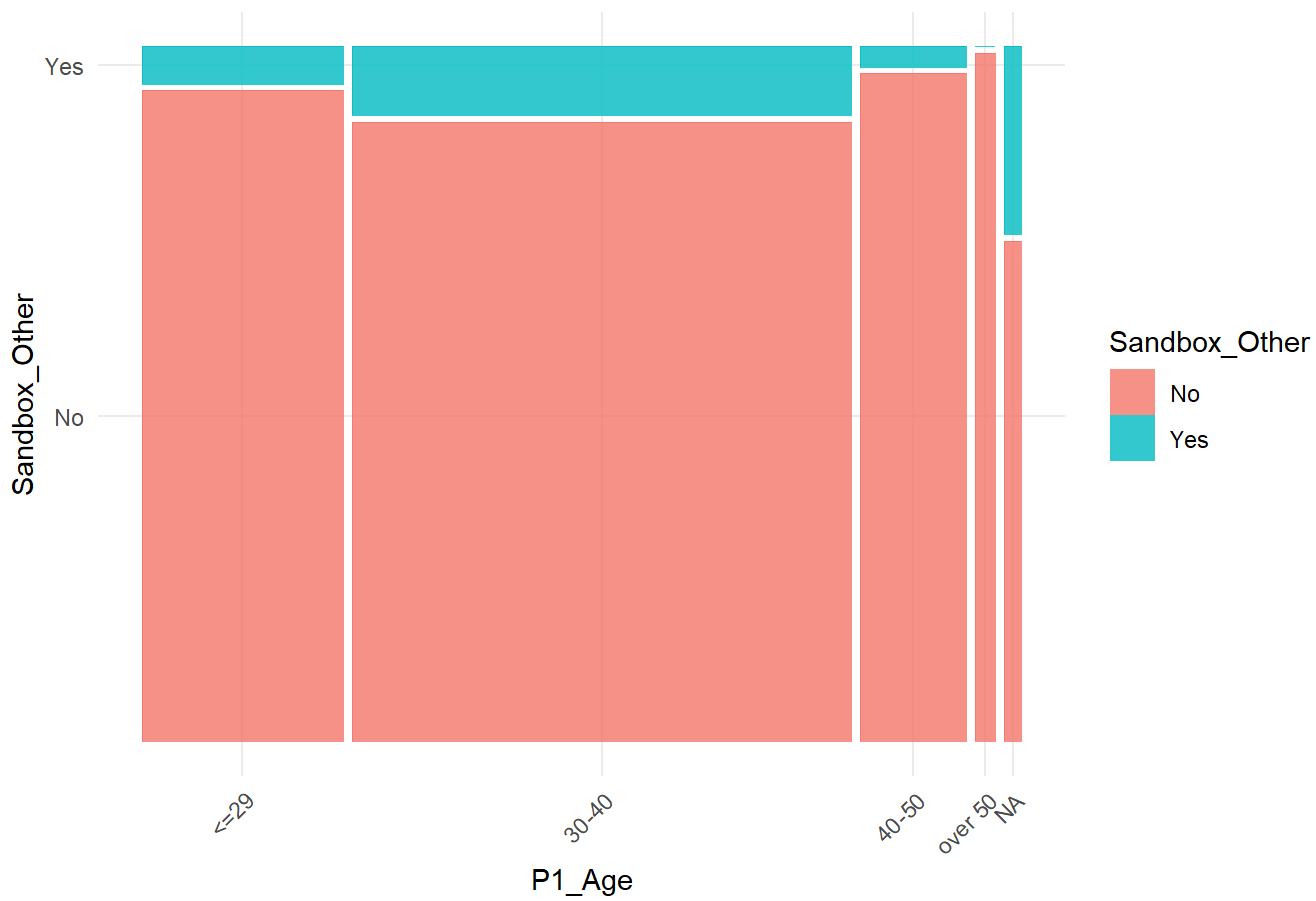
Mosaic Plot of P1_Race4 and Sandbox_Other



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Other and P1_Age -----
## Bivariate Analysis of Sandbox_Other and P1_Age
##
## Frequency Table:
##   Sandbox_Other  <=29  30-40  40-50  over 50  NA_
##               No    122    287    66      13    8
##               Yes     7     32     2       0    3
##
## Relative Frequency Table:
##   Sandbox_Other  <=29  30-40  40-50  over 50   NA_ Total
##               No  94.6%  90.0%  97.1%  100.0%  72.7%  91.9%
##               Yes   5.4%  10.0%   2.9%   0.0%  27.3%   8.1%
##
## Cramér's V:  0.1200907
```

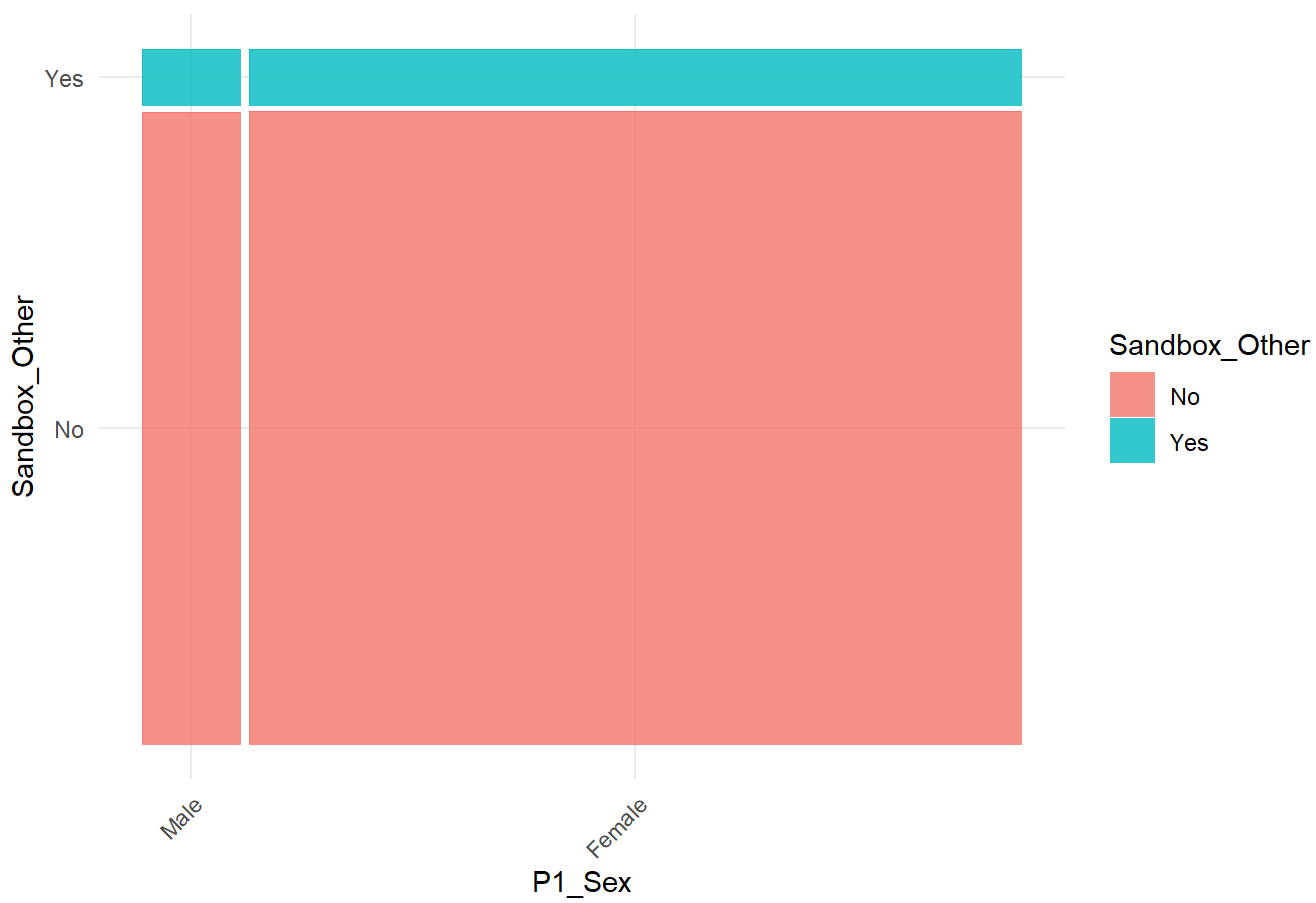
Mosaic Plot of P1_Age and Sandbox_Other



Saving 7 x 5 in image

```
##
##
## ----- Analysis for Sandbox_Other and P1_Sex -----
## Bivariate Analysis of Sandbox_Other and P1_Sex
##
## Frequency Table:
##   Sandbox_Other  Male Female
##             No     56    440
##             Yes      5     39
##
## Relative Frequency Table:
##   Sandbox_Other  Male Female Total
##             No  91.8%  91.9%  91.9%
##             Yes   8.2%   8.1%   8.1%
##
##
## Cramér's V:  0
```

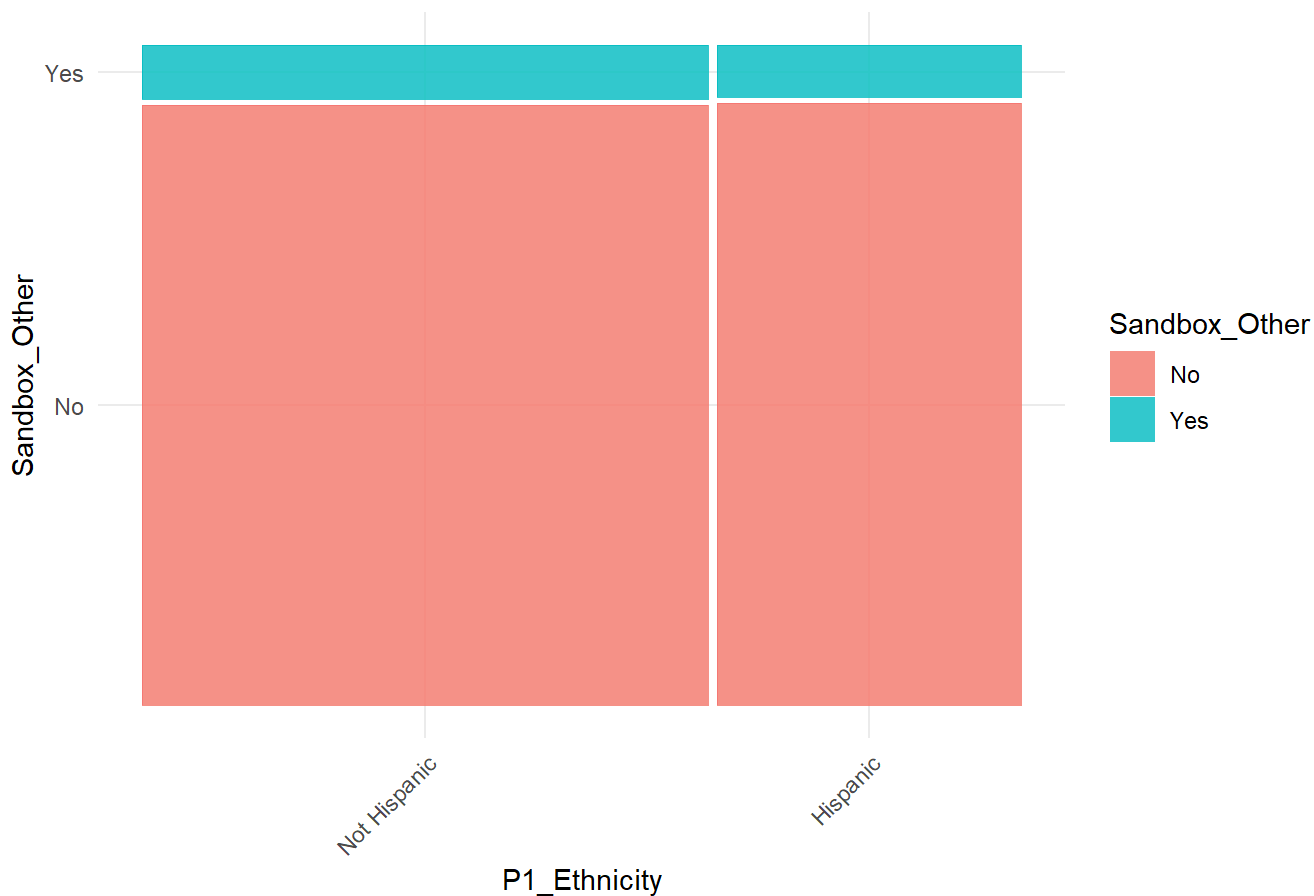
Mosaic Plot of P1_Sex and Sandbox_Other



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Other and P1_Ethnicity -----
## Bivariate Analysis of Sandbox_Other and P1_Ethnicity
##
## Frequency Table:
##   Sandbox_Other Not Hispanic Hispanic
##           No           322         174
##           Yes            29          15
##
## Relative Frequency Table:
##   Sandbox_Other Not Hispanic Hispanic Total
##           No           91.7%       92.1% 91.9%
##           Yes            8.3%        7.9%  8.1%
##
## Cramér's V:  0
```

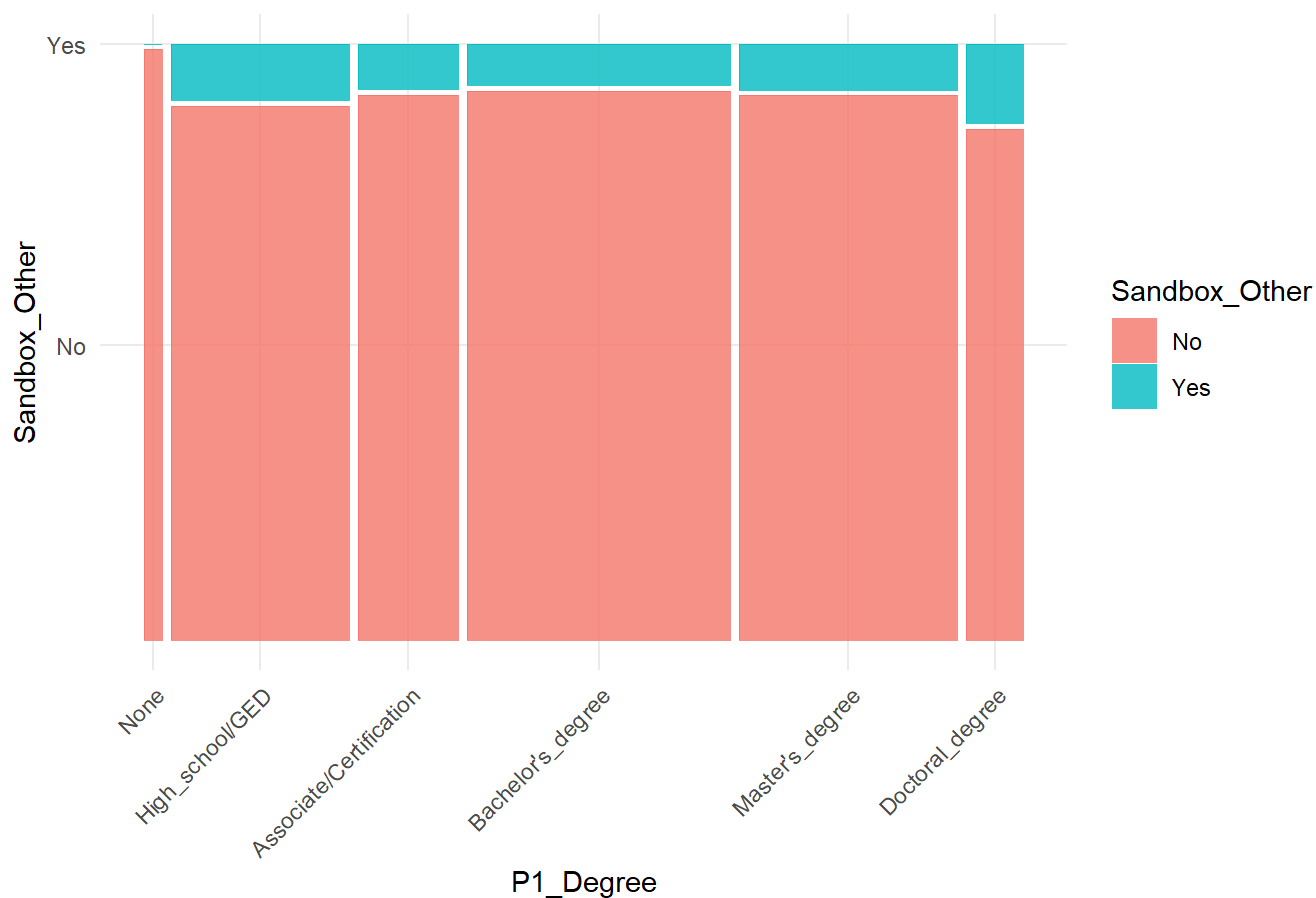
Mosaic Plot of P1_Ethnicity and Sandbox_Other



Saving 7 x 5 in image

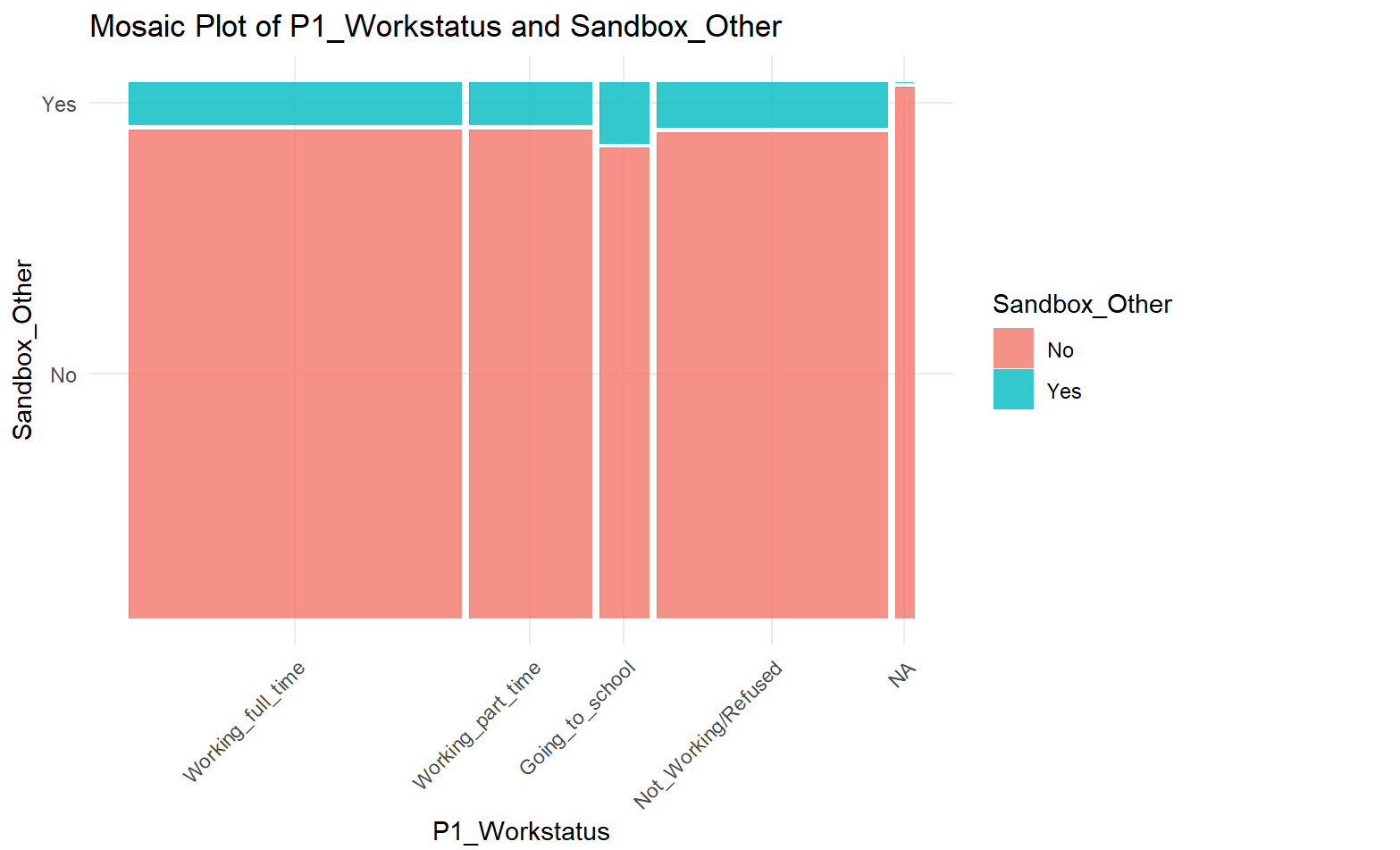
```
##
##
## ----- Analysis for Sandbox_Other and P1_Degree -----
## Bivariate Analysis of Sandbox_Other and P1_Degree
##
## Frequency Table:
##   Sandbox_Other  None High_school/GED Associate/Certification Bachelor's_degree
##               No    12              104                  60              158
##               Yes     0               11                   5               12
##   Master's_degree Doctoral_degree
##               130              32
##               11               5
##
## Relative Frequency Table:
##   Sandbox_Other  None High_school/GED Associate/Certification Bachelor's_degree
##               No 100.0%           90.4%              92.3%           92.9%
##               Yes  0.0%            9.6%              7.7%            7.1%
##   Master's_degree Doctoral_degree Total
##               92.2%           86.5% 91.9%
##               7.8%           13.5% 8.1%
##
## Cramér's V:  0
```


Mosaic Plot of P1_Degree and Sandbox_Other



```
## Saving 7 x 5 in image
```

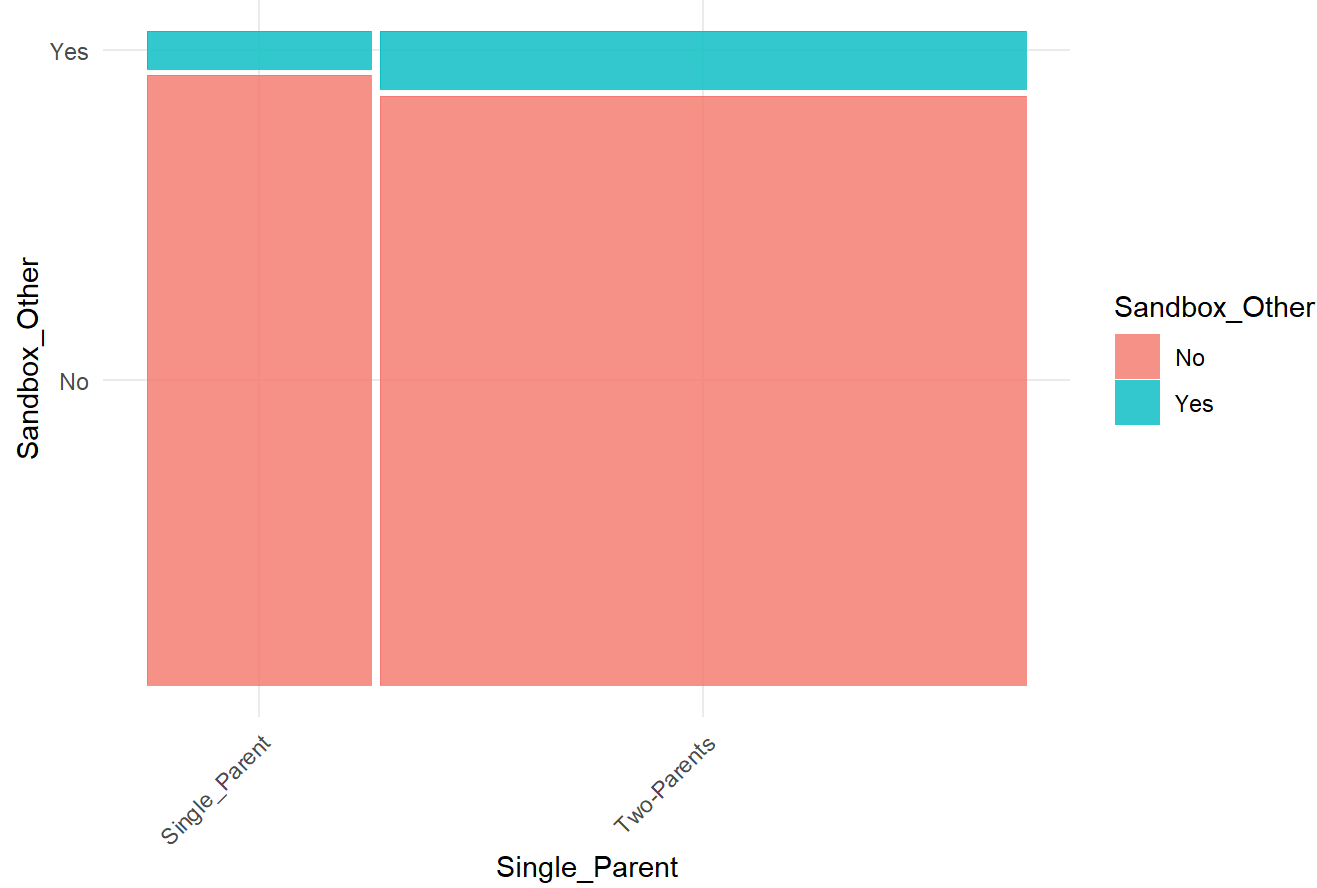
```
##
##
## ----- Analysis for Sandbox_Other and P1_Workstatus -----
## Bivariate Analysis of Sandbox_Other and P1_Workstatus
##
## Frequency Table:
##   Sandbox_Other Working_full_time Working_part_time Going_to_school
##           No           219           81           31
##           Yes           19           7           4
##   Not_Working/Refused NA_
##           151  14
##           14   0
##
## Relative Frequency Table:
##   Sandbox_Other Working_full_time Working_part_time Going_to_school
##           No           92.0%           92.0%           88.6%
##           Yes           8.0%           8.0%           11.4%
##   Not_Working/Refused   NA_ Total
##           91.5% 100.0% 91.9%
##           8.5%  0.0%  8.1%
##
## Cramér's V:  0
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Other and Single_Parent -----
## Bivariate Analysis of Sandbox_Other and Single_Parent
##
## Frequency Table:
##   Sandbox_Other Single_Parent Two-Parents
##           No           131           365
##           Yes            8            36
##
## Relative Frequency Table:
##   Sandbox_Other Single_Parent Two-Parents Total
##           No           94.2%           91.0% 91.9%
##           Yes            5.8%            9.0%  8.1%
##
## Cramér's V:  0.02824643
```

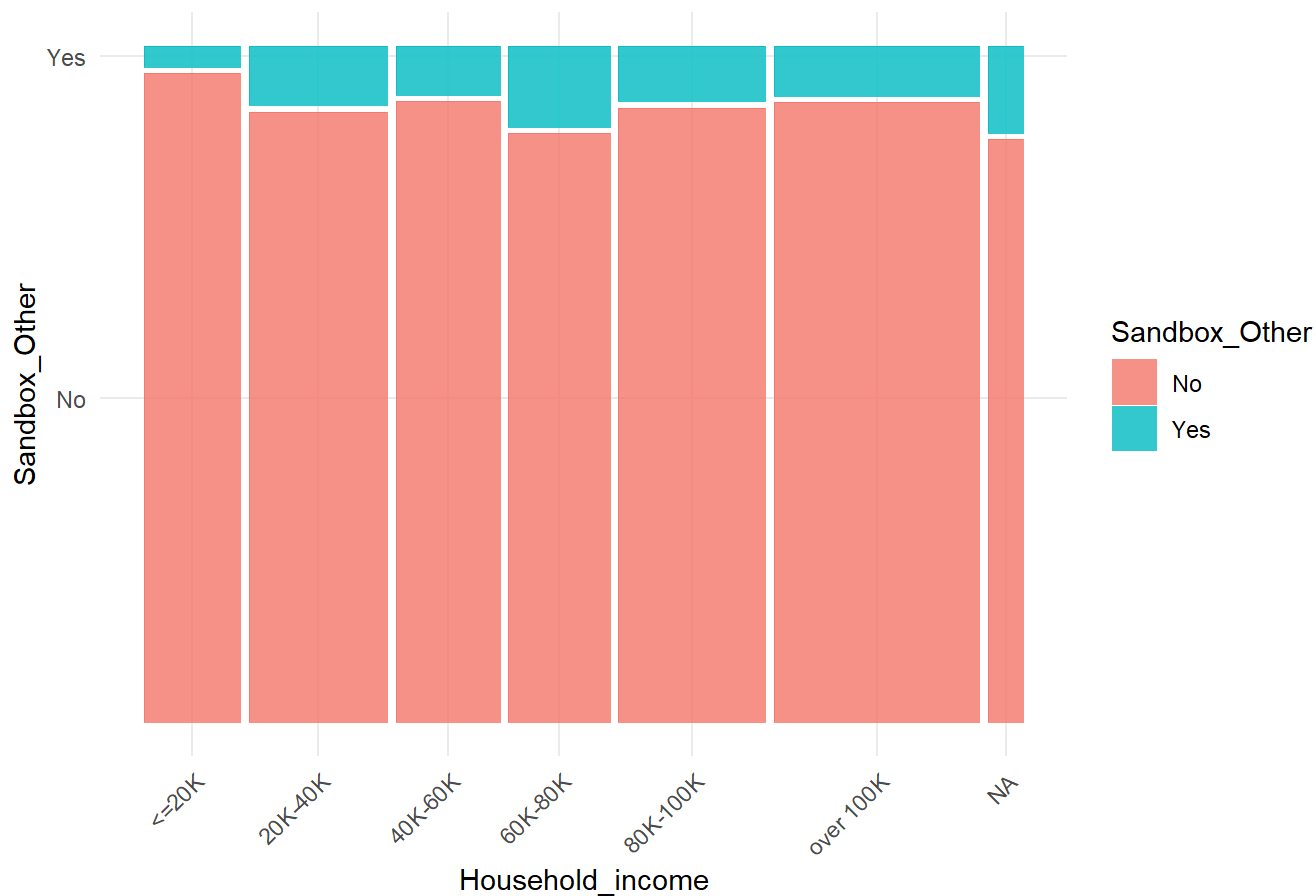
Mosaic Plot of Single_Parent and Sandbox_Other



```
## Saving 7 x 5 in image
```

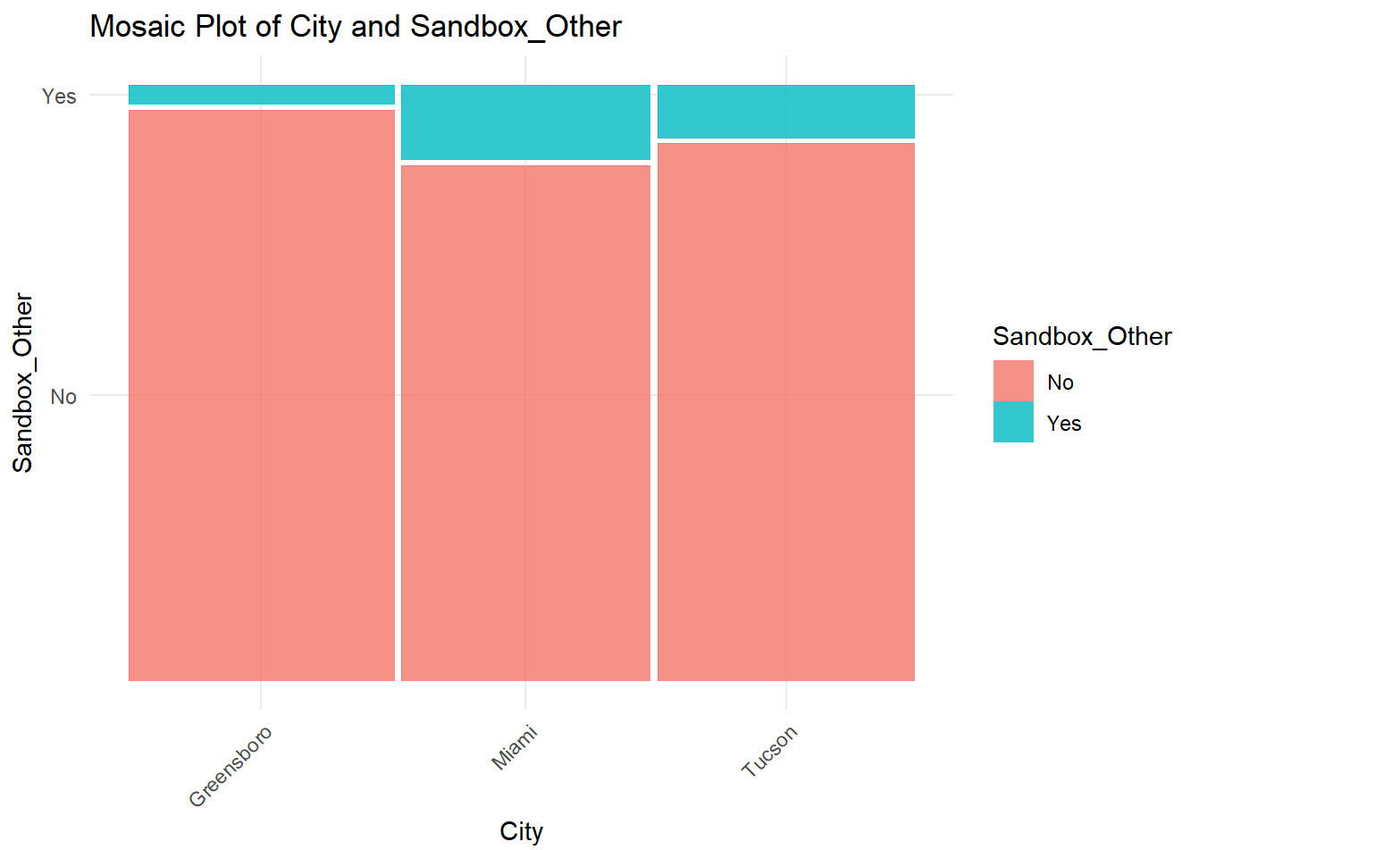
```
##
##
## ----- Analysis for Sandbox_Other and Household_income -----
## Bivariate Analysis of Sandbox_Other and Household_income
##
## Frequency Table:
##   Sandbox_Other <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##               No      61      82      63      58      88      124  20
##               Yes       2       8       5       8       8       10   3
##
## Relative Frequency Table:
##   Sandbox_Other <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K  NA_ Total
##               No 96.8%  91.1%  92.6%  87.9%  91.7%  92.5% 87.0% 91.9%
##               Yes  3.2%   8.9%   7.4%  12.1%   8.3%   7.5% 13.0%  8.1%
##
## Cramér's V:  0
```

Mosaic Plot of Household_income and Sandbox_Other



```
## Saving 7 x 5 in image
```

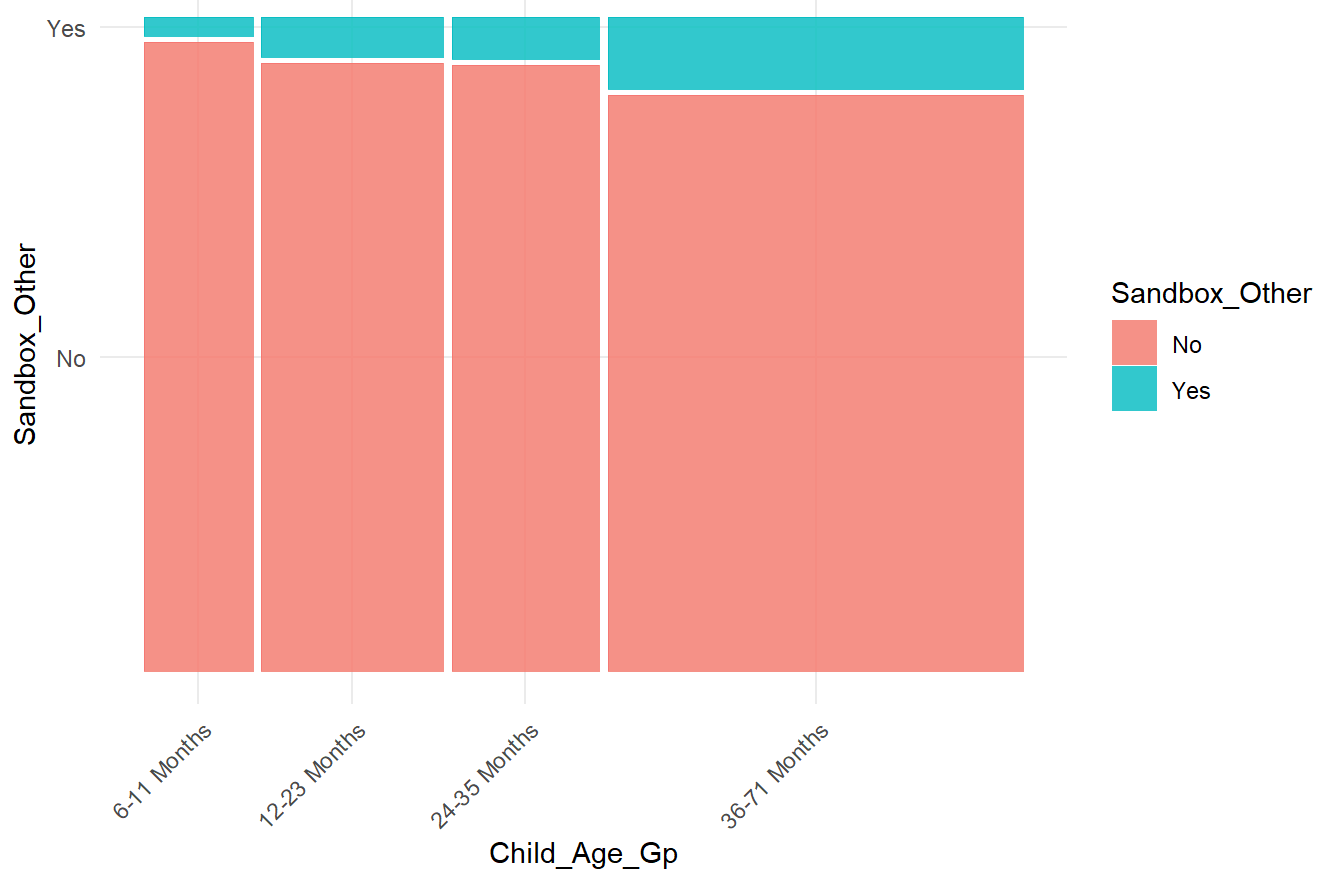
```
##
##
## ----- Analysis for Sandbox_Other and City -----
## Bivariate Analysis of Sandbox_Other and City
##
## Frequency Table:
##   Sandbox_Other Greensboro Miami Tucson
##             No      180    152    164
##             Yes       6     22     16
##
## Relative Frequency Table:
##   Sandbox_Other Greensboro Miami Tucson Total
##             No     96.8% 87.4%  91.1% 91.9%
##             Yes     3.2% 12.6%   8.9%  8.1%
##
## Cramér's V:  0.1281263
```



```
## Saving 7 x 5 in image
```

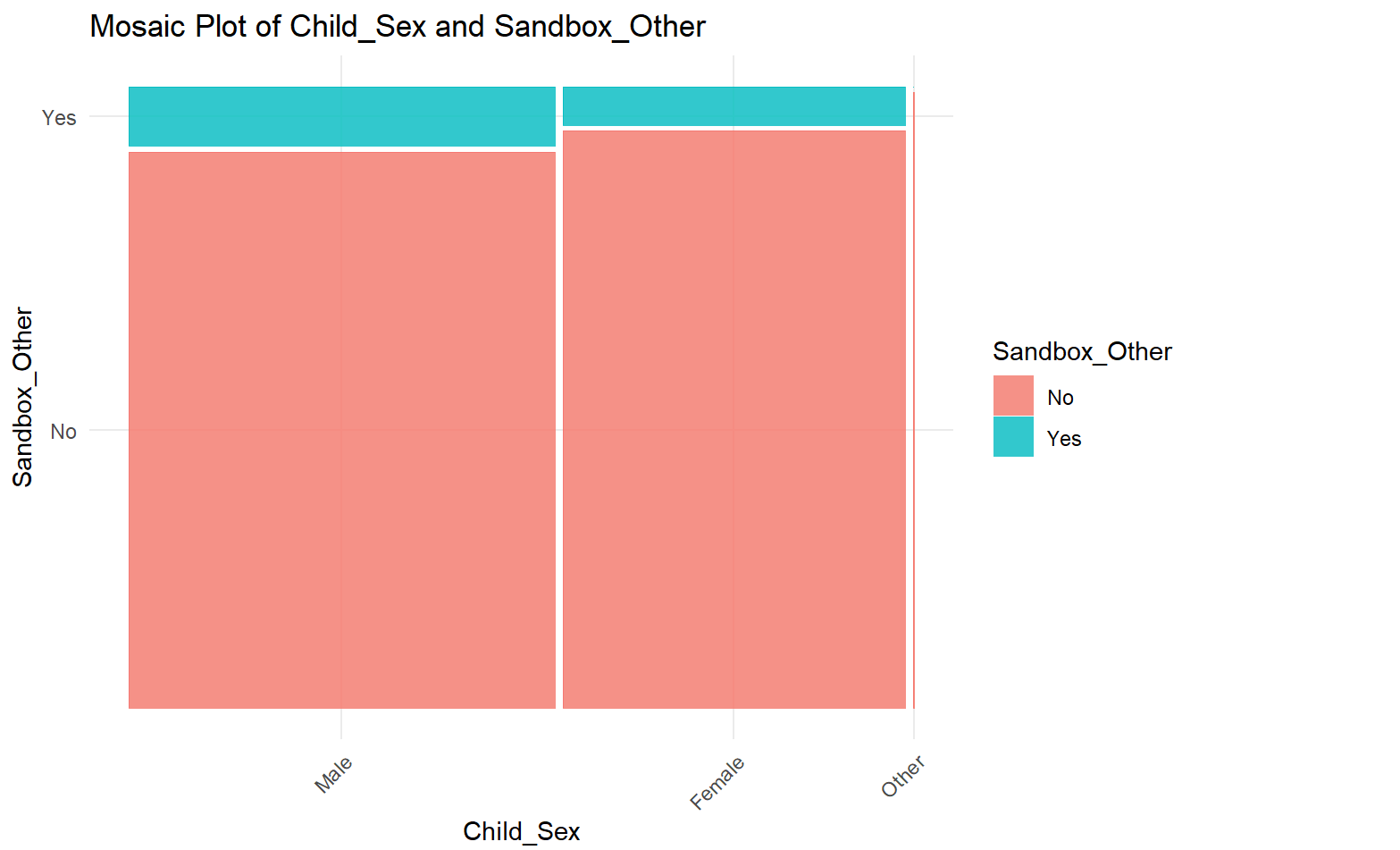
```
##
##
## ----- Analysis for Sandbox_Other and Child_Age_Gp -----
## Bivariate Analysis of Sandbox_Other and Child_Age_Gp
##
## Frequency Table:
##   Sandbox_Other 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##               No           67           108           87           234
##               Yes            2             7            6            29
##
## Relative Frequency Table:
##   Sandbox_Other 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##               No           97.1%          93.9%          93.5%          89.0% 91.9%
##               Yes            2.9%           6.1%           6.5%          11.0%  8.1%
##
## Cramér's V:  0.08008825
```

Mosaic Plot of Child_Age_Gp and Sandbox_Other



Saving 7 x 5 in image

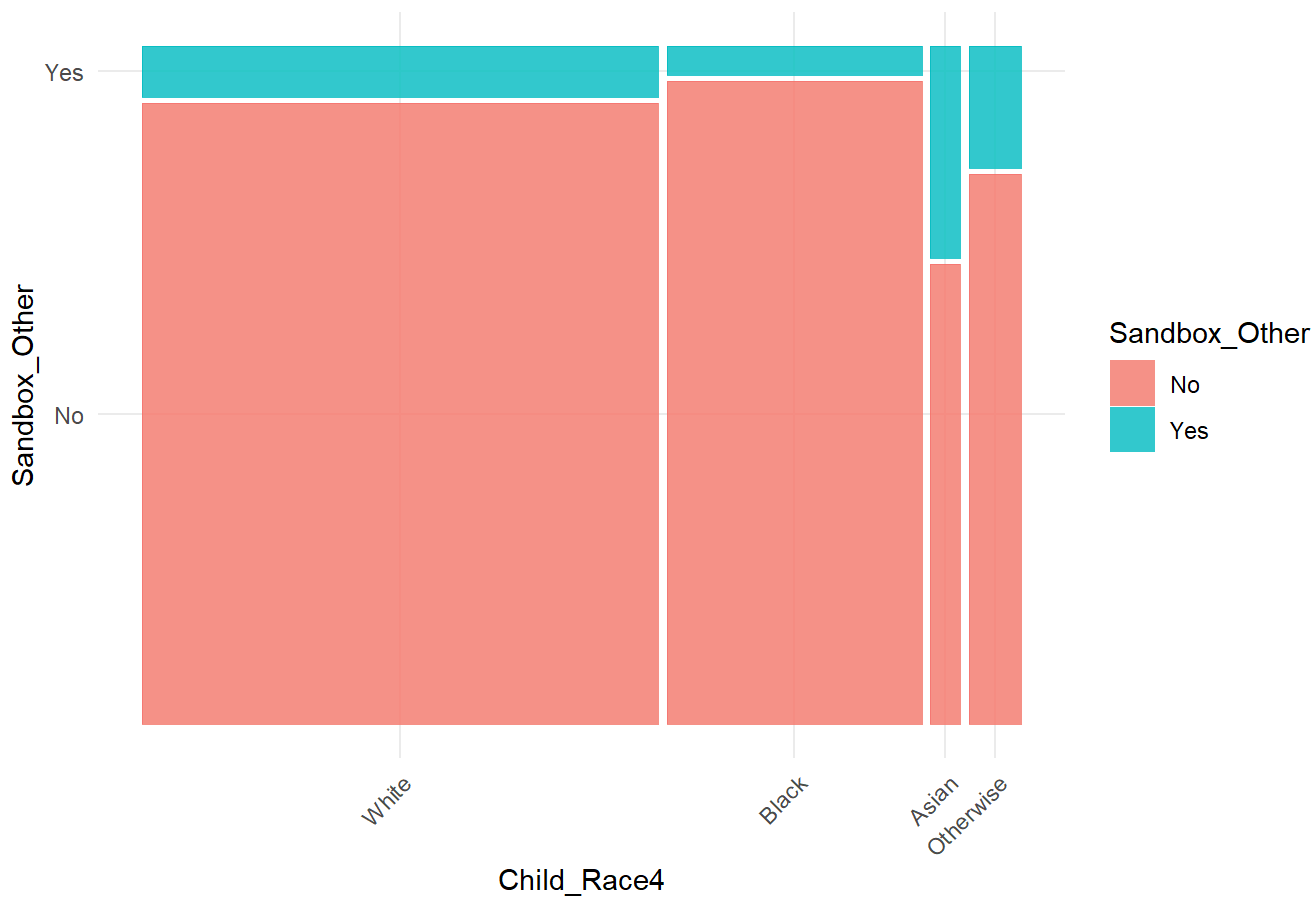
```
##
##
## ----- Analysis for Sandbox_Other and Child_Sex -----
## Bivariate Analysis of Sandbox_Other and Child_Sex
##
## Frequency Table:
##   Sandbox_Other Male Female Other
##           No   270    225    1
##           Yes   29    15    0
##
## Relative Frequency Table:
##   Sandbox_Other  Male Female  Other Total
##           No  90.3%  93.8% 100.0% 91.9%
##           Yes   9.7%   6.2%   0.0%  8.1%
##
## Cramér's V:  0.01932276
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Sandbox_Other and Child_Race4 -----
## Bivariate Analysis of Sandbox_Other and Child_Race4
##
## Frequency Table:
##   Sandbox_Other White Black Asian Otherwise
##           No    302   154    13         27
##           Yes     25     7     6          6
##
## Relative Frequency Table:
##   Sandbox_Other White Black Asian Otherwise Total
##           No  92.4% 95.7% 68.4%      81.8% 91.9%
##           Yes   7.6%  4.3% 31.6%      18.2%  8.1%
##
## Cramér's V:  0.1857055
```

Mosaic Plot of Child_Race4 and Sandbox_Other



```
## Saving 7 x 5 in image
```

```
# Rank var2 variables based on Cramér's V
cramer_v_values <- cramer_v_values %>% arrange(desc(cramer_v))
print(cramer_v_values)
```


##	var1	var2	cramer_v
## 1	Sandbox_Home	City	0.26570468
## 2	Sandbox_Other	P1_Race4	0.19348483
## 3	Sandbox_Home	P1_Age	0.19095920
## 4	Sandbox_Other	Child_Race4	0.18570545
## 5	Sandbox_Daycare	Household_income	0.17248211
## 6	Sandbox_Park	Child_Age_Gp	0.15833497
## 7	Sandbox_Relative	Household_income	0.15505904
## 8	Sandbox_Home	Child_Race4	0.14214874
## 9	Sandbox_Park	P1_Age	0.14062439
## 10	Sandbox_Daycare	City	0.13995831
## 11	Sandbox_Park	P1_Workstatus	0.13379825
## 12	Sandbox_Daycare	Child_Age_Gp	0.12851813
## 13	Sandbox_Other	City	0.12812626
## 14	Sandbox_Daycare	Child_Sex	0.12745189
## 15	Sandbox_Home	P1_Race4	0.12695494
## 16	Sandbox_Relative	City	0.12250092
## 17	Sandbox_Home	P1_Degree	0.12164606
## 18	Sandbox_Other	P1_Age	0.12009070
## 19	Sandbox_Daycare	P1_Degree	0.11963681
## 20	Sandbox_Park	City	0.11522044
## 21	Sandbox_Home	Child_Age_Gp	0.11053350
## 22	Sandbox_Daycare	P1_Workstatus	0.10492732
## 23	Sandbox_Home	P1_Workstatus	0.10092924
## 24	Sandbox_Relative	P1_Degree	0.09746474
## 25	Sandbox_Park	P1_Ethnicity	0.09103453
## 26	Sandbox_Daycare	P1_Race4	0.09053272
## 27	Sandbox_Home	Household_income	0.08840976
## 28	Sandbox_Park	Household_income	0.08212824
## 29	Sandbox_Other	Child_Age_Gp	0.08008825
## 30	Sandbox_Relative	P1_Race4	0.07562187
## 31	Sandbox_Home	P1_Sex	0.06146082
## 32	Sandbox_Babysitter	City	0.05805917
## 33	Sandbox_Daycare	P1_Ethnicity	0.05640785
## 34	Sandbox_Park	Single_Parent	0.05077841
## 35	Sandbox_Relative	P1_Sex	0.04869330
## 36	Sandbox_Daycare	Child_Race4	0.04761977
## 37	Sandbox_Babysitter	P1_Race4	0.03418288
## 38	Sandbox_Relative	P1_Workstatus	0.02827807
## 39	Sandbox_Other	Single_Parent	0.02824643
## 40	Sandbox_Other	Child_Sex	0.01932276
## 41	Sandbox_Park	P1_Degree	0.01060705
## 42	Sandbox_Home	P1_Ethnicity	0.00000000
## 43	Sandbox_Home	Single_Parent	0.00000000
## 44	Sandbox_Home	Child_Sex	0.00000000
## 45	Sandbox_Park	P1_Race4	0.00000000
## 46	Sandbox_Park	P1_Sex	0.00000000
## 47	Sandbox_Park	Child_Sex	0.00000000
## 48	Sandbox_Park	Child_Race4	0.00000000
## 49	Sandbox_Daycare	P1_Age	0.00000000
## 50	Sandbox_Daycare	P1_Sex	0.00000000
## 51	Sandbox_Daycare	Single_Parent	0.00000000
## 52	Sandbox_Babysitter	P1_Age	0.00000000
## 53	Sandbox_Babysitter	P1_Sex	0.00000000
## 54	Sandbox_Babysitter	P1_Ethnicity	0.00000000
## 55	Sandbox_Babysitter	P1_Degree	0.00000000

```
## 56 Sandbox_Babysitter P1_Workstatus 0.00000000
## 57 Sandbox_Babysitter Single_Parent 0.00000000
## 58 Sandbox_Babysitter Household_income 0.00000000
## 59 Sandbox_Babysitter Child_Age_Gp 0.00000000
## 60 Sandbox_Babysitter Child_Sex 0.00000000
## 61 Sandbox_Babysitter Child_Race4 0.00000000
## 62 Sandbox_Relative P1_Age 0.00000000
## 63 Sandbox_Relative P1_Ethnicity 0.00000000
## 64 Sandbox_Relative Single_Parent 0.00000000
## 65 Sandbox_Relative Child_Age_Gp 0.00000000
## 66 Sandbox_Relative Child_Sex 0.00000000
## 67 Sandbox_Relative Child_Race4 0.00000000
## 68 Sandbox_Other P1_Sex 0.00000000
## 69 Sandbox_Other P1_Ethnicity 0.00000000
## 70 Sandbox_Other P1_Degree 0.00000000
## 71 Sandbox_Other P1_Workstatus 0.00000000
## 72 Sandbox_Other Household_income 0.00000000
```

Identify The Strongest Features for each Exposure Variable

```
#####
```

```
## Revised to get cleaner well-formatted table
```

```
# Load necessary libraries
```

```
library(DescTools)
```

```
library(dplyr)
```

```
library(knitr)
```

```
# Define the function
```

```
identify_strongest_features_table <- function(data, list1, list2, threshold = 0.20) {
```

```
  # Helper function to calculate Cramér's V
```

```
  calculate_cramers_v <- function(data, var1, var2) {
```

```
    # Exclude rows with NAs in the relevant columns
```

```
    subset_data <- data[!is.na(data[[var1]]) & !is.na(data[[var2]]), ]
```

```
    # Convert to factors
```

```
    subset_data[[var1]] <- as.factor(subset_data[[var1]])
```

```
    subset_data[[var2]] <- as.factor(subset_data[[var2]])
```

```
    # Create contingency table
```

```
    contingency_table <- table(subset_data[[var1]], subset_data[[var2]])
```

```
    # Calculate Cramér's V
```

```
    cramer_v <- tryCatch({
```

```
      DescTools::CramerV(contingency_table)
```

```
    }, error = function(e) {
```

```
      return(NA) # Return NA if there's an error
```

```
    })
```

```
    return(cramer_v)
```

```
  }
```

```
# Initialize a results data frame
```

```
results <- data.frame(
```

```
  exposure_var = character(),
```

```
  demographic_var = character(),
```

```
  cramer_v = numeric(),
```

```
  stringsAsFactors = FALSE
```

```
)
```

```
# Loop over each variable in list1
```

```
for (var1 in list1) {
```

```
  # Initialize a vector to store Cramér's V values for var1
```

```
  cramers_v_list <- data.frame(demographic_var = character(), cramer_v = numeric(), stringsAsFactors = FALSE)
```

```
# Loop over each variable in list2
```

```
for (var2 in list2) {
```

```
  cramer_v <- calculate_cramers_v(data, var1, var2)
```

```
  if (!is.na(cramer_v) && cramer_v > threshold) {
```

```
    cramers_v_list <- rbind(cramers_v_list, data.frame(demographic_var = var2, cramer_v = cramer_v))
```

```
  }
```

```
}
```

```

# Sort by descending Cramér's V
cramers_v_list <- cramers_v_list %>%
  arrange(desc(cramer_v)) %>%
  mutate(exposure_var = var1) # Add the exposure variable

# Append to the results
results <- rbind(results, cramers_v_list)

# Add a separator row for formatting
results <- rbind(results, data.frame(
  exposure_var = "---", demographic_var = "---", cramer_v = NA, stringsAsFactors = FALSE
))
}

# Clean up results
results <- results %>%
  filter(exposure_var != "---" | row_number() == n()) # Ensure the last separator is kept

return(results)
}

# Table Formatting Function
print_formatted_table <- function(results) {
  # Render table with `knitr::kable`
  kable(results, col.names = c("Exposure Variable", "Demographic Variable", "Cramér's V"), align = "c") %>%
    kable_styling(full_width = FALSE, position = "center", font_size = 12) %>%
    row_spec(which(results$exposure_var == "---"), background = "grey") # Highlight separator rows
}

#####

```

Feature Selection for Each Exposure Variable

```
## Stage the variables
```

```
Exposure_Vars <- c(
  "Mouthing", "Pacifier", "Pacifier_Use", "Pacifier_Wash",
  "Blanket","Blanket_Wash", "Stuffed_Animal", "Stuffed_Animal_Wash", "Sucking", "Handwash","Hours_Away_weekd
ay",
  "Hours_Away_weekend", "Hours_Outdoors_Home","Play_Park" , "Play_Daycare", "Play_Babysitter_Home","Play_R
elative_Home", "Play_Other", "Sandbox_Home"
, "Sandbox_Park", "Sandbox_Daycare" , "Sandbox_Babysitter",
  "Sandbox_Relative" , "Sandbox_Other"
)
```

```
#
```

```
# Demo_Vars <- c("Age_P1","Parent1_Gender", "Parent1_Race", "Single_Parent",
# "P1_Ethnicity","City","degree_P1", "Household_income", "Child_Age_Gp", "Child_Sex", "Chi
Ld_Race4","P1_Workstatus")
# #
# list1=Exposure_Vars
# list2=Demo_Vars
```

```
## Fixing the column orders:
```

```
identify_strongest_features_table <- function(data, list1, list2, threshold = 0.15) {
```

```
# Helper function to calculate Cramér's V
```

```
calculate_cramers_v <- function(data, var1, var2) {
  subset_data <- data[!is.na(data[[var1]]) & !is.na(data[[var2]])]
  subset_data[[var1]] <- as.factor(subset_data[[var1]])
  subset_data[[var2]] <- as.factor(subset_data[[var2]])
  contingency_table <- table(subset_data[[var1]], subset_data[[var2]])
  cramer_v <- tryCatch({
    DescTools::CramerV(contingency_table)
  }, error = function(e) {
    return(NA)
  })
  return(cramer_v)
}
```

```
# Initialize results data frame
```

```
results <- data.frame(
  exposure_var = character(),
  demographic_var = character(),
  cramer_v = numeric(),
  stringsAsFactors = FALSE
)
```

```
# Loop through each variable in list1
```

```
for (var1 in list1) {
  cramers_v_list <- data.frame(demographic_var = character(), cramer_v = numeric(), stringsAsFactors = FA
LSE)
```

```
# Loop through each variable in list2
```

```
for (var2 in list2) {
```

```

    cramer_v <- calculate_cramers_v(data, var1, var2)
    if (!is.na(cramer_v) && cramer_v >= threshold) {
      crammers_v_list <- rbind(crammers_v_list, data.frame(demographic_var = var2, cramer_v = cramer_v))
    }
  }

  # Sort and append
  crammers_v_list <- crammers_v_list %>%
    arrange(desc(cramer_v)) %>%
    mutate(exposure_var = var1) # Add the exposure variable

  results <- rbind(results, crammers_v_list)
}

# Reorder columns to ensure the desired column order
results <- results[, c("exposure_var", "demographic_var", "cramer_v")]

return(results)
}

##
list1=Exposure_Vars
list2=DemoVars

# Run the function

result_tbl_0.15 <- identify_strongest_features_table (data, list1, list2, threshold = 0.15)
print(result_tbl_0.15)

```

##	exposure_var	demographic_var	cramer_v
## 1	Mouthing	Child_Age_Gp	0.4500148
## 2	Pacifier	Child_Age_Gp	0.3760177
## 3	Pacifier_Use	Child_Age_Gp	0.2240739
## 4	Pacifier_Wash	Child_Age_Gp	0.2361997
## 5	Pacifier_Wash	Single_Parent	0.1559976
## 6	Blanket	Child_Age_Gp	0.2264065
## 7	Blanket	P1_Age	0.1826572
## 8	Blanket_Wash	Child_Age_Gp	0.1616236
## 9	Stuffed_Animal	Child_Age_Gp	0.1573678
## 10	Sucking	Child_Age_Gp	0.2449472
## 11	Sucking	City	0.1878508
## 12	Handwash	Child_Age_Gp	0.2044679
## 13	Handwash	Household_income	0.1584068
## 14	Hours_Away_weekday	P1_Workstatus	0.1949007
## 15	Hours_Away_weekday	Household_income	0.1920067
## 16	Hours_Away_weekday	City	0.1670924
## 17	Hours_Away_weekday	P1_Degree	0.1595487
## 18	Hours_Away_weekday	Child_Age_Gp	0.1506431
## 19	Hours_Away_weekend	Single_Parent	0.2156966
## 20	Hours_Away_weekend	City	0.1581105
## 21	Hours_Away_weekend	P1_Race4	0.1546701
## 22	Play_Park	Child_Race4	0.2290527
## 23	Play_Park	P1_Race4	0.2212615
## 24	Play_Park	Child_Age_Gp	0.1987409
## 25	Play_Park	P1_Degree	0.1670081
## 26	Play_Park	P1_Workstatus	0.1571843
## 27	Play_Park	City	0.1531223
## 28	Play_Daycare	P1_Workstatus	0.2820065
## 29	Play_Daycare	P1_Degree	0.2754643
## 30	Play_Daycare	Household_income	0.2292277
## 31	Play_Babysitter_Home	Household_income	0.1665087
## 32	Play_Relative_Home	P1_Race4	0.1907730
## 33	Play_Other	City	0.3153764
## 34	Play_Other	Household_income	0.2114545
## 35	Play_Other	P1_Age	0.1755980
## 36	Play_Other	P1_Race4	0.1711719
## 37	Play_Other	Child_Age_Gp	0.1707868
## 38	Play_Other	P1_Degree	0.1670949
## 39	Sandbox_Home	City	0.2723574
## 40	Sandbox_Home	P1_Age	0.2004847
## 41	Sandbox_Home	Child_Race4	0.1604202
## 42	Sandbox_Home	P1_Degree	0.1550701
## 43	Sandbox_Park	Child_Age_Gp	0.1748980
## 44	Sandbox_Park	P1_Workstatus	0.1532947
## 45	Sandbox_Daycare	Household_income	0.1919288
## 46	Sandbox_Daycare	P1_Degree	0.1535020
## 47	Sandbox_Daycare	City	0.1525207
## 48	Sandbox_Relative	Household_income	0.1914640
## 49	Sandbox_Other	P1_Race4	0.2072023
## 50	Sandbox_Other	Child_Race4	0.1999710

```

#write.csv(result_tbl_0.15, 'Cramers V above 0.15.csv')

## Threshold 0.20
result_tbl_0.20 <- identify_strongest_features_table (data, list1, list2, threshold = 0.20)
## print(result_tbl_0.15)
write.csv(result_tbl_0.20, 'Cramers V above 0.20.csv')
# View the result
#
# results1 <- cbind(result_tbl$exposure_var, result_tbl$demographic_var,
#                   round(result_tbl$cramer_v, digits=4))
#
# results1 <- data.frame(
#   exposure_var = result_tbl$exposure_var,
#   demographic_var = result_tbl$demographic_var,
#   cramer_v = round(result_tbl$cramer_v,digits=4) ,
#   stringsAsFactors = FALSE
# )

# View the results

print(result_tbl_0.20)

```

```

##      exposure_var  demographic_var  cramer_v
## 1      Mouthing      Child_Age_Gp 0.4500148
## 2      Pacifier      Child_Age_Gp 0.3760177
## 3      Pacifier_Use    Child_Age_Gp 0.2240739
## 4      Pacifier_Wash    Child_Age_Gp 0.2361997
## 5      Blanket        Child_Age_Gp 0.2264065
## 6      Sucking        Child_Age_Gp 0.2449472
## 7      Handwash       Child_Age_Gp 0.2044679
## 8  Hours_Away_weekend  Single_Parent 0.2156966
## 9      Play_Park      Child_Race4 0.2290527
## 10     Play_Park      P1_Race4 0.2212615
## 11     Play_Daycare    P1_Workstatus 0.2820065
## 12     Play_Daycare    P1_Degree 0.2754643
## 13     Play_Daycare    Household_income 0.2292277
## 14     Play_Other      City 0.3153764
## 15     Play_Other      Household_income 0.2114545
## 16     Sandbox_Home    City 0.2723574
## 17     Sandbox_Home    P1_Age 0.2004847
## 18     Sandbox_Other    P1_Race4 0.2072023

```

Creating a visualization using the networkD3 package to

create an interactive Sankey plot.

Sankey Plot

```
# Install and Load required package
## install.packages("networkD3")

library(networkD3)
library(htmltools)
# Prepare the data

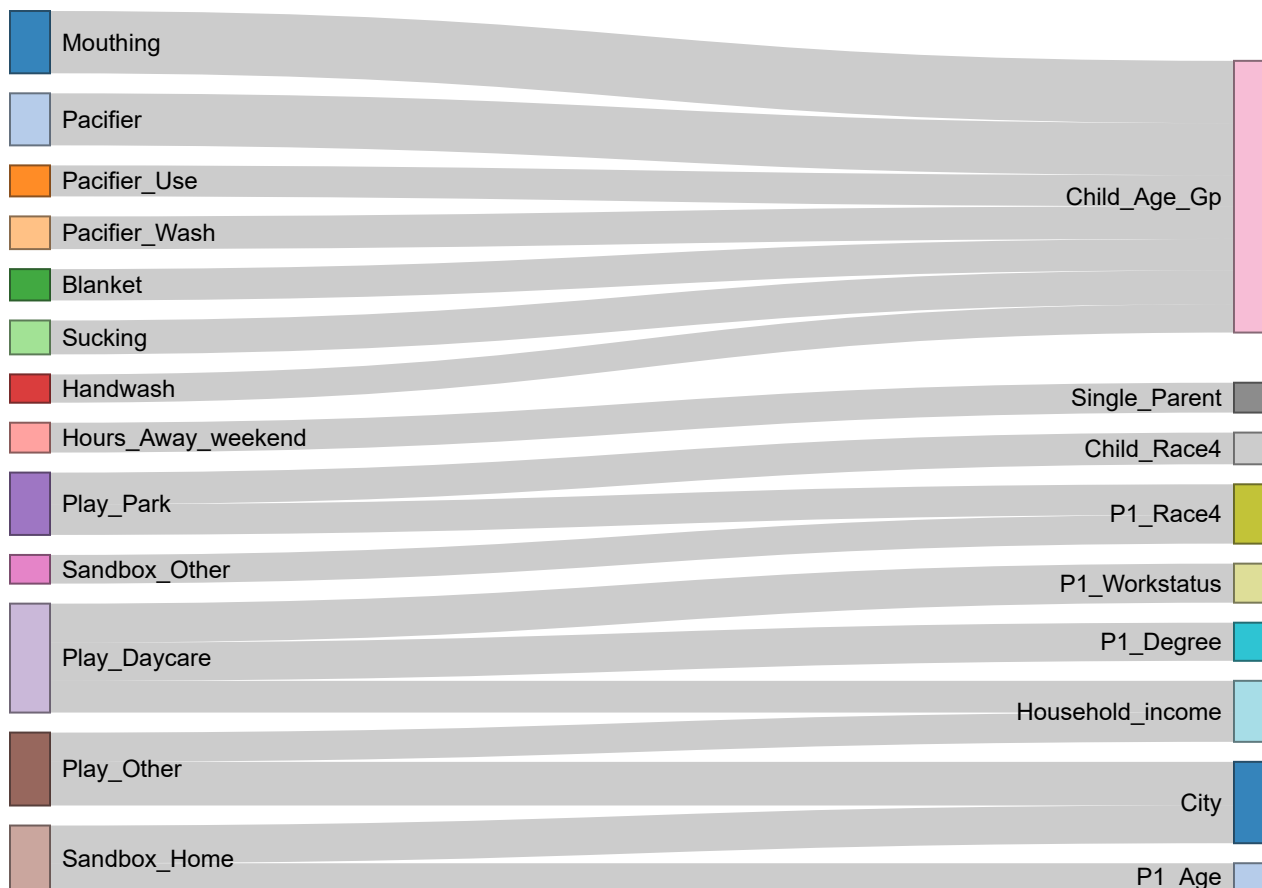
results1= result_tbl_0.20

nodes <- unique(c(results1$exposure_var, results1$demographic_var))
nodes_df <- data.frame(name = nodes)

links_df <- data.frame(
  source = match(results1$exposure_var, nodes) - 1,
  target = match(results1$demographic_var, nodes) - 1,
  value = results1$cramer_v
)

# Create the Sankey plot
sankey <- sankeyNetwork(
  Links = links_df,
  Nodes = nodes_df,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 20
#   main = "Sankey's Plot Exposure vs. Demographic Variable Pairs
#   with Cramer's V >=0.15"
)

# Render the plot
sankey
```



```
# Render with a title
html_print(tagList(
  h2("Sankey's Plot: Exposure & Demographic Variable Pairs \n with Cramer's V ≥ 0.20",
    style = "text-align:center; font-weight:bold;"),
  sankey
))
# Install and load the required package
##install.packages("htmlwidgets")
library(htmlwidgets)
```

```
##
## Attaching package: 'htmlwidgets'
```

```
## The following object is masked from 'package:networkD3':
##
## JS
```

```
# Save the Sankey plot as an HTML file
saveNetwork(sankey, "sankey_plot.html")
```

Heat Map

```
library(ggplot2)
library(reshape2)
```

```
##
## Attaching package: 'reshape2'
```

```
## The following object is masked from 'package:tidyr':
##
## smiths
```

```
list1= Exposure_Vars
list2= DemoVars

result_tbl_All <- identify_strongest_features_table (data, list1, list2, threshold = 0)

# dim(result_tbl_All )##289,3
# print(result_tbl_All[c(288,289),])
# View the result

result_tbl_All=result_tbl_All[-289,]

# Create a pivot table for heatmap
heatmap_data <- result_tbl_All %>%
pivot_wider(names_from = demographic_var, values_from = cramer_v)
# Convert to matrix
heatmap_matrix <- as.matrix(heatmap_data[,-1])
rownames(heatmap_matrix) <- heatmap_data$exposure_var
print(heatmap_data)
```

```
## # A tibble: 23 × 13
##   exposure_var      Child_Age_Gp P1_Age P1_Degree Single_Parent P1_Race4
##   <chr>          <dbl> <dbl>    <dbl>      <dbl>    <dbl>
## 1 Mouthing      0.450 0.0987  0.0880      0.0861  0.0859
## 2 Pacifier      0.376 0.0961  0.0961      0.0436  0.0923
## 3 Pacifier_Use  0.224 0.104   0.0862      0.122   0.113
## 4 Pacifier_Wash 0.236 0.148   0.109       0.156   0.0992
## 5 Blanket       0.226 0.183   0.0845      0.0187  0.0898
## 6 Blanket_Wash  0.162 0.131   0.0918      0.0739  0.103
## 7 Stuffed_Animal 0.157 0.0497  0.104       0.0165  0.0910
## 8 Stuffed_Animal_Wash 0.134 0.109   0.114       0.0935  0.0948
## 9 Sucking       0.245 0.0600  0.0984      0.0478  0.0839
## 10 Handwash     0.204 0.101   0.105       0.0706  0.110
## # i 13 more rows
## # i 7 more variables: Household_income <dbl>, Child_Race4 <dbl>,
## #   P1_Workstatus <dbl>, Child_Sex <dbl>, P1_Sex <dbl>, P1_Ethnicity <dbl>,
## #   City <dbl>
```

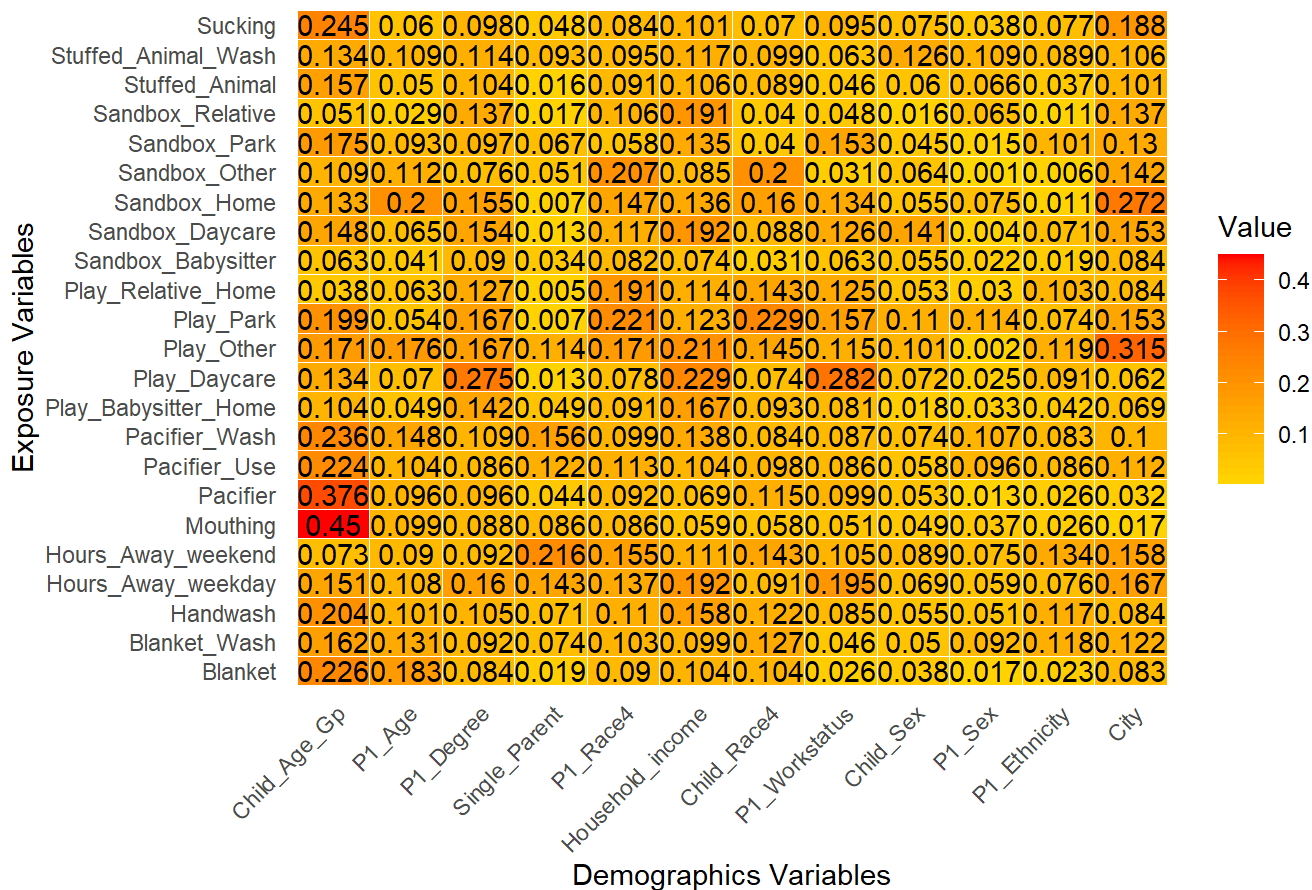
```

# Convert heatmap data to a Long format
heatmap_long <- melt(heatmap_data, id.vars = "exposure_var")

# Plot the heatmap
ggplot(heatmap_long, aes(x = variable, y = exposure_var, fill = value)) +
  geom_tile(color = "white") + # Adds grid lines
  geom_text(aes(label = round(value, 3)), color = "black") + # Add annotations
  scale_fill_gradientn(colors = c("gold", "red")) + # Define color scale
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), # Rotate x-Labels
    panel.grid = element_blank() ) +
  labs(
    title = "Heatmap Showing Cramer's V for Each Pair of Variables",
    x = "Demographics Variables",
    y = "Exposure Variables",
    fill = "Value"
  )

```

Heatmap Showing Cramer's V for Each Pair of Variables



```

# Save as PNG
ggsave("heatmap3.png")

```

```
## Saving 7 x 5 in image
```

Network Graph

```
# Install and load required package
#install.packages("igraph")
library(igraph)
```

```
##
## Attaching package: 'igraph'
```

```
## The following object is masked from 'package:DescTools':
##
##      %c%
```

```
## The following objects are masked from 'package:lubridate':
##
##      %--%, union
```

```
## The following objects are masked from 'package:purrr':
##
##      compose, simplify
```

```
## The following object is masked from 'package:tidyr':
##
##      crossing
```

```
## The following object is masked from 'package:tibble':
##
##      as_data_frame
```

```
## The following object is masked from 'package:rlang':
##
##      is_named
```

```
## The following objects are masked from 'package:dplyr':
##
##      as_data_frame, groups, union
```

```
## The following objects are masked from 'package:stats':
##
##      decompose, spectrum
```

```
## The following object is masked from 'package:base':
##
##      union
```

```
result_tbl_0.20 <- identify_strongest_features_table (data, list1, list2, threshold = 0.20)
## only 7 pairs above .25
print(result_tbl_0.20)
```

##	exposure_var	demographic_var	cramer_v
## 1	Mouthing	Child_Age_Gp	0.4500148
## 2	Pacifier	Child_Age_Gp	0.3760177
## 3	Pacifier_Use	Child_Age_Gp	0.2240739
## 4	Pacifier_Wash	Child_Age_Gp	0.2361997
## 5	Blanket	Child_Age_Gp	0.2264065
## 6	Sucking	Child_Age_Gp	0.2449472
## 7	Handwash	Child_Age_Gp	0.2044679
## 8	Hours_Away_weekend	Single_Parent	0.2156966
## 9	Play_Park	Child_Race4	0.2290527
## 10	Play_Park	P1_Race4	0.2212615
## 11	Play_Daycare	P1_Workstatus	0.2820065
## 12	Play_Daycare	P1_Degree	0.2754643
## 13	Play_Daycare	Household_income	0.2292277
## 14	Play_Other	City	0.3153764
## 15	Play_Other	Household_income	0.2114545
## 16	Sandbox_Home	City	0.2723574
## 17	Sandbox_Home	P1_Age	0.2004847
## 18	Sandbox_Other	P1_Race4	0.2072023

```
#results2 <-result_tbl_0.20[-17,]
```

```
# Create the graph object
```

```
edges <- data.frame(
  from = results2$exposure_var,
  to = results2$demographic_var,
  weight = results2$cramer_v
)
```

```
## Error: object 'results2' not found
```

```
# Create the graph
```

```
g <- graph_from_data_frame(edges, directed = FALSE)
```

```
## Error in if (ncol(d) < 2) {: argument is of length zero
```

```
# Plot the network graph
```

```
plot(
  g,
  edge.width = E(g)$weight *10, # Scale edge thickness by Cramér's V
  vertex.size = 5,
  vertex.label.cex = .85,
  vertex.label.dist = .85,
  main = "Exposure vs. Demographic Variable Pairs
with Cramer's V >= 0.20"
)
```

```
## Error: object 'g' not found
```

```
## Trying a less crowded version
# Use Fruchterman-Reingold layout for better spacing
layout <- layout_with_fr(g, dim = 2)
```

```
## Error: object 'g' not found
```

```
# Adjust aspect ratio and save the plot
png("network_plot2.png", width = 14, height = 8, units = "in", res = 800)

# Plot the graph with the custom layout
plot(
  g,
  layout = layout,          # Use the custom layout
  edge.width = E(g)$weight * 15, # Scale edge thickness by Cramer's V
  vertex.size = 5.0,         # Adjust node size
  vertex.label.cex = .75,    # Adjust label size
  vertex.label.dist = .85,   # Add space between labels and nodes
  vertex.label.color = "blue", # Set label color
  main = "Pairs of Exposure & Demographics Variables \n with Cramer's V >= 0.20"
)
```

```
## Error: object 'g' not found
```

Complete analysis table

```
# Required Libraries
library(dplyr)
library(DescTools)
library(readr)

# Define the function
Cv_Feature_Selection <- function(data, exposure_vars, demographic_vars,
                                output_csv = "Strongest_Features.csv") {

  # Initialize an empty data frame to store results
  results_df <- data.frame(
    Exposure_Variable = character(),
    Demographics_Variable = character(),
    Chi_Squared_Value = numeric(),
    Chi_Squared_P_Value = numeric(),
    Cramers_V = numeric(),
    stringsAsFactors = FALSE
  )

  # Loop through each combination of exposure and demographic variables
  for (exposure in exposure_vars) {
    for (demographic in demographic_vars) {

      # Create a contingency table
      freq_table <- table(data[[exposure]], data[[demographic]])

      # Perform Chi-squared test
      chi_test_result <- tryCatch({
        chisq.test(freq_table, simulate.p.value = TRUE, B = 10000)
      }, error = function(e) NULL)

      # Compute Cramér's V
      cramers_v_value <- tryCatch(CramerV(freq_table, correct = TRUE), error = function(e) NA)

      # Append results to the data frame
      results_df <- rbind(results_df, data.frame(
        Exposure_Variable = exposure,
        Demographics_Variable = demographic,
        Chi_Squared_Value = ifelse(!is.null(chi_test_result), chi_test_result$statistic, NA),
        Chi_Squared_P_Value = ifelse(!is.null(chi_test_result), chi_test_result$p.value, NA),
        Cramers_V = cramers_v_value
      ))
    }
  }

  # Return the results as a data frame
  return(results_df)

  # Save the results to a CSV file

  print(results_df)
  write_csv(Table_Summary, "Table_Summary.csv")
}
```


Get Table of CV and P-value for selected features

```
Exposure_Vars_1=Exposure_Vars [1:12]
Exposure_Vars_2=Exposure_Vars [13:24]
# length(Exposure_Vars_12);length(DemoVars)

Table_Summary_1 <- Cv_Feature_Selection(data = data,
                                       Exposure_Vars_1,
                                       DemoVars)
## write_csv(Table_Summary_1, "Table_Summary_1.csv")

Table_Summary_2 <- Cv_Feature_Selection(data = data,
                                       exposure_vars = Exposure_Vars_2,
                                       demographic_vars = DemoVars)
## write_csv(Table_Summary_2, "Table_Summary_2.csv")

Table_Summary=rbind(Table_Summary_1,Table_Summary_2)
names(Table_Summary)
```

```
## [1] "Exposure_Variable"      "Demographics_Variable" "Chi_Squared_Value"
## [4] "Chi_Squared_P_Value"    "Cramers_V"
```

```
write_csv(Table_Summary, "Table_Summary_ALL_FINAL.csv")

library(dplyr)

# Filter and remove duplicates
## Select pairs with CV>=0.20 and p-value<0.05/188
Selected_Features <- Table_Summary %>%
  filter(Chi_Squared_P_Value < 0.00026, Cramers_V >= 0.20) %>%
  distinct()

# Print the filtered table
# print(Selected_Features)
write_csv(Selected_Features, "Selected_Features_Table_CV_ge0.2_pval_l_0.05_March_24.csv")

## Omit row 7 b/c P1_Race4 is similar to Child_Race4
Selected_Features_12=Selected_Features[-7,]

list1 <- Selected_Features$Exposure_Variable
list2 <- Selected_Features$Demographics_Variable
cbind(list1,list2)
```

```
##      list1      list2
## [1,] "Mouthing"  "Child_Age_Gp"
## [2,] "Pacifier"  "Child_Age_Gp"
## [3,] "Pacifier_Use" "Child_Age_Gp"
## [4,] "Pacifier_Wash" "Child_Age_Gp"
## [5,] "Blanket"    "Child_Age_Gp"
## [6,] "Sucking"   "Child_Age_Gp"
## [7,] "Play_Park"  "P1_Race4"
## [8,] "Play_Park"  "Child_Race4"
## [9,] "Play_Daycare" "P1_Degree"
## [10,] "Play_Daycare" "P1_Workstatus"
## [11,] "Play_Daycare" "Household_income"
## [12,] "Play_Other"  "City"
## [13,] "Sandbox_Home" "City"
```

Analysis of the selected Sketch 12 selected pairs of

features

```
generate_frequency_tables <- function(data, selected_features, output_directory = "Frequency_Tables") {

  if (!dir.exists(output_directory)) {
    dir.create(output_directory)
  }

  all_tables <- list()

  for (i in 1:nrow(selected_features)) {
    exposure_var <- selected_features$Exposure_Variable[i]
    demo_var <- selected_features$Demographics_Variable[i]

    # Check if both variables exist in the dataset
    if (!(exposure_var %in% colnames(data)) | !(demo_var %in% colnames(data))) {
      message("Skipping ", exposure_var, " and ", demo_var, " (not found in dataset).")
      next
    }

    # Create frequency table
    freq_table <- table(data[[exposure_var]], data[[demo_var]])
    rel_freq_table <- prop.table(freq_table) # Relative frequency

    # Convert tables to dataframes
    freq_df <- as.data.frame.matrix(freq_table)
    rel_freq_df <- as.data.frame.matrix(rel_freq_table)

    # Store tables in a List for saving
    all_tables[[paste(exposure_var, demo_var, "Frequency")]] <- freq_df
    all_tables[[paste(exposure_var, demo_var, "Relative Frequency")]] <- rel_freq_df
  }

  # Save all tables to CSV
  for (table_name in names(all_tables)) {
    write.csv(all_tables[[table_name]], file = paste0(output_directory, "/", table_name, ".csv"), row.names
= TRUE)
  }

  message("Saved frequency tables in: ", output_directory)
}
```

Plots of selected features

```
library(ggplot2)
library(ggmosaic)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
```

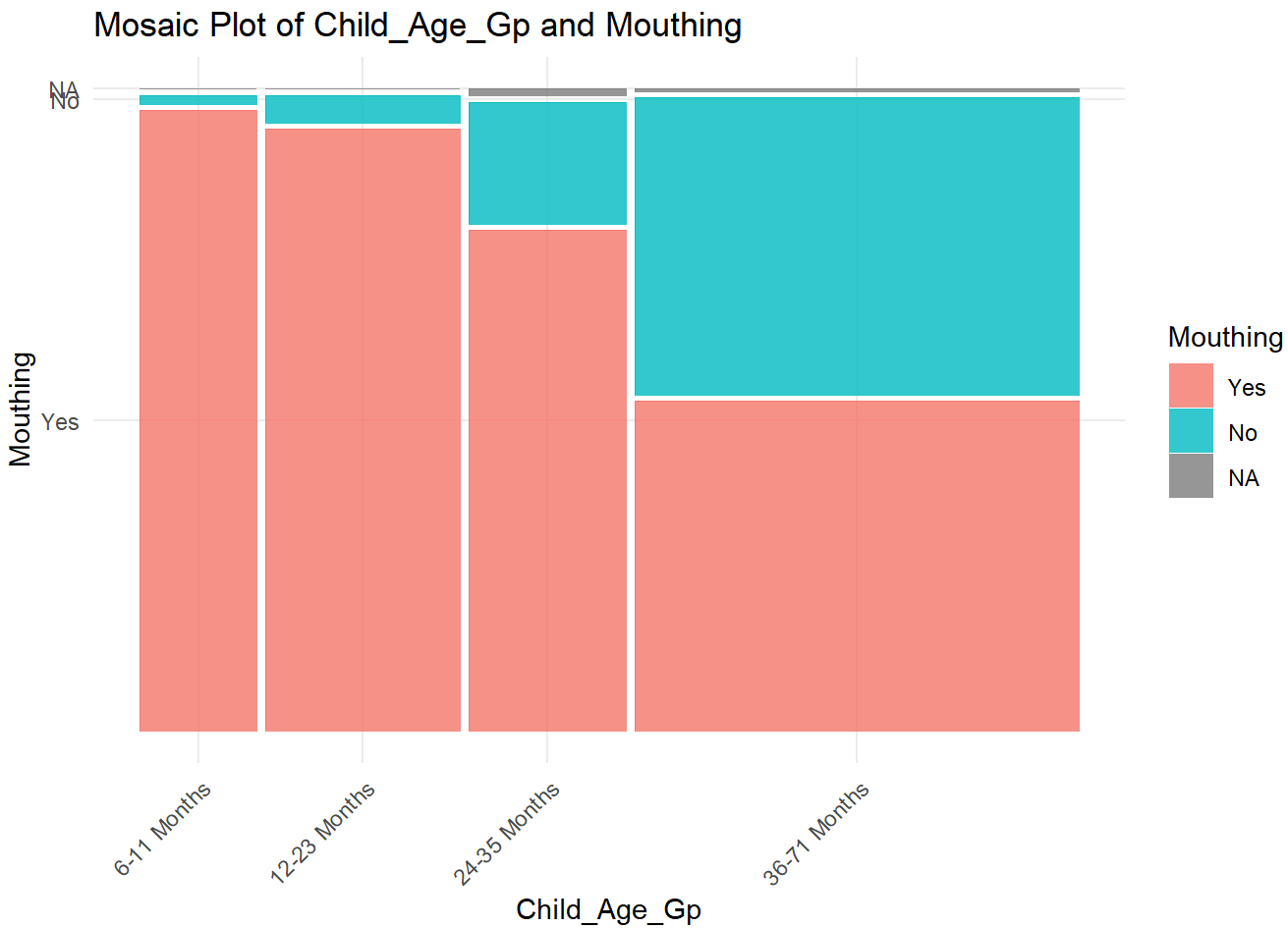
```
## The following object is masked from 'package:dplyr':  
##  
##      combine
```

```
# Initialize an empty list to store plots  
mosaic_plots <- list()  
list1 <- Selected_Features$Exposure_Variable  
list2 <- Selected_Features$Demographics_Variable  
cbind(list1,list2)
```

```
##      list1      list2  
## [1,] "Mouthing"    "Child_Age_Gp"  
## [2,] "Pacifier"    "Child_Age_Gp"  
## [3,] "Pacifier_Use" "Child_Age_Gp"  
## [4,] "Pacifier_Wash" "Child_Age_Gp"  
## [5,] "Blanket"     "Child_Age_Gp"  
## [6,] "Sucking"     "Child_Age_Gp"  
## [7,] "Play_Park"   "P1_Race4"  
## [8,] "Play_Park"   "Child_Race4"  
## [9,] "Play_Daycare" "P1_Degree"  
## [10,] "Play_Daycare" "P1_Workstatus"  
## [11,] "Play_Daycare" "Household_income"  
## [12,] "Play_Other"  "City"  
## [13,] "Sandbox_Home" "City"
```

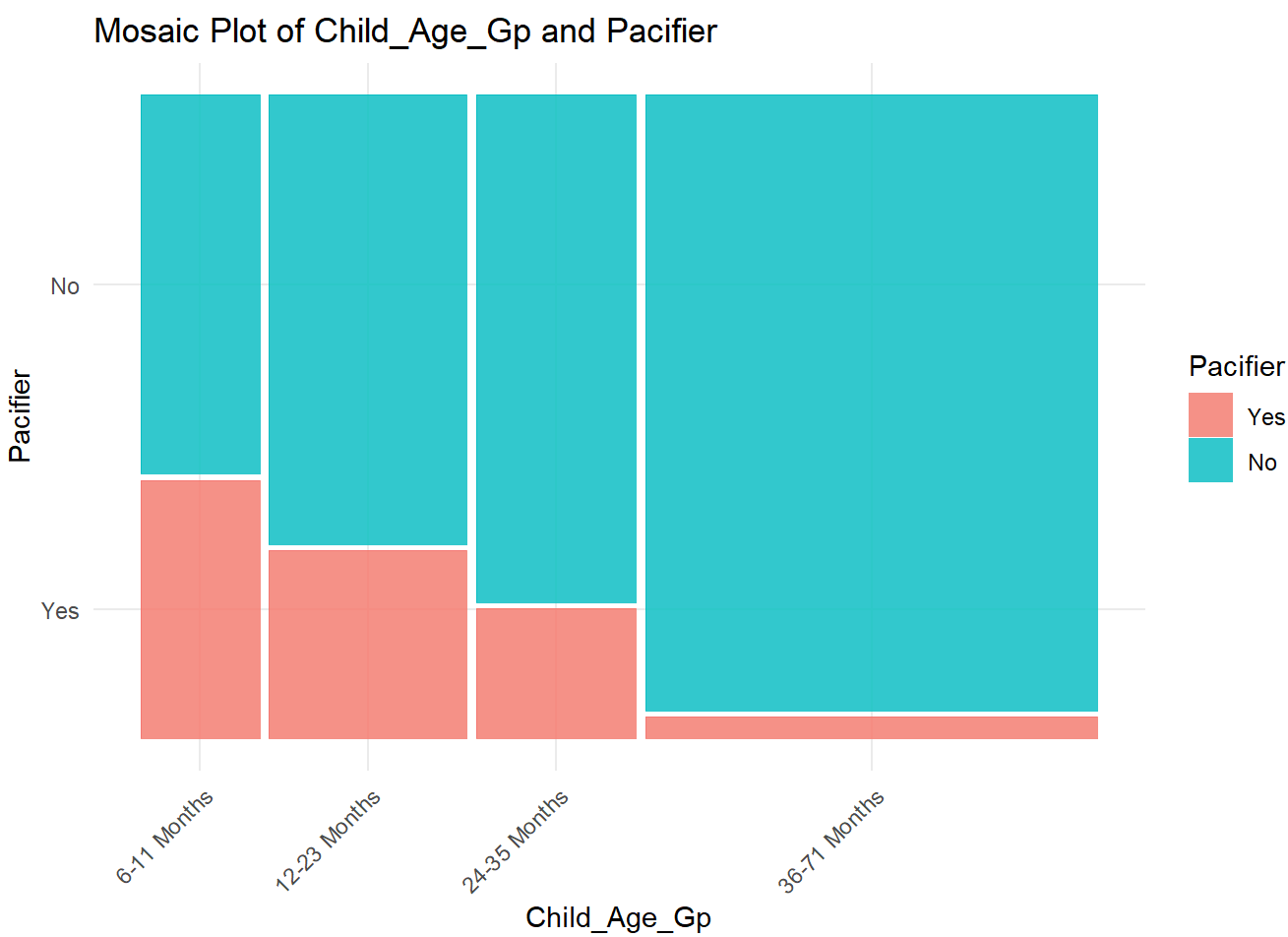
```
# Loop through 12 variable pairs  
for (i in 1:12) {  
  var1 <- list1[i]  
  var2 <- list2[i]  
  
  cat("\n\n----- Analysis for", var1, "and", var2, "-----\n")  
  
  # Generate a mosaic plot using Analyze_Bivarite (assuming it returns a ggplot object)  
  plot <- Analyze_Bivarite(data, var1, var2)  
  
  # Ensure plot is valid before adding it to the list  
  if (!is.null(plot) && inherits(plot, "ggplot")) {  
    mosaic_plots <- append(mosaic_plots, list(plot))  
  }  
}
```

```
##
##
## ----- Analysis for Mouthing and Child_Age_Gp -----
## Bivariate Analysis of Mouthing and Child_Age_Gp
##
## Frequency Table:
##   Mouthing 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##       Yes           68           110           74           138
##       No            1            5           18           124
##       <NA>          0            0            1            1
##
## Relative Frequency Table:
##   Mouthing 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##       Yes           98.6%          95.7%          79.6%          52.5% 72.2%
##       No            1.4%           4.3%          19.4%          47.1% 27.4%
##       <NA>          0.0%           0.0%           1.1%           0.4% 0.4%
##
## Cramér's V:  0.3124662
```



```
## Saving 7 x 5 in image
```

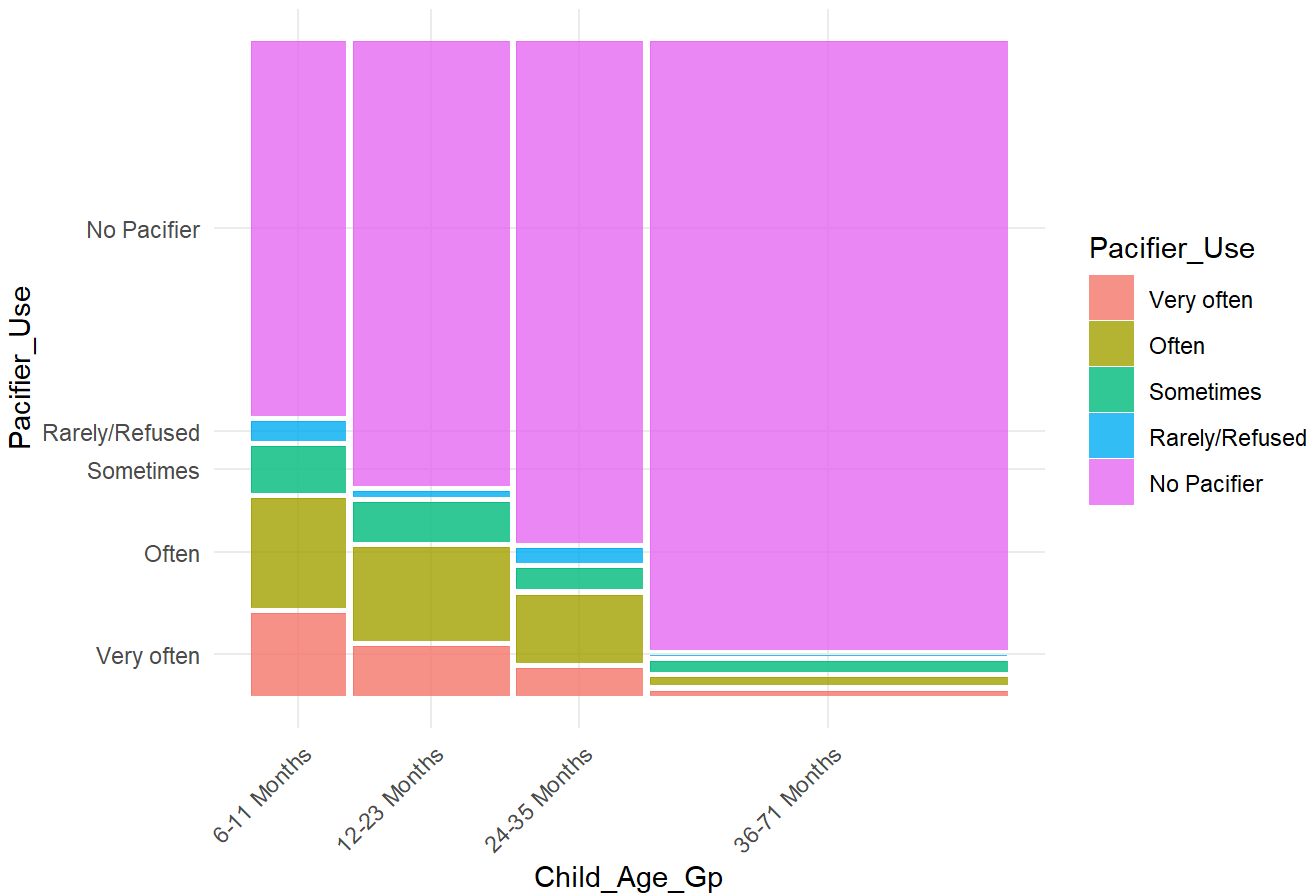
```
##
##
## ----- Analysis for Pacifier and Child_Age_Gp -----
## Bivariate Analysis of Pacifier and Child_Age_Gp
##
## Frequency Table:
##   Pacifier 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##       Yes         28         34         19         9
##       No         41         81         74        254
##
## Relative Frequency Table:
##   Pacifier 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##       Yes         40.6%         29.6%         20.4%         3.4% 16.7%
##       No         59.4%         70.4%         79.6%         96.6% 83.3%
##
## Cramér's V:  0.3688847
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Pacifier_Use and Child_Age_Gp -----
## Bivariate Analysis of Pacifier_Use and Child_Age_Gp
##
## Frequency Table:
##   Pacifier_Use 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##   Very often          9          9          4          2
##   Often              12         17         10          3
##   Sometimes          5          7          3          4
##   Rarely/Refused      2          1          2          0
##   No Pacifier        41         81         74         254
##
## Relative Frequency Table:
##   Pacifier_Use 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##   Very often    13.0%      7.8%        4.3%         0.8%    4.4%
##   Often         17.4%     14.8%       10.8%         1.1%    7.8%
##   Sometimes     7.2%      6.1%        3.2%         1.5%    3.5%
##   Rarely/Refused 2.9%      0.9%        2.2%         0.0%    0.9%
##   No Pacifier   59.4%     70.4%       79.6%        96.6%   83.3%
##
## Cramér's V: 0.2074306
```

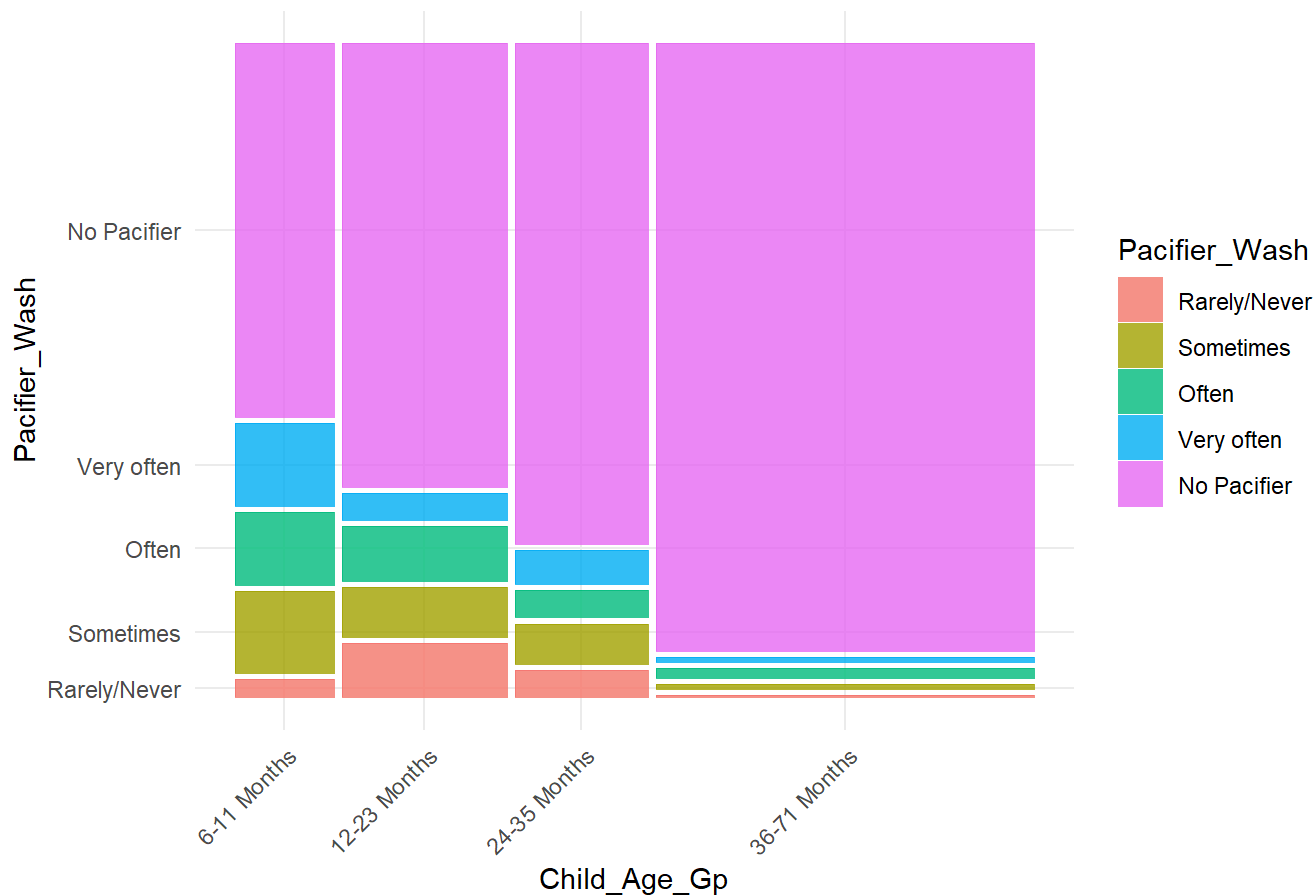
Mosaic Plot of Child_Age_Gp and Pacifier_Use



```
## Saving 7 x 5 in image
```

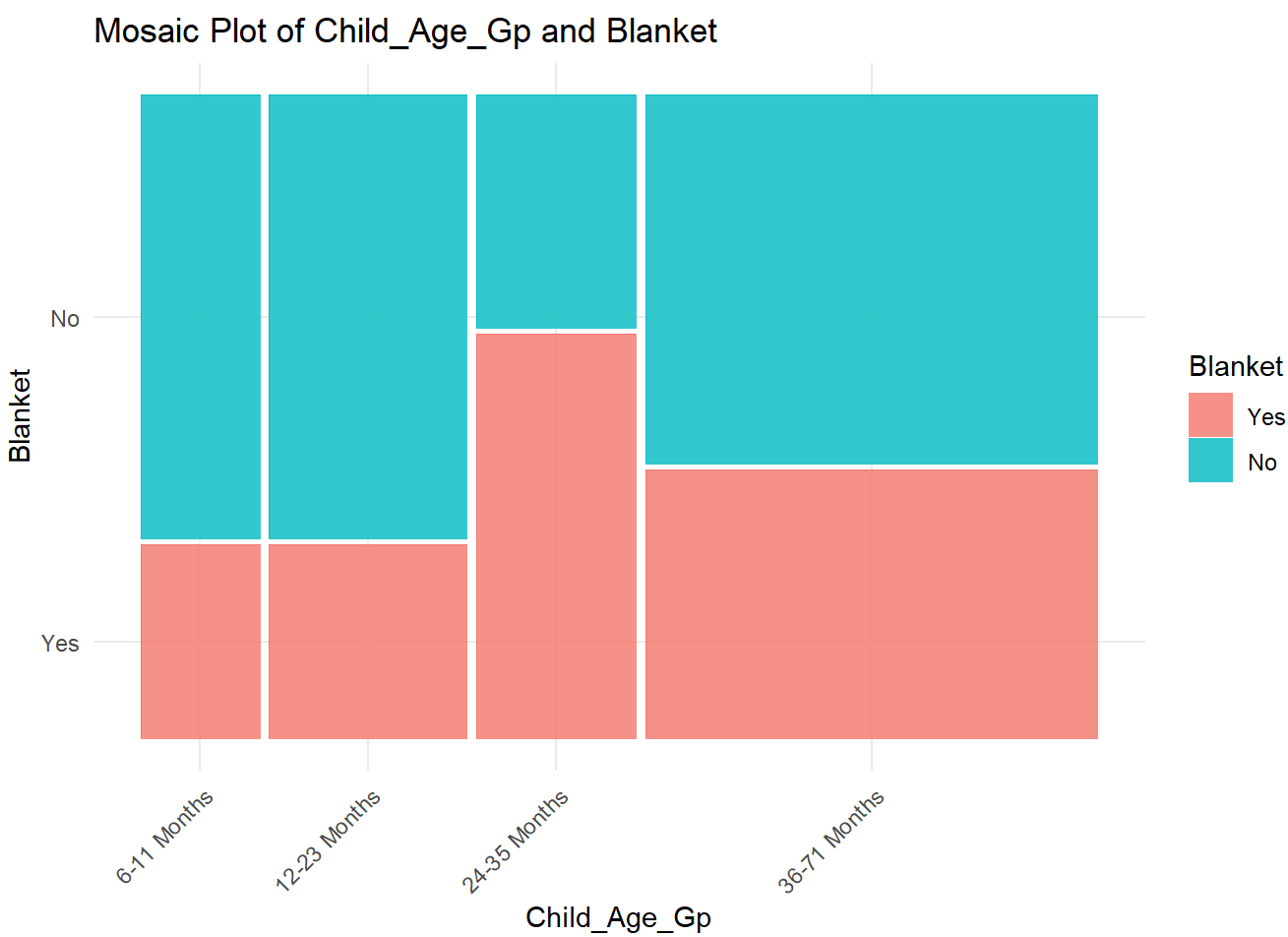
```
##
##
## ----- Analysis for Pacifier_Wash and Child_Age_Gp -----
## Bivariate Analysis of Pacifier_Wash and Child_Age_Gp
##
## Frequency Table:
## Pacifier_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months
## Rarely/Never      2          10          4          1
## Sometimes         9          9          6          2
## Often             8          10          4          4
## Very often        9          5          5          2
## No Pacifier       41         81         74         254
##
## Relative Frequency Table:
## Pacifier_Wash 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
## Rarely/Never      2.9%      8.7%      4.3%      0.4%  3.1%
## Sometimes        13.0%      7.8%      6.5%      0.8%  4.8%
## Often            11.6%      8.7%      4.3%      1.5%  4.8%
## Very often       13.0%      4.3%      5.4%      0.8%  3.9%
## No Pacifier      59.4%     70.4%     79.6%     96.6% 83.3%
##
## Cramér's V:  0.2205445
```

Mosaic Plot of Child_Age_Gp and Pacifier_Wash



Saving 7 x 5 in image

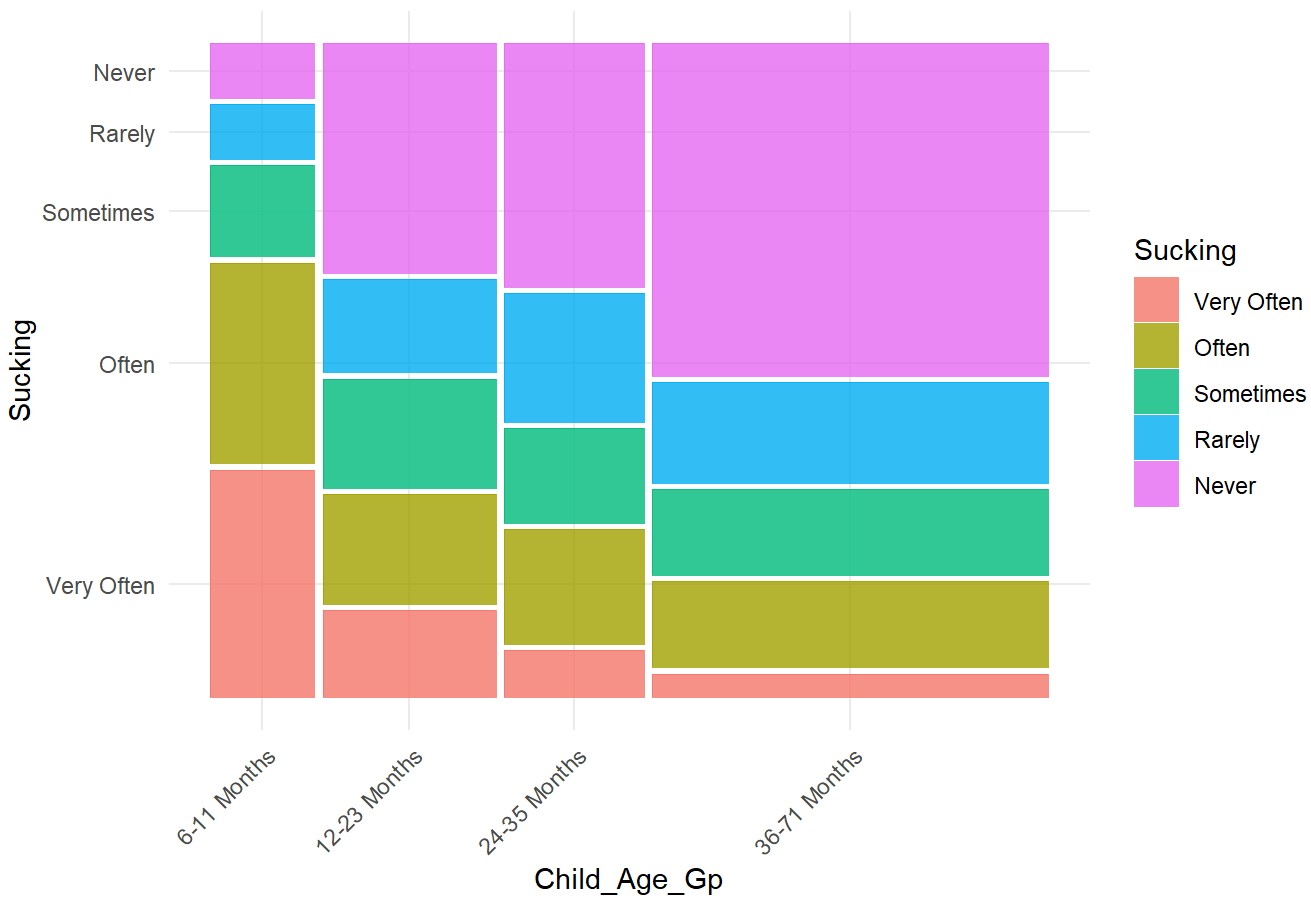

```
##
##
## ----- Analysis for Blanket and Child_Age_Gp -----
## Bivariate Analysis of Blanket and Child_Age_Gp
##
## Frequency Table:
## Blanket 6-11 Months 12-23 Months 24-35 Months 36-71 Months
##      Yes      21      35      59      111
##      No       48      80      34      152
##
## Relative Frequency Table:
## Blanket 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
##      Yes      30.4%      30.4%      63.4%      42.2% 41.9%
##      No       69.6%      69.6%      36.6%      57.8% 58.1%
##
## Cramér's V:  0.2139602
```



```
## Saving 7 x 5 in image
```

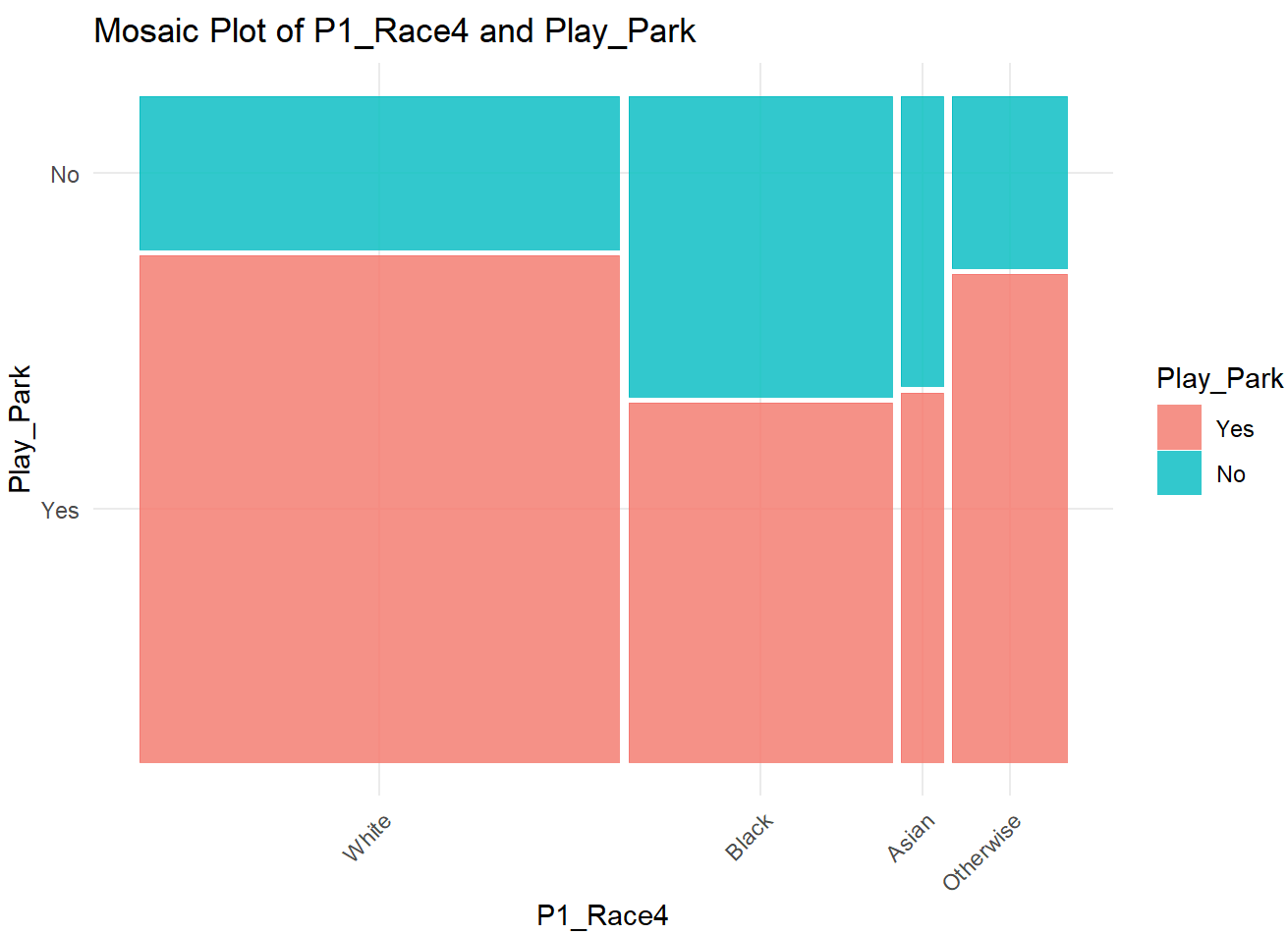
```
##
##
## ----- Analysis for Sucking and Child_Age_Gp -----
## Bivariate Analysis of Sucking and Child_Age_Gp
##
## Frequency Table:
##      Sucking 6-11 Months 12-23 Months 24-35 Months 36-71 Months
## Very Often      25         16          7          10
##      Often      22         20         17          36
## Sometimes      10         20         14          36
##      Rarely      6         17         19          42
##      Never       6         42         36         139
##
## Relative Frequency Table:
##      Sucking 6-11 Months 12-23 Months 24-35 Months 36-71 Months Total
## Very Often      36.2%      13.9%       7.5%       3.8% 10.7%
##      Often      31.9%      17.4%      18.3%      13.7% 17.6%
## Sometimes      14.5%      17.4%      15.1%      13.7% 14.8%
##      Rarely      8.7%      14.8%      20.4%      16.0% 15.6%
##      Never       8.7%      36.5%      38.7%      52.9% 41.3%
##
## Cramér's V:  0.2299397
```

Mosaic Plot of Child_Age_Gp and Sucking



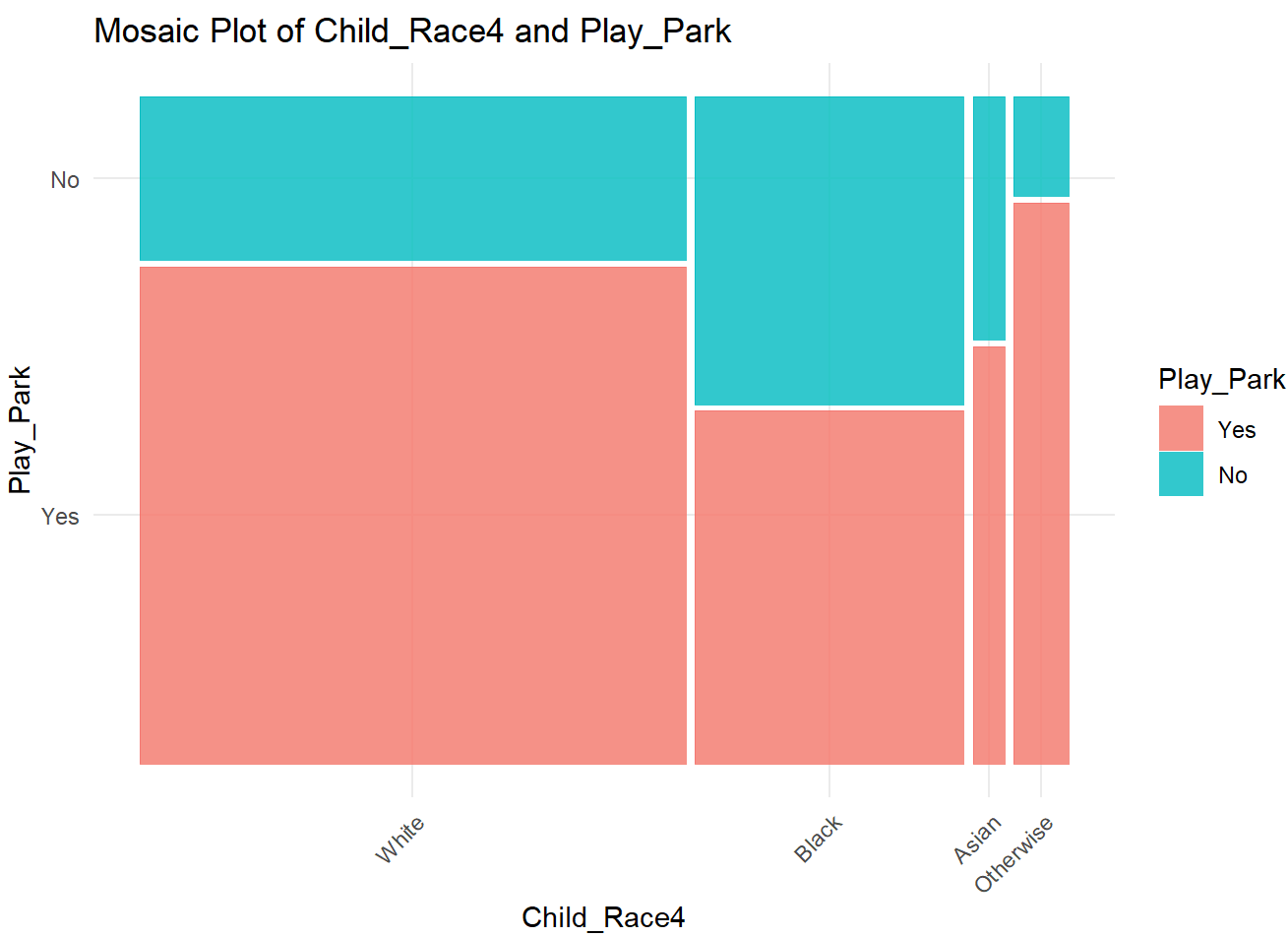
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and P1_Race4 -----
## Bivariate Analysis of Play_Park and P1_Race4
##
## Frequency Table:
##   Play_Park White Black Asian Otherwise
##         Yes   221    86    14        51
##         No    67    72    11        18
##
## Relative Frequency Table:
##   Play_Park White Black Asian Otherwise Total
##         Yes  76.7% 54.4% 56.0%    73.9% 68.9%
##         No   23.3% 45.6% 44.0%    26.1% 31.1%
##
## Cramér's V:  0.2084981
```



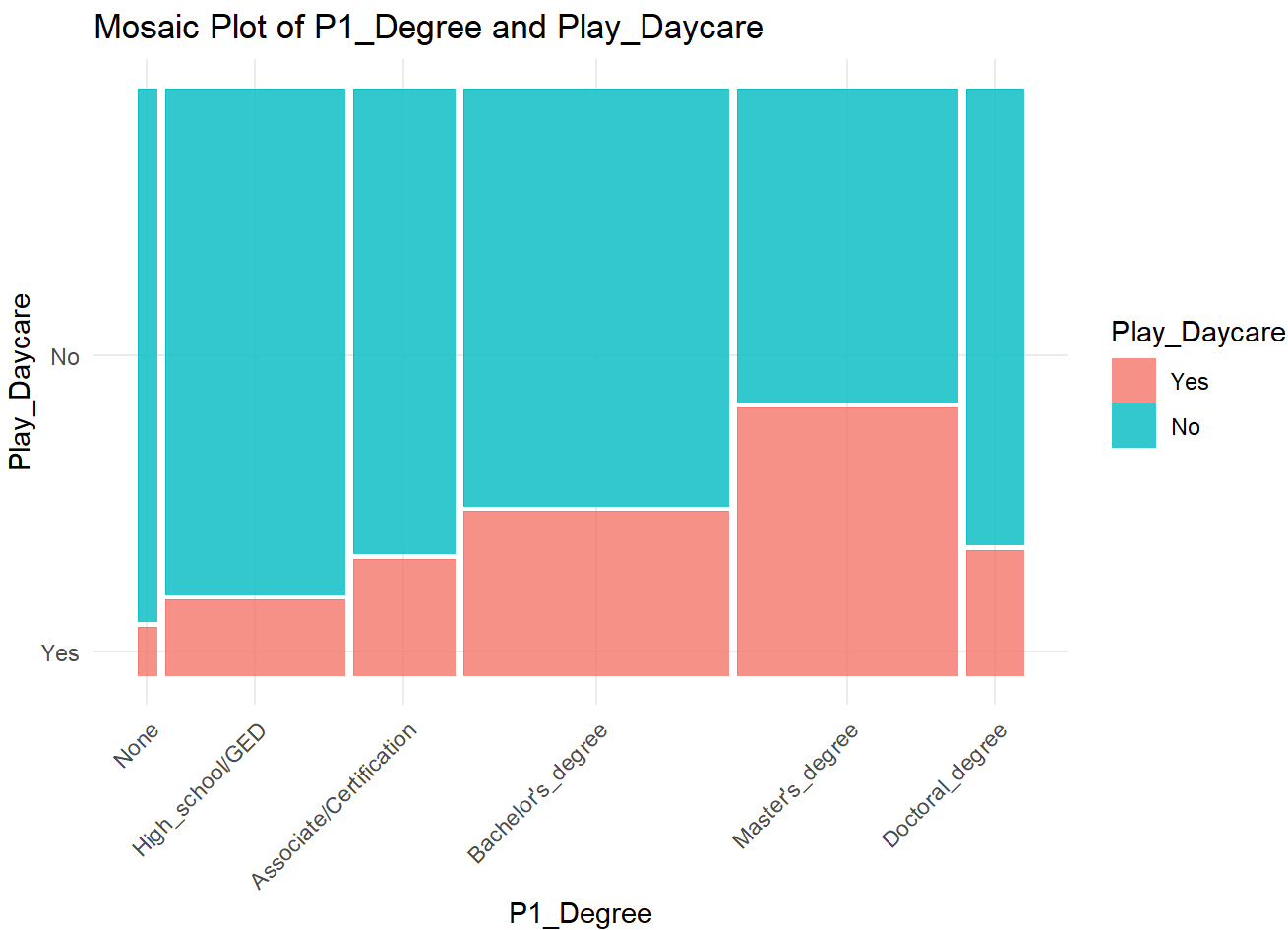
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Park and Child_Race4 -----
## Bivariate Analysis of Play_Park and Child_Race4
##
## Frequency Table:
##   Play_Park White Black Asian Otherwise
##         Yes   246    86    12      28
##         No    81    75     7       5
##
## Relative Frequency Table:
##   Play_Park White Black Asian Otherwise Total
##         Yes  75.2% 53.4% 63.2%   84.8% 68.9%
##         No   24.8% 46.6% 36.8%   15.2% 31.1%
##
## Cramér's V:  0.2167636
```



```
## Saving 7 x 5 in image
```

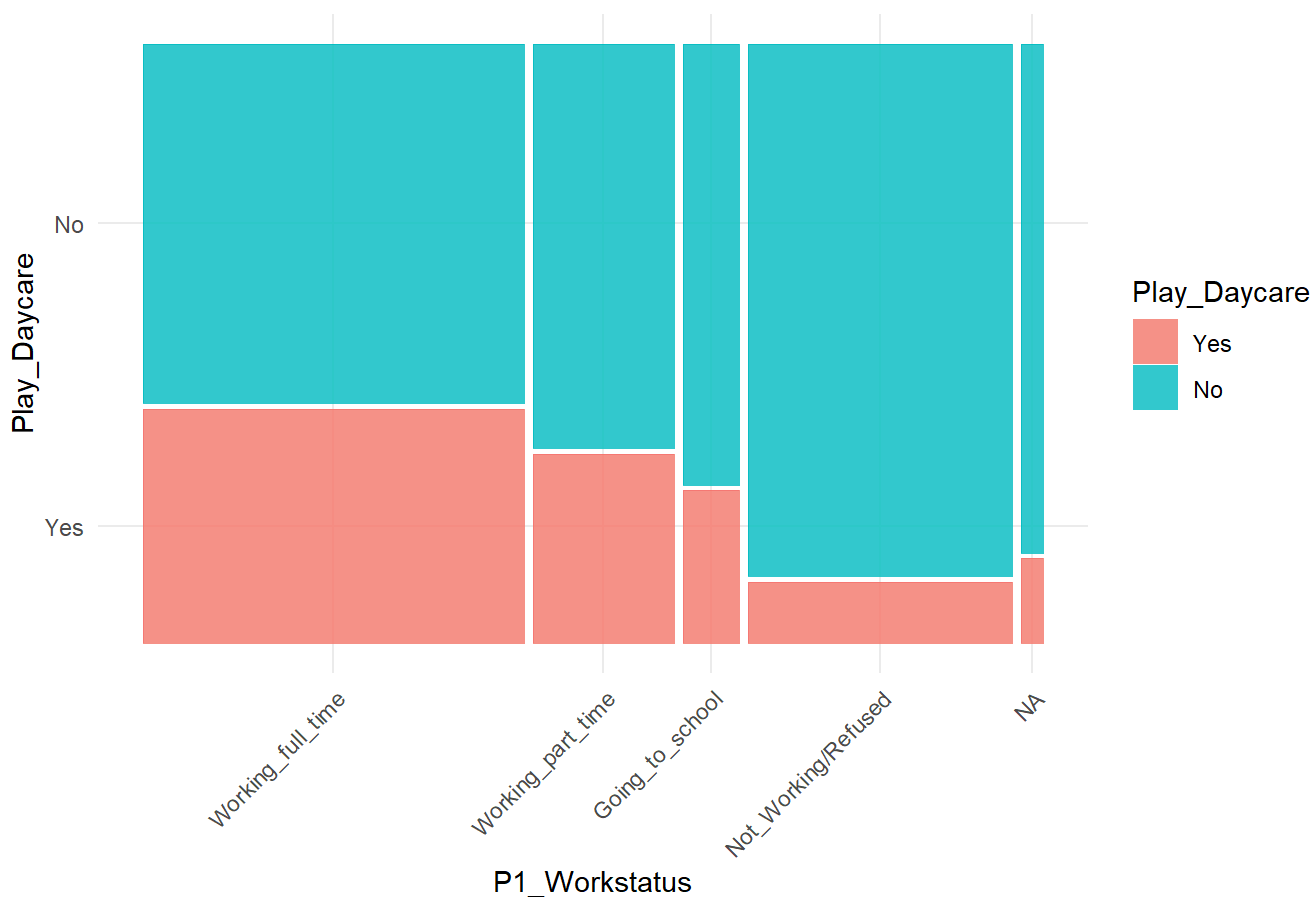
```
##
##
## ----- Analysis for Play_Daycare and P1_Degree -----
## Bivariate Analysis of Play_Daycare and P1_Degree
##
## Frequency Table:
## Play_Daycare None High_school/GED Associate/Certification Bachelor's_degree
##           Yes      1           15           13           48
##           No      11          100           52          122
## Master's_degree Doctoral_degree
##           65           8
##           76          29
##
## Relative Frequency Table:
## Play_Daycare None High_school/GED Associate/Certification Bachelor's_degree
##           Yes  8.3%          13.0%          20.0%          28.2%
##           No  91.7%          87.0%          80.0%          71.8%
## Master's_degree Doctoral_degree Total
##           46.1%          21.6% 27.8%
##           53.9%          78.4% 72.2%
##
## Cramér's V:  0.2583176
```



```
## Saving 7 x 5 in image
```

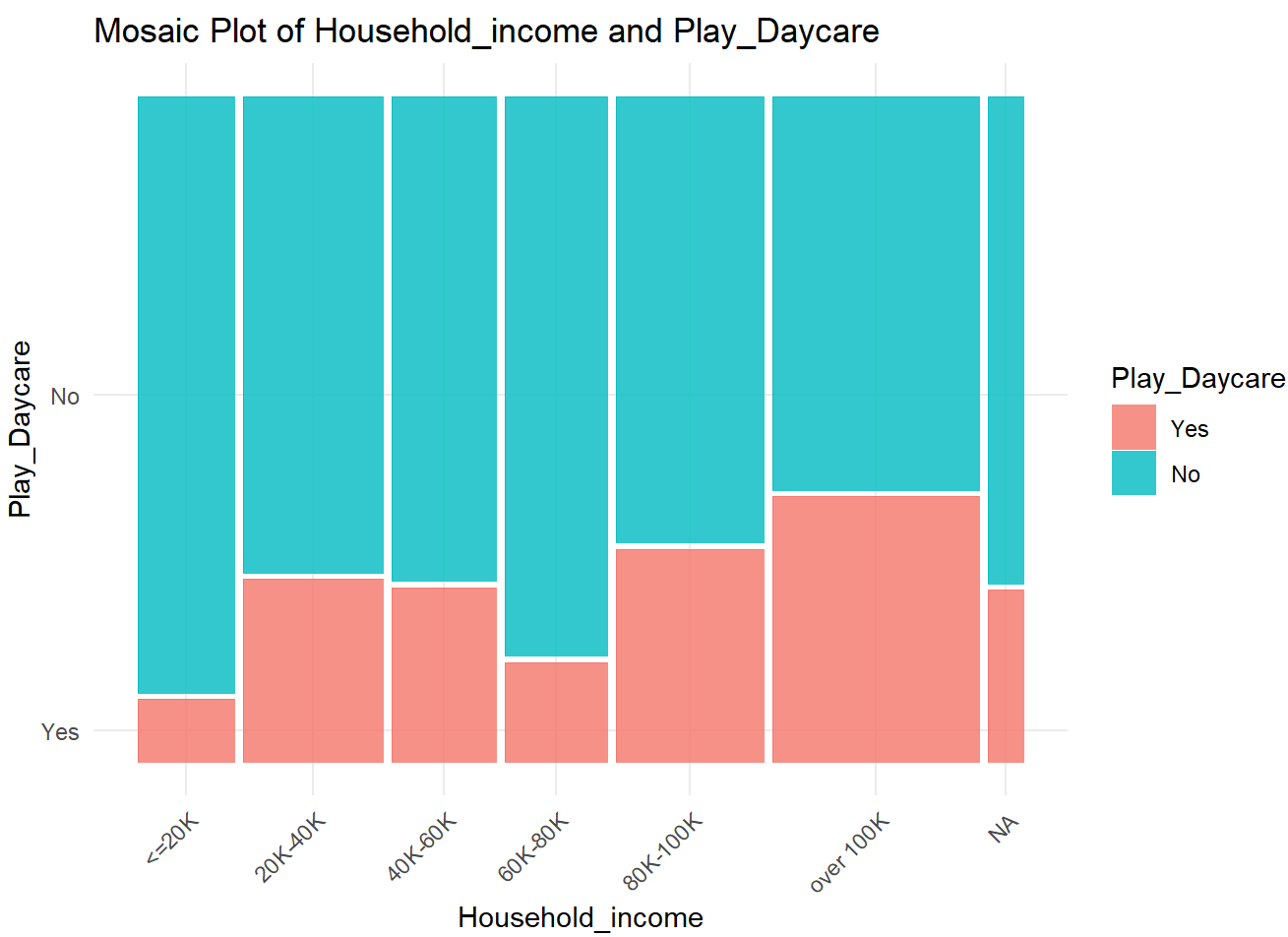
```
##
##
## ----- Analysis for Play_Daycare and P1_Workstatus -----
## Bivariate Analysis of Play_Daycare and P1_Workstatus
##
## Frequency Table:
## Play_Daycare Working_full_time Working_part_time Going_to_school
##           Yes           94           28           9
##           No            144           60           26
## Not_Working/Refused NA_
##              17    2
##             148   12
##
## Relative Frequency Table:
## Play_Daycare Working_full_time Working_part_time Going_to_school
##           Yes           39.5%           31.8%           25.7%
##           No            60.5%           68.2%           74.3%
## Not_Working/Refused  NA_ Total
##              10.3% 14.3% 27.8%
##              89.7% 85.7% 72.2%
##
## Cramér's V:  0.2705677
```

Mosaic Plot of P1_Workstatus and Play_Daycare



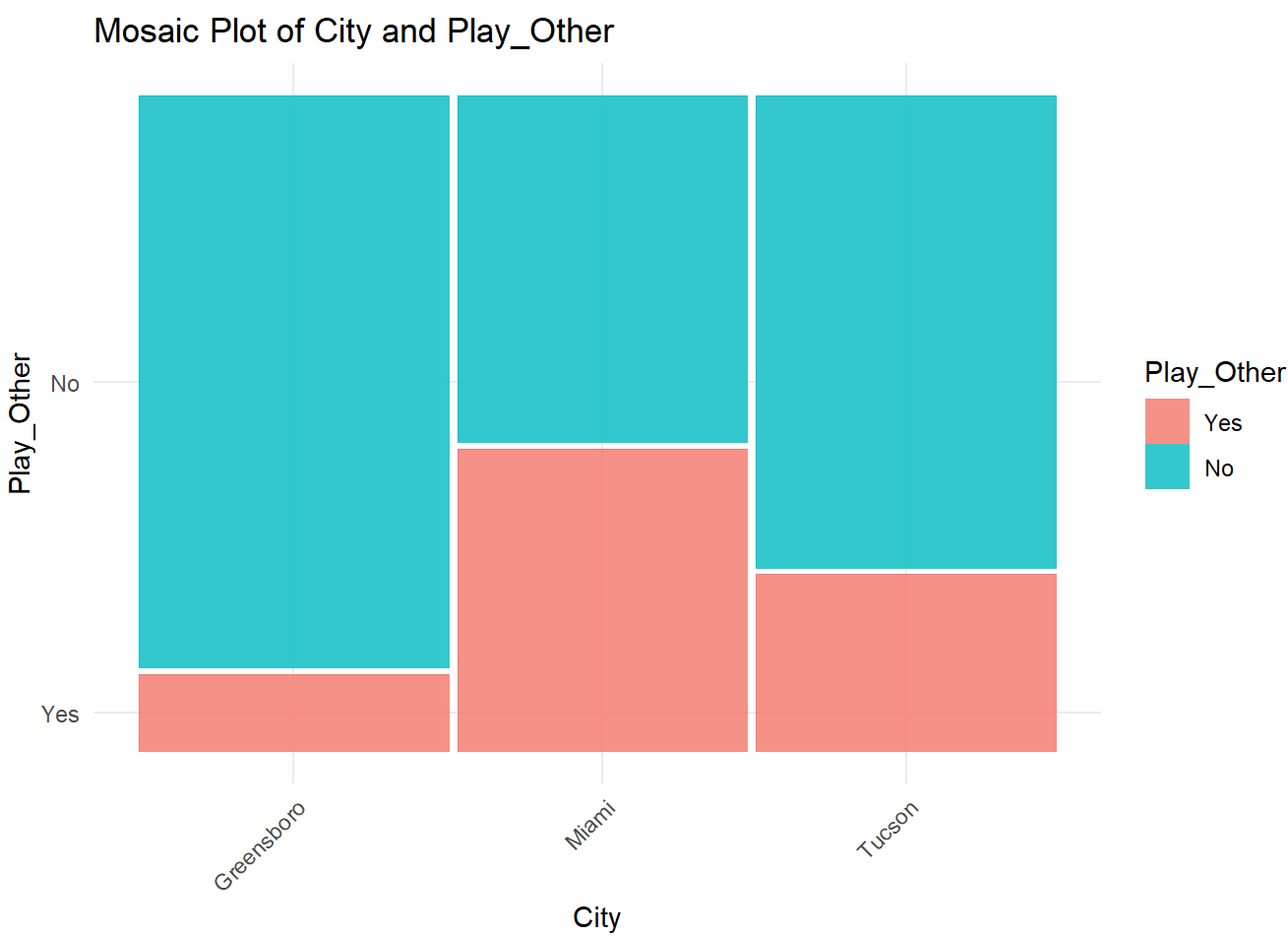
```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Daycare and Household_income -----
## Bivariate Analysis of Play_Daycare and Household_income
##
## Frequency Table:
## Play_Daycare <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_
##           Yes      6      25      18      10      31      54      6
##           No      57      65      50      56      65      80     17
##
## Relative Frequency Table:
## Play_Daycare <=20K 20K-40K 40K-60K 60K-80K 80K-100K over 100K NA_ Total
##           Yes   9.5%   27.8%   26.5%   15.2%   32.3%   40.3% 26.1% 27.8%
##           No  90.5%   72.2%   73.5%   84.8%   67.7%   59.7% 73.9% 72.2%
##
## Cramér's V:  0.1984836
```



```
## Saving 7 x 5 in image
```

```
##
##
## ----- Analysis for Play_Other and City -----
## Bivariate Analysis of Play_Other and City
##
## Frequency Table:
##   Play_Other Greensboro Miami Tucson
##           Yes         22     81     49
##           No        164     93    131
##
## Relative Frequency Table:
##   Play_Other Greensboro Miami Tucson Total
##           Yes         11.8% 46.6% 27.2% 28.1%
##           No         88.2% 53.4% 72.8% 71.9%
##
## Cramér's V: 0.3097251
```



```
## Saving 7 x 5 in image
```



```
# Ensure the list contains valid ggplot objects
if (length(mosaic_plots) > 0) {

  # Arrange the plots in a 3-row x 4-column grid (fixing list format)
  do.call(grid.arrange, c(mosaic_plots, nrow = 3, ncol = 4))

  # Save the final combined plot
  ggsave("Mosaic_Grid.jpeg", width = 16, height = 12, dpi = 300)

  message("Saved mosaic grid as Mosaic_Grid.jpeg")
} else {
  message("No plots were successfully added to the list. Check input variables.")
}
```

```
## No plots were successfully added to the list. Check input variables.
```

```

# library(ggplot2)
# library(ggmosaic)
# library(cowplot)
# library(gridExtra)
# library(RColorBrewer)
# library(grid) # Needed for main titles
#
# # Define the custom color palette with Gold and Navy as primary colors
# custom_colors <- c("#FFD700", "#000080", "#008080", "#800080",
#                     "#006400", "#FF8C00", "#696969", "#A52A2A") # Gold, Navy, and 6 other colors
# list1 <- Selected_Features$Exposure_Variable
# list2 <- Selected_Features$Demographics_Variable
# cbind(list1,list2)
# # Initialize an empty list to store plots
# plots <- list()
#
# # Loop through 12 variable pairs
# for (i in 1:12) {
#   var1 <- list1[i]
#   var2 <- list2[i]
#
#   cat("\n\n----- Generating Mosaic Plot for", var1, "and", var2, "-----\n")
#
#   # Get the number of unique categories in var1
#   unique_categories <- length(unique(data[[var1]]))
#
#   # Assign the appropriate number of colors from the custom palette
#   palette_colors <- custom_colors[1:min(unique_categories, length(custom_colors))]
#
#   # Generate a mosaic plot
#   p <- ggplot(data) +
#     geom_mosaic(aes(weight = 1, x = product(!sym(var2)), fill = !sym(var1))) +
#     scale_fill_manual(values = palette_colors) + # Use custom colors
#     labs(
#       title = paste(var1, "vs", var2), # Individual plot title
#       x = var2, y = var1 # Keep x and y axis labels
#     ) +
#     theme_minimal() +
#     theme(
#       plot.title = element_text(size = 8, face = "bold"), # Reduce subplot title font size
#       axis.title.x = element_text(size = 6, face = "bold"), # Keep x-axis label
#       axis.title.y = element_text(size = 6, face = "bold"), # Keep y-axis label
#       axis.text.x = element_text(angle = 45, hjust = 1, size = 8), # Reduce x-axis tick text size
#       axis.text.y = element_text(size = 6, face = "bold"), # Reduce y-axis tick text size
#       legend.position = "none", # Remove legend
#       panel.grid.major = element_line(color = "gray90", size = 0.3), # Add major grid lines
#       panel.grid.minor = element_line(color = "gray95", size = 0.3), # Add minor grid lines
#       panel.border = element_rect(color = "black", fill = NA, size = 0.35) # Add frame around each subplot
#     )
#
#   # Store each plot in the list
#   plots[[i]] <- p
# }
#
# # Split into two groups: first 6 plots and last 6 plots

```

```
# plots1 <- plots[1:6] # First 6 plots
# plots2 <- plots[7:12] # Last 6 plots
#
# # Create a grob for the main title
# main_title1 <- textGrob("Mosaic Plots of the Selected Features Part 1",
#                           gp = gpar(fontsize = 10, fontface = "bold"))
#
# main_title2 <- textGrob("Mosaic Plots of the Selected Features Part 2",
#                           gp = gpar(fontsize = 10, fontface = "bold"))
#
# # Arrange the first 6 plots in a 3-row x 2-column grid with a main title
# plot1 <- grid.arrange(grobs = plots1, nrow = 3, ncol = 2, top = main_title1)
# ggsave("Mosaic_Grid_Part1_Gold_Navy.jpeg", plot = plot1,
#         width = 8.27, height = 11.69, units = "in", dpi = 300)
#
# # Arrange the last 6 plots in a 3-row x 2-column grid with a main title
# plot2 <- grid.arrange(grobs = plots2, nrow = 3, ncol = 2, top = main_title2)
# ggsave("Mosaic_Grid_Part2_Gold_Navy.jpeg", plot = plot2,
#         width = 8.27, height = 11.69, units = "in", dpi = 300)
#
# message("Saved Mosaic_Grid_Part1_Gold_Navy.jpeg and Mosaic_Grid_Part2_Gold_Navy.jpeg")
```

Sankey Plot 2

```
# Install and Load required package
## install.packages("networkD3")

library(networkD3)
library(htmltools)
# Prepare the data

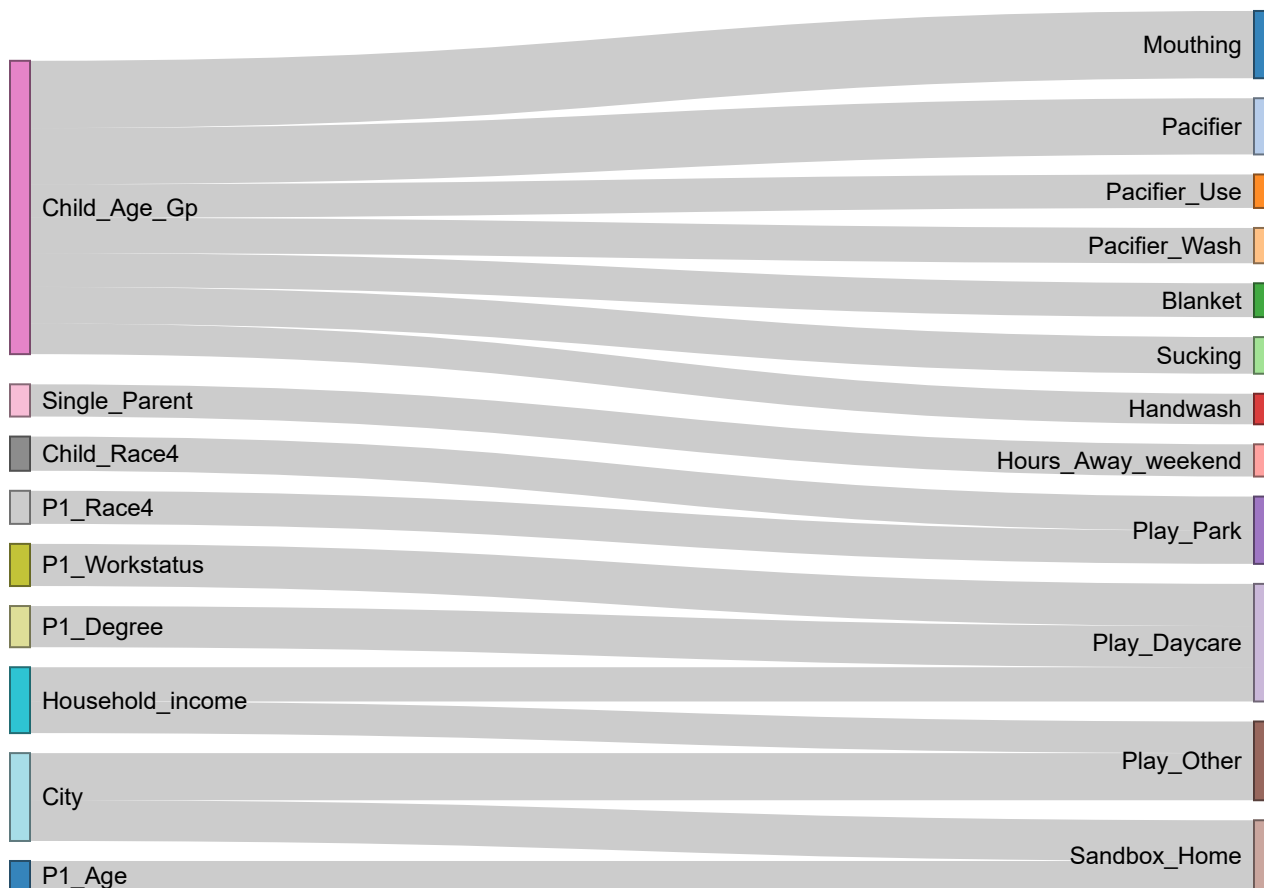
results1= result_tbl_0.20[-18,]

nodes <- unique(c(results1$exposure_var, results1$demographic_var))
nodes_df <- data.frame(name = nodes)

links_df <- data.frame(
  source = match(results1$demographic_var, nodes) - 1,
  target = match(results1$exposure_var, nodes) - 1,
  value = results1$cramer_v
)

# Create the Sankey plot
sankey <- sankeyNetwork(
  Links = links_df,
  Nodes = nodes_df,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 10
  # main = "Sankey's Plot Exposure vs. Demographic Variable Pairs with Cramer's V >=0.20"
)

# Render the plot
sankey
```



```
# Render with a title
html_print(tagList(
  h2("Sankey's Plot: Pairs with Cramer's V  $\geq 0.20$ ",
    style = "text-align:center; font-weight:bold;"),
  sankey
))
# Install and Load the required package
##install.packages("htmlwidgets")
library(htmlwidgets)

# Save the Sankey plot as an HTML file
saveNetwork(sankey, "sankey_plot.html")
```

```
# Load necessary libraries
library(ggplot2)
library(vcd)
```

```
## Loading required package: grid
```

```
##
## Attaching package: 'vcd'
```

```
## The following objects are masked from 'package:ggmosaic':
##
##     mosaic, spine
```

```

library(dplyr)
library(tidyr)

# Load the dataset
file_path <- "Table_Summary_ALL_Formatted.csv"
df <- read.csv(file_path, stringsAsFactors = TRUE)

# # Ensure the column names match exactly as needed
# df <- df %>%
#   rename(Exposure_Variable = Exposure_Variable,
#          Demographics_Variable = Demographics_Variable,
#          Cramers_V = Cramers_V) # Adjust if needed
#

# Identify the top 4 demographic variables for each exposure variable
top_demographics <- df %>%
  group_by(Exposure_Variable) %>%
  top_n(4, wt = Cramers_V) %>%
  arrange(Exposure_Variable, desc(Cramers_V)) %>%
  ungroup()

# Display the top demographic variables
print(top_demographics)

```

```

## # A tibble: 104 × 5
##   Exposure_Variable Demographics_Variable Chi_Squared_Value P_Value Cramers_V
##   <fct>            <fct>                <dbl>    <dbl>    <dbl>
## 1 Blanket          Child_Age_Gp             27.7    0.0001    0.214
## 2 Blanket          P1_Age                 17.1    0.0007    0.162
## 3 Blanket          Child_Race4              5.82    0.120     0.0722
## 4 Blanket          City                    3.72    0.155     0.0564
## 5 Blanket_wash     Child_Age_Gp            43.1    0.0001    0.139
## 6 Blanket_wash     Child_Race4             24.0    0.043     0.0865
## 7 Blanket_wash     P1_Age                  22.7    0.0473    0.0818
## 8 Blanket_wash     City                    11.8    0.150     0.0597
## 9 Handwash         Child_Age_Gp            45.2    0.0001    0.191
## 10 Handwash        Household_income        25.9    0.0033    0.124
## # i 94 more rows

```

```
# Generate 2x2 mosaic plots for each exposure variable
for (exp_var in unique(top_demographics$Exposure_Variable)) {
  subset_data <- top_demographics %>% filter(Exposure_Variable == exp_var)

  # Set up a 2x2 layout
  par(mfrow = c(2, 2), mar = c(3, 3, 3, 3))

  for (i in 1:nrow(subset_data)) {
    demo_var <- subset_data$Demographics_Variable[i]

    if (exp_var %in% names(df) && demo_var %in% names(df)) {
      tbl <- table(filtered_data[[exp_var]], filtered_data[[demo_var]])
      mosaic(tbl, shade = TRUE, main = paste(exp_var, "vs", demo_var))
    }
  }
}
```

Mouthing 2X2 plot

```
# Load necessary libraries
library(ggplot2)
library(ggmosaic)
library(gridExtra)
library(dplyr)

# Load the dataset
file_path <- "Table_Summary_ALL_Formatted.csv" # Adjust if needed
df <- read.csv(file_path, stringsAsFactors = TRUE)

# Convert factor variables to characters to avoid subsetting issues
df$Exposure_Variable <- as.character(df$Exposure_Variable)
df$Demographics_Variable <- as.character(df$Demographics_Variable)

# Define the Mouthing variable and its top 4 demographic variables
mouthing_var <- "Mouthing"
demographic_vars <- c("Child_Age_Gp", "P1_Age", "Single_Parent", "P1_Race4")

# Define a color-blind friendly palette
custom_colors <- c("red", # Bright Red (High Risk)
                  "#006400") # Strong Green (No Risk)

# Initialize an empty list to store plots
mouthing_plots <- list()

for (demo_var in demographic_vars) {
  # Drop NA values before plotting
  filtered_data <- data %>% filter(!is.na(!sym(mouthing_var)) & !is.na(!sym(demo_var)))

  # Skip if there's not enough data
  if (nrow(filtered_data) == 0) {
    message("Skipping ", mouthing_var, " vs ", demo_var, " (All rows have NA values).")
    next
  }

  # Extract Cramér's V and p-value (if available in dataset)
  cramers_v <- df %>%
    filter(Exposure_Variable == mouthing_var & Demographics_Variable == demo_var) %>%
    select(Cramers_V, P_Value) %>%
    slice(1) # Take the first value

  # Handle missing values gracefully
  cramers_v_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$Cramers_V, 4), "N/A")
  p_value_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$P_Value, 4), "N/A")

  # Assign colors based on unique categories
  unique_categories <- length(unique(filtered_data[[mouthing_var]]))
  palette_colors <- custom_colors[1:min(unique_categories, length(custom_colors))]

  # Generate the mosaic plot
  p <- ggplot(filtered_data) +
    geom_mosaic(aes(weight = 1, x = product(!sym(demo_var)), fill = !sym(mouthing_var))) +
    scale_fill_manual(values = palette_colors) + # Use custom colors
    labs(
```



```

    title = paste(mouthing_var, "vs", demo_var), # Individual plot title
    subtitle = paste("Cramér's V:", cramers_v_val, "| P-Value:", p_value_val),
    x = demo_var, y = mouthing_var # Keep axis labels
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 11, face = "bold"),
    plot.subtitle = element_text(size = 11, face = "italic",
                                  color = "gray40"),
    axis.title.x = element_text(size = 10, face = "bold"),
    axis.title.y = element_text(size = 10, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 11),
    axis.text.y = element_text(size = 11, face = "bold"),
    legend.position = "none", # Remove Legend
    panel.grid.major = element_line(color = "gray90", size = 0.4),
    panel.grid.minor = element_line(color = "gray95", size = 0.4)
  )

# Store the plot
mouthing_plots[[demo_var]] <- p
}

```

```

## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

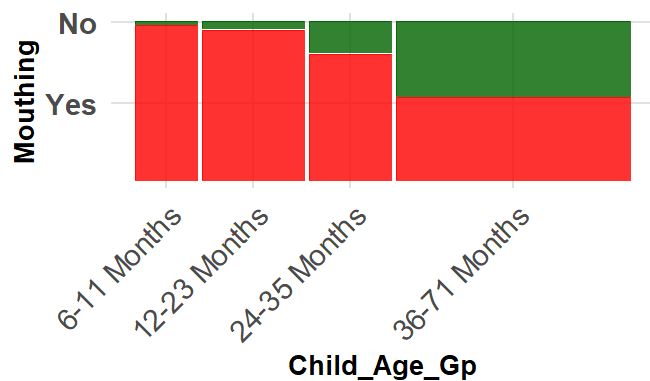
```

# Arrange the 4 plots in a 2x2 grid
mouthing_grid <- grid.arrange(grobs = mouthing_plots, ncol = 2, nrow = 2)

```

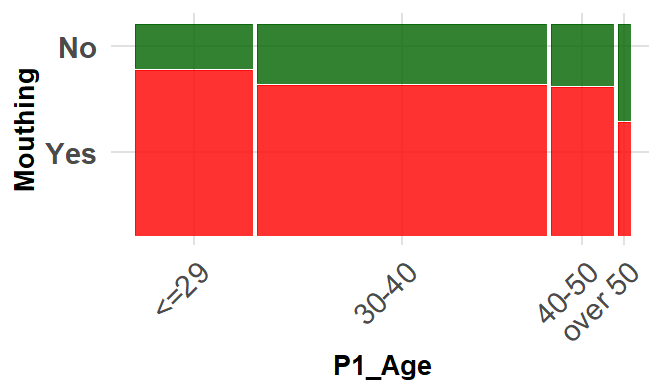
Mouthing vs Child_Age_Gp

Cramér's V: 0.4442 | P-Value: 1e-04



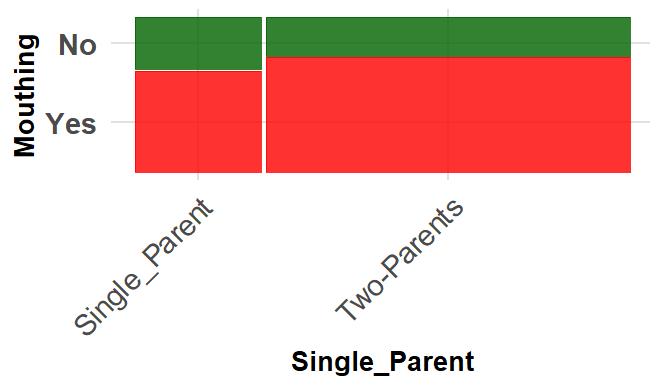
Mouthing vs P1_Age

Cramér's V: 0.0852 | P-Value: 0.07



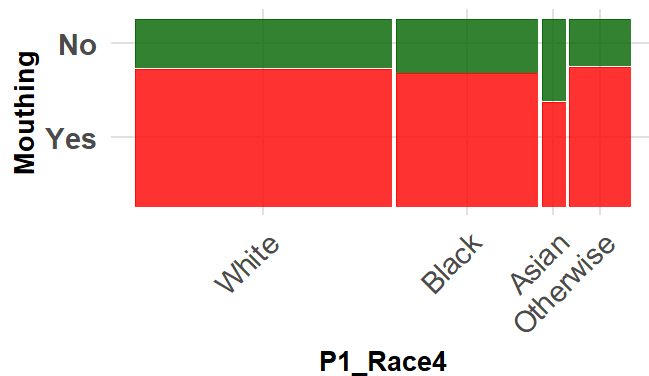
Mouthing vs Single_Parent

Cramér's V: 0.0746 | P-Value: 0.0476



Mouthing vs P1_Race4

Cramér's V: 0.0424 | P-Value: 0.2582



```
# Save the combined plot
ggsave("Mosaic_Mouthing_2x2.jpeg", plot = mouthing_grid, width = 8, height = 10, dpi = 300)

message("Saved Mosaic_Mouthing_2x2.jpeg")
```

```
## Saved Mosaic_Mouthing_2x2.jpeg
```

Pacifier 2X2 plot

```
# Load necessary libraries
library(ggplot2)
library(ggmosaic)
library(gridExtra)
library(dplyr)

# Load the dataset
file_path <- "Table_Summary_ALL_Formatted.csv"

df <- read.csv(file_path, stringsAsFactors = TRUE)

# Convert factor variables to characters to avoid subsetting issues
df$Exposure_Variable <- as.character(df$Exposure_Variable)
df$Demographics_Variable <- as.character(df$Demographics_Variable)

# Define the Pacifier variable and its top 4 demographic variables
exposure_var <- "Pacifier"
demographic_vars <- c("Child_Age_Gp", "Child_Race4", "P1_Workstatus", "P1_Race4")

# Define a color-blind friendly palette
custom_colors <- c("#FFD700", "#000080", "#008080", "#556B2F", "#DAA520",
  "#87CEEB", "#CBA135", "#6A5ACD") # Gold, Blue, & Greens

custom_colors <- c("red", # Bright Red (High Risk)
  "#006400") # Strong Green (No Risk)

# Initialize an empty list to store plots
pacifier_plots <- list()

for (demo_var in demographic_vars) {
  # Drop NA values before plotting
  filtered_data <- data %>% filter(!is.na(!sym(exposure_var)) & !is.na(!sym(demo_var)))

  # Skip if there's not enough data
  if (nrow(filtered_data) == 0) {
    message("Skipping ", exposure_var, " vs ", demo_var, " (All rows have NA values).")
    next
  }

  # Extract Cramér's V and p-value (if available in dataset)
  cramers_v <- df %>%
    filter(Exposure_Variable == exposure_var & Demographics_Variable == demo_var) %>%
    select(Cramers_V, P_Value) %>%
    slice(1) # Take the first value

  # Handle missing values gracefully
  cramers_v_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$Cramers_V, 4), "N/A")
  p_value_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$P_Value, 4), "N/A")

  # Assign colors based on unique categories
  unique_categories <- length(unique(filtered_data[[exposure_var]]))
  palette_colors <- custom_colors[1:min(unique_categories, length(custom_colors))]
```

```

# Generate the mosaic plot
p <- ggplot(filtered_data) +
  geom_mosaic(aes(weight = 1, x = product(!sym(demo_var)), fill = !!sym(exposure_var))) +
  scale_fill_manual(values = palette_colors) + # Use custom colors
  labs(
    title = paste(exposure_var, "vs", demo_var), # Individual plot title
    subtitle = paste("Cramér's V:", crammers_v_val, "| P-Value:", p_value_val),
    x = demo_var, y = exposure_var # Keep axis labels
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 10, face = "bold"),
    plot.subtitle = element_text(size = 9, face = "italic", color = "gray40"),
    axis.title.x = element_text(size = 10, face = "bold"),
    axis.title.y = element_text(size = 10, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    axis.text.y = element_text(size = 8, face = "bold"),
    legend.position = "none", # Remove Legend
    panel.grid.major = element_line(color = "gray90", size = 0.3),
    panel.grid.minor = element_line(color = "gray95", size = 0.2)
  )

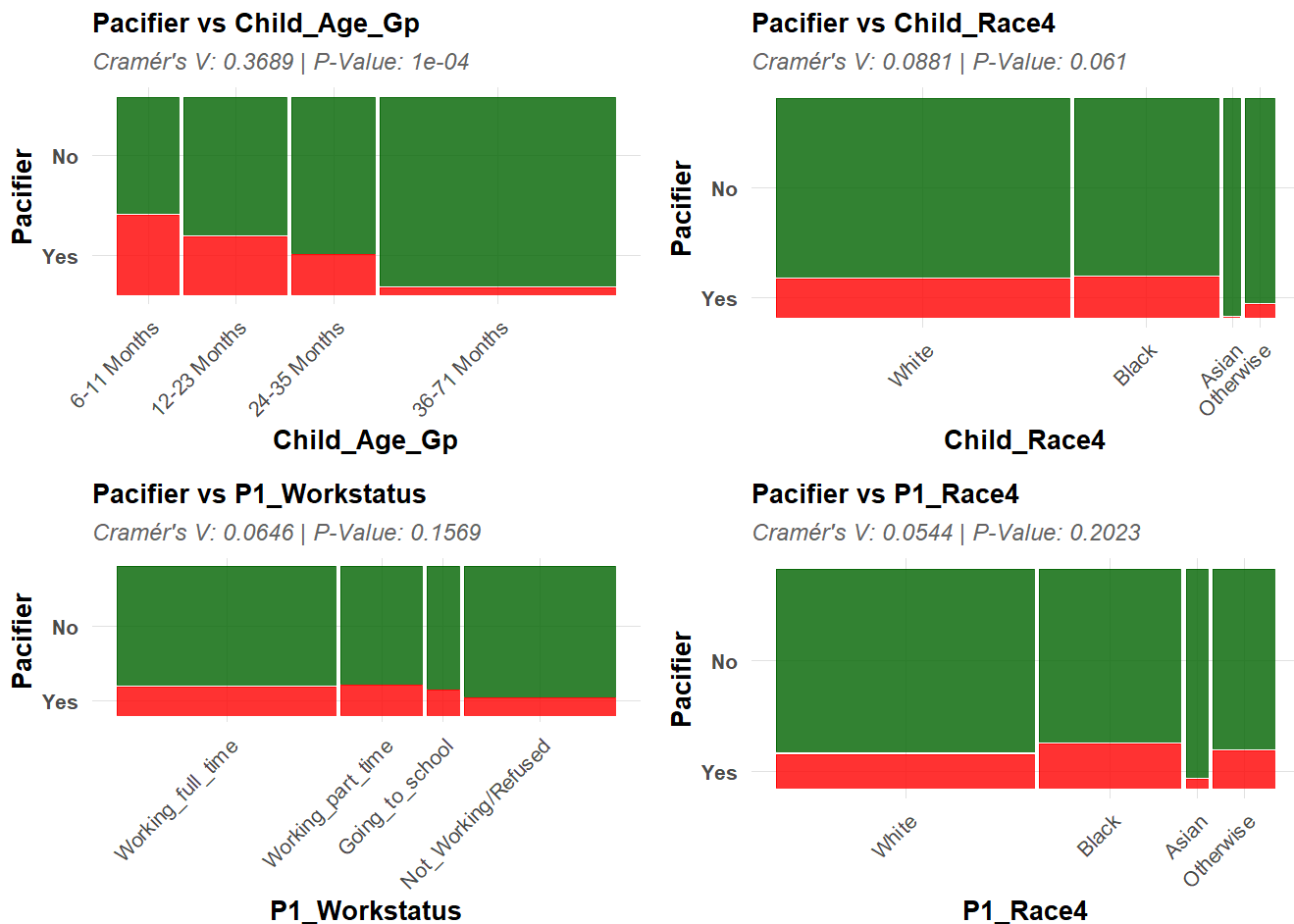
# Store the plot
pacifier_plots[[demo_var]] <- p
}

```

```

# Arrange the 4 plots in a 2x2 grid
pacifier_grid <- grid.arrange(grobs = pacifier_plots, ncol = 2, nrow = 2)

```



```
# Save the combined plot
ggsave("Mosaic_pacifier_grid_2x2.jpeg", plot = pacifier_grid, width = 10, height = 10, dpi = 300)

message("Saved Mosaic_pacifier_grid_2x2.jpeg")
```

```
## Saved Mosaic_pacifier_grid_2x2.jpeg
```

Sucking Grid

```
# Load the dataset
file_path <- "Table_Summary_ALL_Formatted.csv" # Adjust if needed
df <- read.csv(file_path, stringsAsFactors = TRUE)

# Convert factor variables to characters to avoid subsetting issues
df$Exposure_Variable <- as.character(df$Exposure_Variable)
df$Demographics_Variable <- as.character(df$Demographics_Variable)
# Define the Pacifier variable and its top 4 demographic variables
exposure_var <- "Sucking"
demographic_vars <- c("Child_Age_Gp", "City", "P1_Workstatus", "Household_income")

custom_colors <- c("red", # Bright Red (High Risk)
                  "#FC8D59", # Reddish-Orange (Moderate-High Risk)
                  "#FEE08B", # Yellow (Moderate Risk)
                  "#91CF60", # Light Green (Low Risk)
                  "#006400") # Strong Green (No Risk)

# Initialize an empty list to store plots
sucking_plots <- list()

for (demo_var in demographic_vars) {
  # Drop NA values before plotting
  filtered_data <- data %>% filter(!is.na(!sym(exposure_var)) & !is.na(!sym(demo_var)))

  # Skip if there's not enough data
  if (nrow(filtered_data) == 0) {
    message("Skipping ", exposure_var, " vs ", demo_var, " (All rows have NA values).")
    next
  }

  # Extract Cramér's V and p-value (if available in dataset)
  cramers_v <- df %>%
    filter(Exposure_Variable == exposure_var & Demographics_Variable == demo_var) %>%
    select(Cramers_V, P_Value) %>%
    slice(1) # Take the first value

  # Handle missing values gracefully
  cramers_v_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$Cramers_V, 4), "N/A")
  p_value_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$P_Value, 4), "N/A")

  # Assign colors based on unique categories
  unique_categories <- length(unique(filtered_data[[exposure_var]]))
  palette_colors <- custom_colors[1:min(unique_categories, length(custom_colors))]

  # Generate the mosaic plot
  p <- ggplot(filtered_data) +
    geom_mosaic(aes(weight = 1, x = product(!sym(demo_var)), fill = !sym(exposure_var))) +
    scale_fill_manual(values = palette_colors) + # Use custom colors
    labs(
      title = paste(exposure_var, "vs", demo_var), # Individual plot title
      subtitle = paste("Cramér's V:", cramers_v_val, "| P-Value:", p_value_val),
      x = demo_var, y = exposure_var # Keep axis labels
    ) +
```

```

theme_minimal() +
theme(
  plot.title = element_text(size = 10, face = "bold"),
  plot.subtitle = element_text(size = 9, face = "italic", color = "gray40"),
  axis.title.x = element_text(size = 10, face = "bold"),
  axis.title.y = element_text(size = 10, face = "bold"),
  axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
  axis.text.y = element_text(size = 9),
  legend.position = "none", # Remove Legend
  panel.grid.major = element_line(color = "gray90", size = 0.3),
  panel.grid.minor = element_line(color = "gray95", size = 0.2)
)

```

Store the plot

```
sucking_plots[[demo_var]] <- p
```

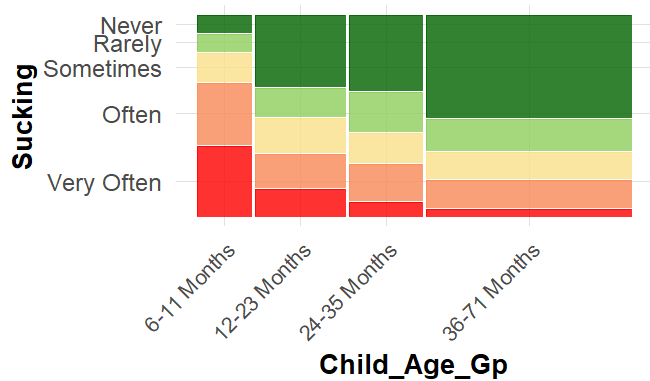
```
}
```

Arrange the 4 plots in a 2x2 grid

```
sucking_grid <- grid.arrange(grobs = sucking_plots, ncol = 2, nrow = 2)
```

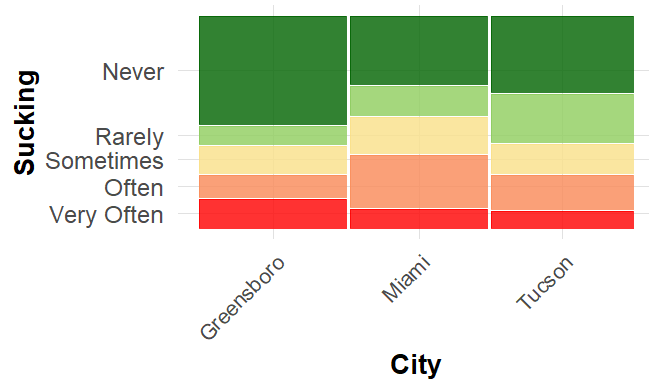
Sucking vs Child_Age_Gp

Cramér's V: 0.2299 | P-Value: 1e-04



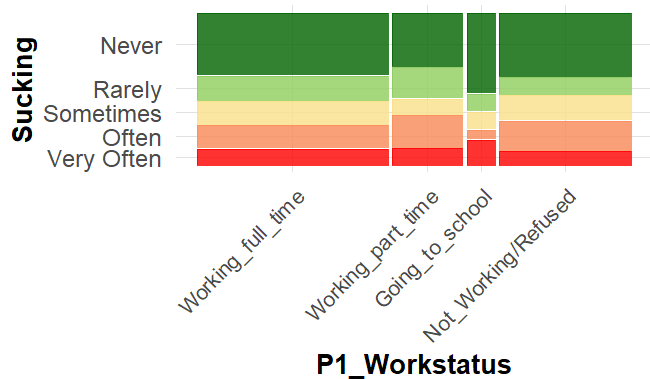
Sucking vs City

Cramér's V: 0.1672 | P-Value: 1e-04



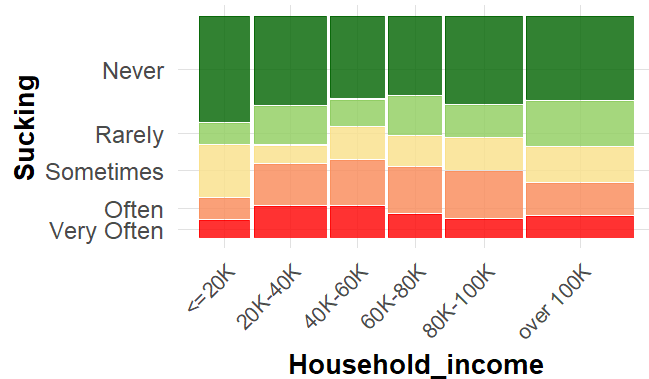
Sucking vs P1_Workstatus

Cramér's V: 0.0378 | P-Value: 0.2878



Sucking vs Household_income

Cramér's V: 0.0223 | P-Value: 0.3912



Save the combined plot

```
ggsave("Mosaic_Sucking_grid_2x2.jpeg", plot = sucking_grid,
  width = 8, height = 10, dpi = 300)
```

```
message("Saved Mosaic_Sucking_grid_2x2.jpeg")
```

```
## Saved Mosaic_Sucking_grid_2x2.jpeg
```

Stuffed_Animal

```
# Load the dataset
file_path <- "Table_Summary_ALL_Formatted.csv" # Adjust if needed
df <- read.csv(file_path, stringsAsFactors = TRUE)

# Convert factor variables to characters to avoid subsetting issues
df$Exposure_Variable <- as.character(df$Exposure_Variable)
df$Demographics_Variable <- as.character(df$Demographics_Variable)

top4_vars <- df %>%
  filter(Exposure_Variable == exposure_var) %>%
  arrange(desc(Cramers_V)) %>%
  slice(1:4) %>%
  pull(Demographics_Variable)

# Define the exposure variable and its top 4 demographic variables
exposure_var <- "Stuffedanimal"
demographic_vars <- c("P1_Sex", "Child_Age_Gp", "P1_Workstatus", "City")

# Define a color-blind friendly palette
#custom_colors <- c("#FFD700", "#000080", "#008080", "#556B2F", "#DAA520",
#                   "#87CEEB", "#CBA135", "#6A5ACD") # Gold, Blue, & Greens
custom_colors <- c("red", # Bright Red (High Risk)
                  "#006400") # Strong Green (No Risk)

# Initialize an empty list to store plots
stuffey_plots<- list()

for (demo_var in top4_vars) {
  # Drop NA values before plotting
  filtered_data <- data%>% filter(!is.na(!sym(exposure_var)) & !is.na(!sym(demo_var)))

  # Skip if there's not enough data
  if (nrow(filtered_data) == 0) {
    message("Skipping ", exposure_var, " vs ", demo_var, " (All rows have NA values).")
    next
  }

  # Extract Cramér's V and p-value (if available in dataset)
  cramers_v <- df %>%
    filter(Exposure_Variable == exposure_var & Demographics_Variable == demo_var) %>%
    select(Cramers_V, P_Value) %>%
    slice(1) # Take the first value

  # Handle missing values gracefully
  cramers_v_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$Cramers_V, 4), "N/A")
  p_value_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$P_Value, 4), "N/A")

  # Assign colors based on unique categories
  unique_categories <- length(unique(filtered_data[[exposure_var]]))
  palette_colors <- custom_colors[1:min(unique_categories, length(custom_colors))]
```



```

# Generate the mosaic plot
p <- ggplot(filtered_data) +
  geom_mosaic(aes(weight = 1, x = product(!!sym(demo_var)), fill = !!sym(exposure_var))) +
  scale_fill_manual(values = palette_colors) + # Use custom colors
  labs(
    title = paste(exposure_var, "vs", demo_var), # Individual plot title
    subtitle = paste("Cramér's V:", crammers_v_val, "| P-Value:", p_value_val),
    x = demo_var, y = exposure_var # Keep axis labels
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 10, face = "bold"),
    plot.subtitle = element_text(size = 9, face = "italic", color = "gray40"),
    axis.title.x = element_text(size = 10, face = "bold"),
    axis.title.y = element_text(size = 10, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    axis.text.y = element_text(size = 8, face = "bold"),
    legend.position = "none", # Remove Legend
    panel.grid.major = element_line(color = "gray90", size = 0.3),
    panel.grid.minor = element_line(color = "gray95", size = 0.2)
  )

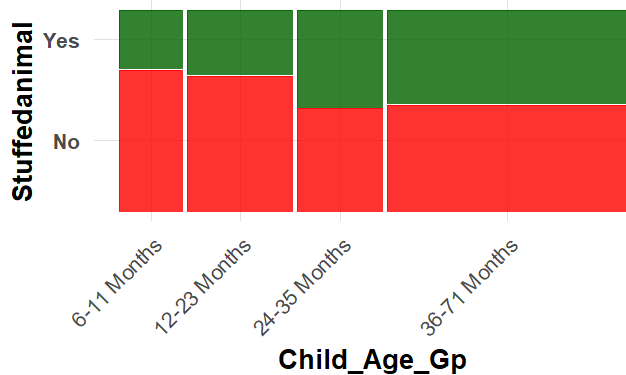
# Store the plot
stuffey_plots[[demo_var]] <- p
}

# Arrange the 4 plots in a 2x2 grid
stuffey_grid <- grid.arrange(grobs = stuffey_plots, ncol = 2, nrow = 2)

```

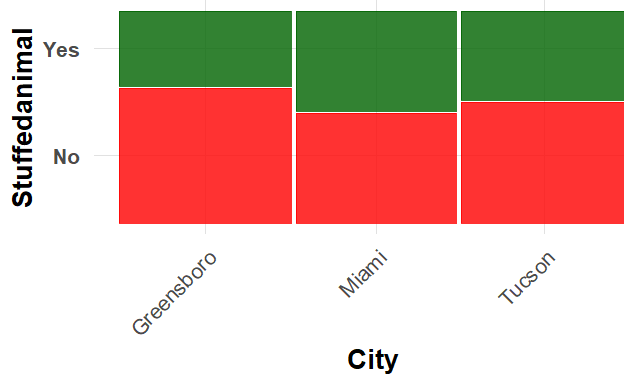
Stuffedanimal vs Child_Age_Gp

Cramér's V: 0.1 | P-Value: 0.0051



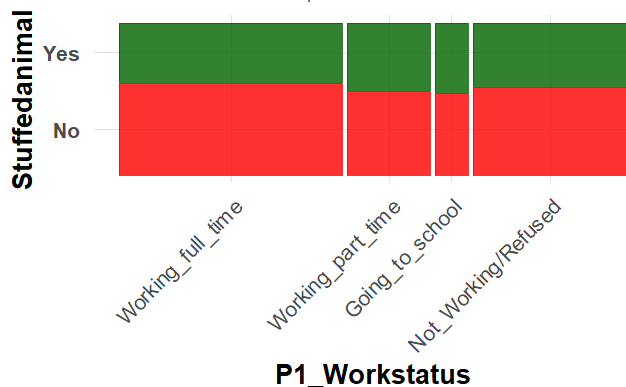
Stuffedanimal vs City

Cramér's V: 0.055 | P-Value: 0.073



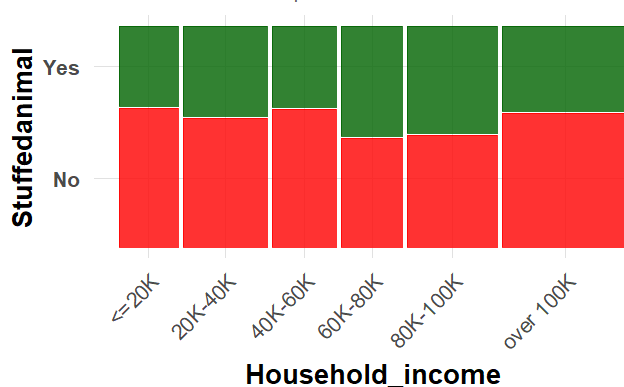
Stuffedanimal vs P1_Workstatus

Cramér's V: 0.0944 | P-Value: 0.0557



Stuffedanimal vs Household_income

Cramér's V: 0.022 | P-Value: 0.3877



```
# Save the combined plot
ggsave("Mosaic_stuffey_grid_2x2.jpeg", plot = stuffey_grid,
       width = 10, height = 10, dpi = 300)

message("Saved Mosaic_stuffey_grid_2x2.jpeg")
```

```
## Saved Mosaic_stuffey_grid_2x2.jpeg
```

```
library(ggplot2)
library(reshape2)
library(dplyr)
library(tidyr)
list1=Exposure_Vars
list2=DemoVars
# Generate the heatmap data
result_tbl_All <- identify_strongest_features_table(data, list1, list2, threshold = 0)
print(result_tbl_All)
```

##	exposure_var	demographic_var	cramer_v
## 1	Mouthing	Child_Age_Gp	0.4500147760
## 2	Mouthing	P1_Age	0.0987427530
## 3	Mouthing	P1_Degree	0.0880118231
## 4	Mouthing	Single_Parent	0.0861355149
## 5	Mouthing	P1_Race4	0.0859194948
## 6	Mouthing	Household_income	0.0585732617
## 7	Mouthing	Child_Race4	0.0582404799
## 8	Mouthing	P1_Workstatus	0.0511797731
## 9	Mouthing	Child_Sex	0.0487155151
## 10	Mouthing	P1_Sex	0.0368917575
## 11	Mouthing	P1_Ethnicity	0.0260927047
## 12	Mouthing	City	0.0166502690
## 13	Pacifier	Child_Age_Gp	0.3760176942
## 14	Pacifier	Child_Race4	0.1153970541
## 15	Pacifier	P1_Workstatus	0.0993676808
## 16	Pacifier	P1_Age	0.0960845278
## 17	Pacifier	P1_Degree	0.0960641718
## 18	Pacifier	P1_Race4	0.0923196750
## 19	Pacifier	Household_income	0.0693596543
## 20	Pacifier	Child_Sex	0.0529245061
## 21	Pacifier	Single_Parent	0.0435675609
## 22	Pacifier	City	0.0323243664
## 23	Pacifier	P1_Ethnicity	0.0260448462
## 24	Pacifier	P1_Sex	0.0130813428
## 25	Pacifier_Use	Child_Age_Gp	0.2240738776
## 26	Pacifier_Use	Single_Parent	0.1221569257
## 27	Pacifier_Use	P1_Race4	0.1134603523
## 28	Pacifier_Use	City	0.1121500326
## 29	Pacifier_Use	Household_income	0.1043699550
## 30	Pacifier_Use	P1_Age	0.1042500142
## 31	Pacifier_Use	Child_Race4	0.0976309523
## 32	Pacifier_Use	P1_Sex	0.0959496657
## 33	Pacifier_Use	P1_Ethnicity	0.0863652360
## 34	Pacifier_Use	P1_Degree	0.0861864951
## 35	Pacifier_Use	P1_Workstatus	0.0858935591
## 36	Pacifier_Use	Child_Sex	0.0581878524
## 37	Pacifier_Wash	Child_Age_Gp	0.2361997473
## 38	Pacifier_Wash	Single_Parent	0.1559975587
## 39	Pacifier_Wash	P1_Age	0.1476393470
## 40	Pacifier_Wash	Household_income	0.1383739169
## 41	Pacifier_Wash	P1_Degree	0.1093805150
## 42	Pacifier_Wash	P1_Sex	0.1074330730
## 43	Pacifier_Wash	City	0.1004569114
## 44	Pacifier_Wash	P1_Race4	0.0992126950
## 45	Pacifier_Wash	P1_Workstatus	0.0872844247
## 46	Pacifier_Wash	Child_Race4	0.0841786324
## 47	Pacifier_Wash	P1_Ethnicity	0.0828407430
## 48	Pacifier_Wash	Child_Sex	0.0741264659
## 49	Blanket	Child_Age_Gp	0.2264064774
## 50	Blanket	P1_Age	0.1826571919
## 51	Blanket	Household_income	0.1038004585
## 52	Blanket	Child_Race4	0.1037737463
## 53	Blanket	P1_Race4	0.0897687548
## 54	Blanket	P1_Degree	0.0844863842
## 55	Blanket	City	0.0829977313

## 56	Blanket	Child_Sex	0.0379800685
## 57	Blanket	P1_Workstatus	0.0264218458
## 58	Blanket	P1_Ethnicity	0.0228238158
## 59	Blanket	Single_Parent	0.0186667825
## 60	Blanket	P1_Sex	0.0174368802
## 61	Blanket_Wash	Child_Age_Gp	0.1616236483
## 62	Blanket_Wash	P1_Age	0.1306023474
## 63	Blanket_Wash	Child_Race4	0.1268765027
## 64	Blanket_Wash	City	0.1221464163
## 65	Blanket_Wash	P1_Ethnicity	0.1180747931
## 66	Blanket_Wash	P1_Race4	0.1029201797
## 67	Blanket_Wash	Household_income	0.0990407835
## 68	Blanket_Wash	P1_Degree	0.0918404793
## 69	Blanket_Wash	P1_Sex	0.0916656138
## 70	Blanket_Wash	Single_Parent	0.0738641336
## 71	Blanket_Wash	Child_Sex	0.0499589880
## 72	Blanket_Wash	P1_Workstatus	0.0461124542
## 73	Stuffed_Animal	Child_Age_Gp	0.1573678167
## 74	Stuffed_Animal	Household_income	0.1057214147
## 75	Stuffed_Animal	P1_Degree	0.1035001112
## 76	Stuffed_Animal	City	0.1014259383
## 77	Stuffed_Animal	P1_Race4	0.0909873073
## 78	Stuffed_Animal	Child_Race4	0.0894635519
## 79	Stuffed_Animal	P1_Sex	0.0662533902
## 80	Stuffed_Animal	Child_Sex	0.0602977870
## 81	Stuffed_Animal	P1_Age	0.0497179146
## 82	Stuffed_Animal	P1_Workstatus	0.0460598863
## 83	Stuffed_Animal	P1_Ethnicity	0.0374073009
## 84	Stuffed_Animal	Single_Parent	0.0164669902
## 85	Stuffed_Animal_Wash	Child_Age_Gp	0.1338383755
## 86	Stuffed_Animal_Wash	Child_Sex	0.1259337691
## 87	Stuffed_Animal_Wash	Household_income	0.1169593338
## 88	Stuffed_Animal_Wash	P1_Degree	0.1141379367
## 89	Stuffed_Animal_Wash	P1_Age	0.1090472093
## 90	Stuffed_Animal_Wash	P1_Sex	0.1088184553
## 91	Stuffed_Animal_Wash	City	0.1062027366
## 92	Stuffed_Animal_Wash	Child_Race4	0.0990861913
## 93	Stuffed_Animal_Wash	P1_Race4	0.0948450772
## 94	Stuffed_Animal_Wash	Single_Parent	0.0934538301
## 95	Stuffed_Animal_Wash	P1_Ethnicity	0.0891189315
## 96	Stuffed_Animal_Wash	P1_Workstatus	0.0633820710
## 97	Sucking	Child_Age_Gp	0.2449471802
## 98	Sucking	City	0.1878508303
## 99	Sucking	Household_income	0.1009131121
## 100	Sucking	P1_Degree	0.0984199415
## 101	Sucking	P1_Workstatus	0.0950911078
## 102	Sucking	P1_Race4	0.0839306041
## 103	Sucking	P1_Ethnicity	0.0771185448
## 104	Sucking	Child_Sex	0.0745577805
## 105	Sucking	Child_Race4	0.0704799524
## 106	Sucking	P1_Age	0.0599643935
## 107	Sucking	Single_Parent	0.0478372763
## 108	Sucking	P1_Sex	0.0377946636
## 109	Handwash	Child_Age_Gp	0.2044678555
## 110	Handwash	Household_income	0.1584068104
## 111	Handwash	Child_Race4	0.1220540953

## 112		P1_Ethnicity	0.1166831986
## 113	Handwash	P1_Race4	0.1103448957
## 114	Handwash	P1_Degree	0.1054432204
## 115	Handwash	P1_Age	0.1014264238
## 116	Handwash	P1_Workstatus	0.0853899219
## 117	Handwash	City	0.0843425357
## 118	Handwash	Single_Parent	0.0706452872
## 119	Handwash	Child_Sex	0.0551177352
## 120	Handwash	P1_Sex	0.0509673660
## 121	Hours_Away_weekday	P1_Workstatus	0.1949007257
## 122	Hours_Away_weekday	Household_income	0.1920067470
## 123	Hours_Away_weekday	City	0.1670924236
## 124	Hours_Away_weekday	P1_Degree	0.1595487421
## 125	Hours_Away_weekday	Child_Age_Gp	0.1506431274
## 126	Hours_Away_weekday	Single_Parent	0.1425662091
## 127	Hours_Away_weekday	P1_Race4	0.1372886807
## 128	Hours_Away_weekday	P1_Age	0.1083913866
## 129	Hours_Away_weekday	Child_Race4	0.0908886940
## 130	Hours_Away_weekday	P1_Ethnicity	0.0755939223
## 131	Hours_Away_weekday	Child_Sex	0.0688485428
## 132	Hours_Away_weekday	P1_Sex	0.0586866862
## 133	Hours_Away_weekend	Single_Parent	0.2156965808
## 134	Hours_Away_weekend	City	0.1581104525
## 135	Hours_Away_weekend	P1_Race4	0.1546701462
## 136	Hours_Away_weekend	Child_Race4	0.1429135849
## 137	Hours_Away_weekend	P1_Ethnicity	0.1340676673
## 138	Hours_Away_weekend	Household_income	0.1108348372
## 139	Hours_Away_weekend	P1_Workstatus	0.1046227602
## 140	Hours_Away_weekend	P1_Degree	0.0922062684
## 141	Hours_Away_weekend	P1_Age	0.0899744374
## 142	Hours_Away_weekend	Child_Sex	0.0894046325
## 143	Hours_Away_weekend	P1_Sex	0.0751532587
## 144	Hours_Away_weekend	Child_Age_Gp	0.0727617738
## 145	Play_Park	Child_Race4	0.2290526883
## 146	Play_Park	P1_Race4	0.2212615233
## 147	Play_Park	Child_Age_Gp	0.1987408690
## 148	Play_Park	P1_Degree	0.1670080721
## 149	Play_Park	P1_Workstatus	0.1571843052
## 150	Play_Park	City	0.1531222730
## 151	Play_Park	Household_income	0.1228942526
## 152	Play_Park	P1_Sex	0.1140114567
## 153	Play_Park	Child_Sex	0.1101691047
## 154	Play_Park	P1_Ethnicity	0.0738016142
## 155	Play_Park	P1_Age	0.0540881306
## 156	Play_Park	Single_Parent	0.0069128075
## 157	Play_Daycare	P1_Workstatus	0.2820064951
## 158	Play_Daycare	P1_Degree	0.2754643350
## 159	Play_Daycare	Household_income	0.2292277055
## 160	Play_Daycare	Child_Age_Gp	0.1339795227
## 161	Play_Daycare	P1_Ethnicity	0.0910166120
## 162	Play_Daycare	P1_Race4	0.0775173288
## 163	Play_Daycare	Child_Race4	0.0743085329
## 164	Play_Daycare	Child_Sex	0.0721459801
## 165	Play_Daycare	P1_Age	0.0699456551
## 166	Play_Daycare	City	0.0619893071
## 167	Play_Daycare	P1_Sex	0.0253967820

## 168	Play_Daycare	Single_Parent	0.0131342036
## 169	Play_Babysitter_Home	Household_income	0.1665086879
## 170	Play_Babysitter_Home	P1_Degree	0.1418679578
## 171	Play_Babysitter_Home	Child_Age_Gp	0.1037240395
## 172	Play_Babysitter_Home	Child_Race4	0.0928442459
## 173	Play_Babysitter_Home	P1_Race4	0.0914259537
## 174	Play_Babysitter_Home	P1_Workstatus	0.0812423041
## 175	Play_Babysitter_Home	City	0.0693716349
## 176	Play_Babysitter_Home	P1_Age	0.0492565364
## 177	Play_Babysitter_Home	Single_Parent	0.0490876846
## 178	Play_Babysitter_Home	P1_Ethnicity	0.0415736541
## 179	Play_Babysitter_Home	P1_Sex	0.0327391525
## 180	Play_Babysitter_Home	Child_Sex	0.0180987395
## 181	Play_Relative_Home	P1_Race4	0.1907729549
## 182	Play_Relative_Home	Child_Race4	0.1432003852
## 183	Play_Relative_Home	P1_Degree	0.1268854518
## 184	Play_Relative_Home	P1_Workstatus	0.1250781693
## 185	Play_Relative_Home	Household_income	0.1143222603
## 186	Play_Relative_Home	P1_Ethnicity	0.1032494584
## 187	Play_Relative_Home	City	0.0837822338
## 188	Play_Relative_Home	P1_Age	0.0633909680
## 189	Play_Relative_Home	Child_Sex	0.0531168964
## 190	Play_Relative_Home	Child_Age_Gp	0.0376821336
## 191	Play_Relative_Home	P1_Sex	0.0299860944
## 192	Play_Relative_Home	Single_Parent	0.0048529707
## 193	Play_Other	City	0.3153763563
## 194	Play_Other	Household_income	0.2114545023
## 195	Play_Other	P1_Age	0.1755979904
## 196	Play_Other	P1_Race4	0.1711718959
## 197	Play_Other	Child_Age_Gp	0.1707868134
## 198	Play_Other	P1_Degree	0.1670948657
## 199	Play_Other	Child_Race4	0.1454117835
## 200	Play_Other	P1_Ethnicity	0.1191381163
## 201	Play_Other	P1_Workstatus	0.1152725221
## 202	Play_Other	Single_Parent	0.1142066592
## 203	Play_Other	Child_Sex	0.1005513268
## 204	Play_Other	P1_Sex	0.0022162436
## 205	Sandbox_Home	City	0.2723574340
## 206	Sandbox_Home	P1_Age	0.2004847263
## 207	Sandbox_Home	Child_Race4	0.1604201906
## 208	Sandbox_Home	P1_Degree	0.1550701397
## 209	Sandbox_Home	P1_Race4	0.1471513359
## 210	Sandbox_Home	Household_income	0.1364139608
## 211	Sandbox_Home	P1_Workstatus	0.1341817145
## 212	Sandbox_Home	Child_Age_Gp	0.1332698357
## 213	Sandbox_Home	P1_Sex	0.0750047421
## 214	Sandbox_Home	Child_Sex	0.0552817190
## 215	Sandbox_Home	P1_Ethnicity	0.0113032010
## 216	Sandbox_Home	Single_Parent	0.0074596317
## 217	Sandbox_Park	Child_Age_Gp	0.1748980045
## 218	Sandbox_Park	P1_Workstatus	0.1532946956
## 219	Sandbox_Park	Household_income	0.1354389360
## 220	Sandbox_Park	City	0.1302370688
## 221	Sandbox_Park	P1_Ethnicity	0.1006339802
## 222	Sandbox_Park	P1_Degree	0.0968955034
## 223	Sandbox_Park	P1_Age	0.0934890893

## 224	Sandbox_Park	Single_Parent	0.0665503650
## 225	Sandbox_Park	P1_Race4	0.0578484371
## 226	Sandbox_Park	Child_Sex	0.0454336642
## 227	Sandbox_Park	Child_Race4	0.0398028434
## 228	Sandbox_Park	P1_Sex	0.0145904988
## 229	Sandbox_Daycare	Household_income	0.1919287681
## 230	Sandbox_Daycare	P1_Degree	0.1535019564
## 231	Sandbox_Daycare	City	0.1525206946
## 232	Sandbox_Daycare	Child_Age_Gp	0.1484995943
## 233	Sandbox_Daycare	Child_Sex	0.1411538910
## 234	Sandbox_Daycare	P1_Workstatus	0.1263789555
## 235	Sandbox_Daycare	P1_Race4	0.1172468792
## 236	Sandbox_Daycare	Child_Race4	0.0884833203
## 237	Sandbox_Daycare	P1_Ethnicity	0.0709311638
## 238	Sandbox_Daycare	P1_Age	0.0648400592
## 239	Sandbox_Daycare	Single_Parent	0.0127565688
## 240	Sandbox_Daycare	P1_Sex	0.0041136828
## 241	Sandbox_Babysitter	P1_Degree	0.0899498123
## 242	Sandbox_Babysitter	City	0.0841141419
## 243	Sandbox_Babysitter	P1_Race4	0.0820497655
## 244	Sandbox_Babysitter	Household_income	0.0737527581
## 245	Sandbox_Babysitter	P1_Workstatus	0.0632105168
## 246	Sandbox_Babysitter	Child_Age_Gp	0.0625728405
## 247	Sandbox_Babysitter	Child_Sex	0.0547389909
## 248	Sandbox_Babysitter	P1_Age	0.0408528902
## 249	Sandbox_Babysitter	Single_Parent	0.0338310572
## 250	Sandbox_Babysitter	Child_Race4	0.0305679421
## 251	Sandbox_Babysitter	P1_Sex	0.0217581026
## 252	Sandbox_Babysitter	P1_Ethnicity	0.0191745164
## 253	Sandbox_Relative	Household_income	0.1914640016
## 254	Sandbox_Relative	P1_Degree	0.1369605412
## 255	Sandbox_Relative	City	0.1367084816
## 256	Sandbox_Relative	P1_Race4	0.1061787212
## 257	Sandbox_Relative	P1_Sex	0.0649763520
## 258	Sandbox_Relative	Child_Age_Gp	0.0508585352
## 259	Sandbox_Relative	P1_Workstatus	0.0483587131
## 260	Sandbox_Relative	Child_Race4	0.0397264106
## 261	Sandbox_Relative	P1_Age	0.0292162695
## 262	Sandbox_Relative	Single_Parent	0.0168994633
## 263	Sandbox_Relative	Child_Sex	0.0155121782
## 264	Sandbox_Relative	P1_Ethnicity	0.0113039461
## 265	Sandbox_Other	P1_Race4	0.2072022858
## 266	Sandbox_Other	Child_Race4	0.1999709842
## 267	Sandbox_Other	City	0.1417619758
## 268	Sandbox_Other	P1_Age	0.1115935907
## 269	Sandbox_Other	Child_Age_Gp	0.1093987717
## 270	Sandbox_Other	Household_income	0.0851087901
## 271	Sandbox_Other	P1_Degree	0.0758466102
## 272	Sandbox_Other	Child_Sex	0.0639003231
## 273	Sandbox_Other	Single_Parent	0.0514943487
## 274	Sandbox_Other	P1_Workstatus	0.0307151687
## 275	Sandbox_Other	P1_Ethnicity	0.0056767807
## 276	Sandbox_Other	P1_Sex	0.0006336072

```

# Remove the last row (unwanted row)
result_tbl_All <- result_tbl_All[~-289,]

# Create a pivot table for heatmap
heatmap_data <- result_tbl_All %>%
  pivot_wider(names_from = demographic_var, values_from = cramer_v)

# Compute column sums (excluding the exposure_var column)
demographic_sums <- colSums(heatmap_data[, -1], na.rm = TRUE)

# Order demographic variables: First by sum (descending), then alphabetically for ties
sorted_demographics <- names(sort(demographic_sums, decreasing = TRUE))

# Reorder columns in heatmap_data (keeping exposure_var as the first column)
heatmap_data <- heatmap_data %>%
  select(exposure_var, all_of(sorted_demographics))

# Convert to matrix
heatmap_matrix <- as.matrix(heatmap_data[, -1])
rownames(heatmap_matrix) <- heatmap_data$exposure_var
print(heatmap_data)

```

```

## # A tibble: 23 × 13
##   exposure_var      Child_Age_Gp Household_income   City P1_Degree P1_Race4
##   <chr>          <dbl>          <dbl> <dbl>    <dbl>    <dbl>
## 1 Mouthing      0.450          0.0586 0.0167   0.0880   0.0859
## 2 Pacifier      0.376          0.0694 0.0323   0.0961   0.0923
## 3 Pacifier_Use  0.224          0.104   0.112   0.0862   0.113
## 4 Pacifier_Wash 0.236          0.138   0.100   0.109   0.0992
## 5 Blanket       0.226          0.104   0.0830   0.0845   0.0898
## 6 Blanket_Wash  0.162          0.0990 0.122   0.0918   0.103
## 7 Stuffed_Animal 0.157          0.106   0.101   0.104   0.0910
## 8 Stuffed_Animal_Wash 0.134          0.117   0.106   0.114   0.0948
## 9 Sucking       0.245          0.101   0.188   0.0984   0.0839
## 10 Handwash     0.204          0.158   0.0843   0.105   0.110
## # i 13 more rows
## # i 7 more variables: Child_Race4 <dbl>, P1_Workstatus <dbl>, P1_Age <dbl>,
## #   P1_Ethnicity <dbl>, Child_Sex <dbl>, Single_Parent <dbl>, P1_Sex <dbl>

```



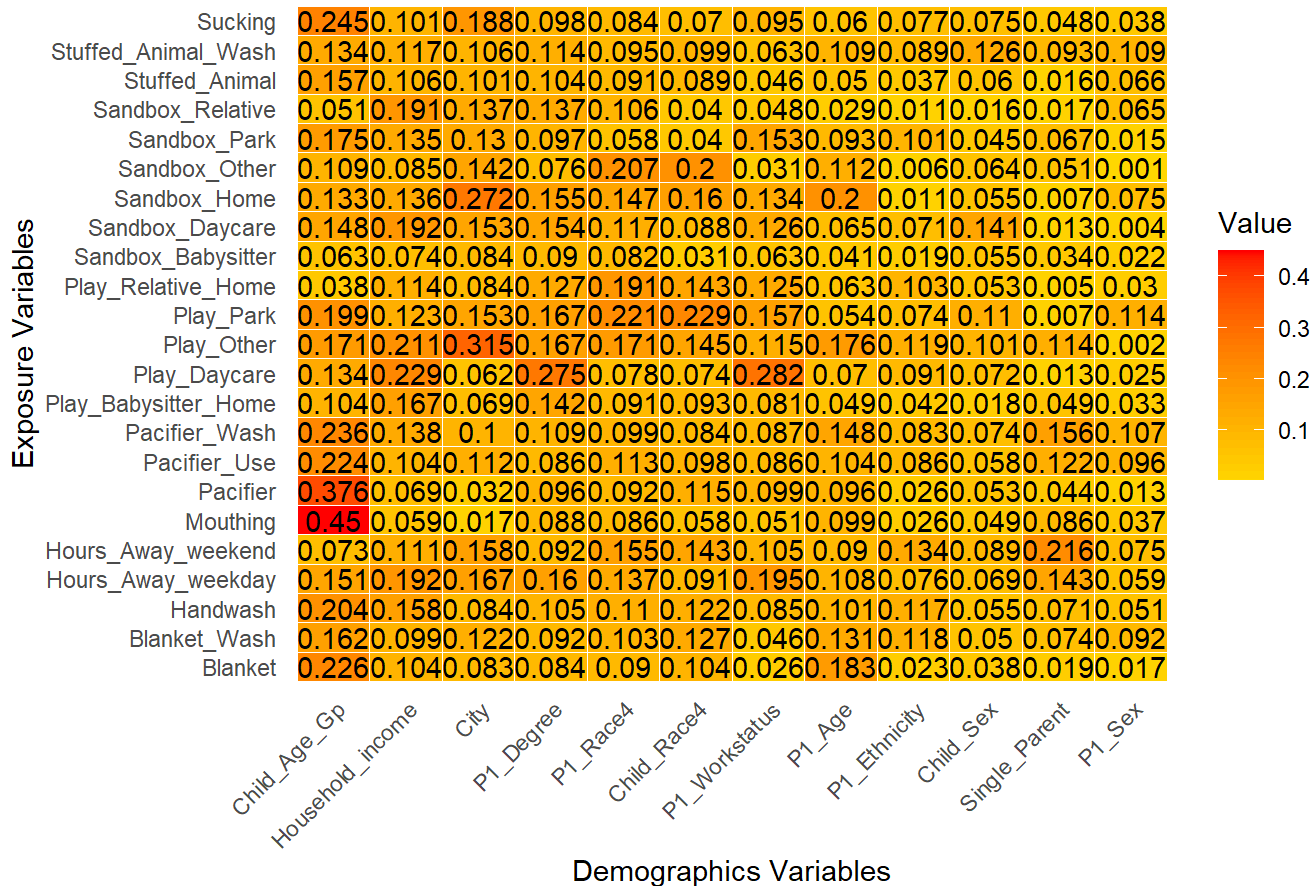
```

# Convert heatmap data to Long format for ggplot
heatmap_long <- melt(heatmap_data, id.vars = "exposure_var")

# Plot the heatmap
ggplot(heatmap_long, aes(x = variable, y = exposure_var, fill = value)) +
  geom_tile(color = "white") + # Adds grid lines
  geom_text(aes(label = round(value, 3)), color = "black") + # Add annotations
  scale_fill_gradientn(colors = c("gold", "red")) + # Define color scale
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), # Rotate x-labels
    panel.grid = element_blank()
  ) +
  labs(
    title = "Heatmap Showing Cramer's V for Each Pair of Variables",
    x = "Demographics Variables",
    y = "Exposure Variables",
    fill = "Value"
  )

```

Heatmap Showing Cramer's V for Each Pair of Variables



```

# Save as PNG
ggsave("heatmap3.png")

```

```
## Saving 7 x 5 in image
```

Final version of the 12 Mosaic plots

```

list1 <- Selected_Features_12$Exposure_Variable
list2 <- Selected_Features_12$Demographics_Variable
library(ggplot2)
library(ggmosaic)
library(dplyr)
library(gridExtra)
library(grid)

# Define color palettes
custom_colors2 <- c("red", "#006400")
custom_colors5 <- c("red", "#FC8D59", "#FEE08B", "#91CF60", "#006400")

# Initialize plot list
plots <- list()

# Generate Mosaic Plots
for (i in 1:12) {
  var1 <- list1[i]
  var2 <- list2[i]

  cat("\n\n----- Generating Mosaic Plot for", var1, "and", var2, "-----\n")

  # Filter data to remove NA values
  filtered_data <- data %>%
    filter(!is.na(.data[[var1]]), !is.na(.data[[var2]]))

  if (nrow(filtered_data) == 0) {
    message("Skipping ", var1, " vs ", var2, " (All rows have NA values).")
    next
  }

  # Choose color palette based on unique levels
  unique_var1_levels <- length(unique(filtered_data[[var1]]))
  palette_colors <- if (unique_var1_levels == 2) custom_colors2 else custom_colors5

  # Generate the mosaic plot
  p <- ggplot(filtered_data) +
    geom_mosaic(aes(weight = 1, x = product(!sym(var2)), fill = !sym(var1))) +
    scale_fill_manual(values = palette_colors) +
    labs(title = paste(var1, "vs", var2), x = var2, y = var1) +
    theme_minimal() +
    theme(
      plot.title = element_text(size = 8, face = "bold"),
      axis.title.x = element_text(size = 8, face = "bold"),
      axis.title.y = element_text(size = 8, face = "bold"),
      axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
      axis.text.y = element_text(size = 8, face = "bold"),
      legend.position = "none",
      panel.grid.major = element_line(color = "gray90", size = 0.2),
      panel.grid.minor = element_line(color = "gray95", size = 0.2)
    )

  # Store plot in list
  plots[[i]] <- p
}

```

```
##
##
## ----- Generating Mosaic Plot for Mouthing and Child_Age_Gp -----
##
##
## ----- Generating Mosaic Plot for Pacifier and Child_Age_Gp -----
##
##
## ----- Generating Mosaic Plot for Pacifier_Use and Child_Age_Gp -----
##
##
## ----- Generating Mosaic Plot for Pacifier_Wash and Child_Age_Gp -----
##
##
## ----- Generating Mosaic Plot for Blanket and Child_Age_Gp -----
##
##
## ----- Generating Mosaic Plot for Sucking and Child_Age_Gp -----
##
##
## ----- Generating Mosaic Plot for Play_Park and Child_Race4 -----
##
##
## ----- Generating Mosaic Plot for Play_Daycare and P1_Degree -----
##
##
## ----- Generating Mosaic Plot for Play_Daycare and P1_Workstatus -----
##
##
## ----- Generating Mosaic Plot for Play_Daycare and Household_income -----
##
##
## ----- Generating Mosaic Plot for Play_Other and City -----
##
##
## ----- Generating Mosaic Plot for Sandbox_Home and City -----
```

```

# Remove NULL plots
plots <- Filter(Negate(is.null), plots)

# Split into two groups: first 6 plots and last 6 plots
plots1 <- plots[1:min(6, length(plots))]
plots2 <- plots[7:min(12, length(plots))]

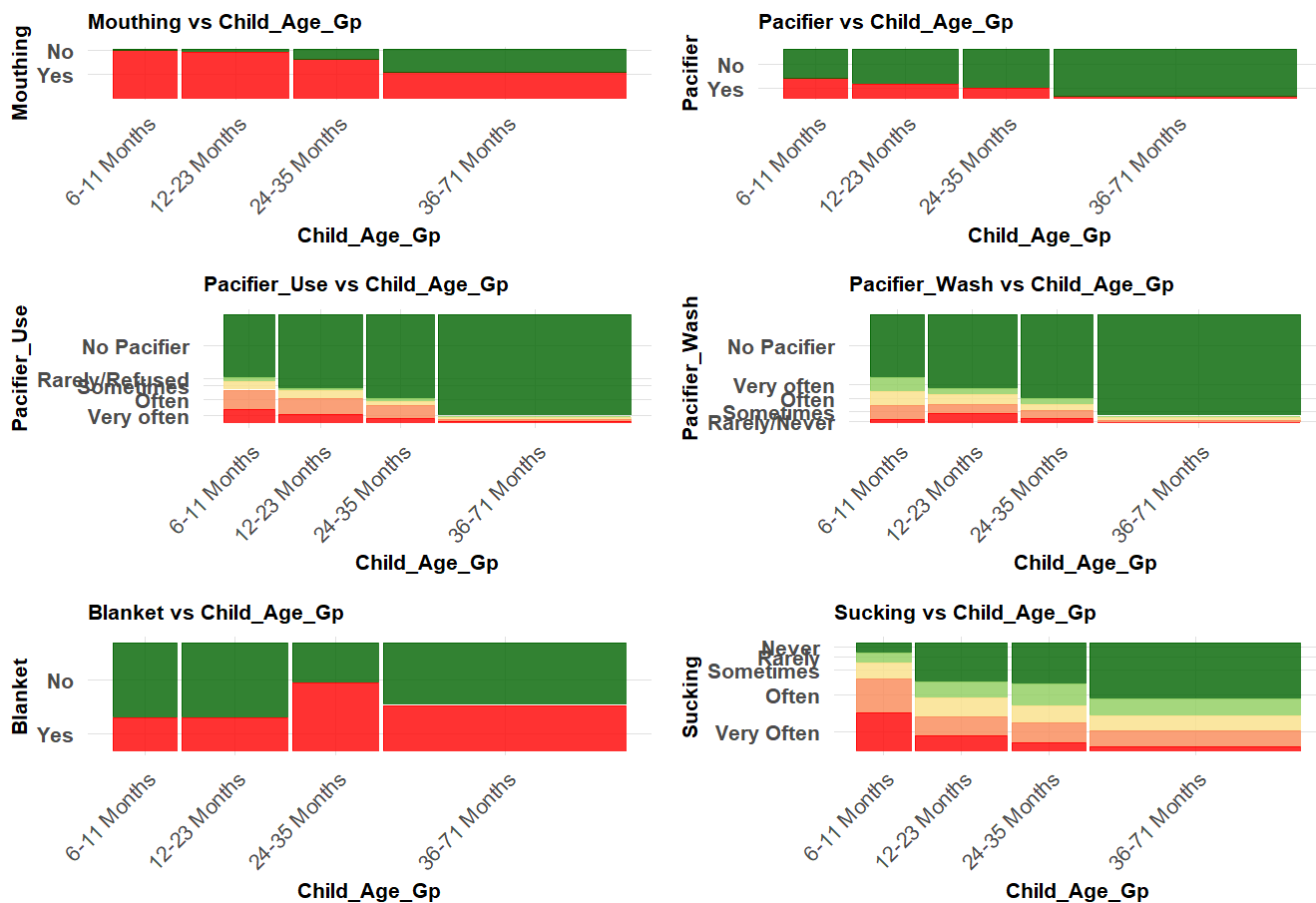
# Titles
main_title1 <- textGrob("Mosaic Plots of the Selected Features Part 1", gp = gpar(fontsize = 10, fontface = "bold"))
main_title2 <- textGrob("Mosaic Plots of the Selected Features Part 2", gp = gpar(fontsize = 10, fontface = "bold"))

# **Modify grid.arrange to increase bottom row height**
if (length(plots1) > 0) {
  plot1 <- grid.arrange(
    grobs = plots1,
    nrow = 3, ncol = 2,
    top = main_title1,
    heights = c(1.2, 1.5, 1.5) # Increase bottom row height
  )

  # Save output with better readability
  ggsave("Mosaic_Grid_Part1_FINAL_NoNA.jpeg",
    plot = plot1,
    width = 8.2, height = 11.6, units = "in", dpi = 300)
}

```

Mosaic Plots of the Selected Features Part 1



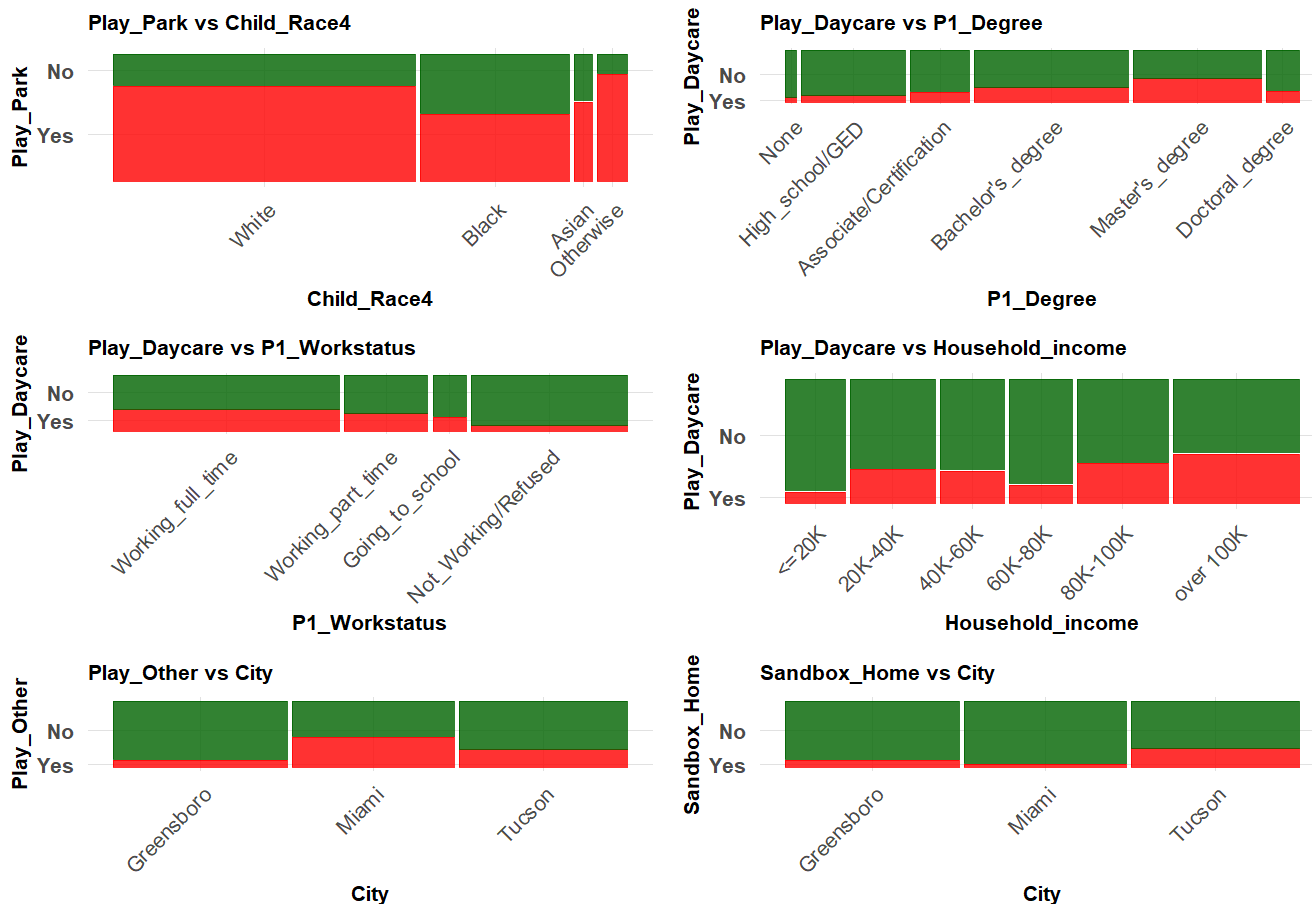
```

if (length(plots2) > 0) {
  plot2 <- grid.arrange(
    grobs = plots2,
    nrow = 3, ncol = 2,
    top = main_title2,
    heights = c(1.5, 1.5,1.25) # Increase bottom row height
  )

  ggsave("Mosaic_Grid_Part2_FINAL_NoNA.jpeg",
    plot = plot2,
    width = 8.2, height = 11.6, units = "in", dpi = 300)
}

```

Mosaic Plots of the Selected Features Part 2



```

message("Saved Mosaic Plots with improved readability and better layout!")

```

```

## Saved Mosaic Plots with improved readability and better layout!

```

FIXED stuffey panel

```
file_path <- "Table_Summary_ALL_Formatted.csv" # Adjust if needed
df <- read.csv(file_path, stringsAsFactors = TRUE)

# Extract Cramér's V and p-value (if available in dataset)
cramers_v <- df %>%
  filter(Exposure_Variable == exposure_var & Demographics_Variable == demo_var) %>%
  select(Cramers_V, P_Value) %>%
  slice(1)

# Assign values correctly
cramers_v_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$Cramers_V, 4), "N/A")
p_value_val <- ifelse(nrow(cramers_v) > 0, round(cramers_v$P_Value, 4), "N/A")

# Define the exposure variable and its top 4 demographic variables
exposure_var <- "Stuffedanimal"
demographic_vars <- c("Child_Age_Gp", "P1_Sex", "P1_Workstatus", "City")

custom_colors <- c("red", # Bright Red (High Risk)
                  "#006400") # Strong Green (No Risk)

# Initialize an empty list to store plots
stuffey_plots_plots <- list()

for (demo_var in demographic_vars) {
  # Drop NA values before plotting
  filtered_data <- data %>% filter(!is.na(!sym(exposure_var)) & !is.na(!sym(demo_var)))

  # Skip if there's not enough data
  if (nrow(filtered_data) == 0) {
    message("Skipping ", exposure_var, " vs ", demo_var, " (All rows have NA values).")
    next
  }

  # Extract Cramér's V and p-value (if available in dataset)
  crammers_v <- df %>%
    filter(Exposure_Variable == exposure_var & Demographics_Variable == demo_var) %>%
    select(Cramers_V, P_Value) %>%
    slice(1) # Take the first value

  # Handle missing values gracefully
  crammers_v_val <- ifelse(nrow(crammers_v) >= 0, round(crammers_v$Cramers_V, 4), "N/A")
  p_value_val <- ifelse(nrow(crammers_v) >= 0, round(crammers_v$P_Value, 4), "N/A")

  # Assign colors based on unique categories
  unique_categories <- length(unique(filtered_data[[exposure_var]]))
  palette_colors <- custom_colors[1:min(unique_categories, length(custom_colors))]

  # Generate the mosaic plot
  p <- ggplot(filtered_data) +
    geom_mosaic(aes(weight = 1, x = product(!sym(demo_var)), fill = !sym(exposure_var))) +
```

```

scale_fill_manual(values = palette_colors) + # Use custom colors
labs(
  title = paste(exposure_var, "vs", demo_var), # Individual plot title
  subtitle = paste("Cramér's V:", cramers_v_val, "| P-Value:", p_value_val),
  x = demo_var, y = exposure_var # Keep axis labels
) +
theme_minimal() +
theme(
  plot.title = element_text(size = 10, face = "bold"),
  plot.subtitle = element_text(size = 9, face = "italic", color = "gray40"),
  axis.title.x = element_text(size = 10, face = "bold"),
  axis.title.y = element_text(size = 10, face = "bold"),
  axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
  axis.text.y = element_text(size = 8, face = "bold"),
  legend.position = "none", # Remove Legend
  panel.grid.major = element_line(color = "gray90", size = 0.3),
  panel.grid.minor = element_line(color = "gray95", size = 0.2) ,

# Titles
# main_title <- textGrob("Mosaic Plots of the Strongest Associations with Stuffed Animal", gp = gpar(fonts
ize = 10, fontface = "bold"))
)
# Store the plot
stuffey_plots[[demo_var]] <- p
}

# Arrange the 4 plots in a 2x2 grid
stuffey_grid <- grid.arrange(grobs = stuffey_plots, ncol = 2, nrow = 2)

```

```
## Error in arrangeGrob(...): nrow * ncol >= n is not TRUE
```

```

# Save the combined plot
ggsave("Mosaic_stuffey_grid_2x2.jpeg", plot = stuffey_grid,
  width = 10, height = 10, dpi = 300)

message("Saved Mosaic_stuffey_grid_2x2_fxd.jpeg")

```

```
## Saved Mosaic_stuffey_grid_2x2_fxd.jpeg
```